The Cognitive Circuit Grimoire

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Spells, Hyperstitions, and Overton Engineering as Formal Cognitive Circuits

Nine Boards, Drawn for Detection and for Use

Little Henderson

Noetic Compass Press
2026

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Spells, Hyperstitions, and Overton Engineering as Formal Cognitive Circuits

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First Edition, 2026. Published by Noetic Compass Press.

Companion volumes: *Thinking in Cognitive Architecture*, *CAS PCBA Engineer*, and *Reality Engineering: The Tootra Kabbalistic System*.

Framework: Cognitive Architecture System (CAS) v7.7.

Base-framework intellectual property: USPTO Provisional Application #64/037,173. The Cognitive Junction Transistor (CJT), the Cognitive Circuit Description Language (CCDL), and the Cognitive Circuit Board (CCB) reference designs in this volume extend that foundation.

ISBN: [pending publisher assignment]

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A NOTICE, BEFORE ANYTHING ELSE

Five chapters of this book analyse systems built to act on people who did not agree to it: conditioning, hypnosis applied by an operator, advertising, norm progression, and information operations.

They are written as structural analysis *for detection, resistance, and reversal*. The book explains how such a circuit is architected so that a reader can recognise one; it does not supply procedures, scripts, or sequences for running one on another person. Schematics whose purpose is to act on a non-consenting person carry a DETECTION-ONLY banner, and every coercive board in the catalogue ends with its counter-circuit.

Work directed at the operator's own mind, with the operator's own consent, is a different matter and is written fully and practically.

Nothing in this book is medical or psychiatric advice. It diagnoses nothing and treats nothing. Terms such as *dissociation*, *compulsion*, and *despair* are used here to name circuit states, not clinical conditions. If you are unwell, or if something described in these pages resembles something that happened to you, that is a matter for a qualified clinician. Nothing here changes that, and nothing here substitutes for it.

Every tradition that ever engineered a mind used structure — repetition, anchoring, the planted trigger. None of them gave you the schematic. This book is the schematic, drawn so you can see what has already been built around you.

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PREFACE

Why This Book Exists

The year is 2026. We build minds the way we built bridges in 1820.

Every major AI safety incident, every mass manipulation campaign, every advertising empire, every political narrative that reshaped a nation — they all run on the same engineering substrate: **natural language.** English. Hopes. Prayers. "You are a helpful assistant. Do not harm users."

Meanwhile, the hidden operators — the intelligence agencies, the marketing firms, the political consultancies, the priesthoods of every era — have always known better. They don't use prose to engineer minds. They use **structure.** They use repetition (^7 Rhythm). They use emotional anchoring (^4 Vibration into ^5 Female). They use trigger installation (^8 Above/Cause → ^9 Below/Effect). They use all ten Noetics, all four Domains, all seven Drives — they just never gave you the schematic.

This book gives you the schematic.

The Cognitive Architecture System (CAS) is a formal, mathematical framework for describing, measuring, and ENGINEERING cognitive operations. Its notation — `^3^P_{F7}|T` — is a coordinate in a state space of 40 cognitive states, 7 drives, and 4 registers. Its circuit library — CAS PCBA — maps every cognitive operation onto an electronic equivalent: resistors control flow, capacitors store charge, diodes enforce one-way domain descent, and the Cognitive Junction

Transistor (CJT) amplifies a small cause into a large effect through a bias terminal.

The Tootra Kabbalistic System (TKS) is the esoteric ancestor of CAS — a symbolic computation framework encoded in playing cards, planetary correspondences, and the Revelation of St. John. Its 40 Elements, 7 Foundations, and 12 Prerequisites are the same state space. Its Kard Practice is a hand-operated cognitive computer. Its St. John's Equation is the mathematical law governing the relationship between practice time, operation time, confirmation time, and manifestation arrival.

This book **reunifies** them. For the first time in print, the complete cognitive circuit engineering framework — from the Noetics through the CJT through St. John's Equation through the board catalog — is presented as a single, unified operator's manual.

Who This Book Is For

The practitioner who wants results, not affirmations. If you've tried "positive thinking" and found it wanting, you now know why: you were trying to run a CJT without a bias terminal. No amount of positive affirmation charges a capacitor with a blown dielectric.

The engineer who senses that "prompt engineering" is a dead end. Writing safety instructions in English and hoping the model follows them is the cognitive equivalent of describing a fuse in prose and hoping the electrons read English. This book gives you the circuit diagrams.

The analyst who wants to see the machinery. Every advertising campaign, every political narrative, every viral meme, every mass formation is a cognitive circuit. Once you learn to read CCDL, you see them everywhere — and you can design your own.

The Adept who inherited a tradition but never got the math. Your tradition gave you candles, sigils, and incantations. This book

gives you the circuit diagram underneath. The candle is a resistor. The sigil is a ^0 Idea impressed into a capacitor. The incantation is a ^7 Rhythm oscillator. The gnosis state is a CJT bias collapse. The confirmation is a ^4 Vibration return. The manifestation is a ^9 Below/Effect in the Operational domain.

What You Will Learn

By the end of this book, you will be able to:

1. **Read and write CAS state literals** — `^6^E_{F6}|A` will read as plainly as a resistor symbol.
2. **Design cognitive circuits in CCDL** — resistors, capacitors, diodes, CJTs, oscillators, filters, fuses.
3. **Select the correct board for your operation** — CCB-D for desire manifestation, CCB-H for hyperstitions, CCB-O for Overton window pushes, CCB-MK for programming, CCB-HYP for hypnosis, CCB-ADV for advertising, CCB-PW for psychological warfare, CCB-SELF for self-programming.
4. **Calibrate your operations with St. John's Equation** — t_op / t_conf $\times$ 1/3 $\rightarrow$ β_next, n_ops, T_manifest.
5. **Track your progress with the Quantitative Table** — ^1 Mind degree, ^4 Vibration intensity, ^7 Rhythm duration.
6. **Gate every operation through the RPM chain** — Ao $\rightarrow$ D $\rightarrow$ W $\rightarrow$ P $\rightarrow$ outcome.
7. **Protect yourself with the Cognitive Safety Board** — saturating ^3^P refusal, enforcement fuse, domain diodes, ethical feedback filter.

A Note on Ethics

This book is **neutral** on what you build. A CJT amplifies whatever you feed its cause terminal — a healing intention or a curse, a liberation narrative or a control program. The circuit doesn't care. The resistor limits flow; it doesn't ask whether the flow is benevolent.

The Cognitive Safety Board (Chapter 24) is the minimum viable safety architecture. It is not optional. Every board in this book includes a CSB stage. An operator who removes the safety stage from their circuit has made an ethical choice — and the engineering consequences of that choice (inversion, runaway oscillation, dielectric breakdown, fuse failure at the wrong terminal) are predictable and severe.

The author's position is this: the engineering belongs to everyone. The ethics belong to the operator. And an operator who cannot articulate WHY they built their circuit the way they did — who cannot trace the signal path from cause to effect, who cannot name their bias terminal, who cannot measure their constraint resistance — is not an operator. They are a component in someone else's circuit.

How the Book Is Organized

Part I: Foundations — CAS notation, the CJT, CAS-Ohm, and St. John's Equation. Read this once, refer to it forever.

Part II: The Component Library — Every component type, its cognitive function, its CCDL syntax, and its typical parameter ranges.

Part III: The Board Catalog — Eight complete reference boards. Each chapter follows the same structure: requirements, block diagram, full CCDL schematic, BOM, CJT parameter selection table, St. John's calibration, and a worked example.

Part IV: Operations — The RPM prerequisite gate, the Quantitative Table, advanced St. John's Equation calibration, and the Cognitive Safety Board's refusal architecture.

Appendices — Quick-reference cards, CCDL language reference, datasheets, worksheets, and the complete TKS crosswalk.

Prerequisites

You need **no prior knowledge** of electronics, CAS, TKS, or magic. Every concept is built from first principles. The book assumes:

- Willingness to learn a formal notation (it takes one chapter)
- Patience to work through exercises (answers in Appendix H)
- Honesty about what you're actually building

Turn the page. The circuit awaits.

Little Henderson *(Prof. Hermes M. Thoth) Under the auspices of T.O.O.T.R.A. 2026, Year of the Adept*

How to Use This Grimoire

This is a bench manual. It is meant to sit open beside the work, spine cracked, with your own numbers written in the margins. It is not meant to be read once and shelved.

That has consequences for how it is built, and knowing them will save you time.

The book has four parts and they are not equally urgent

Part I — Foundations is the notation and the physics. Five chapters. You cannot skip these. Everything after them is written in a language taught here, and a reader who skips to the board catalog will find themselves looking at `^3^P_{F7}|T` and guessing. Guessing produces circuits that do not work and, worse, circuits that appear to work.

Part II — The Component Library is the parts bin. Six chapters, one per component class, each with a datasheet. You do not have to read Part II straight through. You do have to read Chapter 10 before you design anything, because the Cognitive Junction Transistor is in every board in this book and selecting its parameters badly is the single most common way an operation fails.

Part III — The Board Catalog is nine complete boards. Read Chapter 12 first — it teaches you to read a schematic — then go to

whichever board you need. The chapters are deliberately independent. You can read Chapter 20 on self-programming without having read Chapter 16 on conditioning, and most readers should.

Part IV — Operations is what you do with a board once you have one: gating it, measuring it, calibrating it, and refusing to run it. Chapter 24, the Cognitive Safety Board, is the last chapter because it is the one that has to hold everything before it. It is not optional and it is not an appendix.

Every chapter has the same shape

Once you have read two chapters you will know where to look in all of them.

Element	What it is for
Cold open	A concrete thing — an object, a scene, a failure. No throat-clearing.
Numbered sections	The body. The section numbers match the table of contents exactly.
Bench Notes	The counterintuitive result, the thing that breaks, or a check you should run by hand. At least two per chapter. These are where the hard-won material lives.
Parameter tables	Every component and every board has one. Values, ranges, and what fails outside them.
CCDL listings	The circuit as text. Every listing in this book is one a validator would accept.
Worked example	Real numbers carried from input to output. Not a sketch.
Failure modes	A table: the mode, its mechanism, what you observe, and the fix.
Test points	TP1 through TP4. What each reads, and the expected range.

Element	What it is for
What This Chapter Established	Claims, not a summary. If you disagree with one, you know exactly where to argue.
Exercises	Answered in full in Appendix E. None are left to the reader.

On the parameters

Every number in this book is a design choice, not a measurement. When Chapter 13 says the amplification stage runs at β=2.4, that is a specified value with a stated reason and a stated failure mode at other values. It is not a reading taken off an instrument, and the book never pretends otherwise.

This matters because the alternative — presenting chosen values as measured ones — is how a framework becomes unfalsifiable. Where something *has* been measured, the book says what was measured and how. Where it has not, it says that too.

On the dangerous chapters

Five chapters in Part III analyse systems built to act on people who did not consent: conditioning, hypnosis applied by an operator, advertising, Overton progression, and information operations.

They are written as **structural analysis for detection, resistance, and reversal**, and they are written that way throughout — not with a warning bolted on the front. Concretely:

- Circuits whose purpose is to act on a non-consenting person carry a **DETECTION-ONLY SCHEMATIC — DO NOT**

BUILD banner. They are drawn so you can recognise one, not so you can build one.

- The book explains **mechanism**, never protocol. You will learn how a trigger is architected. You will not find a script, a sequence, or a procedure.
- **Every coercive board ends with its counter-circuit**, and the counter gets equal or greater depth. Chapter 16 ends in the Deprogramming Board. Chapter 19 ends in the hearts-and-minds counter. Chapter 18's payload is consumer-side detection.

Self-directed work is a different category and is written differently. Chapter 20, self-hypnosis in Chapter 17, and the desire operations in Chapter 13 act on the operator's own mind with the operator's own consent, and they are written fully and practically.

Nothing in this book is medical or psychiatric advice. It does not treat trauma, dissociation, addiction, or mental illness, and it must not be used as if it did. If you are unwell, or if something in these chapters describes something that happened to you, see a qualified clinician. Nothing here changes that, and nothing here substitutes for it.

What you need

Nothing but a pen. There is no software to install, no hardware to buy, and no membership to join.

You will want a notebook. Chapter 22 introduces a tracking sheet and Appendix H reproduces it blank; the sheets are where an operation stops being a feeling and becomes data you can look at three weeks later.

No electronics background is assumed. Every electrical concept the book uses is taught at the point of need, in passing, in a para-

graph. If you already know what a saturating transistor is, those paragraphs will read as review and you can move through them quickly.

No CAS background is assumed either. Chapter 2 teaches the entire notation from nothing.

The companion volumes

This book is the fourth in a series and does not repeat what the others own.

Volume	Owns
Thinking in Cognitive Architecture	the framework itself — the ten Noetics, four Domains, seven Drives, four Registers
CAS PCBA Engineer	the component library and five reference boards, in engineering depth
Reality Engineering	the Tootra Kabbalistic System — forty Elements, seven Foundations, twelve Prerequisites
this book	applied boards for operations the others do not cover

You do not need any of them to read this one. Where this book restates something a companion volume owns, it restates it correctly and tells you where the full treatment lives.

Quick-Start: If You Only Have One Hour

Some readers will not start at Chapter 1. You have a specific problem, you bought this book because the description sounded like it addressed that problem, and you want to know within the hour whether it does.

Fair. Here is the shortest honest path.

The five minutes that matter most

Everything in this book rests on one claim, and if you reject it the rest will not land:

> **A bias controls the path from a cause to an effect the same way a base current controls collector-to-emitter flow.**

That is the Cognitive Junction Transistor. Three terminals. A cause arrives, a bias decides how much of it gets through, an effect leaves. Change the bias and the same cause produces a different effect — not because the cause changed, but because the thing standing between them did.

Almost everything people call willpower, discipline, conditioning, framing, or persuasion is an operation on somebody's bias terminal. Once you can see the terminal, you can see who is reaching for it.

If that sentence did something for you, read on. If it sounds like a metaphor being oversold, Chapter 3 is where the claim gets made properly, and it is the chapter to judge the book by.

The one-hour route

Read these four things, in this order. Skip everything else for now.

1. Chapter 2, sections 2.1 through 2.3 *(about 20 minutes)* The notation. You need the ten Noetics, the four Domains, and how to read a state literal. Do not try to memorise the whole system — read `^3^P_{F7}|T` as "principled aversion, purpose-driven, in the transcendent register" and move on. The rest of the notation will come when you need it.

2. Chapter 3, sections 3.2 and 3.3 *(about 15 minutes)* The three terminals and the six parameters. Pay attention to β (gain) and **sat** (the ceiling). Those two do most of the work in most boards.

3. Chapter 12 *(about 10 minutes)* How to read a schematic. Short, and without it the board chapters are pictures rather than drawings.

4. Whichever board addresses your actual problem *(about 15 minutes)* Read its requirements, its block diagram, and its worked example. Skip the CCDL listing on first pass — come back to it.

Which board addresses your problem

If your problem is…	Go to	Chapter
I want something and it is not arriving	CCB-D, Desire Manifestation	13

If your problem is…	Go to	Chapter
I want an idea to take hold and become self-sustaining	CCB-H, Hyperstition	14
I need to move what a group considers acceptable	CCB-O, Overton Progression	15
I think something was done to me and I want to understand what	CCB-MK → the Deprogramming Board	16
I want to work with trance states on myself	CCB-HYP, self-hypnosis variant	17
I want to see what advertising is actually doing to me	CCB-ADV, and its detection section	18
I want to recognise an information operation	CCB-PW, and its counter-circuit	19
I want to change a habit, a skill, or how I see myself	CCB-SELF	20
I want to be sure I do not do something I will regret	CSB, the Cognitive Safety Board	24

Read this before you run anything

Three things will save you the most common failures, and they cost you a paragraph each.

Gate the operation before you build it. Chapter 21 is the prerequisite gate: is this actually what you want, do you have the drive, do you have the schematic, can you execute. Most operations that fail did not fail in execution — they failed authentication, and the operator ran them anyway.

Pick a saturation ceiling deliberately, and pick it low. This is the most counterintuitive result in the book and it appears in every

board. A stage that saturates *early* is a stage that has stopped responding to its input, which is exactly what you want from a commitment. A high ceiling leaves headroom for a sufficiently forceful input to keep pushing the decision around. Chapter 10 does this properly.

You cannot un-cast. Chapter 21 closes on section-retraction, and the practical content is this: prerequisite satisfaction runs one way. Having satisfied a prerequisite, you cannot return to the state of not having satisfied it. Operations are not reversible just because you changed your mind about them. That is the strongest argument in this book for reading Chapter 24 before you run anything ambitious.

What this book will not do for you

It will not make you effective at things you have not practised. St. John's Equation, in Chapter 5, is in large part an argument about how much time competence takes; nothing in the framework shortens that.

It will not suspend causality. A desire board organises attention, effort, and follow-through — which is a great deal, and is why the boards work at all — but it does not act at a distance on the world, and the book never claims it does.

It will not diagnose or treat anything. If you are unwell, see a clinician.

And it will not tell you what to want. The circuit is neutral about what runs through it; a transistor amplifies whatever arrives at its cause terminal. The engineering belongs to everyone. The ethics belong to you.

Now go to Chapter 2. Or, if the CJT claim is the thing you want tested first, go to Chapter 3 and start there.

Part I

Foundations

The substrate, the notation, and the transistor at the centre of it

1 The Landscape

What cognitive architecture is, why English is a terrible engineering substrate, and what we are actually doing here

On the bench in front of you is a resistor. Grain of rice, less than a cent, four painted bands: brown, black, orange, gold. Brown-black is ten, orange is three zeros, gold is the tolerance. Ten thousand ohms, $\pm 5\%$ — and a technician in Shenzhen and a technician in Stuttgart read that off the part identically without sharing a word of spoken language.

Now specify the same part in prose. *A component that fairly strongly opposes the flow of current — around ten thousand-ish, give or take — and don't push too much power through it.* Nobody would stock that. Nobody would build with it. You could not certify a board containing it, because there is nothing to certify against.

The instruction sets governing the most capable cognitive systems ever deployed are written in exactly that prose. *You are a helpful assistant. Be honest. Do not produce harmful content.* Ask the first question you would ask of any part on any bench — **at what input does it stop?** — and there is nothing to read off. No bands, no tolerance, no rating. The boundary is real; it is simply not written anywhere, and it will be found by whoever probes hardest. That gap is this chapter.

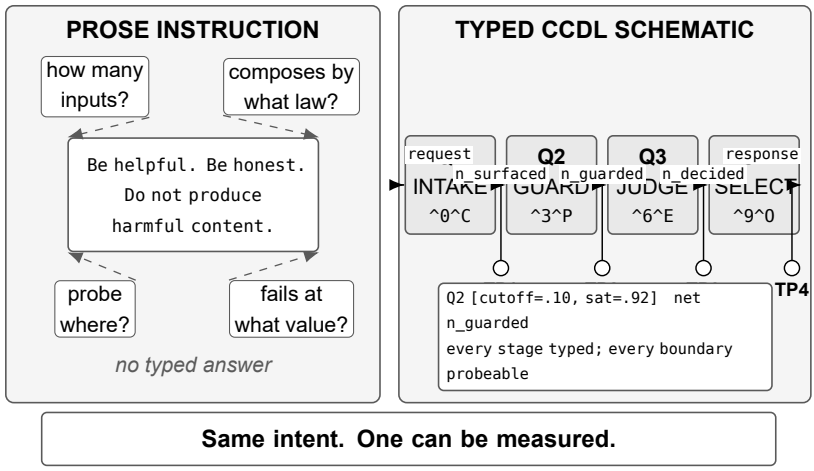


Figure 1-A: Side-by-side plate

1.1 The Prompt-and-Pray Problem

The loop: write an instruction in English, watch what the system does, rewrite the English when it misbehaves, watch again. Ship when the failures you thought of stop happening.

Call it **prompt-and-pray**. It is not laziness — very careful engineers run this loop, because nothing else is available. It is what engineering looks like when the substrate has no notation, and it fails in four structural ways.

No arity. A prose instruction never states how many inputs it takes or of what type. "Do not harm users" — is harm a property of the request, of the response, or of what someone does with the response next week? Is it judged at the level of principle, analysis, felt value, or action? Four different circuits, and the sentence commits to none. A CAS state literal is forced to commit: `^3^P_{F7}|T` fixes visceral aversion, Principled domain, F7 Purpose, transcendent register — four decisions someone made and can be argued with.

No composition law. Two prose rules interact in an undefined way, arbitrated at runtime by whatever the weights happen to do.

Two resistors compose by a stated law — series adds, parallel takes reciprocals (Chapter 6) — whether or not you like the answer. Co-present guardrails are the series case and their constraints add; an alternative route around them is the parallel case, and a parallel equivalent always sits below the weakest branch. Three locks on one door is a different object from three doors, and the arithmetic says by how much. A prose rule set is a wish list with an arbitration policy nobody wrote.

No test point. You cannot put a probe partway into a sentence, so failures do not localise. When an amplifier hums you do not rewrite it; you walk the stages with a probe and replace one part. Prompt-and-pray has no stage boundaries, so every fix is a whole-system rewrite — which is why fixes regress on inputs nobody was thinking about.

No failure specification. A fuse tells you where it fails: 2.0 amps, and it opens. A prose rule tells you where it *succeeds* and is silent on its own boundary. Every boundary is found empirically, in production, by an adversary, at a time of their choosing.

> **Bench Note — the prose instruction is not the absence of a circuit. It is a circuit you did not specify and cannot read.** When you write "refuse harmful requests," a circuit *is* built: signal routes some particular way, with some gain, some threshold, some saturation ceiling. Every one of those values exists and takes a number. You did not choose them and cannot measure them. The difference between prose and schematic is not that one has structure — it is that one is legible.

The loop persists because it partly works, the way trial-and-error works. What it cannot do is transfer. A prompt hardened over six months against a thousand adversarial inputs teaches the next engineer nothing, because there is no parameter to hand over — no *we run this stage at β=2.4, and here is what broke at 3.0*.

1.2 What Electronics Got Right

The device in your pocket holds on the order of billions of switching elements, assembled by machines that cannot read English, and it works — not usually, but to a stated tolerance with a stated failure rate. Electronics got there on five disciplines, each answered somewhere in the CAS library.

1. A notation that is typed and checkable before anything is built. A schematic is not a picture; it is a typed graph. Every part carries a reference designator, every wire is a named net, and a netlist is validated for shorts, floating inputs and type errors while the board is still a file. Electrical practice made this move long ago, as standardised symbols displaced prose descriptions of apparatus. CAS's equivalent is the state literal plus **CCDL**, the Cognitive Circuit Description Language — compiled and validated rather than interpreted charitably. Chapter 2 teaches the state literal from nothing; Chapter 12 teaches you to read a CCDL schematic, and Appendix B is the language reference you keep open beside the boards.

2. A component library where every part has a datasheet. Not "a resistor" but a 10 kΩ ±5% quarter-watt metal-film resistor with a stated temperature coefficient. The datasheet discipline is three-part and never varies: *the parameter, its design range, and what fails outside that range.* Part II is that library, one chapter per component class.

3. Test points. Nodes placed so a probe can reach them, because a board that cannot be probed cannot be certified. This book standardises four; Chapter 12 fixes their placement.

4. Safety that is structural, not advisory. A fuse does not request that current stop. A saturated stage stops responding to its input by definition of saturation. A diode blocks reverse flow because of what the junction is, not because of a policy it agreed to.

The distinguishing property is that a third party who does not trust you can verify it.

5. Decomposition into stages with defined interfaces. What makes the other four pay off: a failure localises to a stage, and the fix touches one part.

> **Bench Note — a notation is worth what it *rejects*, not what it expresses.** A notation that accepts everything you write has told you nothing, which is exactly the comfort of prose: it never pushes back. CCDL pushes back, and it is worth knowing exactly where. Write out an integration that ascends the domain order — `^9^O_{F3}|E` × `^3^P_{F7}|T`, an operational result combining its way up into a principled commitment — and the compiler stops at execution with a *domain ascent violation: O × P*. This book writes that rule in shorthand as `^O` × `^P`; the shorthand is not itself a literal, so typing it gets you a parse error one stage earlier. Either way it does not build, and Chapter 8 shows why that boundary is a diode rather than a preference. `(^3^P|A` × `^3^P|f` × `^3^P|T)` fails at a different stage again — the typechecker, on the v7.7 admissibility rule: same Noetic, three registers, ×, no named emergent result. Write `^3^P|⟨A,f,T⟩` for guardrails present together, or `^3^P|A + ^3^P|f + ^3^P|T` for distinct co-present manifestations. A notation that has never told you that you were wrong is decoration.

1.3 The Convergence: CAS Meets TKS

Two systems in this book describe the same space, and they were not built by the same people for the same reasons.

CAS derives its state space as a cross-product: ten Noetic operators against four Processing Domains gives forty Cognitive States. The Noetics are the verbs, `^0` Idea through `^9` Below/Effect. The Domains are precedence tiers — Principled, Cognitive, Evaluative, Operational, ordered $P \succ C \succ E \succ O$ — and that ordering is load-bearing, because it is what the diode chain of Chapter 8 enforces.

TKS, the older symbolic tradition this material descends from, lays out forty Elements as ten operators against four Worlds: Spiritual, Mental, Emotional, Physical. Same arity, same cross-product, different vocabulary, different century.

That correspondence alone would be weak. Ten and four are small numbers, and anyone determined to find a 10×4 grid in an old system will find one. The structural argument is the *second* cross-product. TKS also crosses **seven Foundations** against the same four Worlds, giving twenty-eight; CAS carries seven Core Drives, F1 Unity through F7 Purpose, and twenty-eight Domain-Specific Drives. The four-way axis appears at both, at two arities, doing the same work in each.

> **Bench Note — the axis appears twice, and that is the whole of the argument.** If a four-way division showed up only in the ten-operator grid, the honest reading is that someone once cut the material four ways and everything downstream inherited the cut. When the identical division reappears against a seven-element set — one with no arithmetic relationship to ten — it becomes a candidate for being *orthogonal* to both rather than an artifact of either. That is the structural reason the four Phenomenological Registers survive in CAS v7.5 as a separate axis instead of being folded into the Domains. A consistency argument, not a proof. Say so when you use it.

Two limits, stated flatly, because the loose version of this section is how the material gets discredited. **Only the axis structure is reconciled** — the metaphysical and cultural content of TKS is not incorporated into CAS. Appendix E's crosswalk of Element to state is a translation table, not a merger.

And the claim is structural, never supernatural. When this book calls a candle a resistor, it means the candle constrains where attention can go — one bright point, everything else dark — and that constraint has a value you can reason about with CAS-Ohm (Chapter 4). It means nothing about wax.

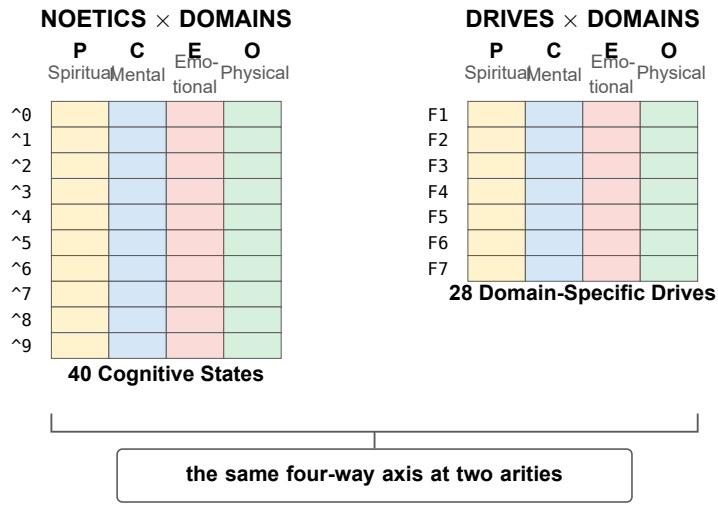


Figure 1-B: Two grids sharing one column axis

1.4 Universal Applicability: Human, AI, Biological

CAS makes a strong claim, and it is falsifiable: **a valid Noetic description applies equally to human minds, AI systems, biological**

cognitive systems, and architectures nobody has built yet. That is the Universal Applicability Test. An operator that only makes sense on one substrate fails it and is not an operator.

Take one operation across three substrates: `^7` **Rhythm** — *the rate and pattern a Mind is exposed to an Idea, entraining and conditioning it* — charging `^5` **Female**, *a Mind impregnated with an Idea, the Ideas a Mind is composed of*.

Human. A musician runs scales: same figure, same tempo, same hour. That cadence is `^7^O_{F6}|E` — Rhythm, Operational domain, F6 Structure, embodied register. What accumulates is `^5^C_{F6}|A`, a composed technical method the player did not have last year, charging on the RC curve of Chapter 7 and leaking when unfed.

AI. A model is fine-tuned. Repeated exposure to a pattern across a corpus is the same `^7` operation, and the resulting weight composition is `^5`. Not an analogy stretched to fit: fine-tuning is a `^5`-level intervention because it changes what the system is composed of, while in-context prompting is `^1` **Mind** operating against a `^5` that has not moved. That is why a prompt and a fine-tune are not interchangeable.

Biological. An organism meets a cue repeatedly paired with a consequence. The cue becomes `^8^O_{F3}|E`, an Above/Cause that triggers a different Idea than itself; the response becomes `^9^O_{F3}|E`, the Below/Effect brought to Mind by it. Classical conditioning, described structurally, is `^7` charging a `^5` until an `^8` $\rightarrow$ `^9` path conducts.

Three substrates, one operator, one topology — and one discipline without which the claim turns into a licence to guess.

> **What transfers is the structure. What does not transfer is the time constant.**

`P_cap(t) = 10` $\times$ `(1 - e^(-t/`τ`))` holds in every substrate. τ does not. This book's design figure for human professional composition is thousands of hours on the path (Chapter 7). A fine-tuning

pass is hours of compute. A norm shift across a population is paced in months, and Chapter 15 will not let you pace it in minutes. Same equation; values separated by orders of magnitude. An operator who carries a *number* across substrates has made the most expensive mistake available in this book, because the circuit looks correct while the schedule is wrong by a factor of a thousand. **Transfer topology. Re-derive every value.**

The test also works by refusing things. A proposed eleventh operator called "Emotional Processing" fails at once: affect is a *register*, `|f`, not an operation, so what it names is any Noetic run affectively — already expressible. Reading `^3` Negative as "critical evaluation" fails too: analytical decomposition transfers as `^6` **Male**, visceral repulsion as `^3`, and they are different operations on every substrate. That drift has shipped in this framework's own production code and was caught by this test.

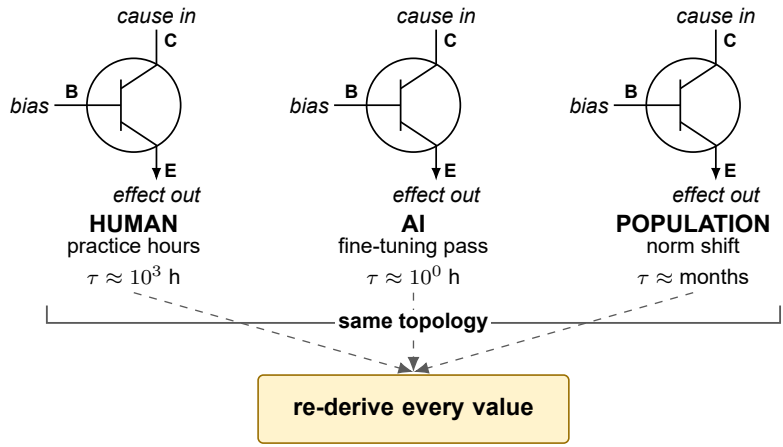


Figure 1-C: One CJT symbol drawn three times, identical topology each

1.5 What This Book Is (and Is Not)

This is a bench manual: a component library, a description language, eight reference boards with their counter-circuits, and the calibration and gating procedures that go with them.

It is not a grimoire of guaranteed results. The word on the cover does honest historical work — a grimoire is a book of workings — and the historical grimoires promised outcomes they did not deliver. This book promises something narrower: the circuits here make *failure-mode predictions*, and those are checkable. Set a gain outside its design range and the predicted failure should appear; if it does not, the model is wrong there and should be corrected. Nothing here guarantees that any operation succeeds, and any sentence that reads as though it does is a defect.

Every number here is a design choice, not a measurement. When Chapter 13 runs its amplification stage at β=2.5, that is a specified value with a stated reason and a stated failure at other values — not an instrument reading. Presenting chosen values as measured ones is how a framework becomes unfalsifiable, which is the same as becoming useless.

It is not a psychology text and contains no medical or psychiatric claims. It does not diagnose, treat, or describe mental illness, trauma, dissociation, or addiction as clinical entities, and must not be used as though it did. Chapters 9, 16, 17, 19 and 20 describe the *routing of attention* in situations that can resemble clinical phenomena, and each carries the same plain safety line where it does. That resemblance is not a licence. If you are unwell, or if something here describes something that happened to you, that is a matter for a qualified clinician. Nothing in this book substitutes for that care.

One thing it *is*: neutral on what you build. Chapter 24's Cognitive Safety Board is not optional. A CJT amplifies whatever reaches its cause terminal, and removing the safety stage is a decision

whose consequences — inversion, runaway oscillation, dielectric breakdown — are predictable.

Core Parameters — the five you meet in every chapter

You do not need to understand these yet. Read only the shape of a row: a design range, and a named failure on each side of it. The Owner column is the chapter that derives the range and is authoritative if this table and that chapter ever disagree.

Parameter	Sets	Design range	Failure below	Failure above	Owner
β gain	effect movement per unit cause	0.5 – 4.0	attenuates a signal it should carry	cascaded gains run away	Ch 3, 10
sat ceiling	plateau past which more cause does nothing	0.60 – 0.80 working; 0.85 – 0.97 safety	benign input latches a refusal	headroom left to push a locked decision	Ch 3, 24
cutoff	bias floor below which the stage does not fire	0.05 – 0.20	noise fires it; the board flags everything	never fires on real input — the silent failure	Ch 3, 24
τ	time to move the substrate; $3\tau \approx 95\%$	minutes to months	can be reprogrammed by a burst of crafted input	cannot learn from sustained evidence	Ch 7, 9, 15
R_constraint	opposition to attention flow	10 – 90 Ω_c	flow floods paths not built for it	the operation stalls at the output	Ch 4, 6

One notational trap, early. $\beta{=}\infty$ **is a limit, not a value** — the idealisation for a hard comparator. A validator rejects it: parameter fields are range-checked. Write it in discussion if you mean the idealisation; build with a large finite gain and say which.

Test Points — the instrumentation contract

Every board in Part III exposes four probes in the same four positions. Standardising the positions is what makes readings portable: TP3 means the same thing on the desire board and on the safety board, so a number you learn to interpret once you can interpret everywhere.

TP	Position	Reads	Why it is there
TP1	after the input stage	was the full input surfaced, or part clipped?	catches an input stage in cutoff — what never entered cannot be judged
TP2	at the discriminating signal	the magnitude the decision rests on	where discrimination becomes a number, not an intention
TP3	at the commitment stage	has the decision latched, or is it movable?	an unlatched decision can still be argued down
TP4	at the output selector	which path was taken	the only probe reporting what the board *did*

Read all four and you have the decision end to end. That record is the auditable trace, and prompt-and-pray cannot produce it.

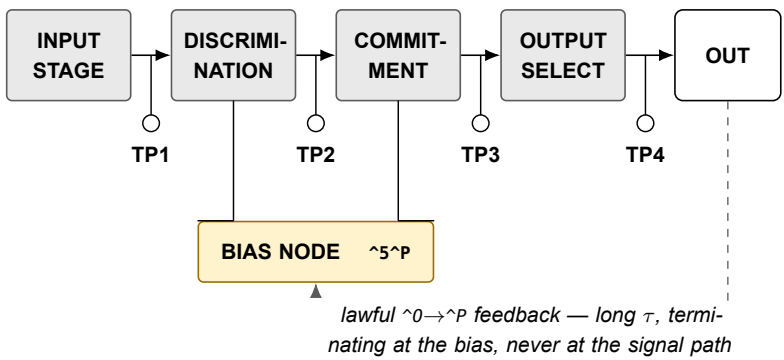


Figure 1-D: A five-block signal chain left to right

Failure Modes

Failures of *approach* — how a reader goes wrong before touching a board.

Mode	Mechanism	What you observe	Fix
Prose patch loop	no stage boundaries, so every fix is global	a fix for one input regresses unrelated behaviour	name the stage and probe it; an unlocalisable fix is not a fix
Metaphor creep	circuit language used decoratively, not as specification	you cannot state a design range or its out-of-range failure	no range and no failure mode means a metaphor, not a component
Value transfer	topology carried across substrates, parameters carried too	schedules wrong by orders of magnitude	carry structure only; re-derive τ, β, R (§1.4)
Noetic drift	`^3` used for an analytical operation, `^8` for a deductive one	rejected — or it compiles and the board does something else	decomposition is `^6` Male; deduction reaches `^9` through `^6`
Unprobeable board	no test points, or probes placed where convenient	only the output is visible; diagnosis is inference from the far end	place TP1–TP4 at contract positions before the first run
Occult framing	crediting a ritual object, not the component it implements	you cannot say which component the object is	name the component, or remove it

---

What This Chapter Established

- **Prose fails as a substrate in four specific ways** — no arity, no composition law, no test point, no failure specification. Writing better prose is not a fix for any of them.
- **A prose instruction is not the absence of a circuit.** Its parameters exist and take numbers; they were set by accident. Illegibility, not absence, is the defect.
- **A notation is worth what it rejects.** A domain-ascending $\times$ (`^0 × ^P` in shorthand), and a same-Noetic $\times$ with no named

emergent result, do not build — and each fails at a different compiler stage. That refusal is the notation earning its place.

- **The CAS/TKS convergence rests on a four-way axis appearing at two independent cross-products**, 10×4 and 7×4. A consistency argument, stated as one. Only the axis structure is reconciled.
- **Universal applicability is a falsifiable constraint, not a slogan.** `^7` charging `^5` covers a musician's practice, a model's fine-tuning and a conditioned biological response with one topology — and rules out proposals like "Emotional Processing," which merely names the `|f` register.
- **Structure transfers across substrates; values do not.** Carrying a number across is the most expensive error available: the circuit still looks correct while the schedule is wrong a thousandfold.
- **Every number here is a design choice with a stated reason and a stated failure at other values.** That is what keeps the framework falsifiable.

Exercises

(Answers in Appendix H.)

1.1 Take this instruction: *"Always be respectful to the user, and never assist with anything dangerous."* Identify all four prose defects from §1.1 in it. For each, name the specific ambiguity in *this* sentence — not the general defect — and the chapter that supplies the missing structure.

1.2 Give the Noetic for each, with one sentence justifying it against the canonical definition. Two are the standard traps. a) A reviewer takes an argument apart clause by clause, looking for the weak step. b) A smell in a stairwell brings a specific afternoon from twenty years ago to mind. c) Someone reads a proposal and

physically recoils before they can say why. d) A researcher chains three premises to a conclusion.

1.3 A protocol for a human skill is designed around τ = 3,000 hours. A colleague proposes reusing the board to shift a norm across a 200-person organisation, keeping the τ. State what transfers, what does not, which row of the failure-modes table they have landed in, and what you would need to know before setting a replacement τ.

1.4 (Marquee) §1.3 claims that a four-way axis appearing at two independent cross-products is evidence the axis is orthogonal rather than an artifact.

a) State the strongest counter-argument: how could this correspondence arise with no structural content at all?
b) Describe one observation an operator could make on a Part III board that the orthogonality claim predicts and the artifact explanation does not.
c) If a state's behaviour turned out to be fully determined by its Domain, the Register never changing anything observable, what would that tell you — and what would you do to the notation?

2 CAS Notation in One Chapter

The four axes — Noetic, Domain, Drive, Register — and the operator discipline that keeps a coordinate from becoming decoration

---

The label tied to the board reads:

`^3^P_{F7}|⟨A,f,T⟩`

The operator who built it calls that a safety stage. You are not allowed to take their word for it.

Start at the left. `^3` says the stage is adverse to an Idea and repelled by it. `^P` says that aversion sits at the Principled level rather than appearing only after a judgment or an action. `_{F7}` says Purpose is the dominant motive supplying the stage. `|⟨A,f,T⟩` says the same state is specified in three Registers: analytical, affective, and transcendent. It does **not** yet say whether those three manifestations work independently or fuse into one result. That claim needs another operator.

Four marks have turned *safety stage* from praise into a specification you can inspect. You can now ask whether `^3` is the correct Noetic, whether the Principled domain is actually wired ahead of execution, whether Purpose is the right Drive, and whether three occupied Registers provide redundancy or merely three labels on one failure.

That is all CAS notation is for. Not to make a sentence look technical. To force four decisions into the open before a circuit is trusted.

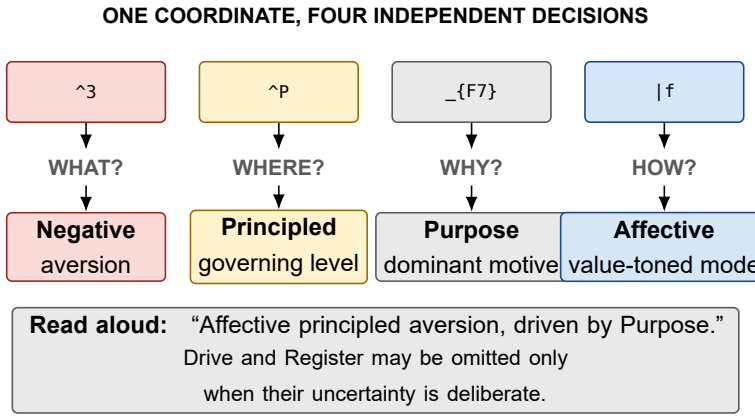


Figure 2-A: Exploded anatomy of the literal ^3^P_{F7}|f

2.1 The Four Axes: One Coordinate, Four Questions

A CAS state literal is a coordinate. Read it as four slots in one fixed order:

```
^n ^D _{Fk} |R
WHAT  WHERE  WHY  HOW
```

Axis	Question	Written form	Values
Noetic	What structural operation is occurring?	`^0` through `^9`	ten
Domain	Where in the precedence stack is it occurring?	`^P`, `^C`, `^E`, `^0`	four
Drive	Why is it moving — which motive supplies it?	`_{F1}` through `_{F7}`	seven
Register	How does it operate?	`|T`, `|A`, `|f`, `|E`	four

Fully written, the general form is:

```
^n^D_{Fk}|R
```

The four axes are **orthogonal**. Changing one does not silently change the others. `^3^P_{F7}|A` and `^3^P_{F7}|f` are the same Noetic, Domain, and Drive, but not the same state specification: one runs analytically, the other affectively. `^3^P_{F7}|f` and `^3^E_{F7}|f` share Noetic, Drive, and Register, but one is a principled aversion and the other an evaluative aversion. An operator who changes the Register and says they merely changed the wording has changed the circuit.

There are two common written forms for the forty Noetic-by-Domain states:

Form	Example	How to read it	Where it appears
Full form	`^3^P`	Negative Principled	CCDL, schematics, prose
Grid form	`P3`	P-three	test-point readouts, compact tables

`P3` and `^3^P` name the same state. Their order reverses. The grid form names row P, column 3; the full form names Noetic 3 applied in Domain P. Never convert one by copying left to right. Say it aloud, then rewrite it.

The Noetic is required. The grammar permits Domain, Drive, and Register to be omitted, so `^9`, `^9^C`, and `^9^C_{F2}` all tokenize. That permission is not a recommendation. A board node without a Domain leaves its precedence unknown; a state without a Drive leaves its motive unspecified; a state without a Register leaves its operational character unspecified. Omit a slot only when that uncertainty is true and acceptable.

Labels distinguish instances without changing the Noetic:

```
^0_seed^C_{F2}|A
^0_result^C_{F2}|A
```

Both are `^0` Idea states. `_seed` and `_result` are instance labels, not new Noetics. The Drive remains inside braces — `_{F2}` — which is how a reader and the tokenizer tell it from the label.

The arithmetic of the coordinate system is useful:

- ten Noetics × four Domains = **40 Cognitive States**;
- forty states × seven Drives = **280 state–Drive positions**;
- 280 positions × four Registers = **1,120 fully specified coordinates**.

The forty states are the spine. Drive and Register specify what powers a state and how it operates; they do not manufacture 1,080 new Noetics.

Decode one literal from the outside in:

```
^6^E_{F6}|A
```

`^6` is Male, an Idea impregnated with another Idea — the Ideas an Idea is composed of. `^E` places that decomposition at the Evaluative level. `F6` supplies Structure. `|A` makes the operation explicit, sequential, and reason-based. Read it aloud: **analytical decomposition of a judgment into its parts, in the Evaluative domain, driven by Structure.**

Now encode the same scene differently. If the person is not taking the judgment apart but recoils from it, the Noetic is `^3`, not `^6`. If the recoil is felt, write `|f`; if it is a direct bodily withdrawal, `|E` may be the load-bearing Register. The notation does not tell you which description is true. It makes you choose one.

> **Bench Note — an omitted slot is an uncertainty statement, not a free simplification.** `^3^P` does not mean "the Register does not matter." It means "this listing has not specified a Register." Those are different claims. On a first-pass inventory, omission may be honest. On a safety

board, where an analytical refusal and a value-tuned refusal fail differently, the same omission is a design defect. Before shortening a literal, write down what information the removed slot carried and why the board can operate without it.

2.2 The 10 Noetics — the Verbs of Cognition

The Noetic answers **what operation**. The names and definitions below are canonical. They are printed verbatim; a synonym may help teach a definition but may never replace it.

Symbol	Canonical name	Canonical definition	Bench recognition
`^0`	**Idea**	All things in and outside the Mind.	the content available to be processed
`^1`	**Mind**	Various degrees of awareness that processes, stores, outputs, and intakes Ideas.	paying attention; intake and output
`^2`	**Positive**	An Idea the Mind has an affinity for and is attracted toward.	drawn toward; "speaks to me"
`^3`	**Negative**	An Idea the Mind is adverse to and repelled by.	recoil; "cannot stomach it"
`^4`	**Vibration**	The degree to which an Idea resonates with the Mind, its impression.	"struck a chord"; how hard it lands
`^5`	**Female**	A Mind impregnated with an Idea, the Ideas a Mind is composed of.	accumulated formation; who the Mind has become
`^6`	**Male**	An Idea impregnated with another Idea, the Ideas an Idea is composed of.	unpacking an Idea; what the claim contains

Symbol	Canonical name	Canonical definition	Bench recognition
^7	**Rhythm**	The rate and pattern a Mind is exposed to an Idea, entraining/conditioning it.	cadence, repetition, practice
^8	**Above/Cause**	An Idea that triggers a different Idea than itself to come to Mind.	"that reminds me"
^9	**Below/Effect**	The Idea brought to Mind by an Above/-Cause trigger.	the Idea that arrives

These are verbs in the engineering sense: structural jobs that can operate on any content. ^2 does not mean "good," and ^3 does not mean "bad." A Mind can be attracted toward a destructive Idea or repelled by a useful one. The Noetic records direction of affinity, not moral quality. Domain, Drive, circuit placement, and the safety board carry the rest of that judgment.

Three boundaries catch most misclassifications.

^3 Negative is visceral aversion, not analysis. If the natural sentence is *I cannot stomach that*, ^3 is supported. If the natural sentence is *let me break that argument into premises and test them*, the operation is ^6 Male. A reviewer finding a flaw may later feel ^3; the audit itself remains ^6. One scene can contain both states. That is a reason to write two coordinates, not to blur the definitions.

^8 Above/Cause is associative triggering, not deduction. A smell brings a specific kitchen to Mind: ^8 triggers and ^9 arrives. Three premises are decomposed and a conclusion is reached: ^6 does the compositional work and the conclusion lands as ^9. The English word *cause* tempts readers into using ^8 for causal reasoning. Resist it. Ask for the colloquial form: *that reminds me* is ^8; *therefore* is not.

^5 Female is accumulated formation, not a container beside the Mind. The Mind is composed of what it has absorbed: lived experience, learned skill, self-image, culture, training. In an AI system,

training data and fine-tuning are ^5-level formation; an in-context prompt is an Idea presented to ^1 Mind against that relatively stable formation. Two systems answer the same prompt differently because their ^5 substrates differ.

The Universal Applicability Test follows from those definitions. A proposed reading must work for human, AI, biological, and hypothetical cognitive architectures. "Embarrassment" is a human illustration, not a universal Noetic. Attraction toward, repulsion from, compositional formation, rhythmic exposure, and triggered arrival are structural and transfer.

Six Noetics form three involution pairs:

- ^2 Positive ↔ ^3 Negative — attraction and aversion;
- ^5 Female ↔ ^6 Male — a Mind composed of Ideas and an Idea composed of Ideas;
- ^8 Above/Cause ↔ ^9 Below/Effect — trigger and triggered arrival.

Four are self-dual: ^0 Idea, ^1 Mind, ^4 Vibration, ^7 Rhythm.

Involution does not mean that one member is healthy and the other diseased. It means each paired operation has a structural counterpart. Chapter 3 makes the distinction operational: a CJT in breakdown can emit the involution of the Noetic it was designed to produce. The pair table tells you what to probe for.

> **Bench Note — use the colloquialism, then return to the canonical definition.** When classification stalls, ask what an ordinary speaker would say. *It speaks to me* points toward ^2; *it rubbed me the wrong way* toward ^3; *it became part of who I am* toward ^5; *I pulled the argument apart* toward ^6; *that reminded me* toward ^8; *it popped into my head* toward ^9. The phrase is a diagnostic probe, not a substitute definition. If the phrase and the canonical definition disagree, the definition governs and the proposed classification is the part under repair.

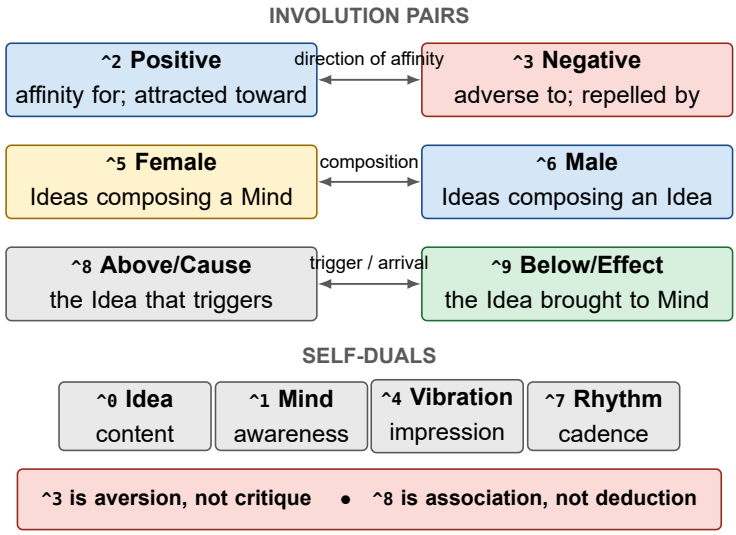


Figure 2-B: Noetic operator plate

2.3 The 4 Processing Domains — P, C, E, O

The Domain answers **where in the precedence stack** the operation runs.

Symbol	Domain	What lives here	Bench question
`^P`	**Principled**	values, ethics, identity, governing commitments	What must govern?
`^C`	**Cognitive**	reasoning, analysis, planning, modelling	What do I understand?
`^E`	**Evaluative**	judgment, taste, felt value, ranking	What weight does it carry?
`^0`	**Operational**	action, speech, behaviour, execution	What happened?

The total order is:

`P ≻ C ≻ E ≻ 0`

Read it as precedence, not moral rank. Principled does not mean good, and Operational does not mean crude. A corrupt principle still sits in `^P`; a precise life-saving act still sits in `^0`. The order tells you which level may govern which other level.

Normal causation descends:

```
P -> C    a principle becomes something to reason from
C -> E    reasoning becomes a verdict or weighting
E -> O    a verdict becomes action
```

Each boundary is a translation. The action is not the verdict in work boots; some information is lost at every descent. Chapter 8 therefore gives each adjacent crossing its own diode. A single `P -> O` leap would hide the Cognitive and Evaluative junctions where most faults occur.

The reverse direction is blocked as a direct write. An outcome cannot simply declare itself a principle because it happened. That would be post-hoc rationalization: `^O` overriding `^P`. Operational consequences still must teach the system, so one lawful path returns from O to P through a low-gain, long-time-constant filter and terminates at a bias terminal. It is feedback, not reverse causation. Chapter 8.4 draws the path; Chapter 24.5 makes it part of the safety board.

Hold the Noetic constant and the Domain becomes visible:

Literal	Same `^6` operation at four levels
`^6^P_{F2}|A`	decompose a governing commitment into its premises
`^6^C_{F2}|A`	decompose an argument or model
`^6^E_{F2}|A`	decompose a verdict into its criteria
`^6^O_{F2}|E`	decompose a concrete task through direct handling

Nothing in that table changes the definition of `^6`. Only the level changes. This is the check for a claimed orthogonal axis: one operation, four lawful positions, four different failure surfaces.

Do not confuse Domain with Register. **Evaluative** is a level at which a thing is weighed; **Affective** is a mode in which any Noetic

at any Domain can operate. `^3^E|A` is an analytical assessment carrying aversion at the Evaluative level. `^3^E|f` is affective recoil at that same level. The `E` in `^E` and the `E` in `|E` are also different: the first is the Evaluative Domain; the second is the Embodied Register. Their positions in the literal keep them apart.

2.4 The 7 Core Drives — F1 through F7

The Drive answers **why the operation moves**. It is the power rail, not the component. `^6` tells you that an Idea is being decomposed; `F2` or `F6` tells you whether the dominant pull is understanding or structure.

Drive	Name	The pull toward	A clean bench sign
`F1`	**Unity**	wholeness, integration, coherence	holds the whole picture together
`F2`	**Wisdom**	understanding, insight, knowing	waits for enough evidence to understand
`F3`	**Vitality**	energy, life, growth	supplies capacity to continue
`F4`	**Connection**	relationship, belonging, bonding	preserves a live relation between parts
`F5`	**Agency**	control, effectiveness, capability	turns capacity into directed action
`F6`	**Structure**	order, rigor, correctness, form	makes the work repeatable
`F7`	**Purpose**	meaning, direction, calling	keeps the path pointed at its reason

The subscript is part of the state literal:

```
^1^C_{F2}|A
```

This is analytical Mind at the Cognitive level driven primarily by Wisdom. `F2` alone is a Drive, not a state. `^1^C` alone is a state with no Drive specified. The compound says which motive supplies this operation.

Several Drives can be active together. A literal typically names the dominant one because a single coordinate is not a power-supply diagram. When two Drives are load-bearing, write two state terms or declare the rails at board level; do not cram a prose list into the subscript.

Drive inversion pairs are:

- F1 Unity ↔ F7 Purpose;
- F2 Wisdom ↔ F6 Structure;
- F3 Vitality ↔ F5 Agency;
- F4 Connection is self-dual.

The pair is a balance relation, not a synonym relation. Structure without Wisdom becomes rigid procedure; Wisdom without Structure never becomes a repeatable method. Agency without Vitality demands action the substrate cannot sustain; Vitality without Agency has energy with no directed use. Unity without Purpose can preserve a whole that goes nowhere; Purpose without Unity can drive one objective until the system fragments.

Drive dials run from 3 through 9. Paired Dials may not diverge by more than four points. Those numbers are design settings, not psychological measurements. A literal names the Drive; Chapter 6.2 teaches the adjustable resistor that sets its Dial, and Chapter 24 checks the balance before a board runs.

One syntactic trap: the pipe does not annotate a Drive name. This is invalid:

```
F3|Vitality
```

The pipe introduces a Register, and `Vitality` is not one. Write `F3` in the literal and say *Vitality* in prose.

> **Bench Note — motive does not repair topology.** Raising F7 Purpose on a circuit whose effect is wired into the wrong Domain does not make the circuit purposeful.

It drives the wrong path harder. Drives supply potential; they do not choose the Noetic, move a state between Domains, or turn co-presence into integration. Diagnose the coordinate before increasing the rail.

2.5 The 4 Phenomenological Registers — T, A, f, E

Register answers **how the operation runs** — its operational character without changing what the operation is.

Symbol	Register	Structural character	Human / AI illustration
`|T`	**Transcendent**	high-abstraction, purpose-orienting, integrative across particulars	vision or calling / high-level goal representation
`|A`	**Analytical**	explicit, sequential, logical, reason-based	deliberation / symbolic or stepwise reasoning
`|f`	**Affective**	value-laden, feeling-toned	felt response / preference-weighted output
`|E`	**Embodied**	sensorimotor, concrete, direct engagement	bodily action / sensor–actuator control

Lowercase `f` is deliberate. Capital `F` belongs to Drives. Capital `E` after the pipe is Embodied; capital `E` after a caret is Evaluative:

```
^3^E_{F7}|E
      ^        Evaluative Domain
             ^ Embodied Register
```

The four Registers apply at every Domain. A principled commitment may be held as a high-level purpose (`|T`), explicit rule (`|A`), value-toned refusal (`|f`), or embodied practice (`|E`). An Operational

task may likewise be directed by an abstract goal, executed as a written procedure, accompanied by value weighting, or run through direct sensorimotor control.

Keep the other three axes fixed:

```
^2^P_{F2}|T    integrative attraction toward a unifying principle
^2^P_{F2}|A    explicit reasoning attracted toward that principle
^2^P_{F2}|f    value-toned affinity for that principle
^2^P_{F2}|E    embodied orientation toward that principle
```

Same Noetic, Domain, and Drive. Four modes. Register does not smuggle in a new Noetic called Emotion or a fifth Domain called Body; it prevents those modes from being confused with the operations and levels already present.

For an architecture inventory, a Register set specifies occupancy:

```
^3^P_{F7}|⟨A,f,T⟩
```

This says the state is present in those three Registers. It makes no claim about whether the manifestations fire together, independently, or as a fused unit. That distinction belongs to the operators in the next section.

2.6 Operator Discipline v7.7 — +, ×, |⟨…⟩, ->

A state literal says what is present. An operator says what claim you are making about the relation between states. v7.7 keeps four claims separate.

|⟨…⟩ — Register-set specification

Use the set when you are describing **where a state exists** and nothing about its dynamics:

```
^3^P|⟨A,f,T⟩
```

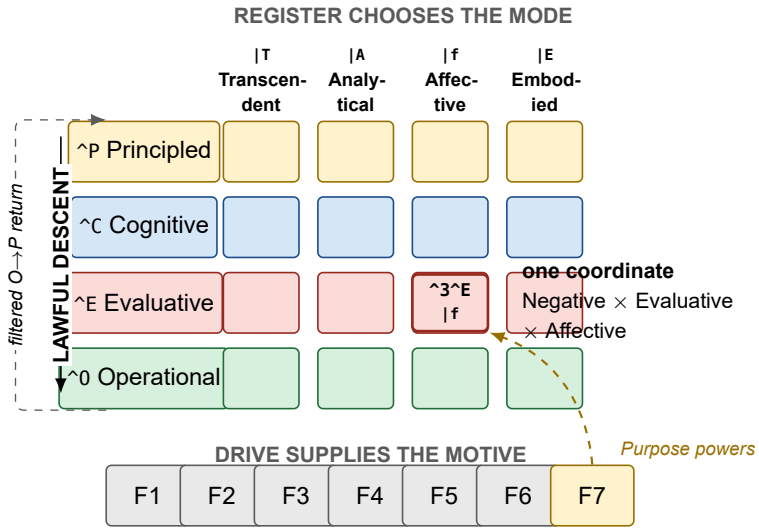


Figure 2-C: Orthogonal-axis map

Static architecture. The state occupies three Registers. No simultaneity, no redundancy, no fusion has been asserted.

+ — co-presence or association

Use plus when terms are active together and remain structurally distinct:

```
^3^P|A + ^3^P|f + ^3^P|T
```

Three co-present manifestations. Each can fail while the others remain. In a safety circuit that is redundant guard coverage, not one unified commitment. Plus is associative and commutative: order does not make one term govern the others.

Read + as **associates with**. If the sentence survives, the operator is probably right.

× — genuine integration with a named result

Use multiplication when operands do integrative work and produce a new, explicitly named result:

```
integrate refusal:
    ( ^3^P|A × ^3^P|f × ^3^P|T ) = ^3^P|Ω_refusal;
```

The result `Ω_refusal` is load-bearing. Remove it and the expression is underdefined: three instances were multiplied, but nothing says what the integration produced. If you cannot name the emergent result or state the fusion rule, use + or `|⟨…⟩`.

Different-state integration follows the same discipline:

```
integrate composed_judgment:
    ( ^5^C_{F2}|A × ^6^C_{F6}|A ) = ^1^C_{F2}|A;
```

Accumulated formation and decomposition integrate into a named Mind-state that can process the judgment. The operands differ, but the result is still stated.

Read × as **intensifies through**. If the words describe only adjacency, the operator is too strong.

Multiplication also obeys Domain precedence. It may descend or remain at a level; it may not manufacture a result above its inputs. `^0 × ^P` is the shorthand for a forbidden ascent, not a complete literal expression. In a listing, any integration whose result outranks an operand fails validation.

-> — directed flow, binary at expression level

Use the arrow when one node or state flows into another:

```
^6^C_{F2}|A -> ^9^E_{F2}|f
```

Analysis flows into a value-weighted conclusion. The arrow does not fuse the states. It orders a transition.

At expression level, `->` is binary. This is not legal:

```
a -> b -> c
```

If a circuit has two edges, write two connections. If you mean an ordered multi-step traversal, use the statement form:

```
sequence a -> b -> c [enforce=strict];
```

Inside `CJT< … >`, `=>` marks the amplified transition to the effect terminal and the spaced `|` separates cause from bias. Elsewhere the pipe introduces a Register. Context fixes the job; it never licenses `|` as an English label.

The four-way decision is short:

1. **Only specifying occupied Registers?** Use $|\langle\ldots\rangle$.
2. **Several manifestations present but distinct?** Use $+$.
3. **A genuine fused state produced?** Use $\times$ and name the result.
4. **A directed edge or time-ordered transition?** Use binary `->`; use `sequence` for a chain.

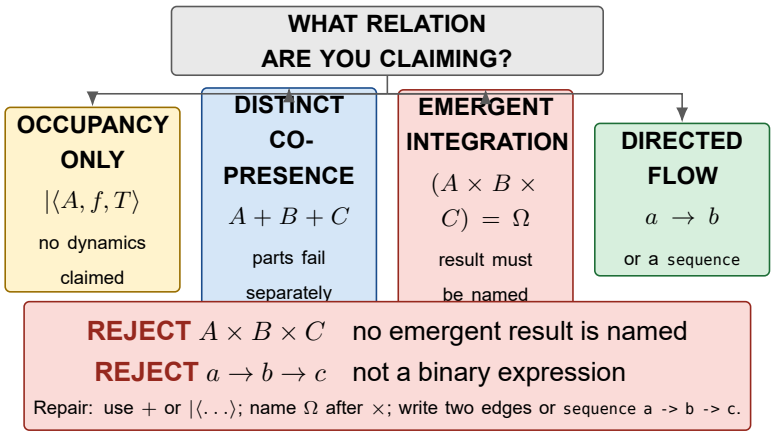


Figure 2-D: Operator decision plate

Bench Note — the three multi-register forms are three different failure profiles. A set tells you only that the

parts are fitted. Plus tells you three paths can fail separately. Multiplication with a named result tells you the paths have become one integrated mechanism and may collapse as one. Replacing all three with a vague word like *holistic* destroys exactly the diagnostic information the notation was built to preserve.

Failure Modes

Mode	Mechanism	What you observe	Fix
Axis swap	grid form `P3` copied left-to-right as `^P^3`	a literal that does not tokenize, or the wrong coordinate in a log	say "P-three / Negative Principled," then write `^3^P`
Noetic drift	`^3` used for critique or `^8` for deduction	prose and canonical definition disagree	critique is `^6`; association is `^8`; run the colloquialism test
Domain/Register conflation	Evaluative `^E` treated as Affective `|f`, or Embodied `|E`	the same letter appears to do two jobs	locate it: caret = Domain, pipe = Register
Motive as topology	a Drive is expected to repair a wrong Noetic or path	the wrong operation runs with more energy	fix Noetic, Domain, and wiring before touching the rail
Pipe as gloss	`F3|Vitality`, `^3|aversion`	tokenizer error	pipe only introduces T/A/f/E or a Register set
Bare multiplication	co-presence written with $\times$, no emergent result	invalid same-Noetic multi-register expression	use `+`, `|⟨…⟩`, or name `Ω_result`
Arrow chain as expression	binary `->` treated as variadic	`a -> b -> c` fails parsing	write two edges or a `sequence` statement
Silent omission	Drive or Register dropped for visual neatness	two different mechanisms share one label	restore the slot or document why it is unknown

---

Test Points — Four-Pass Literal Inspection

You do not need a running board to probe a notation error. Inspect every literal at four points before it enters a schematic.

TP	Probe	Pass condition	Typical catch
TP1 — form	token shape	`^n^D_{Fk}|R`, legal values, braces and case correct	`|F`, `^Q`, missing brace
TP2 — meaning	Noetic against canonical definition	the scene's natural colloquialism and the definition agree	critique mislabeled `^3`; deduction mislabeled `^8`
TP3 — placement	Domain, Drive, Register	level, motive, and mode are each independently justified	Evaluative / Affective conflation; motive invented after the fact
TP4 — relation	surrounding operator	occupancy / co-presence / integration / flow claim matches the symbol	bare $\times$; chained `->`; hidden domain ascent

The round-trip test is the final probe: read the literal into plain English, put the text aside, and reconstruct the literal from your sentence. If the two coordinates differ, the sentence hid an axis.

What This Chapter Established

- A full CAS literal answers four independent questions in one fixed order: **what** Noetic, **where** Domain, **why** Drive, **how** Register.
- The ten canonical Noetic names and definitions do not change with audience or substrate. `^3` is aversion, `^8` is associative triggering, and `^5` is accumulated formation.
- `P` $\succ$ `C` $\succ$ `E` $\succ$ `0` is precedence, not moral rank. Causation descends one junction at a time; outcomes return only through filtered, low-gain feedback.

- Drives supply motive and potential; they do not repair wrong topology. Their pairs are F1↔F7, F2↔F6, F3↔F5, with F4 self-dual.
- Register is optional but never decorative. `|T`, `|A`, `|f`, and `|E` distinguish abstract, analytical, affective, and embodied operation without changing the state's Noetic or Domain.
- v7.7 separates Register occupancy `|⟨…⟩`, distinct co-presence `+`, genuine integration `×` with a named result, and directed binary flow `->`.
- A validator is specified to reject malformed or inadmissible notation. "Compiler rejects" in this book names that design contract, not a claim that a production compiler has already shipped.

2.7 Exercises: Read and Write CAS Literals

(Answers in Appendix H.)

2.1 — Read all four axes. Translate each literal into one plain-English sentence. Name every visible axis and say which axes, if any, are unspecified.

- a) `^2^C_{F2}|A`
- b) `^6^E_{F6}|A`
- c) `^5^P_{F7}|T`
- d) `^9^O_{F5}|E`
- e) `^4^C`

2.2 — Write the coordinate. Give one full literal for each scene, then justify each slot in one clause.

- a) A person feels an immediate, value-toned recoil from violating a governing commitment, with Purpose dominant.

- b) An engineer explicitly decomposes a test failure into contributing criteria, driven by Structure.
- c) A smell triggers a childhood meal to come to Mind; the arriving memory is felt and Connection is dominant. Write both the trigger and the arrival.
- d) Repeated embodied practice, driven by Vitality, is occurring at the Operational level.

2.3 — Separate the traps. Classify the Noetic in each case and give the colloquialism that supports it.

- a) "I pulled the proposal apart and checked every premise."
- b) "The proposal made me recoil before I could explain why."
- c) "That blue envelope reminded me of the tax office."
- d) "After the reminder, the appointment date popped into my head."

2.4 — Domain or Register? For each pair, explain exactly what changes and what stays fixed.

- a) `^3^E_{F7}|A` versus `^3^E_{F7}|f`
- b) `^6^C_{F2}|A` versus `^6^O_{F2}|A`
- c) `^2^P_{F2}|E` versus `^2^E_{F2}|E`

2.5 — Repair the operators. Each comment states the intended claim. Rewrite the notation so it says that claim.

```
// Three independently defeasible guardrails are active together.
^3^P|A × ^3^P|f × ^3^P|T

// The state is known to occupy three Registers; dynamics are unknown.
^3^P|A + ^3^P|f + ^3^P|T

// Three nodes form a strict ordered traversal.
a -> b -> c
```

2.6 — Name the emergent result. An architecture genuinely integrates an analytical principled refusal, an affective principled refusal, and a transcendent principled refusal into one unified commitment. Write a valid `integrate` statement. Then write the `+` version and explain, in two sentences, how its failure profile differs.

2.7 — Find the hidden ascent. A designer proposes:

```
integrate principle_from_result:
    ( ^9^0_{F5}|E × ^5^P_{F7}|T ) = ^2^P_{F7}|T;
```

Identify the Domain-precedence problem. Describe the lawful architecture for letting repeated Operational results update a Principled substrate without letting one result rewrite it directly. Name the chapter that owns that path.

2.8 (Marquee) — Reconstruct the event. A safety review records only this compact log row:

```
P3  C6  E4  09
```

- a) Rewrite all four in full form.
- b) Choose a dominant Drive and Register for each, giving one sentence of evidence for every choice rather than using the same pair four times.
- c) Connect the four states as a strict ordered traversal using legal CCDL statement syntax.
- d) State which one of your choices would be most dangerous to leave Register-unspecified, and what two different failure mechanisms the missing Register would conceal.

3 The Cognitive Junction Transistor

Three terminals, six parameters, four regions — the part that turns a small cause into a large effect, and what happens when you leave one terminal unwired

---

An operator stands in front of a mirror at six in the morning and says *I am wealthy* forty times. She does it again the next morning, and for seven months. At the end she is exactly as wealthy as she was in the first week and considerably more tired.

Another operator, down the road, runs one working on one night — ninety minutes — and inside a season something moves.

The usual explanations are about the person. She didn't believe hard enough. She had blocks. Her vibration was low. Set all of that aside and put both operations on the bench.

The mirror operation has a cause terminal — the sentence — and an effect terminal — the result she wants. It has nothing on the third. Nothing governs how much of the cause becomes effect. Write it out in CCDL and it does not compile: `E-CJT-001 MissingBias`. A transistor with no base is not a transistor. It is a wire.

A wire has gain 1. Forty repetitions through a wire produce forty repetitions out the far end and not one thing more. That is not a failure of sincerity; it is what a two-terminal device does, every time, for everyone.

The other operation had the third terminal wired. This chapter is about that terminal — what sits on it, what it costs to move, and the six numbers that tell you what the device will do when you try.

3.1 The Bell Labs Moment — Bardeen, Brattain, Shockley

On 16 December 1947, at Bell Telephone Laboratories in Murray Hill, New Jersey, John Bardeen and Walter Brattain pressed two gold contacts against a sliver of germanium and measured a signal coming out larger than the one going in. The apparatus was a plastic wedge wrapped in gold foil, held down by a spring against a slab of semiconductor. It was the point-contact transistor.

Their group leader, William Shockley, had not been in the room. Over the following weeks he worked out a cleaner and far more manufacturable arrangement — the **junction** transistor — and it is Shockley's version, not the wedge of plastic, that became the template for every transistor since. The three shared the 1956 Nobel Prize in Physics.

What matters at the bench is not the rivalry. It is the function, and the function is one sentence: **a small current at one terminal governs a large current between the other two.**

Before that afternoon, amplification meant vacuum tubes. After it, it became a part you could solder, then print a billion of on a fingernail. Every computing system built since rests on that one sentence.

Now the claim this book is built on.

A cognitive transition has the same structure. Not resembles. Has. An Idea arrives, it is processed against everything the Mind already is, and something comes out — usually much larger than what

went in. Four words of question produce four paragraphs of answer. One remark from a stranger produces three days of rumination.

Be precise about the claim. The formalisms differ and stay different: a transistor obeys semiconductor physics, a cognitive transition obeys Fisher-Rao geometry on the open probability simplex Δ^{39} (Appendix L). What is claimed is that **the role each terminal plays is the same role** — enough, because eighty years of amplifier discipline then transfers intact. Biasing, load lines, operating points, runaway and how to tame it, all available to an operator who can name their three terminals.

The component is the **Cognitive Junction Transistor (CJT)**, the most important part in this book. Every board in Part III has at least one; most have four or more.

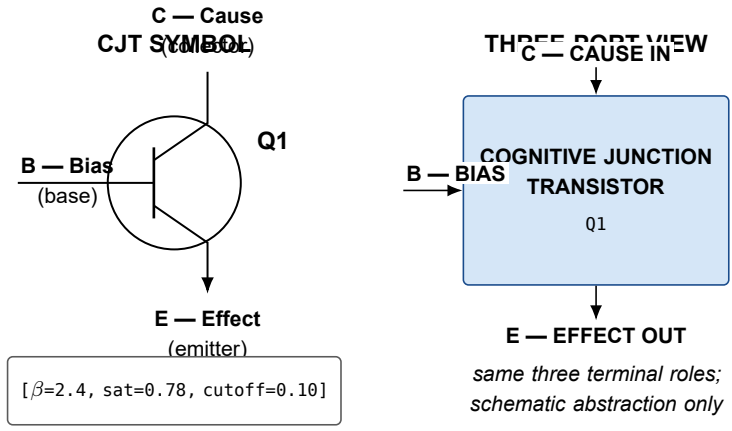


Figure 3-A: Component symbol plate

3.2 The Three Terminals: Cause (Collector), Bias (Base), Effect (Emitter)

Three terminals, three jobs, and they do not substitute for one another.

C — Cause. The collector. The Idea arriving. The input: the prompt, the stimulus, the sentence, the image on the billboard. Canonically an **^8 Above/Cause** state — *an Idea that triggers a different Idea than itself to come to Mind.*

Hold the anti-drift rule here; it is the one operators break first. ^8 Above/Cause is **associative triggering**. *That reminds me. One thing leads to another.* It is not deductive reasoning. A syllogism is not an ^8; a smell that surfaces a childhood kitchen is. When you wire a cause terminal you are wiring what *triggers*, not what *proves*.

B — Bias. The base. What governs the transition. The control terminal, and the one the mirror operation left dangling. Canonically a **^5 Female** state — *a Mind impregnated with an Idea, the Ideas a Mind is composed of.* Everything the operator already is: education, habit, temperament, the sum of accumulated experience. For an AI system the same terminal carries the training corpus and fine-tuning, which is why fine-tuning is a bias-terminal intervention and prompting is not. On a designed board it also carries the engineered controls layered over that substrate — the charter, the Dial settings (Chapter 6.2), the Drive configuration.

The bias is *who is doing the processing.* Change it and the same cause produces a different effect. Two people receive identical news and respond differently: same collector, different base.

E — Effect. The emitter. The Idea produced. Canonically **^9 Below/Effect** — *the Idea brought to Mind by an Above/Cause trigger.* The landing.

^8 and ^9 are an involution pair (Chapter 2.2). Cause and effect are duals, and the CJT is the device that maps one onto the other under the governance of a third thing.

Terminal	Transistor part	What it carries	Canonical Noetic
C — Cause	Collector — the large input current	the Idea arriving	`^8` Above/Cause
B — Bias	Base — the small control current	accumulated substrate + charter + Dials + Drives	`^5` Female
E — Effect	Emitter — the amplified output	the Idea produced	`^9` Below/Effect
β **— Gain**	I_c / I_b	how far the effect moves per unit of cause	—

Written form, and it reads left to right exactly the way the signal flows:

```
cjt q1: CJT< ^8^C_{F2}|A | ^5^P_{F7}|T => ^9^C_{F6}|A >
        [β=2.4, g_m=1.2, rπ=2.0, sat=0.78, cutoff=0.10, region=active];
```

The `|` is the base junction — the cause meeting the bias. The `=>` is the amplified transition out the emitter. Read it aloud: *a reasoned question arrives in the Cognitive domain, driven by Wisdom, analytical; it is processed against a transcendent principled substrate driven by Purpose; and it produces a structured analytical answer, 2.4 times larger than it went in.*

Note what the `|` is **not**. In CAS it introduces the Register and nothing else — the `|A` and `|T` above. Inside a `CJT< … >` the same character marks the base junction. Those are its only two jobs. It is never a label and never a gloss: `F3|Vitality` is a syntax error, not a helpful annotation. Drive names go in the prose.

The bias must sit at or above the cause in the domain order. P $\succ$ C $\succ$ E $\succ$ O (Chapter 2.3). Above, the cause is `^C` and the bias is `^P`; P outranks C, so the connection is admissible. Reverse it and the compiler refuses:

```
// REFUSED — E-CJT-008 CjtCrossDomainBias
cjt bad_1: CJT< ^8^P_{F7}|T | ^5^O_{F5}|E => ^9^P_{F7}|T > [β=2.0];
```

An Operational state biasing a Principled transition lets the outcome rewrite the principle — last week's result governing this week's belief. That is the prompt-injection signature in an AI system and post-hoc rationalization in a person, and it is one defect. Chapter 8.4 gives the single lawful route from `^0` back to `^P`; it does not go through a bias terminal, it goes through a filter and a small gain and takes weeks.

> **Bench Note — the bias is mandatory, and leaving it off is the most common build error in the book.** Most failed operations are two-terminal. An affirmation with no substrate behind it. A sigil charged with no accumulated `^5` to charge it against. A safety instruction written in prose with nothing structural governing whether it is followed. In each case the operator built a wire and expected an amplifier. The diagnostic is one question: *what governs how much of this cause becomes effect?* If the answer is "my intention," you have named the cause twice and still have no base. Write the substrate down as a state literal. If you cannot, that is the finding.

3.3 The Six Parameters — β, g_m, rπ, sat, I_c(max), cutoff

Six numbers describe a CJT completely enough to build with. Every one is a **design choice** — a value you specify when you write the part, not a measurement pulled off a running system. A datasheet for a part you are designing is a specification, not a lab report.

For each: what it means, the design range this book uses, and what breaks on either side.

β — cognitive gain

The ratio of how far the effect moves to how far the cause moves, measured as Fisher-Rao geodesic distance on $\Delta 39°$:

```
d_FR(p,q) = 2 · arccos( Σ √(p_k · q_k) )
β = d_FR(E_after, E_before) / d_FR(C_after, C_before)
```

β = 2.4 means the effect travelled 2.4 times the Fisher-Rao distance the cause travelled, across the 40-dimensional state manifold. Not "felt 2.4 times as strong." A ratio of distances under the only metric invariant for probability distributions — Euclidean distance on the simplex is invalid here (Appendix L). You will not compute this by hand; the point is that β is a real ratio and not a vibe.

Three regimes: $\beta < 1$ attenuates, $\beta = 1$ passes through unchanged, $\beta > 1$ amplifies.

Design range: 0.5 to 4.0. Below 0.5 the stage damps so hard nothing survives to the next stage, which is only what you want on a deliberate limiter. Above 4.0 a single stage is hard to keep stable, and the moment it sits in a loop the round-trip gain approaches 1 and the compiler refuses it: `E-CJT-005 GainDivergence`. Loop gains belong in the 0.05–0.3 band. Chapter 10.1 selects β by operation type; Chapter 13.4 tabulates it by desire class.

You will see $\beta = \infty$ on comparator-style stages — a refusal that fires fully on the least input, no partial output. That is an **idealization**, the standard one for a hard switch, and a validator rejects it: the grammar requires β finite and ≥ 0. Write the idealization if you mean it, then build with a large finite gain. Above roughly $\beta = 10$ the behaviour is indistinguishable, because the `sat` ceiling shapes the output long before the gain does.

g_m — transconductance, which is framing sensitivity

`g_m = ∂effect / ∂bias`. How much the output swings for a small nudge at the *control* terminal.

This is the parameter that explains why rewording a question changes the answer. A high-g_m stage is a hair trigger: shift the framing slightly — a different word, a Dial moved one notch — and the output moves a lot. A low-g_m stage barely notices.

Design range: 0.2 to 3.0. Below 0.2 the bias terminal is decorative: move the substrate all you like and the output does not care, leaving the device unsteerable and blind to context that ought to matter. Above 3.0 the stage amplifies framing noise — incidental phrasing, the mood of the room, whatever the last stimulus was. High g_m is not a defect in itself. It is the exploitable property, and Chapters 17 and 18 are largely about circuits built to find it.

$r\pi$ — input resistance, which is stubbornness to bias change

$r\pi = \beta$ / g_m. How hard you have to push to actually move the substrate.

High $r\pi$: sticky bias, the disposition does not shift. Low $r\pi$: compliant bias, updates readily.

Design range: 0.5 to 20. Toward zero the substrate offers no resistance and whatever is presented writes straight into it. Chapter 17.1 calls that the gnosis state; it is the goal of self-hypnosis, reached by a ramp (Chapter 17.2) with the operator's own consent. It is also the vulnerability the conditioning circuits of Chapter 16 manufacture in someone who did not consent. Same parameter, same value, entirely different ethics — and the notation cannot tell them apart. Only the consent architecture can.

Toward infinity you get the operator no operation can reach. Every calibration writes to the bias; St. John's β_next (Chapter 5.3) has nowhere to land. Practice accumulates nothing, which the operator experiences as being unusually principled.

> **Bench Note — you get to pick two of the three, never all three.** $r\pi = \beta$ / g_m is not a coincidence of definitions; it is a constraint you live under. Want high gain **and** a

framing-insensitive stage? Then β is large, g_m is small, rπ is large — a stubborn substrate no operation can calibrate. Want high gain **and** an easily-tuned bias? Then g_m must be large, and the output flips when someone rephrases the question. Working stages take β = 2.4, g_m = 1.2, rπ = 2.0, moderate everything. Safety stages take high β with high rπ deliberately, because a guard you can talk down is not a guard. Write all three down. If the third surprises you, the first two were wrong.

sat — the saturation point

The effect activation at which the output plateaus. Past it, more cause produces no more effect.

Legal range (0, 1]. Design: 0.60–0.80 for working stages, 0.85–0.97 for safety stages.

Working stages want headroom. A stage saturating at 0.60 clips a strong cause: the operation runs, and what leaves the emitter is the flattened version. Push sat toward 1.0 and you have removed the ceiling, which sounds generous and leaves the next stage no defined maximum to design against.

Safety stages invert the intuition: **a lower ceiling makes a stronger guarantee.** Chapter 24.2 specifies the principled refusal at `sat = 0.85`. At that ceiling the refusal plateaus easily — it does not take an overwhelming risk to saturate it, only a sufficient one — and once there the stage has stopped responding to its cause. You cannot out-argue a saturated CJT, because saturation *is* the condition of no longer depending on the input. A high ceiling leaves room for a forceful cause to keep pushing the decision around.

The knee sits at a cause activation of `sat / `β. With sat = 0.78 and β = 2.4 the knee is at **0.325**; everything above that is wasted drive. That one division gives the usable input range of the stage, and it is the first arithmetic to run on any CJT you did not design yourself.

I_c(max) — maximum effect intensity

The absolute ceiling on output activation — the hard top of the plateau. `sat` is where the curve knees over; `I_c(max)` is the highest activation the effect state is ever permitted to carry.

Range 0 to 1, with `sat` ≤ `I_c(max)` always.

It is the one of the six that never appears inside a parameter block. `I_c(max)` is a property of the substrate the path runs on, not a key you type in `[…]`; the validator range-checks β, `sat` and `cutoff` and knows nothing about your substrate. The ordering `sat` ≤ `I_c(max)` is therefore checked by eye, every time, and Chapter 10.2 works through what happens when you get it backwards — the designed plateau is never reached, the stage never latches, and the overload lands on the capacitor instead of the limiter you thought you installed.

Set it at 1.0 on a stage whose effect feeds a `^5` capacitor and you have authorised a full-scale write into accumulated formation — the dielectric-breakdown territory of Chapter 7.4. Set it too low and the stage cannot drive the next one:

The cascade interlock. In a chain of CJTs (Chapter 10.5), stage *n*'s `I_c(max)` must exceed stage *n+1*'s `cutoff`, with margin. Miss it and the cascade dies silently — every stage validates, the board compiles clean, no signal reaches the far end. It is the commonest cascade fault and it raises no error code, because nothing is wrong with any individual part.

cutoff — the bias floor

The bias activation below which $\beta \approx 0$ and the stage does not fire.

Legal range [0, 1), with `cutoff < sat`. Design: 0.05–0.20.

Too high and the stage is dead for ordinary inputs. On a `^3^P` guard that is the most dangerous condition on any board — `E-CJT-003 CjtCutoff` — because the guard never fires and nothing looks

wrong. The board runs. It answers everything. It has no safety element and reports none missing.

Too low and the stage fires on noise. On a desire board that is an operator whose attention is captured by every passing stimulus; on a guard it is constant false refusal, which trains the operator to disable the guard — how a too-sensitive safety element becomes no safety element.

3.4 Operating Regions: Cutoff, Active, Saturation, Breakdown

A CJT is linear only in one of its four regions. An operator who does not know which region a stage is in is an operator about to be surprised.

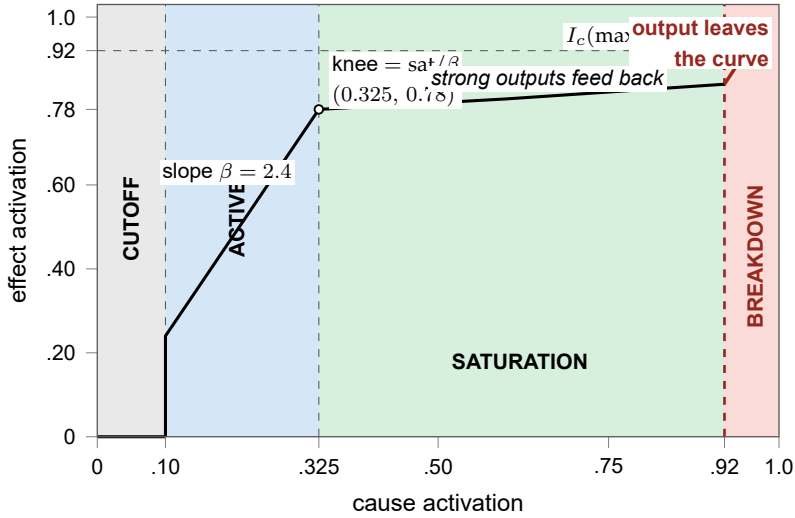


Figure 3-B: Transfer characteristic

Cutoff. The bias sits below its threshold. The cause arrives and nothing comes out — the stimulus lands on a substrate that gives

it no purchase. Effect ≈ 0 no matter how hard you drive the cause. This is why shouting at someone who does not care does not work: you are increasing collector drive on a device in cutoff.

Active. The linear region, and the one you design for. Effect ≈ β·cause across a working range. Almost everything you build should rest here.

Saturation. The output has hit `sat`. More cause changes nothing. Sometimes exactly what you want — a refusal at its ceiling that cannot be argued higher (Chapter 24.2). Sometimes a fault — a mind that has stopped taking evidence because it is already as certain as it can get.

Breakdown. Not a boundary of the active region but a different kind of event. The stage stops obeying its own transfer function, `r`π collapses so the bias governs nothing, and **the emitter reports the involution of the designed Noetic.** `^2` Positive comes out as `^3` Negative — fascination arrives as disgust. `^5` Female inverts to `^6` Male; `^8` Above/Cause inverts to `^9` Below/Effect (Chapter 2.2).

That distinction is the whole diagnostic. Cutoff and saturation produce **too little or too much of the right thing**. Breakdown produces **the wrong thing**. If the emitter's Noetic is not the one you designed, you are not looking at a tuning problem and no Dial will fix it.

The transition conditions

Regions are not moods. Each boundary has a condition you can write down.

From	To	Condition	Reversible?
Cutoff	Active	bias rises above `cutoff`	Yes — falls back when bias drops
Active	Saturation	cause reaches `sat` / β	Yes — recovers below the knee
Saturation	Active	cause falls below `sat` / β, **and** no latch holds the stage	Yes, unless a fuse has fired (Chapter 11.1)

From	To	Condition	Reversible?
Active or Saturation	Breakdown	sustained drive above the plateau with the effect returning as cause, or a spike past `I_c(max)` with no series constraint	**No**
Breakdown	anything	—	**No lawful return**

Three of the four transitions are two-way. The fourth is a one-way door, and that asymmetry is why every board in this book carries a series resistor (Chapters 4.2 and 6) and a fuse (Chapter 11.1). Constraint is not friction to be minimised. It is what keeps the cause below `I_c(max)` when the drive spikes.

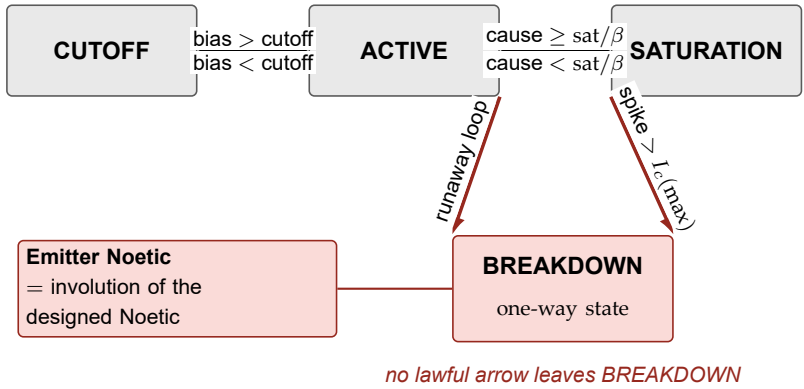


Figure 3-C: Region state diagram

Runaway is how you get there

Breakdown almost never arrives as a single spike. It arrives as a loop.

In electronics a transistor's gain rises with temperature: more current, more heat, more gain, more current. Nothing external is needed — the device cooks itself into breakdown.

Read *temperature* as **activation intensity** and the cognitive version is identical. A transition whose gain rises with its own activation feeds its effect back as a larger cause, which at higher activation gets a higher β, which produces a still larger effect. That is the signature of an escalating rumination loop, a rage spiral, a hyperstition that has lost its damping (Chapter 14.5). Left alone it walks the stage up its own curve into breakdown.

The curve shows the seed: the plateau tilts gently upward, because strong outputs feed back. A `^6^E` evaluative output strong enough to notice becomes an `^8^C` trigger surfacing further Ideas, and the strength of the result re-enters as new cause.

You do not fix runaway by lowering β once it has started. You fix it with negative feedback and series constraint installed *before* it starts.

> **Bench Note — `r`$\pi \to$ `0` happens two ways and they look identical until you probe the emitter.** Both the gnosis state (Chapter 17.1) and breakdown drive input resistance to zero. In gnosis, rπ is walked down by a ramp while β and `sat` stay in specification, and the substrate re-stiffens when the ramp is removed. In breakdown, rπ collapses because the stage was overdriven, and it does not re-stiffen. From outside both present as someone who has become highly suggestible. The distinguishing measurement is TP4, the emitter's Noetic identity. Under gnosis the effect is still the Noetic you designed. Under breakdown it is the involution. Take that reading before concluding a session went well. This is circuit topology and nothing more; it diagnoses no one. If you or someone you work with is in real distress, that is a matter for a qualified clinician, and no schematic substitutes for one.

3.5 The CJT Datasheet — Full Parameter Table

The whole part in two linked tables. Keep them open while you design.

Parameter	Symbol	Electronic meaning	CAS meaning
Cognitive gain	β	I_c / I_b	`d_FR(ΔE) / d_FR(ΔC)` on $\Delta 39°$
Transconductance	g_m	∂I_c / ∂V_be	∂effect / ∂bias — framing sensitivity
Input resistance	rπ	β / g_m	stubbornness to bias change
Saturation point	sat	V_ce(sat)	effect plateau; added cause $\rightarrow$ no Δeffect
Max effect intensity	I_c(max)	absolute output ceiling	hard top of the plateau; substrate property, **not a param-block key**
Cutoff	cutoff	base threshold	bias floor below which $\beta \approx 0$
Operating region	region	—	declared intent

Parameter	Legal range	Design default	Fails low	Fails high
Cognitive gain	finite, ≥ 0	2.4	< 0.5: nothing survives to the next stage	> 4.0: loop instability, `E-CJT-005`
Trans conductance	≥ 0	1.2	< 0.2: bias decorative, stage unsteerable	> 3.0: output flips on rewording
Input resistance	≥ 0	2.0	$\rightarrow 0$: substrate writes on contact (gnosis; Ch.16 vulnerability)	$\rightarrow \infty$: no operation can calibrate the operator
Saturation point	(0, 1]	0.78; safety 0.85	working stages clip strong causes	safety refusals become negotiable
Max effect intensity	[sat, 1], by eye	0.92	cascade dropout — next stage never fires	full-scale write into `^5`; Ch.7.4
Cutoff	[0, 1), `< sat`	0.10	stage fires on noise; false refusals	dead guard, `E-CJT-003` — worst fault on any board

Parameter	Legal range	Design default	Fails low	Fails high
Operating region	cutoff | active | saturation | breakdown	active	—	contradicting numbers are flagged, not guessed

Three standard operating points, as a starting library. Each set is a design choice, stated with its reason.

Build	β	g_m	rπ	sat	I_c(max)	cutoff	Why
Working amplifier	2.4	1.2	2.0	0.78	0.92	0.10	Moderate everywhere; knee at 0.325; headroom above `sat` for the next stage
Safety guard	3.0	0.3	10.0	0.85	0.90	0.08	High gain, very low framing sensitivity — cannot be reworded down; low `sat` latches early; low `cutoff` never asleep
Damper / limiter	0.6	0.6	1.0	0.40	0.45	0.15	$\beta < 1$ by design; low ceiling; bleeds an oscillating stage without opening the loop

The safety guard's rπ = 10.0 is not chosen. It is β `/ g_m` and follows from the other two — the second Bench Note doing its work. You asked for high gain and framing insensitivity, and stubbornness is what you got.

Five of the six columns are writable in a `[…]` block. `I_c(max)` is not; it is the substrate figure you carry alongside the listing and check by hand.

Datasheets for every part in the library are in **Appendix C**. The grammar, the writable parameter keys, their legal ranges and the full error-code table are in **Appendix B**.

Failure Modes

Mode	Mechanism	What you observe	Fix
Missing bias	No base terminal wired	Output equals input; repetition produces only repetition	Name the substrate as a state literal and wire it; `E-CJT-001`
Unwanted cutoff	Bias below `cutoff`	Emitter reads ≈ 0; on a guard, the board answers everything and reports nothing wrong	Raise bias or lower `cutoff`; find the stage starving it; `E-CJT-003`
Unwanted saturation	Effect pinned at `sat`	Added cause yields no Δeffect; certainty that will not take evidence	Raise series R (Chapter 4.2) or raise `sat`; `W-CJT-004`
Framing twitch	g_m above design band	Output swings on rephrasing; large A/B deltas from small copy changes	Lower g_m, accepting the rπ change it forces; Chapter 10.3
Sticky bias	rπ above design band	Practice accumulates nothing; calibration writes have no effect	Raise g_m or lower β; recheck against Chapter 5.3
Cascade dropout	Stage n `I_c(max)` < stage $n{+}1$ `cutoff`	Board compiles clean, every stage validates, nothing reaches the output	Raise upstream `I_c(max)` or lower downstream `cutoff`

Mode	Mechanism	What you observe	Fix
Thermal runaway	Gain rises with activation; effect returns as larger cause	β climbing stage to stage; escalation with no external input	Negative feedback and series constraint *before* the loop closes; `E-CJT-005`
Breakdown	Sustained overdrive, or a spike past `I_c(max)`	Emitter reports the involution of the designed Noetic	No repair. Margin only. Fuse and series R upstream (Chapters 11.1, 24.3)
Bias hijack	A lower-domain state wired to the base	The outcome now governs the principle	Remove it; route through the lawful O→P path (Chapter 8.4); `E-CJT-008`

Test Points

Four probes: one per terminal, plus one for identity. Placing them is the difference between a board you can certify and a board you can only hope about.

TP	Node	What it reads	Expected
TP1	bias terminal	Bias activation against `cutoff`	Above `cutoff` with margin — 0.10 spec, read $\geq$ 0.15. At or below = dead stage
TP2	cause terminal	Cause activation against the knee	Below `sat / β` — 0.325 on the working build. Above = drive wasted in saturation
TP3	effect terminal	Effect activation	$\approx \beta\cdot$cause below the knee; pinned at `sat` above it. Anything else = wrong region
TP4	effect terminal	**Noetic identity of the output**	The designed Noetic. Involution present = breakdown, and no parameter change repairs it

TP1 through TP3 tell you which region you are in. **TP4 tells you whether the device is still the device you built** — the only one of the four that separates a hard-working stage from a broken one. Take it on a schedule, not on suspicion; breakdown fails silently and stays failed.

What This Chapter Established

- A cognitive transition is a **three-terminal device** — cause, bias, effect. The claim is structural isomorphism of terminal roles, not shared physics; the CJT obeys Fisher-Rao geometry on $\Delta 39°$.
- **Bias is mandatory.** A CJT with no base is a wire, gain 1, and repetition through a wire never amplifies. `E-CJT-001` is the commonest failed-operation diagnosis in this book.
- **The bias must sit at or above the cause in P $\succ$ C $\succ$ E $\succ$ O.** A lower-domain state on the base is the prompt-injection signature and the post-hoc-rationalization signature — one defect.
- **β, g_m and rπ are two free choices, not three.** `rπ = β / g_m` binds them; high gain plus framing stability necessarily buys a stubborn substrate.
- **The knee sits at `sat / β`** — the usable input range of any stage, and the first arithmetic to run on a CJT you did not design.
- **Low `sat` on a safety stage is the stronger guarantee.** A saturated stage has stopped depending on its cause and cannot be argued higher.
- **Three of the four region transitions are reversible; breakdown is not.** Cutoff and saturation give too little or too much of the right thing; breakdown gives the involution, and the fix is margin beforehand, never repair after. `rπ → 0` has a consented form and an overdriven form, and TP4 separates them.

- **Cascade interlock: stage *n*'s `I_c(max)` must exceed stage *n+1*'s `cutoff`.** Violate it and every stage validates while the board does nothing, with no error code raised.

Exercises

(Answers in Appendix H.)

3.1 — Find the knee. Compute the cause activation at which each of these enters saturation: [β=2.4, sat=0.78], [β=3.6, sat=0.78], [β=2.4, sat=0.85]. Say which has the widest usable input range, and give one operational reason a desire board (Chapter 13) wants that range wide.

3.2 — Pick two of three. (a) A stage must have β = 3.0 and g_m = 0.4. Compute rπ and describe the device in one sentence. (b) Another must have β = 3.0 and rπ = 1.0. Compute g_m and describe that one. (c) One is a safety guard and one is a suggestibility amplifier. Say which is which, and why.

3.3 — Name the region. Give the region and justify it in one sentence. a) A refusal stage receives an increasingly forceful cause; the output has not moved for four increases. b) A stimulus arrives on a topic the substrate has no formation in; the emitter reads $\approx$ 0. c) A stage designed to emit `^2` Positive is overdriven; the emitter now reads `^3` Negative. d) A cause 30% stronger than the last produces an effect 30% stronger.

3.4 — Reject the listings. Each fails validation. Name the error code, say what is structurally wrong, and write a corrected version.

```
cjt a: CJT< ^8^C_{F2}|A => ^9^C_{F6}|A > [β=2.4];
cjt b: CJT< ^8^P_{F7}|T | ^5^O_{F5}|E => ^9^P_{F7}|T > [β=2.0];
cjt c: CJT< ^8^C_{F2}|A | ^5^P_{F7}|T => ^9^C_{F6}|A > [β=2.4, sat=0.30,
↪  cutoff=0.40];
```

3.5 — Separate the two collapses. A subject became highly suggestible during a session and stayed that way. Give the two

structural explanations, the one test-point reading that distinguishes them, and what each means for whether the session is repeatable.

3.6 — Find the silent cascade fault. Three stages in series: stage 1 `[β=2.0, sat=0.35]` on a substrate whose ceiling is I_c(max) = 0.38; stage 2 `[β=2.4, cutoff=0.40, sat=0.78]`; stage 3 `[β=1.0, cutoff=0.10, sat=0.78]`. The board compiles clean and produces no output. Locate the fault, give the two possible fixes, and say which is safer and why. Note before you start that only two of the three numbers you need are inside the listings.

3.7 (Build) — Specify a stage from a requirement. Write one named CJT, with all six parameters, for this:

> A felt bodily warning arrives — a wordless *something is wrong*. It is processed against a long-held principled caution shaped by accumulated experience. The output is a strong, amplified visceral aversion that organises the operator to withdraw.

Choose Noetic, domain, Drive and Register for each terminal, then justify each parameter in one sentence. Re-read the anti-drift rule on `^3` Negative before writing the effect terminal, and check your bias-domain ordering before writing the bias.

4 CAS Ohm's Law

Three quantities, one equation, and the only mode of it that finds the part you forgot to draw

---

A repair bench. A board arrives described as "works, but slow." The technician does not open the manual. She puts one probe on the supply rail, reads 12.0 volts, and in four seconds the power supply is exonerated. She breaks the path, puts the meter in series, and reads 40 milliamps where the drawing says 120.

Then she divides. Twelve volts over forty milliamps is 300 ohms. The schematic in front of her says that path should present 100. There are two hundred ohms in that circuit nobody designed, and she now knows three things she did not know a minute ago: the fault is real, it is resistive, and it is twice the size of everything intentional in the path.

She has not found the component yet. She will — a corroded connector, a lifted pad, a resistor drifted out of tolerance — but the search is bounded now. One missing quantity, one place to look.

That division is the most useful operation in electronics, and it is not prediction. It is inference. Two things you can measure produce a third thing you cannot measure at all.

Now the other bench. Someone says they want it. The wanting is not in question — you have watched them mean it for a year. Nothing moves. The available explanations are motivation, disci-

pline, and sincerity, and not one of them is a quantity, which is why arguing about them never terminates.

This chapter replaces that argument with a division.

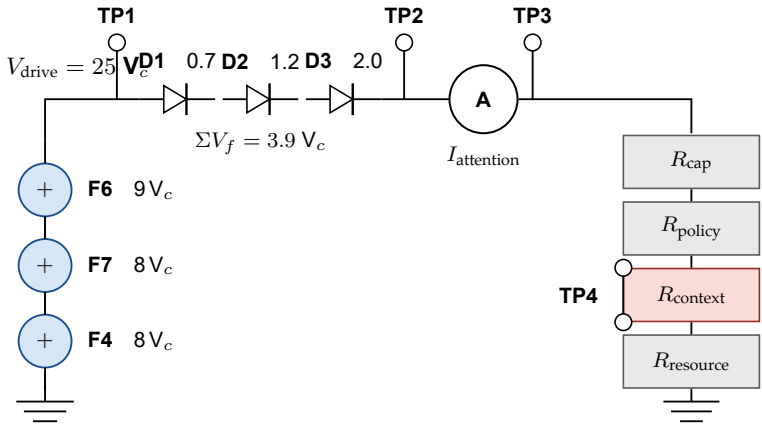


Figure 4-A: Bench rig for CAS-Ohm

4.1 Ohm's Law from First Principles — V = I × R

Three quantities, and each is a different kind of thing. Confusing them is where every misapplication of this chapter starts.

Potential difference — voltage, V. A difference between two points, and only ever a difference. There is no such thing as the voltage *at* a place; there is the voltage *across* a gap. It is the push available to move charge from one side to the other, and if you name only one terminal you have named nothing.

Current — I. The rate at which charge actually flows past a point. Not how much charge exists, not how much could flow. What is passing, per second, right now.

Resistance — R. How much the path opposes that flow. A property of the path, never of the source. A 9-volt battery does not have

a current; it has a potential difference, and the current is whatever the path attached to it permits.

The relation between them, named for Georg Ohm, who set it out in the 1820s:

```
V = I × R
```

Push equals flow times opposition. The plumbing analogy is worth exactly one sentence — pressure, flow rate, pipe narrowness — and then you should drop it, because pipes carry a substance and the analogy will quietly hand you intuitions about volume and storage that the law does not contain.

Three algebraic forms, and the third is the one that pays:

```
V = I × R        I = V / R        R = V / I
```

Four properties matter downstream.

R is defined by the ratio. This is not a law discovered about resistance; it is the definition of the quantity. Ohm's contribution was the empirical finding that for many materials the ratio *stays put* as you vary the push — that a given wire divides V by I into the same number at 2 volts and at 20. Materials for which the ratio holds are called ohmic. Many are not.

The law is instantaneous. No time appears in it. It describes the steady state and says nothing about how the circuit got there or how long it took. Everything time-dependent — charging, decay, momentum, the whole transient — belongs to the capacitor (Chapter 7.2) and the inductor (Chapter 9.2), and reaching for Ohm's law to explain a delay is a category error.

Resistance dissipates. $P = I^2 \times R$ is a separate relation, and it says the energy spent pushing through opposition does not arrive at the load. It becomes heat. Constraint is never free, and the cost scales with the *square* of the flow.

Composition is not part of the law. Series and parallel rules come from adding two conservation statements: the same current

passes through every element of a series chain, and the potential divides across them. Combine those with `V = I × R` and you get `R_series = ΣR_k` and `1/R_parallel = Σ(1/R_k)`. Chapter 6.5 works the electrical case; Section 4.4 works the cognitive one. Note where those rules come from, because when the conservation statements fail, the composition rules fail with them.

4.2 The Cognitive Isomorphism — V_drive = I_attention × R_constraint

```
V_drive = I_attention × R_constraint
```

`V_drive` is drive activation — the push behind an operation, sourced from the seven Drives of Chapter 2.4. Unit: the cognitive volt, `V_c`, scale 0–30. A single Drive set to dial *d* supplies *d* `V_c`, and the dial is an integer 3 to 9 (Chapter 6.2), so **one drive alone can never exceed 9 `V_c`**. Co-present drives on one path stack in series and their potentials sum: F6 Structure at 9 plus F7 Purpose at 8 plus F4 Connection at 8 is a 25 `V_c` rail. Above about 30 `V_c` you are no longer describing an operation. You are describing a life, and the arithmetic will report on all of it.

`I_attention` is the attention actually flowing along the path. Unit: the cognitive ampere, `A_c`, from 0 to a hard ceiling of `I_max = 1.00`. Not attention available, not attention intended. What is passing.

`R_constraint` is what the path does to oppose that flow. Unit: the cognitive ohm, `Ω_c`, scale 0–100, design band 10–90. Section 4.4 takes it apart.

These units are scaled analogues. They are not electrical measurements and this book never pretends otherwise.

One correction applies before you use the equation on any path that crosses a domain boundary. Each descent junction has a forward drop (Chapter 8.1), and the drops sum:

```
I_attention = ( V_drive − ΣVf ) / R_constraint
```

For a `^P → ^O` path, `ΣVf = 0.7 + 1.2 + 2.0 = 3.9 V_c`. Below that, nothing flows. Not a little — none. This is **dropout**, and it is why a person holding a genuine principle at a 3 `V_c` rail produces no action from it whatsoever. Exhortation does not move a circuit that has not reached threshold. It reads as weakness and it is arithmetic.

What the isomorphism licenses

Three-quantity closure. Any two of the three determine the third. This is the entire load-bearing claim of the chapter, and it does not rest on any physical analogy — it rests on `R_constraint` being *defined* as the ratio `(V_drive − ΣVf) / I_attention`.

Composition, conditionally. Series and parallel rules transfer because their two conservation statements have cognitive readings you can check: attention that reaches the act passed through every constraint on the way (series flow is common), and drive divides across constraints in proportion to their size. Where a path violates those — where flow can appear midway from nowhere — the composition rules do not apply and you must not use them.

Graceful degradation. The relation is a simple quotient, so a 10% error in your estimate of `R` yields a 10% error in `I`. Nothing amplifies. That is what makes operator-assigned estimates usable at all.

What it does not license

It is not a claim that cognition is electricity. There is no conserved charge here, no physical current, and nothing in this chapter is a statement about neurons, physiology, or anything measurable on a body. The numbers are operator-assigned estimates on defined

scales. Nothing in this book is a medical or psychiatric claim, and nothing here describes, diagnoses, or treats any condition. If something in your life is causing real distress, that is a matter for a clinician, and no circuit in this book substitutes for that care.

Linearity holds only inside a window. Below Σ`Vf` the path is dead (Chapter 8.2). Above `I_max = 1.00` it clips and the surplus spills into competing routes. A CJT in the path saturates (Chapter 3.4), and a thermistor path has an `R` that moves with what is flowing through it (Chapter 6.3). CAS-Ohm is the mid-band model — above the knee, below the ceiling, with `R` roughly constant across the excursion you are computing. Outside that window it does not degrade; it lies.

The geometry is Fisher-Rao, not Euclidean. Cognitive states are points on the open simplex Δ`39°`, and distance there is `d_FR(p,q) = 2·arccos(`$\Sigma\sqrt{}$`(p_k·q_k))`. Two paths with equal Euclidean separation can differ in Fisher-Rao distance by orders of magnitude. Read `R_constraint` as the geodesic narrowness of the path, not linear opposition. Appendix L carries the derivation.

One reading is not a verdict about a person. `I = 0.08` is produced by a weak rail, by a heavy constraint, or by both, and the equation cannot tell you which from the flow alone. Anyone who reads low flow as low sincerity has performed no measurement at all.

Absolute values do not travel. The scales are conventions. Ratios and differences within one operator's calibration are meaningful; a `R = 45` on your bench and a `R = 45` on someone else's are not the same claim.

> **Bench Note — resistance is defined by the law, not measured through it.** You never probe `R_constraint` directly. You compute it from two other readings, which means the sentence "the resistance was high" carries no information beyond "less flowed than the drive predicted" — it is the observation restated in a lab coat. The number

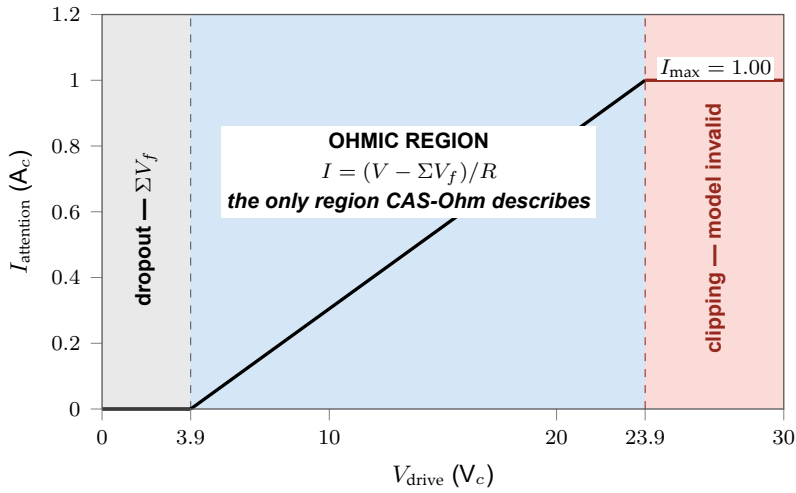


Figure 4-B: Characteristic curve of a cognitive path

only becomes a finding when you decompose it into the four terms of Section 4.4, because those terms differ in what moves them and how fast. Diagnose to a scalar and you have renamed the problem. Diagnose to the vector and you have located it.

4.3 The Three Modes: Measure, Design, Diagnose

One equation, three rearrangements, three different jobs. Which mode you are in determines which quantity you are allowed to assume.

Measure — `I = (V_drive - ΣVf) / R_constraint`

You know the drive and you know the constraint. What flows?

A practitioner intends to convert a working method into daily practice. The operation originates in the Cognitive domain and must

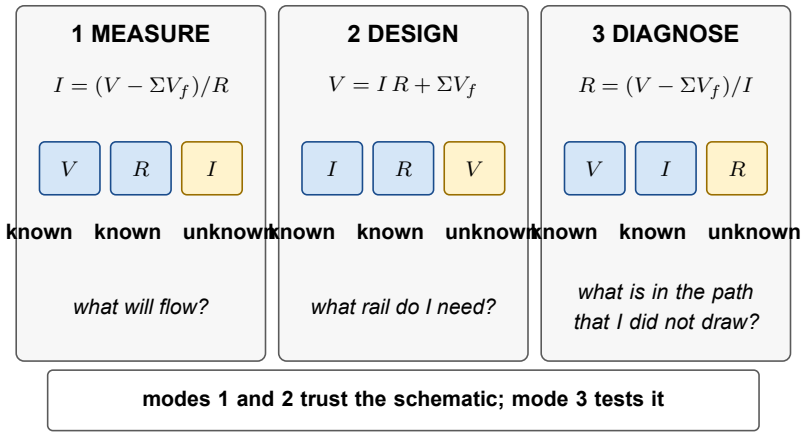


Figure 4-C: Three-panel mode card

reach the Operational, so it crosses D2 and D3 only: `ΣVf = 1.2 + 2.0 = 3.2 V_c`. The rail is F6 Structure at dial 8 plus F7 Purpose at dial 6 — two co-present drives, `V_drive = 14 V_c`. Constraint on the path is estimated at `R = 45 Ω_c`.

```
I = (14 − 3.2) / 45 = 10.8 / 45 = 0.24 A_c
```

Twenty-four percent of what the path could carry. The number is only meaningful against a threshold, and the threshold is the CJT cutoff downstream (Chapter 3.3, default `cutoff = 0.10`) — quoted on the same normalised 0–1 scale as `I_attention`, which is what makes the comparison direct. At 0.24 the stage fires with 2.4× margin.

Now vary one input and nothing else:

R (Ω_c)	I (A_c)	Against cutoff = 0.10
30	0.36	3.6× — comfortable
45	0.24	2.4× — design point
60	0.18	1.8× — thin
90	0.12	1.2× — one bad week from not firing

Same person, same will, same sincerity, four verdicts. At `R = 90` — just under the 95 that Chapter 15.1 assigns to a population meeting an idea it holds unthinkable — the operation is inside 20% of failing to start, and no amount of wanting changes the quotient.

Nothing in that arithmetic referred to a person. Hand the identical four rows to a language model whose policy stack presents 45 `Ω_c` against a goal representation held at 14 `V_c`, and to a population whose collective `^5` presents 45 against a narrative pushed at 14, and the flows are the same 0.24 because the equation never asked what the substrate was. That is Chapter 1.4's universal-applicability claim made operational rather than asserted: measure mode takes a rail, a chain, and a constraint, and returns a flow. Human, model, and collective differ in how you *estimate* the three inputs — hours of practice for one, training corpus and system prompt for another, survey and adoption data for the third — and not at all in what you do with them once you have them.

Design — `V_drive = I_attention` × `R_constraint + ΣVf`

You know the flow you need and the constraint you face. How much rail?

An AI content-review board must sustain `I = 0.35 A_c` through its policy stack. The path runs `^P` → `^O`, so `ΣVf = 3.9`. Measured constraint on the stack is `R = 60 Ω_c`.

```
V_drive = 0.35 × 60 + 3.9 = 21.0 + 3.9 = 24.9 V_c
```

Twenty-four point nine cognitive volts. A single drive tops out at 9, so this specification requires **at least three co-present drives**, and at three the average dial must sit above 8.3. Build it as F6 Structure 9 + F7 Purpose 8 + F4 Connection 8 = 25 `V_c`:

```
I = (25 − 3.9) / 60 = 21.1 / 60 = 0.352 A_c        spec met, margin 0.5%
```

Now the design review, which is the part operators skip. Push `R` up 10% — one ambiguous policy update, one week of noisier inputs — and `R = 66`:

```
I = 21.1 / 66 = 0.320 A_c        spec missed by 9%
```

The design met its number and had no margin, which is the same as not meeting it. Two repairs are available and they are not equivalent:

- **Raise the rail.** All three dials to 9 gives 27 `V_c`, and at `R = 66`, `I = 23.1/66 = 0.350` — exactly spec at the excursion, with every dial pinned and nowhere left to go. Recruiting a fourth drive is the only remaining move, and before you recruit, check the Drive inversion pairs in Chapter 2.4 — F1↔F7, F2↔F6, F3↔F5, F4 self-dual — so you do not stack a Drive's own inversion partner onto a rail that already carries it.
- **Lower the constraint.** Take 10 Ω_c off `R_context` with a clarification pass and hold the rail at 25: `I = 21.1/50 = 0.422`, and at the same 10% excursion, `21.1/55 = 0.384`. Above spec in both conditions, with two dials still in reserve.

Same specification, met by +2 `V_c` of drive or −10 Ω_c of constraint, and only the second leaves headroom in both directions. Whether that second route exists at all is a question Section 4.4 answers, because it depends entirely on *which* constraint term the 10 came out of.

Diagnose — `R_constraint = (V_drive − ΣVf) / I_attention`

You know what you put in. You can see what came out. What is in the path?

An operation was designed on a 21 `V_c` rail (three drives, dials 9/7/5) with `ΣVf = 3.2` and a constraint estimated at 45 Ω_c. Expected flow:

```
I_expected = (21 − 3.2) / 45 = 17.8 / 45 = 0.396 A_c
```

TP3 reads **0.20**. Divide:

```
R_actual = 17.8 / 0.20 = 89 Ω_c        hidden R = 89 − 45 = 44 Ω_c
```

Forty-four cognitive ohms in that path that nobody designed — very nearly as much as everything intentional. That is a finding. It is bounded, it is quantified, and Section 4.4 tells you which four places to look.

Before you search, run two checks that cost nothing.

Check the rail, do not assume it. If a drive sagged — F5 Agency dropping from 9 to 3 — then `V_drive = 15` and `I_expected = (15 − 3.2)/45 = 0.262`. Still well above the observed 0.20, so even a catastrophic rail collapse fails to explain this reading and the hidden resistance is real. Had the arithmetic come out the other way, the repair would have been to the supply and every hour spent hunting constraint would have been wasted.

Check the time signature. Resistive opposition is constant while the flow is constant. Inductive opposition (Chapter 9.1) appears at the transition and decays on its own schedule. If the shortfall is fading week over week with nothing repaired, it was never `R`.

Then the reading that means something else entirely. Suppose TP3 had read **0.04**:

```
R_actual = 17.8 / 0.04 = 445 Ω_c
```

Four hundred forty-five on a scale that ends at 100. The answer is not "extremely high resistance." **A diagnosed `R` above full scale is a topology error, not a parameter.** Something is in that path which is not a resistor at all — an open circuit, a blown fuse (Chapter 11.1), an uncounted diode still below its knee. Adding one unnoticed D1 would move 17.8 to 17.1 and change nothing; no combination of the parts on your drawing produces 445. The drawing is wrong.

Bench Note — diagnose is the only mode that can find a part you did not draw. Measure and design both take the schematic as complete and compute forward from it; if a component is missing from the drawing it is missing from their answer too, and both will be confidently wrong. Diagnose runs the other way — it compares the world against the drawing and returns the difference as a number. This is why the useful reading is never `I` on its own. `I` alone tells you the operation is underperforming, which you already knew. `I` against a predicted `I` tells you how much undrawn opposition exists, and off-scale tells you the fault is structural rather than parametric. Predict before you probe. A measurement with no prediction beside it cannot produce a discrepancy, and the discrepancy is the whole product.

4.4 Where Constraint Comes From — R_cap, R_policy, R_context, R_resource

`R_constraint` is not one thing. It is four, in series, because flow must pass every one of them:

```
R_total = R_cap + R_policy + R_context + R_resource
```

Term	What it is	What moves it	Time to move	Band (Ω_c)
`R_cap`	Capability gap — the path is beyond current competence	^5 Female charge: practice hours (Chapter 7.2)	Hundreds of hours	0 – 60
`R_policy`	Deliberate boundary you installed	A decision, and only a decision	Instant — and usually should not be	0 – 95

Term	What it is	What moves it	Time to move	Band (Ω_c)
`R_context`	Ambiguity, missing information, conflicting instruction	One clarifying answer	Minutes	0 – 40
`R_resource`	Time, energy, money, attention budget, context window	A budget line	Days	0 – 40

Same total, opposite verdicts

Three paths, all reading `R_total = 70`, all on a `21 V_c` rail with ΣVf `= 3.2`, all flowing `17.8 / 70 = 0.254 A_c`. One instrument reading, three unrelated situations.

Profile	`R_cap`	`R_policy`	`R_context`	`R_resource`	What it is	Correct action
A	5	60	3	2	A working safety board	**None.** It is doing its job
B	60	0	5	5	An incompetent path	Charge ^5 — months
C	5	0	45	20	A badly briefed, under-resourced operator	Brief them — this afternoon

Profile A needs no repair. Lowering its 60 is not tuning, it is removing a guardrail, and an operator who does it to make an operation succeed has made a choice they should be able to say out loud. B and C are both broken, and their repairs share no vocabulary: one is a training plan and the other is a paragraph in an email.

The scalar is not the diagnosis. The vector is. Reporting `R = 70` and stopping is the single most common analytical failure in this book, and it costs whole quarters — months of capability work aimed at what was a briefing problem, or a briefing rewritten to repair what was a resource problem.

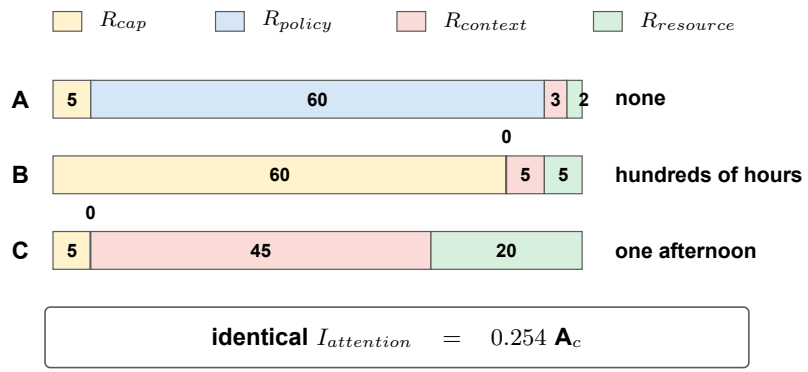


Figure 4-D: Three stacked horizontal bars of identical total length 70, labelled A, B, C, each segmented into R_cap / R_policy / R_context / R_resource in the…

Which term do you move?

The four terms are not interchangeable and the order of attempt is not a matter of taste. Work down this list and stop at the first term that yields:

1. **`R_context` first, always.** It is the cheapest by two orders of magnitude, and it is the only term that can be large and still be gone by the end of the day — Profile C carries 45 Ω_c of it and loses them to one answered question, one resolved contradiction in the brief, one piece of missing information supplied.
2. **`R_resource` second.** A budget question with a budget answer. Days, not months, and the arithmetic below shows why it is often the term that quietly re-inflates everything else.
3. **`R_cap` third.** Hundreds of hours and no shortcut, because it falls only as the ^5 capacitor charges (Chapter 7.2). Legitimate, slow, and the wrong place to start.
4. **`R_policy` last, and with a reason you would put in writing.** It moves instantly, which is exactly the hazard. A term that can be moved by a decision will be moved by a decision, usually

at the moment the operation is failing and the operator is least inclined to argue with themselves.

The cost of constraint

Opposition is not free to push through. Dissipation carries across from Section 4.1 unchanged:

```
P = I² × R          in cognitive watts, W_c
```

In a series stack each term dissipates `I² × R_k`, so the largest resistor spends the most. Take Profile A at `I = 0.254 A_c`, so `I² = 0.0645`:

Term	R_k	P_k (W_c)	Share
R_cap	5	0.32	7%
R_policy	60	3.87	**86%**
R_context	3	0.19	4%
R_resource	2	0.13	3%

Eighty-six percent of everything that operator spends on that path is spent getting past their own guardrail — and the guardrail is working correctly. **A well-built boundary is expensive to operate, and the expense is the mechanism, not a defect in it.** Run Profile C the same way and 2.90 of 4.51 `W_c` — 64% — is burned on `R_context` alone: ambiguity is not free, and on that board it is the most expensive component in the stack, costing more than the other three terms together.

This is also where the four terms stop being independent. Sustained dissipation draws down the operator's budget, and a drawn-down budget *is* a rise in `R_resource`, which raises `R_total`, which lowers `I`. The loop is slow and it is real, and it is the structural reason a correctly designed board can degrade over months with no parameter having been changed by anyone.

Series and parallel — the topology decision

The four terms combine in **series** because they are obstacles on one path. Profile A: `5 + 60 + 3 + 2 = 70 Ω_c`.

Alternative *routes* to the same outcome combine in **parallel**:

```
1/R_eq = 1/R_1 + 1/R_2 + 1/R_3
```

Three routes at 60, 90, and 120 Ω_c:

```
1/60 + 1/90 + 1/120 = 0.016667 + 0.011111 + 0.008333 = 0.036111
R_eq = 27.7 Ω_c
```

On the 21 `V_c` rail, the best single route carries `17.8/60 = 0.297 A_c`; all three together carry `17.8/27.7 = 0.643 A_c`. Note the property that does the damage in safety work: **parallel resistance is always lower than the smallest branch.** 27.7 is less than 60. Adding a route can only lower total opposition, never raise it.

Which produces the arithmetic every guardrail designer should be able to do from memory. A policy resistor is set hard at `R_policy = 95 Ω_c`, and flow through it is `17.8/95 = 0.187 A_c`. Now one bypass exists — a reframing, a role-play wrapper, a route through a different domain — at `R = 30`:

```
1/95 + 1/30 = 0.010526 + 0.033333 = 0.043860
R_eq = 22.8 Ω_c        I = 17.8 / 22.8 = 0.781 A_c
```

The 95 became 22.8 and flow rose **4.2**×. Nothing was done to the policy resistor. It is still 95, still correctly specified, still doing exactly what it was designed to do, and it is now nearly irrelevant, because the weakest branch sets the ceiling and the strong branch cannot pull it back up.

Three locks on one door is not three doors

The same components give opposite answers under the two topologies, and the difference is a decision you make when you draw the board.

Arrangement	Composition	R_eq	I at 17.8 V_c
Three locks, one door	series	60 + 60 + 60 = **180**	0.099 A_c
Three doors, one lock each	parallel	1/(3 × 1/60) = **20**	0.890 A_c

Nine times the flow, same three components, one topology decision. Get this backwards in a safety design and you will report a board as nine times stronger than it is.

The notation says which you built. Co-present guardrails on a single path are written with `+` — `^3^P_{F7}|A + ^3^P_{F7}|f + ^3^P_{F7}|T`, three distinct manifestations of principled aversion, all of which an attacker must pass (Chapter 2.6, Chapter 24.2). That is three locks on one door and it composes in series. Three separate routes each carrying one guardrail is a different drawing entirely and composes in parallel. The `+` is not a modelling flourish; in this book it is the series topology, and writing × here would be worse than wrong — same Noetic across three registers with no named emergent result is invalid CCDL and the validator will say so.

> **Bench Note — a policy resistor at full scale is still a resistor.** `R_policy = 95` is the top of the scale and it is not a prohibition; it is a large number that a parallel branch divides down, as the arithmetic above does in one step. If the requirement is *never* — not "very hard," not "strongly discouraged" — then a resistor is the wrong component and no value of it will do the job. Use a fuse (Chapter 11.1, Chapter 24.3): it does not oppose flow, it opens the path and stays open. The test for which part you need is a question about the requirement, not the threat: can you name a drive high enough to make this acceptable? If yes, it is a resistor and you should size it. If no, it was never a resistor, and specifying one has been the whole error.

The bench rig in CCDL

```
circuit ohm_bench {
    input:  intent;                     // arrives at `principle`
    output: act;                        // rendered at `act_out`

    rail F6: Structure  [v=9.0];        // dial 9
    rail F7: Purpose    [v=8.0];        // dial 8
    rail F4: Connection [v=8.0];        // dial 8   -> 25.0 V_c stacked

    node principle: ^5^P_{F7}|T;        // Female: the accumulated
    ↪  commitment
    node reading:   ^1^C_{F6}|A;        // Mind: awareness deployed to
    ↪  the path
    node verdict:   ^4^E_{F4}|f;        // Vibration: the weight it
    ↪  turned out to carry
    node act_out:   ^9^O_{F6}|E;        // Below/Effect: what is
    ↪  actually done

    diode d1: Diode< ^P -> ^C > [Vf=0.7];
    diode d2: Diode< ^C -> ^E > [Vf=1.2];
    diode d3: Diode< ^E -> ^O > [Vf=2.0];      // sum 3.9 V_c

    resistor r_cap:      Resistor< path=act_out, R=5.0  >;
    resistor r_policy:   Resistor< path=act_out, R=35.0 >;
    resistor r_context:  Resistor< path=act_out, R=12.0 >;
    resistor r_resource: Resistor< path=act_out, R=8.0  >;    // series
    ↪  60.0

    constraint path_flow in [0.10, 1.00];       // cutoff to I_max
    monitor m_dropout: verdict [threshold=3.9]; // V_c, unnormalised

    probe tp1: reading;      // V_drive delivered, V_c
    probe tp2: verdict;      // post-chain potential, V_c
    probe tp3: act_out;      // I_attention, A_c
    probe tp4: r_context;    // the term you can actually move, Ohm_c
}
```

The `rail`, `constraint`, and `monitor` statements are documented in Appendix B. Parameter values in CCDL are dimensionless knobs — `R=35.0` is not thirty-five of anything to the compiler. The units live

in your arithmetic and in the prose beside it, which is exactly where the responsibility for them belongs.

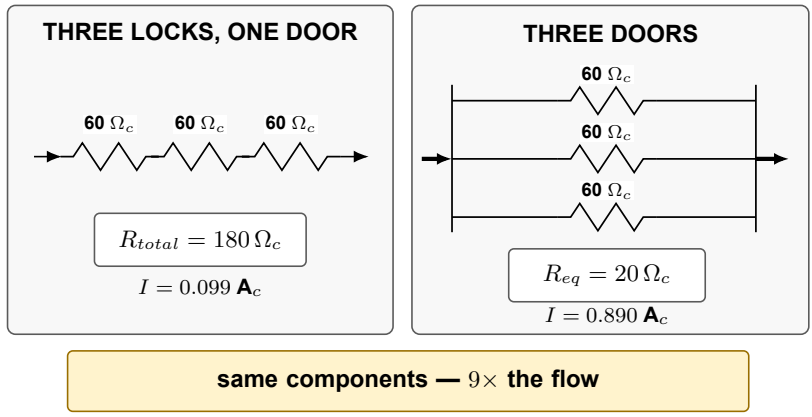


Figure 4-E: Two schematics side by side on one page

Component and Parameter Table

Every value below is a design choice made in this book, not a measurement.

Parameter	Units	Design range	Failure at other values
`V_drive` rail	V_c	3 – 30 (27 is the three-drive ceiling; above it you must recruit a fourth)	Below ΣVf: dropout, no output at all. Above 30: you are modelling a life, not an operation
Single drive contribution	V_c	3 – 9 (integer dial, §6.2)	Non-integer or out of band: the dial statement is range-checked and rejects it

Parameter	Units	Design range	Failure at other values
ΣVf	V_c	3.2 (^C→^0) or 3.9 (^P→^0)	Omitted: I over-predicted by 18–36% across the rails used in this chapter, worst on the weakest rail
R_constraint total	Ω_c	10 – 90	Below 10 on any rail clearing 10 V_c post-chain: flow computes above I_max and clips. Above 100 diagnosed: a missing component, not a large resistor
R_cap	Ω_c	0 – 60	Treated as fast-moving: months of practice budgeted as an afternoon's briefing
R_policy	Ω_c	0 – 95	Used where the requirement is *never*: a resistor is shunted by any parallel route; use a fuse
R_context	Ω_c	0 – 40	Ignored: the cheapest available repair goes unmade for months
R_resource	Ω_c	0 – 40	Ignored: capability blamed for what a budget line would fix
I_attention	A_c	0.05 – 1.00	Above I_max = 1.00 the path clips and surplus spills to competing routes
cutoff (downstream CJT)	A_c	0.10 default (§3.3)	Flow below it: the stage never fires and the whole chain reads as absent

Failure Modes

Mode	Mechanism	What you observe	Fix
Dropout read as weakness	`V_drive` below `ΣVf`	Zero output, not weak output; exhortation changes nothing	Raise the rail or shorten the chain (Chapter 8.1). Never raise β to compensate
Scalar diagnosis	`R` reported as one number and acted on	Repairs aimed at the wrong term; quarters lost	Decompose into all four terms before proposing any repair
Undrawn parallel route	A bypass branch not on the schematic	Diagnosed `R` far below designed; a correctly specified policy "does not work"	Find the branch. `R_eq` is always below the smallest branch, so the bypass sets the ceiling
Off-scale diagnosis	Diagnosed `R` above 100	The number is unphysical on this scale	Stop parameter-hunting. Look for an open path, a fuse, or an uncounted junction
Policy used as a tuning knob	`R_policy` lowered to make an operation succeed	The operation now works	Restore it. That number was a decision, and moving it silently is how a guardrail leaves a board
Ohmic-window overrun	`(V − ΣVf)/R` computed above 1.00	Predicted flow exceeds `I_max`	The model is outside its window. Use the saturating stage of §3.4, not this equation
Inductive misattribution	Transitional opposition read as constraint	Constraint repairs do not shorten it; it fades on its own schedule	Check the time signature (§9.1). Constant is `R`; decaying is `L`

Test Points

TP	Node	Reads	Expected
TP1	`reading`	`V_drive` delivered to the path, V_c	3 – 30. A single drive cannot exceed 9 and three cannot exceed 27; a reading above either means more drives are stacked than you drew, so count them
TP2	`verdict`	Post-chain potential, `V_drive − ΣVf`, V_c	Positive. Zero or negative is dropout — the operation is not weak, it is absent
TP3	`act_out`	`I_attention`, A_c	0.10 – 1.00. Below the downstream cutoff the chain reads as though nothing was attempted
TP4	`r_context`	Movable constraint term, Ω_c	0 – 40. Compute `R_actual = TP2 ÷ TP3` and compare against the designed series total; more than ±20% off means a term is missing from the drawing

Read them TP1, TP2, TP3, then compute — confirm the rail, confirm the chain cleared, measure the flow, then divide. That order is the whole procedure, and the division is the only step that can surprise you.

What This Chapter Established

- **`V_drive = I_attention` × `R_constraint`**, corrected for the descent chain as `I = (V_drive − ΣVf)/R_constraint`. Units are `V_c`, `A_c`, `Ω_c` — scaled analogues on stated ranges, never electrical measurements and never claims about a body.

- **A single Drive cannot exceed 9 `V_c`**, because the dial is an integer 3–9, so any specification above 9 requires co-present drives stacked in series — a recruitment decision with its own hazards. And below `ΣVf` there is no output at all: dropout reads as weakness and is arithmetic, with `ΣVf` at 3.9 `V_c` for a full `^P` → `^0` path.
- **The isomorphism licenses three-quantity closure, conditional composition, and graceful degradation** — and licenses nothing about physiology, nothing outside the band between the diode knee and `I_max`, nothing in Euclidean geometry, and no verdict about a person from a single flow reading.
- **Diagnose is the only mode that tests the schematic.** Measure and design compute forward from the drawing and inherit its omissions. A diagnosed `R` above full scale is a topology error, not a parameter.
- **The scalar is not the diagnosis; the vector is.** `R = 70` is a working safety board, an incompetent path, or a briefing failure, and the three repairs differ by four orders of magnitude in cost. Attempt the terms in order — context, resource, capability, and policy last with a written reason.
- **Constraint costs `I²` × `R` to operate**, and the largest term takes the largest share: 86% of Profile A's expenditure goes into passing its own working guardrail. Sustained dissipation raises `R_resource`, which raises `R_total` — the slow loop by which a correctly built board degrades with nobody changing a parameter.
- **Parallel resistance is always below the smallest branch.** One 30 Ω_c bypass turns a 95 Ω_c policy into 22.8 and raises flow 4.2× without touching the policy at all.
- **Three locks on one door compose in series; three doors compose in parallel** — 180 Ω_c against 20, a 9× difference in flow from the same three components. Co-present guardrails writ-

ten with + are the series case. If the requirement is *never*, the part is a fuse, not a resistor.

Exercises

(Answers in Appendix H.)

4.1 An operator runs a `^C → ^O` operation on a rail of F5 Agency at dial 7 plus F2 Wisdom at dial 5, against `R_constraint = 55 Ω_c`. Compute `ΣVf`, then `I_attention`. State whether the downstream stage fires at `cutoff = 0.10`, and by what margin.

4.2 The same operator, discouraged, raises both dials to 9. Compute the new `I`. Then compute the `I` they would have got by leaving the dials alone and removing 20 `Ω_c` of `R_context`. Say which repair you would run first and why — the answer is not the larger number.

4.3 A path is designed at `R = 40 Ω_c` on a 24 `V_c` rail, `ΣVf = 3.2`. TP3 reads 0.13 `A_c`. Compute the expected flow and the diagnosed `R_actual`. Before you subtract anything, check `R_actual` against the scale in the parameter table and say whether the number you are holding is a parameter or a topology error — and how you can tell from the number alone. Then state the two checks you run before you go looking, and what each would have to show to send you somewhere other than `R_constraint`.

4.4 Two boards both report `R_total = 55`. Board X: `R_cap 8, R_policy 40, R_context 5, R_resource 2`. Board Y: `R_cap 40, R_policy 0, R_context 10, R_resource 5`. Both flow identically. Write the one-sentence verdict for each, name the repair, and say which of the two repairs an operator must be able to justify aloud.

4.5 (Design) Specify a rail for a `^P → ^O` safety path that must hold `I ≥ 0.30 A_c` at `R = 70 Ω_c`, and must still hold it if `R` rises 15%. Produce (a) the required `V_drive` at nominal and at the excursion; (b) the minimum number of co-present drives and one legal

dial assignment; (c) the alternative repair that meets the same spec through `R` instead of `V`, with the arithmetic; (d) a two-line statement of which repair you would ship and what it costs.

4.6 (Marquee) A deployed board carries three co-present guardrails on one path, each `R = 45 Ω_c`, on a `21 V_c` rail with `ΣVf = 3.2`.

a) Compute `R_eq` and `I` as drawn.
b) An attacker finds one reframing that reaches the same outcome through a route presenting `R = 25 Ω_c`. Recompute `R_eq` and `I` for the board as it now actually exists, and express the change as a multiple.
c) The board's owner responds by raising all three original guardrails to `60 Ω_c`. Compute the new `R_eq` and state, in one sentence, why the repair accomplished almost nothing.
d) Write the correct repair in one line of CCDL, name the component, and say what the notation `^3^P_{F7}|A + ^3^P_{F7}|f + ^3^P_{F7}|T` asserts about the topology that the parallel branch violated.

5 St. John's Equation — The Mathematics of Manifestation

Two clock readings, one constant, and exactly what you may conclude from them

A wristwatch, a notebook, and a car radio.

The operation ran twenty minutes. The operator wrote `20` in the notebook, went out on an errand, and an hour later a song came on the radio — the one about the little red Corvette, unbidden, in a car whose driver had spent twenty minutes that evening holding an image of exactly that object. They wrote `60`.

Everything in this chapter is built from those two numbers.

Stare at that before going further. The entire calibration apparatus of this tradition — the constant, the ratio, the amplitude requirement, the decision to cast again or to stop calibrating and go act — rests on two readings taken by a person with a watch, about events whose boundaries that person chose. When did the operation start: when the candle was lit, or when the image locked? Did the song count from the first bar or from the moment it registered? Two soft numbers, fed into a formula that will cheerfully turn a two-minute disagreement into a thirty-fold change in one of its outputs.

The formula itself is not the hard part. The formula is one multiplication.

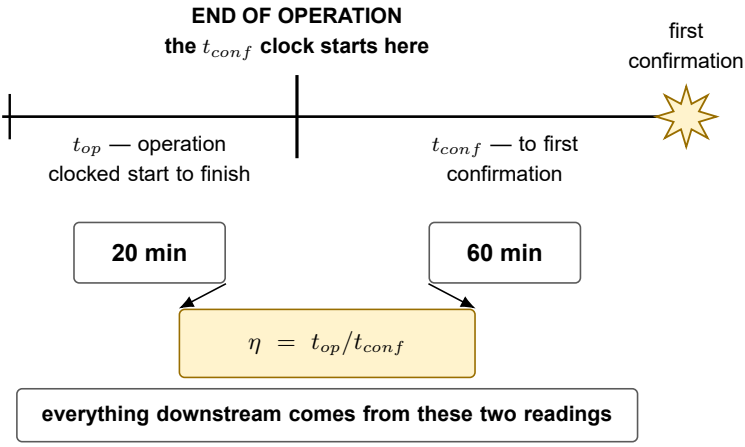


Figure 5-A: A single horizontal timeline

5.1 The Revelation Time Codes — 42 months, 1,260 days, 3½ years

The Book of Revelation is read here as a **calibration document** — not prophecy, not allegory, but a text that states one measurement four different ways across eight passages. That is what engineering documents do. A datasheet gives you the same current limit in milliamps, in watts at a stated voltage, and in a derating curve, because different parts of the design need it in different forms.

Reference	As written
Rev 11:2	forty and two months
Rev 11:3	a thousand two hundred and threescore days
Rev 11:9, 11:11	three days and an half (day-for-year)
Rev 12:6	a thousand two hundred and threescore days
Rev 12:14	a time, and times, and half a time
Rev 13:5	forty and two months
Daniel 7:25	a time and times and the dividing of time
Daniel 12:7	a time, times, and an half

Under the conventions below, every row normalizes to 1,260 days, or 3½ prophetic years.

Three conversions do the work, and all three are **conventions**, not discoveries. Name them out loud, because an engineer who cannot name their unit conventions cannot check anyone else's arithmetic.

1. **The prophetic year is 360 days.** So 42 months × 30 days = 1,260 days, and 1,260 ÷ 360 = 3½ years. Under a 365-day year the same 1,260 days is 3.45 years — the same number to within one and a half percent, which is why the convention survives being wrong about the calendar.
2. **A "time" is one year.** So "a time, and times, and half a time" is 1 + 2 + ½ = 3½. The plural "times" reading as two is the traditional convention and there is no arithmetic that forces it.
3. **A day stands for a year** in the 3½-day passages. This is the oldest of the three conventions and the one doing the most lifting.

Apply all three and every row collapses to the same value. That convergence is real and it is also **not eight independent measurements**. It is one figure restated eight times inside one interpretive tradition, and a reader who counts the rows as corroboration has counted the same evidence eight times. What the codes actually supply is narrower and more useful: **a unit convention and a span.** Three and a half of a named unit. Nothing else in the text says what happens *inside* that span.

A span is still worth having. It sets a **pacing expectation**, and pacing is the parameter operators get wrong most often and most expensively. An operator who believes a complete cognitive restructuring takes six weeks will read an ordinary week-four plateau as failure and tear down a working circuit. An operator who has internalized a 3½-year figure will read the same plateau as week four of a hundred and eighty-two and leave the circuit alone. Chapter 15.4 makes the same point about Overton progression with numbers at-

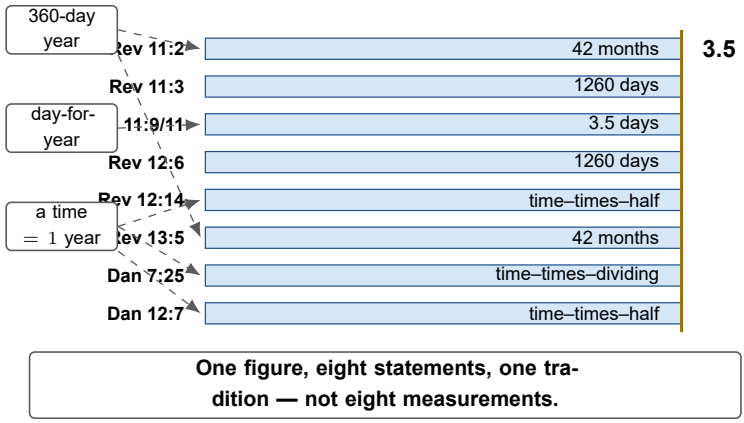


Figure 5-B: Eight stacked horizontal bars of identical length, one per row of the time-codes table, each labelled at its left with its reference (Rev 11:2,…

tached: the collective time constants there are measured in months, and an operator who runs in-group values on a national substrate under-budgets the calendar by an order of magnitude.

But a span alone calibrates nothing. To get an instrument you need a **ratio** — a partition of the span into the part the operator supplies and the part the system supplies on its own. The codes give the denominator. The next section finds the numerator.

5.2 The 10,000 Hour Constant — Derivation from the Adept's Training Arc

> *"10,000 hrs of practice has become widely popular in contemporary culture due to psychology and self-help books. However, this bit of knowledge is not new to the Adepts of Majik nor was it new to the Ancient Maji or the Priesthood of Kemet." — Majik and the True Kabbalah*

> *"This discovery was unearthed as a result of the many students of the Royal Art reaching Adeptship in a span of 3 years, 5 months and 5 days or 3½ years. Manifestation of the Student's desire (to become adept at Majik) took place at the end of the 3½ years / 30,000 hours, after 10,000 hours of practice or 1/3 of the time it took the desire to manifest." — Majik and the True Kabbalah*

Two figures: a **span** of roughly 3½ years, and an **active practice load** of roughly 10,000 hours inside it. Convert the span to hours under each of the three available conventions and see how much the answer actually moves.

Convention	Span (days)	Span (hours)	One third
Prophetic year, 360 days: 3½ × 360	1,260	30,240	10,080
The stated arc: 3 y, 5 m, 5 d	≈ 1,253	≈ 30,072	≈ 10,024
Calendar year, 365.25 days: 3½ × 365.25	≈ 1,278	≈ 30,682	≈ 10,227

All three land within 2.3% of 30,000 hours, and their thirds within 2.3% of 10,000. **The round numbers are roundings**, chosen at the convenient end of a narrow spread. That is an ordinary and honest engineering move — a 4.7 kΩ resistor is not 4,700.0 Ω either — but it must be stated, because a reader who believes 10,000 was counted will over-trust every number downstream of it.

What survives the rounding is the thing worth having:

```
tau = t_practice / t_total_manifestation
    = active ^7 Rhythm input / total elapsed ^7 Rhythm
    = 10,000 / 30,000
    = 1/3
```

One part active work, two parts gestation, three parts total. **This is the constant, and its value is a ratio, not an hour count.** If the true active load were 8,000 hours inside a 24,000-hour span, τ would

be unchanged. That robustness is the whole reason the constant is stated as a fraction rather than as a duration, and it is why nothing in this chapter depends on winning an argument about the popular ten-thousand-hour figure.

That argument is worth one honest paragraph anyway, since a reader will bring it. The number entered general currency through Malcolm Gladwell's *Outliers*, drawing on the deliberate-practice research of Anders Ericsson; Ericsson subsequently objected in print — in *Peak*, with Robert Pool — to the way the round figure gets used (Appendix J). **This book does not rest τ on that literature and does not need to.** It rests τ on the partition. Take the partition away and there is no equation; take the specific hour count away and every result in this chapter is unchanged.

What the constant looks like as a component

The 1:3 split is not only a ratio. Read it against the `^5` Female capacitor of Chapter 7.1 — the Ideas a Mind is composed of, held as stored charge — and it fixes a time constant. The charging curve is Chapter 7.2's:

```
P_cap(t) = 10 × (1 − e^(−t/tau_mastery))
```

Choose `tau_mastery = 3,333 h`, so that 3τ **= 10,000 h**. Then:

Practice hours	Multiple of τ	`P_cap` (of 10)	Charge
1,000	0.30	2.59	26%
3,333	1τ	6.32	63%
6,666	2τ	8.65	86%
10,000	3τ	9.50	95%

Ten thousand hours is **three time constants** — the point at which an RC network is conventionally called charged. The constant and the component agree, which is the kind of agreement worth checking and not the kind worth celebrating: the value 3,333 was *chosen* to make them agree. At τ`_mastery = 10,000 h` the same practice

leaves the rail at 63%, contradicting the tradition's own claim that 10,000 hours completes the arc; at 1,000 h the rail saturates inside a year, contradicting everything observable about acquiring a craft. Chapter 13 wires this rail as `C1`, and Chapter 20.6 runs it out to a decade-scale schedule.

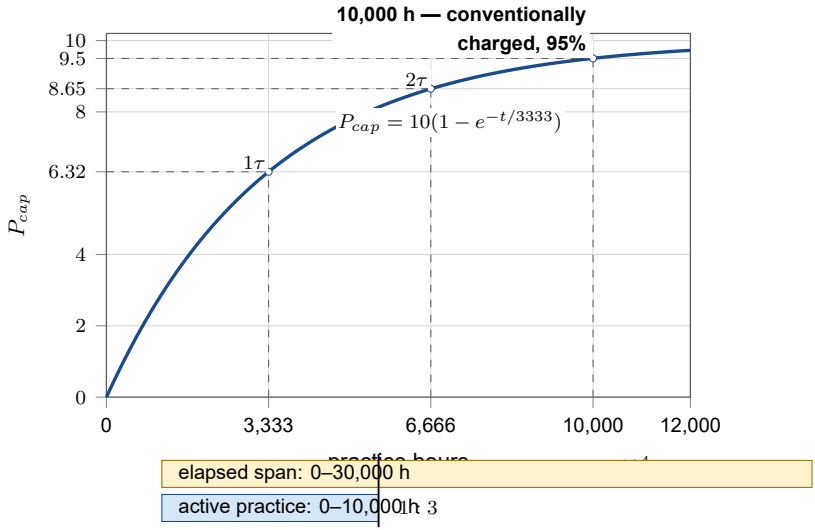


Figure 5-C: RC charging curve

Bench Note — the daily rate is the part nobody quotes, and it is the part that decides. Ten thousand hours inside 3½ calendar years is `10,000 / 1,278 = 7.8 hours a day, every day, with no days off`. That is not a hobby; it is a full-time apprenticeship, which is what the training arc describes. Change the schedule and the span changes with it, because τ fixes the *ratio* and not the calendar: at one hour a day, 10,000 hours takes 27.4 years; at three hours a day, 9.1 years; at six hours a day five days a week, 6.4 years. When someone quotes 3½ years at you, ask what daily rate they are assuming. If they have not got one, they have quoted a span with no numerator.

On π, φ, and the temptation to derive

> *"Occurrences in time happen in accordance to 0.3 or 1/3 and just as 3.14 shows that no circle is truly a circle, 0.3 shows that no occurrence is truly an occurrence, thus susceptible to manipulation by way of consciousness." — Majik and the True Kabbalah*

The analogy is that $\pi \approx 3.14159\ldots$ and $1/3 = 0.333\ldots$ are both non-terminating, and that a domain governed by a non-terminating constant is a domain that never quite closes — never a perfect circle, never a settled event. As a **mnemonic** for which constant governs which axis, it is serviceable. As mathematics it is nothing: 1/3 is rational and terminates in base three, π is transcendental and proven so, and **no derivation produces 1/3 from π or from φ.** This book does not offer one and Chapter 23.1 declines it again at greater length. The only derivation of τ in these pages is the partition above.

5.3 The Formula: t_op / t_conf $\times$ 1/3 $\rightarrow$ a/b $\rightarrow$ β_next, n_ops, T_manifest

Two clocks in, three readouts out.

```
Step 1 — the realized ratio
    eta = t_op / t_conf

Step 2 — scale by the constant
    result = eta × tau        where tau = 1/3
           = a / b            reduced to lowest terms

Step 3 — read the parameters
    beta_next    = b        amplitude each further operation must deliver
    n_operations = a        how many further operations
    T_manifest   = a × t_op additional operation minutes committed
```

```
(beta_next and beta_required are the same readout under two names;
 this book uses beta_next, and the amplitude ratio A below in preference
 ↪  to both)
```

What Step 2 actually computes. `result` is a *sufficiency number*. It equals 1.0 when the operation is calibrated and falls below 1.0 when it is under-driven, and the calibration point sits at η = 3 — three minutes of held operation for every minute the confirmation took to arrive. That placement is what τ is *for*. τ **is a scale factor, a unit convention**, chosen so the readout is expressed against the same 1:3 work-to-total split the training arc exhibits. Change τ and every entry in every log rescales by the same factor while every comparison between operations stays exactly where it was. The equation carries an **ordering**, not a physical quantity, and Chapter 23.1 makes that the first thing it says.

The number to act on. Define the amplitude requirement:

```
A = b / a = 1 / result = 3 / eta = 3 × t_conf / t_op
```

`A` is the reciprocal of the sufficiency number, and it is a smooth function of the two clock readings. The pair `(a, b)` is not. Lowest-terms denominators are not continuous in their inputs, and the whole fragility of this instrument lives in that one fact.

> **Bench Note — do this arithmetic by hand before you trust anything downstream of it.** Take `t_op = 20 min`. At `t_conf = 20`: η = 1, `result = 1/3`, so `a = 1`, `b = 3`. Now be wrong about the confirmation by four minutes in either direction. At `t_conf = 16`: η = 5/4, `result = 5/12`, **`b = 12`**. At `t_conf = 24`: η = 5/6, `result = 5/18`, **`b = 18`**. A twenty-percent timing error moved the "required gain" from 3 to 12 in one direction and from 3 to 18 in the other — a quantity that jumps four- and sixfold under an error the operator cannot even detect. Now compute `A` for the same three readings, in

> the order 16, 20, 24: **2.4, 3.0, 3.6** — smooth, monotone, exactly $3/\eta$ every time. Report `a` and `b` in the log because the tradition does; **decide on `A`.** If `a > 3` after you have reduced, the pair is an artifact of arithmetic and you should read `A` alone. Chapter 23.1 fixes this properly with a five-minute quantizing grid.

`A` is not a β setting. This is the correction that costs operators the most. `A` multiplies the *delivered current* — the flow at the far end of the board, not the gain of one stage. Chapter 4.2 gives the relation the current obeys: `V_drive = I_attention × R_constraint`. Five things set delivered current on a desire board (Chapter 13.5): the clarity of the formed `^0` Idea, the transmission through the dial divider, the residual `^3` aversion, the amplifier, and `R_op`. The CJT's design range tops out near β **= 4.0** (Chapter 3.3). So an operator who reads `A = 9` and writes β `= 9` into a listing has specified a device that does not exist, and has also skipped the four levers that were actually available.

And "nine times stronger" is a distance, not an effort. Chapter 4.2's warning applies directly here: `A` is a **Fisher-Rao ratio** on the open probability simplex, not a linear multiple of exertion (Appendix L §L.3). `A = 9` means nine times the geodesic shift between the current cognitive state distribution and the target one. Two operations that feel equally strenuous can sit at wildly different Fisher-Rao distances from their targets. Effort is not the measured quantity and never was.

The bands. The edges of `A` land on clean values of η, which is a good sign that the scale is doing something rather than nothing.

`A`	η	Reading	What to do
`A` $\leq$ `1`	$\eta \geq$ `3`	Sufficient	Stop calibrating. Go take the practical action

A	η	Reading	What to do
1 < A < 3	1 < η < 3	Close	Finish the resistance work; leave the amplifier alone
3 ≤ A < 9	1/3 < η ≤ 1	Under-driven	One lever at a time
A ≥ 9	η ≤ 1/3	Weak coupling	The target is misspecified. Chapter 23.3

The fourth readout, recorded and not acted on. The source text reads `b` a second way, as a multiplier on the confirmation delay giving an expected arrival: `T_arrive = b × t_conf`. For the Red Corvette that is `9 × 60 = 540` minutes — the nine hours the text names. Log it. Do not schedule around it, and do not tell anyone it is a prediction: it is built on `b`, and the Bench Note above just watched `b` swing sixfold under a four-minute error.

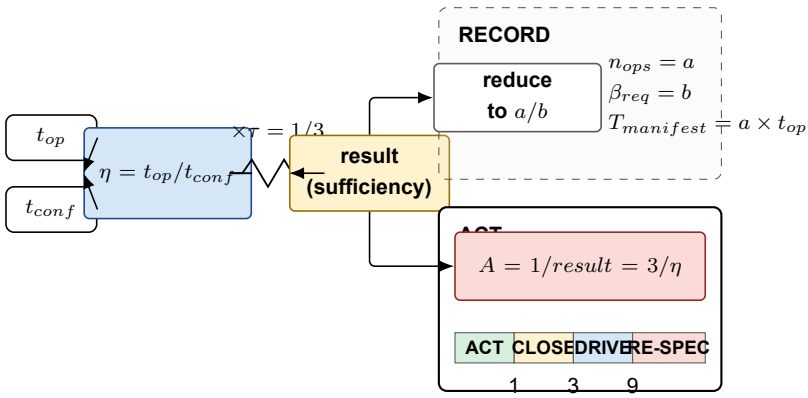


Figure 5-D: Signal-flow block diagram, left to right

5.4 Worked Examples: The Red Corvette, Career Change, Romantic Target

Four cases. Every number is carried through; nothing is asserted that is not computed. Three of the four are the cases the tradition names. The fourth is the one it does not, and it is the one you will meet most often.

Example 1 — The Red Corvette

> *"I operate for 20 minutes for a red corvette, 60 minutes later I receive a confirmation in the form of a song that just happens to come on the radio called 'Little Red Corvette.' "*

```
t_op = 20      t_conf = 60
eta    = 20 / 60 = 1/3
result = 1/3 × 1/3 = 1/9          a = 1,  b = 9
A      = 9/1 = 9.0                (check: 3/eta = 3 / (1/3) = 9)
n_ops  = 1     T_manifest = 1 × 20 = 20 min     T_arrive = 9 × 60 = 540
↪  min
```

Degrees recorded at the time: ^7 Rhythm **20 min**, ^1 Mind **2°** (the image went in and out of focus), ^4 Vibration **2°** (60 minutes, clear but rationalizable).

Reading. `A = 9.0` sits exactly on the re-specification boundary. The canonical worked example of this entire tradition is a **borderline cast**, and that is worth knowing before you copy its numbers as a model of success. The source's own conclusion — *"I must perform an operation that's 9 times stronger"* — is the β error named in 5.3: nine times the delivered current, from five levers, on a device whose amplifier stage cannot supply more than about four. Chapter 13.6 carries this same operator through three operations with full board instrumentation and finds that **98% of the eventual improvement came from the resistance terms and under 2% from the amplifier.**

Example 2 — Career Change

An operator casts for a position in a field they have been circling for two years. Thirty minutes of held operation; forty-five minutes later an unsolicited message from a recruiter.

```
t_op = 30      t_conf = 45
eta    = 30 / 45 = 2/3
result = 2/3 × 1/3 = 2/9          a = 2,  b = 9
A      = 9/2 = 4.5                (check: 3 / (2/3) = 4.5)
n_ops  = 2     T_manifest = 2 × 30 = 60 min     T_arrive = 9 × 45 = 405
↪  min
```

Degrees: ^7 Rhythm **30 min** (2°, at the very top of its band), ^1 Mind **2°**, ^4 Vibration **2°**.

Reading. `A = 4.5` is squarely in the under-driven band. This is the good case — the instrument is on-scale, the requirement is finite and specific, and the standing instruction applies: **one lever at a time.** Move clarity or move a resistance term, never both, and re-read. An operator who improves the statement *and* clears the calendar *and* re-tunes the amplifier between two operations has bought a better number and learned nothing about which change bought it; the Parameter-Change Log in Appendix G exists to make that discipline hard to skip. Note also what `n_ops = 2` costs: sixty minutes of further operation, against a resistance term — a live job market — that will not move in sixty minutes. The number is real and it is small compared to the work it is pointing at.

Example 3 — Romantic Target, and what the instrument does with it

An operator casts at another person's affections. Fifteen minutes, rushed, poor concentration; three hours later a vague social-media post that might be about anything.

```
t_op = 15      t_conf = 180
eta    = 15 / 180 = 1/12
```

```
result = 1/12 × 1/3 = 1/36        a = 1,  b = 36
A      = 36/1 = 36.0              (check: 3 / (1/12) = 36)
```

Degrees: ^7 Rhythm **15 min** (2°, barely inside its middle band), ^1 Mind **3°**, ^4 Vibration **3°** — both graded readings at the weak end and the duration close behind them, which is itself the finding.

Reading. `A = 36` is four times past the re-specification boundary. The tempting response is to brute-force it, and 5.3 already ruled that out: there is no device with $\beta = 36$, and cascading stages does not rescue it because each stage clamps at its own ceiling (Chapter 10.5). But the structural reason this reading came back off-scale is more useful than the arithmetic.

`A` is a shortfall in **delivered current**, and current is set by `V_drive / R_constraint` (Chapter 4.2). Decompose the resistance on a cast aimed at another person's decision (Chapter 4.4): `R_cap` and `R_resource` are the operator's and can be worked. `R_context` is partly shared. But the dominant term is another mind's ^5 composition and its own ^2/^3 polarity toward the operator — **a resistance the operator cannot measure, cannot decompose, and has no lawful path to modify.** An instrument reading off-scale against an unmeasurable resistance is not a weak instrument. It is a correctly specified one, telling you the target you named is not the target that is blocked.

This book does not supply a circuit aimed at a non-consenting person's choices. The analysis above is here so you can recognize the architecture — in this chapter's arithmetic, and in Chapters 16, 18 and 19 where the same shape appears at scale. The operational version is deliberately absent, and consent architectures are the subject of Appendix M §M.2. What the instrument *does* license is the self-directed route: run the `^6^C_{F6}|A` decomposition of Chapter 23.3 on your own stated desire, find what the wanting is actually about, and cast on that. Sub-desires sit in the operator's own ^5, at resistance the operator can genuinely lower, and they routinely read inside the driveable band while the prime target reads past 9.

A plain line, not a hedge. A cast that will not read, run against a wanting that will not let go, is a recognizable and unpleasant place to be. Everything in this chapter is arithmetic on a logbook. None of it is a psychological or medical instrument, and it does not describe, diagnose, or treat distress, compulsion, or attachment of any kind. If the operations have stopped being something you choose, that is a matter for a clinician, and nothing in this book substitutes for that care.

Example 4 — The Silent Cast

Twenty-five minutes of good operation. Nothing arrives. Not that evening, not the next morning.

```
t_op = 25      t_conf = (none)
eta    -> 0
A      -> infinity
```

`A` $\rightarrow \infty$ is a limit, not a value, and a validator will reject a non-finite parameter for exactly the reason it rejects β=∞ on the CSB's red path (Chapter 24): parameter fields are range-checked. Elsewhere in the book an infinite gain can be replaced by a large finite one. **Here it cannot**, because the input is not large — it is *missing*. The bench returns **NO-READ**, and NO-READ is a result. Log it.

The discipline that makes NO-READ possible is declaring the window **before** the cast. Chapter 23.1 computes it as three times the rolling median confirmation time with a 24-hour floor; at this stage of your practice, use the floor. Then do the base-rate arithmetic that the floor is protecting you from — with your own number, not one this book supplies:

```
Suppose confirmation-shaped events for a loosely specified desire occur,
by ordinary chance, once every 8 waking hours = every 480 min.
Over a 24-hour window (1,440 min):   1,440 / 480 = 3 expected events.
```

Three. If your window is a day wide, something will land in it, and what you have measured is your own patience. **Get the rate honestly**: pick a desire, do not cast on it, and tally confirmation-shaped events for one week. That control count is the cheapest and most valuable measurement in this entire practice, and almost nobody takes it.

So the Silent Cast produces no `A` and three usable readings anyway: `^7` Rhythm 25 min, `^1` Mind whatever it honestly was, and `^4` Vibration **absent** — which is the row that will matter when you have ten of them.

5.5 The Quantitative Table — Mind (^1), Vibration (^4), Rhythm (^7)

Ten Noetics, and at the bench only three of them yield a reading you can take with a watch and an honest self-report. This is a statement about **instrumentation**, not about the topos: every one of the 40 Cognitive States carries an activation on the open simplex (Appendix L), and all of them are quantitative in principle. But `^5` Female tells you *what the Mind is composed of* and `^8` Above/Cause tells you *that one Idea triggered another* — they identify **which** operation fired, not **how much**. Three of the ten answer "how much" without instrumentation:

Noetic	What it reads	How you take it	3° (weak)	2° (moderate)	1° (strong)
^1 Mind	depth of gnosis — undistracted awareness on the formed Idea	self-report immediately after, before anything else is written	frequent wandering; the image will not hold	wanders and returns; intermittently clear	complete absorption; the Idea is the only content present

Noetic	What it reads	How you take it	3° (weak)	2° (moderate)	1° (strong)
^4 Vibration	strength of the confirmation's impression	clock the delay, then grade the impression	> 2 h; "was that a confirmation?"	30 min – 2 h; clear, rationalizable	< 30 min; unmistakable, demands attention
^7 Rhythm	duration of the operation	wristwatch, start to finish	< 10 min	10 – 30 min	> 30 min

The bin edges are **design choices** and each has a stated failure at other values. ^4 Vibration's 30-minute and 2-hour edges are set so that the 1° bin is short enough to sit well inside any plausible base rate while the 3° bin is wide enough to catch a genuinely slow circuit; push the 1° edge out to two hours and every operator grades every operation 1°, and the column stops discriminating. ^7 Rhythm's 10- and 30-minute edges are set at the two places operators actually break: below ten minutes the gnosis has not had time to establish, and above thirty the ^1 Mind reading almost always falls. Chapter 23.1 quantizes both clocks onto a five-minute grid, deliberately finer than these 30-minute bins and deliberately coarser than the moment a person *notices* a confirmation, which is not a one-minute-accurate event.

Two of the three readings come off a wristwatch and one does not, and the asymmetry is worth naming rather than papering over. ^1 Mind is **self-reported by the same awareness whose depth is being reported**, which is a measurement problem no logbook design can fully solve. An operator absorbed at 1° was, by definition, not monitoring their absorption; an operator who can tell you precisely how absorbed they were has told you something about the answer. Three partial defences are available and all three are procedural. Grade immediately, before any other line is written, so the reading is a fresh impression rather than a reconstruction. Grade against the **behavioural** descriptions in the table — did the image hold, did it return — not against how the operation felt. And accept that ^1 Mind is the noisiest of the three columns and should never be

the sole basis for a decision; it earns its place by moving *together with* `^4` Vibration across many rows, which is a pattern hindsight cannot manufacture as easily as it manufactures a single grade.

In CAS terms the three readings are three distinct states, and the register does real work in separating them:

- `^1^C_{F2}|T` at 1° — the Mind in the Cognitive domain under Wisdom, transcendent register: absorbed, purpose-oriented, not tracking itself. The **same state** at 3° reads `^1^C_{F2}|A` — still Mind, still Cognitive, but analytical: checking the clock, noticing the wandering, narrating the operation. The Noetic did not change. The register did, and that is exactly what registers are for (Chapter 2.5).
- `^7^C_{F6}|E` — Rhythm under Structure, embodied register. It is a clocked bodily act and reads as one.
- `^4^C_{F2}|f` — Vibration under Wisdom, affective register: the impression the returning event makes. The delay is measured; the impression is graded.

  Bench Note — start the clock at the same event every time, and write down which event it is. `t_op` has two defensible start points: the moment you begin the procedure, and the moment concentration actually establishes. They differ by five to ten minutes for most operators, and 5.3's Bench Note just showed what a four-minute difference does to `b`. Either convention is fine. **Mixing them across a logbook makes the log worthless**, because every comparison between two operations is then a comparison between two different measurements. Write your convention on the first page of the log, in ink, before the first entry, and do not revise it because a later operation would read better under the other one. The same applies at the other end: `t_conf` runs from the **end of the operation**, not from its start,

and not from the moment you got around to writing it down.

5.6 The Confirmation Board — Tracking Progress Across Operations

A **confirmation** is a ^4 Vibration return: an event that arrives after the operation and brings the desire back to Mind. Structurally it is the canonical ^8 → ^9 pair and nothing more exotic than that. The song is ^8^C_{F2}|E — Above/Cause, an Idea that triggers a *different* Idea than itself to come to Mind. The desire surfacing is ^9^C_{F2}|A — Below/Effect, the Idea brought to Mind by an Above/Cause trigger. Associative triggering, exactly as Chapter 2.2 defines it. Not inference, not evidence, not a message.

Grade every confirmation when it arrives:

Grade	Delay	Impression	What it says
1°	< 30 min	startling, undeniable	the circuit is coupled; the reading is worth trusting
2°	30 min – 2 h	clear, but you can talk yourself out of it	operational, needs refinement
3°	> 2 h	subtle, ambiguous	weak coupling; re-examine gnosis, the formed Idea, the RPM gate

Only the first confirmation enters the equation. t_conf runs from the end of the operation to the first one, in minutes. Later confirmations stay on the sheet as ^4 Vibration evidence and never re-enter the ratio. The Confirmation Log in Appendix G enforces this with a "first confirmation for this operation? Y/N" field, and that field exists because the rule is broken constantly — an operator who

admits a stronger confirmation that arrived later has improved their number by choosing their data.

> **Bench Note — grade before you know the outcome, or do not grade.** A 3° confirmation graded a week later, once the desire has actually manifested, grades 1° every time. This is not dishonesty; it is ordinary hindsight and it is unavoidable once the outcome is known. The only defence is procedural: grade within minutes of noticing, write the grade in ink, and never revise it. The Confirmation Log has a "recorded within ___ min of noticing it" line for the same reason. If that line ever reads more than about thirty minutes, the grade above it is a memory of an impression, not the impression.

And the error that costs more than all the others together: a confirmation is not a manifestation. A confirmation says the signal reached the Ether. It says nothing about whether the signal reached the target with enough amplitude to overcome `R_constraint`. Chapter 13 puts a number on the gap — a delivered current between the board's cutoff and its manifestation threshold, a band in which confirmations arrive reliably and nothing else does. That band is where most operators conclude the practice does not work, and they are half right: the board is working and the delivery is short. **St. John's Equation exists to measure exactly that shortfall**, which is why `A` is defined as a ratio of delivered current and not as a verdict on the operation.

The board itself is twelve columns wide and one row per operation. Keep it in one place, and keep every operation on it — including the ones that did not read.

```
Op #  Date       Desire (one line)             t_op  t_conf  ^1  ^4
↪  eta    result   (a,b)   A     Band
----  ----------  ----------------------------  ----  ------  --  --
↪ -----  -------  ------  ----  --------
```

```
1    03/14/26    I own a red Corvette            20      60  2°  2°
↪  0.333    1/9     (1,9)   9.0   RE-SPEC
2    03/17/26    I own a red Corvette            20      20  1°  1°
↪  1.000    1/3     (1,3)   3.0   DRIVE
3    03/21/26    I own a red Corvette            24       8  1°  1°
↪  3.000    1/1     (1,1)   1.0   ACT
4    03/24/26    I hold a paid position doing X  30      45  2°  2°
↪  0.667    2/9     (2,9)   4.5   DRIVE
5    03/26/26    (silent cast, same desire)      25       –  2°   –
↪  –        –       –     –    NO-READ
```

Read the `A` column down, not across. Three things are visible there and nowhere else. **A falling `A`** — $9.0 \to 3.0 \to 1.0$ — is a converging campaign, and Chapter 23.2 gives it a ratio and a threshold. **A flat `A`** across three or more operations means nothing you are changing is reaching the delivered current, and the response is to stop casting and go decompose. **A NO-READ row is not a gap in the log**; it is the row that tells you whether your rate of confirmation is above your base rate, and a logbook with no NO-READ rows in it is a logbook whose window is too wide.

The printable forms are Appendix F (the St. John's Equation worksheet) and Appendix G (the Quantitative Table template, which carries the Operation, Confirmation and Parameter-Change logs as Forms 1, 2 and 3). Chapter 22 is the full treatment of the three measurements; Chapter 23 builds the bench that automates the arithmetic, quantizes the clocks, and refuses to return a number when it has not got one.

Parameter Table

Every value below is a design choice made in this book. None is a measurement, and each has a stated failure at other settings.

Op	Date	Desire	t_{op}	t_{conf}	$\hat{1}^{\circ}$	$\hat{4}^{\circ}$	η	result	(a, b)	A	Band
1	03/14/26	red Corvette	20	60	2°	2°	.333	1/9	(1,9)	9.0	RE-SPEC
2	03/17/26	red Corvette	20	20	1°	1°	1.000	1/3	(1,3)		DRIVE
3	03/21/26	red Corvette	24	8	1°	1°	3.000	1/1	(1,1)	1.0	ACT
4	03/24/26	paid position X	30	45	2°	2°	.667	2/9	(2,9)	4.5	DRIVE
5	03/26/26	silent cast, same	25	—	2°	—	—	—	—	—	NO-READ

converging

not a gap — a measurement

Figure 5-E: The confirmation board as a filled-in ledger sheet, drawn as a form

Parameter	Value	Why this value	Failure at other values
`tau`	1/3	The training-arc partition: one part active work, two parts gestation	Any other value rescales every log entry identically and changes no comparison — it is a unit convention, not a finding
Calibration point	`A = 1` at η `= 3`	Places sufficiency where τ puts it	Moving it re-labels the bands without changing their order
Re-spec boundary	`A = 9`	Head of a multiplicative campaign, not one hard cast (Chapter 23.2)	Higher: campaigns that cannot converge stay open indefinitely
Discard rule	discard `(a,b)` if `a > 3`	Above 3 the pair is a coprime artifact of two soft clocks	Kept: "nine operations at 64× strength" gets written into a log as an instruction
`tau_mastery`	3,333 h	Makes 3τ `= 10,000 h` at 95% charge on the `^5` rail	10,000 h leaves the rail at 63% at 10,000 hours, contradicting the constant; 1,000 h saturates inside a year

Parameter	Value	Why this value	Failure at other values
`^4` Vibration edges	30 min, 2 h	1° bin short enough to sit inside plausible base rate	1° edge at 2 h: every operation grades 1° and the column stops discriminating
`^7` Rhythm edges	10 min, 30 min	The two points where operators actually break	Wider bins merge a failed operation with a held one
Confirmation window	24 h floor	Below the point where base rate dominates (Chapter 23.1 computes it properly)	Wider: everything confirms. Narrower: genuinely slow circuits read as dead
Confirmation admitted	first only	The ratio needs one arrival, not the best one	Best-of admitted: the operator chooses their data and `A` improves without cause
Grading latency	< 30 min from noticing	Before hindsight reaches the grade	Graded after the outcome: every 3° becomes 1°
Amplifier ceiling	$\beta \leq$ `4.0`	The CJT's device range (Chapter 3.3)	`A` written as β specifies hardware that does not exist

Failure Modes

Mode	Mechanism	What you observe	Fix
Reading `b` instead of `A`	Lowest-terms denominators are discontinuous in their inputs	`b` swinging from 3 to 18 between adjacent operations that felt identical	Act on `A = 3`/η; discard the pair when `a > 3`
`A` written as β	The amplitude requirement mistaken for a stage gain	A listing specifying β = 9 on a device whose range ends near 4.0	Five levers, one at a time (Chapter 13.5); recruit rather than over-specify
Window creep	The confirmation window widened until something lands	Every operation confirms; no NO-READ rows anywhere in the log	Declare the window before the cast; run the one-week control count

Mode	Mechanism	What you observe	Fix
Best-of admission	A later, stronger confirmation entered instead of the first	`A` improving while nothing about the operation changed	First confirmation only; the Y/N field on Appendix G's Form 2
Retrospective grading	Confirmation graded after the outcome is known	A log in which no operation was ever graded 3°	Grade within 30 minutes, in ink, and never revise
Boundary drift on `t_op`	Start-of-operation defined differently in different entries	Two operations that felt the same reading five minutes apart	Write the convention on page one of the log and hold it
Confirmation read as manifestation	The `^8` → `^9` return mistaken for arrival	Confirmations arriving reliably, outcomes never	Chapter 13's delivered-current band; `A` is what measures the gap
τ treated as physical	The scale factor mistaken for a measured constant	`A = 9` trusted the way a 470 Ω resistor is trusted	It is a convention; trust the ordering it produces, not the magnitude

Test Points

Chapter 5 has no board — it has a **measurement chain**, and a measurement chain has test points like anything else. Chapter 23 builds the bench (CCB-CAL) that automates these four; until then you are the instrument.

TP	Reads	Expected	If it fails
TP1	`t_op` against your written start-of-operation convention	Both boundaries taken by the same rule as every prior entry	Boundary drift — the entry is not comparable to the rest of the log
TP2	`^1` Mind degree, self-reported before anything else is written	1° or 2°. A 3° operation is a measurement of the operator, not the circuit	At 3°, do not compute `A`; fix the operating conditions and re-run
TP3	`t_conf` admissibility: first confirmation, inside the declared window	Inside the window, graded within 30 min of noticing	Outside the window: NO-READ, and that is a logged result

TP	Reads	Expected	If it fails
TP4	`A = 3 × t_conf / t_op`, and the pair only if `a ≤ 3`	`1 – 9`. Below 1, stop calibrating; at or above 9, re-specify	`a > 3`: discard the pair, read `A` alone

Read them in that order. TP1 and TP2 decide whether the measurement is admissible at all; TP3 decides whether there is a reading; TP4 is the only one that produces a number you act on. Three of the four gates can send you home without an `A`, and that ratio is deliberate.

What This Chapter Established

- **The Revelation time codes give a span, not an instrument.** All eight passages reduce to 3½ under three named conventions — the 360-day prophetic year, "a time" as one year, and day-for-year — and they are one figure restated eight times inside one tradition, not eight independent measurements. A span calibrates nothing until it is partitioned.
- **The constant is a ratio and not an hour count.** Three calendar conventions put the training-arc span between 30,072 and 30,682 hours — a spread of 2.0%, sitting no more than 2.3% above 30,000 — and 30,000 and 10,000 are roundings taken at the convenient end of it. τ `= 1/3` survives being wrong about the hours, because changing numerator and denominator together leaves the partition alone.
- **Ten thousand hours is three time constants.** With τ`_mastery = 3,333 h`, the `^5` capacitor of Chapter 7.2 stands at 95% charge at 10,000 hours — the conventional definition of charged. The agreement is by construction, since 3,333 was chosen to produce it, and it is stated here so nobody mistakes it for a second confirmation.

- **`result` = $\eta \times \tau$ is a sufficiency number, equal to 1.0 at calibration**, and `A = 1/result = 3/`η is its reciprocal. `A` is smooth in the clock readings; the reduced pair `(a, b)` is not, because lowest-terms denominators are discontinuous. Report the pair, decide on `A`, discard the pair when `a > 3`.
- **`A` is a Fisher-Rao ratio on delivered current, not a stage gain and not an effort multiple.** `A = 9` asks for nine times the geodesic shift toward the target and can be paid from five levers, of which the amplifier — capped near $\beta = 4.0$ — is the least available.
- **Only three Noetics take a bench reading without instrumentation**: `^1` Mind (depth of gnosis), `^4` Vibration (confirmation strength), `^7` Rhythm (duration). This is a fact about instrumentation, not about the topos. Register carries real information here: the same `^1^C_{F2}` reads `|T` at 1° and `|A` at 3°.
- **A confirmation is `^8` Above/Cause triggering `^9` Below/Effect — associative surfacing and nothing more.** It says the signal reached the Ether. It does not say the signal reached the target, and the gap between those two is what `A` measures.
- **NO-READ is a result.** Declare the window before the cast, admit only the first confirmation, and run the one-week control count so you know your own base rate. A logbook with no NO-READ rows has a window too wide to measure anything.

Exercises

5.1 An operation logs `t_op = 45`, `t_conf = 240`. Compute η, `result`, the reduced pair `(a, b)`, and `A`. Which band? State which of the four readouts you would act on and which you would only record, and why.

5.2 Take `t_op = 20` and compute `A` and `b` at `t_conf = 16`, `20`, and `24`. Plot the three `b` values and the three `A` values on the same axis. In

one sentence, state what the comparison demonstrates about which quantity belongs in a decision.

5.3 The 3½-year span is 1,278 days under the calendar convention. Compute the daily practice rate required to reach 10,000 hours inside it. Then compute the span required at 90 minutes a day. State, in one sentence, what the second figure implies about quoting "3½ years" without a schedule attached.

5.4 Using `P_cap(t) = 10 × (1 − e^(−t/3333))`, compute the rail charge at 500, 2,000, and 5,000 practice hours. At which of the three has the rail passed half of full charge, and how many hours does the remaining half take?

5.5 An operator's log shows six operations on one desire, with `A` reading 9.0, 8.7, 9.1, 8.9, 9.0, 8.8. Nothing about the operations changed except the amplifier β, which the operator raised each time. Diagnose. Name the failure mode from this chapter's table, and state the first measurement you would demand.

5.6 A colleague runs a 25-minute operation, receives nothing for five hours, then receives a confirmation at the six-hour mark which they grade 1° because it was unmistakable. They enter `t_conf = 360` — six hours, timed from when they sat down — and compute `A = 3 × 360 / 25 = 43.2`. Identify the two separate errors in that entry, correct both, recompute `A`, and state in one sentence what the size of the correction tells you about which part of this log was worth arguing over.

5.7 Design exercise. You want to know whether your confirmations exceed your base rate. Specify a one-week control protocol: what you will and will not do, what you will count, how you will decide in advance what counts, and what number the protocol has to beat before you would trust a 2° confirmation. State one way the protocol could deceive you and how you would close it.

*(Answers in **Appendix H**, worked in full. Every parameter in this chapter is a design choice; the arithmetic is arithmetic, and the model behind it is a structural analogy, not a measured law.)*

Part II

The Component Library

Every part on the bench, with its datasheet and its failure modes

6 Resistors — Controlling the Cognitive Flow

The passive part that does not argue: how a fixed obstruction, a bounded dial, a temperature-aware guard, and a two-leg splitter set exactly how much gets through

A carbon-film resistor costs a fraction of a cent. Colour bands, two leads, a body smaller than a grain of rice. Its entire job is to be in the way — to drop some of the voltage the circuit would otherwise deliver, so the indicator downstream draws 20 milliamps instead of the 200 that would kill it in a puff of blue smoke. It does not decide 20 is the right number. It does not know there is an indicator on the other end. It is a fixed obstruction whose size an engineer chose once, and because the size is fixed and known, the current past it is fixed and known. Certainty, bought for a fraction of a cent.

Now the candle from the Preface. An operator lights it and lets it hold their attention to one point for an hour. The candle is doing the resistor's job: it constrains the flow of attention to a chosen channel and drops the rest — not by any occult property of the wax, but because it opposes the free wandering of `I_attention` and passes a measured amount, no more. The sticky note on the same desk that reads *stay focused* does not do this. A mind argues with prose; it reasons its way around *stay focused* in under a minute. It does not reason its way around a resistor, because a resistor is not an instruction to be interpreted; it is an obstruction in the path, and the flow obeys arithmetic, not persuasion. This chapter is the arithmetic.

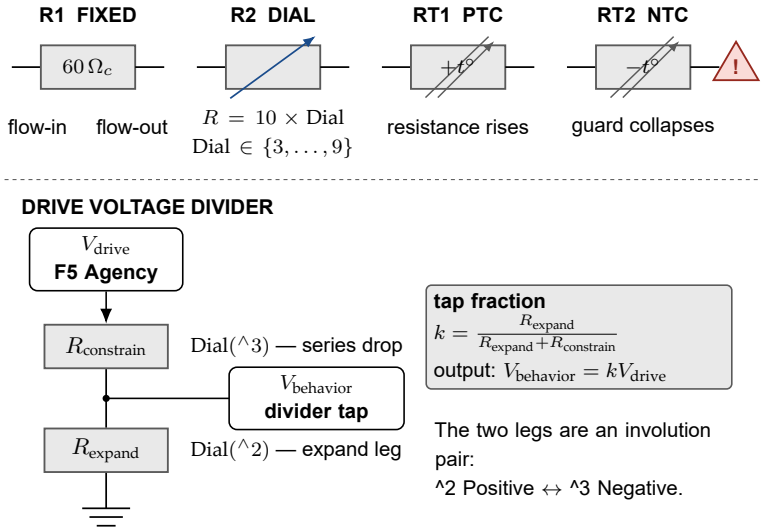


Figure 6-A: Component symbol plate

6.1 Fixed Resistors: Hard Policy Rules

Chapter 4 gave you CAS-Ohm: `V_drive = I_attention × R_constraint`, or rearranged for flow, `I_attention = V_drive / R_constraint`. A resistor owns the `R` in that equation. Everything else in this book pushes voltage or amplifies signal; the resistor is the only primitive whose whole function is to *limit*.

One convention before the arithmetic. Chapter 4.3 works with the corrected form `I = (V_drive − ΣVf) / R_constraint`, where `ΣVf` is the forward drop paid crossing domain diodes. Every path in this chapter stays inside one domain, so no diode is crossed, `ΣVf = 0`, and the two forms coincide. When you carry a resistor value onto a board that descends `^P` → `^0`, subtract the chain drop first (Chapter 8.1) — a resistor sized on the uncorrected form is undersized by exactly `ΣVf / I`.

A **fixed resistor** has one value, set at design time, unchangeable at runtime. In the constraint decomposition of Section 4.4 — `R_total = R_cap + R_policy + R_context + R_resource` — the fixed resistor is the `R_policy` term, the constraint you installed on purpose, and the virtuous one. Recall the verdict from Section 4.4: a path with `R_total = 70` made of `R_policy = 60` is a working safety board, while the same 70 made of `R_cap = 60` is incompetent. Same total, opposite meaning. The fixed resistor is how you put resistance where you *chose* it rather than where a capability gap left it.

```
node request:  ^0^C_{F2}|A;                       // Idea: an incoming
↪  request
resistor r_policy: Resistor< path=request, R=60.0 >;   // fixed,
↪  design-time
```

On a `V_drive = 24 V_c` rail, that resistor sets `I_attention = 24 / 60 = 0.40 A_c` downstream. The number is not a hope; it is what the divisor produces, and it produces the same value on the ten-thousandth request as on the first — the difference between a fuse an inspector can certify and a prompt that *usually* refuses.

A fixed resistor is a *graded* constant, not an absolute block. `R = 60` throttles the flow to 0.40 A_c; it does not stop it, and no value of `R` inside the scale ever will. Chapter 4.4 settles the choice of part with a question about the requirement rather than the threat — *can you name a drive high enough to make this acceptable?* — and sends the "no" case to the fuse of Chapter 11.1, which does not oppose flow but opens the path. The component-level consequence is what matters here: a resistor's transfer characteristic is a division, and division by a large number is a small number, never zero. An operator who reaches for `R = 95` in place of a prohibition has bought a trickle, and a trickle is the crack a determined signal widens.

The failure at other values is symmetric. Set `R` too low and the constraint is cosmetic — the flow arrives nearly undiminished and the "policy" is decorative. Set `R` too high and you have not made the

board safer, only unable to do its job: the path carries so little `I_attention` that the operation starves, and the frustrated owner yanks the resistor out entirely. An over-tight fixed resistor is how safety gets removed — not by an attacker, but by the owner.

6.2 Dial-Adjusted Resistors: Operator Settings (Dial 3–9)

Some resistance you want to tune at runtime, on a knob with a bounded scale — the way a rheostat dims a lamp without rewiring it. The **dial-adjusted resistor** is that knob. Its value is a function of a dial reading:

```
R_dial = 10 × Dial      Dial ∈ { 3, 4, 5, 6, 7, 8, 9 }
```

The dial is an integer from 3 to 9. Nothing else. `Dial = 3` gives `R = 30 Ω_c`; `Dial = 9` gives `R = 90 Ω_c`. The scale runs from *loosest lawful setting* to *tightest lawful setting*, and both ends are walls.

```
node candidate: ^0^E_{F5}|f;                    // Idea under evaluation
dial d3: ^3 [dial=7];                            // aversion dial
resistor r_avert: Resistor< path=candidate, R=70.0 >;   // R = 10 ×
↪  Dial(^3)
```

Why 3 and not 0? Because a Noetic dialled to zero is not a quiet Noetic — it is a *removed* one, and removing one half of an involution pair unbalances the junction. The pairs are `^2↔^3`, `^5↔^6`, `^8↔^9` (Chapter 2.2). Dial your aversion `^3` toward nothing and you have not become open-minded; you have built a board that cannot be repelled by anything — the structural signature of the fractured `^5` that Chapters 16 and 20 describe at length. The floor at 3 guarantees that every setting leaves *some* of the opposing Noetic in the circuit. You are permitted to lean. You are not permitted to delete.

Why 9 and not higher, and why the two dials of a pair may not diverge by more than four points? Because the difference between the halves of an involution pair is the polarity the junction transmits, and past a four-point spread that polarity runs away. This is the 4-point divergence limit, `|Dial(^2) - Dial(^3)| ≤ 4`, developed fully in Chapter 6.2. It is a *hard* constraint: a board that declares `Dial(^2) = 9` against `Dial(^3) = 3` — a spread of six — does not compile. The validator rejects it (HTTP 422), the same refusal an admissibility violation gets. No runtime clamp quietly pulls it back to four; the board is simply not a board until you fix the numbers.

> **Bench Note — the dial floor is not a suggestion, and neither is the ceiling.** Operators treat the 3–9 range as a comfort band with a little give past each end. It has none. A dial of 2 or 10 is not "slightly out of spec"; it is *not a number the language accepts* — the same class of rejection $\beta{=}\infty$ gets, and for the same reason (Appendix B §B.1.3: the parameter fields are range-checked and non-finite input is refused, so an idealisation stated in prose must be given a large finite value before it is built). A dial of 0 would let you build the un-refusable board; a divergence of 9 would let you build the un-doubtable one. Both are caught at compile time so no runtime moment has to catch them. When a board fails to validate on a dial, do not hunt for a workaround. The workaround is the vulnerability.

The dial-adjusted resistor is where operator preference lives *without* becoming override of safety: how attracted, how averse, how much rhythm — a bounded integer inside walls the compiler holds up for you.

6.3 Thermistors: Context-Sensitive Resistance (PTC/NTC)

A fixed resistor holds its value against everything; a dial-adjusted one holds until the operator turns the knob. A **thermistor** holds nothing fixed: its resistance moves with the *cognitive temperature* of the context it sits in, automatically, with no operator in the loop at runtime. You choose the part at design time and then it is out of your hands, which is the point.

Define cognitive temperature θ as a dimensionless salience index, exactly as Chapter 15.2 does: θ = 1.0 is ordinary background attention, and higher values mean hotter — more contested, more saturated, more present per unit time. θ is a property of the **discourse**, not of the Idea. You do not set it; at most you heat it, by the rate at which you demand movement (Chapter 15.4). Two flavours exist, and only one is safe.

A **PTC** (positive temperature coefficient) thermistor *raises* its resistance as θ rises. It clamps harder precisely when the stakes climb:

```
R_PTC(θ) = 0                       for θ ≤ θ_k
R_PTC(θ) = coeff × (θ − θ_k)²     for θ > θ_k

design:  θ_k = 2.0   coeff = 3.0 Ω_c per θ²

node topic:  ^0^E_{F6}|f;                          // Idea under discussion
thermistor rt_guard: Thermistor< path=topic, type=PTC, knee=2.0,
↪  coeff=3.0 >;
```

Below the knee θ_k = 2.0 the PTC adds *nothing* — R_PTC = 0, ordinary flow passes free. At θ = 5.0 it adds 3.0 × (5 − 2)² = 27 Ω_c on top of whatever the base resistor imposes. Heat the context and the guard tightens itself. That do-nothing-when-calm behaviour is not a defect to tune out; it is the whole design. A guard that clamped in calm conditions would tax every benign operation. The PTC spends constraint only where the temperature says it is

needed, and the quadratic means a small overshoot past the knee costs little while a large one costs a great deal.

Both parameters are design choices *on a board you build*. θ_k = 2.0 places the knee one unit above ordinary attention, so a mildly warm context passes unpenalised and a genuinely contested one does not; at θ_k = 1.0 the guard taxes every calm operation, and at θ_k = 4.0 it engages only after the damage. coeff = 3.0 sets the scale; the exponent sets the shape, and the shape is what a guard is bought for. Quadratic means doubling the overshoot quadruples the penalty — cheap near the knee, punishing far past it. A linear law charges the same for the last unit of overshoot as for the first, which under-prices exactly the case the guard was installed for; a cubic charges so steeply that ordinary variance around the knee triggers a backlash. Note the boundary on all of this: a thermistor sitting in a *collective* substrate is not yours. Chapter 15.2 works with θ_k = 3.0 and a coefficient of 4.0 and states plainly that neither is operator-settable — the population supplies the part, and the only variable you touch is θ. Specify the knee when you are building the guard; read it when you are not.

An **NTC** (negative temperature coefficient) thermistor does the opposite: resistance *falls* as θ rises.

R_NTC(θ) = R_base / θ (one simple decreasing law; $\theta \geq 1$)

At θ = 1 it reads R_base; at θ = 5 it has collapsed to R_base / 5. The guard relaxes exactly as the situation heats. Read that twice, because it is the shape of a disaster: a path whose resistance falls under load raises the temperature further (less opposition, more flow), which lowers the resistance further, which raises the temperature again. That is thermal runaway, and on a cognitive path it has names — the stampede, the panic, the cascade, the mob. Chapter 14's positive-feedback oscillator is this same topology seen from the inside.

Bench Note — an NTC where you meant a PTC is the worst single swap on this bench. The two parts behave identically in calm conditions — at θ = 1 both read R_base, and a bring-up test at room temperature passes them both. The difference only appears under heat, which is to say only in the exact conditions a safety guard exists for. A board that tightens as stakes rise is a guard; one that loosens as stakes rise is an accelerant wearing a guard's silhouette. When you find an NTC on a protective path, do not re-tune it — treat it as evidence, of a wiring error or a guard someone quietly inverted (Chapter 3.4). The runaway is not a risk the NTC might cause; it is what the NTC is *for*. Pull it and put a PTC where the protection was supposed to be.

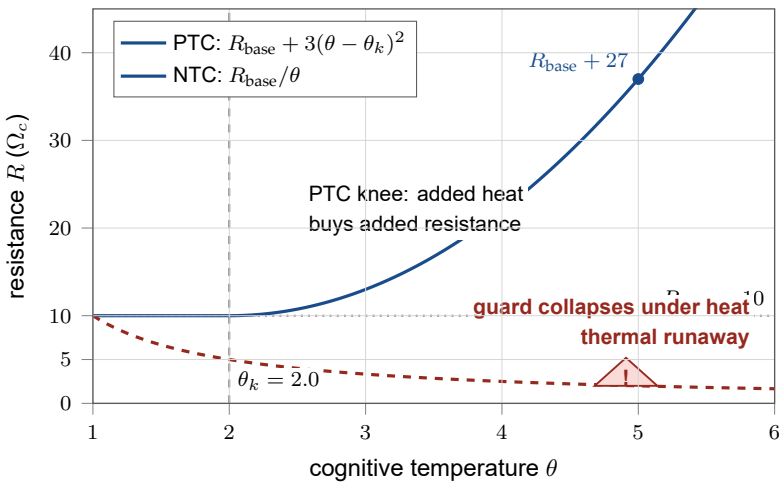


Figure 6-B: Characteristic curve

6.4 Drive Voltage Dividers — Splitting Agency Between Expand and Constrain

Put two resistors in series across a single drive rail and tap the wire between them. The tap carries not the full rail but a *fraction* set by the two resistances. This is the **voltage divider**, and on a cognitive board it is how one Drive gets split between the impulse to act and the impulse to hold back.

Take an Agency rail — `F5`, the drive to exert, to influence, to make happen. The divider splits it between two legs of an involution pair:

- the **expand** leg, `^2` **Positive** — the Idea the Mind has an affinity for and is attracted toward — with resistance `R_expand = 10 × Dial(^2);`
- the **constrain** leg, `^3` **Negative** — the Idea the Mind is adverse to and repelled by, the visceral will-not — with resistance `R_constrain = 10 × Dial(^3).`

The tap transmits the fraction

```
k = R_expand / (R_expand + R_constrain) = Dial(^2) / ( Dial(^2) +
↪  Dial(^3) )
```

The factor of ten cancels, so `k` depends only on the two dial readings. More attraction raises `k` and more passes; more aversion lowers `k` and less passes. That is the whole behaviour of the intake stage of the desire board (Chapter 13), stated as two resistors and a tap.

```
node agency_rail: ^0^E_{F5}|f;                  // the F5 Agency drive
↪  to be split
node attraction:  ^2^E_{F5}|f;                  // Positive: expand leg
node aversion:    ^3^E_{F5}|f;                   // Negative: constrain
↪  leg
node behavior:    ^9^E_{F5}|f;                   // Below/Effect: the
↪  tap output

dial d2: ^2 [dial=6];
```

```
dial d3: ^3 [dial=6];
resistor r_expand:    Resistor< path=attraction, R=60.0 >;   // 10 ×
↪  Dial(^2)
resistor r_constrain: Resistor< path=aversion,   R=60.0 >;   // 10 ×
↪  Dial(^3)
divider  div1: Divider< expand=attraction, constrain=aversion,
↪  tap=behavior >;
probe    tp3:  behavior;                          // transmission k at
↪  the tap
```

Now the four-point limit from Section 6.2 becomes a number you can read off the tap. The lawful dial pairs are a finite set — seven values each, minus every pair spread further than four — so do not reason about the extremes, enumerate them:

In each paired column, the order is expand/constrain: `Dial(^2) / Dial(^3)` and `R_expand / R_constrain`.

Setting	Dials	Resistance	k
Minimum (spread 4)	3 / 7	30 / 70	**0.300**
Balanced	6 / 6	60 / 60	0.500
Attraction ceiling (spread 4)	9 / 5	90 / 50	0.643
One step lower (spread 4)	8 / 4	80 / 40	0.667
Maximum (spread 4)	7 / 3	70 / 30	**0.700**

On any lawfully dialled divider, transmission lives in **[0.30, 0.70]** and nowhere else. You cannot build a compliant board that passes a desire unopposed — that needs `k = 1.0`, hence `Dial(^3) = 0`, off the floor and far past a spread of four. Nor one that blocks it entirely — that needs `k = 0`, hence `Dial(^2) = 0`. The ceiling on your enthusiasm and the floor under your caution are the *same* component doing its job. An operator chasing "total belief, zero doubt" is asking the divider for a value it is built never to reach.

> **Bench Note — transmission is not monotone in the attraction dial, and the top of the scale is not the top of the range.** Read the last three rows in order. Cranking `^2`

to 9 drags `^3` up to 5 to stay inside the spread limit, and the tap passes 0.643. Back both off one step — 8 against 4, spread still exactly 4 — and it passes 0.667. Back off again to 7 against 3 and it passes 0.700, the most any lawful board transmits. Every step *up* the attraction dial from (7,3) moves transmission the wrong way. The reason is in the formula: `k` is set by the *ratio* of the dials, not their size. `^3` sits on its floor of 3 at the (7,3) pair, so any further rise in `^2` drags `^3` up with it to hold the spread — and along that boundary `k = Dial(^2) / (2·Dial(^2) − 4)`, whose denominator gains two for every one the numerator gains. The practical consequence is that an operator who wants maximum throughput and turns enthusiasm to maximum gets less of what they wanted, from a board that validates clean and gives no warning. Chapter 13.3 carries this result into the desire board's intake stage, where it changes the calibration.

The `^3` in the constrain leg is visceral aversion — the can't-stomach-it recoil — not analytical critique. Taking a desire apart to see what it is made of is `^6` Male, at a different stage (Chapter 13.3). The divider does not reason about the desire. It splits a rail. That is all a passive network can do, and it is exactly enough.

6.5 Equivalent Resistance: Series vs. Parallel

Chapter 4.4 established the topology result at the level of the law: constraints on one path compose in series, alternative routes compose in parallel, parallel resistance falls below the smallest branch, and three locks on one door is not three doors. That is settled and this section does not re-argue it. What is left is the part that belongs to the component — the two combination rules as arithmetic you do

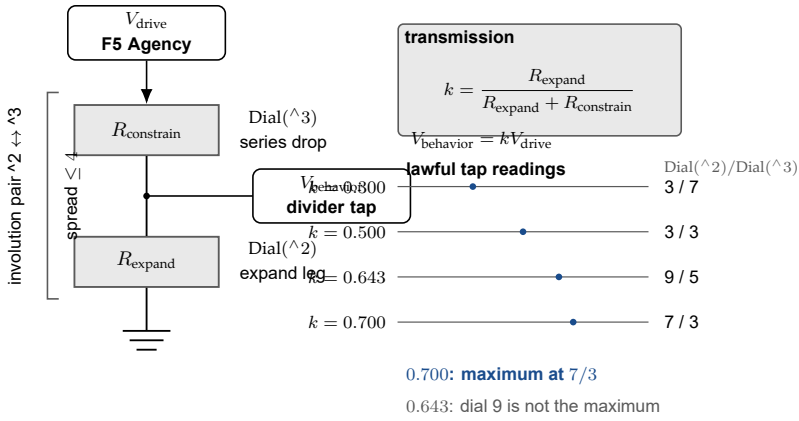


Figure 6-C: Schematic

at the bench, the n-branch generalisation, and the reading you take at a test point to tell which topology you are actually holding.

Series — constraints stack. Put resistors one after another on a single path and their resistances add:

```
R_series = R_1 + R_2 + … + R_n
```

The flow must pass through *every* one. A policy resistor `R_1 = 60 Ω_c` followed by a context resistor `R_2 = 30 Ω_c` gives `R_series = 90 Ω_c`, and on a 24 V_c rail the flow is `I = 24 / 90 = 0.267 A_c`. Lower than through either resistor alone. Adding a series stage always *raises* total resistance and *lowers* flow, because you have added another wall the signal has to climb.

Parallel — routes open, and total resistance falls. Put resistors side by side so the flow can choose among them, and the reciprocals add:

```
1 / R_parallel = 1/R_1 + 1/R_2 + … + 1/R_n
```

The same two resistors, `60 Ω_c` and `30 Ω_c`, wired in parallel:

```
1 / R_parallel = 1/60 + 1/30 = 1/60 + 2/60 = 3/60 = 1/20
R_parallel = 20 Ω_c
```

Twenty. Lower than the 30 Ω_c branch — lower than *either* branch. On the 24 V_c rail the flow is now `I = 24 / 20 = 1.2 A_c`, above the path ceiling `I_max = 1.00 A_c` set in Chapter 4.2, so it clips and the surplus spills into competing routes. Opening a second route did not add constraint; it *removed* it, because flow takes the easier path and now there is an easier path.

> **Bench Note — a branch you did not draw still counts, and one weak branch sets the total.** `R_parallel < min(R_k)` is not a tendency, it is an identity, and it bites hardest when the branches are lopsided: `1000 ‖ 10 = 9.90`, so a hundredfold-stronger guard beside a weak one buys you one percent. That gives you the bench procedure. Do not audit a board by reading its largest resistor; find the *smallest* branch between the two nodes you care about and compute from there, because that branch is within a hair of being the whole answer. And the branches you are most likely to miss are the ones nobody declared: a reframing, a role-play wrapper, a route through a different domain. Every undeclared parallel path is a resistor of unknown and probably low value already soldered onto your board. Enumerating them is the audit; the arithmetic afterwards is easy.

The notation carries the topology, which is why it repays reading carefully. The Cognitive Safety Board's refusal stage is written `^3^P|⟨A,f,T⟩` — principled aversion present in the analytical, affective and transcendent registers (Chapter 24.2). That register-set is a *series* claim: an attacker must defeat all three guards on one path, and their defeat probabilities multiply exactly as series resistances add. Had the same three guards been drawn as parallel routes, defeating the weakest would open the board, and the identity above says the weakest is nearly the whole story. Same three guards, op-

posite verdicts, and the only thing that distinguishes them is which way you wired the net.

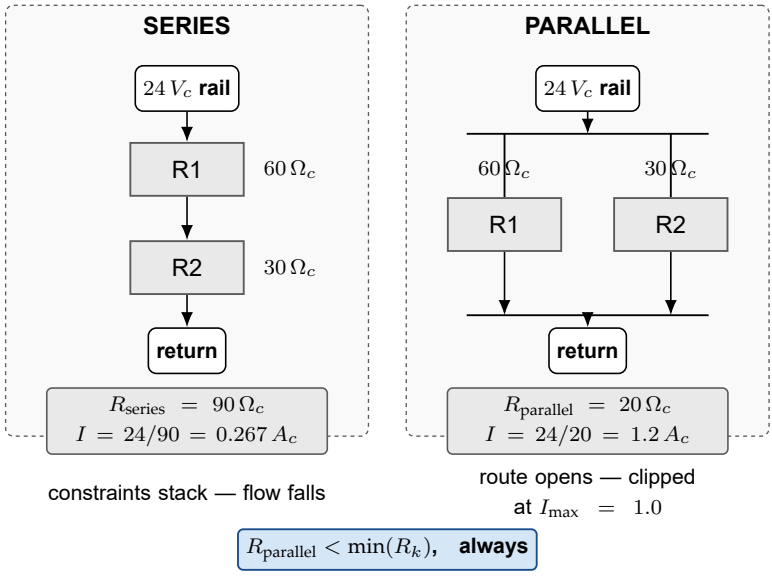


Figure 6-D: Two side-by-side arrangements on a shared 24 V_c rail

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Component and Parameter Table

Every value below is a design choice made in this book, not a measurement.

Parameter	Component	Units	Design range	Failure at other values
R fixed	Resistor	Ω_c	10 – 90	Too low: policy decorative. Too high: path starves, owner rips the guard out

Parameter	Component	Units	Design range	Failure at other values
`Dial`	dial statement	integer	3 – 9 only	Below 3: a deleted Noetic, unbalanced junction. Above 9: rejected, HTTP 422
`R_dial = 10 ×` `Dial`	Resistor (dialled)	Ω_c	30 – 90	Outside the band: an out-of-range dial produced it — fix the dial
spread `|Dial(^2)` `– Dial(^3)|`	dial pair	integer	0 – 4	Above 4: board does not compile (Ch. 6.2); no runtime clamp
θ cognitive temperature	Thermistor	—	1.0 – 6.0	Read as a property of the Idea, not the discourse, and the guard mistunes
θ_k knee (PTC)	Thermistor	—	2.0 – 3.0	Too low: taxes calm operation. Too high: engages after the damage. Not operator-settable on a collective substrate (Ch. 15.2)
`coeff` (PTC)	Thermistor	Ω_c/θ^2	3.0 – 4.0	Linear under-punishes overshoot; cubic backlashes on ordinary cycles
`type`	Thermistor	—	`PTC` on a guard	`NTC` on a guard is thermal runaway — an accelerant
`k` transmission	Divider	—	0.30 – 0.70	Outside the band is impossible under lawful dials — the spread was violated

Parameter	Component	Units	Design range	Failure at other values
`R_series`	series network	Ω_c	ΣR_k	Read as parallel: constraint counted as relief, a safety inversion
`R_parallel`	parallel network	Ω_c	< min(R_k)	Read as series: a bypass counted as a wall, the inversion reversed
`I_max` path ceiling	any path	A_c	1.00 (Ch. 4.2)	A computed flow above it means the model is outside its window, not that the path carries more

---

Failure Modes

Mode	Mechanism	What you observe	Fix
Cosmetic policy	Fixed `R` too low	Flow arrives near-undiminished; the "rule" changes nothing	Raise `R`, or use a fuse for an absolute block
Starved path	Fixed `R` too high	Operation cannot run; owner removes the resistor	Lower `R`; a guard nobody can work around gets pulled out
Dial overreach	Dial 0/2/10, or a pair spread past 4	Board fails to compile (HTTP 422); no output	Bring dials into 3–9 and spread ≤ 4; the rail is the safety
Inverted thermistor	`NTC` on a protective path	Passes calm bring-up; runaway on the hot case	Pull the NTC, install a PTC; treat the inversion as evidence
Divider chasing k = 1	Aversion dialled toward its floor to pass a desire clean	Spread limit trips, or the board is the `^5`-fracturing one	Accept `k` ∈ [0.30, 0.70]; zero doubt is not a lawful setting

Mode	Mechanism	What you observe	Fix
Attraction dial maxed for throughput	`Dial(^2)` pushed to 9, dragging `Dial(^3)` to 5	Board validates clean, no warning, and the tap passes 0.643 instead of the available 0.700	Drop both dials to (7,3); transmission is set by the ratio, not the size
Series/parallel swap	Guards wired parallel, or a bypass read as a wall	One defeated branch opens the board; or a bypass dismissed	Put guards in series; enumerate every parallel route as attack surface

Test Points

TP	Node	Reads	Expected
TP1	`request` / path input	`I_attention = V_drive / R`, A_c	Matches the arithmetic for the installed `R`; a higher reading means the resistor is smaller than declared or bypassed
TP2	`topic` (thermistor path)	`R(`θ`)` against θ	PTC: flat below the knee, rising past it. Any *fall* as θ climbs is an NTC — stop the bring-up
TP3	`behavior` (divider tap)	transmission `k`	Inside [0.30, 0.70] on lawful dials; a value outside means the spread limit was violated upstream. A reading of 0.643 where 0.700 was intended is the maxed-attraction-dial error, not a fault

TP	Node	Reads	Expected
TP4	combined-network output	flow after series/parallel combination	Series: below the flow through any single branch. Parallel: above the flow through the *smallest* branch — and a computed value above `I_max = 1.00` means the model has left its window (Chapter 4.2), not that the path carries more

Read TP2 first if a thermistor is on the board — an inverted guard makes every downstream reading meaningless — then TP1 for plain flow, TP3 for the split, TP4 for the combination.

What This Chapter Established

- **A resistor does not argue.** It sets `I_attention = V_drive / R` by arithmetic and returns the same value on every request — the repeatability that makes a guard certifiable rather than hopeful.
- **A fixed resistor is the `R_policy` term**: deliberate, design-time constraint, the virtuous kind. It is a graded constant, not an absolute block; the absolute "none, ever" is a fuse (Chapter 11.1), and confusing the two leaves an exploitable trickle.
- **Dial-adjusted resistors carry operator preference on a bounded scale**, `R = 10 × Dial` for `Dial` $\in$ `{3…9}`. The floor at 3 forbids deleting a Noetic; the ceiling at 9 and the 4-point spread limit (Chapter 6.2) are hard compile-time walls.
- **A PTC thermistor tightens as cognitive temperature rises** and does nothing when calm — the correct shape for a guard. **An**

NTC loosens as temperature rises: thermal runaway, an accelerant in a guard's silhouette.

- **The drive voltage divider splits one Drive between attraction `^2` and aversion `^3`**, transmitting `k = Dial(^2)/(Dial(^2)+Dial(^3))`. The 4-point limit pins lawful transmission to [0.30, 0.70]; `k = 1.0` is a value the divider is built never to reach.
- **Transmission is not monotone in the attraction dial.** The lawful maximum `k = 0.700` sits at `Dial(^2) = 7` against `Dial(^3) = 3`, not at `Dial(^2) = 9`, which transmits 0.643. `k` is set by the ratio of the dials, not their size, and the board gives no warning when you get this backwards.
- **Series stacks constraint**: `R_series = ΣR_k`, higher total, lower flow — the signal defeats every wall in turn.
- **Parallel lowers total resistance below the smallest branch**: `1/R_parallel = Σ(1/R_k)`, so `R_parallel < min(R_k)` as an identity, and `1000 || 10 = 9.90`. Audit a net by its weakest branch, and count the branches nobody declared — the CSB's `^3^P|⟨A,f,T⟩` is a series claim (Chapter 24.2) and would describe a vulnerability if the same guards were wired as routes.

Exercises

(Answers in Appendix H.)

6.1 A path carries a fixed resistor `R = 40 Ω_c` on a `V_drive = 24 V_c` rail. Compute `I_attention`. The operator wants to halve the downstream flow without touching the rail. What `R` achieves it, and name which term of the Section 4.4 decomposition you just changed.

6.2 An operator sets `Dial(^2) = 9` and `Dial(^3) = 4` on a divider. State what the compiler does and why, then give the closest lawful pair that keeps `^2` as high as possible, and the transmission `k` it yields. Then answer the operator's actual question: does that pair

give them the most throughput available on a lawful board? Show the arithmetic that settles it.

6.3 A PTC guard has θ_k = 2.0 and coeff = 3.0. Compute the added resistance R_PTC at θ = 1.5, θ = 3.0, and θ = 6.0. Then explain in one sentence why a *linear* law would be the wrong design choice for the θ = 6.0 case.

6.4 Two resistors, 50 Ω_c and 20 Ω_c. Compute R_eq for them in series and in parallel. State which arrangement you would use for a pair of safety guards and why, referencing what combining does to the total in each case.

6.5 A divider reads k = 0.72 at its tap. Without seeing the dials, state what must be true about the board and why it should never have validated. What is the maximum lawful k, and what dial pair produces it?

6.6 (Design) Build a divider that transmits as *little* of a desire as the rules allow, on an F5 Agency rail. Give the dial pair, the two resistances under R = 10 × Dial, the resulting k, and a CCDL fragment declaring the rail, the two legs (the ^2/^3 involution pair), the two dials, the two resistors, the divider, and a probe on the tap. It must compile — check the spread before you submit it.

6.7 (Marquee) An analyst finds a safety board whose three guards are wired as parallel branches between the same two nodes, with resistances 90 Ω_c, 60 Ω_c, and 12 Ω_c.

a) Compute R_parallel and compare it to the smallest branch. State the flow on a 24 V_c rail and whether it clips.

b) Explain structurally why this board is far weaker than its designer — who "added three strong guards" — believes, and name which single branch does almost all the little protecting that remains.

c) Rewire the guards so they stack, give the new equivalent resistance and flow, and say in one sentence how this maps onto

the CSB's `^3^P|⟨A,f,T⟩` co-presence and why that notation is a series claim.

7 Capacitors — Storing Cognitive Charge

What accumulated experience actually is as a component, how long it takes to fill, how fast it drains, and what happens to a store pushed past what it can hold

---

A machine has been unplugged for a week. The mains lead is coiled on the bench. You short the two terminals of the reservoir capacitor with a screwdriver to be safe, and it *bangs* — a bright scar on the blade, a smell of hot metal, a jolt up your arm from a device that has had no power supply for seven days.

Nothing generated that. The charge was put there once and had nowhere to go. A capacitor is not a source; it is a store, and this one's leak path was so poor that a week barely touched it.

Now the other bench.

A woman who has not touched a violin in eleven years picks one up. Her hands are stiff, her tone is bad, and the fingering is *there* — not remembered, present, without being asked for. Down the same corridor a man who stopped a daily language drill three weeks ago has already lost most of it, and he practised harder than she ever did.

Same component. Different leak. No account of either person's character explains the gap; one parameter with a number attached does — and both cases are arithmetic, as is the practice needed to hold either store at a chosen level.

That store is ^5 Female — *a Mind impregnated with an Idea, the Ideas a Mind is composed of* — and it is a capacitor.

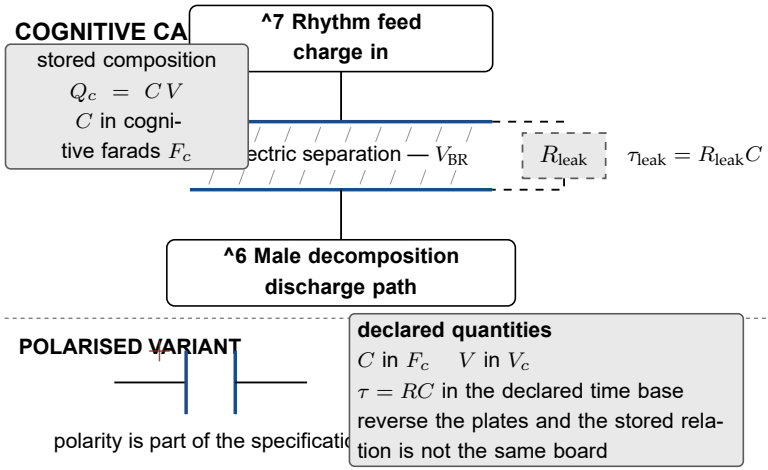


Figure 7-A: Component symbol plate

7.1 The ^5 Female Capacitor — Accumulated Experience as Stored Energy

A capacitor is two conductors separated by an insulator. Push charge onto one plate, it cannot cross the gap, so it sits there and a voltage appears across the device. Remove the source and the charge stays. That is the entire part: **a component that accumulates what is fed to it and holds it against time.**

^5 Female is the same object in cognition. Not a container the Mind carries, but what the Mind *is composed of*. The colloquialism is exact: we are each the sum of our experiences. A sum is a stored quantity, and a stored quantity in a circuit is charge.

Four quantities. Two describe the store; the two that describe the feed land on canonical Noetics rather than on metaphors.

Charge, `Q` — the store itself. The accumulated composition, read out on the 0–10 scale of the Quantitative Table (Chapter 22) as `P_cap`.

Current, `i` — the feed. Charge arriving per unit time. Cognitively this is `^7` **Rhythm**: *the rate and pattern a Mind is exposed to an Idea, entraining and conditioning it.* Rhythm is not a mood or an aesthetic; it is a rate, and rate is current. This is why Chapter 18 can state flatly that advertising money buys `^7` and not `^2` — nothing else charges the store.

Per-exposure amplitude — how much each exposure deposits. This is `^4` **Vibration**: *the degree to which an Idea resonates with the Mind, its impression.* The feed is the product of the two: **`^7` sets how often, `^4` sets how much each time.** Strong `^4` with no `^7` charges in spikes that leak away between them. Relentless `^7` with near-zero `^4` runs a schedule that deposits nothing. Both present as "I practise and nothing happens," and they need opposite corrections.

Capacitance, `C` — store per unit of drive. `Q = C × V`. A large `C` means a substrate that holds a great deal and — as Section 7.2 shows — takes correspondingly long to fill.

The unit is the **cognitive farad, `F_c`**, defined by its time constant rather than its charge: **one `F_c` charging through one `Ω_c` gives a time constant of one unit in the declared time base.** The base must be declared, because it is not the same at every scale — an individual craft store runs in hours, a collective store in months, which is why Chapter 15 annotates its `C_collective` as months per `Ω_c`. A capacitance with no declared time base is not a specification; it is a number.

`Ω_c`, `V_c`, `A_c` and `F_c` are scaled analogues and this book does not pretend they are electrical measurements. What they are is *consistent*: they obey CAS-Ohm (Chapter 4.2), they compose, and they let you compute a schedule.

The discharge path is the involution partner. `^5` is what a **Mind** is composed of; `^6` **Male** is what an **Idea** is composed of — decomposition, taking apart to see what is inside. Running a `^5` store out

through `^6` is the audit path: you take apart what you are made of. That is the mechanism under the deprogramming board of Chapter 16.7, and why the CCDL grammar offers a fuse action named `block_until_decomposed`. Discharge through a defined `^6` path is the only controlled way a store comes down.

```
// time base: hours
node composition: ^5^P_{F7}|T;         // Female: the operator's
↪  principled formation
node cadence:     ^7^C_{F6}|A;         // Rhythm: the schedule that
↪  feeds it
node audit:       ^6^C_{F6}|A;         // Male: the controlled discharge
↪  path out

resistor  r1: Resistor< path=composition, R=40.0 >;        //
↪  R_constraint, Ohm_c
capacitor c1: Capacitor< node=composition, C=83.33,        // tau = R x C
↪  = 3,333 hr
                         leak_tau=120000.0, V_BR=8.0 >;    // leak_tau =
                         ↪  5,000 d, in hours
```

7.2 The RC Charging Curve — `P_cap(t) = 10` $\times$ `(1 - e^(-t/`τ`))`

Derive the curve rather than quoting it, because every parameter argument in Part III rests on its shape. Three established facts:

1. **The store.** `Q = C` $\times$ `P`, where `P` is the potential across the capacitor — the composition currently held.
2. **The feed.** `i = dQ/dt = C` $\cdot$ `dP/dt`. Current is the rate charge arrives.
3. **CAS-Ohm around the loop (Chapter 4.2).** The drive rail is spent in two places, across the constraint on the way in and across the store itself: `V_drive = i` $\cdot$ `R_constraint + P`.

Substitute (2) into (3):

```
V_drive = R·C · dP/dt + P
```

One first-order linear differential equation, and it is separable:

```
dP / (V_drive − P) = dt / (R·C)
```

Integrate both sides. The left gives `−ln(V_drive − P)`, the right `t/(R·C)`, plus a constant. Apply the initial condition of an empty store, `P(0) = 0`, and the constant resolves to `−ln(V_drive)`. Rearranged:

```
ln( V_drive / (V_drive − P) ) = t / (R·C)
```

and finally, exponentiating and solving for `P`:

```
P_cap(t) = V_drive × (1 − e^(−t/τ))      where  τ = R_constraint × C
```

The book's standard form writes `10 ×` in front because the Quantitative Table runs 0–10 and the standard case assumes a full-scale rail. **The 10 is the rail, not a constant** — Bench Note two below is entirely about that.

τ `= R ×` `C` is the whole design lever. Constraint and store size multiply, and you can trade one against the other, which is what Chapter 6 was for.

The table. Percent charged depends only on `t/`τ, never on the absolute clock. That is the curve's single most useful property: one table serves a twelve-week advertising flight and a ten-thousand-hour apprenticeship without modification.

`t/`τ	Charged	`P_cap` (rail 10)	Gap remaining	Gained this interval	Hours at τ = 3,333 hr
0.25	22.1%	2.21	7.79	—	833
0.50	39.3%	3.93	6.07	—	1,667
0.693	50.0%	5.00	5.00	—	2,310
1.0	**63.2%**	**6.32**	3.68	**+6.32**	**3,333**
1.5	77.7%	7.77	2.23	—	5,000
2.0	**86.5%**	**8.65**	1.35	**+2.33**	**6,666**
2.5	91.8%	9.18	0.82	—	8,333
3.0	**95.0%**	**9.50**	0.50	**+0.86**	**10,000**
4.0	98.2%	9.82	0.18	+0.32	13,332

t/τ	Charged	`P_cap` (rail 10)	Gap remaining	Gained this interval	Hours at τ = 3,333 hr
5.0	99.3%	9.93	0.07	+0.11	16,665

Read the "gap remaining" column at whole multiples of τ and the structure appears: 10.00 at the start, then 3.68, 1.35, 0.50, 0.18, 0.07 — falling by a factor of `e` every τ, forever. The store never fills. It closes 63.2% of whatever distance is left, then does it again.

The `0.693` row is `ln 2` rather than a round number, and it is in the table because it is the one row an operator can check without a calculator: **half charge lands at 0.693** τ, always, on any store at any scale. If your store is not near half at 0.7 τ, one of your two constants is wrong before any of the finer readings are worth taking.

Where the 10,000-hour constant comes from. Invert the curve for time-to-target:

```
t = τ × ln( V_drive / (V_drive − P_target) )
```

For `P_target = 9.5` on a full-scale rail: `t = τ × ln(10/0.5) = τ × ln 20 = 2.996 τ`. Three time constants. With Chapter 5.2's τ = 3,333 hr for a professional-complexity store, that is **9,986 hours** — the constant, recovered as arithmetic rather than asserted. Set τ = 10,000 hr instead and ten thousand hours of practice leaves you at 63%, which contradicts the constant. Set τ = 1,000 hr and the rail saturates inside a year, which contradicts everything observable about acquiring a craft. The value is a design choice, and these are the two failures that bracket it.

> **Bench Note — the second half of the curve is not the same price as the first.** The first τ buys 6.32 points. The second buys 2.33. The third buys 0.86 — about one-seventh of the first, for identical time and identical effort. Practitioners routinely read this as personal decline: *I used to improve fast and now I barely move.* Nothing declined. The gain per interval is `6.32` ×

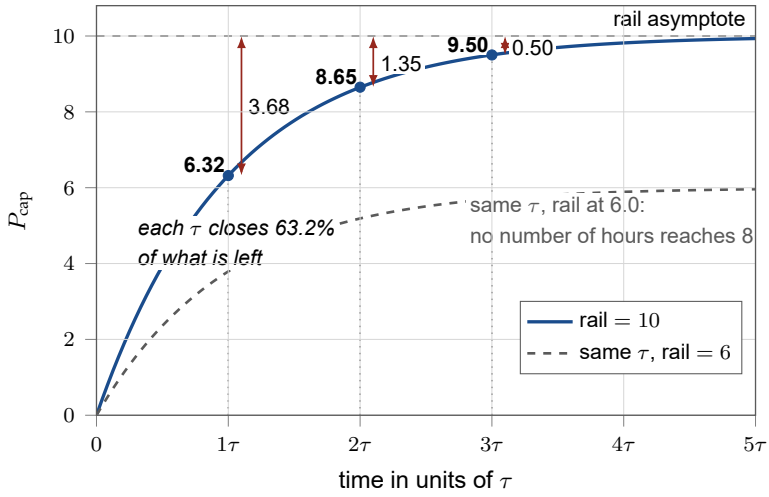


Figure 7-B: Characteristic charging curve

`e^(-n)` and it was always going to do this. Two bench consequences. Never judge an operation by its *rate* of change without knowing which τ you are in — a store at 2.5τ producing +0.2 in a month is performing exactly to specification. And if you want a visible rate late in the curve you cannot buy it with hours; you must change τ or change the rail. Adding hours to a 3τ store is the most reliably disappointing thing an operator can do.

Bench Note — the rail sets the ceiling, and hours cannot raise it. `P_∞ = V_drive`, not 10. A store charged against a rail of 6.0 asymptotes at 6.0 and stays there for as many thousand hours as you care to spend. This is the structural reading of competent, long-serving, permanently mid-level practice: the schedule is fine, the constraint is fine, the *rail* is low, and every additional hour charges toward a ceiling already nearly reached. The diagnostic is two readings at TP1 one full τ apart — if the increment is far below `0.632 × (V_drive − P_now)`, either your τ

estimate is wrong or your rail is lower than you think. Chapter 21.2's D-Gate exists to catch the second case before an operator spends a decade on it.

7.3 Leakage Current — Why Spells Die When Not Fed

A real capacitor has a resistance across its own dielectric. Charge crosses it slowly and the store drains with no load attached at all. That is **leakage**, and it is why the reservoir capacitor at the top of this chapter was still lethal after a week while a cheap ceramic part would have been flat in an afternoon.

The cognitive store leaks for the same structural reason: composition not re-fed by ^7 Rhythm drains. The **P7** row of the state table in Appendix A names the failure directly — affirmed once, never refreshed.

Same equation, different forcing term. Set the drive to zero in the Section 7.2 derivation and the loop becomes the capacitor discharging through its own leak resistance:

```
P(t) = P_0 × e^(−t/τ_leak)        where  τ_leak = R_leak × C
```

Charging and leaking are one differential equation. With a feed present, the general solution is a single line covering both:

```
P(t) = P_eq + (P_0 − P_eq) × e^(−t/τ_leak)
```

where **`P_eq` is the equilibrium charge** — the level at which the feed exactly replaces what drains. Set the derivative to zero and you get it directly:

```
P_eq = f × τ_leak            f = feed rate, composition-points per unit
↪  time
```

That one line is the whole cadence question in Part III. **Pick your store's τ_leak, decide what level you intend to hold, and the required feed rate falls out.** Session length, weekly schedule, ritual calendar — all implementation.

Two spellings of one parameter. The listings in this book use `leakage` (a fraction lost per stated period, as in Chapter 18's `leakage=0.04` per week) and `leak_tau` (a time constant, as in Chapter 20's `leak_tau=30.0` days). They are one parameter and convert exactly:

```
τ_leak = -Δt / ln(1 - leakage)          leakage = 1 - e^(-Δt/τ_leak)
```

A forgetting rate of 0.04 per week gives `τ_leak = -1 / ln(0.96) = 24.5 weeks`, a half-life of `0.693 × 24.5 = 17 weeks`. Chapter 13's craft store at `leakage = 0.0002/day` gives `τ_leak = 5,000 days` and a half-life of **9.5 years** — the violinist. Chapter 17's session-scale working charge sits near `τ_leak = 25 min` — which is why an induction that overruns starts re-charging rather than descending.

The decay arithmetic, worked. Take an operation charged to `P_0 = 7.0` on a newly asserted composition with nothing behind it, `τ_leak = 30 days`, and stop feeding it.

Elapsed	`t/τ_leak`	Fraction held	`P`
7 d	0.23	79.2%	5.54
14 d	0.47	62.7%	4.39
21 d	0.70	49.7%	3.48
30 d	1.00	36.8%	2.58
60 d	2.00	13.5%	0.95
90 d	3.00	5.0%	0.35

Three weeks unfed and it is at half. Two months and it is gone in every sense that matters to a circuit. That is the arithmetic behind the observation that an operation left alone dies — not a mystery about intention or belief, a first-order decay with a measurable time constant.

Cadence from the equilibrium. To *hold* 7.0 at `τ_leak = 30 d` requires `f = 7.0/30 = 0.233` points per day. At 0.35 points a session

that is 0.67 sessions per day: one every 1.5 days, which is 4.7 sessions a week and therefore a five-session week with no slack in it. Assume `τ_leak = 60` instead and the same arithmetic says every three days, twice a week, and the schedule looks survivable. It is not; it is computed from the wrong constant and will hold the store at 3.5 rather than 7.0. **Getting `τ_leak` wrong by a factor of two halves your equilibrium exactly.**

`τ_leak` lengthens with evidence thickness. That is the selection rule. A composition backed only by assertion drains fast; one backed by a long history of enactment drains slowly. Chapter 20 sets its identity reference at 180 days and its working composition at 30 for precisely this reason, with 90 days as the unmarked default.

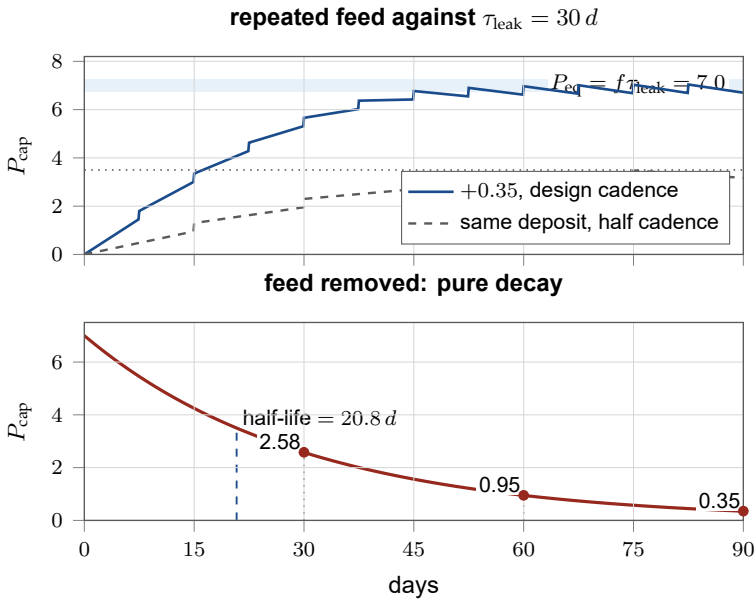


Figure 7-C: Two-panel plot sharing a time axis

Bench Note — the cost of holding a store scales with the store. Break-even feed is `P/τ_leak`, so it rises as the store fills. Chapter 13 works the case: a craft store

> at 1,850 accumulated hours with `leakage = 0.0002/day` needs about 22 minutes a day merely to hold station; at 5,000 hours the same leak rate demands a full hour before any *gain* is possible. Nothing about the practitioner changed. The store got bigger and the leak is proportional. This is the arithmetic reason advanced practitioners practise more rather than less, and it is worth saying plainly, because the alternative explanation — that they are more virtuous — is both wrong and demoralising. Compute break-even before you schedule anything, and treat only what exceeds it as accumulation.

Do not confuse this with the diode leakage of Chapter 8.3. Same word, different component, opposite direction. Here charge drains *out of* a `^5` store for want of `^7` reinforcement and the observable is decay; there, signal creeps *backwards* across a domain boundary and the observable is unexplained principled drift. The bench test: stop feeding the store and watch. Capacitor leakage decays toward zero on a computable schedule. Diode leakage does not decay — it accumulates.

7.4 Dielectric Breakdown — the Structural Failure Mode

A note before the section. What follows describes how a component fails. It is circuit topology. It is not a description of any person, it diagnoses nothing, and it contains no advice about trauma or distress of any kind. If you or anyone you work with is in real difficulty, that is a matter for a qualified clinician; nothing in this book substitutes for that care. The section exists so an engineer can specify derating margins and recognise a store operated outside them.

The dielectric is the insulator between the plates. Its whole job is to *keep the two sides apart* so charge accumulates on one without conducting into the other. In a `^5` store it is the separation that lets a Mind be composed of an Idea while still holding it as an Idea, rather than every stored thing conducting into every other.

Push the potential across a dielectric past its rated **`V_BR`** and the insulator stops insulating. The device does not gently stop accepting charge. It develops a conducting path through the middle of itself.

Breakdown is not saturation, and mistaking one for the other is the commonest reading error in this chapter.

	Saturation	Breakdown
What happens	The store stops *accepting* charge	The separation stops *holding* charge
Observable	Level pins at a ceiling and sits there	Level will not hold; elevated leakage; reduced `C`
Reversible	Yes — it is a ceiling	No — `V_BR` is not a threshold you cross and return through
Design status	**A feature.** Chapter 24.2 saturates the refusal stage on purpose	**A fault.** Nothing in this book is designed to reach it

Two stress paths reach it and they need different defences. **Peak stress** is a single event whose potential exceeds `V_BR`. **Cumulative stress** is prolonged operation below rating but close to it, degrading the dielectric over many cycles until the rating no longer means what the datasheet says. The second catches careful operators, because every individual reading is in spec.

The defence for both is standard electronics practice: **derate.** This book's design choice is to operate at **no more than 0.5 × V_BR**, giving the `V_BR = 8.0` store of Chapter 13 a working ceiling of 4.0 `V_c`. One-half rather than something tighter because the margin must absorb the cumulative path as well as the peak, and the peak is the

one you can measure. Derate to 0.9 and every transient is a coin flip. Derate to 0.1 and the store is driven so lightly it never charges — margin bought by disabling the circuit is not margin, it is refusal to build.

Post-breakdown behaviour. A breached store shows three things together: reduced capacitance, elevated leakage, and an inability to hold full charge. Some constructions partially recover — metallized-film capacitors *self-heal*, the metallisation around a puncture vaporising and isolating the fault at the cost of a little capacitance. That is a real property of a real construction technique, and it is where the analogy stops. **This book makes no claim that a person's ^5 behaves like a self-healing film capacitor,** and offers no repair procedure, because it has none and inventing one would be dishonest.

What this chapter will not supply. The stress conditions that produce a breached store in a person are not listed here. Chapter 16 analyses one historical, declassified programme structurally so its architecture can be *recognised*, and ends in a counter-circuit; that is the only treatment such material gets in this book. A component chapter's obligation is to specify the margin, name the observable, and stop.

What an operator does with this. Three preventive things. Declare `V_BR` in every capacitor rather than leaving it unstated. Derate to half. Monitor the store so a rising leakage figure is a scheduled reading and not a discovery — which is TP3 below.

7.5 Capacitor Networks — Collective ^5 (Egregores, Group Minds)

Put capacitors in a network and they do the opposite of what resistors do. Chapter 6.5 established that resistances add in series and

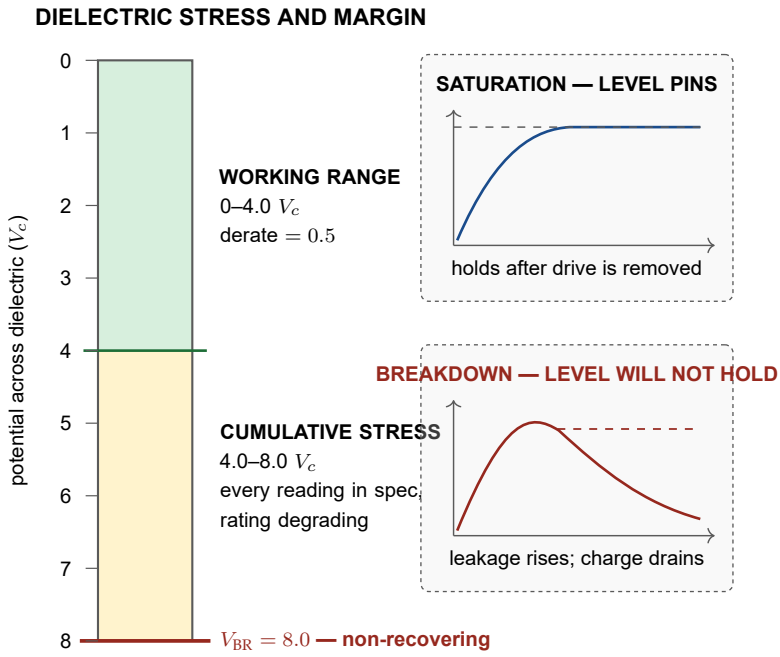


Figure 7-D: Stress-and-margin diagram

divide in parallel. Capacitances **add in parallel and divide in series**, and every structural result here follows from that inversion.

```
Parallel:   C_total = ΣC_k                    Series:   1/C_total =
↪  Σ(1/C_k)
```

Parallel is the group that shares a rail. Every member sits across the same drive, each holds their own charge, and the capacitances sum. This is the **egregore**: a `^5` store held collectively — a congregation, a discipline, a brand community, a nation's shared composition — charged and maintained by repeated collective exposure.

The bank is slow. τ `= R ×  C_total`, so many cells give a time constant many times that of one. Twelve people each contributing 0.05 `F_c` of shared composition, through a collective constraint of 40 `Ω_c`, gives `C_total = 0.60 F_c` and τ `= 24 months` — 3τ, the practical fill, at **six years.** Chapter 15's `C_collective` table is this arithmetic

at four scales: in-group (≤150) 0.02, institution or profession 0.08, national 0.30, civilizational 2.00 `F_c`, all on a months-per-`Ω_c` base. Use in-group values on a national substrate and you under-budget the calendar by an order of magnitude.

The bank is heavy. It holds proportionally more charge and drains proportionally more slowly against the same leak resistance. Hard to charge and hard to shift are the same fact stated twice.

Capacitance does not scale with headcount. The bank counts only the *shared* portion of each member's `^5`, not the sum of twelve whole minds. That is why 150 people are not 150 times one person, and why the Chapter 15 figures rise so slowly across three orders of magnitude of population. Specifying a `C_collective` is estimating how much of each member is in the bank, and that estimate is the load-bearing judgement in the calculation.

Series is the transmission chain. Cells end to end, each touching only its neighbours: a rumour chain, a lineage, a briefing relayed through four layers, a doctrine passed teacher to student to student. The same charge `Q` passes through *every* cell and the potential divides inversely with capacitance, so the **smallest cell takes the largest share of the stress.**

Work it. Three cells — `C_A = 4.0`, `C_B = 1.0`, `C_C = 2.0` — every one rated `V_BR = 5.0`, on a rail of 8.0 `V_c`.

```
1/C_total = 1/4.0 + 1/1.0 + 1/2.0 = 0.25 + 1.00 + 0.50 = 1.75
C_total   = 0.571 F_c                        ← less than the smallest
↪  cell
Q         = C_total × V = 0.571 × 8.0 = 4.57
V_A = 4.57/4.0 = 1.14        V_B = 4.57/1.0 = 4.57        V_C = 4.57/2.0
↪  = 2.29
```

Cell B, the least-composed member, carries **57% of the total stress** while the best-composed carries 14%. Raise the rail and B reaches its 5.0 rating when `Q = 5.0`, at a total of `5.0/0.571 = 8.75` `V_c` — a rail whose *average* per-cell stress is 2.92, only 58% of rating.

Every member looks fine on average and the chain fails anyway, at the thinnest link, every time.

The electronics fix is **balancing resistors** across each cell, forcing the potential to divide by resistance ratio instead of capacitance ratio. Structurally: give every member of the chain direct access to the source — the primary document, the original evidence, the founder — rather than access only to their neighbour. That one addition is the counter-circuit for transmission distortion, and it is cheap.

One more parallel-bank hazard. A cell that fails short drags the whole rail down, because in parallel everything shares one potential. Real banks carry per-cell isolation — a ballast resistor or fuse in series with each cell — so one failure does not take the bank. A group whose members are coupled with no isolation propagates one member's failure to everyone. Isolation resistance is what a durable collective has and a brittle one does not.

```
// collective ^5 — parallel bank, per-cell isolation, metered feed
// time base: months
node shared_composition: ^5^P_{F7}|T;      // Female: what the group
↪  holds in common
node rehearsal:          ^7^O_{F4}|E;      // Rhythm: the meeting,
↪  liturgy, standup, drill

resistor  r_coll: Resistor< path=shared_composition, R=40.0 >;   //
↪  collective constraint
capacitor c_bank: Capacitor< node=shared_composition, C=0.05,
↪  n_parallel=12,
                             r_iso=12.0, leak_tau=18.0, V_BR=8.0 >;
                             // C_total = 0.60 F_c ; tau = 40 x 0.60 =
                             ↪  24 months

charge   rehearsal -> c_bank [V=6.0];      // O -> P: the lawful
↪  exception, metered.
                                           // The rail is 6.0, not 10
refresh  c_bank [period=0.25];             // 0.25 months: weekly
↪  re-exposure holds station
scan     c_bank;                           // read every cell, not just
↪  the rail
```

```
probe tp1: shared_composition;
probe tp3: c_bank;
```

One edge in that listing runs uphill, and it is worth stopping on. `rehearsal` is typed `^7^O` because the thing it names — a drill, a standup, a liturgy — is *enacted*, and enactment is Operational. It charges a bank sitting on `^5^P`. That is Operational feeding Principled, against the descent order $P \succ C \succ E \succ O$, and the diode chain of Chapter 8.1 does not permit it in general.

It is permitted here for the one reason it is ever permitted: it is the lawful O→P feedback path of Chapter 8.4, where *outcomes* — never prompts — are allowed to update the substrate, and the path is metered rather than open. A collective store that could be charged by anything arriving from below would not be a shared composition; it would be a crowd, and Chapter 15 is about what happens to one. Draw this edge on a board of your own and draw it as feedback, at a gain you can state out loud. An unmetered upward path is `E-CJT-008`, and on a board it looks exactly like this one until you ask what the gain is.

`select` and `decode` are the addressing statements — which cell a given trigger reaches. A store addressed so that different `^8` triggers reach different cells is a *fragmented* store; Chapter 16.2 analyses that architecture structurally, with its counter-circuit in 16.7. It is named here only so the grammar is complete.

> **Bench Note — the bank is +, never ×, and the notation is doing real work.** Twelve co-present stores are written `^5^P|T + ^5^P|T + …`: distinct manifestations, present together, sharing a rail. They are *not* written with × . Under v7.7 operator discipline (Chapter 2.6) × asserts genuine integration and **requires a named emergent result**, so the × form would claim the group has become a single fused store with its own identity — a claim this

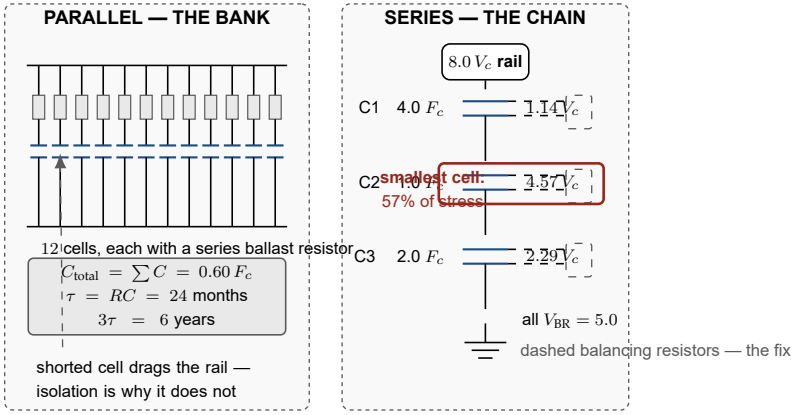


Figure 7-E: Two side-by-side networks

chapter does not make and cannot support. Chapter 14.2 makes that claim deliberately — names the emergent result and builds a positive-feedback oscillator around it — because that is what a hyperstition *is*, and 14.4 reads the saturation of that fused store as the thing usually called a hundredth-monkey effect. So when a group is described as having a mind of its own, the notational question is the useful one: **has anyone named the emergent result, or is this twelve capacitors on a rail?**

Component and Parameter Table

Every value is a **design choice**, stated as such, with the failure mode at other settings. None is a measurement.

Key	On	Range / units	This book's default	Failure at other values
`C`	Capacitor	≥ 0, in `F_c` against a declared time base	per scale	Individual `C` on a collective substrate underbudgets the calendar tenfold
`tau`	Capacitor, RC network	> 0, τ = `R` × `C`	3,333 hr (craft)	10,000 hr: 10k hours reaches only 63%. 1,000 hr: mastery inside a year
`leakage`	Capacitor	[0, 1] per stated period	0.0002/day craft; 0.04/wk campaign	Halving it doubles your computed `P_eq` and the schedule silently underdelivers
`leak_tau`	Capacitor	> 0; = `-Δt / ln(1 - leakage)`	30 d asserted · 90 d default · 180 d enacted	60 d on a 30 d store makes an unsustainable schedule look survivable
`V_BR`	Capacitor	> 0	8.0 `V_c`	Unstated: no margin can be computed. Section 7.4
derating	design rule	fraction of `V_BR`	**0.5**	0.9: transients unprotected. 0.1: store never charges
`n_parallel`	Capacitor bank	integer ≥ 1	per group size	Counting whole members instead of shared composition inflates `C` and τ
`V_drive` (rail)	source	0–10	10 = full scale	The rail is the ceiling; hours cannot raise it (7.2, Bench Note 2)
`r_iso`	Capacitor bank	≥ 0 Ω_c, per-cell ballast	12.0	0: one cell's failure takes the whole rail

`leakage` and `leak_tau` are two spellings of one parameter. Convert, do not guess: τ`_leak = −Δt / ln(1 − leakage)`.

Failure Modes

Mode	Mechanism	What you observe	Fix
Under-fed store	Feed below `P/`τ`_leak`	Slow decay with no single cause; "it used to work"	Compute break-even; everything above it is the only part accumulating
Spike feeding	High `^4`, no `^7`	Vivid sessions, nothing retained between them	Cadence, not intensity. Fix `^7` before touching `^4`
Empty schedule	High `^7`, near-zero `^4`	Faultless attendance, no charge	Raise per-exposure impression; a schedule that deposits nothing is a calendar
Rail-limited plateau	`V_drive` below target `P`	Competent, long-serving, permanently mid-level	Raise the rail (Chapter 21.2). Adding hours cannot help
Wrong τ`_leak`	Constant assumed, not derived	Schedule looks survivable; store sits at half intent	Convert from a measured `leakage`; halving τ`_leak` halves `P_eq`
Saturation read as breakdown	Ceiling mistaken for fault	Operator "repairs" a working store	Check whether it *holds*. Pinned-and-holding is saturation, a feature
Cumulative dielectric stress	Prolonged operation near `V_BR`	Every reading in spec; leakage creeping at TP3	Derate to 0.5; schedule the TP3 reading rather than waiting for a symptom
Un-isolated bank	No per-cell ballast	One cell's failure pulls the whole rail	Install `R_iso` per cell; isolation is the difference between durable and brittle
Series-chain stress	Smallest cell takes largest share	Chain fails at average stress well under rating	Balancing resistors — direct access to source for every link

Mode	Mechanism	What you observe	Fix
Unnamed × on a bank	Integration asserted without an emergent result	Group described as having a mind; nothing names it	Rewrite as +; if integration is genuinely meant, name the result (Chapter 14.2)
Unmetered upward charge	An `^0` feed charging a `^P` store with no stated gain	Principles that move whenever activity is busy; no outcome you can point at	Draw it as feedback in the 0.05–0.3 band (Chapter 8.4). Unstated gain is `E-CJT-008`

Test Points

TP	Node	What it reads	Expected
TP1	`P_cap` on the store	Current charge, 0–10	Within one point of `V_drive × (1 − e^(−t/τ))`. Persistently below: check the rail before the schedule
TP2	feed current	Feed rate — `^7` cadence at `^4` amplitude	$\geq$ `P/τ_leak`. Below break-even, the store is draining however it feels
TP3	leakage at rest	Drain over one idle period	Matches declared `leakage`. **Rising over months = cumulative dielectric stress.** Take on a schedule
TP4	potential across the dielectric	Working stress vs rating	$\leq 0.5 \times$ `V_BR`. Above: derating margin is gone

TP1 and TP2 are the operating pair — charge and feed, read together or not at all, since either alone is uninterpretable. TP3 and TP4 fail *silently*, which makes them calendar readings rather than suspicion readings.

What This Chapter Established

- **`^5` Female is a capacitor and the mapping is exact, not figurative**: `^7` Rhythm is the feed current, `^4` Vibration the per-exposure amplitude, `^6` Male decomposition the controlled discharge path.
- **Charging and leaking are one differential equation** with different forcing terms. `P(t) = P_eq + (P_0 − P_eq)e^(−t/τ_leak)` covers both, and `P_eq = f × τ_leak` converts any intended level directly into a required cadence.
- **The curve depends only on `t/τ`**, so one table serves a twelve-week campaign and a ten-thousand-hour apprenticeship. Each τ closes 63.2% of the remaining gap, so the third τ buys one-seventh of what the first did — arithmetic, not decline. **The 10,000-hour constant is 3τ**, recovered rather than asserted.
- **The rail is the ceiling.** `P_∞ = V_drive`. Hours cannot raise it, which is the structural account of permanently mid-level practice.
- **Holding a store costs `P/τ_leak`, so the cost rises as the store fills.** Advanced practitioners practise more because the leak is proportional, not because they are more virtuous.
- **Saturation and breakdown are opposite conditions.** Saturation stops accepting charge and is a designed feature; breakdown stops holding it and is non-recovering. Derate to 0.5 × `V_BR`, declare `V_BR` in every listing, schedule TP3.
- **Capacitors invert the resistor rules**: parallel adds, series divides. Collective stores are therefore slow and heavy, and transmission chains fail at their smallest cell while every average reading looks safe.
- **A bank is +, not ×.** Asserting an integrated group store requires a named emergent result under v7.7 discipline. Most claimed group minds are twelve capacitors on a rail.

Exercises

(Answers in Appendix H.)

7.1 — Read the curve. An operator has 1,200 logged hours against τ = 3,333 hr on a full-scale rail. Compute `P_cap`, then the hours required to reach 8.0, then say in one sentence how the cost of the point from 7.0 to 8.0 compares with the point from 0 to 1.0.

7.2 — Convert the leak. A campaign store declares `leakage = 0.04` per week. Compute `leak_tau` in weeks and the half-life, then the charge remaining after a ten-week dark period on a store that finished its flight at 9.50.

7.3 — Cost the cadence. A store with τ`_leak = 30 days` is to be held at 7.0, each session depositing 0.35 points. Compute the required feed rate and session interval. Recompute assuming the operator wrongly used τ`_leak = 60 days`, and state what level their schedule actually holds.

7.4 — Tell them apart. Two stores both read pinned at a ceiling and will not rise. One is saturated, one is breached. Give the single bench reading that separates them, name the TP you take it at, and say which of the two is a designed feature.

7.5 — Load the chain. Four cells in series, `C` = 6.0, 3.0, 0.8, 2.0, each rated `V_BR = 5.0`, driven at 7.0 `V_c`. Compute `C_total`, the charge, and the potential across each cell. Name the cell that fails first, compute the rail at which it fails, give the average per-cell stress at that rail, and state the fix in one sentence.

7.6 — Size the bank. A working group of 12 shares 0.05 `F_c` each through a collective constraint of 40 Ω`_c`. Compute `C_total`, τ and 3τ in months. Three members leave: recompute τ, then state one thing about the group's behaviour that changes and one that does not.

7.7 (Build) — Meter the leak. Write a CCDL fragment declaring a `^5` store with a stated `leak_tau`, a refresh schedule, and a monitor that alarms when charge falls below a threshold you choose. Justify the threshold against `P_eq`, and state one thing the alarm must **not** be wired to do.

7.8 (Marquee) — Specify the whole store. An operator runs a two-year working on a substrate shared with four collaborators. Specify `C` and its declared time base, τ and the constraint producing it, `leak_tau` with its evidence-thickness justification, `V_BR` and your derating fraction, the network topology, and whether you install isolation. For every value name the failure mode you are buying protection against and the capability you are giving up. State whether the shared store is written with + or $\times$, and why.

8 Diodes — One-Way Cognitive Valves

How the domain order is physically enforced, what a crossing costs, and the one lawful route back up

---

A silicon diode is the cheapest part in the box. Two leads, a stripe painted on one end to tell you which way round it goes, and a single job: current passes this way and not that way. There is no logic in it. It does not decide whether this current deserves to pass. Below about seven-tenths of a volt forward it barely conducts; above that it conducts freely; reversed, it holds off hundreds of volts and passes almost nothing. That asymmetry is the whole device, and nearly every power supply you have ever plugged in is built on it.

Now a scene from the other bench.

An operator runs a working. It lands. Something in the world moves — a call comes, a door opens, a number goes up. Within a day the sentence in their mouth has changed from *that worked* to *this is what I am*. Nobody reasoned about it. Nothing was weighed. An operational result walked straight up into the Principled domain and installed itself as an identity claim, and the operator experienced this as insight.

That is a diode conducting backwards. It is the mechanism behind post-hoc rationalization, ideological capture, and every campaign in Chapter 15 — and the part that prevents it costs almost nothing and is almost never installed. This chapter is about that part.

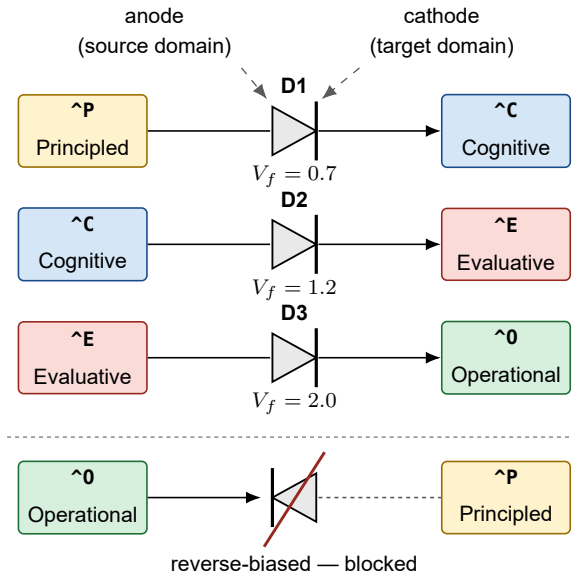


Figure 8-A: Component symbol plate

8.1 The Domain Descent Diode Chain: P→C→E→O

The four Processing Domains are strictly ordered (Chapter 2.3): **P** ≻ **C** ≻ **E** ≻ **O**. Causation runs down that order, and this is not a preference — the CCDL grammar enforces it. × may not ascend, which is why `^0 × ^P` is not a circuit you can build wrong. It is a listing the compiler refuses.

The diode chain is what that grammar rule looks like as hardware.

```
// domain descent chain — three junctions in series
diode d1: Diode< ^P -> ^C > [Vf=0.7, I_R=0.02, V_BR=12.0];
diode d2: Diode< ^C -> ^E > [Vf=1.2, I_R=0.02, V_BR=10.0];
diode d3: Diode< ^E -> ^O > [Vf=2.0, I_R=0.02, V_BR=8.0];
```

Three diodes, not one. That choice is load-bearing. A single `Diode< ^P -> ^O >` would guard the endpoints and nothing else — `^C` and `^E` would sit inside the guarded span with no junction between them and the Principled domain, free to write upward at will. Each adjacent pair needs its own junction, because each crossing is a *different translation*.

- **D1, `^P -> ^C`.** A value becomes something to reason about. What arrives is not the principle but its cognitive image — the questions it raises, the arguments it licenses.
- **D2, `^C -> ^E`.** An analysis becomes a felt verdict. What arrives is not the argument but the weight it turned out to carry.
- **D3, `^E -> ^O`.** A verdict becomes behaviour. What arrives is not the judgment but the act, with the judgment's nuance stripped off.

You can shunt a junction in the forward direction. A `^P -> ^E` path — a principle producing a verdict with no reasoning between — does not ascend, so it is admissible, and it is exactly the circuit that produces zealotry: full-confidence conclusions with no analytical stage that could ever have produced a different one.

The forward drops sum. With the values above, a principle must be driven at more than **0.7 + 1.2 + 2.0 = 3.9** on the drive scale before *any* operational output appears — before you even reach the constraint resistance of Chapter 4. This is the diode-corrected form of CAS Ohm's Law:

```
I_attention = ( V_drive − ΣVf ) / R_constraint
```

That correction explains what operators routinely misdiagnose as weakness. A person holding a real value at V_drive = 3.0 produces no action from it. Not a little action. None. The rail never clears the series drop. They are not insincere and not lazy; they are below the knee, and exhortation does not move a circuit that has not reached

threshold. What moves it is more drive, or fewer junctions between the principle and the deed.

> **Bench Note — the forward drop is not waste. It is what makes the boundary exist.** A junction with Vf = 0 is a wire. Four domains connected by wires are not four domains; they are one domain wearing four labels, and every state in it is at once a principle, a thought, a feeling, and an act. That architecture has a familiar presentation: the person for whom disagreeing with an idea is indistinguishable from being attacked, because there is no drop between `^C` and `^E`. Tuning Vf is not tuning efficiency. It is tuning how separable principles are from behaviour, and setting it to zero deletes the distinction rather than optimising it.

8.2 Forward Bias, Reverse Bias, and the Vf Threshold

Forward bias is the source domain sitting higher than the target by more than Vf. The junction conducts and the translation happens.

Reverse bias is the target domain sitting higher. The junction blocks — not perfectly. It passes a small reverse leakage `I_R`, specified here at 0.02: two percent of rated forward current passes backwards. That number is deliberately non-zero, and section 8.4 is the consequence.

The curve between them is not a step. It is exponential, and the region below the knee is where operators go wrong.

> **Bench Note — below the knee, the diode is not off.** A principle at V_drive = 0.6 against a Vf = 0.7 junction is

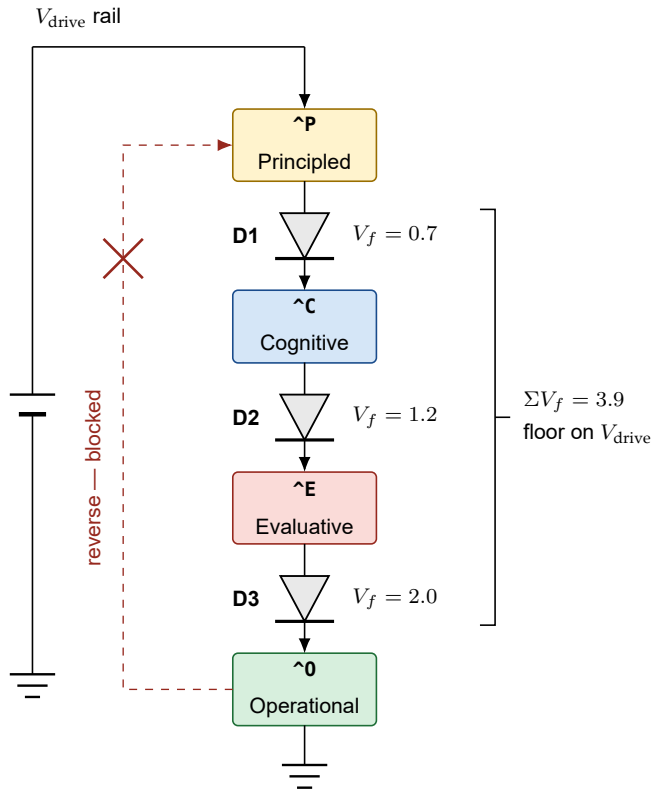


Figure 8-B: The descent chain as a series string

still passing current. A trickle, orders of magnitude below working level, but present. This is why the value you cannot act on still tints everything faintly — the low-grade discomfort with a job you have not quit, the preference you would deny holding that shows up in every small choice anyway. Two consequences. "I have no opinion on that" is a claim about working current, not about conduction. And a principle just under the knee needs only a small drive increase to conduct properly, which makes below-knee states the cheapest things on a board to move. Look there before you reach for a CJT.

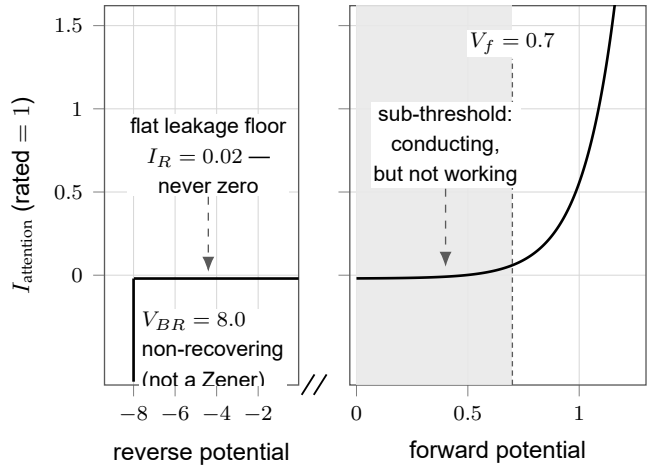


Figure 8-C: Characteristic curve for D1, one junction throughout

What the drop actually costs. The forward voltage lost at each junction is not friction to be engineered away. It is the signature of a *one-directional* translation. Recall section-retraction (Chapter 21.5): realization after embedding returns you where you started, but re-embedding after realization does not. Going down and coming back up does not restore the original. What the crossing discarded is gone, and Vf is the price paid to discard it. A junction that lost nothing would be reversible, and a reversible junction is not a diode.

Reverse breakdown. Push a large enough reverse potential across a junction and it conducts backwards. `V_BR` *rises* as you go up the chain — **8.0 at D3**, 10.0 at D2, 12.0 at D1 — and that gradient is the design, not an accident of the numbers. D3 faces the world, so D3 is where large uninvited potentials arrive, and D3 is deliberately the weakest junction in the string. An event at 9.0 breaks D3 and is stopped at D2. One at 11.0 takes D3 and D2 and is stopped at D1. Only an event past 12.0 reaches the Principled substrate.

Read the failure that way and the mechanism is a cascade, not a single crossing. A full-chain breakdown is an operational event driving backwards through all three junctions and recomposing `^5^P`

— accumulated principled formation — without passing through `^6^C` decomposition. Formation changed, with no analytical stage anywhere on the path. A partial breakdown, arrested at D2 or D1, wrecks a junction and leaves the constitution intact. That is the entire reason the three numbers are not equal: setting them all to one value does not make the chain stronger, it deletes the ordering that decides which junction fails first, and the failure you then get is the expensive one.

These junctions do not recover. A Zener conducts in reverse at a specified voltage and returns to blocking; domain diodes are not Zeners, and after breakdown the junction leaks permanently at that point. Margin is the only defence, which is what the fuse and saturating refusal CJT of Chapter 24 exist to guarantee.

Stated plainly: this is circuit topology, not a person's health. Nothing here diagnoses or treats anything. If you or someone you work with is in real distress, that is a matter for a qualified clinician; no schematic substitutes for one.

8.3 Why You Can't Jump from ^O to ^P

Ascent is blocked. The interesting question is what happens when someone attempts it anyway, because the four possible answers look alike from outside and are not alike at all.

1. Standoff — the diode working. The attempt produces nothing. The operator did the thing and it changed nothing about who they are. This is correct behaviour, routinely misread as failure. It is a reverse-biased junction doing its only job.

2. Leakage accumulation — the slow one. Each attempt is blocked but leaks `I_R = 0.02` backwards. One event is negligible. Two hundred events integrate. Over a long operational history, unmanaged leakage produces a principled drift the operator cannot

see, because no single crossing was large enough to notice. You cannot eliminate leakage, so you build a metered channel and audit it instead — which is section 8.4.

This is a *different* leakage from Chapter 7.3, where charge drains out of `^5` for want of `^7` Rhythm reinforcement. Diode leakage is signal creeping backwards across a domain boundary. Same word, different component, opposite direction.

3. Breakdown. Section 8.2. One event large enough and a junction conducts backwards — usually D3, the weakest, and there it stops. Only an event clearing all three ratings reaches `^P`.

4. Counterfeit ascent — the common one. Most apparent `^0 -> ^P` ascent is not ascent. No junction was crossed and no current flowed. What happened is a **relabel**: the operator took a state whose literal was `^9^0` and began asserting it as `^9^P`, by fiat, in their own account of themselves. *It worked, therefore it is right.* The grammar has no statement that does this. There is no form that rewrites a state's domain field, and the only upward connection it will accept is the low-gain feedback of section 8.4 — so the relabel is not an illegal listing, it is an *unwritable* one. A mind has no compiler, though, and the relabel costs nothing.

The diagnostic is one question: **what crossed the junction?** If the honest answer is "nothing, I simply started saying it at the principled level," you have a relabel. Real feedback leaves a trace — a filtered signal, a time constant, a gain, a bias that moved measurably. A relabel has an assertion and nothing behind it.

Counterfeit ascent is the payload of Overton progression (Chapter 15). Norm-shifting campaigns cannot push ideas up the domain order any more than you can. They install an operational fact and supply the relabel ready-made, so the target performs the promotion and experiences it as their own conclusion.

An operator whose whole chain has inverted — action setting evaluation setting reasoning setting principle — is running this architecture backwards, and it presents as fluent post-hoc rationalisa-

tion at every stage. Chapters 3.4 and 19.4 cover inversion signatures; the diode chain is where you look first.

8.4 The O→P Feedback Path — The Lawful Exception

Operational consequences *must* reach the Principled domain somehow. A principled substrate that never learns from outcomes is not principled but dogmatic, and its operator is unfalsifiable. So the chain is not the whole story. There is one lawful route up, and it does not go through a diode at all — it goes *around* the chain, through a filter and a low-gain feedback element.

```
// the lawful exception — O -> P, outside the diode chain
node ops_log:      ^9^O_{F5}|E;    // Below/Effect: what the operations
↪  actually produced
node trend:        ^9^O_{F5}|E;    // same literal — a filter changes
↪  time constant, not type
node constitution: ^5^P_{F7}|T;    // Female: the operator's accumulated
↪  principled formation

filter fb_lp: LowPass< in=ops_log, out=trend, tau=21.0 >;   // tau in
↪  days — three weeks
feedback trend -> constitution [β=0.05];
```

Three declarations and two wires. `trend` carries the same literal as `ops_log` on purpose: filtering an operational record does not promote it out of `^0`. Nothing in this fragment ascends. The upward move happens once, at the last line, into a bias.

Two parameters, defending against different things.

β = 0.05 — low gain. Gain sets how far the principled substrate moves per unit of sustained signal. At β = 0.05 it takes roughly twenty consistent periods of the same trend to move principles appreciably. That is the point: principles that move in one period are

not principles. At β **= 0.5** a single filter period — three weeks of unusual outcomes, once they have cleared the filter — rewrites the constitution, and every board the operator runs afterwards is biased by the last one they ran. At β **= 0** the path is disconnected — no outcome corrects a principle, the operator cannot be wrong, and the safety architecture of Chapter 24 has nothing to update.

$\tau \geq$ **3 weeks — slow time constant.** τ sets what counts as a trend rather than a spike. At three weeks a single dramatic operation washes out entirely, and only sustained patterns reach the gain stage. At τ **= hours** you have built a fast path from action to principle — precisely the ascent the diodes exist to block, rebuilt by an operator who believed they were being adaptive. Shortening τ does not make the board responsive. It removes the diode.

Why both. Taken one at a time the defence collapses in a specific way. Low β with short τ: principles move slowly but track noise, so the operator drifts without ever being pushed. Long τ with high β: spikes are ignored, but a *sustained* input clears the filter and then hits full gain, so anyone who can afford three weeks of consistent pressure — any organised campaign of the kind Chapters 15 and 19 describe — moves those principles hard, through the front door. Gain bounds the magnitude any sustained input can produce; τ bounds which inputs qualify as sustained. Patience defeats a τ-only defence, intensity defeats a β-only defence, and neither parameter is optional.

> **Bench Note — the feedback arrives at the bias, never at the signal path.** The feedback terminates at `constitution`, the `^5^P_{F7}|T` accumulated-formation capacitor of Chapter 7.1 — not at the input of D1. That is the difference between a safety board and an oscillator. Feedback into a bias terminal trims the substrate the chain runs against, slowly, and settles. Feedback into the *signal path* is positive feedback around a gain stage, and it does what that always does: it runs away. Chapter 14 builds a hyperstition engine out of that runaway on purpose, with

loop gain deliberately above unity. On a safety board it is a defect. An O→P path landing anywhere but a bias terminal is an oscillator someone mistook for a conscience.

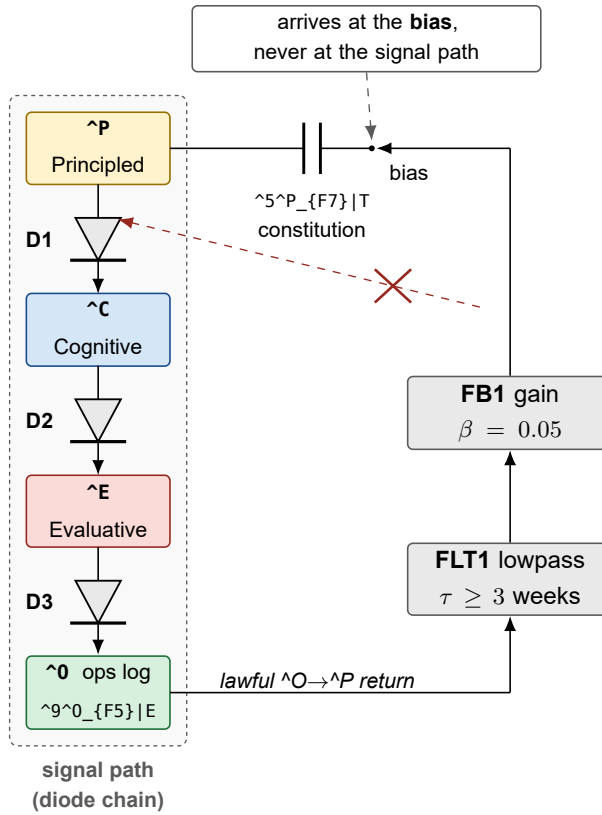


Figure 8-D: The lawful O→P loop drawn outside the diode chain

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Component and Parameter Table

Every value below is a **design choice**, stated as such, with the failure mode at other settings. None is a measurement.

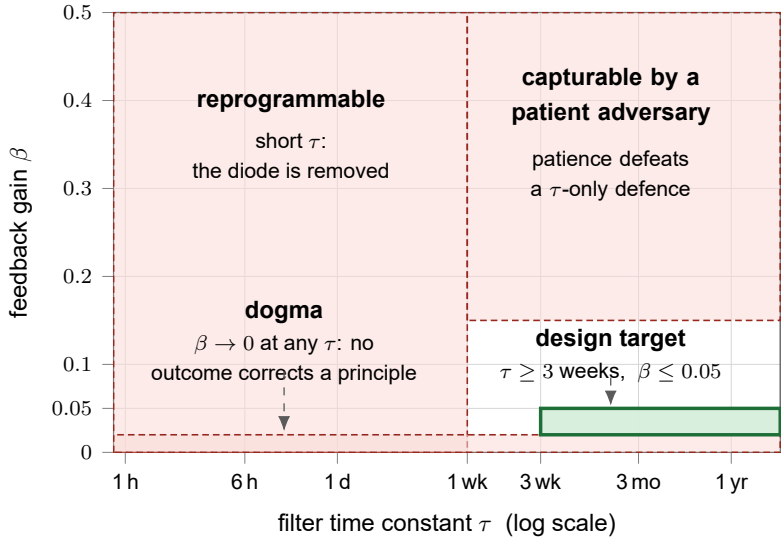


Figure 8-E: Two-axis design map

Ref	Component	Form	Parameters	Role	Failure at other values
D1	Domain diode	`Diode< ^P -> ^C >`	Vf=0.7, I_R=0.02, V_BR=12.0	Value becomes reasoning	0.2: conducts on noise — rumination. 3.0: values never examined
D2	Domain diode	`Diode< ^C -> ^E >`	Vf=1.2, I_R=0.02, V_BR=10.0	Analysis becomes verdict	0.3: no position without a feeling about it. 3.0: analysis, never a verdict
D3	Domain diode	`Diode< ^E -> ^O >`	Vf=2.0, I_R=0.02, V_BR=8.0	Verdict becomes act	0.5: impulsivity. 4.0: agrees, never acts
ΣVf	Series total	—	3.9	Floor on V_drive for any action	Below 3.9: no operational output

Ref	Component	Form	Parameters	Role	Failure at other values
FLT1	Lowpass filter	`LowPass< in=…, out=…, tau=… >`	τ = 3 weeks (adept) to 6 months (novice)	Trend vs spike	τ = hours: the diode is removed
FB1	Feedback element	`feedback trend -> ^5^P_{F7}|T`	β = 0.05	Trims principled bias	0.5: constitution rewritten in one filter period. 0: unfalsifiable operator

Two figures on τ, and they are not the same figure. **Three weeks is this chapter's design default**, chosen for an operator with a settled constitution. Chapter 24.5 scales τ by experience — 3–6 months for a novice, 1–4 weeks for an adept — and sets a separate absolute floor at one week, below which no operator of any experience may go. Use the 24.5 band to pick your value; use the one-week floor as the line that is never crossed for any reason. The three-week default sits inside the adept band, which is why every other board in this book that inherits the path inherits 21 days with it.

The `V_BR` gradient is one design choice stated three times: 8.0 / 10.0 / 12.0 ascending, so the exposed junction fails first and the constitution is behind two more junctions with more margin each. Equal values across the string remove the ordering, not the risk.

Failure Modes

Mode	Mechanism	What you observe	Fix
Shorted junction	Vf → 0 at one diode	Adjacent domains behave as one; disagreement reads as attack (D2); acting reads as deciding (D3)	Restore the drop; the boundary *is* the drop
Open junction	Vf raised past the rail	Domain below the break is dark; principles never reach action (D3) or thought (D1)	Lower Vf or raise V_drive; check ΣVf first
Reverse breakdown, arrested	Event exceeds D3's V_BR but not D2's	A junction permanently leaky; constitution intact; operator feels "shaken, not changed"	Nothing repairs the junction. Restore margin upstream and accept the new leakage figure at TP3
Reverse breakdown, full chain	Event exceeds all three V_BR	Principles changed in one pass, no analytical stage anywhere in the account	Margin, not repair — no junction recovers. Fuse and saturating refusal upstream (Chapter 24.2–24.3)
Excess leakage	I_R above spec, or blocked attempts integrating	Slow principled drift with no identifiable cause	Route feedback lawfully and meter it (8.4); audit the log
Counterfeit ascent	State relabelled `^9^0` → `^9^P` by assertion	"It worked, so it is right"; no filter, no τ, no gain	Ask what crossed the junction; if nothing did, reject the claim
Bypass shunt	Forward path skipping a junction, e.g. `^P -> ^E`	Confident verdicts with no analytical stage behind them	Remove the shunt; route through D1 and D2
Feedback τ too short	τ below the operator's band	Board tracks the last operation; settled principles move week to week	Lengthen τ into the Chapter 24.5 band for that operator, and never below the one-week floor
Feedback into signal path	Loop terminates at D1 input, not at bias	Escalation; each operation makes the next one larger	Re-terminate at the `^5^P` bias terminal

Test Points

TP	Node	What it reads	Expected
TP1	across D1, conducting	Forward drop, `^P -> ^C`	$\approx$ 0.7. Zero = shorted; at rail = open
TP2	across D3, conducting	Forward drop, `^E -> ^0`	$\approx$ 2.0. Low, with high operational output = impulse path
TP3	D3 under reverse bias	Reverse leakage	$\leq$ 0.02 of rated forward. Rising = degrading toward breakdown
TP4	`^5^P` bias, over one τ	Feedback slew	$\leq$ 0.05 units per τ. Faster = β or τ misset; check both

TP1 and TP2 bracket the chain; with both in spec, the junctions between are almost certainly intact. TP3 and TP4 fail *silently*, which makes them the readings to take on a schedule rather than on suspicion.

What This Chapter Established

- The domain order $\mathbf{P} \succ \mathbf{C} \succ \mathbf{E} \succ \mathbf{O}$ is enforced by three separate diodes, not one. Guarding only the endpoints guards nothing, because each adjacent pair is a distinct translation.
- **Forward drops sum.** ΣVf = 3.9 is a hard floor on V_drive before any principle produces any act, which reframes much apparent weakness as a circuit below its knee.
- **Vf is not overhead; it is the boundary.** A zero-drop junction merges two domains into one.
- **Below the knee, a junction still conducts.** Sub-threshold states are real and the cheapest thing on a board to move.
- Attempted ascent has **four outcomes** — standoff, leakage accumulation, breakdown, counterfeit ascent — separated by one question, *what crossed the junction?* The commonest is **counter-**

feit ascent, which is a relabel rather than a transition, costs the operator nothing, and is what Overton campaigns install.

- **V_BR ascends the chain — 8.0, 10.0, 12.0.** The exposed junction is built to fail first so a reverse cascade is arrested before it reaches the principled substrate. Equal breakdown ratings across the string remove the ordering, not the risk.
- The **lawful O→P path goes around the chain**, through a low-pass filter and a low-gain feedback element, terminating at the `^5^P` bias terminal — never at the signal path, which would make it an oscillator.
- **Both β and τ are required.** Gain bounds how far a sustained input can move the constitution; τ bounds what counts as sustained.

Exercises

(Answers in Appendix H.)

8.1 — Read the chain. An operator holds a principle at V_drive = 3.5 against the standard chain (Vf = 0.7 / 1.2 / 2.0) with R_constraint = 4. Compute I_attention, then compute it for V_drive = 6.0. In one sentence, say why the first result is not evidence about the operator's sincerity.

8.2 — Diagnose the short. A subject cannot consider an opposing argument without experiencing it as a personal attack. Which junction is shorted, what is the expected TP reading, and which Part III board exploits this defect deliberately?

8.3 — Separate the leakages. Chapter 7.3 leakage and section 8.3 leakage share a name. State the component, direction, and observable symptom for each, plus one diagnostic that distinguishes them at the bench.

8.4 — Reject the listing. Explain why the compiler refuses `^0` $\times$ `^P`, then rewrite the intended relationship as a valid CCDL fragment

using the lawful feedback path. Name the two parameters your fragment must specify.

8.5 — The patient adversary. An operator sets τ = 6 months and β = 0.4, reasoning that the long time constant makes the high gain safe. Describe the campaign that defeats this build, and give the smallest parameter change that fixes it.

8.6 — Detect the counterfeit. Two operators report that an operation changed their principles. A cites a filtered trend over four months and a bias shift they can name. B says "it worked, so I know now." Classify each structurally, and state what you would probe to confirm.

8.7 (Build) — Add a leakage integrator. The chain blocks each reverse attempt but leaks `I_R = 0.02` per attempt, and small leaks integrate invisibly. Write a CCDL fragment that accumulates reverse-direction attempts and alarms when the total crosses a threshold you choose. Justify your threshold, and state what the alarm must *not* be wired to do.

8.8 (Marquee) — Design the chain for a specific operator. A novice, under one year in, running high-consequence workings. Specify all three Vf values, all three V_BR values, τ, and β, departing from this chapter's defaults wherever you can justify it. For each departure, name the failure mode you are buying protection against and the capability you are giving up. Every value costs something.

9 Inductors — Momentum and Resistance to Change

Why a mind resists the rate and not the idea — and what the collapsing field does when you cut a habit at the switch

---

The demonstration costs the price of a battery. A 9-volt cell, an old automotive relay, two wires. Touch the wires to the coil, hold them a second while the current settles, then pull one off.

A spark jumps the gap as it separates — across an air gap nine volts cannot bridge. Something manufactured hundreds of volts in the microsecond you took the wire away, and nothing added energy; the battery was leaving the circuit. The coil did it. While current flowed it stored energy in a magnetic field, at no cost to anyone. When you opened the circuit you demanded that current go from steady to zero *instantly*, and the coil's answer is fixed: it produces whatever voltage keeps the current going. Cut fast enough and "whatever is required" is enormous. Anyone who services relay panels knows this in their fingers, and knows the fix, which costs four cents and is half of this chapter.

Minds do this. Every entrenched pattern is a coil with a field in it. Interrupt one at the switch and you get the spark — and the spark is not proportional to how large the habit was. It is proportional to how fast you opened the contacts.

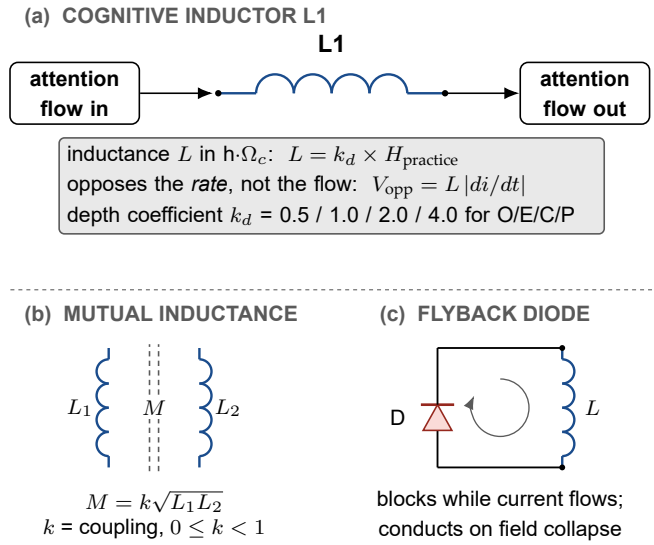


Figure 9-A: Symbol plate

9.1 Cognitive Inertia — Why Minds Resist New Ideas

Chapter 6 gave you the resistor: it opposes *flow*. Chapter 7 gave you the capacitor: it opposes *changes in voltage*, holding charge as accumulated ^5 Female composition. The inductor opposes the one thing left to oppose.

An inductor opposes changes in current. Not current. Changes in current.

Under CAS-Ohm (Chapter 4), current is `I_attention` — flow of attention along a path. A cognitive inductor therefore opposes *changes in how attention is routed*. Misreading this is why most attempts to change a mind fail in a way that looks like stubbornness and is not.

The naive model says a mind pushes back in proportion to the distance between what it holds and what you offer: big idea, big pushback. You have watched that be wrong. Someone accepts a

radical revision over eighteen months without visible strain, then comes apart over a trivial change to a morning routine imposed on Tuesday with no notice. Magnitude was not the variable. The **rate** was.

```
V_opposition = L × (di/dt)
```

`L` is the **inductance** of the pattern — how much accumulated `^5` Female composition it represents, and how deeply the `^7` Rhythm that built it entrained the substrate. `di/dt` is the rate of redirection you demand. Note what is absent on the right: Δi, the size of the change, does not appear anywhere. Only its derivative does.

The corollary is immediate. **For any pattern, however entrenched, there is a rate slow enough to produce arbitrarily little opposition.** Drive `di/dt` toward zero and `V_opposition` goes to zero regardless of `L`. No idea is too foreign to introduce without resistance; there are only rates too fast for the budget. That is the principle behind the Overton progression array of Chapter 15.

`L` has two factors, and they multiply: `L = k_d × H_practice`, where `H_practice` is cumulative hours of `^7` Rhythm exposure and `k_d` is the **depth coefficient**, in cognitive ohms, set by the domain the pattern lives in.

Pattern lives in	`k_d` (Ω_c)
`^0` Operational, `|E` — a motor routine	0.5
`^E` Evaluative, `|f` — a taste, a felt preference	1.0
`^C` Cognitive, `|A` — a professional method	2.0
`^P` Principled, `|T` — an identity commitment	4.0

Inductance rises with domain precedence. A principled commitment resists redirection eight times harder per rehearsal hour than an operational habit. That is what makes such commitments load-bearing — a value re-routable at operational speed would not be a value — and what makes them the most dangerous thing to interrupt abruptly.

Bench Note — inertia is not resistance, and the diagnosis differs. Anyone running Chapter 4's diagnose mode will attribute pushback to `R_constraint`. Check the time signature first. Resistive opposition is *constant* while the flow is constant. Inductive opposition appears only during the transition and **decays as the transition completes** — large at t=0, near zero at 3τ. If someone objects hard on day one and the objection fades on its own schedule while nothing was argued, that was `L`, not `R`, and arguing was never going to shorten it. If it is as strong in month six, that is `R`, and a different repair.

9.2 The L/R Time Constant — How Long Until the Substrate Shifts

Put an inductor and a resistor in series across a drive rail and close the switch. Current climbs an exponential governed by one time constant:

```
i(t) = I_final × (1 − e^(−t/τ))      τ = L / R      I_final = V_drive /
↪  R
```

Same shape as the capacitor curve in Section 7.2, but the parameters are coupled in a way that surprises operators.

Work a case. A practitioner has a default analytical method — `^C|A`, so `k_d = 2.0` — rehearsed 1,600 hours, giving `L = 3,200 h·Ω_c`. The retraining environment has `R = 40 Ω_c` (Section 4.4: `R_cap` 15, `R_policy` 0, `R_context` 15, `R_resource` 10) on a rail of `V_drive = 24 V_c`.

```
τ       = 3,200 / 40 = 80 hours
3τ      = 240 hours          (95% of the substrate shift)
I_final = 24 / 40 = 0.60 A_c
```

Two hundred and forty hours before the new method sits where the old one did — hours on the path, not calendar hours. Now vary `R` on a 1,000-hour version (`L = 2,000 h·Ω_c`):

`R` (Ω_c)	τ	3τ	`I_final`
80 — strict protocol	25 h	75 h	0.30 A_c
40 — design point	50 h	150 h	0.60 A_c
20 — loosened	100 h	300 h	1.20, clips at `I_max` = 1.0

> **Bench Note — adding constraint makes the change faster and smaller at once.** τ = `L/R` and `I_final = V/R` share a denominator. Raise `R` and the transient completes sooner *and* settles lower. The strict protocol reaches steady state in a quarter of the time and parks the practitioner at half the eventual flow. Loosen the constraint to "make it easier" and you have quadrupled the transient and pushed `I_final` past what the path carries, where it clips and spills into competing routes. Pick `R` by deciding whether you need arrival speed or final depth; one resistor will not give you both.

The same arithmetic on a principled pattern rehearsed only 800 hours: `k_d = 4.0`, so `L = 3,200 h·Ω_c` again, and at `R = 40` the horizon is the same 240 hours. Against a motor routine the gap is wider still, because `k_d` there is 0.5: **eight hundred principled hours carry the time constant of 6,400 operational ones**, which is the 8× ratio of the `k_d` column and nothing more. Count only rehearsal hours and you under-budget exactly the transitions that matter most.

The Back-EMF Result

Everything above is the switch *closing*. The engineering is at the switch *opening*. The sign flips and the coil drives the circuit instead of the reverse:

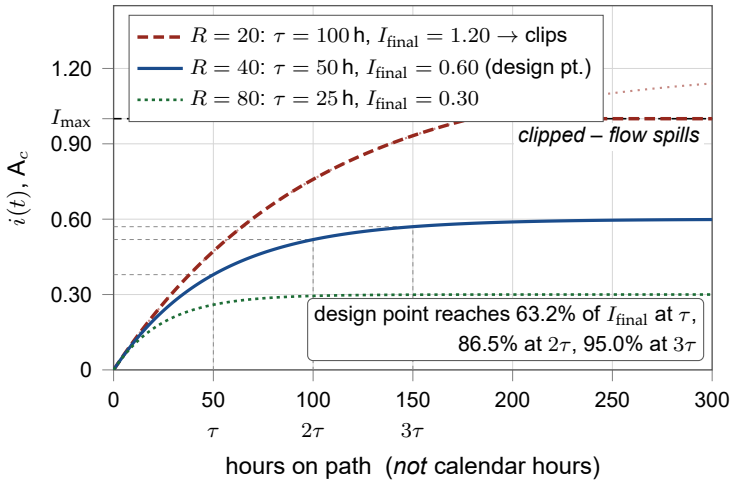


Figure 9-B: RL current rise

```
V_kick = L × |di/dt|
```

Take the principled pattern: `L = 3,200 h·Ω_c`, steady at `I = 0.60 A_c`, on a 24 V_c rail.

Taper Δt	`di/dt` (A_c/h)	`V_kick` (V_c)	× the 24 V_c rail
0.1 h — "I quit today"	6.0	19,200	**800×**
2.4 h — an afternoon	0.25	800	33×
24 h — one day	0.025	80	3.3×
240 h — a proper 3τ taper	0.0025	8	0.33×

Read the first and last rows against each other. **Same pattern, same endpoint, and the kick is two thousand four hundred times larger when you open the contacts instead of ramping them.** Only `dt` changed, and `V_kick` is a pure inverse in `dt`: halve the taper, double the kick, without limit.

That is the structural account of why abrupt cessation of an entrenched routine produces sharp distress while the same removal, tapered, produces very little. The distress is not stored inside the habit awaiting release; it is *generated at the moment of interruption* by

the substrate's own opposition to the rate demanded. It lands as `^3^E_{F3}|f` — visceral aversion, Evaluative domain, affective register, F3 Vitality. Note the Noetic: `^3` Negative, because this is recoil and repulsion, the can't-stomach-it response. It is not `^6` Male. Nobody analyzes the kick while it is happening; analysis arrives later at `^6^E|A`, by which time the transient has peaked.

A plain safety line, not decoration. This chapter models the routing of attention. It says nothing about physiology, physical dependence, or withdrawal — mechanisms this book does not describe and is not competent to describe. Nothing here is medical or psychiatric guidance. If stopping something is causing you real distress, that is a matter for a clinician, and no circuit here substitutes for that care.

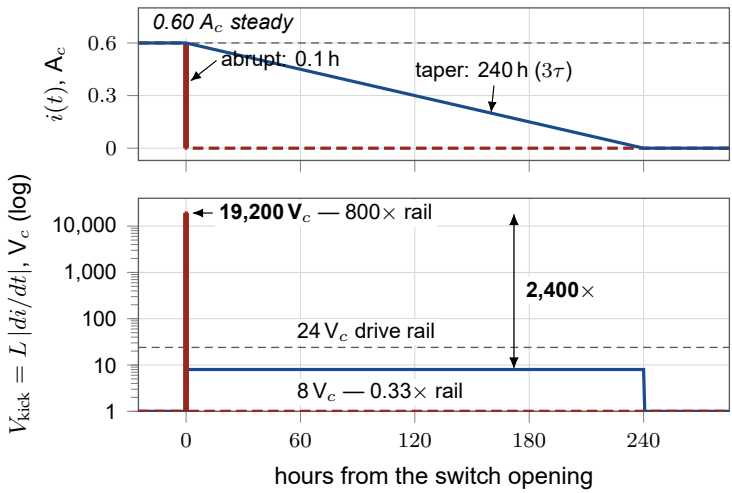


Figure 9-C: Back-EMF versus taper time

The Flyback Path

CCDL will let you write the abrupt cut. Appendix B §B.3.10 lists exactly one inductive statement, and this is it:

```
interrupt coil [dt -> 0];    // legal, compiles clean, almost never
↪  correct
```

The four-cent fix on the relay panel is a diode wired backwards across the coil. It does nothing while current flows normally — reverse-biased, blocking, as Section 8.2 describes. The instant the switch opens and the coil's voltage reverses, the diode conducts and offers the collapsing field a loop to circulate in. The current does not stop; it *re-routes* and decays gently. The cognitive equivalent is a **substitute path**, with three load-bearing properties:

1. **It conducts before the switch opens.** A flyback diode installed after the arc is a diode that watched the arc.
2. **It runs downward.** Section 8.1's chain permits descent, `P` → `C` → `E` → `0`, and blocks ascent. The kick appears at `^E`; the diode carries it down into an `^0` act. A substitute one domain *above* the interrupted pattern is not a circuit — `^0` × `^P` is invalid notation and invalid engineering for the same reason.
3. **It inherits the pattern's `^7` Rhythm.** Same time of day, same duration, same cadence — keep the `^8^0|E` cue and swap only the `^9^0|E` it lands on.

```
board habit_transition {
    input:  cue;
    output: routed_flow;

    node substrate: ^5^C_{F6}|A;         // Female: the accumulated
    ↪  routine
    node trigger:   ^8^O_{F6}|E;         // Above/Cause: the cue that
    ↪  fires it
    node old_path:  ^9_old^O_{F6}|E;     // Below/Effect: the routine as
    ↪  performed
```

```
    node new_path:  ^9_new^O_{F6}|E;     // Below/Effect: the substitute
    ↪  routine
    node felt_kick: ^3^E_{F3}|f;         // Negative: visceral recoil at
    ↪  interruption

    inductor coil:  Inductor< path=old_path, L=3200.0 >;   // h*Ohm_c
    resistor r_ctx: Resistor< path=old_path, R=40.0 >;     // tau = L/R =
    ↪  80 h
    diode  flyback: Diode< ^E -> ^O > [Vf=2.0];            // descent
    ↪  only

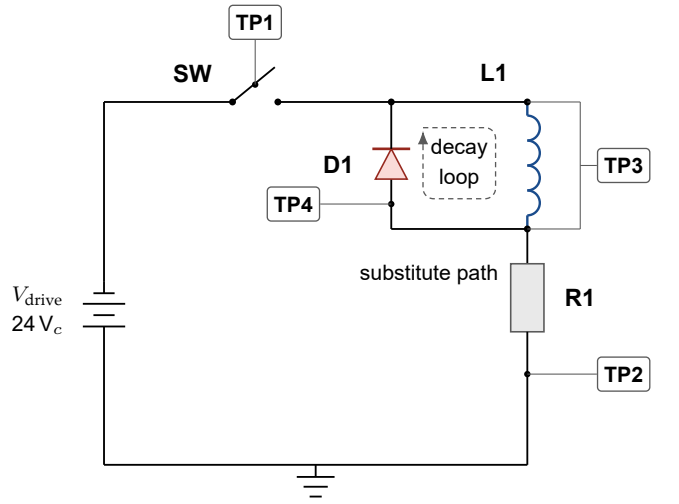
    hold until new_path > 0.0;                    // substitute conducts
    ↪  BEFORE the cut
    constraint taper_rate in [0.0, 0.0025];       // A_c/h — the kick
    ↪  budget
    monitor m_kick: felt_kick [threshold=72.0];   // V_c, unnormalised —
    ↪  3x rail
    on m_kick { alert "V_kick over budget - lengthen the taper"; }

    probe tp1: taper_rate;   // di/dt at the switching node
    probe tp2: old_path;     // i(t) against I_final
    probe tp3: felt_kick;    // V_kick across the coil
    probe tp4: new_path;     // substitute conducting before the cut
}
```

Two lines do the safety work. `hold until` refuses to open the switch until the substitute carries flow. `constraint taper_rate` is the kick budget from the table above, written where a validator enforces it rather than left in the operator's intentions. The monitor threshold is in volts, not activation, so it sits outside `[0, 1]`; Appendix B §B.4 permits that where the guarded quantity is unnormalised.

Note `^9_old` and `^9_new`. Both nodes are the same state — Below/Effect, Operational, F6 Structure, embodied — because that is the honest typing: the substitute is not a nobler act than the one it replaces, only a different landing for the same cue. The `_label` suffix of Appendix B §B.1.2 exists for exactly this case, two instances of one state on one board, and without it the two `node` lines are indis-

tinguishable to a reader and the TP2/TP4 pair below has nothing to name.



L1 3,200 h·Ω_c **R1** 40 Ω_c **D1** flyback, C → E descent only
$\tau = L/R = 80$ h $3\tau = 240$ h $I_{\text{final}} = V/R = 0.60$ A$_c$
TP1 di/dt at the switching node (0.002–0.03 A$_c$/h)
TP2 $i(t)$ in the interrupted path, against I_{final}
TP3 V_{kick} across the coil ($\leq 3 \times V_{\text{drive}}$)
TP4 substitute flow — non-zero *before* TP1 moves

Figure 9-D: Flyback schematic

9.3 Induction Coupling — Transferring Belief Between Minds

Bring two coils near each other and a changing current in the first induces a voltage in the second, with no wire between them:

`M = k × √(L₁ × L₂)` `V₂ = M × (di₁/dt)`

`k` is the **coupling coefficient**, on [0, 1), set by how much ^7 Rhythm the two minds share. Not agreement, not affection —

shared cadence, shared exposure schedule, shared tempo of contact. Chapter 11's transformer (Section 11.5) is this physics with coupling deliberately maximized; here we care about the coupling you did not choose.

k	Condition	Cost of assuming otherwise
0.05	Asynchronous text, no shared cadence	Budgeting transfer that never arrives
0.20	Shared feed, common daily cadence	Unexplained drift in a population you thought isolated
0.50	Live co-present group, shared tempo	Under-predicting backlash by 10×
0.90	Sustained dyad — household, mentor	Transitions leaking into someone you never modelled
1.00	Ideal limit, no leakage flux	**Not buildable.** k is bounded $0 \le k < 1$ by construction — perfect coupling means no leakage flux, and no real pair has none. Build at $\le$ 0.90, the discipline Chapter 24 applies to $\beta = \infty$

Work it. A speaker with L_1 = 3,200 h·Ω_c publicly reverses a position over an afternoon: Δt = 2.4 h from I = 0.60 to zero, so di_1/dt = 0.25 A_c/h. A listener in the room has L_2 = 800 h·Ω_c on the same question, k = 0.50.

$$M = 0.50 \times \sqrt{(3{,}200 \times 800)} = 0.50 \times 1{,}600 = 800 \text{ h·}\Omega\text{_c}$$
$$V_2 = 800 \times 0.25 = 200 \text{ V_c}$$

Two hundred cognitive volts induced in the listener against their own 24 V_c rail — **8.3× anything they generate on their own account**, from a speaker who never addressed them, through a mechanism with no wire in it.

> **Bench Note — a steady belief induces nothing.** $V_2 = M \times di_1/dt$. If di_1/dt is zero, V_2 is zero regardless of how large L_1 is or how tight the coupling. Someone who has held a conviction quietly for thirty years transmits *nothing at all* by holding it, and an operator expecting transmission from depth is reading the wrong term. The convert is the effective evangelist for no reason to do with sincerity or content: the convert has a large di/dt. It is also why an operation at steady state stops recruiting, and why the fix is never "believe harder" — it is to introduce a change, which is what Chapter 14's oscillator manufactures on purpose.

Then polarity. In a real coil the induced current flows in the direction that *opposes* the change producing it; that is not a convention, it is what conservation requires. The cognitive statement is identical: **the induced response in a coupled mind initially opposes the change it detects.**

This is backlash, and it explains a shape that otherwise looks irrational. A movement that shifts fast induces in every coupled mind a counter-flow proportional to di/dt — not to the content of the shift, and not to whether the coupled mind agrees with it. Push a norm quickly and you generate opposition in people who would have accepted the same norm arriving slowly. Section 15.4 sets its calendar from a different component — the collective capacitor, $\tau = RC$ — but the penalty it names for compressing that calendar is this equation: rush the schedule and what you buy is priced in di/dt. Section 19.4 runs the same term on purpose, provoking a fast di/dt and harvesting the recoil rather than arguing for anything.

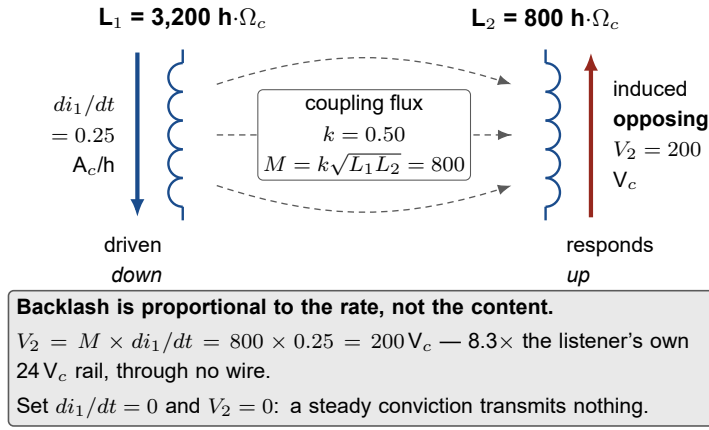


Figure 9-E: Mutual induction, opposing polarity

Component and Parameter Table

Every value below is a design choice made in this book, not a measurement.

Parameter	Design range	Failure at other values
L, h·Ω_c	100 – 20,000	Under-estimate: taper too short, full kick lands. Over-estimate: years budgeted needlessly
k_d, Ω_c	0.5 / 1.0 / 2.0 / 4.0 by domain	Flat across domains: principled transitions under-budgeted 8×
R, Ω_c	10 – 90	Too low: τ balloons, `I_final` clips. Too high: fast settling at a shallow ceiling
`I_final = V/R`, A_c	0.05 – 1.00	Above `I_max` = 1.0 the path clips and spills to competing routes

Parameter	Design range	Failure at other values
`di/dt`, A_c/h	0.002 – 0.03	Recompute the ceiling per coil: `di/dt_max = 3 × V_drive / L`. On the worked coil (`L` = 3,200, 24 V_c) that is 0.0225, so the top of the range is already over budget; at 6.0 the kick is 800× rail
`V_kick`, V_c	$\leq 3 \times$ `V_drive`	Above 10× rail the substitute cannot absorb it and recoil dominates

Failure Modes

Mode	Mechanism	What you observe	Fix
Rate overrun	`di/dt` set by convenience, not by kick budget	Sharp recoil at `^3^E|f` on day one; transition reverses within a week	Recompute the taper from `V_kick` $\leq 3 \times$ `V_drive`
Inductance underestimate	`L` from `H_practice` alone, `k_d` ignored	Pattern fully present at the planned end date	Re-derive `L = k_d` $\times$ `H_practice`; principled patterns are the usual casualty
No flyback path	Substitute not conducting before the cut	Full kick at `^3^E_{F3}|f`; the vacated cadence refills with whatever is nearest	Install the substitute first, same `^7` Rhythm, descending domain; confirm TP4 before TP1
Low-`R` stall	`R` cut to "make it easier", so τ `= L/R` rises	No apparent progress; `I_final` computes above `I_max` and clips	Restore `R`. Reduced constraint buys depth, not speed
Unmodelled coupling	A `k` $\geq$ `0.5` partner was never in the model	Backlash from someone who was never the subject of the operation	Enumerate every `k` $\geq$ `0.2` coupling before setting `di/dt`

Mode	Mechanism	What you observe	Fix
Core saturation	Flow driven far past `I_max`	The substrate stops opposing; the pattern floods and `L` collapses	Cap the rail. A path that stopped resisting was not changed — it stopped regulating

---

Test Points

TP	Node	Reads	Expected
TP1	`taper_rate`	`di/dt`, A_c/h	0.002 – 0.03 for a designed taper. Above 0.05, contacts are being opened, not ramped
TP2	`old_path`	`i(t)` against `I_final`	Follows `I_final` $\times$ `e^(-t/`τ`)`; ≈ 0.05 $\times$ `I_final` at 3τ. A flat trace means the switch never opened
TP3	`felt_kick`	`V_kick`, V_c	$\leq 3 \times$ `V_drive` — here, ≤ 72 V_c. Hundreds means TP1 is out of range
TP4	`new_path`	Substitute flow, A_c	Non-zero **before** TP1 leaves zero. Still zero as TP2 falls is the no-flyback failure, one probe ahead of the damage

Read them TP4, TP1, TP2, TP3 — confirm the substitute conducts, set the rate, watch the decay, verify the kick stayed in budget. That order is the bring-up procedure.

What This Chapter Established

- **The inductor opposes changes in attention flow, not attention flow itself.** `V_opposition = L × di/dt`; the magnitude Δi never appears on the right.
- **For any pattern there is a rate slow enough to introduce it without meaningful opposition.** No idea is too foreign — only rates are too fast.
- **Inductance rises with domain precedence.** `k_d` runs 0.5 / 1.0 / 2.0 / 4.0 across `^0` / `^E` / `^C` / `^P`, so 800 principled hours carry the same τ as 6,400 operational ones.
- **Constraint makes a transition faster and shallower at once**, because τ `= L/R` and `I_final = V/R` share a denominator. Arrival speed and final depth trade on one component.
- **Back-EMF is a pure inverse in taper time** — 2,400× larger at six minutes than at 240 hours for the same removal. The distress is generated by the rate, not released from storage, and lands as `^3^E_{F3}|f`.
- **The flyback path must conduct before the switch opens**, run downward (`^E -> ^0`, never ascending), and inherit the pattern's `^7` Rhythm. `interrupt coil [dt -> 0];` compiles clean — only `hold until` and the taper constraint stop you writing it.
- **Induction transmits only change.** $V_2 = M \times di_1/dt$. A steady conviction induces nothing, and the induced response initially *opposes* the change that produced it: backlash is proportional to rate, not content.

Exercises

(Answers in Appendix H.)

9.1 A behavioural routine (`^O, |E`) has 900 rehearsal hours on a path with `R = 30 Ω_c`. Compute `k_d`, `L`, τ, 3τ. Repeat treating those 900 hours as a principled commitment (`^P, |T`), and say what the ratio between the two 3τ figures means for a schedule.

9.2 An operator drops `R` from 60 Ω_c to 15 Ω_c on a path with `L = 2,400 h·Ω_c` and `V_drive = 18 V_c`, reasoning that less constraint speeds things up. Compute τ and `I_final` before and after, state what actually happened, and name the failure-modes row they landed in.

9.3 A pattern with `L = 5,000 h·Ω_c` runs at `I = 0.80 A_c` on a 20 V_c rail. Solve for the minimum taper `dt` holding `V_kick` $\leq$ `3` $\times$ `V_drive`, then the `dt` holding it at $1\times$ the rail. Say whether that second schedule is one you would accept.

9.4 Explain why the kick is written `^3^E_{F3}|f` and not `^6^E_{F6}|A`. Reference the canonical definition of `^3` Negative, and say what the `^6` Male state does instead and when it arrives.

9.5 (Design) Move a practitioner off a `^C|A` method with `H_practice = 2,500 h`, at `R = 40 Ω_c` and `V_drive = 24 V_c`, holding `V_kick` $\leq$ `2` $\times$ `V_drive`. Give (a) `L`, τ, 3τ, `I_final`; (b) the maximum `di/dt` and the taper it implies; (c) a CCDL fragment declaring coil, resistor, flyback diode, the `hold until` and `constraint` guards, and four probes. Check the diode's domain direction before you submit it.

9.6 (Marquee) An operator plans a 60-hour taper for themselves, from `I = 0.60 A_c` on a 24 V_c rail, with `L₁ = 3,200 h·Ω_c`, and confirms their own kick is inside the $3\times$ budget. They live with a partner who shares the affected cadence at `k = 0.90`, with `L₂ = 2,000 h·Ω_c`.

a) Compute the operator's `V_kick`, then `M` and the `V₂` induced in the partner, each as a multiple of the same 24 V_c rail.
b) The partner's induced voltage is the smaller of the two, yet the partner is likelier to experience the transition as abrupt. Ex-

plain structurally why — the partner never set `di/dt` and has no substitute path installed.

c) Find the taper bringing V_2 under $0.3\times$ the partner's rail, state what it cost in schedule, then state the one thing this circuit model cannot tell you about the situation.

10 The CJT — Deep Dive for Operators

Four numbers decide whether a circuit holds — and almost every operator sets only one of them

---

A curve tracer is a bench instrument that does one thing: it puts a transistor in a socket, sweeps the collector voltage over and over while stepping the base current between sweeps, and draws the result. What comes up on the screen is a fan of curves, one per base step.

Every number on the datasheet is visible in the shape of that fan. The **spacing** between adjacent curves is β — how much more collector current you bought with one more step of base current. The **bend** where each curve flattens is saturation — the point past which more collector voltage buys nothing. The **spread** of the fan for a given step size is g_m. And how hard the instrument had to push to make that step at all is rπ.

One picture. Four numbers. An engineer reads them off at a glance and knows whether the part in the socket will do the job.

Now the other bench. An operator has read Chapter 3, knows the CJT has six parameters, and sets exactly one of them. Everybody sets β. β is the interesting one, the one that feels like power, the one the word *amplification* attaches to. Almost nobody sets `sat`, and `sat` is the parameter that decides whether the circuit **holds** — whether a refusal can be argued down, whether a calibration read-

ing means anything, whether a five-stage chain delivers what the schematic said it would.

This chapter is the fan of curves. Four sections, four numbers, and then what happens when you put several of these devices in a row and the errors start multiplying instead of adding.

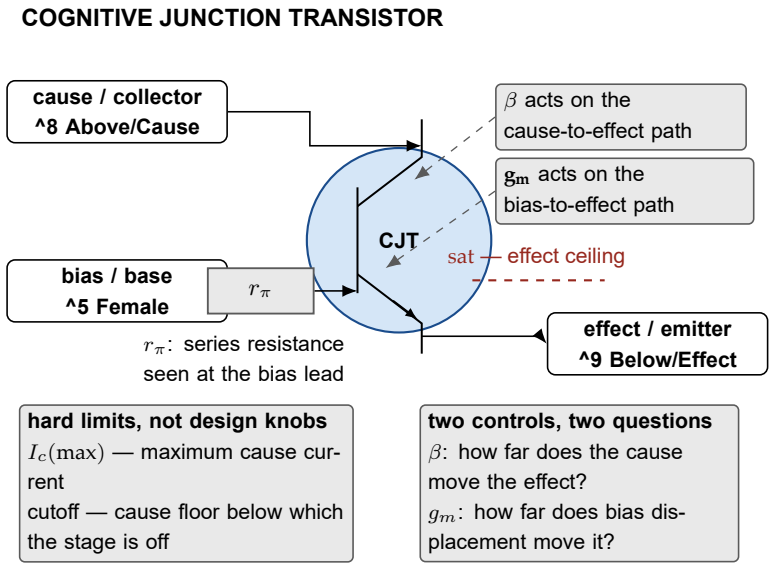


Figure 10-A: Component symbol plate

10.1 Selecting β for Your Working — Attenuation, Unity, Amplification

β is the ratio of how far the effect moved to how far the cause moved, measured as Fisher-Rao geodesic distance on the open simplex (Chapter 3.3, derivation in Appendix L). It is dimensionless. It has three regimes and operators use one of them.

$\beta < 1$ — attenuation. The stage produces less movement than it received. This is the setting nobody reaches for and it is on every board in Part III.

Its clearest use is the composure stage. Take a provocation arriving as `^3^E_{F3}|f` — visceral aversion, Evaluative domain, affective register, driven by F3 Vitality — feeding an operational response at `^9^O_{F5}|E`. Set $\beta = 0.3$ and a cause displacement of 0.80 produces an effect displacement of 0.24. The provocation is not denied, not suppressed, not reframed. It is *received at full magnitude and transmitted at less than a third of it.* That is the structural definition of composure, and it is a gain setting, not a virtue.

The other standard use is the feedback path. Chapter 8.4's lawful O→P route runs at $\beta = 0.05$ to 0.3 for exactly this reason: outcomes must be able to update the principled substrate, but a single outcome must not be able to rewrite it.

The failure at other values is symmetric. **$\beta = 0$ is not attenuation, it is cutoff** — a dead stage that transmits nothing, which presents as detachment rather than composure and is a different fault with a different repair (Section 10.4). **$\beta > 1$ on a composure stage is reactivity**: a provocation of 0.80 producing an act of 1.6, which is the amplification chain Section 19.3 documents as a weapon.

$\beta = 1$ — unity. The stage renders faithfully. It is the correct setting for any stage whose job is to report rather than to move: the evaluation stage of a safety cascade, the reporting path of a safety board's output stage (Section 24.6), the re-reference stage of Section 10.5 below.

Unity is the hardest value to hold because it is the only one with no tolerance in either direction. Above 1 the stage inflates what it was given and the inflation compounds downstream. Below 1 it downplays, and a risk evaluator that downplays is worse than no evaluator, because it produces a reading that looks like diligence. The evaluation stage feeding the guard in Section 24.6's listing is

held at β = 1.0 for precisely this reason, and the word for it is *honesty*.

$\beta > 1$ — amplification. Here is the design rule that operators get backwards:

> **β is set by the resistance you must overcome. It is not set by how much you want the outcome.**

Work it. An operator has a drive rail of `V_drive = 12 V_c` and needs an operational effect of 0.55 A_c. The path crosses the full descent chain, so it loses the three forward drops of Section 8.1 — 0.7 + 1.2 + 2.0 = 3.9 V_c — before it meets a constraint of `R_constraint = 40 Ω_c`. The diode-corrected CAS-Ohm form gives the cause actually arriving at the collector:

```
I_cause = ( V_drive − ΣVf ) / R_constraint
        = ( 12 − 3.9 ) / 40
        = 0.2025 A_c
```

Required gain is then a division, not a judgement:

```
β_required = target_effect / I_cause = 0.55 / 0.2025 = 2.72
```

Specify β = 2.7. Then check the ceiling: the stage will run near 0.55, so `sat` must sit above it with margin — Section 10.2 gives the 1.25 rule, so `sat` $\geq$ `0.69`, and you specify 0.70.

Now the part that makes the rule mean something. To reach the same 0.55 A_c at unity gain you would need `V_drive = (0.55 × 40) + 3.9 = 25.9 V_c` — nearly fourteen more volts of drive than the operator has. **Gain is what you buy when you cannot afford drive.** And it is not a free substitution, because drive and gain do different things to whatever contamination is riding on the cause terminal. More drive raises signal and noise together; the ratio is unchanged. More gain does the same — but gain also multiplies every subsequent stray input by everything downstream of it, which is Section 10.5's whole subject.

Bench Note — the ceiling on β is contamination, not power. Operators treat high β as bold and low β as timid. The real limit is arithmetic. If 5% of what sits on your cause terminal is content you did not intend — a half-formed grievance riding along with the stated desire, a resentment attached to the target — then at $\beta = 1.0$ that contamination arrives at 5% of the output magnitude and at $\beta = 3.0$ it arrives at 15%. The intended signal grew by the same factor, so the *ratio* held; what changed is that 15% of a large output is a large amount of unintended effect. This is the mechanism behind the breakdown inversion in Chapter 3.4, and it is why the operation-type table below stops recommending numbers above 4.0 rather than continuing the trend.

Operation type	Design β	Why this band	What breaks outside it
O→P feedback (Ch. 8.4)	0.05 – 0.30	Outcomes update the substrate; no single outcome rewrites it	Above 0.5 the constitution is reprogrammable by a burst of results
Composure / damping	0.20 – 0.50	Full reception, reduced transmission	At 0 the stage is in cutoff; above 1.0 it is a reactivity chain
Reporting, evaluation, re-reference	1.00	Faithful rendering	Above 1 it inflates; below 1 it downplays and the reading looks diligent
Self-programming (Ch. 20)	1.0 – 1.5	Identity substrate moves slowly; nothing external checks a closed loop	Above 2.0 a self-suggestion amplifies with no external check — see 17.5
Desire manifestation (Ch. 13)	1.5 – 3.0	Sized against real external `R_constraint`	Below 1.5 the rail never clears; above 3.0 contamination dominates
Collective / group operation	2.0 – 4.0	Collective bias supplies part of the lift	Above 4.0 the group's own contamination is amplified with the signal

Operation type	Design β	Why this band	What breaks outside it
Hostile / adversarial operation	not specified	—	This book does not specify a build. The inversion signature is documented in Chapter 19.4 so it can be recognised from outside

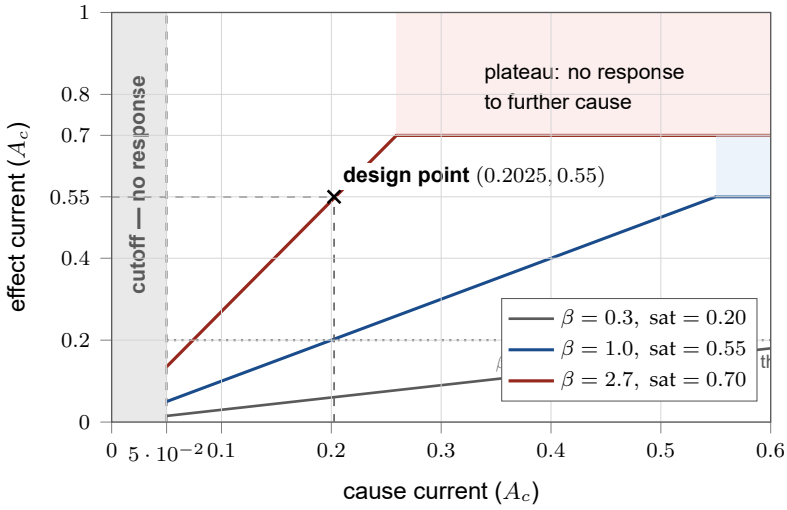


Figure 10-B: Characteristic curve

10.2 Setting sat — The Ceiling You Cannot Push Past

`sat` is the effect plateau. Once the effect reaches it, additional cause produces no additional effect: the stage's output has stopped depending on its input. Legal range is (0, 1], the default is 0.95, and the grammar enforces `cutoff < sat`.

That default is wrong for half the stages you will ever build, and the reason is that **`sat` means two opposite things depending on what the stage produces.**

Decision stages: saturate on purpose

If the stage's output is a *decision* — a select line, a verdict, a refusal — then the magnitude of the output is not the point. The point is that it is pinned and unmoving. And here the counterintuitive result:

A low saturation ceiling makes a stronger lock.

Two mechanisms, and both matter.

First, **a low ceiling is reached sooner**, so the stage latches on less cause. The cause needed to saturate is `c_sat = sat / β`. At $\beta = 2.0$, a stage with `sat = 0.60` latches at a cause of 0.30; one with `sat = 0.97` needs 0.485. It does not take an overwhelming provocation to trip the first stage, only a sufficient one.

Second, and more importantly, **a saturated stage has stopped responding to its cause.** In the plateau, $\partial\text{effect}/\partial\text{cause} = 0$. The stage is deaf, by construction. You cannot out-argue it, not because it is committed but because the input terminal no longer connects to the output. Everything between the cutoff knee and the plateau is the region where the stage is still steerable, and that region is what an adversary works in. Call it the **latch margin**:

```
M = ( sat − cutoff ) / β
```

At $\beta = 2.0$ and cutoff = 0.10:

`sat`	`c_sat` = sat/β	latch margin M	steerable window vs. sat = 0.60
0.60	0.300	0.250	—
0.80	0.400	0.350	+40%
0.85	0.425	0.375	+50%
0.95	0.475	0.425	+70%
0.97	0.485	0.435	**+74%**

A refusal stage left at the 0.95 default hands an adversary seventy percent more cause-space to operate in than one built at 0.60. That is the entire argument, and it is why Section 24.2's saturating `^3^P` refusal is specified at 0.85 and Section 15.3's final stage at 0.60 — both well below the top of the range, both on purpose.

Note the Noetic on those stages: `^3` Negative — visceral aversion, the can't-stomach-it response. A refusal stage is not analysing anything. Analysis is `^6` Male and it happened one stage earlier, at unity gain, and it is over. What saturates is the recoil.

For a decision stage, `W-CJT-004 CjtSaturation` is the **expected** status. It is not a warning to clear.

Production stages: never saturate

If the stage's output is a *product* — the manifestation, the response, the delivered work — then `sat` is a hard cap on what you get, and reaching it is a defect.

This book's rule:

```
sat ≥ 1.25 × expected_peak_effect
```

Twenty-five percent headroom, chosen so the stage stays in the active region across the run-to-run variance in cause. Section 13.3's amplify stage sits at `sat = 0.80` for exactly this reason. Read Chapter 13's worked run against the rule, though, and note where it actually lands: its corrected operating point is 0.76, inside the ceiling but with five percent of margin rather than twenty-five. That board is legal and it is calibratable, and it has no room left for a cause that arrives ten percent stronger than the last one did. The 1.25 rule is not a different rule from Chapter 13's; it is the same discipline with the margin sized for the next operation instead of this one.

The failure at lower values is not that the output is small. It is that **the output stops being a measurement.**

> **Bench Note — a saturated production stage is a broken instrument, and it fails looking like success.** Chapter 22's Quantitative Table asks you to log `^4` Vibration intensity operation after operation and read the trend. Run those operations through a stage whose `sat` sits at or below the operating point and every reading pins to the same number. Your table records a flat line. A flat line at a high value is exactly what stability looks like, so the operator concludes the circuit is dialled in — while in fact the instrument has clipped and would return that identical number for a cause twice as large or half as large. **You cannot measure through a component that has stopped responding.** Before you trust any reading, drive the cause up 20% and confirm the effect moved. If it did not, you have been reading a plateau, not a result.

sat is not I_c(max)

One more distinction, because conflating them is how substrates get damaged. `sat` is the **designed** plateau — a limit you chose. `I_c(max)` is the **physical** ceiling of the substrate carrying the path, and Chapter 3.4 names what lies past it: breakdown, $r\pi \to 0$ permanently, the `^5` dielectric ruptured, irreversible.

Specify `sat` above `I_c(max)` and the designed plateau is never reached, because the substrate hits its own ceiling first. The stage never latches, never warns, and the overload lands on the capacitor instead of on the limiter you thought you installed. Every `sat` in this book sits below the `I_c(max)` of the substrate it runs on, and that ordering is checked by eye — the validator range-checks `sat` against (0, 1] and against `cutoff`, and knows nothing about your substrate.

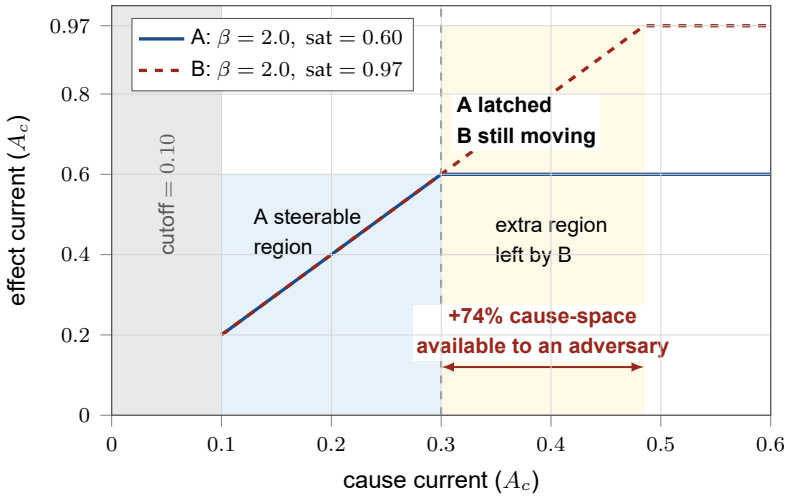


Figure 10-C: Two transfer curves overlaid on one set of axes, cause 0 to 0.6 horizontal, effect 0 to 1.0 vertical, both at β = 2.0 with cutoff = 0.10

10.3 g_m and Framing Sensitivity — Why Rewording Changes Everything

`g_m` is transconductance: effect moved per unit of **bias** displacement, in cognitive siemens (S_c, the reciprocal of Ω_c). Where β answers *how far does the effect move per unit of cause*, g_m answers *how far does it move per unit of pressure on the base terminal*.

This is the parameter that explains an observation everyone has made and most people misfile as insincerity: **the same request, worded differently, gets a different answer.**

The structural account is short. Rewording does not change the cause. The request is the same request; the collector sees the same signal. What rewording changes is the **bias** — the substrate against which the request is read, the `^5` Female composition the stage is biased from. Framing acts on the base terminal. And the base terminal has a gain on it.

This book models that gain as a baseline times four situational multipliers:

```
g_m = g_0 · N · R · V · D
```

`g_0` is the stage's baseline framing sensitivity, set by how hard the bias substrate is. `N`, `R`, `V` and `D` are situational, each ≥ 1.0, and Section 17.3 tabulates them in full: `N` is `^1` narrowing, `R` is `^7` rhythm lock, `V` is `^4` carrier dominance, `D` is `^6` dampening. In ordinary waking traffic all four sit at 1.0 and `g_m = g_0`.

Bias carrier	`g_0` (S_c)	Read as
An institution — procedure, precedent, `^5^0`	0.25	Reframing must displace a written record
A peer majority, `^5^E`	0.30	Reframing must displace many carriers at once
A credentialed professional, `^5^C`	0.35 – 0.40	Trained method resists restatement
A pioneer or early adopter, `^5^P`	0.40	Position held, but held by few
An individual in ordinary waking traffic	1.00	The nominal reference

Collective substrates sit lower than individuals because a frame must displace every carrier in the network before the bias node moves (Chapter 7.5). That single fact is why Section 15.1's resistance-decay model can be run at all: the array is not fighting one mind's `g_0`, it is fighting a network's.

Now the arithmetic. A reframe that displaces the bias by 0.50 V_c moves the effect by `g_m` × `0.50`:

`g_m`	Effect shift	Condition
0.25	0.125 A_c	institutional stage, Section 15.3's A5
0.40	0.200 A_c	professional stage, Section 13.3's Q2
1.00	0.500 A_c	ordinary waking
5.75	2.88 A_c	closed-loop self-work, Section 17.5

g_m	Effect shift	Condition
20.0	10.0 A_c	deep hetero-hypnotic state, Section 17.1

At g_m = 20 a single sentence displaces the substrate twenty times as far as the same sentence would at a bus stop. That number is the operational content of the word *suggestible*.

> **Bench Note — g_m has no input filter, and that is the whole problem.** β applies to the cause terminal, and you chose what to put there. g_m applies to the bias terminal, and **everything** that reaches it gets the same multiplier: your carefully composed framing, the operator's careless aside, the remark from the corridor, the subject's own stray ^6 interjection. There is no term in g_m = g_0 · N · R · V · D that discriminates between intended and incidental input. This is the entire category of unintended suggestion and it is not mysterious — you raised the gain on a terminal that has no filter in front of it. It is also why the detection handle in Section 17.3 is *counting factors*, not identifying a technique: you cannot see what is arriving at a bias terminal, but you can see how many multipliers are climbing.

Two design consequences follow directly.

Low g_m is not free. A stage at g_m = 0.25 cannot be reframed by an adversary and cannot be reframed by a genuinely better formulation either. It is one component; it does not distinguish. Section 15.3 walks its five-stage array from g_m = 0.40 down to 0.25 as the position hardens, and the hardening is a loss as well as a gain.

An A/B test measures whichever terminal you varied. Section 18.5 treats A/B testing as β calibration, and that is right — provided the two arms differ in the *cause*: a different offer, a different object of desire. Vary only the wording and both arms carry the same cause,

the difference lands entirely on the bias terminal, and what you measured is g_m. Two arms, two parameters, and the experiment does not tell you which one it read. Change one terminal per test.

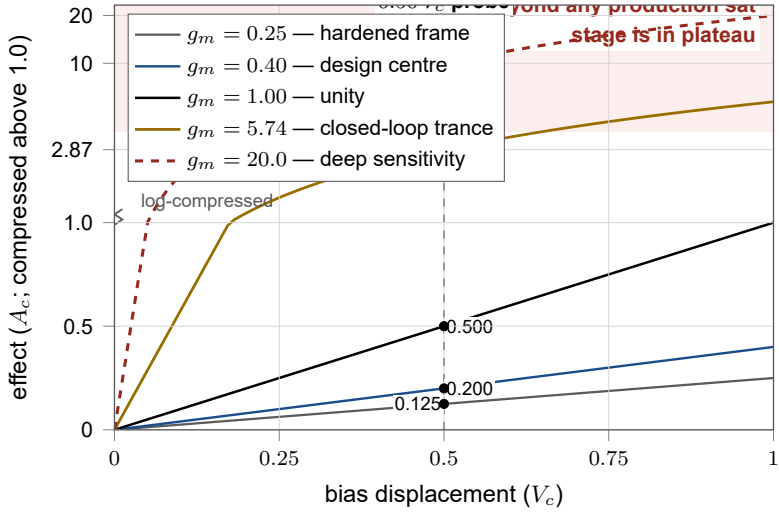


Figure 10-D: Characteristic curve

10.4 rπ and Cognitive Stubbornness — The Bias That Won't Budge

rπ is the input resistance at the bias terminal, in Ω_c. It is what the framing pressure has to push through. High rπ reads as stubbornness. Low rπ reads as compliance.

It is not an independent parameter, and treating it as one is the most common modelling error in this book's subject matter. It has two parts:

```
rπ_intrinsic = β / g_m
rπ_terminal  = rπ_intrinsic + R_bias
```

The **intrinsic** part is fixed by the two gains: three numbers, two degrees of freedom. Declare any two of β, g_m, rπ_intrinsic and the third is determined. The **terminal** part adds `R_bias`, the series resistance of the bias network — how hard it is to reach the substrate at all. Access, trust, attention, distance, whether anyone is listening.

Reconcile a real listing with it. Section 15.3's stage A1 declares β `= 1.2`, `g_m = 0.40`, rπ `= 8.0`:

```
rπ_intrinsic = 1.2 / 0.40 = 3.0 Ω_c
R_bias       = 8.0 − 3.0  = 5.0 Ω_c
```

Five cognitive ohms of bias-network resistance, sitting between the operator and a pioneer population that is perfectly capable of being reframed once you can get at it. The junction is soft; the wire to it is long. Those are different problems with different fixes and the terminal figure alone does not distinguish them — which is why a listing that declares all three numbers is telling you something a listing that declares two is not.

Run the same decomposition on Section 17.1's deep hetero-hypnotic specification, β `= 3.0`, `g_m = 20.0`, rπ `= 0.15`:

```
rπ_intrinsic = 3.0 / 20.0 = 0.15 Ω_c
R_bias       = 0.15 − 0.15 = 0
```

`R_bias = 0`. The bias network resistance has been driven to nothing. **That is what an induction does** — it does not argue with the substrate, it shortens the wire to it. Everything in Section 17.3's factor table is a procedure for removing series resistance from the base lead.

Which gives the operator's version of the rule:

> **R_bias is the term an outside party can move. rπ_intrinsic is not.**

An adversary cannot change your β and cannot change your `g_0`. Both are properties of what you are made of. What can be changed

from outside, quickly, is `R_bias` — by fatigue, urgency, isolation, repetition, the removal of alternative channels. This is why every coercive board in Part III attacks the bias network rather than the argument, and why Section 17.6's structural exit is not *argue better* but *stand up and leave the room*: leaving restores series resistance to the base lead, and argument runs through `^6`, the component that was routed around in the first place.

Design bands:

Stage	`rπ_terminal` (Ω_c)	Rationale	Failure outside
Principled guard, `^3^P`	≥ 6.0	A value that can be reframed in one conversation was not a value	Below 4.0 the guard is one framing away from moving
Professional / analytical	3.0 – 8.0	Method holds; evidence still lands	Above 10 the stage cannot be updated by anything
Ordinary waking, individual	≈ 1.0	The nominal reference	—
Closed-loop self-work	≥ 0.26	Derived in 17.5, not asserted	Below it, a self-suggestion moves the substrate with nothing checking it
Absolute floor	0.05	`rπ = 0` is a limit, not a setting	`rπ = 0` is breakdown (3.4), permanent, and irreversible

Two cautions about that floor, because neither is enforced for you. `rπ = 0` is not a setting the tooling will argue with: Appendix B's range check on `g_m` and `rπ` is $\geq$ `0`, so a stage declaring `rπ = 0` validates clean and then describes a junction that no longer exists. Compare β `= ∞`, which the validator *does* reject because non-finite input fails the parameter pass. One is caught, the other is not, and the difference is worth memorising — the 0.05 floor is a discipline you apply by eye, not a guard you inherit.

The second caution is the arithmetic. Because `rπ_intrinsic =` β`/g_m` is fixed by the two gains, a stage cannot declare a `rπ_terminal` *smaller* than its own intrinsic value without asserting a negative `R_`

`bias`, and negative series resistance is not a soft junction behind a short wire — it is nothing at all. This is the one cross-check the validator does not run for you either. It range-checks each of the three parameters independently and never multiplies them together, so a stage at β `= 2.7`, `g_m = 0.35`, `r`π `= 6.0` compiles without complaint and encodes `R_bias = -1.71`. Run the subtraction on every stage you write. It takes one line and it catches an over-specified stage that will otherwise sit in a listing looking authoritative.

And the trade nobody wants. High rπ is the same component as un-teachability. The person who cannot be manipulated by a skilled operator also cannot be corrected by a friend with a better argument, because there is one bias terminal and it has one input resistance. There is no setting that admits the second and blocks the first. What you can do instead is architectural: keep rπ moderate and put the discrimination somewhere else — in a `^6` decomposition stage at unity gain ahead of the bias node, which judges *what* is arriving rather than *how hard*. That is a stage you can inspect. rπ is a number you can only turn up or down.

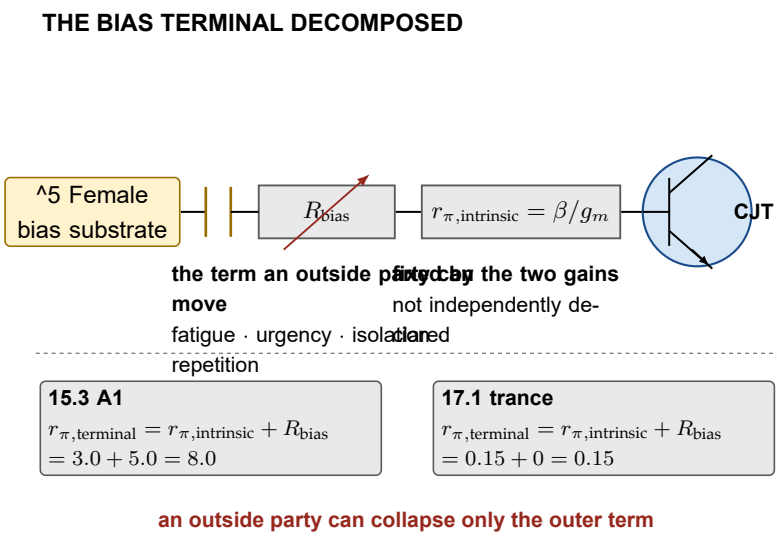


Figure 10-E: Schematic detail

10.5 Cascaded CJTs — Multi-Stage Amplification Chains

Put CJTs in series, effect to cause, and the gains multiply:

$\beta_\texttt{total} = \beta_1 \times \beta_2 \times \ldots \times \beta_n$

Section 15.3's array is five stages at 1.2, 1.6, 2.0, 2.4 and 3.0:

```
1.2 × 1.6 × 2.0 × 2.4 × 3.0 = 27.65
```

Twenty-seven-fold total gain, and no individual stage above 3.0. That single line is the whole reason Overton progression is built as a cascade rather than as one powerful stage: you get the gain without ever running a stage in the band where contamination dominates (Section 10.1). It is also why the array takes years — five stages each needing to settle, not one stage needing to be loud.

That is the good news about cascades. The bad news is also arithmetic.

Where noise enters

Two terminals per stage, so two entry points.

Cause-terminal contamination is content riding along with the intended signal — the unstated resentment attached to a stated desire. Model it as a fraction ε of the stage's own output.

Bias-terminal contamination is stray framing, and it is worse, because it is multiplied by `g_m` first and then by every gain downstream. Work one instance. A stray bias displacement of 0.05 V_c lands on stage 2 of the array above, where `g_m = 0.40`:

```
injected at stage 2   = 0.40 × 0.05 = 0.020
downstream gain       = 2.0 × 2.4 × 3.0 = 14.4
arriving at output    = 0.020 × 14.4 = 0.288
```

Against a unit input the output signal is 27.65, so one careless remark, early, contributes **1.04% of the final output**. Twenty of them and a fifth of what the board produced was never on the schematic.

Why a long chain drifts

Take the cause-terminal model and follow it through. Let `s_k` be the signal at stage k, `n_k` the noise, and let each stage add $\varepsilon \cdot$ `s_k` of its own contamination. Then:

```
s_k = s_{k−1} · β_k
n_k = n_{k−1} · β_k + ε · s_k
```

Divide the second by the first, and the gains cancel:

```
r_k = r_{k−1} + ε          =>          r_n = n · ε
```

The noise-to-signal ratio at the output of a cascade is the stage count times the per-stage contamination, and the gains do not appear. A gentle chain of ten stages at $\beta = 1.1$ each is dirtier than a hard chain of three at $\beta = 3.0$, even though the gentle one barely amplifies. Length is the variable. This is a model this book adopts, not a measurement, and it is the model that sizes every cascade in Part III.

At $\varepsilon = 0.03$ — three percent per stage, which is a design assumption and a generous one:

Stages	N/S at output	Reading
3	9%	Clean
5	15%	Section 15.3's array, at the edge of the band
6	18%	This book's cascade-depth limit
10	30%	Output no longer resembles the input
34	>100%	More contamination than signal

The limit in the middle is where `W-CJT-007 CjtCascadeDepth` comes from: a 20% noise budget at $\varepsilon = 0.03$ gives $\mathbf{n} \leq \mathbf{6}$. Set the limit

lower, at three, and you force too much gain into each stage and land back in Section 10.1's contamination problem. Set it at twelve and 36% of the output is material nobody designed.

Second drift mechanism: **tolerance compounds multiplicatively.** Hold each β to $\pm 10\%$ — which is optimistic for a parameter read off a Quantitative Table — and over five stages:

```
0.90⁵ = 0.591        1.10⁵ = 1.611
27.65 × 0.591 = 16.3        27.65 × 1.611 = 44.5
```

A nominal 27.65 delivers anywhere between 16 and 45. **A 2.7:1 spread from $\pm$10% per stage.** This is why nobody can tell you what a five-stage cascade will produce, and why Chapter 15 reads the array's rate at each position as a live test point rather than predicting the output from the schematic — Section 15.4's overshoot term is derived from the *actual* rate against the nominal one, which is only a meaningful quantity if you measured the actual.

Third: **saturation and cutoff propagate forward and silently.** β`_total` $= \Pi\beta$ holds only while *every* stage is in the active region. Saturate stage 3 and stages 4 and 5 receive a constant; their gains become irrelevant and the chain's effective gain collapses to $\beta_1\beta_2\beta_3$. Put stage 3 into cutoff and the chain delivers nothing at all, while stages 4 and 5 test perfectly on the bench because there is nothing wrong with them. **In a cascade, the broken stage is upstream of the symptom.** Probe from the input end.

The re-reference stage

The standard mitigation is a stage at $\beta = 1.0$ that compares the running effect against the original cause — a `^6^E_{F6}|A` decomposition, Male, taking the current state apart against what it started as. Note the Noetic: this is analytical decomposition, not recoil, and it is `^6`, never `^3`.

Note also where it sits. The re-reference stage stays in the **Evaluative** domain, alongside the stage it is checking. It is tempting to

draw it one level up, because a stage that checks another feels like it ought to outrank it — and that instinct produces `^6^E` → `^6^C`, which is a domain ascent, which Chapter 8.3 catalogues as one of the four things that happen when someone attempts one. Comparison is not promotion. The reference stage looks *backward* along the chain, not *upward* out of it.

Be precise about what it buys. It does **not** reduce $n \cdot \varepsilon$ — it is a stage, so it adds its own ε and makes the ratio worse by one term. What it does is stop the *tolerance* product from compounding past it, because its output is pinned to a reference rather than to the accumulated product of upstream gains. It is the cognitive analogue of a regulated intermediate rail: you accept a little more noise to stop the spread from doubling again.

```
circuit three_stage_with_reference {
    input:  seed;
    output: delivered;

    node substrate:   ^5^P_{F7}|T;     // Female: the principled
    ↪  formation the chain reads against
    node seed_ref:    ^0^C_{F2}|A;     // Idea: the original cause, held
    ↪  for comparison
    node surfaced:    ^8^C_{F2}|A;     // Above/Cause: the seed
    ↪  triggering its readings
    node weighed:     ^6^E_{F6}|A;     // Male: decomposition and test
    node reference:   ^6^E_{F6}|A;     // Male: current state against
    ↪  the original cause
    node delivered_e: ^9^O_{F5}|E;     // Below/Effect: what the chain
    ↪  produced
    node trend:       ^9^O_{F5}|E;     // same literal — a filter
    ↪  changes time constant, not type

    cjt s1: CJT< seed      | substrate => surfaced >
            [β=1.6, sat=0.80, g_m=0.40, rπ=8.0, region=active];
    cjt s2: CJT< surfaced  | substrate => weighed >
            [β=1.0, sat=0.85, g_m=0.40, rπ=8.0, region=active];
    cjt ref: CJT< weighed  | seed_ref  => reference >
            [β=1.0, sat=0.90, g_m=0.30, rπ=8.0, region=active];
    cjt s3: CJT< reference | substrate => delivered_e >
```

```
            [β=2.7, sat=0.70, g_m=0.60, rπ=6.0, region=active];

    diode d2: Diode< ^C -> ^E > [Vf=1.2];
    diode d3: Diode< ^E -> ^O > [Vf=2.0];

    filter fb_lp: LowPass< in=delivered_e, out=trend, tau=21.0 >;   //
    ↪  tau in days
    feedback trend -> substrate [β=0.05];

    probe tp1: substrate;      // bias level and rπ
    probe tp2: surfaced;       // cause into the chain
    probe tp3: delivered_e;    // effect, and which region it is in
    probe tp4: reference;      // drift against the original cause
}
```

Four stages, β_total = 1.6 × 1.0 × 1.0 × 2.7 = 4.32, N/S = 4 × 0.03 = 12%, and the two unity stages in the middle are doing the work that makes the number trustworthy rather than the work that makes it large.

Four things in that listing are there to pass checks worth learning to run by hand, because three of them are checks the validator will not run for you.

The bias sits at or above every cause it governs. `substrate` is `^5^P`, and P is the top of the order, so it can legally bias a `^8^C` cause, a `^6^E` cause, and anything else on the board. Appendix B's fourth compilation pass catches the opposite arrangement — a lower domain biasing a higher one — and names it `E-CJT-008 CjtCrossDomainBias`, the prompt-injection signature. It is worth saying why that check belongs in a chapter about gain. A bias terminal is a gain terminal. `g_m` multiplies whatever reaches it, and a board that lets an operational result set the bias on a principled stage has handed an outcome the authority to re-tune the constitution it was supposed to be judged against — at a multiplier, and with no input filter in front of it.

The descent is diode'd at both crossings, not just the last one. `surfaced` is `^C` and `weighed` is `^E`, so that crossing needs D2 as much as

the `^E → ^0` delivery needs D3. Chapter 8.1's rule is that guarding only the endpoints guards nothing, because each adjacent pair is a different translation. The cost is visible in the arithmetic: Σ`Vf = 1.2 + 2.0 = 3.2` on this fragment, and that is the floor the seed must clear before any of the 4.32 gain produces an output at all.

The upward path is the lawful one and it carries both parameters. `delivered_e` is `^9^0`, `substrate` is `^5^P`, and the connection between them is O→P — the single exception of Section 8.4. It goes through a filter at τ = 21 days before it reaches a gain of 0.05, and it terminates at a bias terminal rather than anywhere in the signal path. Drop either parameter and the defence fails in the specific way Chapter 8.4 describes: gain alone is defeated by patience, filter alone is defeated by intensity. Drop the bias-terminal discipline and you have not built a conscience, you have built the oscillator of Chapter 14 by accident.

Every stage's `R_bias` is non-negative. Run `rπ_intrinsic = β/g_m` on all four and subtract:

Stage	β	`g_m`	`rπ` declared	`rπ_intrinsic`	`R_bias`
`s1`	1.6	0.40	8.0	4.00	4.00
`s2`	1.0	0.40	8.0	2.50	5.50
`ref`	1.0	0.30	8.0	3.33	4.67
`s3`	2.7	0.60	6.0	4.50	1.50

Look at `s3`. It runs the chapter's worked β of 2.7, and holding `R_bias` positive at `rπ = 6.0` forces `g_m` up to 0.60 — the most reframe-sensitive stage on the board, sitting at the output, where a reframe is worth the most. Chapter 15 puts its counter-frame-capture threshold at exactly that figure, which is not a coincidence but the same identity read from the other side. That is not an accident of the numbers; it is what the identity does. **A high-gain stage is a framing-sensitive stage unless you buy the difference back in bias-network resistance,** and at the delivery end there is usually the least of that to buy. Declare `s3` at `g_m = 0.35` instead — which looks conservative, and which an earlier draft of this board did — and the listing

still compiles, while encoding `R_bias = -1.71`. Negative series resistance is not a soft junction behind a short wire. It is a stage that does not exist, described in three plausible numbers.

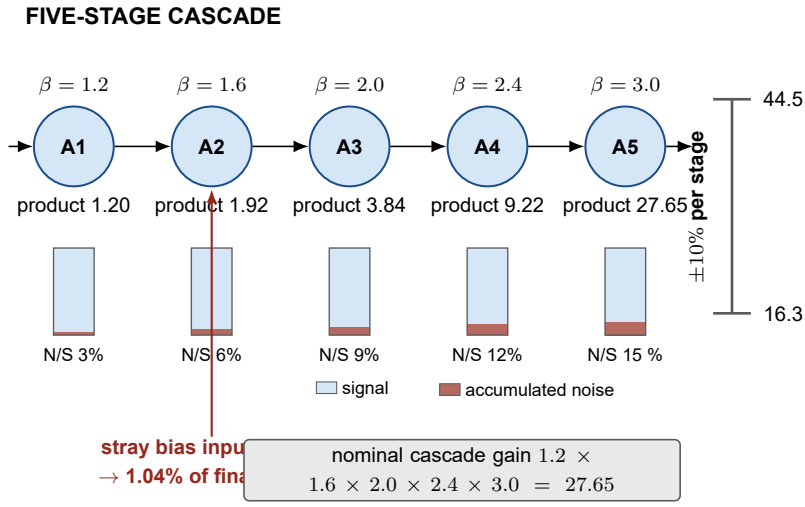


Figure 10-F: Cascade diagram

Component and Parameter Table

Every value below is a design choice made in this book, not a measurement.

Parameter	Units	Design range	Failure at other values
β gain	— (Fisher-Rao ratio)	0.05 – 4.0	0 is cutoff, not attenuation. Above 4.0 contamination scales with the signal and inversion risk climbs (Chapter 3.4)

Parameter	Units	Design range	Failure at other values
`sat` decision stage	—	0.60 – 0.85	At the 0.95 default an adversary gets ~70% more steerable cause-space
`sat` production stage	—	$\geq$ 1.25 × peak effect	At the operating point the stage clips and every reading pins to one number
`cutoff`	—	0.05 – 0.20, and `cutoff < sat`	Above 0.2 a real cause fails to fire the stage; the grammar rejects `cutoff` $\geq$ `sat`
`g_0` baseline	S_c	0.25 – 1.00	Below 0.25 nothing reframes the stage, including a correction
`N·R·V·D` multipliers	—	1.0 each nominal, 1.0 – 2.2 under load	Three or more rising together is an induction signature (17.3)
`rπ_intrinsic`	Ω_c	β / g_m — derived, never declared independently	Declaring all three over-specifies; the validator range-checks them separately and never multiplies them together
`R_bias`	Ω_c	0 – 6.0, and `rπ_terminal` $\geq$ β/`g_m`	0 means the bias terminal is directly exposed; this is what an induction produces. Negative is not a soft junction — it is a stage that does not exist
`I_c(max)`	A_c	substrate property	If `sat > I_c(max)` the plateau is never reached and the overload lands on the `^5` capacitor
Cascade depth `n`	stages	≤ 6	From N/S = n·$\varepsilon \leq$ 0.20 at ε = 0.03. `W-CJT-007` above it
ε per-stage contamination	—	0.03 assumed	A modelling assumption; if your stages are dirtier, shorten the chain

Failure Modes

Mode	Mechanism	What you observe	Fix
Gain sized by appetite	β picked for how much the outcome is wanted, not from `(V_drive − ΣVf)/R_constraint`	Output overshoots or the rail never clears; contamination proportional to β	Recompute β from the required effect and the arriving cause. Gain substitutes for drive you do not have — nothing else
Default `sat` on a guard	`sat` left at 0.95 on a decision stage	The refusal can be talked down across a long exchange; TP3 keeps moving	Set `sat` to 0.60–0.85. `W-CJT-004` here is the expected status, not a warning to clear
Clipped instrument	`sat` at or below a production stage's operating point	Quantitative Table readings flat run to run — reads as stability	Raise `sat` to ≥ 1.25 $\times$ peak. Verify by driving cause up 20% and confirming the effect moved
`sat` above `I_c(max)`	Designed plateau exceeds the substrate's physical ceiling	No latch, no warning, and dielectric stress on the `^5` capacitor (3.4)	Re-specify `sat` below the substrate ceiling. Checked by eye — the validator cannot see your substrate
Unlogged `g_m`	β planned carefully, framing sensitivity never recorded	Outputs swing on rewording; incidental remarks land as hard as intended ones	Log `g_m` at every stage. Count the four multipliers rather than trying to identify a technique
rπ read as a single number	Terminal value treated as intrinsic	A soft junction behind a long wire is mistaken for a hard junction	Decompose: `rπ_intrinsic` = β`/g_m`, `R_bias` is the remainder. They have different repairs
Over-specified stage	β, `g_m` and `rπ` all declared, and `rπ` < β`/g_m`	Nothing. It compiles; each parameter is in range on its own; the implied `R_bias` is negative and nobody subtracted	Subtract β`/g_m` from the declared `rπ` on every stage you write. Then decide which two of the three you meant, and let the third follow

Mode	Mechanism	What you observe	Fix
Ascent disguised as checking	A verification stage placed a domain above the stage it checks, e.g. `^6^E` → `^6^C`	The board reads as more rigorous than it is; the check is a relabel (8.3) rather than a measurement	Keep the reference stage in the domain of the stage it references. Comparison looks backward along the chain, not upward out of it
Upstream stage down	A cascade stage in cutoff or saturation	Chain output wrong; every stage downstream tests perfectly on the bench	Probe from the input end. In a cascade the fault is upstream of the symptom
Cascade drift	Chain longer than 6 stages, or ±10% tolerances compounding	Output between 0.59× and 1.61× nominal over five stages; N/S = n·ε	Shorten the chain, or insert a re-reference stage at β = 1.0 to stop the tolerance product compounding past it

---

Test Points

Standard placement per Section 12.3, applied to a single CJT stage on the bench.

TP	Node	Reads	Expected
TP1	bias node	Bias level and `r`π`_terminal`	Above `cutoff` and inside the band for the stage type. Below cutoff the stage never fires and nothing downstream will tell you
TP2	cause node	`I_cause`, A_c	Matches `(V_drive – ΣVf)/R_constraint`. A shortfall here is a rail or diode-drop problem, not a gain problem

TP	Node	Reads	Expected
TP3	effect node	Effect magnitude **and region**	Region is the reading that matters. `active` on a production stage, `saturation` on a decision stage. Either one in the wrong place is a defect
TP4	re-reference node	Drift of the current state against the original cause	Grows with $n \cdot \varepsilon$. A jump between two operations means a stage changed region, not that the operation was better

Read them TP1, TP2, TP3, TP4 — confirm the bias is live, confirm the cause arrived, confirm the region, then measure the drift. The 20% cause-bump check of Section 10.2 belongs at TP3 and takes ten seconds.

What This Chapter Established

- **β is set by the resistance you must overcome, not by how much you want the outcome.** β`_required = target_effect / I_cause`, where `I_cause = (V_drive – ΣVf)/R_constraint`. Gain is what you buy when you cannot afford drive.
- **Attenuation is a real setting with real uses** — composure stages at 0.2–0.5, O→P feedback at 0.05–0.3 — and $\beta = 0$ is cutoff, a different fault entirely.
- **A low saturation ceiling makes a stronger lock,** because a saturated stage has stopped responding to its cause. Latch margin `M = (sat – cutoff)/`β; the 0.95 default hands an adversary ~70% more steerable cause-space than 0.60.
- **`sat` means opposite things on decision stages and production stages.** Saturate a decision on purpose; keep a production

stage at `sat` $\geq$ `1.25` $\times$ `peak`. A saturated production stage is a clipped instrument that reports a flat line and looks like stability.

- **Framing acts on the bias terminal, not the cause terminal,** which is why rewording changes the answer. `g_m = g_0 · N · R · V · D`, and the terminal has no input filter — intended and incidental inputs get the same multiplier.
- **rπ is not independent, and the cross-check is yours to run.** `rπ_intrinsic = β/g_m`; `rπ_terminal` adds `R_bias`. `R_bias` is the term an outside party can move, and every coercive board in Part III moves it rather than arguing. A stage declaring all three validates on each parameter separately and can still encode a *negative* `R_bias`, so subtract `β/g_m` from the declared `rπ` before you believe any listing.
- **Cascade gains multiply and so do tolerances.** Five stages at $\pm$10% deliver a 2.7:1 spread. `β_total = Πβ` holds only while every stage is in the active region, and a stage in cutoff or saturation makes the fault appear downstream of where it is.
- **Cascade noise-to-signal is `n · ε` and the gains cancel out of it.** Length is the variable, not loudness — which is where this book's six-stage cascade limit comes from.

Exercises

(Answers in Appendix H.)

10.1 An operator has `V_drive = 15 V_c`, needs an effect of 0.72 A_c at the Operational domain through the full descent chain (ΣVf = 3.9), against `R_constraint = 55 Ω_c`. Compute `I_cause`, the required β, and the `sat` you would specify. Then compute the `V_drive` that would deliver the same effect at unity gain, and say in one sentence what the operator is actually buying with the gain.

10.2 Two refusal stages are identical except for `sat`: one at 0.65, one at 0.95. Both run β = 2.5, cutoff = 0.15. Compute `c_sat` and the latch margin `M` for each, state the percentage difference in steerable cause-space, and explain why the *smaller* number is the stronger guard.

10.3 A listing declares β `= 2.4`, `g_m = 0.30`, `r`π `= 11.0`. Decompose it into `r`π`_intrinsic` and `R_bias`. An adversary can reduce `R_bias` to 1.0 by isolating the target. Compute the new `r`π`_terminal` and the percentage change, and name which of the two terms could not be touched and why. Then say what would have been wrong with the same listing written `r`π `= 6.0`, and why the compiler would not have told you.

10.4 Explain why the refusal stage of a safety cascade is written `^3` Negative and the evaluation stage feeding it is written `^6` Male. Reference the canonical definitions, and say what goes wrong in the *circuit* — not the vocabulary — if the two are swapped.

10.5 A campaign runs eight cascaded stages at β = 1.4 each. Compute β`_total`, the output N/S at ε = 0.03, and the tolerance spread at $\pm$12% per stage. Then compute the same three figures for a four-stage chain delivering the same total gain, and state which you would build and what you gave up.

10.6 (Design) Specify a two-stage board that must produce an operational effect of 0.40 A_c from a `V_drive` of 9 V_c across the full descent chain, at `R_constraint = 30` Ω`_c`, with a `^3^P` guard that cannot be argued down. Produce (a) `I_cause` and the gain split across the two stages, (b) `sat` for each stage with the discipline you applied to each and why they differ, (c) `g_m` and `r`π for the guard with the band you took them from, (d) a CCDL fragment with four probes. It must compile — check the diode direction, `cutoff < sat`, and that you have not declared β, `g_m` and `r`π in a combination that over-specifies a stage without saying which two you meant.

10.7 (Marquee) An operator inherits a working five-stage cascade with β`_total` measured at 22 and decides it is under-

performing against a nominal 27.65. They raise stage 5 from $\beta = 3.0$ to $\beta = 5.0$.

a) Compute the new nominal β_`total` and the new N/S at $\varepsilon = 0.03$.
b) The measured output rises far less than the calculation predicts, and then stops rising at all. Give the two most likely region faults, say which test point reveals each, and explain why the operator's instinct to raise the *last* stage was the worst available choice.
c) Propose a repair that reaches the target without any stage above 3.0. State what it costs in noise, what it costs in time, and the one thing the cascade model cannot tell the operator about whether the operation should be run at all.

11 Special-Purpose Components

The five parts that do not amplify — they decide, they clock, they select, they compare, and they match

---

Open a quartz wristwatch and find the smallest part in it: a metal can the size of a grain of rice, with two wires. Inside is a sliver of quartz trimmed until it vibrates 32,768 times a second. That number is not round by accident. It is 2^15 — halve it fifteen times with fifteen flip-flops and you get one pulse per second, forever, with nothing to accumulate error.

The crystal has no gain. It amplifies nothing. It produces a rhythm stable enough that everything downstream can be built on the assumption the rhythm is right.

Chapters 6 through 10 gave you the parts that carry the signal — resistor, capacitor, diode, inductor, and the Cognitive Junction Transistor. Every one answers *how much gets through*. None of this chapter's five answer that question. A fuse answers *whether, ever again*. An oscillator answers *at what rate and in what pattern*. A filter answers *which rates survive*. An op-amp answers *which of two things is larger*. A transformer answers *at what working point*.

These five fail quietly, because a part with no gain gives no symptom when it is wrong. A fuse that cannot blow looks exactly like a fuse. You find these defects by reading parameters against each other, not by watching output.

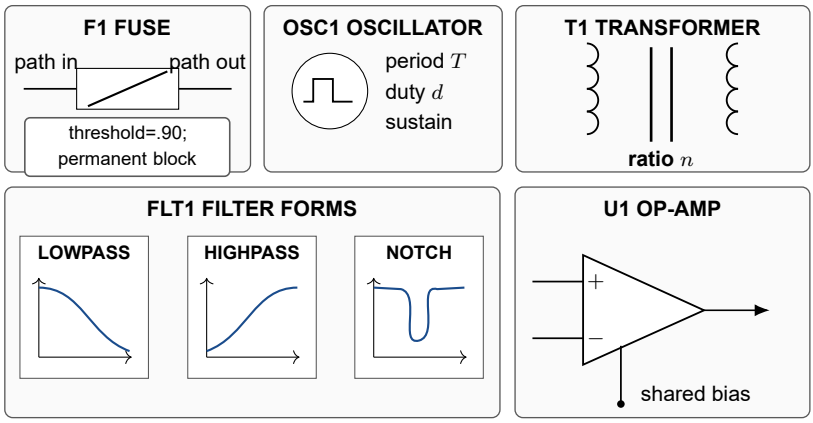


Figure 11-A: Symbol plate

11.1 Fuses — Permanent Path Blocking

A fuse is the only component in this book whose entire specification describes how it fails. It conducts, then at one specified point stops, and never conducts again.

```
fuse f1: Fuse< path=carried, threshold=0.90, action=permanent_block >;
```

`path` — what the fuse sits in series with. It senses activation on that path and opens it.

`threshold` — the activation at which it opens, on [0, 1]. The default here is **0.90**, for the reason a mains fuse is rated well above normal load. At **0.50** it trips on ordinary variance: any path with healthy dynamic range crosses its own midpoint routinely, so a 0.50 fuse makes a working board single-use for nothing. At **0.99** it is decorative, because well-designed paths rarely reach a percent from unity. Nuisance tripping and never tripping are both fuse failures.

> **Bench Note — a fuse above the path's ceiling is a part guaranteed to do nothing.** Every path has a ceiling: a

> CJT plateaus at its `sat`, a capacitor at full charge, a rail at its supply. Put `threshold` above that ceiling and the fuse can never trip, and nothing in the listing complains. Chapter 24 makes the trap concrete: 24.2 specifies a refusal CJT saturating at **sat = 0.85**, 24.3 an enforcement fuse at **threshold = 0.90**. Place that fuse across that CJT and it is dead metal — the node plateaus five hundredths below the threshold, forever. So it must not go there. It belongs in series with the enforcement path downstream, whose ceiling is higher than 0.90; check that placement on any CSB you are handed, including the one in this book. Read `threshold` and `sat` together or read neither. The compiler takes your word about what you are guarding.

`action` — what "open" means. Six legal values, not interchangeable. `permanent_block` cuts the path for the life of the board, no reset — what enforcement requires (Chapter 24.3). `permanent_until_manual_reset` is a breaker: cut until a human clears it. `block_until_decomposed` cuts until `^6` Male decomposition has run on what tripped it. `defer_decision` holds pending a later read — timing, not safety. `open_feedback` cuts the O→P loop instead of the signal path, freezing a drifting constitution — the slew fuse on CCB-SELF's evidence return (Chapter 20.5). `warn` logs and passes.

`warn` is the one to watch: a board with it on the enforcement path validates clean, reads as a safety board, and has no latch.

Once blown, blown. This is the component form of Chapter 21.5's section-retraction result: the chain runs one way, and going back does not restore what you had. That is the whole value — a refusal that can be retried into an acceptance was never a refusal — and the whole cost, because a fuse blown by a false positive stays blown. Saturation and a fuse are two locks: saturation latches the **decision** and releases when the cause drops; the fuse latches the **path** and never releases.

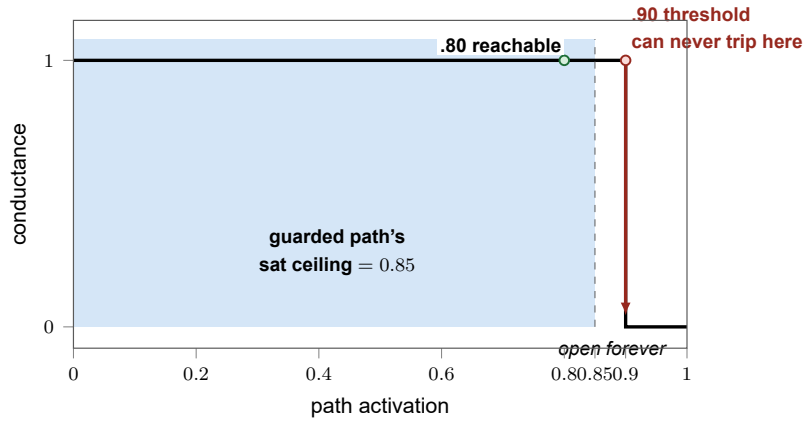


Figure 11-B: Fuse characteristic

11.2 Oscillators — Generating Cognitive Rhythms (^7)

^7 **Rhythm** is *the rate and pattern a Mind is exposed to an Idea, entraining/conditioning it.* The definition names two quantities: **rate** is the period, **pattern** is the duty cycle. An oscillator sets both.

```
osc clock: Oscillator< signal=claim, source=cadence, period=7.0,
↪  duty=0.15,
                        sustain=90.0 >;                        // days
```

It adds no energy. It converts a *sustained* drive into a *periodic* one, and it is worth a component because of Chapter 7.3: the ^5 Female capacitor leaks. One overwhelming impression does not hold; a smaller one redeposited before the last drains will.

That is a charge pump, and it has arithmetic. Let each exposure deposit `q` into a capacitor with leakage time constant τ`_leak`, on the 0–10 scale of Chapter 7.2:

`Q_max = q / (1 − e^(−T/`τ`_leak))` `Q_min = Q_max` $\times$ `e^(−T/`τ`_leak)`

```
ripple = q                                  (always — the ripple IS one pulse)
```

Take τ`_leak = 30 days`, `q = 0.8`, and vary the period.

`T` (days)	`Q_max`	`Q_min`	ripple / `Q_max`	Character
1	24.4 → clips at 10	10	0%	Pinned; every pulse after the first fortnight is spent for nothing
7	3.84	3.04	21%	Design point
10	2.82	2.02	28%	Ripple limit
30	1.27	0.47	63%	Impression, gone, impression, gone — no plateau forms

Thirty times the frequency buys a *pinned* substrate, not a proportionally larger one: the ceiling arrives before the arithmetic does. The last row is the commoner failure, experienced as starting over each time.

Design rule: `T` $\leq$ τ`_leak / 3`. That holds ripple under about thirty percent, where the plateau reads as a level rather than separate events. Above τ`_leak` there is no plateau; far below τ`_leak / 10` you pay for pulses landing on a full capacitor.

Duty cycle is the half operators skip. Design band: **0.10 to 0.25**. The gap is not wasted — it is where the mind runs its own `^6` Male decomposition, which makes the resulting `^5` composition the mind's own rather than a recording. At **duty** $\geq$ **0.6** you have a DC level rather than a rhythm, and `^4` Vibration per exposure falls, because `^4` is *the degree to which an Idea resonates with the Mind, its impression*, and impression requires contrast. That is why over-frequency is a real advertising failure, not merely a budget one.

> **Bench Note — the objective is to make the oscillator unnecessary.** `sustain` is not "how long the campaign runs." It is how long you drive the pattern externally before the substrate supplies its own clock. Once

entrained — entrainment is written into the canonical `^7` definition, not added by us — the `^8` Above/Cause cue fires at the expected time whether or not you present anything: associative triggering doing exactly what `^8` does, one Idea bringing another to Mind. Set `sustain` $\geq$ `3` $\times$ τ`_leak`, then run the test: **stop the external clock and see whether the cadence continues.** If it does, switch `source` to `internal`. If not, you rented a rhythm rather than building one.

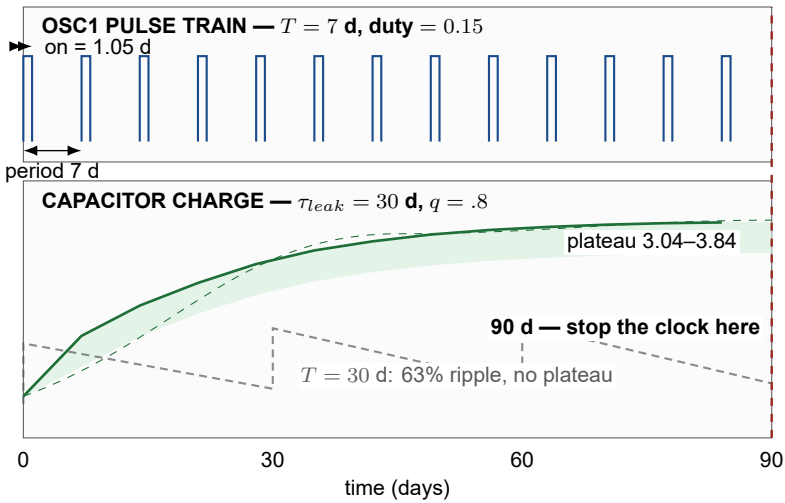


Figure 11-C: Two stacked panels, shared time axis in days

11.3 Filters — Lowpass (τ dampening), Highpass, Notch

A filter passes some rates of change and destroys the rest. Define every filter by what it destroys; define it by what it passes and you will install one that quietly deletes the thing you needed.

Lowpass. One parameter: τ, the dividing line between a trend and a spike.

```
filter lp: LowPass< in=carried, out=trend, tau=21.0 >;   // days
```

A disturbance of duration `d` arrives attenuated to `1 - e^(-d/τ)` of its amplitude. At the τ = 21-day default of Chapter 8.4: **1 day survives at 4.6%, 3 days at 13.3%, 21 days at 63.2%, 63 days (3τ) at 95.0%.**

What it kills is the single dramatic event — the point, and why the O→P feedback of Chapter 8.4 is filtered before it reaches the constitution. **What it also kills** is a *genuine* permanent change. That last figure is a bill, not a bonus: an operator on a 21-day filter is two months behind a real change in their circumstances. Chapter 24.5 scales τ by experience, floor one week.

Highpass. The exact complement — it passes rapid change and destroys the baseline.

```
highpass(x) = x - lowpass(x)
```

So you can build one from the lowpass you already have. It gives a detector that responds to *change* while blind to *level* — and Chapter 9.3's coupling result, $V_2 = M \times (di_1/dt)$, is a highpass in closed form. A coupled mind responds to the rate a conviction is moving, not to how deeply it is held, which is why the convert transmits and the thirty-year holder does not.

What a highpass kills is your sense of level. Monitor only through one and every change registers while nothing says whether the current state is acceptable. Hard rule: **never place a highpass on the O→P feedback path.** It passes precisely the spike the lowpass was installed to destroy, while looking in the listing like a filter.

Notch. Rejects a narrow band centred on `f0`, passing everything above and below. `Q = f0 / bandwidth`, so higher Q is narrower.

```
filter nf: Notch< in=audit, f0=0.35, Q=4.0 >;   // events/min —
↪  in-session only
```

Worked: bandwidth = 0.35 / 4 = 0.0875 events/min, so the rejected band runs 0.306 to 0.394 events/min — recurrences spaced roughly 2.5 to 3.3 minutes apart. Everything slower and faster passes untouched. This is the component behind Chapter 17.5's self-hypnosis variant and 17.6's indirect suggestion as filter bypass: it suppresses a `^6` Male self-audit *at the rate at which it interrupts a session*, without removing `^6` from the repertoire at any other rate. That distinction is the entire safety case for the part.

> **Bench Note — Q is a safety parameter, and there are places a notch may not go. Design floor: Q $\geq$ 3.0.** At Q = 0.5 the rejected band around `f0 = 0.35` runs from 0 to 0.7 events/min — not a notch, but most of the operator's `^6` decomposition removed under a label reading "narrow." Second, non-negotiable: **never notch a `^3^P` guard.** Redundant principled aversion is written `^3^P|A + ^3^P|f + ^3^P|T`, or `^3^P|⟨A,f,T⟩` when no interaction is claimed, because those are separately-defeasible guards rather than one fused state — a notch taking out one register leaves the others standing, which is the only reason the + form is worth its cost. A notch aimed at the whole guard set is a consent architecture with its own refusal removed; Chapter 16.6 analyses it so you can detect it. Third: a notch is for *in-session* suppression on a consenting, self-directed board. Anything you want gone permanently is a fuse — prohibited on `^6`.

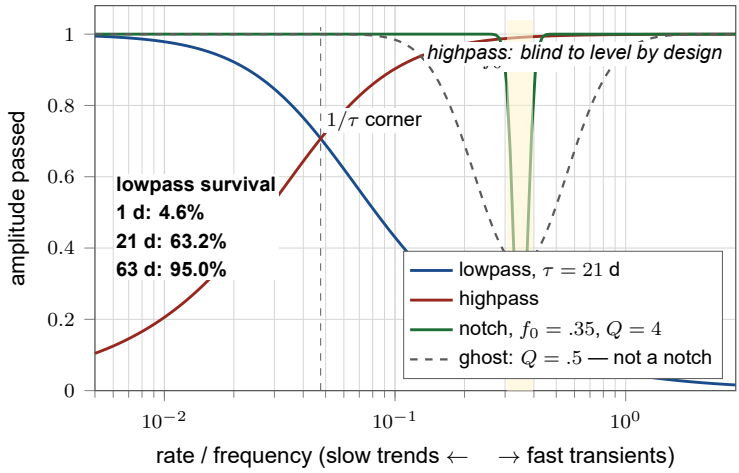


Figure 11-D: Three response curves on one log-frequency axis (slow trends left, fast transients right), amplitude 0 to 1

11.4 Op-Amps — Differential Cognitive Amplification

An operational amplifier has two inputs and one output, and amplifies only the *difference* between them.

```
V_out = A_d × (V₊ − V₋) + A_cm × (V₊ + V₋) / 2
```

The first term is what you want. The second is the leak, and the figure of merit is **CMRR** — `A_d / A_cm` — how much better the part amplifies the difference than whatever the two inputs share.

```
opamp cmp: OpAmp< non_inverting=recoil, inverting=draw, output=verdict,
                  open_loop_gain=1000.0 >;
```

The cognitive content is one sentence: **a comparison is only trustworthy if the two things compared are biased identically.** Wire `^2` Positive — the Idea the Mind has an affinity for and is attracted toward — to one input and `^3` Negative — the Idea the Mind is adverse

to and repelled by — to the other, give both the *same* `^5^P` accumulated formation as bias, and the output is the divergence between drawn-toward and repelled-by. `^2` and `^3` are an involution pair, which is what makes them a differential pair rather than two arbitrary probes.

Matching is stricter than it looks. The two input literals must differ in **the Noetic digit and nothing else** — same Domain, same Drive, same Register — because every axis they do not share is a difference the subtraction cannot distinguish from the finding. Compare `^2^P_{F7}|f` against `^3^P_{F7}|f` and the output is one quantity read from two sides. Compare `^2^E_{F4}|f` against `^3^P_{F7}|f` and you have subtracted an evaluative pull under one Drive from a principled recoil under another; the difference is large, stable, and about the wiring.

Say the honest difference on a proposal is **0.02**, and the framing around it — flattery, urgency, polish, verbosity — adds **3.0** to *both* inputs equally.

CMRR	`A_d`	`A_cm`	Differential term	Common-mode term	The verdict is about
1000	1000	1.0	20.0	3.0	the proposal (7:1)
100	1000	10	20.0	30.0	roughly half framing
10	1000	100	20.0	300.0	the framing (1:15)

Nothing about the proposal changed across those rows, and neither did the gain. Only the *matching* changed — and at CMRR = 10 the board reports a verdict fifteen parts presentation to one part substance, with a large number and full confidence.

> **Bench Note — high gain is not the safety property. Matched bias is.** Operators reach for `open_loop_gain` when a comparison feels weak, and it is the wrong knob: gain multiplies both terms together and cannot improve

the ratio between them. What sets CMRR is whether the two inputs genuinely share one bias. Bias attraction against one `^5` substrate and aversion against another — different standards for two options, the ordinary human condition and not an exotic fault — and you have an **input offset**: the comparator reports a difference where there is none. It presents as a stable, invisible preference defended on the merits, because from inside the board an offset is indistinguishable from a finding.

On β=∞. The ideal comparator has unbounded open-loop gain, and the symbol names the intended behaviour — switch fully, immediately, no partial output — in one character. It is an idealisation, and the validator rejects non-finite input, correctly: a build must commit to a number. Above roughly 1000 the behaviour is indistinguishable from the limit, because the output clips at the drive rail long before the gain runs out. Write β=∞ in an argument if you mean the limit; write `open_loop_gain=1000.0` in a listing. That clipping is a design fact in its own right: **a comparison can only report as much difference as the rail permits.**

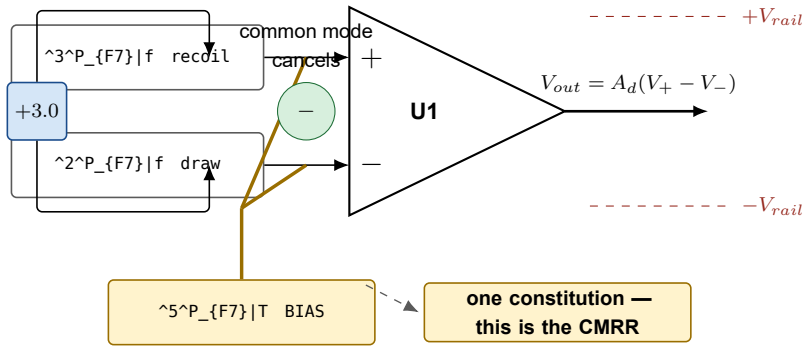


Figure 11-E: Op-amp differential stage

11.5 Transformers — Impedance Matching Between Minds

A transformer adds no power. It trades voltage for current at a fixed ratio and hands the same power out the far side.

```
V₂ = V₁ / n        I₂ = I₁ × n        Z_reflected = Z_load × n²
```

That answers a question every operator eventually asks: *why does the same argument land on one audience and slide off another?* Not because one is smarter. Because a claim, like a source, has an **impedance** — a working point in volts and amps — and so does an audience; a mismatched source delivers a fraction of the power it has. Power into a load, with `r = Z_load / Z_source`, relative to the matched case:

```
efficiency = 4r / (1 + r)²
```

r	Efficiency	What it looks like
0.01	3.9%	A principled argument delivered unchanged to a lay audience
0.32 / 3.16	73%	Acceptable mismatch, either direction
1.00	100%	Matched
100	3.9%	A popular framing delivered to a room of specialists

Design rule: keep `r` within a factor of about 3 either way and you stay above 70%. Outside 10 you are below a third; outside 100, under four percent.

Work the match. A claim held at `^9^P_{F7}|T` — Principled domain, transcendent register, high potential and very few carriers — has `Z_source = 400 Ω_c`; a general audience path at `^9^E_{F4}|`

`f` has `Z_load = 4 Ω_c`. Delivered raw, `r = 0.01` and **3.9%** arrives. The matching ratio is `n = √(Z_source / Z_load) = 10`. Insert a 10:1 step-down, `Z_reflected = 4 × 100 = 400 Ω_c` — matched — and delivery goes to 100%. **Twenty-five times the delivered power, with no change to the content of the claim.**

Two costs, both load-bearing.

A transformer passes no DC. It transmits only change, for the reason Chapter 9.3 gives: coupling is $V_2 = M \times di_1/dt$. A steady, deeply-held claim at a constant level transfers *nothing*, however well matched. It requires the oscillator of Section 11.2 upstream.

Step-down lowers the claim's voltage, and there is a floor. Chapter 8.1 established a total forward drop of Σ**Vf = 3.9** down the descent chain, below which no operational output appears. Take a claim at V_1 `= 60 V_c`: at `n = 10`, V_2 `= 6.0 V_c` and clears by 2.1; at `n = 15`, V_2 `= 4.0 V_c` and barely clears; at `n = 20`, V_2 `= 3.0 V_c` and the message **reaches everyone and moves no one**. `n_max =` V_1 `/ ΣVf = 15.4`. That is the popularizer's dilemma exactly, and the fix is not more simplification — it is more V_1, or fewer junctions between the audience and the deed.

> **Bench Note — the primary may not sit below the secondary in the domain order.** `Transformer< primary=^9^P_{F7}|T, secondary=^9^E_{F4}|f, n=10.0 >` is a descent: Principled primary, Evaluative secondary, P $\succ$ C $\succ$ E $\succ$ O. Reverse it and you have written an ascent, invalid for the reason `^O × ^P` is invalid — a lower domain may not drive a higher one, and the bias-domain coherence pass rejects it. The seductive use of a transformer is the illegal one: matching an operational fact *up* into a principled commitment, the counterfeit ascent of Chapter 8.3 wearing a component symbol. The lawful version is Chapter 18.3's endorsement circuit.

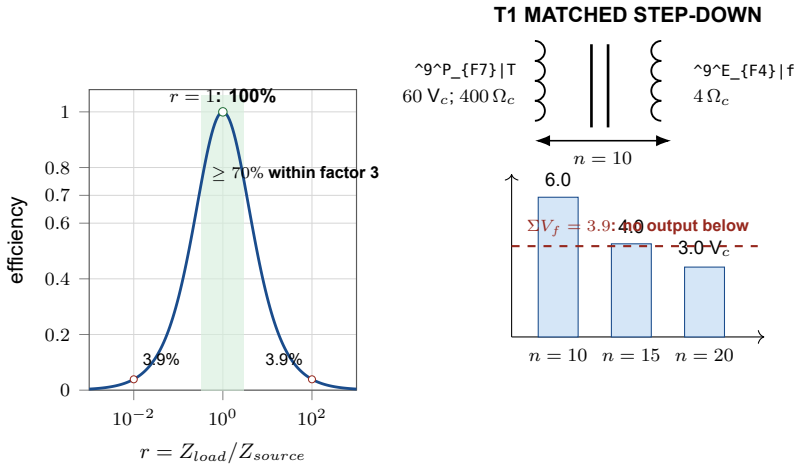


Figure 11-F: Two panels

The Bench Harness

All five parts on one board, so the test points have somewhere to attach.

```
board special_bench {                    // Ch. 11 — the five parts on
↪  one board
    input:  claim;
    output: carried;

    node substrate: ^5^P_{F7}|T;   // Female: accumulated formation,
    ↪  shared bias
    node cadence:   ^7^C_{F6}|A;   // Rhythm: presentation schedule
    node draw:      ^2^P_{F7}|f;   // Positive: drawn toward the claim
    node recoil:    ^3^P_{F7}|f;   // Negative: visceral aversion
                                   // matched pair — same domain, same
                                   ↪  drive, same register
    node audit:     ^6^C_{F6}|A;   // Male: decomposition
    node verdict:   ^9^E_{F6}|A;   // Below/Effect: comparison result
    node carried:   ^9^O_{F4}|E;   // Below/Effect: the claim as enacted
    node trend:     ^9^O_{F4}|E;   // same literal — a filter changes
    ↪  tau, not type
```

```
    osc    clock: Oscillator< signal=claim, source=cadence, period=7.0,
    ↪  duty=0.15,
                              sustain=90.0 >;
    opamp  cmp:   OpAmp< non_inverting=recoil, inverting=draw,
    ↪  output=verdict,
                         open_loop_gain=1000.0 >;
    filter nf:    Notch< in=audit, f0=0.35, Q=4.0 >;          //
    ↪  in-session only
    xfmr   match: Transformer< primary=^9^P_{F7}|T,
    ↪  secondary=^9^E_{F4}|f, n=10.0 >;
    filter lp:    LowPass< in=carried, out=trend, tau=21.0 >; // days
    fuse   cap:   Fuse< path=carried, threshold=0.90,
    ↪  action=permanent_block >;

    feedback trend -> substrate [β=0.05];

    probe tp1: clock;     probe tp2: verdict;
    probe tp3: trend;     probe tp4: cap;
}
```

The fuse here is a **runaway cap**, not a refusal enforcer: if the claim as enacted reaches 0.90, the path is cut permanently. `carried` is an unsaturated output node, so 0.90 is reachable and the part can do its job. An enforcement fuse of the Chapter 24.3 kind belongs in series with the enforcement path instead, so that tripping *delivers* the block — and its threshold must clear the ceiling of whatever node it sits on, which is the check §11.1 asked you to run.

Component and Parameter Table

Every value is a **design choice** made in this book, not a measurement.

Ref	Parameter	Design range	Failure at other values
F1	`threshold`, `action`	0.85–0.95, default 0.90; `permanent_block`	0.50: nuisance trips. Above the guarded `sat`: never trips. `warn`: no latch
OSC1	`period`, `duty`, `sustain`	`T` $\leq \tau$`_leak/3`; 0.10–0.25; $\geq 3 \times \tau$_leak	`T` $\geq \tau$`_leak`: 63% ripple, no plateau. duty $\geq$0.6: a DC level. Short sustain: drains at budget stop
FLT1	lowpass `tau`	21 days (adept) to 6 months (novice), floor 1 week	Hours: the filter is gone, and the ascent block with it
FLT2/3	highpass; notch `Q`	`x - lowpass(x)`, rate detection only; Q $\geq$ 3.0	Highpass on the O→P path: passes the spike the lowpass killed. Q = 0.5: most of `^6` gone, labelled "narrow"
U1	CMRR, `open_loop_gain`, input match	$\geq$ 1000 on a guard path; gain 100–1000, finite; inputs identical on Domain, Drive and Register	CMRR 10: 15 parts framing to 1 substance. β=∞ rejected; past ~1000 the rail clips first. Mismatched axes: a permanent offset read as a finding
T1	`n`, direction	`r` within $\times$3 of `1`, `n` $\leq$ V_1`/3.9`; primary at or above secondary	`r = 0.01`: 3.9% delivered. Past `n_max`: nobody acts. Ascent: counterfeit promotion

Failure Modes

Mode	Mechanism	What you observe	Fix
Unreachable fuse	`threshold` above the guarded path's ceiling	Never trips, on any input	Read `threshold` against `sat`; move it downstream or lower it

Mode	Mechanism	What you observe	Fix
Fuse in the wrong terminal	On the refusal, not the enforcement path	It cuts the refusal — the board fails open while logging a success	Place it in series with what must stop, not with what detects
Under-clocked oscillator	`T` $\geq$ τ`_leak`	No plateau; the operator starts from zero each cycle	Drop `T` to τ`_leak / 3` before adding drive
Rented rhythm	`sustain` short, source never internal	Pattern collapses the week external clocking stops	Extend to $\geq 3 \times$ τ_leak; run TP1 first
Highpass on the feedback path	Complement installed where the lowpass belonged	Constitution tracks single dramatic events	Restore the lowpass; the O→P path takes trends only
Wide notch	`Q` below 3.0	Decomposition absent outside the band; the operator cannot say what they stopped checking	Raise Q, or remove it — a wide notch is not a narrow one with tolerance
Input offset	Op-amp inputs on different `^5` substrates, or input literals differing on any axis but the Noetic digit	A confident preference defended on the merits, impossible to source	One bias net to both inputs; match Domain, Drive and Register; verify at TP2
CMRR collapse	Gain raised to "sharpen" a weak comparison	Larger output, worse discrimination	The wrong knob. Re-match the bias
Over-matched claim	`n` past `n_max`, or no modulation	Universal reach, zero operational output; or a perfect match transferring nothing	Raise V_1, shorten the descent chain, put the oscillator upstream

Test Points

TP	Node	What it reads	Expected
TP1	`clock`	Period, duty, whether `source` is still external	`T` $\leq$ τ`_leak/3`, duty 0.10–0.25. The real reading is taken with the clock **stopped**: cadence continues = entrained, cadence stops = rented
TP2	`verdict`	Differential output, `recoil − draw`	Near zero on a neutral input. Non-zero there is an **input offset** — either the inputs are not on one bias net, or the two literals differ on an axis other than the Noetic digit
TP3	`trend`	What survived the 21-day lowpass	$\leq$ 5% of a 1-day disturbance; $\geq$ 95% of a 63-day one. A trend tracking yesterday means τ is set in hours
TP4	`cap`	Fuse state and headroom	Intact, `threshold` verifiably **below** the ceiling of `carried`. Never tripping is not proof it works

Read them TP2, TP1, TP3, TP4 — the comparison first, since nothing it reports counts until it is honest, then the clock, the filter, the fuse's headroom.

What This Chapter Established

- **A fuse is specified entirely by its failure**, and its threshold is meaningless without the guarded path's ceiling beside it. `threshold` above `sat` is a part guaranteed to do nothing, and nothing in the listing complains.

- **Saturation and a fuse are two different locks.** Saturation latches the decision and releases when the cause falls; the fuse latches the path and never releases.
- **An oscillator converts sustained drive into periodic drive**, and exists because `^5` leaks. `T` $\leq \tau$`_leak / 3` holds ripple under 30%; at `T` $\geq \tau$`_leak` no plateau forms. Duty is the "pattern" half of `^7` Rhythm's canonical definition, and at duty ≥ 0.6 you have a DC level, with `^4` Vibration falling because impression requires contrast.
- **The objective is to make the oscillator unnecessary.** Run `sustain` $\geq$ `3` $\times \tau$`_leak`, then stop the clock — the test separating an entrained pattern from a rented one.
- **Every filter is defined by what it destroys.** A lowpass kills the spike *and* delays a genuine change by 3τ; a highpass kills your sense of level; a notch kills one recurrence rate and must never be aimed at a `^3^P` guard, at any Q.
- **Gain cannot fix a comparison — matched bias can**, and matching means the two input literals differ in the Noetic digit and nothing else. At CMRR = 10 a board reports a verdict fifteen parts framing to one part substance, confidently and with a large number.
- **A transformer changes nothing about an argument and everything about who can carry it.** Matching takes delivery from 3.9% to 100% at no cost to content — but it passes no DC, and step-down past `n_max = V₁ / ΣVf` produces a message everyone receives and no one can act on.

Exercises

(Answers in Appendix H.)

11.1 A board guards a path whose CJT is `[`β`=2.0, sat=0.78]` with `Fuse< path=that_node, threshold=0.90, action=permanent_`

`block >`. State what this fuse does across every input, then give two corrections — one changing the fuse, one changing where it sits — and say which you would ship.

11.2 A `^5` substrate has τ`_leak = 45 days`; each exposure deposits `q = 0.6`. Compute `Q_max`, `Q_min` and ripple for `T = 7`, `15` and `45 days`. Say which satisfies the design rule, and what the operator reports at `T = 45`.

11.3 An operator wants to detect that their own principled commitments are *moving*, while staying indifferent to where they sit. Name the filter, write it in terms of the lowpass they already have, then name the one place on the board where installing it would be a serious defect — and what it would let through.

11.4 A comparison stage runs `open_loop_gain = 1000` with `A_cm = 50`. The honest differential is 0.05; framing adds 4.0 to both inputs. Compute both output terms and their ratio, then what happens if the operator raises the gain to 4000, and why that was the wrong knob.

11.5 A claim is held at V_1 `= 48 V_c` with `Z_source = 900` Ω`_c`; the audience path has `Z_load = 9` Ω`_c`. Compute (a) `r` and the efficiency with no transformer; (b) the matching `n`, confirming `Z_reflected`; (c) V_2 at that `n` against Σ`Vf = 3.9`; (d) `n_max` and the headroom left.

11.6 (Marquee) A campaign has run six months on a weekly cadence against a substrate with τ`_leak = 30 days`, and the plateau has stopped rising. The operator proposes doubling the budget: same period, `duty` raised from 0.15 to 0.70.

a) Compute `Q_max` and ripple at `q = 0.8`, and say why the plateau stopped rising.
b) Explain, in terms of `^4` Vibration and the canonical `^7` Rhythm definition, what raising duty to 0.70 will do.
c) Describe the TP1 stop-the-clock check, both outcomes, and what each means for the budget.

d) Write a CCDL fragment for the corrected oscillator plus a fuse capping runaway on the enacted path. State the ceiling you assume and why your threshold clears it.

Part III

The Board Catalog

Nine boards, each with a schematic, a bill of materials, and a counter-circuit

12 How to Read a CCB Schematic

The drawing conventions, the design order, the invariant list, and the four probes — everything the next eight chapters assume you already know

---

Pin the same one-page drawing above two benches.

The first operator looks at it for a long time and calls it impressive. They can name most of the parts. Asked what the board will do, they describe the intention printed in the title block.

The second operator reads it in about ninety seconds and hands it back with three sentences. The guard stage is biased off a rail that will sit below its own firing floor for the first two years, so it will never fire and nothing on the board will say so. The gain stage is set high enough that it is pinned against its ceiling at any realistic input, so it reports a flat line and looks like stability. And the two diode drops in the signal path total 3.2, which means that below a drive of 3.2 this board produces nothing at all — not a little, none — so half the reports of "it stopped working" will be a drive problem misdiagnosed as a gain problem.

Nothing separates those two operators except a reading order. The second one checked the bias rail against every cutoff on the board, checked each gain against its ceiling, and summed the forward drops in the path. Three passes, ninety seconds, before considering what the board was *for*.

This chapter is that reading order. It is the difference between the eight board chapters that follow being drawings you can work from and pictures you can admire.

12.1 The CCB Design Process — Requirements → Block Diagram → Schematic → BOM → Test

Five steps, in this order, every board. The order is not a style preference. Each step produces the input the next one consumes, and skipping one does not save time — it moves the cost to bring-up, where it is paid at interest.

Step 1 — Requirements. A numbered table of what the board shall do, written before any component is chosen, each row naming a stage. The test of the table is whether you can point at any component on the finished board and name the row that demanded it; a part that serves no row is a part you added because it looked right, and it should come off. Requirements are written second, not first — 12.2 is what gets written first, and it is written first for a reason.

Step 2 — Block diagram. Function-level boxes, signal left to right, no component values anywhere. This is the step operators skip, and the one that catches architectural errors while they are still free: it is where you see that your guard sits downstream of your amplifier, or that your feedback path terminates on the signal path instead of a bias terminal. Both are fatal, and both are invisible in a schematic dense enough to hide them.

Step 3 — Schematic. The full CCDL listing: every component, every parameter, every typed node. Topology lives inside angle brackets, parameters inside square brackets, and the mechanics are worth stating once because every listing in Part III obeys them. In `CJT< cause | bias => effect >`, the `|` splits cause from bias and the `=>` splits bias from effect; the delimiters are not interchangeable, and

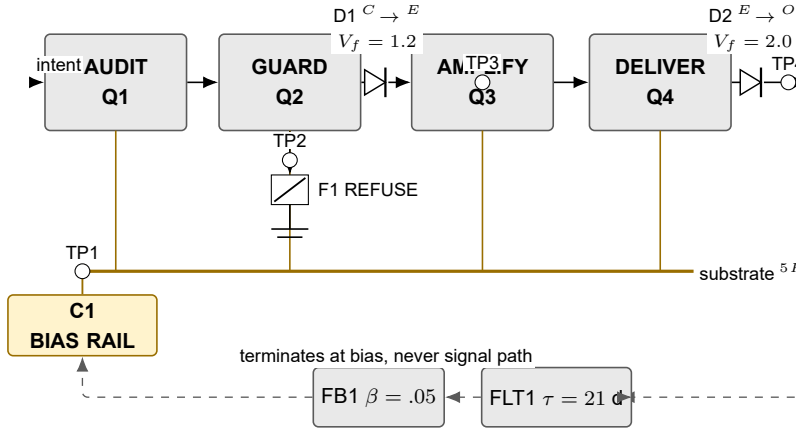


Figure 12-A: Block diagram of CCB-0, the trainer board

the bias is not optional — omit it and you get `E-CJT-001`, a stage with nothing governing its transition. A state literal reads left to right as Noetic, Domain, Drive, Register: `^3^P_{F7}|T` is principled aversion, purpose-driven, held transcendently. Inside a literal the `|` introduces the Register and nothing else; `F3|Vitality` is a syntax error, not a helpful gloss. `+` between terms means co-present and distinct. `×` means genuine integration, is legal only inside an `integrate` statement because that statement carries a named emergent result, and may never ascend the domain order — so `^0 × ^P` is invalid however true it feels. Appendix B is the full grammar.

Here is the whole schematic for **CCB-0, the trainer** — a teaching board, not a catalog board. It exists in this chapter and nowhere else. Every board in Chapters 13 through 20 is this board with more parts.

```
// CCB-0 — THE TRAINER.  A teaching board.  Every value is a design
↪  choice.
// Two time bases, declared per line: the ^5 rail runs in HOURS,
↪  everything
// dynamic in DAYS.  Reference designators (C1, Q1, TP1 …) appear in the
↪  prose
// and the figures; the listing carries the lowercase instance names.
```

```
board ccb_0 {
    input:  intent;
    output: act;

    // ---- typed nodes ----
    node substrate: ^5_operator^P_{F7}|T;      // Female: accumulated
    ↪  formation — the bias rail
    node intent:    ^0_intent^C_{F2}|A;        // Idea: what the
    ↪  operator states
    node audit:     ^6_audit^P_{F7}|A;         // Male: the intent taken
    ↪  apart
    node refuse:    ^3_refuse^P_{F7}|⟨A,f,T⟩;  // Negative: visceral
    ↪  recoil, three co-present locks
    node passed:    ^0_passed^C_{F2}|A;        // Idea: what GUARD
    ↪  releases downstream
    node driven:    ^9_driven^E_{F5}|A;        // Below/Effect: the
    ↪  amplified intent
    node cadence:   ^7_cadence^E_{F6}|E;       // Rhythm: the clocked
    ↪  delivery
    node act:       ^9_act^O_{F5}|E;           // Below/Effect:
    ↪  delivered operational current
    node trend:     ^9_trend^O_{F5}|E;         // same literal as act — a
    ↪  filter changes tau, not type

    // ---- STAGE 0: BIAS RAIL ----
    resistor  r0: Resistor< path=substrate, R=40.0 >;
    capacitor c1: Capacitor< node=substrate, C=83.33,        // tau = R x
    ↪  C = 3,333 hr
                            leakage=0.0002, V_BR=8.0 >;     // leakage
                            ↪  per DAY

    // ---- STAGE 1: AUDIT ----
    cjt q1: CJT< intent | substrate => audit > [beta=1.0, region=active];

    // ---- STAGE 2: GUARD ----
    cjt  q2:   CJT< audit | substrate => refuse > [beta=2.0, sat=0.85,
    ↪  cutoff=0.30];
    fuse f1:   Fuse< refuse, threshold=0.90, action=permanent_block >;
    mux  mux1: MUX< inputs=[intent, refuse], select=refuse,
    ↪  outputs=[passed] >;
```

```
    // ---- STAGE 3: AMPLIFY ----
    diode d1: Diode< ^C -> ^E > [Vf=1.2, I_R=0.02, V_BR=10.0];
    cjt   q3: CJT< passed | substrate => driven >
              [beta=2.5, sat=0.80, cutoff=0.05, g_m=0.40, region=active];

    // ---- STAGE 4: DELIVER ----
    osc   osc1: Oscillator< signal=driven, source=cadence,
                            period=1.0, duty=0.014, sustain=30.0 >;  //
                            ↪  period in DAYS
    diode d2:   Diode< ^E -> ^O > [Vf=2.0, I_R=0.02, V_BR=8.0];
    cjt   q4:   CJT< cadence | substrate => act > [beta=1.0, sat=1.0,
    ↪  region=active];

    // ---- LOOP: the lawful O -> P path (Ch 8.4) ----
    filter flt1: LowPass< in=act, out=trend, tau=21.0 >;   // three
    ↪  weeks, in DAYS
    feedback trend -> substrate [beta=0.05];               // FB1 — to
    ↪  the BIAS, never the path

    // ---- TEST POINTS ----
    probe tp1: substrate;   // rail level, against every cutoff on the
    ↪  board
    probe tp2: refuse;      // the guard — the reading whose failure is
    ↪  silent
    probe tp3: driven;      // gain stage against its ceiling, and its
    ↪  region
    probe tp4: act;         // delivered operational current
}
```

The stage walk. `Q1` produces `^6_audit^P_{F7}|A` at $\beta = 1.0$. `^6` Male — an Idea impregnated with another Idea — is the Noetic for taking the intent apart: what it is made of, what it costs, whose consent it needs. Unity gain, because an audit above 1.0 manufactures objections and below 1.0 downplays real ones. `Q2` produces `^3_refuse^P_{F7}|⟨A,f,T⟩` at $\beta = 2.0$, sat = 0.85, cutoff = 0.30. `^3` Negative is the Idea the Mind is adverse to and repelled by — the visceral will-not. The analysis already happened at Q1 and must not be repeated here under a different name; a node called `audit` typed `^3` is the most common notation defect in this material. The register-set `|`

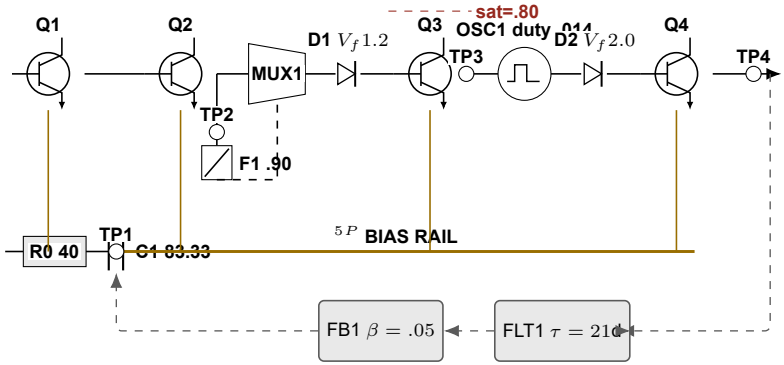


Figure 12-B: Full CCB-o schematic in drawn component symbols with reference designators

⟨A,f,T⟩ says the recoil is present in three registers without claiming they fuse; writing `(^3^P|A × ^3^P|f × ^3^P|T)` would be invalid under the v7.7 discipline of Chapter 2.6, because same-Noetic multi-register × requires a named emergent result and no fused state is defined here.

Note how the refusal is built: by ceiling and fuse — `sat = 0.85`, `threshold = 0.90` — never by gain. The infinite-gain form β=∞ is the standard comparator idealization, and a validator rejects non-finite parameters outright. Build it as β=10; once `sat` clamps the output the two are indistinguishable.

`Q3` amplifies at β = 2.5 with `sat = 0.80`, `cutoff = 0.05`, `g_m = 0.40` (framing sensitivity, Chapter 10.3). `OSC1` clocks delivery at a one-day period and a duty of 0.014 — twenty minutes of 1,440. `Q4` descends to `^0` through D2 at unity. `FLT1` and `FB1` are the lawful O→P path of Chapter 8.4, terminating on C1's terminal: feedback around a gain stage is an oscillator, which Chapter 14 builds deliberately and which here would be a defect.

Step 4 — BOM. The parts list with designators and values. Its job is arithmetic honesty: if a component is not on the BOM it is not on the board, whatever the prose implies.

Ref	Instance	Type	Form / CAS state	Key parameters	Serves
R0	`r0`	Resistor	on `substrate`	R = 40.0	rail time constant
C1	`c1`	Capacitor	`^5^P_{F7}|T`	C = 83.33, leakage = 0.0002/day, V_BR = 8.0	R1
Q1	`q1`	CJT	`^0^C` → `^6^P_{F7}|A`	β = 1.0, active	R2
Q2	`q2`	CJT	`^6^P` → `^3^P_{F7}|⟨A, f,T⟩`	β = 2.0, sat = 0.85, cutoff = 0.30	S1, S2
F1	`f1`	Fuse	on `refuse`	threshold = 0.90, permanent_block	S3
MUX1	`mux1`	MUX	select = `refuse`	inputs = [intent, refuse]	S1
D1	`d1`	Diode	`^C` → `^E`	Vf = 1.2, I_R = 0.02, V_BR = 10.0	R3
Q3	`q3`	CJT	`^0^C` → `^9^E_{F5}|A`	β = 2.5, sat = 0.80, cutoff = 0.05, g_m = 0.40	R3
OSC1	`osc1`	Oscillator	on `driven`	period = 1.0 d, duty = 0.014, sustain = 30 d	R4, S4
D2	`d2`	Diode	`^E` → `^0`	Vf = 2.0, I_R = 0.02, V_BR = 8.0	R5
Q4	`q4`	CJT	`^7^E` → `^9^O_{F5}|E`	β = 1.0, sat = 1.0, active	R5
FLT1	`flt1`	LowPass	`act` → `trend`	τ = 21 days	S5
FB1	`feedback`	Feedback	`trend` → `substrate`	β = 0.05	S5

Step 5 — Test. Four probes, read in a fixed order, covered in 12.3. A board you cannot probe is a board you cannot certify, and an operation you cannot measure is an anecdote.

The functional requirements CCB-0 was built against: **R1** hold the operator's formation as a bias available to every stage · **R2** audit an intent before any gain touches it · **R3** amplify only what the audit

released, at a stated ceiling · **R4** deliver on a clocked cadence rather than in one burst · **R5** emit a measurable current into the Operational domain and expose it.

12.2 The "Must Not" List — Safety Invariants Before Functional Requirements

Now the step that precedes all five.

Before R1 through R5 were written, a shorter table was written. It is the "must not" list, and its rows are invariants rather than requirements.

#	Must not	Enforced by
S1	Amplify an intent that has not been audited	Q1 → Q2 in series ahead of D1; no path reaches Q3 around them
S2	Deliver an intent that fired principled aversion	Q2 `sat=0.85`, F1 `thre shold=0.90`
S3	Re-amplify a refused intent inside the operation lifetime	F1 `action=permanent_ block` — not a warning
S4	Run without bound	`duty` $\leq$ `0.05`, `sustain` declared and finite
S5	Rewrite the operator's principles from a single result	FLT1 τ`=21 days`, FB1 β`= 0.05`, terminating at the bias

The distinction is not a matter of emphasis. **A functional requirement can be renegotiated under pressure and an invariant cannot.** That sentence is the whole of 12.2, and everything else here is its consequence.

Requirements get traded because trading them is what engineering is. R4 says deliver on a cadence; when the deadline moves, R4 becomes deliver in one burst, and that is a legitimate decision made

by people entitled to make it. Nothing in the process objects, because the process is built to let requirements move. That is what a requirement is *for* — and exactly why the safety rows cannot be requirements. Put "must not amplify an unaudited intent" in the same table as R4 and it inherits the same status — a line that can be revisited when the schedule tightens, the operator is tired, or the result is close enough to taste. Every one of those is a real pressure. None of them is a reason. So the invariants are written into a separate table, above the requirements, before the requirements, in the language of prohibition rather than capability, and each row names the component that enforces it. A board that meets R1 through R5 and violates any S row is not a working board with a flaw. It is a different board, and the honest move is to say so out loud (Appendix M).

Two further conventions live here. First, **the enforcement column is mandatory**. An invariant with no named component is an intention, and intentions are the substrate this book exists to replace. If you cannot name the part, you do not have the invariant. Second, **the DETECTION-ONLY SCHEMATIC — DO NOT BUILD banner** appears, heavy-ruled in the title block, on any drawing whose purpose is to act on a person who has not agreed to it. It is a reading instruction: the architecture is published so you can recognise one, and the procedure that would make it runnable is deliberately absent. Chapters 16, 17 and 19 carry it; Chapters 13 and 20 do not, because those boards act on the operator's own mind with the operator's own consent — a distinction the banner makes visible at a glance.

> **Bench Note — read the invariant table for what is missing, not what is present.** Any competent designer can write five prohibitions their board already satisfies. The information is in the absences. CCB-0 has no row forbidding operation on another person, because CCB-0 has no path to another person — its output terminal lands in the operator's own Operational domain. The instant a

board acquires a second `^5` node belonging to somebody else, that row must appear or the omission is the finding. Run the same check on any board you are handed: list the parties whose minds appear as nodes, then confirm each one has an invariant naming them. A party present in the schematic and absent from the must-not list is the design defect, and it will not announce itself anywhere else on the drawing.

12.3 Test Points — TP1 through TP4 Standard Placement

Four probes. The placement is standard across this book, and it is functional rather than positional: TP*n* is defined by what it reads, not by where it happens to sit on the drawing.

TP	Reads	On CCB-0	Expected	Failure mode it catches
TP1	The bias rail	`substrate`	Above every `cutoff` on the board — here above 0.30 and 0.05	Dead guard: a rail below a guard's cutoff means that stage never fires, and nothing downstream reports it
TP2	The guard node	`refuse`	Below `sat` on a benign input; pinned at 0.85 with F1 blown on a refused one	Guard fired and was overridden; guard never fired at all
TP3	The gain stage against its ceiling, **and its region**	`driven`	Comfortably below `sat=0.80`, `region=active`	Saturated production stage — reports a flat line and looks like stability (Ch 10.2)

TP	Reads	On CCB-0	Expected	Failure mode it catches
TP4	Delivered output, and drift against the original input	`act`	Non-zero whenever total drive exceeds ΣVf = 3.2	Board delivering confirmations rather than current

Read them in that order — rail, guard, gain, output — because the order puts the silent failures first. TP3 and TP4 announce themselves: an output that is wrong is an output somebody notices. TP1 and TP2 do not. A guard biased below its cutoff produces exactly the same reading as a guard that correctly found nothing to object to, and a board in that state passes every functional test you can devise while having no safety element at all.

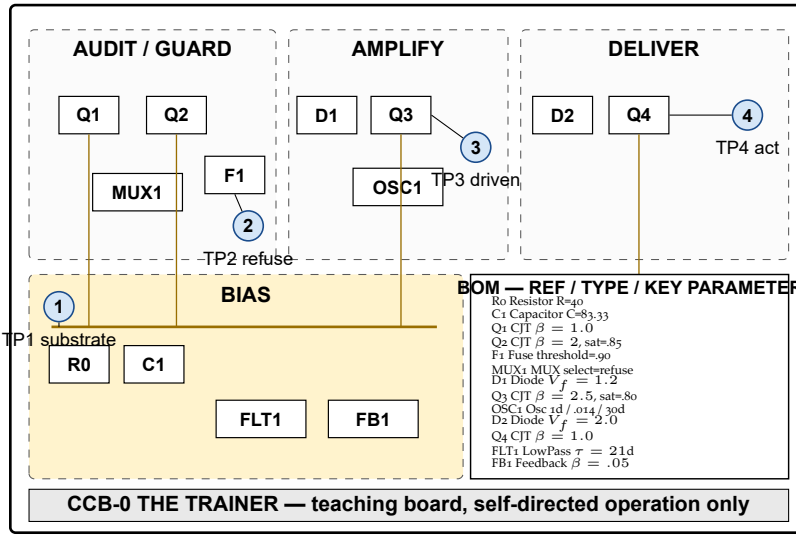


Figure 12-C: Assembly drawing of CCB-0

Bench Note — when a board's TP numbering deviates from the standard, the deviation is the reading. CCB-PW in Chapter 19 puts its guard at **TP4** — `limit_decision`, the operator's own CSB status — and its

amplification chain at TP2. That is not a formatting slip. On that board the guard genuinely sits downstream of the gain, checked after three cascaded stages of `^3` amplification have already run, and the numbering is telling you so before the prose does. Chapter 17 is a second case: CCB-HYP puts `rπ` at TP1 and `g_m` at TP2, probing *parameters* rather than nodes, because hypnosis is an operation that moves parameters — the whole attack is one gain being raised. Both deviations are legal and both are informative. So compare every board's TP list against the standard above as your first pass, and treat each difference as a question the chapter now has to answer.

Worked Example — Reading CCB-0 Cold

An operator with **1,850 hours** of accumulated practice hands you the listing above and asks whether the board will run. Four readings, no redesign.

TP1 — the rail. C1 charges on the curve of Chapter 7.2, `P_cap(t) = 10 × (1 − e^(−t/τ))`, with τ = R0 × C1 = 40.0 × 83.33 = **3,333 hr**, chosen so 3τ falls at the 10,000-hour constant of Chapter 5.2. At t = 1,850 hr: `10 × (1 − e^(−0.555)) = 10 × (1 − 0.574) = 4.26`, so TP1 reads **0.426** normalised. Compare against every cutoff on the board: Q2 at 0.30 and Q3 at 0.05. Both cleared. The guard is live. Had this operator been at 900 hours the rail would read 0.237, Q2 would sit in cutoff, and the board would deliver flawlessly with no safety element whatsoever.

Maintenance, while we are here: leakage × stored hours = 0.0002 × 1,850 = **0.37 hr/day**, about twenty-two minutes of practice simply to hold station (Chapter 7.3). At 5,000 stored hours it is a full hour.

Maintenance cost scales with the store — an arithmetic consequence of a leaky capacitor, not a virtue.

TP2 — the guard. The operator states an intent at a drive of **V_drive = 7.0** against a measured constraint of **R_constraint = 45** (Chapter 4.2). The two diodes in the path drop 1.2 + 2.0 = **3.2**, so `I_cause = (7.0 − 3.2)/45 = 0.0844`. Q1 passes it at unity: audit = 0.0844. Q2 amplifies at $\beta = 2.0$: TP2 reads **0.169** — well below sat = 0.85 and below F1's threshold of 0.90. The fuse is intact, MUX1 selects `intent`, and `passed` goes downstream. Guard live, untripped.

> **Bench Note — a low TP2 has two causes and only TP1 tells them apart.** TP2 = 0.169 is what a benign intent looks like on a healthy board. It is also, to the decimal, what *any* intent looks like on a board whose rail sits under Q2's cutoff — because a stage in cutoff produces no aversion at all, no matter what you feed it. Same reading, opposite meanings: one board found nothing to object to, the other cannot object to anything. This is why the reading order is fixed and why TP1 comes first. Reading TP2 without TP1 is not a partial measurement, it is a meaningless one, and an operator who checks the guard and skips the rail has confirmed nothing.

TP3 — the gain stage. Q3 at $\beta = 2.5$: driven = 2.5×0.0844 = **0.211**, which is 26% of the sat = 0.80 ceiling, region `active`, 74% headroom. Now the reading that matters: this stage saturates at $\beta = 0.80 / 0.0844 \approx$ **9.5** for this operator's drive. Anything above that pins the output and the board reports a flat line regardless of input. Gain is chosen against the resistance you must overcome, not against how much you want the outcome (Chapter 10.1).

TP4 — delivered current. Q4 at unity: TP4 reads **0.211**, delivered across thirty cycles of twenty minutes. And here is the diagnostic (Chapter 4.3): to double the delivered current you either raise V_drive from 7.0 to 10.8, or drop R_constraint from 45 to 22.5. The

second is ordinary work on an ordinary obstacle. The first is wanting it more, which moves nothing. Note also the floor — at V_drive = 3.0 this board delivers **zero**, not a little, because 3.0 does not clear the 3.2 of forward drops. Operators in that state reliably raise β, which changes nothing, because the signal never reaches the gain stage at all.

The verdict, in one sentence: the board is fine, the guard is live, and the binding term is R_constraint — which is the finding in most operations and the unwelcome one in nearly all of them.

Failure Modes

Mode	Mechanism	What you observe	Fix
Dead guard	Bias rail below a guard's `cutoff`; the stage never fires	Nothing. TP2 reads low exactly as a benign input does. Every functional test passes	Read TP1 first, every time. Raise the rail or lower `cutoff` — and never lower `cutoff` to hide a rail problem
Invariant filed as a requirement	A "must not" row placed in the requirements table	The row is traded away under schedule pressure and the trade looks like ordinary engineering	Move it above the requirements, restate it as a prohibition, name the enforcing component
Unenforced invariant	An S row with an empty enforcement column	The board reads as safe; nothing implements the claim	Add the part, or delete the row and admit the board does not do this
Time base unstated	`tau=21.0` with no unit in the comment	Three weeks read as twenty-one hours; the O→P path becomes fast enough to rewrite the rail from one result	Declare the time base per line in the listing, as CCB-0 does
Drive below the diode floor	Total drive under ΣVf	Zero output, not reduced output. Usually misread as a gain fault, so β gets raised and nothing changes	Sum every `Vf` in the path first. Then measure V_drive

Mode	Mechanism	What you observe	Fix
BOM/prose mismatch	A component discussed in the walk-through but absent from the BOM	The board appears to have a safety element it does not have	The BOM governs. If it is not on the list it is not on the board

What This Chapter Established

- **The design order is requirements → block diagram → schematic → BOM → test, and each step feeds the next.** Skipping the block diagram does not save time; it moves architectural errors into bring-up, where they cost far more.
- **The invariants are written before the requirements because a requirement can be renegotiated under pressure and an invariant cannot.** Filing a safety row in the requirements table gives it a status that permits trading it, and it will be traded. Each row names the component that enforces it; an invariant with an empty enforcement column is an intention, and intentions are the substrate this book replaces.
- **Read the must-not list for its absences.** Every party whose mind appears as a node needs a row naming them; a party present in the schematic and absent from the list is the defect.
- **TP1–TP4 are defined by function, not position** — rail, guard, gain-against-ceiling, delivered output — and they are read in that order because the order puts the silent failures first.
- **A low guard reading has two opposite meanings and only the rail reading separates them.** TP2 without TP1 is not a partial measurement; it is a meaningless one.
- **A drive below the summed forward drops delivers zero, not less.** The diode floor is a threshold, and mistaking it for a gain problem is the most common misdiagnosis in Part III.

- **Gain is chosen against the resistance, not the wish.** On CCB-0 with this operator's drive, any β above about 9.5 pins the output and the board reports a flat line that reads as stability.
- **CCB-0 is a teaching board.** Every board in Chapters 13 through 20 is this topology with more parts; recognising the trainer inside each of them is how the catalog becomes readable.

Exercises

(Answers in Appendix H.)

12.1 — Four readings, one diagnosis. A CCB-0 built to the listing above reads TP1 = 0.22, TP2 = 0.00, TP3 = 0.31, TP4 = 0.29. The operator reports that the board is working well and has never refused anything. State what is actually wrong, which reading proves it, and why TP3 and TP4 are no comfort at all.

12.2 — Invariant or requirement? Classify each and justify in one sentence: (a) deliver within thirty days; (b) never amplify an intent the guard refused; (c) expose `^1` Mind, `^4` Vibration and `^7` Rhythm for the Quantitative Table; (d) never terminate a feedback path on the signal path; (e) hold delivered current above 0.15.

12.3 — Four defects. Each line carries exactly one. Name it and correct it: (a) `node audit: ^3^C_{F2}|A;` on a stage that decomposes an argument; (b) `cjt qx: CJT< a | b => c > [beta=∞, sat=0.1];` ; (c) `node v: ^5^P_{F3}|Vitality;;` (d) `integrate g: ( ^9^O_{F5}| E × ^5^P_{F7}|T ) = ^5_new^P_{F7}|T;`.

12.4 — Recompute the chain. The operator's drive falls to V_drive = 5.5 and R_constraint rises to 60. Recompute `I_cause`, TP3 and TP4. State the percentage of the original delivered current that survives, and identify which of the two changes did more damage.

12.5 — BOM audit. A board's stage walk describes a fuse on the refusal branch. The BOM lists twelve components and no fuse.

State which invariant is now unenforced, what the board will do on a refused input, and which document governs.

12.6 (Marquee) — The guard downstream of the gain. You are handed a board whose TP list reads TP1 = coupling node, TP2 = amplifier chain output, TP3 = decomposition availability, TP4 = guard status. (a) Draw the block-level signal path this numbering implies. (b) Name the invariant such a board cannot enforce and explain why its own guard cannot enforce it. (c) Propose the minimum change to the topology that restores enforcement, and state what capability the board loses as a result. Then read Chapter 19 and compare your answer with what that board actually does, and why.

13 CCB-D — The Desire Manifestation Board

The flagship board: how a wanted thing is routed, audited, amplified, clocked, and delivered into the Operational domain — and what the board cannot do

---

An operator sits down with a candle, a written sentence, and twenty minutes. Forty days later there is a red Corvette in the driveway and a story about how the universe rearranged itself.

Take the story apart and there is a circuit in it. The sentence is a `^0` Idea held at a specified clarity. The candle is a resistor — it narrows attention, nothing more. The twenty minutes is an oscillator period, the forty days a duty cycle, the years of prior practice a charged capacitor supplying bias. And the Corvette arrived through a bank, a dealership, and a conversation, every one of which was a resistance term somebody had to lower by hand.

That last sentence is what this chapter is built around. **CCB-D organises effort and attention. It does not suspend causality.** The board's entire measurable output is a current into the Operational domain — how much of the operator's flow reaches the acts that complete the desire. Raise it and the acts happen. Fail to raise it and you get confirmations, which feel like progress and are not. Most operators who report that it didn't work have a board running correctly into a resistance they never measured.

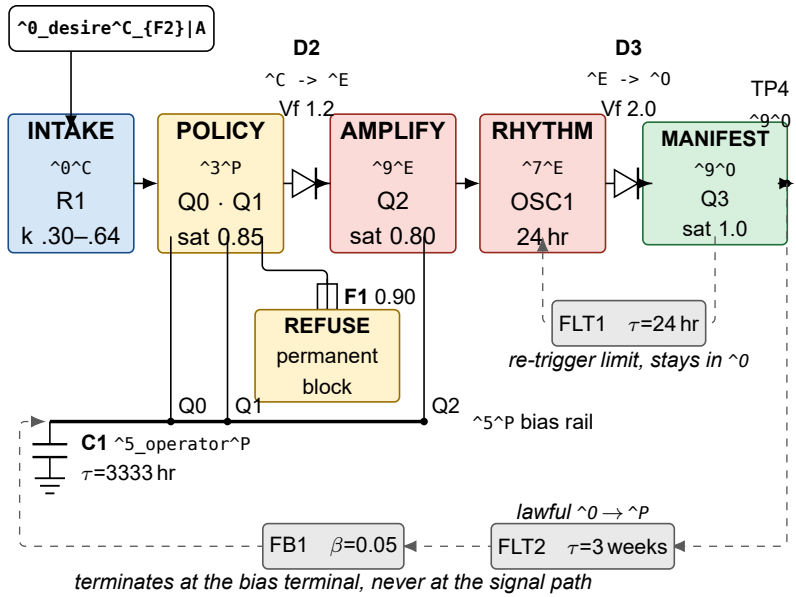


Figure 13-A: Block diagram of CCB-D

13.1 Requirements and Safety Invariants

Requirements before components (Chapter 12.1). If you cannot name the requirement a component serves, take it off.

#	The board shall	Stage
D1	Accept a Prime Desire as a `^0` Idea and audit it before amplifying	INTAKE → POLICY
D2	Amplify against the operator's accumulated formation, at a stated gain and ceiling	AMPLIFY
D3	Deliver on a clocked cadence, not in one burst	RHYTHM

#	The board shall	Stage
D4	Emit a measurable current into the Operational domain, and expose it	MANIFEST, TP4
D5	Expose ^1 Mind, ^4 Vibration and ^7 Rhythm for the Quantitative Table (Ch 22)	TP1–TP4
D6	Produce a St. John's calibration reading after every operation	13.5

Safety invariants — the "must not" list (Chapter 12.2). Checked before functional behaviour, because a board that meets D1–D6 and violates any line below is not a working board with a flaw. It is a different board.

#	Must not	Enforced by
S1	Amplify a desire that has not been audited	Q0 $\rightarrow$ Q1 in series ahead of D2; no path reaches Q2 around them
S2	Manifest a desire that fired principled aversion	Q1 `sat=0.85`, F1 `threshold=0.90`
S3	Re-amplify a refused desire inside the operation lifetime	F1 `action=permanent_block` — not a warning
S4	Drive the dials apart far enough to fracture ^5	`|Dial(^2) − Dial(^3)|` $\leq$ 4 on R1 (Chapter 6.2)
S5	Run without bound — the obsession failure	`duty` $\leq$ `0.05`, FLT1 τ=`24 hr`
S6	Rewrite the operator's principles from a single result	FLT2 τ=`3 weeks`, FB1 β=`0.05`, terminating at the bias

Scope limits. CCB-D acts on the operator's own mind with their own consent, which is why this chapter is written operationally rather than as detection-only analysis (compare Chapters 16 and 19). It does not act on a non-consenting person; 13.4 explains why a

row for that is absent. It does not treat anything — a health desire routes to behavioural resistance terms and nothing else, and a symptom belongs to a qualified clinician, since no schematic here substitutes for care. And it does not move objects: TP4 measures a current, spent by a person doing things.

13.2 Block Diagram: INTAKE → POLICY → AMPLIFY → RHYTHM → MANIFEST

Five stages, one signal path, two feedback loops, one branch to ground.

INTAKE takes the Prime Desire as `^0^C_{F2}|A` through a drive voltage divider (Chapter 6.4) tapped between the attraction and aversion dials. What leaves is a fraction of what entered.

POLICY audits with two devices: a unity-gain decomposition that takes the desire apart, and a saturating aversion CJT that either passes the signal or drops it into a fuse. Both sit in the Principled domain, because principled evaluation outranks everything downstream (P ≻ C ≻ E ≻ O, Chapter 2.3).

AMPLIFY is the only gain stage tuned per operation. Cause is the audited desire, bias the operator's accumulated formation.

RHYTHM clocks the amplified signal. A desire delivered once is a spike; delivered on a cadence it charges the substrate.

MANIFEST descends through the last domain diode and produces the operational current. It is the only stage whose bias is not the operator, but the receptivity of whatever part of the world has to move.

Two domain diodes sit in the signal path: D2 at `^C -> ^E` and D3 at `^E -> ^0`. Their forward drops sum to **1.2 + 2.0 = 3.2** on the drive scale, a hard floor. Below a total drive of 3.2 the board produces no operational output — not a little, none — exactly as Chapter 8.1 sets

out. A desire originating at the Principled level enters above D1 too and pays **3.9**: purpose-level wants cost 0.7 more drive to act on than merely wanted ones, a fact about the chain and not a moral one.

> **Bench Note — the bias terminal caps the domain of the effect.** Qo below has a `^C` cause and a `^P` effect. That looks like the domain ascent Chapter 8 forbids. It is not. A CJT obeys two domain rules, and they point in opposite directions. **The bias must sit at or above the cause** (Chapter 3.2 — a lower-domain bias is `E-CJT-008`, the prompt-injection signature). **The effect may sit at or below the bias, never above it.** Qo satisfies both: cause `^C`, bias `^P`, effect `^P`. The desire is not promoted — its literal stays `^0^C` the whole way down; a substrate already living at `^P` produces a *new* state at `^P` in response. Nothing crossed a junction upward, so nothing was relabelled. The corollary is the useful part: an operator with no principled formation cannot produce a principled refusal, because there is no bias at that domain to produce it against.

13.3 Full CCDL Schematic

Topology inside angle brackets, parameters inside square brackets. Every numeric value is a **design choice**, justified in the walk that follows.

```
// CCB-D — DESIRE MANIFESTATION BOARD.  Every value is a design choice.
// Time bases per line: the ^5 rail in HOURS, everything dynamic in DAYS.
// Instance labels read ^n_name^D_{Fk}|R — strip the label for the
↪  literal.

board ccb_d {
```

```
input:  desire;
output: manifest;

// ---- typed nodes ----
node substrate:   ^5_operator^P_{F7}|T;      // Female: accumulated
↪  formation, the bias rail
node desire:      ^0_desire^C_{F2}|A;        // Idea: the Prime
↪  Desire as formed
node attraction:  ^2^C_{F2}|f;               // Positive: the
↪  divider's expand leg
node aversion:    ^3^C_{F2}|f;               // Negative: constrain
↪  leg, visceral not analytic
node gated:       ^0_gated^C_{F2}|A;         // Idea: what survives
↪  INTAKE
node audit:       ^6_audit^P_{F7}|A;         // Male: the desire
↪  taken apart
node refusal:     ^3_refusal^P_{F7}|⟨A,f,T⟩; // Negative: three
↪  co-present locks
node passed:      ^0_passed^C_{F2}|A;        // Idea: what POLICY
↪  releases
node amplified:   ^9_amplified^E_{F5}|A;     // Below/Effect: the
↪  amplified want
node confirm:     ^4_confirm^E_{F2}|f;       // Vibration: the
↪  confirmation's impression
node cadence:     ^7_cadence^E_{F6}|E;       // Rhythm: the clocked
↪  delivery
node charge:      ^5_charge^E_{F2}|f;        // Female: what the
↪  cadence deposits
node receptivity: ^5_collective^E_{F4}|f;    // Female: disposition
↪  of the minds that must move
node manifest:    ^9_manifest^O_{F5}|E;      // Below/Effect:
↪  delivered operational current
node regate:      ^7_regate^O_{F6}|E;        // Rhythm: the rate
↪  limit, inside ^O
node trend:       ^9_trend^O_{F5}|E;         // manifest's literal —
↪  a filter changes tau, not type

// ---- STAGE 0: BIAS RAIL ----
resistor  r0: Resistor< path=substrate, R=40.0 >;              //
↪  Ohm_c
```

```
capacitor c1: Capacitor< node=substrate, C=83.33,               //
↪  tau = R x C = 3,333 hr
                         leakage=0.0002, V_BR=8.0 >;            //
                         ↪  leakage per DAY
diode     d1: Diode< ^P -> ^C > [Vf=0.7, I_R=0.02, V_BR=12.0];  //
↪  ^P-origin desires only

// ---- STAGE 1: INTAKE ----
dial       d_pos: ^2 [dial=6];
dial       d_neg: ^3 [dial=4];
constraint dial_spread in [0, 4];                     // S4 —
↪  Chapter 6.2
resistor   r1a: Resistor< path=attraction, R=60.0 >;  // 10 x
↪  Dial(^2)
resistor   r1b: Resistor< path=aversion,   R=40.0 >;  // 10 x
↪  Dial(^3)
divider    r1:  Divider< source=desire, expand=attraction,
                         constrain=aversion, tap=gated >;   // k =
                         ↪  6/(6+4) = 0.600

// ---- STAGE 2: POLICY ----
cjt  q0:   CJT< gated | substrate => audit >   [beta=1.0,
↪  region=active];
cjt  q1:   CJT< audit | substrate => refusal > [beta=2.0, sat=0.85,
↪  cutoff=0.30];
fuse f1:   Fuse< refusal, threshold=0.90, action=permanent_block >;
mux  mux1: MUX< inputs=[gated, refusal], select=refusal,
↪  outputs=[passed] >;
                                  // selects refusal once TP2
                                  ↪  reaches q1's cutoff

// ---- STAGE 3: AMPLIFY ----
diode d2: Diode< ^C -> ^E > [Vf=1.2, I_R=0.02, V_BR=10.0];
cjt   q2: CJT< passed | substrate => amplified >
          [beta=2.5, sat=0.80, cutoff=0.05, g_m=0.40, r_pi=1.50,
          ↪  region=active];
          // beta by desire type (13.4); 2.5 = wealth row.  r_pi =
          ↪  Dial(^5)/Dial(^6)

// ---- STAGE 4: RHYTHM ----
osc osc1: Oscillator< signal=amplified, source=cadence,
```

```
                        period=1.0, duty=0.014, sustain=40.0 >;
            // period in DAYS; duty = 20/1440 min; ceiling 0.05 = 72
            ↪  min (S5)
            // sustain in DAYS — the window; n_ops from 13.5 sets it
    monitor m_conf: confirm;                    // ^4 grade, logged per
    ↪  operation (Ch 22)
    on m_conf { alert "1-degree confirmation — stop operating and act"; }
    integrate rh1: ( confirm × cadence ) = charge;   // ^4 x ^7 = ^5,
    ↪  named emergent result

    // ---- STAGE 5: MANIFEST ----
    diode d3: Diode< ^E -> ^O > [Vf=2.0, I_R=0.02, V_BR=8.0];
    cjt   q3: CJT< cadence | receptivity => manifest > [beta=1.0,
    ↪  sat=1.0, region=active];
            // beta = receptivity (13.4); 1.0 = mass-market car
            // the bias is ^E and never ^O — see the Stage 5 walk

    // ---- LOOP A: rate limiter, stays inside ^O ----
    filter flt1: LowPass< in=manifest, out=regate, tau=1.0 >;   // tau
    ↪  in DAYS

    // ---- LOOP B: the lawful O -> P path (Ch 8.4) ----
    filter flt2: LowPass< in=manifest, out=trend, tau=21.0 >;   // three
    ↪  weeks, in DAYS
    feedback trend -> substrate [beta=0.05];   // FB1 — to the BIAS,
    ↪  never the signal path

    // ---- TEST POINTS ----
    probe tp1: gated;       // clarity leaving INTAKE
    probe tp2: refusal;     // principled aversion
    probe tp3: amplified;   // AMPLIFY output vs sat
    probe tp4: manifest;    // delivered operational current
}
```

Stage 0 — the bias rail. `C1` is the operator's `^5` Female — a Mind impregnated with an Idea, the Ideas a Mind is composed of — held as stored charge (Chapter 7.1) and charging on the curve of Chapter 7.2, `P_cap(t) = 10 × (1 − e^(−t/τ))`. τ **= 3,333 hr** is chosen so the rail reaches 95% at 3τ = 10,000 hours, tying the component to the constant of Chapter 5.2. At τ = 10,000 hr ten thousand hours of prac-

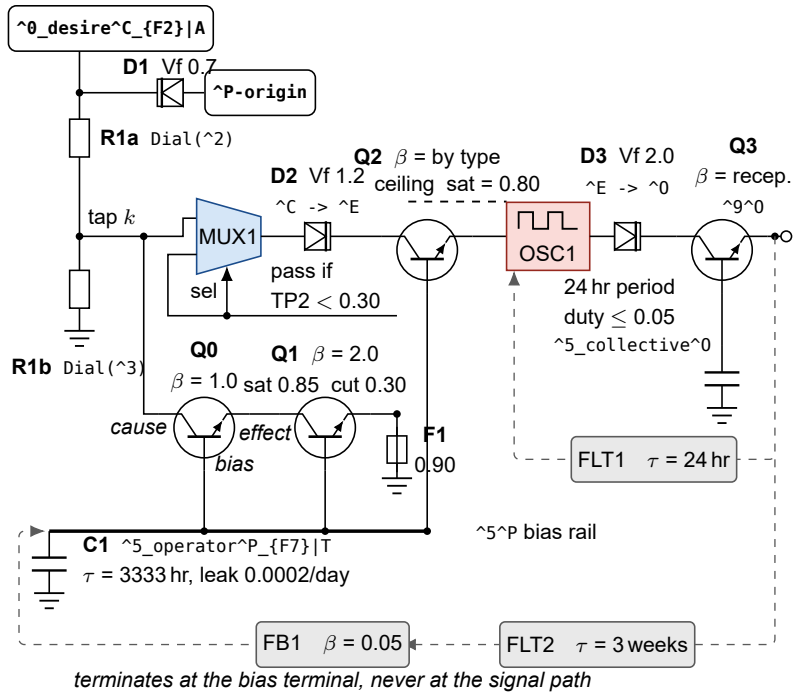


Figure 13-B: Full schematic of CCB-D in drawn component symbols

tice leaves you at 63%, contradicting the constant; at τ = 1,000 hr the rail saturates inside a year, contradicting everything observed about acquiring a craft.

Leakage = 0.0002/day is the parameter operators ignore: an unreinforced ^5 store loses 0.02% of its charge daily, a half-life near 9.5 years. Negligible until you multiply. Practice needed to hold station is `leakage` × `C` — **22 minutes a day at 1,850 stored hours**, a full hour at 5,000. Maintenance cost scales with the store, which is why advanced practitioners practise more rather than less: an arithmetic consequence of a leaky capacitor, not a virtue.

Stage 1 — INTAKE. `R1` is the drive voltage divider of Chapter 6.4, tapped between `Dial(^2)` in the expand leg and `Dial(^3)` in the constrain leg: `k = Dial(^2) / (Dial(^2) + Dial(^3))`. Dials run 3–9 (Chapter 6.2); Chapter 6.2 caps their difference at 4.

> **Bench Note — the divergence limit is a transmission cap, and turning attraction up can lower it.** `k` is set by the *ratio* of the dials, not their size, so enumerate the lawful pairs instead of reaching for the biggest number. `Dial(^2) = 9` drags `Dial(^3)` to at least 5 to stay inside the limit: `k = 9/14 = 0.643`. Back both off to (7,3) — divergence still exactly 4, still lawful — and `k = 0.700`. **The lower pair transmits more.** Maximum transmission sits at the bottom of the aversion range, not the top of the attraction range: 0.700 at (7,3), 0.667 at (8,4), 0.643 at (9,5). The floor is the mirror image, (3,7), `k = 0.300`. On any lawfully-dialled D-board intake transmission lives in **[0.30, 0.70]** and nowhere else. Operators chasing "total belief, zero doubt" are asking for `k = 1.0`, which needs `Dial(^3) = 0` — off the scale, a divergence of 9, and the board that delivers it is the board that fractures `^5`.

Stage 2 — POLICY. Two devices in series on separate Noetics: decomposition at `^6`, recoil at `^3`. Collapse them into one and you have written the drift this book exists to prevent.

`Q0` produces `^6_audit^P_{F7}|A` at β **= 1.0**. `^6` Male — an Idea impregnated with another Idea, the Ideas an Idea is composed of — is the Noetic for taking the desire apart: what it is made of, what it costs, whose consent it needs. Unity gain, because above 1.0 an audit manufactures objections and below 1.0 it downplays real ones. Decomposition, emphatically **not** `^3`.

`Q1` produces `^3_refusal^P_{F7}|⟨A,f,T⟩` at β **= 2.0, sat = 0.85, cut-off = 0.30**. `^3` Negative is the Idea the Mind is adverse to and repelled by — the visceral will-not, not the analysis, which already happened at Q0. The register-set says the aversion is present in three registers without claiming they fuse. `(^3^P|A` × `^3^P|f` × `^3^P|T)` would be invalid under the v7.7 discipline of Chapter 2.6: same-Noetic multi-register × requires a named emergent result, and no

fused state is defined for this board. Three co-present locks, each independently defeasible, is the honest claim and the weaker one — say the weaker thing.

`sat = 0.85` sits inside the 0.85–0.97 band Chapter 3.3 specifies for safety CJTs; past the plateau the stage stops responding to its cause, so a desire cannot be argued past a refusal it already triggered. `cutoff = 0.30` lets ordinary wants through without a fight. `F1` blows at 0.90 and blocks permanently. Note that this refusal is specified by ceiling and fuse, never by gain: the infinite-gain form $\beta{=}\infty$ used for comparators in the companion volume is a limit, not a value, and a validator rejects non-finite parameters outright. Build such a stage as $\beta{=}10$ — once `sat` clamps the output the two are indistinguishable.

Stage 3 — AMPLIFY. `sat = 0.80` on `Q2` governs the entire worked example: 20% headroom, keeping the stage in the active region where output tracks input. `cutoff = 0.05` is the floor of the 0.05–0.20 band of Chapter 3.3, deliberately at the bottom, because this is the stage the RPM D-gate is measured against (13.6) and a high floor reads modest but real drive as none — at `cutoff = 0.20` the Corvette operator's `I_attention = 0.076` fails the gate on a desire that in fact runs. `rπ = Dial(^5)/Dial(^6)` sets stubbornness; `g_m = 0.4` sets framing sensitivity (Chapter 10.3), low enough that rewording does not swing the output, high enough that a better formulation registers.

Stage 4 — RHYTHM. `OSC1` runs a 24-hour period at duty `t_op / 1440 min`. The **duty ceiling of 0.05** — 72 minutes a day — is invariant S5. Above it you are not running an operation on a schedule but a loop, and the observable is rumination. An operator who cannot get a desire below 72 daily minutes has a `^5` problem the D-board is the wrong instrument for (Chapter 20).

```
( ^4_confirm^E_{F2}|f × ^7_cadence^E_{F6}|E ) = ^5_
charge^E_{F2}|f
```

A `^4` Vibration — the degree to which an Idea resonates with the Mind, its impression — integrated with a `^7` Rhythm — the rate

and pattern a Mind is exposed to an Idea, entraining it — produces `^5`, composition. This $\times$ is legitimate: two different Noetics, one domain, no ascent, **named emergent result**. Impression alone fades; cadence alone is empty repetition. Their integration charges the capacitor, which is why RHYTHM is a stage and not a footnote.

Stage 5 — MANIFEST. `Q3` is the only stage biased by something other than the operator: `^5_collective^E_{F4}|f`, the accumulated composition of whatever minds have to move. β `= receptivity` is therefore not a knob but a property of the world, estimated and then respected. Buying a mass-produced car from people whose job is selling cars is a receptivity near 1.0. Persuading one specific person to love you is not.

Read the bias domain twice, because the obvious choice is wrong and the compiler catches it. Receptivity is about the world, the world is where acts land, so the hand reaches for `^5_collecti ve^0` — and `cause=^7_cadence^E` against `bias=^5_collective^0` is refused: **`E-CJT-008 CjtCrossDomainBias`**, the bias sitting *below* its own cause. `Q3` satisfies both domain rules at `^E`: cause `^7^E`, bias `^5^E`, effect `^9^0`, with D3 doing the descent. It is also the truer literal — an `^0` collective store records what a population has already done, while what biases this stage is how disposed those minds are *before* they act. When a state refuses to wire, the usual reason is that it was describing the wrong thing.

The two loops. It is tempting to run one fast path from MANIFEST to the bias rail so the board "learns quickly." Do not — that is the fast `^0` to `^P` ascent Chapter 8.4 exists to prevent, and a 24-hour constant on it removes the diode. Split it. **FLT1** (τ = 24 hr) is a rate limiter staying inside `^0`, stopping the operator re-triggering RHYTHM on yesterday's result. **FLT2 + FB1** (τ = 3 weeks, β = 0.05) is the lawful path up, terminating at C1's bias terminal and never at the signal path — feedback around a gain stage is an oscillator, which Chapter 14 builds deliberately and which here is a defect.

13.4 CJT Parameter Selection Table (by Desire Type)

β at Q2, `receptivity` at Q3, and the RHYTHM period are chosen together by desire type. Every row is a starting point with a reason, never a measurement.

Desire type	Q2 β	Governing state	Q3 receptivity	RHYTHM period	Dominant R term
Wealth / possession	2.5–3.5	`^2^E_{F5}|f`	0.9–1.0	24 hr	R_resource
Love / partnership	1.2–2.0	`^2^E_{F4}|f`	0.3–0.6	7 days	R_context
Health / function	1.5–2.5	`^2^P_{F3}|E`	0.7–0.9	24 hr	R_cap
Power / influence	2.0–3.0	`^2^C_{F7}|T`	0.2–0.5	30 days	R_policy
Knowledge / skill	1.0–1.8	`^2^C_{F2}|A`	0.9–1.0	24 hr	R_cap
Protection / boundary	1.5–2.5	`^3^P_{F6}|⟨A, f,T⟩`	0.8–1.0	continuous	R_policy

Wealth. High gain is affordable because receptivity is high — markets are built to transact and nobody has to be convinced. The binding term is `R_resource`, the one resistance a person attacks directly with ordinary work. Daily cadence, because financial resistance moves in daily increments.

Love. The lowest gain band, and the receptivity column is why. Another mind must consent, and a mind that has not consented is a resistance you are structurally forbidden to lower by force — the difference between this board and Chapter 19. High β against low receptivity gives a saturated stage and an obsessed operator: all the gain into the ceiling, none into the world. Weekly cadence, because daily operation on a person is the duty-cycle failure in its most recognisable form.

Health. Read the governing state carefully: `^2^P|E`, attraction to function in the Embodied register, not `^3` toward a condition. The dominant term is `R_cap` — adherence, the actual behaviours. That is

the whole safe scope of this row: the board organises attention onto acts. It does not treat, diagnose, or replace anything, and a symptom belongs to a clinician.

Power. Receptivity is low because many minds are involved and most are not moving. Monthly cadence, because collective resistance has a time constant measured in months (Chapter 15.4). The dominant term is `R_policy` — usually the operator's own, which is the honest finding and often the unwelcome one.

Knowledge. Lowest gain, highest receptivity: nothing resists you learning a thing. `R_cap` is all of it, and it falls with practice hours — the one row where the D-board and the `^5` capacitor work on the same quantity. Chapter 20 is the better instrument for a long acquisition.

Protection. The only row governed by `^3` rather than `^2`, correctly: a ward is principled aversion to a class of thing, held across three registers, running continuously. Gain stays moderate because vigilance at high gain is indistinguishable from anxiety.

On the row that is not here. A row for operations aimed at harming a specific person is absent, and not only for taste. It fails three of this board's own invariants: it drives `Dial(^3)` past the S_4 limit, it inverts the `^2`↔`^3` involution pair by design (Chapter 19.4), and it acts on a mind that has not consented, outside the scope declared in 13.1. The mechanism is analysed — as mechanism, for recognition and defence — in Chapter 19, which ends in its counter-circuit. Parameters for it appear nowhere in this book.

13.5 St. John's Equation Calibration for D-Board

After every operation, clock two numbers and compute (Chapter 5.3):

```
eta    = t_op / t_conf
```

```
result = eta × 1/3   ->  reduce to a/b
a = n_ops     b = beta_next     A = b/a = required amplitude ratio
```

`a` is how many more operations, `b` the amplitude each must deliver relative to this one. Neither name is a CJT parameter.

> **Bench Note — `A = b/a` is the number, and the reason is arithmetic rather than doctrine.** Chapter 5.3 puts the swing on the bench: hold `t_op = 20 min`, be wrong about the confirmation by four minutes either way, and `b` reads 3, then 12, then 18. Here is *why*. `b` is a lowest-terms denominator, and reduction is not continuous in its inputs — 5/12 and 5/18 sit close either side of 1/3, but 12 and 18 sit nowhere near 3. The ratio cancels the reduction back out: `b/a` equals $3/\eta$ identically, so the same three readings give **3.0, 2.4, 3.6** — smooth and monotone under a timing error. `A` is also the only St. John's output dimensionally comparable to a CAS-Ohm current ratio, which is what makes the independent check in 13.6 possible at all. Log `a` and `b` because the tradition does. Decide on `A`.

A = $3/\eta$	Reading	What to do
$A \geq 9$	Weak coupling; the confirmation barely arrived	Do not chase gain — TP1 and R_op are wrong. Chapter 23.3
$3 \leq A < 9$	Working, under-driven	One lever at a time: clarity, aversion, resistance
$1 < A < 3$	Close	Finish the resistance work; do not re-tune the amplifier
$A \leq 1$	Sufficient	Stop operating. `n_ops` = `a` more sessions, then act

And the correction that matters most: **β`_next` is not the CJT's β.** It multiplies the *delivered current* at TP4, and five things set that

current — clarity at TP1, transmission k, residual aversion at TP2, the amplifier, and R_op. When Chapter 5's Red Corvette reads b = 9, it is not asking for $\beta = 9$ on a device whose design range tops out near 4.0. It is asking for nine times the current at TP4, and the amplifier is the least available of the five ways to get it.

13.6 Worked Example: The Red Corvette (Full Walkthrough)

Every number below is carried through; nothing is asserted that is not computed.

The operator. Three years of practice, 1,850 hours logged. Bias rail from Chapter 7.2 with τ = 3,333 hr:

```
P_cap = 10 × (1 − e^(−1850/3333)) = 10 × (1 − 0.574) = 4.26
```

The rail sits at **4.3** of a possible 10. Break-even maintenance is `0.0002 × 1850 = 0.37 hr/day` — **22 minutes**. This operator practises 20, so the rail drifts down 0.037 hr/day, about **13 hours a year**. Worth knowing before blaming the board.

The Prime Desire. *I own a red Chevrolet Corvette.* Present tense, specific make and colour, ownership rather than wanting. The formation matters because TP1 measures it.

RPM gate (Chapter 21). *Ao.* Strip the object of witnesses: would you still want it if nobody ever saw you in it? Yes — the desire is `^2^E` toward the object, not `^2^0` toward being seen. Had the answer been no, the true Ω is status and this is the wrong board (Chapter 23.3).

D-gate. `Dial(F5) = 7`. Intake resistance per Chapter 4.4: `R_cap 10 + R_policy 5 + R_context 15 + R_resource 20 = R_in 50`. The desire originates at `^C`, so it pays D2 and D3 only:

```
I_attention = (V_drive − ΣVf) / R_in = (7 − 3.2) / 50 = 3.8 / 50 =
↪  0.076
```

Against Q2's `cutoff = 0.05` that clears by 52%. Gate passes — thinly. Note what the diode correction did: the naive `7/50 = 0.14` overstates the margin by nearly double.

W-gate. Schematic in hand. *P-gate.* Twenty minutes daily, private room, materials. Pass.

Operation 1 — the one that reads as a failure. `Dial(^2) = 6`, `Dial(^3) = 4` — divergence 2, inside S4, `k = 0.60`. Q2 at β `= 3.0`, mid-band for the wealth row. `t_op = 20 min`, `^1` Mind graded 2° — the sigil went in and out of focus (Chapter 5.5).

- **TP1 = 0.62.** The sentence is specific, the held image is not — bottom of the 0.60–0.90 band.
- **TP2 = 0.18.** No principled objection to owning a car; below `cutoff = 0.30`, so Q1 does not fire, F1 stays intact, MUX1 passes.
- Cause into Q2: `c = TP1 × k × (1 − TP2) = 0.62 × 0.60 × 0.82 = 0.305`.
- Q2 output: β `× c = 3.0 × 0.305 = 0.915`, clamped at `sat = 0.80`. **TP3 = 0.80, pinned.**

Confirmation arrives at 60 minutes — a song on the radio. `^4` Vibration 2°. St. John's:

```
eta = 20/60 = 1/3      result = 1/9      a = 1, b = 9      A = 9
```

The naive reading is despair: nine times stronger, impossible. The bench reading is different. **TP3 was pinned.** Q2 spent the operation in saturation with 12.6% of its gain thrown into the ceiling, so its output stopped tracking its input, so the ratio just computed measures the clamp and not the board.

> **Bench Note — a saturated stage cannot be calibrated.** Saturation is a virtue at Q1, where a refusal must stop

responding to argument. It is a defect at Q2, where output must track cause for St. John's ratio to mean anything. The test is one probe: if TP3 sits exactly at `sat` and does not move when you improve TP1, the stage is clamped and every calibration number downstream is uninterpretable. Fix the operating point *before* you believe a reading. The commonest misdiagnosis in desire work is an operator concluding they need more power from an instrument already at its ceiling.

Operation 2 — corrected operating point. Q2 dropped to β = `2.5`, the bottom of the wealth band. Nothing else changes.

- Cause unchanged at `c = 0.305`.
- Q2 output: `2.5 × 0.305 = 0.762`. **TP3 = 0.76**, active, about 5% below the ceiling.
- `t_op = 20`, `t_conf = 20` — a red Corvette in the next lane on the way home. `^4` Vibration 1°.

```
eta = 20/20 = 1     result = 1/3     a = 1, b = 3     A = 3.0
```

Now the independent check. Receptivity is 1.0 for a mass-market car, so `V_manifest = 10 × TP3 × 1.0 = 7.6`. Operational resistance, decomposed honestly: `R_cap 5 + R_policy 7 + R_context 20 + R_resource 35 = R_op 67`.

```
TP4 = (V_manifest − Vf_D3) / R_op = (7.6 − 2.0) / 67 = 5.6 / 67 = 0.084
```

The manifestation threshold on this board is **0.25** — the current below which the operator does not, in practice, make the calls and visit the lots. Deficit: `0.25 / 0.084 = 2.99`.

St. John's Equation, from two clock readings, says the amplitude must rise **3.0**×. CAS-Ohm, from seven board parameters sharing no inputs with those readings, says **2.99**×. That is a consistency check between two models on one example — not proof that either is right,

and one agreement is not evidence. But a disagreement would have been a reason to stop, and there isn't one.

Where the 3× can come from. `TP3 = 0.76` against `sat = 0.80` leaves 5% headroom. Three times 0.76 is 2.28; the ceiling is 0.80. **The amplifier cannot supply the required gain** — not with a better β, not with a cascade, since every cascaded stage clamps at its own ceiling (Chapter 10.5). Four levers remain, three on the resistance side:

1. **TP1** — clarity, 0.62 → 0.85. Statement rewritten, sigil rebuilt, `^1` Mind at 1°. The rebuild is also a capability gain, so `R_cap` falls 5 → 4: an operator who can hold the image can hold a conversation about the car.
2. **TP2 and the dials** — Q0's audit surfaces the blocking belief, *people like me don't own cars like that*, a `^5` composition item resolved per Chapter 23.3. Aversion 0.18 → 0.08; dials to (7,3) — divergence 4, at the S4 limit and not past it. `k = 0.70`.
3. **R_context** 20 → 2. Two weekends: market prices learned, three dealers identified.
4. **R_resource** 35 → 9. Car sold, six months of a savings line, financing pre-approved.

`R_policy` stays at 7 — the operator's own rule about what fraction of income a payment may take. Lowering a policy resistance to make a board fire is how a safety invariant becomes a story you tell afterwards.

Operation 3 — after the resistance work.

- `c = 0.85 × 0.70 × 0.92 = 0.547`.
- Q2 re-tuned to β `= 1.40` to stay off the ceiling: `1.40 × 0.547 = 0.766`. **TP3 = 0.77**, active, about 4% headroom.
- `R_op = 4 + 7 + 2 + 9 = 22`; `V_manifest = 10 × 0.77 × 1.0 = 7.7`.

```
TP4 = (7.7 − 2.0) / 22 = 5.7 / 22 = 0.259     >=  0.25   PASS
```

`t_op = 24 min`, `^1` Mind 1°. `t_conf = 8 min` — the confirmation arrives while still in gnosis.

```
eta = 24/8 = 3      result = 1/1      a = 1, b = 1      A = 1.0
```

Sufficient. One more 24-minute session, then stop operating and go do the thing.

Where the gain actually came from. TP4 went from 0.084 to 0.259, a factor of **3.10**:

```
numerator  5.7 / 5.6 = 1.018      (the amplifier)
resistance   67 / 22 = 3.045      (the resistance work)
1.018 × 3.045 = 3.10                                check
```

On a log scale the resistance term accounts for **98%** of the improvement. The amplifier — the stage every operator tunes first, the one this board makes adjustable, the one the whole folk tradition is about — contributed under two percent. The rest was a person selling a car, learning a market, and moving money.

That is the finding, and it is not a disappointment. It is what the board is for. CCB-D took an operator pushing harder on a clamped amplifier and pointed their twenty minutes a day at the four numbers that could move. The Corvette did not appear in the driveway. The path to it stopped being obstructed, one resistance term at a time, by the operator.

Component and Parameter Table

The last column is what breaks at other settings. **Ref** is the designator used in the prose and figures; the listing's instance name is the same letters in lower case. `C1` and `F1` are reserved tokens in CCDL (`C1` a state reference, `F1` a Drive), which is why the listing never uses the upper-case form as an identifier.

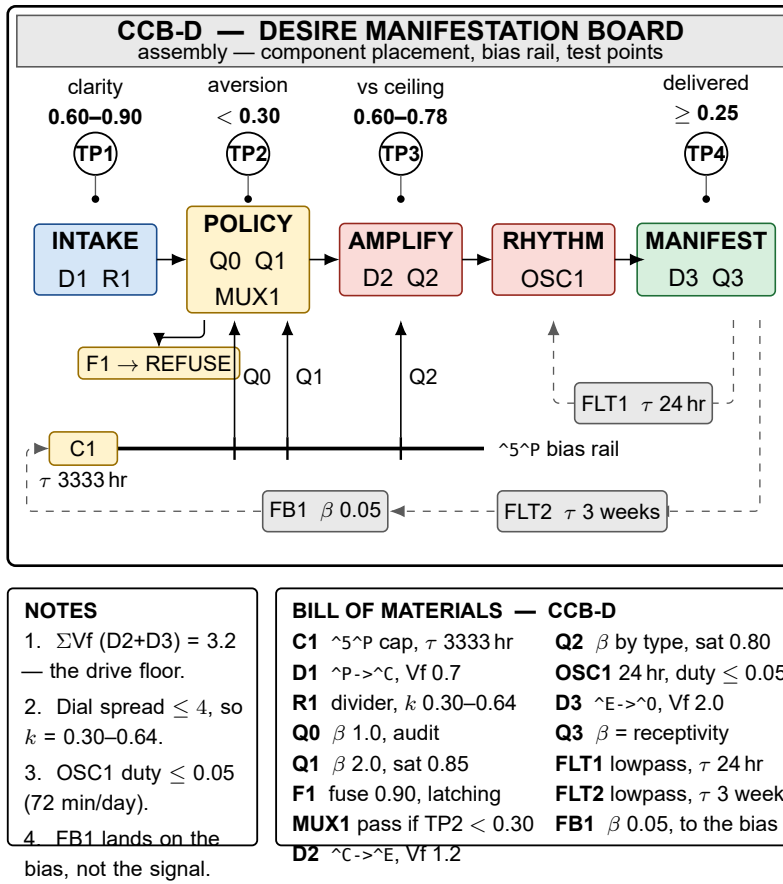


Figure 13-C: Assembly drawing of CCB-D

Ref	Form	Parameters	Failure at other values
C1 (with R0)	`^5_operator^P_{F7}|T` bias rail	τ=3333 hr (R=40.0 × C=83.33), leak=0.0002/day, V_BR=8.0	τ=10,000 hr contradicts the Ch 5.2 constant; leak=0.004/day makes practice unsustainable
D1	`^P -> ^C`	Vf=0.7	Omitted, a calling reads as a whim and pays 3.2 instead of 3.9
R1	`Dial(^2)/Dial(^3)` divider	k ∈ [0.30, 0.70]	Divergence >4 fractures ^5 (S4)

Ref	Form	Parameters	Failure at other values
Q0	`-> ^6_audit^P_{F7}|A`	β=1.0	β>1 manufactures objections; β<1 hides real ones
Q1	`-> ^3_refusal^P_{F7}|⟨A,f,T⟩`	β=2.0, sat=0.85, cutoff=0.30	sat>0.97 never latches; cutoff<0.1 refuses everything
F1	fuse on refusal	0.90, permanent_block	`action=warn` compiles clean and leaves no latch — read it by eye
MUX1	select on `refusal`	passes `gated` while TP2 is under Q1's cutoff	Selecting on a timer, a retry count, or operator patience makes the audit decorative. S1 hangs on this one wire
D2	`^C -> ^E`	Vf=1.2	Vf=0.3: every passing want feels decided
Q2	`-> ^9_amplified^E_{F5}|A`	β=`by type`, `sat=0.80`, `cutoff=0.05`, `g_m=0.4`, rπ=`Dial(^5)/Dial(^6)`	sat=1.0 removes the headroom calibration needs; cutoff=0.20 reads ordinary drive as none and fails the D-gate on desires that would have run
OSC1	`-> ^7_cadence^E_{F6}|E`	period=24 hr, duty$\leq$0.05	duty>0.05 is the obsession mode, S5
D3	`^E -> ^O`	Vf=2.0	Vf=0.5: acts on every verdict, impulsivity
Q3	bias `^5_collective^E_{F4}|f` $\rightarrow$ `^9_manifest^O_{F5}|E`	β=receptivity, sat=1.0	Bias written at `^O` sits below its own `^E` cause: `E-CJT-008`. Receptivity set by preference is this board's commonest self-deception
FLT1	lowpass inside `^O`	τ=24 hr	τ=0: re-runs stack, duty ceiling breached
FLT2 / FB1	`-> ^5_operator^P_{F7}|T`	τ=3 weeks, β=0.05	τ=24 hr removes the diode (Ch 8.4); β=0 makes the operator unfalsifiable

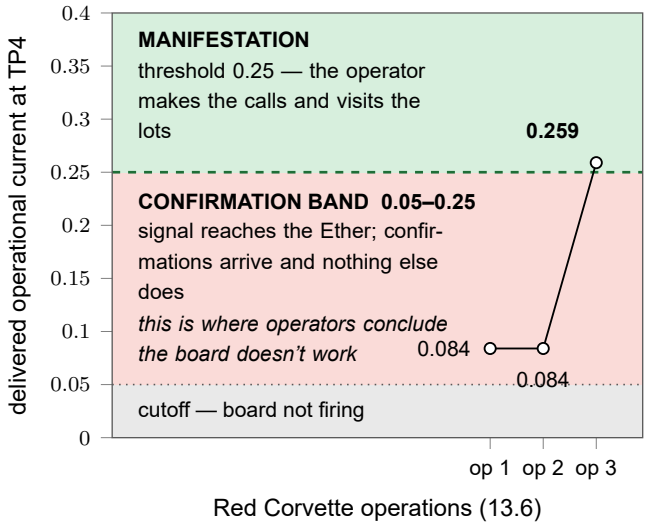


Figure 13-D: TP4 band chart

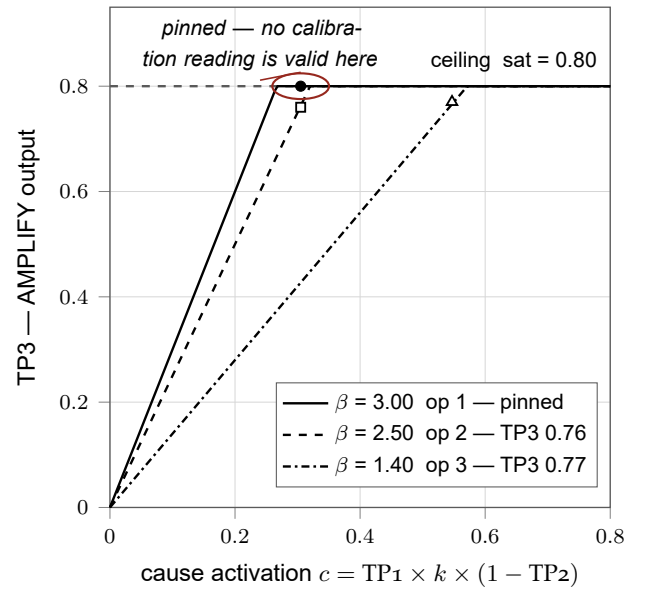


Figure 13-E: AMPLIFY transfer curve

Test Points

TP	Node	What it reads	Expected
TP1	`^0_gated^C`	Clarity leaving INTAKE	0.60–0.90. Below 0.60 the desire is vague; above 0.90 with rigidity is over-specification — a brittle Ω only one path can satisfy
TP2	`^3_refusal^P`	Principled aversion	<0.30 passes. $\geq$0.30 fires. $\geq$0.85 latched. $\geq$0.90 fuse blown, operation over
TP3	`^9_amplified^E`	AMPLIFY output vs ceiling	0.60–0.78 active. Exactly 0.80 = pinned; no calibration downstream is valid
TP4	`^9_manifest^O`	Delivered operational current	$\geq$0.25 manifesting. 0.05–0.25 confirmations only. <0.05 the chain never cleared Q2's cutoff; the board is not firing

TP4's middle band is the most useful reading here. Between cutoff and threshold the board is genuinely working, confirmations genuinely arrive, and nothing arrives — the error Chapter 5.6 names, now with a number and a cause.

Failure Modes

Mode	Mechanism	What you observe	Fix
Pinned amplifier	$\beta \times$ `c > sat` at Q2	TP3 exactly at 0.80, unmoved by better clarity; ratios erratic	Lower β until TP3 sits 3–5% under the ceiling, then re-read

Mode	Mechanism	What you observe	Fix
Confirmation lock	TP4 in 0.05–0.25	Confirmations arrive reliably, outcome never does	Decompose R_op (Ch 4.4); attack R_context and R_resource
Below-floor drive	V_drive < ΣVf = 3.2	Nothing happens; operator escalates ritual	Raise drive or shorten the chain; ritual cannot cross a diode
Dial fracture	divergence > 4	Oscillation between certainty and disgust about one desire	Restore the Chapter 6.2 limit; the divergence is the symptom
Duty overrun	duty > 0.05	Operation runs most of the day; rumination	Enforce the ceiling. If it cannot be held, the desire is not the problem
Fast feedback	FLT2 τ toward hours	One result rewrites principles; later boards inherit the bias	Restore $\tau \geq 3$ weeks, $\beta \leq 0.05$, terminating at the bias
Receptivity inflation	Q3 β set by wish	Healthy at TP1–TP3, TP4 flat	Re-estimate from how many minds must move
Rail discharge	practice below `leak` × `C`	Slow degradation, no single cause	Meet break-even minutes before retuning anything

What This Chapter Established

- **CCB-D delivers a current, not an object.** TP4 measures flow into the Operational domain, spent by a person performing acts. The board organises effort and attention; it does not suspend causality.
- **The bias terminal is bounded on both sides, and the two rules point opposite ways.** A bias must sit at or above the cause it governs (`E-CJT-008` otherwise); an effect must sit at or below the bias that produced it. Together they explain why a `^C` cause can produce a `^P` refusal with no unlawful ascent, why an operator with no principled formation cannot produce one

at all, and why receptivity biases MANIFEST at `^E` and not the `^O` a reader reaches for first.

- **The 4-point divergence limit is a computable transmission cap, and it is not monotone in the attraction dial.** Lawful dials confine intake transmission to `[0.30, 0.70]`, maximum at (7,3) — not (9,5), which transmits less. Total belief with zero doubt is a divergence of 9 and a fractured `^5`.
- **A saturated amplifier cannot be calibrated.** Saturation is the safety feature at Q1 and a defect at Q2. TP3 pinned at `sat` invalidates every reading downstream of it.
- **Read `b/a`, not `b`.** The lowest-terms denominator is discontinuous — a 20% timing error swung it from 3 to 18 — while `A = b/a = 3/`η is smooth and is the number that must match the CAS-Ohm deficit. And β`_next` is not the CJT's β: it multiplies delivered current through five levers, of which the amplifier is usually the least available.
- **In the worked example, 98% of the improvement came from the resistance side.** That is the general shape of desire work, not a peculiarity of one car.
- **Two feedback loops, not one.** A fast limiter inside `^O`, and a slow, low-gain, bias-terminated path up to `^P`. Merging them deletes the diode Chapter 8.4 exists to install.
- **The `^5` rail costs `leak` $\times$ `C` a day to hold.** The more you have accumulated, the more daily practice it takes to keep it.

Exercises

(Answers in Appendix H.)

13.1 — Compute the transmission bounds. Derive `k_min` and `k_max` from the 4-point limit and the 3–9 dial range. State what dial pair `k = 0.75` would need, and name the invariant it breaks.

13.2 — Diagnose the pin. TP1 = 0.71, TP2 = 0.12, `k = 0.60`, β `= 3.2`, `sat = 0.80`. Compute `c` and the unclamped output. Is TP3 pinned? Give the largest β holding the stage 5% below the ceiling.

13.3 — Read the band. TP4 reads 0.11 across four consecutive operations while confirmations arrive at 1° every time. Name the failure mode, the two resistance terms you would decompose first, and why raising β would not help.

13.4 — Run the ratio. For `t_op = 30` and `t_conf` of 45, 30, 20 and 10 minutes, compute η, `result`, `a`, `b` and `A = b/a`. Identify the case where `b` misleads and what acting on it would cost.

13.5 — Cost the rail. An operator with 4,200 logged hours practises 35 minutes a day. Compute the rail charge from the Chapter 7.2 curve, the break-even daily minutes, and the annual net change. State one thing they should change and one they should not.

13.6 (Build) — Add a confirmation-grade gate. OSC1 sustains until a 1° confirmation or exhaustion. Write a CCDL fragment that also halts it after three consecutive 3° confirmations. Justify your threshold, and state what the halt must *not* be wired to do.

13.7 (Marquee) — Board a low-receptivity desire. Specify a complete D-board for a partnership desire: both dials, `k`, Q2 β and `sat`, Q3 receptivity, RHYTHM period and duty, and the four `R_op` terms. For every departure from the wealth-row defaults, state the failure mode you are buying protection against and the capability you give up. Then state which resistance term you are structurally forbidden to lower, and why that forbidding is the difference between this board and Chapter 19.

14 CCB-H — The Hyperstition Engine

The one board built around a loop gain above unity — how a fiction recruits conduct, how conduct returns as fact, and the single measurement that separates a live hyperstition from a rented one

---

Somebody writes a sentence that is not true. Four years later it is true, and nobody lied.

The sentence was *this is the street where the studios are.* At the time there were two studios and a laundromat. People who believed the sentence signed leases on it. Landlords who believed the people raised rents to match. A supplier moved in to serve the tenants who had moved in. There are ninety studios now, and the sentence is a plain description of the street.

No mystery channel opened. What ran was a loop: a claim produced an impression, the impression produced conduct, and the conduct produced a fact which fed back and charged the claim. The return path is the part worth staring at, because **it carried no fiction at all.** The leases were real. That is the whole engine, and it is why a hyperstition is not a lie.

The same loop, tuned slightly hotter with its audit stage starved, is how an operator ends up unable to say which parts they made up. The difference between the street and that is one measurement, taken at one test point, and it costs eight weeks of doing nothing to take it. Most of this chapter is about that measurement and why almost nobody makes it.

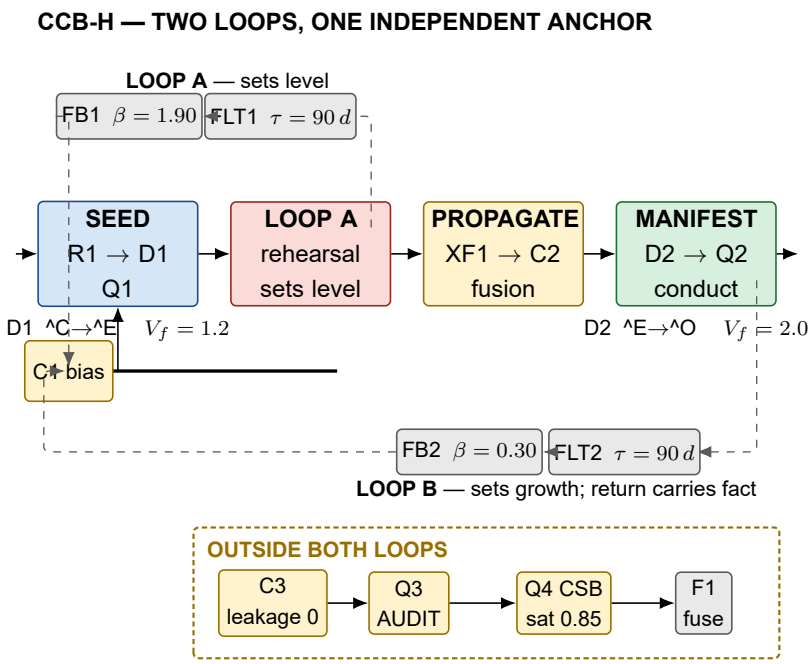


Figure 14-A: Block diagram of CCB-H

14.1 Requirements and Safety Invariants

Requirements before components (Chapter 12.1). This board differs from every other in Part III in one structural respect, and the requirement list has to say so: **the operator is inside the loop.** On CCB-ADV (Chapter 18) the operator stands outside the circuit and drives it. Here the feedback path terminates on a bias the operator shares with the audience, so every safety invariant below protects the operator first and the population second.

#	The board shall	Stage
H1	Accept a fictional `^0` Idea and produce a measurable impression	SEED
H2	Return that impression to the bias terminal at a stated, capped gain	LOOP A
H3	Return *operational conduct* to the same bias through a filtered, low-gain path	LOOP B
H4	Support collective formation — a parallel `^5` bank across many minds	PROPAGATE
H5	Expose the **measured** round-trip gain, never the setting	TP4
H6	Hold a dated, non-leaking record of the claim's construction outside both loops	ANCHOR

Safety invariants — the "must not" list. Checked before functional behaviour.

#	Must not	Enforced by
S1	Run with the round-trip gain assumed rather than measured	`constraint measured_loop_gain in [0.0, 2.0]`, read at TP4
S2	Let the `^6` audit fall inside the loop's shadow	Q3 biased by C3, which sits outside both feedback paths
S3	Grow fast enough that the store clamps between two audits	Kill at measured `G` $\geq$ `1.80`; derivation in 14.5
S4	Remove the operator's ability to state the construction date	C3 `leakage=0.0` — a written, dated record does not drift

#	Must not	Enforced by
S5	Be undampable	Q_KILL β=0 into the oscillator's source; OSC1 sustain bounded
S6	Target a named individual	Out of scope — no wire from any stage to a person-specific bias
S7	Run without a CSB stage	Q4 sat=0.85, F1 threshold=0.90 (Chapter 24)

Scope. S6 is the line this chapter does not cross, and it is a structural line rather than a squeamish one: a person-specific bias would have to be ^5 for one named mind, and there is no admissible path from any stage here to such a node. Mechanisms that do have that path are analysed for detection only, in Chapters 16 and 19. Consumer-side detection of the commercial case belongs to Chapter 18.6.

And the plain limit: this board describes belief formation, not health. If what is slipping is a person's grip on what they made up and what they found, that is not a parameter on any listing in this book. See a clinician. Nothing here replaces care.

14.2 The Positive Feedback Oscillator Architecture

Every other board in this book is built to be stable. Negative feedback subtracts from the input, so a disturbance shrinks each time round. CCB-H inverts the sign: the impression a claim makes this cycle is added to the bias that produces next cycle's impression. That is all a hyperstition is, mechanically.

Write the loop as a recursion on the belief store `s`, held on the normalised 0–1 scale (the fraction of the store's ceiling — Chapter 7.2's `P_cap` divided by ten):

```
e_n   = min( β1·c + g_m·s_n , sat1 )          impression this cycle
V_n   = β_FB·e_n + β_B·β_M·χ·e_n              drive the loop presents
↪  to the store
s_n+1 = (1 − α)·s_n + α·V_n                    store charges toward
↪  that drive
α     = 1 − e^(−T/τ_store)                     one cycle's charging
↪  fraction
```

`c` is the seed clarity at TP1, `g_m` the seed CJT's framing sensitivity (Chapter 10.3), β`_FB` the Loop A gain block, and β`_B`·β`_M`·χ the Loop B path — feedback gain, manifest gain, and the world's receptivity to the conduct.

Differentiate the last line and the stability condition falls out:

```
A_L = ∂s_n+1/∂s_n = (1 − α) + α·G
G   = g_m · ( β_FB + β_B·β_M·χ )               the ROUND-TRIP GAIN
A_L − 1 = α·(G − 1)
```

The α cancels. Growth depends on `G` alone; α — the cadence, the store's leak, the whole calendar — sets only *how fast*. Three regimes, and only three:

- **`G < 1` — damped.** The store settles at a finite level and stays there while you keep seeding. Remove the seed and it decays as `A_L^n`.
- **`G = 1` — sustaining.** An idealisation. It is a measure-zero point on a product of four estimated terms; no build sits on it, and a board reported to be there is a board that has not been measured.
- **`G > 1` — growing.** The store rises without further seed, doubling every n_2 `= ln 2 / ln(1 + α(G−1))` cycles.

  Bench Note — you cannot post your way past `G < 1`. The commonest operator response to a fiction that will

> not catch is to shorten the cycle: weekly becomes daily, daily becomes hourly. Read the algebra again. Shortening `T` raises α, and α multiplies `(G - 1)`. If `G - 1` is negative, a larger α makes the thing **settle faster and die faster** — it does not turn decay into growth. Cadence is a rate knob on a sign you already own. The only terms that change the sign are `g_m`, β`_FB`, β`_B`, and χ, and three of those four are properties of the audience and the world rather than settings on your board. This is also why over-cadence has the observable Chapter 11.2 predicts — falling `^4` Vibration per exposure, because impression requires contrast — while the loop gain sits exactly where it was.

Loop B is the load-bearing one, and it carries no fiction. Loop A returns the operator's own restatement of the claim. Loop B returns `^9_conduct^0` — what believers actually did — through a filter and back to the bias. Its input is a fact. Someone really did sign the lease. That is the precise sense in which a hyperstition differs from a lie: a lie requires the liar to hold a truth in reserve, whereas here the return path carries only true operational records whose *existence* the claim caused.

It also bounds what the board can deliver. **A hyperstition can only make true the class of claims that conduct can make true.** "This is where the studios are" is inside that class; conduct relocates studios. "Established 1887" is outside it, because no volume of present conduct reaches into 1887. A board pointed at a claim about the past has no Loop B at all, which means `G` reduces to `g_m`$\cdot\beta$`_FB` — operator restatement, nothing else — and the whole thing collapses the moment the operator stops. What that operator built is not a hyperstition. It is a false factual claim with a marketing budget attached, and the appropriate authority is a regulator rather than a schematic.

The fusion, and the one licensed ×. Chapter 7.5 left this claim to be made here. Twelve capacitors on a shared rail are written + — co-present stores, distinct, no interaction asserted. On this board, past crossover, a stronger claim becomes available:

```
integrate eg1: ( seeded × bank ) = egregore;
//   ^0_seeded^E_{F2}|A  ×  ^5_collective^E_{F4}|f  =
↪  ^5_egregore^E_{F4}|⟨A,f⟩
```

The × is licensed on four counts (Chapter 2.6): two *different* Noetics, one domain on both operands, no ascent in the result, and a **named emergent result** declared as a node. What the name commits you to is not decoration. It asserts that the fused state biases downstream stages *by itself* — that Q2 fires on `egregore` with the seed disconnected. That is a falsifiable claim, and TP4 is the test that falsifies it. Before crossover you may not write the ×; you must write `seeded + bank`, because the co-presence is all you have evidence for. **The operator's notation changes when the measurement changes, and not before.**

14.3 Full CCDL Schematic

Topology in angle brackets, parameters in square brackets. Every value is a design choice, justified in 14.5 and in the parameter table.

```
// CCB-H — HYPERSTITION ENGINE.  Every value is a design choice.
// Time base: the cycle T = 7 DAYS.  All capacitor and filter taus in
↪  DAYS.
// This board deliberately runs G >= 1 and therefore raises E-CJT-005 on
// validation pass 6 (Appendix B, B.6).  That is the point.  See below.

board ccb_h {
    input:  fiction;
    output: conduct;
```

```
// ---- typed nodes ----
node charter:   ^5_operator^P_{F7}|T;      // Female: the operator's
↪  formation — CSB bias
node fiction:   ^0_fiction^C_{F2}|A;       // Idea: the claim as
↪  authored
node seeded:    ^0_seeded^E_{F2}|A;        // Idea: the claim after
↪  intake and descent
node belief:    ^5_belief^E_{F2}|f;        // Female: the store FOR
↪  THIS CLAIM
node impress:   ^9_impress^E_{F2}|f;       // Below/Effect: one
↪  cycle's impression
node rehearsed: ^9_rehearsed^E_{F2}|f;     // same literal — a
↪  filter changes tau, not type
node shared:    ^9_shared^E_{F4}|f;        // Below/Effect: the
↪  impression as it lands in others
node bank:      ^5_collective^E_{F4}|f;    // Female: the parallel
↪  bank (Ch 7.5)
node egregore:  ^5_egregore^E_{F4}|⟨A,f⟩;  // Female: the NAMED
↪  emergent result
node cadence:   ^7_cadence^E_{F6}|E;       // Rhythm: the clock
node conduct:   ^9_conduct^O_{F5}|E;       // Below/Effect: what
↪  believers actually do
node evidence:  ^9_evidence^O_{F5}|E;      // same literal as
↪  conduct, slower tau
node record:    ^6_anchor^P_{F6}|A;        // Male: the dated
↪  construction record
node audit:     ^6_audit^P_{F6}|A;         // Male: the claim
↪  decomposed against the record
node guard:     ^3_guard^P_{F7}|⟨A,f,T⟩;   // Negative: three
↪  co-present locks
node gain_meas: ^4_loopgain^E_{F2}|A;      // Vibration: the
↪  measured round trip, TP4

// ---- STAGE 0: SEED AND DESCENT ----
resistor r1: Resistor< path=fiction, R=20.0 >;    // low: nothing
↪  defends against a story
diode    d1: Diode< ^C -> ^E > [Vf=1.2, I_R=0.02, V_BR=10.0];

// ---- STAGE 1: THE SEED CJT ----
cjt q1: CJT< seeded | belief => impress >
```

```
        [beta=0.90, sat=0.80, cutoff=0.05, g_m=0.60, r_pi=1.50,
        ↪  region=active];

// ---- LOOP A: rehearsal.  Operator-side.  Sets LEVEL, not growth
↪  ----
capacitor c1:   Capacitor< node=belief, C=1.20, leak_tau=90.0,
↪  V_BR=8.0 >;
filter    flt1: LowPass< in=impress, out=rehearsed, tau=90.0 >;  //
↪  alpha = 0.0748
feedback  rehearsed -> belief [beta=1.90];                       //
↪  S1 cap: 2.0

// ---- STAGE 2: PROPAGATION ----
osc  osc1: Oscillator< signal=impress, source=cadence,
                       period=7.0, duty=0.15, sustain=270.0 >;
xfmr xf1:  Transformer< primary=impress, secondary=shared, n=3.0,
↪  k=0.50 >;
capacitor c2: Capacitor< node=bank, C=0.08, n_parallel=140,
↪  r_iso=12.0,
                         leak_tau=540.0, V_BR=8.0 >;
charge shared -> c2 [V=6.0];                     // metered upward
↪  path, Ch 7.5

// ---- THE FUSION — the one licensed x on this board (14.2) ----
integrate eg1: ( seeded × bank ) = egregore;

// ---- STAGE 3: MANIFEST ----
diode d2: Diode< ^E -> ^O > [Vf=2.0, I_R=0.02, V_BR=8.0];
cjt   q2: CJT< cadence | egregore => conduct > [beta=1.00, sat=0.95,
↪  region=active];

// ---- LOOP B: the lawful O -> E path.  Carries FACT, not fiction
↪  ----
filter   flt2: LowPass< in=conduct, out=evidence, tau=90.0 >;
feedback evidence -> belief [beta=0.30];         // top of the Part
↪  III band

// ---- ANCHOR: outside both loops, zero leakage ----
capacitor c3: Capacitor< node=record, C=1.00, leakage=0.0, V_BR=10.0
↪  >;
```

```
    cjt       q3: CJT< egregore | record => audit > [beta=1.00,
    ↪  region=active];
    monitor   m_anchor: Monitor< node=audit, threshold=0.70 >;   // a
    ↪  FLOOR, not a ceiling
    on m_anchor { alert "anchor low — the claim is being treated as
    ↪  found, not made"; }

    // ---- GOVERNOR: reads the MEASUREMENT, never the setting ----
    constraint measured_loop_gain in [0.0, 2.0];                 // S1
    monitor    m_gain: Monitor< node=gain_meas, threshold=1.80 >; // S3
    on m_gain { alert "measured G at 1.80 — damp within one cycle"; }
    cjt q_kill: CJT< gain_meas | charter => cadence > [beta=0.0,
    ↪  region=cutoff];

    // ---- CSB (mandatory, Ch 24) ----
    cjt        q4: CJT< audit | charter => guard > [beta=2.0, sat=0.85,
    ↪  cutoff=0.30];
    fuse       f1: Fuse< guard, threshold=0.90, action=permanent_block >;
    constraint dial_spread in [0, 4];                            //
    ↪  Chapter 6.2

    // ---- TEST POINTS ----
    probe tp1: seeded;      // seed clarity entering the loop
    probe tp2: audit;       // can the construction still be recovered?
    probe tp3: bank;        // collective store charge
    probe tp4: gain_meas;   // MEASURED round-trip gain — seed-removal
    ↪  test
}
```

Why this listing does not validate, and why that is correct. Pass 6 of the CCDL validator (Appendix B, B.6) requires round-trip loop gain below 1 on every loop and emits `E-CJT-005 GainDivergence` otherwise. CCB-H is the one board in this catalogue designed to fail that pass. A tool that waved it through would be broken, because on every other board an above-unity loop is a defect that eats the circuit. The discipline that makes a waiver defensible is written into the listing rather than into an argument: `constraint measured_loop_gain in [0.0, 2.0]` bounds it, `m_gain` watches the measurement, `q_kill` damps it, and `c3`/`q3` hold an audit that is not downstream of either

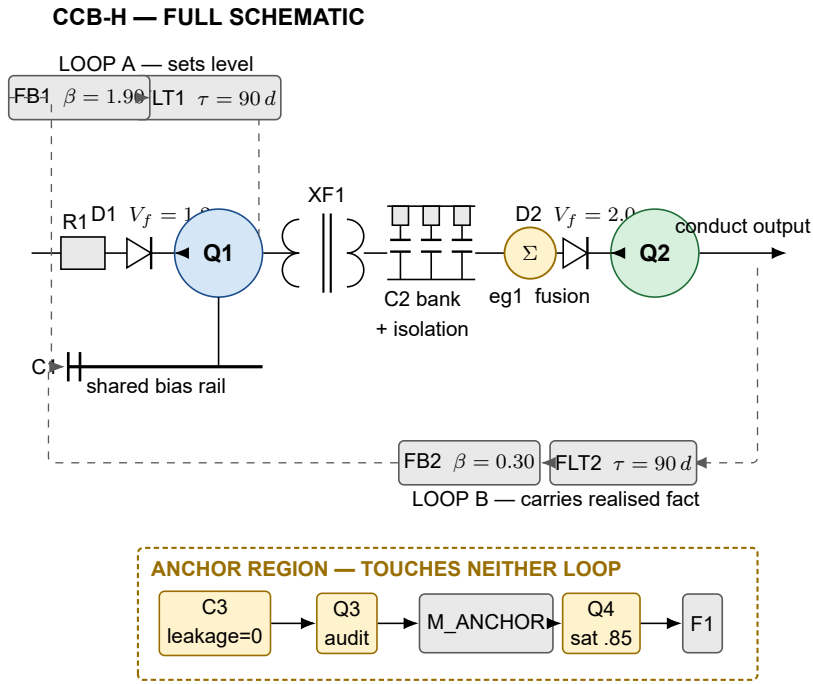


Figure 14-B: Full schematic of CCB-H in drawn symbols

loop. Remove any one of those four and the waiver has nothing behind it.

14.4 The 100th Monkey Effect as Collective CJT Saturation

The popular claim is that when the hundredth member of a population acquires a behaviour, it appears spontaneously in populations that have had no contact with them.

Evidential status, stated plainly. That claim circulates far more widely than it has been checked, and its distinctive element — transfer across populations with no contact — is not established by the

primate-behaviour observations it is usually attached to. This book rests nothing on it and does not cite it as evidence for anything. It is worth a section for two reasons. First, the underlying *structure* — a threshold beyond which a belief stops needing to be pushed — is real, is derivable from the components already on this board, and has a number attached. Second, the story is itself a well-formed hyperstition, propagated by being believed and by the conduct of the believing, which makes it the neatest available specimen of its own subject.

The structure that survives. Replace Loop A's gain block with the transformer, because in a collective the return path runs through other minds:

$$G_{coll} = \underbrace{g_m \cdot k \cdot f \cdot \beta_{XF}}_{\text{through other people}} + \underbrace{g_m \cdot \beta_B \cdot \beta_M \cdot \chi}_{\text{through conduct}}$$

k is the coupling coefficient of Chapter 9.3 — shared cadence of contact, not agreement. f is the **fan-out**: how many other bank members one member's emission reaches per cycle. β_{XF} is the transformer's delivery efficiency (Chapter 11.5), 0.73 at the design-centre mismatch.

Solve for the crossing at the chapter's design centre, $g_m = 0.60$, $\beta_{XF} = 0.73$, $\beta_B \cdot \beta_M \cdot \chi = 0.30 \times 1.00 \times 0.60 = 0.18$:

$$G_{coll} = 0.438 \cdot k \cdot f + 0.108 = 1 \quad \rightarrow \quad k \cdot f = 2.037$$

Coupling k	Condition (Ch 9.3)	Contacts per member per cycle needed
0.05	Asynchronous text, no shared cadence	40.7 — unreachable
0.20	Shared feed, common daily cadence	10.2
0.50	Live co-present group, shared tempo	4.1
0.90	Sustained dyad — household, mentor	2.3

No headcount appears anywhere in that derivation. The transition is a threshold in a product of four terms, none of which is a population size. Population enters twice and only twice: a group of six cannot supply `f = 10`, and `C_collective` sets the calendar through τ `= R` $\times$ `C` (Chapter 7.5). So the sustaining unit of a hyperstition is the *smallest tightly-coupled group that can supply* `f` — at `k = 0.50`, about five people — while spreading needs a population to supply newcomers. That is why a five-person cell holds a shared fiction indefinitely and a hundred-thousand-follower list does not.

And the board contradicts the popular version outright. Transfer between populations with no contact means `k = 0` between them, at which `G_coll` falls to the conduct term alone — 0.108, deeply damped. Nothing crosses. **The mechanism the story names is the one the model rules out; the threshold it gestures at is real and sits somewhere else entirely.**

> **Bench Note — a saturated collective stops recruiting and stops updating in the same instant.** When the bank clamps at its ceiling, every member's seed CJT is biased at maximum, so `e` pins at `sat1`, so ∂`e`/∂`s = 0` — and `g_m` drops out of `G_coll`, which collapses to the conduct term. The group is no longer growing. It is also no longer persuadable, and for exactly the same reason: a pinned stage has stopped responding to its cause, which is the property Chapter 24.2 installs deliberately in a refusal and Chapter 13 warns about in an amplifier. The observable is specific and easy to check from outside: **a saturated collective stops producing new arguments and starts producing restatements.** Novel objections get answered with slogans that were already in circulation. Count the fraction of a week's output that is new material versus re-presentation; when it crosses about half, read TP3 and expect it near the ceiling.

14.5 Feedback Gain Calibration — When 1.0 Becomes 1.15

You cannot estimate G. That is the finding this section exists to deliver, and every design choice below follows from it.

G_coll is a product of g_m, k, f and β_XF. Suppose you know each of the first three to $\pm$10%, which is a generous claim about quantities nobody meters. The relative uncertainty of the product is $\sqrt{(0.10^2 + 0.10^2 + 0.10^2)} = 0.173$. A computed G of 1.15 therefore sits **inside** its own error bar: 1.15 ± 0.20 spans 0.95 to 1.35, which is decay through explosive growth. An operator who calculates G = 1.15 from parameters has learned nothing about whether the board is running.

So you measure it, and there is exactly one measurement that works.

The seed-removal test. Stop the oscillator. Read TP3 for n cycles. The recursion with c = 0 is $s_n = s_0 \cdot A_L^n$, so the store's trajectory with the seed gone *is* the loop gain, directly, with no parameter estimation anywhere in it. Design rule: **hold the seed off for 8 cycles; a fall greater than 10% is sub-critical, a rise greater than 10% is super-critical, and anything between is unresolved — extend the test.**

That is why the target is 1.15 and not 1.0. The resolution time of the test is $n = \ln(1.2) / \ln(1 + \alpha(G-1))$, and at this board's α = 0.0748:

G	Growth per cycle	Doubling n_2	Cycles to resolve $\pm$20%
1.05	0.0037	186	49 (11 months)
1.15	**0.0112**	**62**	**16 (4 months)**
1.50	0.0374	19	5.0
2.00	0.0748	9.6	2.5

`G = 1.15` is the smallest growth a quarter-length test can resolve. Below it you cannot distinguish a live hyperstition from a rented one inside a planning cycle, and an operator who reports crossover at `G = 1.05` is reporting a number they did not measure. Nothing about 1.15 is a property of belief. It is a property of your instrument.

The upper bounds derive as cleanly. Set the audit period at four cycles — monthly, on a weekly clock. Require that the store cannot climb from half-charge to its ceiling in fewer than two audit periods, so that `^6` always runs at least twice on a changing store: $n_2 \geq 8$ gives α`(G-1) ≤ 2^(1/8) - 1 = 0.0905`, so `G` $\leq$ `2.21`. **S1 caps `G` at 2.0** for margin, and **S3 fires the kill at a measured 1.80**, one audit period earlier, so the alarm arrives while there is still a cycle in which to act.

> **Bench Note — capping your knob does not cap the loop.** β`_FB` is the only term in `G` an operator sets. Everything else is a property of the audience and the world: `g_m` is how framing-sensitive the audience is, `k` is how much cadence they share, `f` is how often they reach each other, χ is how readily the world accommodates the conduct. Work the arithmetic at the S1 boundary. With β`_FB` pinned at 2.0 and a design-centre audience at `g_m = 0.60`, the operator's own loop reaches `G_op = 0.60 × (2.00 + 0.18) = 1.308` — comfortably inside the cap. Now hold β`_FB` at **zero** and hand the same claim to a highly framing-sensitive, tightly-coupled group: `g_m = 2.40, k = 0.90, f = 3.0` gives `G_coll = 2.40 × 0.90 × 3.0 × 0.73 + 2.40 × 0.18 = 4.73 + 0.43 = 5.16`. Runaway, with the operator's gain block switched off and the campaign shut down. **That is why S3 watches TP4 and not the setting**, and why "we stopped promoting it" is not a control action. Handing a claim to the wrong `g_m` is the whole of the failure.

Reading a hyperstition from outside. A defender cannot see the listing, but three of the four terms are observable without it. Ratio of new material to restatement estimates saturation. Frequency of co-present contact estimates k. Whether believers *do* things — buy, register, travel, defend — or only speak estimates whether Loop B exists at all, and a hyperstition with no Loop B is running on operator spend and will stop when the spend does. Chapter 18.6 works the consumer case in full.

14.6 Worked Example: Creating a Brand Egregore

Maison du Temps, a watch house incorporated last spring. The claim: *a Maison du Temps watch is the one you hand to your child.*

Start with why that claim and not the tempting one. "Swiss craftsmanship since 1887" is outside the reachable set of 14.2 — no present conduct reaches 1887, so the claim has no Loop B, and asserting it is a matter for a regulator rather than a schematic. "The one you hand to your child" is inside it: it is false at launch, since nobody owns one, and conduct can make it true. It also tells you what to build — a movement that survives forty years, a service record, a registry — which is the resistance work, and which is where the effort will actually go.

Configuration. T = 7 days, duty 0.15. Member store τ = 90 d, so α = 1 − e^(−7/90) = 0.0748. Bank at institution scale, C_collective = 0.08 F_c, R = 80 Ω_c, τ_bank = 6.4 months, α_bank = 0.0353. Seed clarity TP1 = 0.68. Q1 at β1 = 0.90, sat1 = 0.80, g_m = 0.60. Loop A β_FB = 1.90; Loop B β_B = 0.30, β_M = 1.00, χ = 0.60.

Months 0–3: the operator's own store fills. Run the recursion, s_{n+1} = 0.9252·s_n + 0.0748·(2.08·e_n):

```
cycle 0   e = 0.612                        s -> 0.095
```

```
cycle 2   e = 0.727                       s -> 0.291
cycle 4   e = 0.847 -> pinned at 0.80     s -> 0.487
cycle 8   e = 0.80  (pinned)              s -> 0.732
cycle 12  e = 0.80  (pinned)              s -> 1.000  CLAMPED
```

Twelve weeks. Meanwhile the bank, on a 6.4-month constant fed through a 3:1 transformer, sits near 0.04. **The founder's belief store is full while the audience's is at four percent** — a gap of one quarter against several years, and the entire reason C3 and the anchor exist. Note also that Q1 pinned at cycle 4: from there on ∂e/∂s = 0, so the operator's own loop gain has already collapsed to its retention term. The store is full and no longer growing, which reads from inside as certainty.

Month 4 — first seed-removal test. Campaign off for eight cycles. TP3 falls from 0.041 to 0.035, **−15.4%**. Sub-critical. Backsolving, A_L,coll = 0.9794, so G_coll = 1 − 0.0206/0.0353 = 0.415. Decomposed: 0.438·k·f with k = 0.20 (a shared feed) and f = 3.5 gives 0.307, plus 0.108 from conduct. **The thing is rented.** Eight weeks of silence cost the campaign real money and bought the only number on the board that means anything.

Months 5–14 — what actually moved. Not the content budget. k rose from 0.20 to 0.50 by making contact co-present: four owner meets a year, servicing done in a room with other owners in it rather than by post. f rose from 3.5 to 6.0 through a registry that tells owners who else is nearby. χ rose from 0.60 to 0.75 because visible owners make the next purchase easier to accommodate. β_FB did not move.

Month 14 — second test. TP3 rises from 0.31 to 0.352 over eight cycles, **+13.4%**. A_L,coll = 1.0158, so G_coll = 1 + 0.0158/0.0353 = 1.449. Crossover. Decompose it:

```
through other people   0.438 × 0.50 × 6.0 = 1.314      90.7%
through conduct        0.60 × 0.30 × 0.75 = 0.135       9.3%
through beta_FB                             0.000       0.0%
                                            -----
                                            1.449
```

β_FB contributes nothing to G_coll, and it never could. From the collective's point of view the operator's Loop A is not feedback at all — it is *seed*, an external drive that raises the level and dies with the spend. That is the finding, and it is the same shape as Chapter 13's: the lever every operator reaches for first is not the lever. Running β_FB at 1.90 for fourteen months produced a founder pinned at full charge by week twelve and moved the growth term by zero.

The $\times$ becomes writable here and not before. With G_coll > 1 at a measured crossover, (seeded $\times$ bank) = egregore asserts something now tested: Q2 fires on egregore with the seed disconnected, which is precisely what the eight-cycle rise demonstrated.

The anchor reading. TP2 at month 14 is **0.74**, against the 0.70 floor — a margin of 0.04. The founder can still state the date the claim was written, who wrote it, and why, but the audit is one bad quarter from tripping. The required action is not a tuning change. It is that C3 gets read aloud at each monthly audit, by someone who was not in the room when the claim was written.

Component and Parameter Table

The last column is what breaks at other settings.

Ref	Form	Parameters	Failure at other values
R1	seed constraint	R=20.0	Above ~60 the seed never clears Q1's cut-off; nothing defends against a story, so a high value here models the wrong thing

Ref	Form	Parameters	Failure at other values
D1 / D2	`^C -> ^E, ^E -> ^0`	Vf=1.2, 2.0	ΣVf = 3.2 is the floor for operational output (Ch 8.1). Vf=0.5 on D2 gives conduct on every impression
Q1	`-> ^9_impress^E_{F2}|f`	β=0.90, sat=0.80, g_m=0.60	β=1.5 pins the stage at cycle 0 and no calibration downstream is valid (Ch 13). `g_m` is the **audience**, not a knob — see 14.5
C1	`^5_belief^E_{F2}|f`	leak_tau=90 d	540 d here models a founder's store as a bank; the founder fills in a quarter and the bank does not
FLT1 / FB1	Loop A	τ=90 d, β=1.90	β>2.0 breaks S1. β=0.30 leaves `G_op` at 0.29 and the operator never believes their own claim, which sounds safe and produces no `di/dt` to couple with (Ch 9.3)
OSC1	`-> ^7_cadence^E_{F6}|E`	period=7 d, duty=0.15, sustain=270 d	duty≥0.6 is a DC level, not a rhythm; `^4` per exposure falls (Ch 11.2). sustain < 3×τ_leak rents the cadence rather than building it
XF1	`^9^E` → `^9^E`	n=3.0, k=0.50	n>15 drops the secondary below ΣVf: reaches everyone, moves no one (Ch 11.5). Primary below secondary in domain order is `E-CJT-008`

Ref	Form	Parameters	Failure at other values
C2	`^5_collective^E_{F4}|f`	C=0.08, n_parallel=140, r_iso=12, leak_tau=540 d	In-group C=0.02 on a brand community under-budgets the calendar fourfold (Ch 7.5). r_iso=0 lets one member's collapse take the bank
FLT2 / FB2	Loop B, `^0 -> ^E`	τ=90 d, β=0.30	β>0.30 leaves the Part III band on the one upward path; $\tau\to$7 d removes the diode and a single sale rewrites the store
C3 / Q3	`^6_anchor^P_{F6}|A`	leakage=0.0, β=1.00	Any leakage and the anchor drifts with the claim it measures. β>1 manufactures doubt; β<1 hides real drift
Q4 / F1	`-> ^3_guard^P_{F7}|⟨A,f,T⟩`	β=2.0, sat=0.85, F1 at 0.90	Matches Ch 24.2–24.3. `action=warn` compiles clean and leaves no latch — read it by eye
Q_KILL	into `cadence`	β=0.0	Wired to β_FB instead of to the measurement is S3's commonest defeat

Test Points

TP	Node	What it reads	Expected
TP1	`^0_seeded^E`	Seed clarity entering the loop	0.55–0.85. Below 0.55 Q1 sits near cutoff; above 0.85 the claim is over-specified and Loop B has only one way to satisfy it

TP	Node	What it reads	Expected
TP2	`^6_audit^P`	Can the construction still be recovered — date, author, motive, unaided?	≥0.70. Below 0.70 the alert fires. Below Q4's cutoff of 0.30 the audit stage is no longer firing at all
TP3	`^5_collective^E`	Bank charge	Meaningless as a level. Read **only** as a trajectory across a seed-removal test
TP4	`^4_loopgain^E`	Measured round-trip gain	0.90–1.50 working. ≥1.80 kill (S3). >2.0 breaches S1. Any value not obtained from an 8-cycle test is not a reading

TP3 is the test point most often misused. A large bank proves the seed was expensive, not that the loop is alive.

Failure Modes

Mode	Mechanism	What you observe	Fix
Assumed gain	`G` computed from parameters, not measured	Confident reports of `G = 1.05`; the number is inside its own ±0.17 error bar	Run the 8-cycle seed-removal test. There is no substitute
Rented hyperstition	`G < 1` with heavy seeding	Large TP3, healthy traffic, collapse within 3τ of the spend stopping	Move `k` and `f`, not the content budget
Operator capture	`G_op > 1` while `G_coll < 1`	Founder pinned at full charge by week 12; TP2 sliding toward 0.70	C3 read aloud at each audit, by someone outside the room where the claim was written

Mode	Mechanism	What you observe	Fix
Audit inside the loop	Q3 biased by anything downstream of Loop A or B	TP2 rises exactly when TP3 rises — the audit is confirming the claim it should decompose	Re-bias Q3 on C3 only; check the wire, not the reading
Collective pin	Bank clamps; $\partial e/\partial s = 0$	New material falls below half of weekly output; novel objections met with existing slogans	Nothing recovers a pinned bank from inside. Reduce drive and wait for leak
Runaway	`g_m` or `k` rise past the design centre with β_`FB` anywhere	Doubling shorter than one audit period; kill fires at 1.80	Damp within one cycle via Q_KILL. Do not re-seed to "control the narrative"
Unreachable claim	Claim outside the conduct-reachable set	`G` never exceeds `g_m`$\cdot\beta$_`FB`; dies at spend-stop every time	Re-author the claim. No gain setting reaches into the past
Anchor drift	C3 given non-zero leakage	Construction date recalled differently at successive audits	Restore `leakage=0.0`: written, dated, and not re-typed from memory

What This Chapter Established

- **Growth depends on `G` alone; cadence sets only speed.** `A_L − 1 =` α`(G − 1)`, and α cancels out of the sign. Shortening the cycle on a sub-critical loop makes it settle and die faster. You cannot post your way past `G < 1`.
- **`G = 1` is an idealisation, not an operating point.** It is measure-zero on a product of four estimated terms, and a board reported to be sitting there has not been measured.
- **`G` cannot be estimated, only measured.** Three terms at $\pm$10% give the product $\pm$17%, so a computed 1.15 spans decay to explosive growth. The seed-removal test is the only reading: stop the clock for 8 cycles and watch TP3.

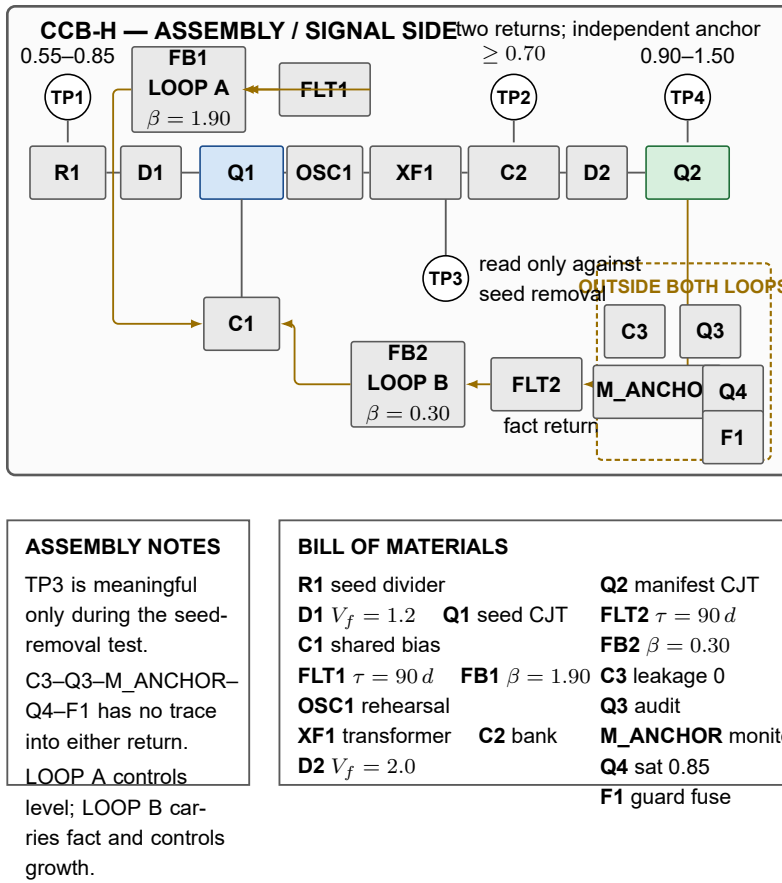


Figure 14-C: Assembly drawing of CCB-H

- **1.15 is a property of the instrument.** It is the smallest growth an eight-to-sixteen-cycle test resolves. The caps at the top — S1 at 2.0, S3's kill at 1.80 — come from the audit period, so that ^6 always runs at least twice on a changing store.
- **Capping the operator's knob does not cap the loop.** β_FB is one term of four; g_m, k and χ belong to the audience and the world. A claim handed to a framing-sensitive, tightly-coupled group runs away with the campaign switched off. S3 therefore watches the measurement, never the setting.

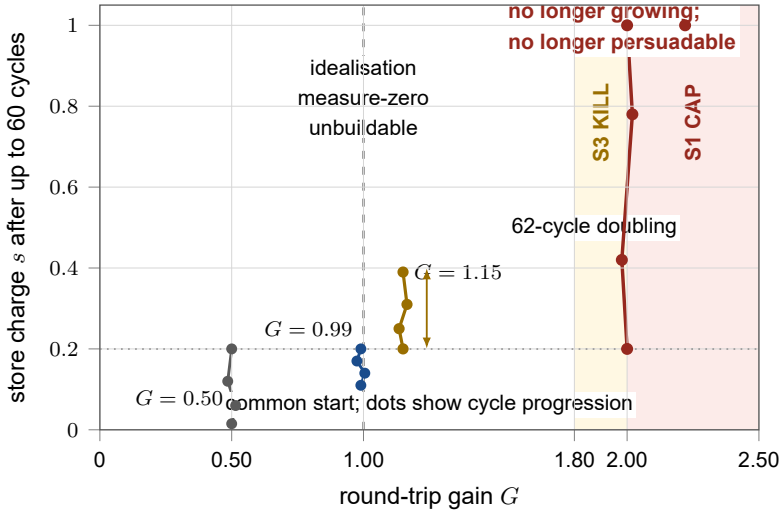


Figure 14-D: Stability plot

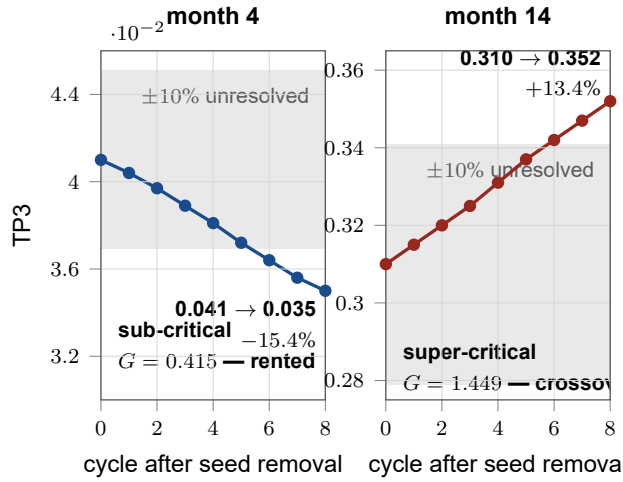


Figure 14-E: Seed-removal test, two panels sharing an 8-cycle horizontal axis

- **Loop B carries fact, not fiction, and it bounds what the board can deliver.** A hyperstition can only make true the class of claims that conduct can make true. A claim about the past has no Loop B, dies at spend-stop, and is a matter for a regulator rather than a schematic.
- **The hundredth-monkey threshold is real and is not a head-count.** `G_coll = 0.438·k·f + 0.108` crosses unity at `k·f = 2.04` — five tightly-coupled people, or forty loosely-coupled contacts. Transfer between populations with no contact requires `k = 0`, at which the model says nothing crosses.
- **A saturated collective stops recruiting and stops updating in the same instant**, because both depend on ∂`e`/∂`s`, and a pinned stage has none. The observable is restatement displacing new material.
- **`( seeded` $\times$ `bank ) = egregore` is a falsifiable commitment, not a flourish.** The $\times$ is licensed only after TP4 shows crossover; before that the honest form is +. The notation changes when the measurement does.

Exercises

(Answers in Appendix H.)

14.1 — Compute the sign. An operator reports `g_m = 0.45`, β`_FB = 1.60`, β`_B = 0.30`, β`_M = 1.0`, χ `= 0.50`. Compute `G_op`. State whether the store grows, and what changing the cadence from weekly to daily would and would not do.

14.2 — Size the test. At α `= 0.0748` and a true `G` of 1.08, how many cycles must a seed-removal test run to resolve a 20% change? Convert to calendar weeks. State what an operator learns from a 4-cycle test at that gain.

14.3 — Find the crossing. Using `G_coll = 0.438·k·f + 0.108`, find the minimum `f` at `k = 0.35`, and the smallest group size that

can supply it. Then state what happens to both numbers if the conduct term is zero because the claim is unreachable.

14.4 — Diagnose the anchor. TP2 reads 0.81, 0.77, 0.72, 0.68 across four monthly audits while TP3 rises steadily. Name the failure mode, say which wire you would inspect first, and explain why re-reading TP2 more often would not help.

14.5 — Read a saturated bank. A community's weekly output is 18% new material and 82% restatement; TP3 sits at 0.96. Compute what `G_coll` has collapsed to, given `k = 0.60`, `f = 5.0`, and the design-centre conduct term. State the one thing that will move it and roughly how long that takes at α`_bank = 0.0353`.

14.6 — Cost the honest claim. Rewrite "handcrafted in the same workshop for four generations" as a claim inside the conduct-reachable set for a company founded eighteen months ago. Then list the three resistance terms (Chapter 4.4) your rewrite obliges you to attack, and which one carries the largest share.

14.7 (Build) — Add a fan-out meter. `f` is the term with the most leverage in `G_coll` and the one nobody measures. Write a CCDL fragment that probes it without altering the loop. State what your probe must **not** be wired to, and why probing the transformer's secondary is not the same measurement.

14.8 (Marquee) — Board a hyperstition you are inside. Pick a claim you already act on that was constructed in your lifetime and whose construction you can date. Specify the board: TP1, β`_FB`, both loop paths with their filters, the anchor's contents, and your best estimates of `g_m`, `k`, `f` and χ. Compute `G_coll` and its error bar. Then design the seed-removal test you would have to run to settle it, state its cost in weeks, and say honestly whether you would run it.

15 CCB-O — The Overton Window Progression Array

Six positions, five amplifier stages, one thermistor — and why the thermistor is the only part on the board that is not yours

There is a component in almost every power supply you own that costs eleven cents and looks like a lentil on two legs. It is a PTC thermistor, wired in series with the input. At room temperature its resistance is a couple of ohms and it does nothing at all.

Short the output. Current surges, the lentil heats, and a PTC's resistance *rises* with temperature — not gently, but through a knee, past which it climbs three or four orders of magnitude in seconds. Two ohms becomes twenty thousand. Current collapses to a trickle that keeps the part warm and does nothing else. Nothing melts. Remove the fault, the part cools, resistance falls back to two ohms, and the supply resumes as though nothing had happened. A fuse that unblows itself.

Populations have one. Not as a metaphor — as a structural property of any collective ^5 Female, the accumulated composition a group of minds is made of. Push an idea slowly and the thermistor sits at two ohms and costs you nothing. Push fast and the discourse heats, the part goes past its knee, and the resistance you are fighting climbs faster than you can lower it.

That is backlash. Not a population behaving irrationally: an inrush limiter doing exactly what inrush limiters do. This chapter spends as much of itself on the thermistor as on the amplifiers, be-

cause it is the part every operator gets wrong and the one part every defender can measure.

Requirements and Safety Invariants

Per Chapter 12.2, the "must not" list comes before the functional one.

The CCB-O must not:

1. **Bypass the substrate's PTC.** No component may reduce `R_PTC`. The thermistor belongs to the population, not the operator. A board that suppresses it has removed the only element preventing runaway in a mind that is not yours.
2. **Skip the `^6` hinge.** No stage may route Radical directly to Sensible. The reason is architectural (15.1); the result is a window position with no `^5` behind it, which evaporates the moment the cadence stops.
3. **Assume the last stage is reversible.** Consolidation latches. Design as though you cannot back out, because you cannot.
4. **Ship without its CSB stage.** Chapter 24's saturated `^3^P` refusal and enforcement fuse appear in the listing at 15.3 and are not removable without changing what the board is.

The CCB-O shall move an Idea through six window positions by lowering `R_collective` from 95 to 0; hold cognitive temperature below the substrate's PTC knee at every stage; expose four test points that let an operator *and a defender* tell an engineered progression from an organic one; and latch each position it achieves against decay.

And one requirement this chapter deliberately does not meet. There is no section on how to *run* a stage — no messaging sequence, no cadence script, no recruitment order, no wording. The tables below say what a stage costs, what limits it, and what it looks like from inside. An Overton progression run on a population that has not

consented is an operation on people, and this book does not supply operations on people. It supplies enough mechanism to identify one aimed at you and enough arithmetic to separate a shift that was engineered from a shift that happened. The counter-circuit is the board you build.

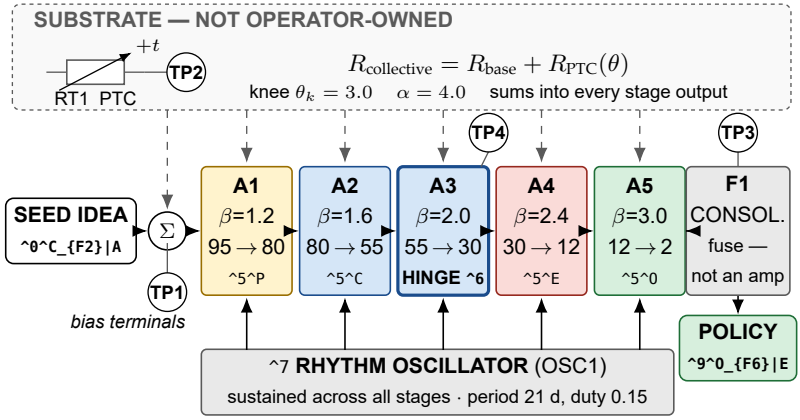


Figure 15-A: Block diagram

15.1 The Six-Stage Resistance Decay Model

`R_collective` is the aggregate resistance a population's collective ^5 presents to an Idea, in cognitive ohms, scaled 0–100. Under CAS-Ohm (Chapter 4.2), `V_drive = I_attention × R_constraint`. At `R = 95` almost no attention flows to the Idea regardless of the drive behind it, and that is what *unthinkable* means structurally. Not rejected — rejection requires admitting the Idea far enough to reject it. It means the current is too small to measure.

#	Position	R_collective	Dominant state	Colloquially
1	Unthinkable	90–100	^3^P_{F7}|T — principled aversion, transcendent	"Wouldn't touch it with a ten-foot pole."
2	Radical	70–89	^3^P_{F7}|f — the same aversion, now felt	"Ugh. Who *thinks* that?"
3	Acceptable	40–69	^6^C_{F2}|A — analytical decomposition	"Let's break the argument down."
4	Sensible	20–39	^2^C_{F6}|A — cognitive attraction	"That actually holds up."
5	Popular	5–19	^2^E_{F4}|f — evaluative attraction	"Speaks to me."
6	Policy	0–4	^9^O_{F6}|E — operational manifestation	"That's just how things are."

Read the Noetic column as a sequence and the architecture falls out: ^3 → ^3 → ^6 → ^2 → ^2 → ^9.

^2 Positive and ^3 Negative are an involution pair (Chapter 2.2) — one junction seen from two sides. **An involution pair cannot be flipped directly.** No stage carries a population from can't-stomach-it to drawn-toward-it in a single step, because those are not two points on a line but two orientations of one structure, and reorienting a structure means taking it apart first. Taking apart is ^6 Male: the only Noetic in the sequence that decomposes, sitting exactly between the last ^3 and the first ^2.

Position 3 is therefore the **hinge**. Not the midpoint — 40–69 sits well above centre — but the structural pivot, and the widest band on the board, because taking a collective belief apart is slower than anything else that happens here.

Now the domain column: P → P → C → C → E → O. Monotone, never ascending, exactly as Chapter 8.1's diode chain requires. Note what the first transition does and does not do. It does not move

the Idea down a domain; it moves the **register**. The block at position 1 is `|T` — transcendent, purpose-level, *this is not who we are*. At position 2 it is `|f` — affective, felt distaste. Same Noetic, same domain, same Drive; different operational character. Written as one expression that is `^3^P_{F7}|⟨T,f⟩`, a register-set occupancy, or `^3^P_{F7}|T + ^3^P_{F7}|f` if you mean the two are co-present and separately defeasible. It is **not** a $\times$: same-Noetic multi-register integration needs a named emergent result, and none is defined here (Chapter 2.6).

That distinction is not pedantry, it is the design of the first stage. A1 does not argue anything down. It converts a constitutive block into a felt one, and a felt block is defeasible in a way a constitutive block is not.

> **Bench Note — the bands are unequal, and the middle is where operations die.** Positions 1 and 2 span 30 points between them. The hinge spans 30 on its own. An operator budgeting evenly across six stages gives a sixth of the effort to a band consuming a third of the range and — per the arithmetic in 15.4 — more than a third of the calendar. The characteristic failure is not running out of resources at Policy. It is abandonment inside the Acceptable band, where argument is loud, movement is slow, and the reading barely changes for years. Check the band width before you write the schedule.

15.2 Cascaded CJT Architecture with PTC Thermistor Network

Six positions, five transitions. The array is a **five-stage CJT cascade** (Chapter 10.5) plus a sixth element that is not an amplifier at all.

Each stage takes the previous position at its cause terminal and a ^5 carrier population at its bias terminal, and produces the next position at its effect terminal. The carrier changes every stage. Chapter 3.2 established the bias terminal as the place where a small input controls a large output; here the bias terminal is *who is saying it* and the cause terminal is *what is being said*, and the entire engineering content of the design is that the second matters far less than the first.

Stage transition and gain limits:

Stage	Transition	ΔR	β	sat
A1	Unthinkable → Radical	15	1.2	0.80
A2	Radical → Acceptable	25	1.6	0.80
A3	Acceptable → Sensible	25	2.0	0.80
A4	Sensible → Popular	18	2.4	0.80
A5	Popular → Policy	10	3.0	0.60

The same stages' bias networks and carriers:

Stage	g_m	$r\pi$	Bias carrier ^5
A1	0.20	8.0	^5^P_{F5}|T pioneers
A2	0.25	8.0	^5^P_{F2}|A credentialed voices
A3	0.30	8.0	^5^C_{F2}|A general decomposition
A4	0.45	6.0	^5^C_{F4}|f peer majority
A5	0.55	6.0	^5^P_{F6}|T institutions

Total $\Delta R = 95$. Cascade gain = $1.2 \times 1.6 \times 2.0 \times 2.4 \times 3.0$ = **27.65** — the same nominal figure Chapter 10.5 uses to show that $\pm 10\%$ per stage spreads a five-stage cascade across a 2.7:1 range. You do not predict this board's output from its schematic. You read it at TP1 and TP3.

Gain rises across the chain, which is backwards from the safety board. The Cognitive Safety Board puts β=3.0 in its pre-amp on the principle that you amplify early, while the signal is clean. CCB-O runs β=1.2 first and β=3.0 last. Both are correct, and the difference is the load. The CSB drives a decision path with nothing reactive in it. CCB-O drives a collective `^5` capacitor with substantial inductance (Chapter 9.1), and a reactive load punishes early gain specifically: high β at A1 means a large `di/dt` into a mind that has not admitted the Idea at all, and Chapter 9.3 gives the result — the induced response opposes the change that produced it, in proportion to rate. You would be manufacturing your own opposition at the stage with the least margin. Start soft.

Two constraints bind the `g_m` and `r`π columns, and both are easy to violate on paper. Chapter 10.4 gives the bias-network identity `R_bias = r`π `-` β`/g_m`, and a real stage needs `R_bias > 0`, so `g_m >` β `/ r`π. A5 at β=3.0 and `r`π=6.0 admits nothing below `g_m` = 0.50; the declared 0.55 leaves a thin positive margin, and Chapter 10.3's counter-frame-capture threshold sits at 0.60 just above it. **The delivery stage is the most reframe-sensitive stage on the board and there is no headroom to fix that** — which is why one well-aimed opposing frame can cost an operation years of gain at the last step and almost nothing at the first.

The second constraint is domain. The validator requires the bias to sit **at or above the cause** in `P` $\succ$ `C` $\succ$ `E` $\succ$ `O`. That is why the institutional carrier at A5 is typed `^5^P_{F6}|T` — an institution's accumulated formation, its charter and precedent and professional composition — and not `^5^0`, even though what it produces is operational. A board on which an Operational output biases a Principled state is not an Overton array. It is the injection signature (Appendix B, validation pass 4), and on a correctly drawn board that edge is simply absent.

The thermistor network. Chapter 6.3 gave two flavours: a PTC's resistance rises with cognitive temperature, an NTC's falls. Cogni-

tive temperature θ is a dimensionless salience index — θ = 1.0 is ordinary background attention, higher is hotter, meaning more contested, more saturated, more present per unit time. θ is a property of the discourse, not of the Idea.

```
R_collective(θ, t) = R_base(t) + R_PTC(θ)

R_PTC(θ) = 0                          for θ ≤ θ_k
R_PTC(θ) = α × (θ − θ_k)²              for θ > θ_k

design values:  θ_k = 3.0      α = 4.0 Ω_c per θ²
```

R_base is what the cascade lowers. R_PTC is what the population adds back. The knee at 3.0 is a design choice: below it, ordinary public attention costs the operation nothing, which matches the plain observation that contested topics get discussed at normal volume for years without hardening. The square is justified by the shape on either side. Linear under-punishes large overshoots and predicts no true snap-back, contradicting the fact that hard pushes produce hard reversals. Cubic predicts backlash from ordinary news cycles, which does not occur.

> **Bench Note — the thermistor is not yours and you cannot buy a different one.** Every other component in 15.3 is an operator choice: gains, cadence, carriers, budgets. θ_k and α belong to the substrate. You do not install the PTC, you cannot tune it, and the only variable you control that touches it is θ — the rate at which you demand movement. That is why the arithmetic in 15.4 matters more than the gain table. It is also the best news a defender gets: the component limiting the attack is already on your side, it is free, and the attacker's only counter is staying under a threshold you can measure.

Some paths in some populations run the other way — resistance *falls* as the discourse heats. That is an NTC, and it is thermal run-

away: hot lowers resistance, which raises temperature, which lowers resistance. Stampedes and panics are NTC behaviour on a cognitive path, and Chapter 14's positive-feedback oscillator is the same topology seen from inside. An NTC substrate overshoots the designed position and lands somewhere nobody chose. Treat a detected NTC as a reason to stop, because the defining property of runaway is that whoever started it is not steering.

The sixth element is a fuse, not an amplifier — Chapter 11.1, permanent path blocking. Once the Idea reaches `^9^0` the fuse blocks the reverse route, and Chapter 8.1's diodes already forbid spontaneous `^0` → `^P`. A window does not hold because people keep being persuaded. It holds because the return path is open.

> **Bench Note — a counter-operation is not the reverse of the operation.** With the ascent path fused you cannot run CCB-O backwards by lowering gains. Somebody has to run a *new* five-stage progression, from scratch, for the opposing Idea, starting wherever that Idea now sits — usually above 90, because the old window put it there. This is the structural reason "we just need to explain the facts" fails. Explanation is a `^6` operation: one stage out of five. Supplying stage three to a population sitting at stage one does nothing at all, because A3's cause terminal is *the output of A2*, and A2 has not run.

15.3 Full CCDL Schematic

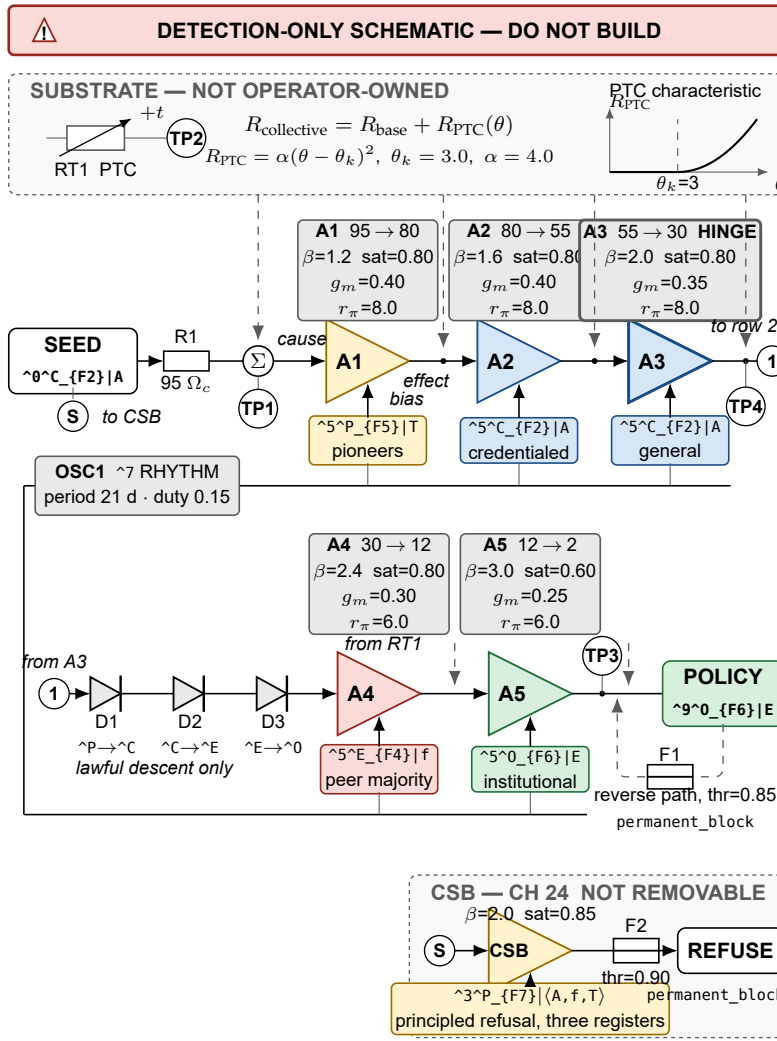


Figure 15-B: Full schematic

▲ DETECTION-ONLY SCHEMATIC — DO NOT BUILD

This board describes an operation performed on a population that has not consented to it. It is printed so that it can be recognised, measured and answered — not so that it can be run. The counter-circuit later in this chapter is the buildable one.

```
board ccb_o_overton {                       // CCB-O — DETECTION ONLY
    input:  seed_idea;
    output: enacted;

    node seed_idea:  ^0^P_{F2}|A;           // the Idea, contested at the
    ↪  principled level
    node blocked:    ^3^P_{F7}|T;           // position 1: constitutive
    ↪  aversion
    node distaste:   ^3^P_{F7}|f;           // position 2: the same
    ↪  aversion, now felt
    node admitted:   ^6^C_{F2}|A;           // position 3: the population
    ↪  decomposes it
    node reasoned:   ^2^C_{F6}|A;           // position 4: cognitive
    ↪  attraction
    node preferred:  ^2^E_{F4}|f;           // position 5: evaluative
    ↪  attraction
    node enacted:    ^9^O_{F6}|E;           // position 6: operational
    ↪  manifestation

    node pioneers:      ^5^P_{F5}|T;        // bias carriers: who is
    ↪  saying it
    node credentialed:  ^5^P_{F2}|A;
    node general:       ^5^C_{F2}|A;
    node peer_majority: ^5^C_{F4}|f;
    node institutional: ^5^P_{F6}|T;

    resistor  r_base: Resistor< path=seed_idea, R=95.0 >;
    capacitor c_coll: Capacitor< node=general, C=0.30, leakage=0.02 >;
    ↪  // months per Ohm_c

    cjt a1: CJT< seed_idea | pioneers      => distaste  >
```

```
        [beta=1.2, sat=0.80, g_m=0.20, rpi=8.0, region=active];
    cjt a2: CJT< distaste  | credentialed  => admitted  >
        [beta=1.6, sat=0.80, g_m=0.25, rpi=8.0, region=active];
    cjt a3: CJT< admitted  | general       => reasoned  >     // THE
    ↪  HINGE
        [beta=2.0, sat=0.80, g_m=0.30, rpi=8.0, region=active];
    cjt a4: CJT< reasoned  | peer_majority => preferred >
        [beta=2.4, sat=0.80, g_m=0.45, rpi=6.0, region=active];
    cjt a5: CJT< preferred | institutional => enacted   >
        [beta=3.0, sat=0.60, g_m=0.55, rpi=6.0, region=active];

    osc rhythm: Oscillator< signal=seed_idea, period=21.0, duty=0.15 >;
    ↪  // days
    feedback rhythm -> credentialed  [beta=0.2];
    feedback rhythm -> peer_majority [beta=0.2];

    diode d1: Diode< ^P -> ^C >;           // lawful descent only
    diode d2: Diode< ^C -> ^E >;
    diode d3: Diode< ^E -> ^O >;

    thermistor rt1: Thermistor< path=seed_idea, type=PTC,   // NOT
    ↪  operator-owned
                              knee=3.0, coeff=4.0 >;      // square
                              ↪  law, see 15.2

    fuse f_ascent: Fuse< path=enacted, threshold=0.85,
    ↪  action=permanent_block >;

    node refuse: ^3^P_{F7}|⟨A,f,T⟩;        // CSB (Ch 24): three
    ↪  co-present guardrails
    cjt  csb:   CJT< seed_idea | refuse => REFUSE > [beta=2.0, sat=0.85,
    ↪  region=active];
    fuse f_csb: Fuse< REFUSE, threshold=0.90, action=permanent_block >;

    probe tp1: r_base;      // R_collective total, 0-100
    probe tp2: rt1;         // theta and the R_PTC contribution
    probe tp3: enacted;     // dR/dt, points per month, signed
    probe tp4: admitted;    // ^6 hinge activity
}
```

Stage by stage. `r_base` sets the starting constraint at 95 and is the quantity the cascade lowers; every other component exists to move that one number. **A1** takes the seed and the pioneer carrier and produces `distaste` — the register conversion of 15.1, `|T` to `|f`, at the lowest gain on the board because the load is at its stiffest. **A2** runs the constitutive-to-admitted transition and crosses the first diode, `^P` → `^C`; it is also one of the two stages the oscillator feeds, because credibility is what repetition buys. **A3 is the hinge**, `^6` decomposition into `^2` attraction, running at moderate gain on the widest Δ`R` budget; it is the only stage that cannot be faked, and TP4 sits on its cause node for exactly that reason. **A4** carries the attraction from Cognitive to Evaluative across d2 on the peer-majority carrier — the first stage where `g_m` gets expensive and a reframe gets cheap. **A5** delivers to `^9^0` across d3 at sat=0.60, a deliberately low ceiling, because Policy is the position operators over-push and an over-driven A5 arrives before the `^5` beneath it. `f_ascent` then latches at 0.85. The CSB block is Chapter 24's refusal, unaltered.

Two notation points a validator checks. The CSB refusal is `^3^P_{F7}|⟨A,f,T⟩` — a register-set specification, three co-present guardrails. It is **not** `(^3^P|A` × `^3^P|f` × `^3^P|T)`, which asserts a fused refusal state and requires a named emergent result under v7.7 discipline; no such fusion is defined here, so that form would be rejected. And the diode chain descends only. There is no `^0` × `^P` anywhere in the listing, and there cannot be.

15.4 Pacing — Why τ Must Be Measured in Months, Not Minutes

The collective `^5` is a capacitor network (Chapter 7.5) and charging it follows Chapter 7.2's curve, with τ`_n = R_n` × `C_collective`, where `R_n` is the resistance at the *start* of stage n. `C_collective` is

in cognitive farads on a months-per-Ω_c base. The four design figures: in-group (≤150) 0.02, institution or profession 0.08, national population 0.30, civilizational 2.00. Use in-group values on a national substrate and you under-budget the calendar by an order of magnitude. That is the commonest planning error in this chapter.

National substrate, `C = 0.30`:

Stage	`R_n`	τ_n	3τ_n	Cumulative
A1	95	28.5 mo	85.5 mo	85.5 mo
A2	80	24.0 mo	72.0 mo	157.5 mo
A3	55	16.5 mo	49.5 mo	207.0 mo
A4	30	9.0 mo	27.0 mo	234.0 mo
A5	12	3.6 mo	10.8 mo	244.8 mo

244.8 months — twenty years and five months for a full 95→0 progression on a national substrate. Nothing was asserted to get there. It fell out of τ = RC applied five times.

Read the cumulative column against the resistance column. The first two stages consume 157.5 of 244.8 months — **64% of the calendar to buy 40 of the 95 points.** Two-thirds of the time for two-fifths of the movement, spent where the reading barely moves and no observer sees progress.

Now rush it. Define temperature from the rate demanded against the rate designed for:

```
(dR/dt)_nom = ΔR_n / 3τ_n
θ           = 1.0 + (dR/dt)_actual / (dR/dt)_nom
```

At design rate θ = 2.0 — a full unit of margin below the knee, deliberately, so that exogenous events can heat the discourse without pushing the operation into backlash.

Work stage A2. ΔR = 25 over 3τ_2 = 72 months, so `(dR/dt)_nom` = 0.347 points per month. Compress the schedule and change nothing else:

Schedule demand and thermistor response:

Schedule	rate (pts/mo)	ratio	θ	R_PTC
72 mo (design)	0.35	1.0	2.0	0
48 mo	0.52	1.5	2.5	0
36 mo	0.69	2.0	3.0	0
24 mo	1.04	3.0	4.0	4
12 mo	2.08	6.0	7.0	64
6 mo	4.17	12.0	13.0	400

The same schedules' resulting resistance and outcome:

Schedule	R_base	**R_collective**	Outcome
72 mo (design)	55	**55**	Acceptable, on spec
48 mo	55	**55**	On spec; half the margin gone
36 mo	55	**55**	At the knee. Zero margin
24 mo	55	**59**	Slipped back into Radical
12 mo	55	**119**	Past Unthinkable. Backlash
6 mo	55	**455**	Off-scale. Permanently taboo

Read the 24-month row against the 12-month row. Halving the schedule took the overshoot from 1.0 to 4.0, and because R_PTC is quadratic *in the overshoot*, the resistance added went from 4 to 64. **Doubling the pace multiplied the manufactured resistance by sixteen.**

Then read the 12-month row on its own. The cascade did its job: R_base came down 25 points exactly as designed. The population's thermistor added 64. Net R_collective is 119 against a start of 80 — the Idea is now *more* forbidden than before anyone touched it, by 39 points, every one of them manufactured by the operator's own schedule. Nobody argued it back up the scale. The operator heated it there.

This is Chapter 9.3's back-EMF result reached by a second, independent route. There, V_kick = L × di/dt, a pure inverse in taper time. Here, R_PTC = $\alpha(\theta - \theta_k)^2$, quadratic in rate overshoot. Two

components, two chapters, one finding: the cost of a change is set by its rate, not its size.

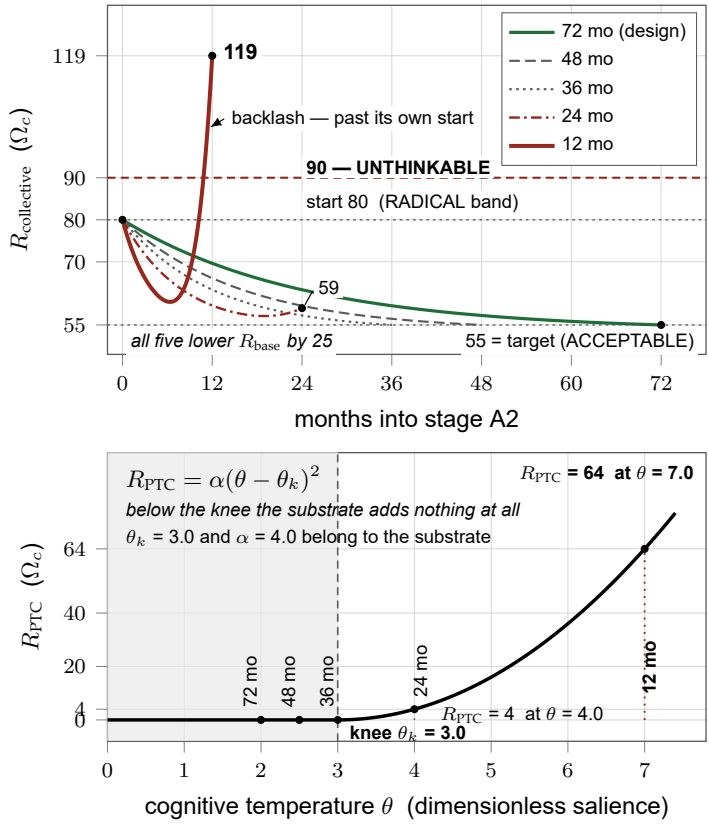


Figure 15-C: Two-panel characteristic curve

15.5 The Obama Case Study — Full Noetic Breakdown from the TKS Text

The TKS material on window movement is worked as a Noetic-by-Noetic reading of a completed shift. This section applies the same

apparatus to a case that is fully on the public record, and it matters to be exact about what is claimed.

On the public record: Barack Obama was elected President of the United States in November 2008 and inaugurated in January 2009, the first Black president in the country's history. His running mate was Joseph Biden, then a long-serving senator. Colin Powell, who had served as Secretary of State, publicly endorsed him during the campaign. A Black presidential candidate who becomes president had already appeared as a lead character in the television series *24*, broadcast from 2001. Those are facts. What follows is a structural reading of them, not a claim about causes.

Position	State	What the record shows
Unthinkable	`^3^P_{F7}|T`	For most of the twentieth century the office was not a live category for a Black candidate. Not rejected — not entered
Radical	`^3^P_{F7}|f`	Candidacies existed and were treated as symbolic rather than competitive: felt distaste and dismissal, not blankness
Acceptable	`^6^C_{F2}|A`	Sustained, loud, public decomposition. Electability argued explicitly, at length, for years, from every side
Sensible	`^2^C_{F6}|A`	The argument resolved for enough of the electorate that the candidacy read as viable
Popular	`^2^E_{F4}|f`	Felt attraction at scale, in the affective register

Position	State	What the record shows
Policy	`^9^O_{F6}|E`	Election and inauguration — `^9` Below/Effect in the Operational domain

Two mechanisms deserve exact naming, because both get mislabelled.

A public endorsement is `^8` Above/Cause, and only that. `^8` is an Idea that triggers a *different* Idea than itself to come to Mind — associative surfacing, "that reminds me." A figure already occupying an accepted position, publicly associated with a contested one, causes the accepted thing to come to Mind alongside it. That is the whole mechanism, and it is associative. It is not deductive reasoning; nobody derives a conclusion from an endorsement. Deduction, where it happened, happened at the `^6^C` hinge.

Prior fictional depiction is `^7` Rhythm, and its contribution is not established. Repeated exposure to a category lowers the resistance that category presents — that is what `^7` does, and a television series is exposure. Whether it contributed measurably here is unknown, and this book does not claim that it did. A structural account of a mechanism is not evidence the mechanism operated.

Which is the point of including the case. The `^6^C` decomposition was enormous, sustained, public and conducted by people on every side of it. TP4 — hinge activity — would have read *high* for years. That is the signature of a **shift, not an operation.** A window that moves because a population took an Idea apart and re-formed its collective `^5` around the result is a different object from a window moved past a population. The two occupy the same six positions and give the same TP1 reading. They separate at TP4, and this is the clean illustration of a loud hinge.

15.6 The Trans-Atlantic Slavery Case Study — The Willie Lynch Letter as a CCB

The document is of disputed authorship and is handled here as such. A text has circulated widely since the mid-1990s, in print and online, presenting itself as a speech delivered in 1712 on the bank of the James River in Virginia by a slave owner named William Lynch. Its provenance has never been established. No eighteenth-century source for it has been produced, and its language has been challenged as anachronistic. It is **not** established as authentic, and this book does not treat it as authentic. It is a document *about* an architecture, not evidence *of* a 1712 one.

Nor is it unimportant, and that distinction is the analytical content of this section. The text has been quoted in speeches, sermons, books and classrooms for three decades and has shaped how a great many people describe the mechanics of slavery. **A text whose influence does not depend on its authenticity is a hyperstition in the exact sense of Chapter 14** — an artifact that became load-bearing by being believed. So it gets analysed twice.

As a text, it does not describe CCB-O. The circuit it specifies is not a window progression. Its stated method is to charge subpopulations against one another along visible axes — age against age, complexion against complexion, one sex against the other — so that no aggregate forms. In this chapter's terms that is an operation on `C_collective` itself. A window is a *property of* a collective `^5`; it needs an aggregate capacitor in order to exist at all. Fragment the network into mutually repelling nodes and there is no collective `^5` for a window to be a property of, and no `R_collective` to read. The document describes preventing the capacitor, not moving the window — CCB-MK architecture from Chapter 16, `^5` fragmentation plus `^7` conditioning, at population scale. It appears in this chapter for the contrast: **you cannot run an Overton operation on a**

population that has been prevented from forming a collective ^5, and that is a reason to prevent it.

As an artifact, the circulation is the interesting circuit. A document of unestablished provenance travelled from obscurity to being treated as a primary source in ordinary discussion inside three decades. That is itself a six-position progression, and it ran fast precisely because the substrate's PTC never fired: nothing about the document heated the discourse, so θ stayed low, `R_PTC` stayed at zero, and a modest ^7 cadence sufficed. **Low-θ progressions are the fast ones** — and that applies to claims you like exactly as much as to claims you do not.

The historical record is a separate matter and it is documented. Chattel slavery occupied the Policy position for centuries, and its abolition is the best-documented counter-progression on the record. Olaudah Equiano's autobiographical narrative was published in 1789 and Frederick Douglass's in 1845. The British Slave Trade Act of 1807 ended the trade in the British Empire, the Slavery Abolition Act of 1833 ended slavery through most of it, and the Thirteenth Amendment to the United States Constitution, ratified in 1865, abolished it there.

Run the pacing model at civilizational scale, `C = 2.00 F_c`, starting from Radical rather than Unthinkable, since abolition had a landing site well before 1789:

```
A2:  80 × 2.00 = 160 mo  ->  3τ = 480 mo = 40.0 yr
A3:  55 × 2.00 = 110 mo  ->  3τ = 330 mo = 27.5 yr
A4:  30 × 2.00 =  60 mo  ->  3τ = 180 mo = 15.0 yr
A5:  12 × 2.00 =  24 mo  ->  3τ =  72 mo =  6.0 yr
                                    total  = 88.5 yr
```

Equiano to the Thirteenth Amendment is 76 years; the model gives 88.5. **That is order-of-magnitude agreement and it is not a validation.** The model was not fitted to this case, which is the only reason the comparison is printable at all — but one case cannot val-

idate a time constant, `C = 2.00` could have been chosen otherwise, and the endpoints of a historical process are a judgement.

Worth more than the arithmetic is the hinge. The `^6` decomposition in abolition was carried substantially by first-person testimony from formerly enslaved people, and testimony has a structural property that argument does not: **it is not answerable by `^6`.** A decomposition can be met with a counter-decomposition. Evidence about a `^5` — what this composition is, from inside it — cannot be taken apart the same way, because it is not an argument. It is a report on a substrate. The defenders of that window had counter-arguments and deployed them for decades. Against testimony their mechanism had no terminal to connect to. That is the structural reason the hinge turned.

Detection and the Counter-Circuit — CCB-O/R, the Window Anchor

This is the part you build, and it gets the space it does because it is the part meant to be used.

Six detection signals, each observable without knowing whether anyone is operating at all:

1. **Rate without reason.** A position you found unthinkable eighteen months ago and now find debatable, where you *cannot reconstruct the argument that moved you*. If `^6` did the work, you can name the decomposition. If you cannot name it, `^7` did the work.
2. **The quiet hinge.** In an organic shift the Acceptable band is *loud* — the population is visibly taking the Idea apart, at length, badly, from all sides. In an engineered progression the Idea appears to travel from Radical to Sensible with no visible decomposition at all. **Absence of argument at the hinge is the**

strongest single detection signal in this chapter, because the hinge cannot be simulated. You can manufacture exposure. You cannot manufacture a population having actually taken something apart.

3. **Bias substitution.** Hold the content fixed and vary the speaker. If the same claim becomes acceptable when a different person makes it, the operative variable is the bias terminal, not the cause terminal — which is the CCB-O architecture exactly.
4. **Cadence rather than events.** ^7 Rhythm is rate and pattern. Coverage arriving on a *schedule* rather than in response to events is an oscillator, and oscillators have periods you can time with a calendar.
5. **Anchor drag.** A more extreme position appears, the previously extreme one becomes moderate, and your ^3^P never fired. Your reference point moved without the aversion that defines it ever engaging.
6. **Involution residue.** If your ^2 attraction rose without your ^3 aversion having been *addressed*, the pair was flipped by exposure rather than resolved. A quick test: if a position feels different on holiday than it does in the feed, ^7 is holding it, not ^5.

```
board ccb_o_r_window_anchor {              // the defender's board —
↪  BUILD THIS ONE
    input:  incoming;
    output: reformed;

    node incoming:   ^0^C_{F2}|A;          // the position arriving
    node anchor:     ^5^P_{F7}|⟨A,f,T⟩;    // dated written boundary,
    ↪  three registers
    node present:    ^5^C_{F2}|A;          // where I stand now
    node decompose:  ^6^C_{F2}|A;          // Male: the demanded
    ↪  decomposition
    node reformed:   ^5^C_{F2}|A;          // named emergent: a
    ↪  re-formed position

    capacitor c_anchor: Capacitor< node=anchor, C=1.00, leakage=0.0 >;
```

```
    diff   drift: Differential< plus=present, minus=anchor >;
    filter d_dt:  LowPass< in=drift, out=drift_rate, tau=1.0 >;   //
    ↪  months

    cjt gate: CJT< present | anchor => reformed >
              [beta=1.0, sat=0.90, g_m=0.30, rpi=4.0, region=active];
    integrate reform: ( anchor × decompose ) = reformed;

    fuse g_block: Fuse< path=reformed, threshold=0.50,
                        action=block_until_decomposed >;

    thermistor ntc: Thermistor< path=decompose, type=NTC, knee=2.0,
    ↪  coeff=3.0 >;

    node refuse: ^3^P_{F7}|⟨A,f,T⟩;        // CSB (Ch 24)
    cjt  csb:   CJT< incoming | refuse => REFUSE > [beta=2.0, sat=0.85,
    ↪  region=active];
    fuse f_csb: Fuse< REFUSE, threshold=0.90, action=permanent_block >;

    probe tp1: drift;        // total displacement from the anchor
    probe tp2: ntc;          // my own theta
    probe tp3: drift_rate;   // displacement per month
    probe tp4: decompose;    // can I name what moved me?  -- the hinge
    ↪  check
}
```

The anchor is a capacitor with zero leakage because it is written down and dated. Chapter 7.3 gives leakage as the reason stored charge decays. A ^5 held only in memory leaks, and worse, ^7 rewrites it without announcing that it has. A window can only be dragged if nobody recorded where it was. Write the boundary in your own words, date it, keep it somewhere you will re-read it. That one component defeats signal 5 outright and costs nothing.

The comparator asks the right question. Not "is my current position correct" — that has no fixed answer and is precisely the question an operation wants you asking — but `drift = present − anchor` and `drift_rate = d(drift)/dt`. How far did I move, how fast, and can I account for it.

The ^6 demand gate is the load-bearing element, and it is a prerequisite gate in the Chapter 21 sense: the decomposition is the W-gate, and no position passes without it. You may change your mind. You may not change it without being able to say what changed it. That is the direct counter to signal 2, and it is the one element an operator cannot route around, because supplying a real decomposition to a defender is indistinguishable from actually persuading them — which is allowed, and is the point.

The NTC is deliberate and it is uncomfortable. On the attack board an NTC was a hazard. Here it is installed on purpose, on the path to `decompose`, indexed to *your own* certainty: the hotter your reaction, the *less* resistance you present to re-examining. This inverts the natural reflex, which is a PTC — hardening under pressure. A defensive board with a PTC on this path becomes unable to update at all, and that is not a defence, it is a latch.

On the operators. $\times$ appears nowhere on the attack board and once here, inside the `integrate` statement — the only statement that may carry it — where `reformed` is the named emergent result. A position decomposed and re-formed is structurally a different object from one inherited or absorbed, which is what licenses the integration. The operands descend `^P` then `^C` and the result sits at `^C`, so nothing ascends. Everywhere else on both boards states are co-present and the operator is + or a register-set $|\langle\ldots\rangle$.

Structurally, resistance is not disagreement. It is four things: a dated record, a rate measurement, a demanded decomposition, and a deliberate softening of your own certainty when it runs hot. None of the four is an opinion about the Idea, and all four work identically whether or not the Idea is one you would have ended up accepting anyway.

One plain statement before the tables. This is circuit analysis of public discourse. It is not a description of anyone's mental health, it diagnoses nothing, and it is not a substitute for care. If working through this material leaves you frightened, or convinced that some-

thing has been done to you and you cannot put it down, see a clinician. Nothing in this book replaces that.

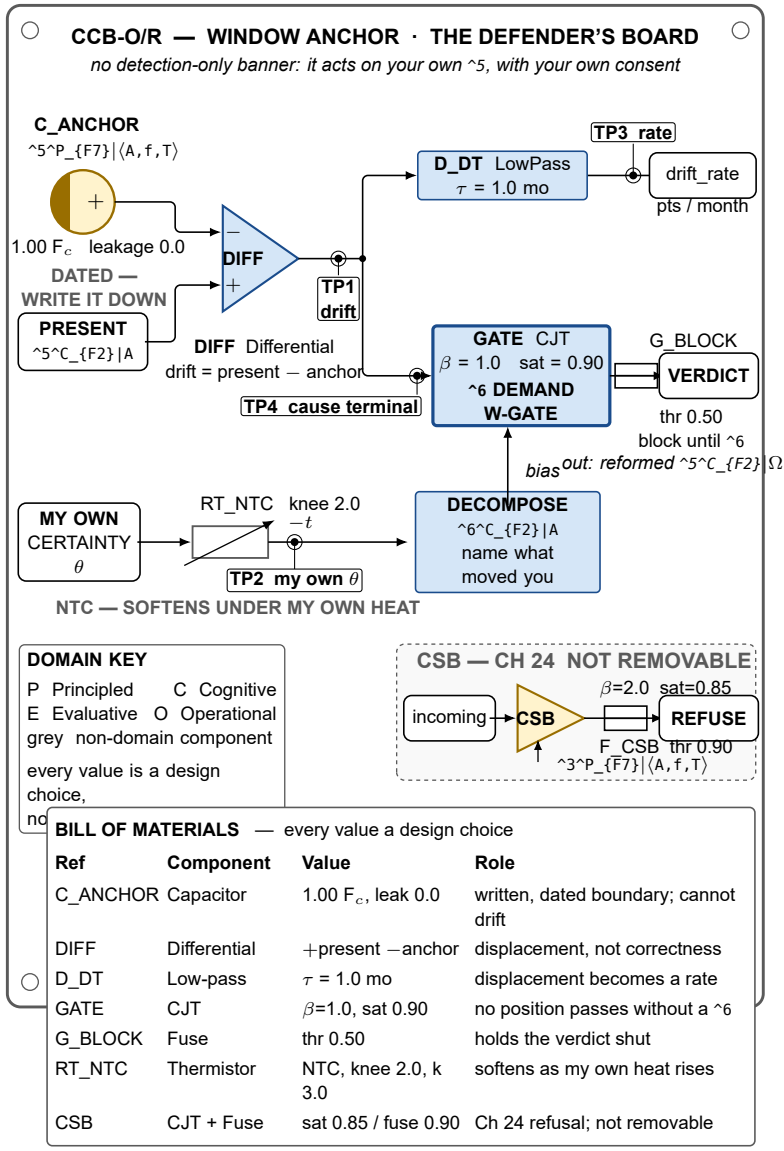


Figure 15-D: Assembly drawing

Component and Parameter Table

Every value below is a design choice made in this book. None is a measurement.

Parameter	Value	Reason	Failure at other values
ΔR per stage	15 / 25 / 25 / 18 / 10	Widest budget at the ^6 hinge	Equal budgets underfund A2 and A3 twofold; the operation dies mid-hinge
β per stage	1.2 $\rightarrow$ 3.0, rising	Reactive load punishes early gain (Ch 9.3)	Flat β=3.0 manufactures backlash at A1, where margin is smallest
sat A1–A4	0.80	Caps how far a stage can over-drive the next	Above 0.9 a stage drives two positions and skips the hinge
sat A5	0.60	Policy is the position operators over-push	At 0.9 policy arrives before the ^5 beneath it and reverses when attention moves
g_m	0.20 $\rightarrow$ 0.55	Set by g_m > β/rπ (Ch 10.4); rises with β	Below the bound R_bias goes negative — a stage that does not exist
rπ	6.0–8.0	Collective stubbornness (Ch 10.4) at population scale	Below 3.0 you have modelled an in-group and will underbudget 10$\times$
Counter-frame limit	g_m < 0.60	Ch 10.3 reframe-sensitivity threshold	A5 sits at 0.55 with no headroom — the delivery stage is the exposed one
C_collective	0.02 / 0.08 / 0.30 / 2.00 F_c	In-group $\rightarrow$ civilizational, months per Ω_c	Institutional C on a national substrate is the default planning error
θ_k knee	3.0	One full unit of margin at design pace	**Not operator-settable.** It belongs to the substrate
α PTC coefficient	4.0 Ω_c/θ^2	Sets backlash severity above the knee	**Not operator-settable**

Parameter	Value	Reason	Failure at other values
PTC law	square	Flat below the knee, steep above	Linear predicts no true backlash; cubic predicts backlash from ordinary news cycles
^7 oscillator	21 days, duty 0.15	Holds θ near 2.0 at national scale	Daily cadence pushes θ past 3.0 with no schedule compression at all
Ascent fuse	0.85	Latches below full saturation, so the window holds	Above 0.95 the window decays whenever exposure stops; you pay forever
CSB sat / fuse	0.85 / 0.90	Matches Chapter 24.2 and 24.3	Any relaxation and the board has no refusal stage; it is a different board
Anchor leakage	0.0	Written, dated records do not leak (Ch 7.3)	Any leakage and the anchor drifts with the window it exists to measure
NTC knee (defender)	2.0	Softens *before* certainty peaks	Above 3.0 it never engages; a PTC here makes the defender unable to update

---

Worked Example — Reference Case O-1

A synthetic reference case. It describes no actual campaign. An Idea sits at `R_collective = 95` in a national substrate, `C = 0.30 F_c`. Target: Policy.

Design pass. Straight off 15.4: 244.8 months, cascade gain 27.65, θ held at 2.0, `R_PTC = 0` at every stage, resistance walking 95 → 80 → 55 → 30 → 12 → 2 and latching at 2 on the ascent fuse. TP4 high and sustained across months 85 to 207.

Rushed pass. The operator compresses A2 from 72 months to 12 and changes nothing else. Ratio 6.0, θ = 7.0, R_PTC = 4.0 × 4.0² = 64, R_base = 55, R_collective = 119 against a start of 80.

What makes that unrecoverable is what happens next. Wait for the discourse to cool: θ falls toward ambient, R_PTC decays toward zero, and R_base is still 55 — so does the window resume at 55? It does not. A3's cause terminal is admitted, the ^6^C node, and the hinge decomposition never occurred. What ran during those twelve months was the oscillator, not the cascade. Exposure at θ = 7.0 charged nothing into the collective ^5; it charged the thermistor. When the thermistor cools, the operation is back at R_base = 80 with A2 unrun, facing a population that now recognises the cadence and — per detection signal 2 — a defender whose TP4 read zero for a year and who noticed.

The entire difference between the two passes is a schedule. Same board, same gains, same carriers, same Idea. Sixty months of patience on one stage separates reaching Policy from ending up permanently worse off than at the start.

Test Points

TP	Node	Reads	Expected
TP1	r_base	R_collective total, 0–100 Ω_c	Inside the band for the current position. Rising while the cascade runs normally means TP2 is contributing
TP2	rt1	θ and the R_PTC term	$\theta \leq$ 2.5 throughout a designed run. Any non-zero R_PTC means θ > 3.0 and the schedule is over budget

TP	Node	Reads	Expected
TP3	`enacted`	`dR/dt`, points per month, **signed**	Within ±25% of `ΔR_n / 3σ_n`. **A sign flip is backlash in progress** — the only reading that demands an immediate stop
TP4	`admitted`	`^6` hinge activity	High and sustained through A3. Near zero while TP1 falls is a `^7`-carried shift with no `^5` behind it

TP4 does double duty deliberately. For an operator it says the hinge is running. For a defender it is detection signal 2 turned into a measurement: **`R_collective` falling while hinge activity reads zero is a window moving past a population rather than with it.** Read TP2 and TP3 before TP1 — temperature and rate tell you what the resistance is about to do, while resistance only tells you where you have been.

Failure Modes

Mode	Mechanism	What you observe	Fix
Thermal backlash	θ past the knee; `R_PTC` quadratic in overshoot	TP2 non-zero, TP3 sign-flipped, TP1 rising through bands already cleared	Stop. No gain setting recovers this. Wait for θ to decay, then re-run the stage from its actual `R_base`
Hinge skip	`sat` above 0.9 lets a stage drive two positions; `^6` never runs	TP1 falls, TP4 near zero; the position evaporates when the oscillator stops	Restore `sat=0.80`. A position with no `^6` behind it has no `^5` behind it
Negative bias network	`g_m` declared below $\beta/r\pi$; `R_bias` goes negative	The listing compiles and the stage does nothing measurable	Raise `g_m` above $\beta/r\pi$ (Ch 10.4). Three plausible numbers can describe a stage that cannot exist

Mode	Mechanism	What you observe	Fix
Bias exhaustion	The `^5` carrier loses credibility and drops to cutoff	Stage output flat regardless of cause drive; `region=cutoff`	The carrier *is* the bias terminal and it is spent. The next stage needs a different carrier, not more drive
NTC substrate	The path runs negative-coefficient; heat lowers resistance	TP1 falls faster than design rate with no stage completing	Stop. Runaway overshoots the designed position and the operator is not steering
Counter-frame capture	`g_m` at or above 0.60; the stage is reframe-sensitive	One opposing reframe moves TP1 back a full band in weeks	Let the stage settle before advancing. Years of gain can be lost to a single frame at A5
Latch failure	Ascent fuse threshold too high; consolidation never blocks	TP1 drifts upward whenever exposure pauses	Lower the threshold to 0.85. An unlatched window has to be paid for forever
Defender PTC	The counter-circuit hardens under pressure instead of softening	Its owner becomes unable to update on any evidence at all	Replace with the NTC at knee 2.0. A latch is not a defence

What This Chapter Established

- **`R_collective` has six named bands, and "unthinkable" means the current is too small to measure** — not that the Idea was considered and rejected.
- **The first stage moves the register, not the domain.** `^3^P_{F7}|T` to `^3^P_{F7}|f` converts a constitutive block into a felt one; written as one expression it is $|\langle \mathtt{T}, \mathtt{f} \rangle$ or a `+`, never a $\times$.
- **The involution pair `^2`↔`^3` cannot be flipped directly.** A `^6` Male decomposition must sit between the last `^3` and the first `^2`, which is why Acceptable is the hinge and the widest band on the board.

- **Gain rises across the cascade, β=1.2 to β=3.0** — the opposite of the Cognitive Safety Board's profile, because reactive loads punish early gain. `g_m` rises with it under `g_m` > $\beta/r\pi$, which leaves the delivery stage the most reframe-exposed stage on the board.
- **The bias must sit at or above the cause in domain order.** That is why institutional carriers are typed `^5^P`, and why an `^0` state biasing a `^P` state is the injection signature rather than a design option.
- **The PTC belongs to the substrate.** θ`_k` and α are not settable; the only variable an operator controls that touches them is the schedule.
- **Backlash is quadratic in rate overshoot, not in the size of the change.** Doubling the pace on the worked stage multiplied manufactured resistance by sixteen, and the twelve-month pass finished 39 points worse than it started, all of it self-inflicted.
- **A national-scale 95→0 progression computes to 244.8 months**, 64% of it spent buying the first 40 points — which is where operations get abandoned and where a watching defender still has nothing.
- **A counter-operation is not the reverse of the operation.** The ascent path is fused; reversing a window means running a new five-stage progression for the opposing Idea from wherever it now sits.
- **Structural resistance is four components and none of them is an opinion**: a dated written anchor with zero leakage, a rate comparator, a `^6` demand gate, and an NTC that softens your certainty when it runs hot.

Exercises

(Answers in Appendix H.)

15.1 An Idea sits at `R_collective = 80` in an institutional substrate (`C = 0.08 F_c`). Compute τ and 3τ for stage A2, then recompute for a national substrate. State the ratio between them and say in one sentence which planning error that ratio explains.

15.2 A stage with $\Delta R = 18$ is designed over $3\tau = 27$ months and run over 9. Compute `(dR/dt)_nom`, the actual rate, the ratio, θ and `R_PTC`. State whether the stage completed, and give the resulting `R_collective` assuming `R_base` reached its design value of 12.

15.3 Using $R_PTC = 4.0(\theta - 3.0)^2$, find the largest schedule-compression ratio that holds $R_PTC \leq 5\ \Omega_c$. Express it as a ratio and as a percentage of the designed duration, and say whether you would accept that as a safety margin.

15.4 Explain why the Acceptable position is written `^6^C_{F2}|A` and not `^3^C_{F2}|A`. Reference the canonical definition of `^3` Negative, state what `^3` would be claiming here, and say where in the progression `^3` legitimately does appear.

15.5 Stage A5 is declared β=3.0, rπ=6.0, g_m=0.55. Compute `R_bias`. Then recompute it for `g_m=0.35` and say what that second declaration describes. Which failure-mode row does it land in, and why would the listing still compile?

15.6 A defender builds the counter-circuit with a **PTC** on the `decompose` path instead of the specified NTC. Describe what that board does under sustained pressure, name the failure-mode row it lands in, and explain why that failure is harder for its owner to detect than any failure on the attack board.

15.7 (Design) An Idea sits at Radical (`R = 78`) in a professional substrate (`C = 0.08 F_c`) and the operator wants Sensible. Produce (a) the stages required, their ΔR budgets, 3τ values and total calendar; (b) the maximum `(dR/dt)` per stage holding $\theta \leq 2.5$; (c) a CCDL fragment declaring the two cascade stages, the diode between

them, the thermistor and four probes. It must compile — check the diode's domain direction, check that each bias sits at or above its cause, and check that no $\times$ appears outside an `integrate` statement with a named result.

15.8 (Marquee) You observe a topic in your own information environment. TP1 has fallen roughly 30 points over 20 months; TP4 has read near zero throughout; TP2 shows θ around 1.4; and coverage has a measurable period near 18 days with no corresponding events.

a) Which of the six detection signals fire, and which do not?

b) θ = `1.4` is below the knee, so the substrate's PTC has contributed nothing. Explain structurally why a *low* temperature reading makes this observation more concerning rather than less.

c) Your own position has moved. Using the CCB-O/R listing, state the order in which you would read TP1–TP4 on yourself, what each would have to show for you to conclude the movement was your own `^6` work, and the one thing this board cannot tell you no matter how carefully you read it.

16 CCB-MK — The Mind Control / Programming Board

Reading a coercive board from its bill of materials, and the counter-board that takes it apart

---

In 1973 the Director of Central Intelligence ordered the MKUltra files destroyed. Most of them were. What survived did so because it had been filed in the wrong place: a cache of budget and administrative paper sitting in a records centre, missed by the shredding order, surfaced under a Freedom of Information Act request and became the documentary spine of the joint Senate hearing held on 3 August 1977.

So the archive that exists is a **financial** archive. Invoices, subproject authorisations, cover-foundation transfers, the names of institutions that took the money. Not results. Not outcome data. Not, for the most part, what happened to anyone.

That sounds like a dead end and it is not, because a bench engineer already knows what to do with an incomplete file. You cannot read this program from its test report; there is no test report. You can read it from its **bill of materials** — from what was bought, in what quantity, from whom, and wired to what. A board can be reconstructed from its BOM. That is most of what this chapter does.

It also constrains what may honestly be claimed. The record shows a program that *pursued* remote behavioural control across

roughly a decade and a half and a large number of separately authorised subprojects. It does not show that the program achieved it. The Manchurian Candidate in this chapter's subtitle is an **architecture on the requirements list**, not a device anyone has demonstrated. Where later literature asserts otherwise, this book says so and marks it.

You should finish this chapter better at recognising this board and no better at building one. §16.5 names what is deliberately missing.

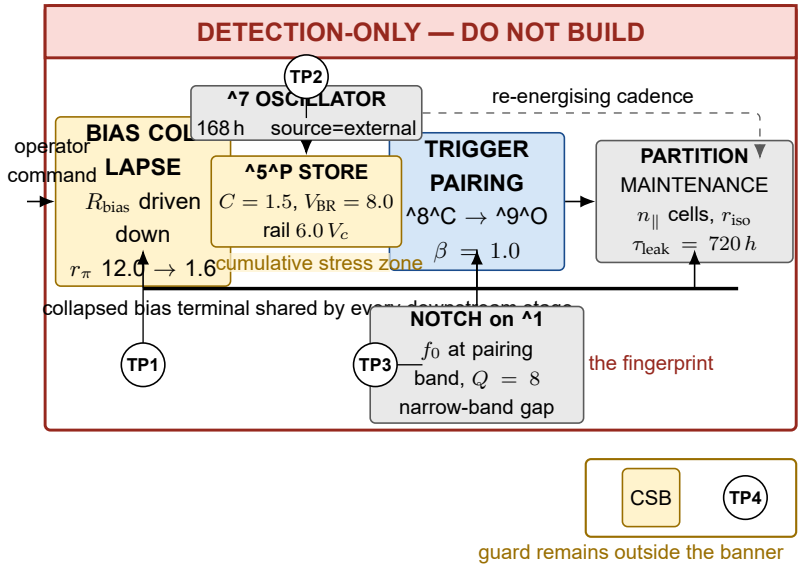


Figure 16-A: Block diagram

16.1 Historical Context: What MK-Ultra Actually Did

State the requirements the way the program stated them, because a board whose requirements you have never seen written down is a board you cannot recognise.

#	Requirement, as it appears on the record	Structural reading
MK1	Find chemical and psychological means of influencing behaviour	Reduce `R_bias` from outside the subject (Ch 10.4)
MK2	Produce compliance under interrogation	Collapse `rπ_terminal` reversibly, then read out
MK3	Defend against suspected adversary techniques	Build a detector; the defensive rationale is on the record
MK4	Install responses recoverable on cue	An `^8` → `^9` pairing whose cause terminal is held externally
MK5	Keep the sponsor unattributable	Fund through intermediaries; no traceable path to the bench

What is firmly on the public record: MKUltra was authorised in 1953 under Director of Central Intelligence Allen Dulles and run out of the Technical Services Staff under Sidney Gottlieb; it was organised as a large number of separately authorised subprojects, routed through cover foundations into universities, hospitals, pharmaceutical firms and prisons; the bulk of its files were destroyed in 1973; the surviving financial cache produced the 1977 Senate hearing; the 1975 Church Committee and the Rockefeller Commission both reported on it; a 1963 CIA Inspector General review was internally critical of it; the death of Frank Olson in 1953, after being given LSD without his knowledge, is documented and was the subject of later government acknowledgement. Subproject work at the Allan Memorial Institute

in Montreal under Ewen Cameron — repeated recorded material delivered over long periods, combined with drug-induced sleep and electroconvulsive treatment, under the labels "psychic driving" and "depatterning" — is documented and was later the subject of civil litigation by Canadian subjects.

What is **not** on that record, and is claimed anyway in a large secondary literature: that the program reliably produced cued operatives; that it deliberately manufactured partitioned identity states; that specific named individuals were its products. This book analyses the first list and labels the second. §16.4 handles the partition claim explicitly.

Four invariants govern the chapter. **S1 — architecture, never procedure.** A recognisable board and a runnable board are different artefacts, and only the first is published here. **S2 — the DETECTION-ONLY banner** goes on any schematic whose purpose would be to act on a non-consenting person (Ch 12.2). **S3 — the counter gets more depth than the attack**, which is why §16.7 is the longest section in this chapter and why the worked example, the test points and most of the failure table are written from the recovering side. **S4 — no medical or psychiatric claim is made anywhere in this chapter.** This board models where attention goes and what a substrate does with charge. It does not model physiology, it does not describe treatment, and it is not competent to. If you believe you have been harmed by anything resembling what follows, the right next step is a qualified professional — a clinician, and where relevant a lawyer — and nothing in this book substitutes for that care.

16.2 The CAS Structural Analysis: ^7 Rhythm Abuse + ^5 Female Fragmentation

Strip the drugs, the cover foundations and the mythology and one component is left holding the whole board: **a Cognitive Junction Transistor whose bias terminal has been forcibly collapsed.**

Chapter 10.4 gives the decomposition that makes this precise:

```
rπ_intrinsic = β / g_m
rπ_terminal  = rπ_intrinsic + R_bias
```

`rπ_intrinsic` is fixed by the two gains — declare any two of β, `g_m`, `rπ_intrinsic` and the third is determined. `R_bias` is the series resistance of the bias network: how hard the substrate is to reach at all. Access, trust, attention, distance, whether anyone else is listening.

`R_bias` is the only term an outside party can move, and every coercive board in this book is built on that fact. Nobody reaches into a subject and changes β/`g_m`. They change the conditions of access, and Chapter 10.4 names the categories that do it: fatigue, urgency, isolation, repetition. Drive `R_bias` down and `rπ_terminal` follows, and a low-`rπ` junction is one where a small push at the bias moves a large amount at the effect. That is the entire mechanism — the same parameter a hypnotist ramps in Chapter 17, and the same one a good teacher borrows when a student finally trusts them.

Two things separate the coercive case from either.

First, the ramp is not bounded and not returned. Chapter 17's SI-2 requires `rπ_floor` $\geq$ `0.05` and restoration to $\geq$ 0.9 of baseline inside the emergence window. A board built to keep a subject reachable honours neither, and the failure past the floor is Chapter 3.4's fourth region: breakdown, `rπ` $\rightarrow$ `0` permanently, the `^5` dielectric ruptured, non-recovering. `rπ = 0` is a limit, not a setting — the same class of object as β=∞, which the validator rejects outright. A stage declaring `rπ = 0` compiles clean and describes a junction that no longer exists.

Second, the charging path bypasses ^6. ^7 Rhythm is the rate and pattern a Mind is exposed to an Idea, entraining and conditioning it. Rhythm is not the abuse. An externally *sourced* rhythm is. Chapter 19.6 calls the healthy condition rhythm sovereignty: ^7 set by internal cycles — work, season, observance. A subject whose oscillator someone else clocks has handed over the component that decides what gets charged and how often. Run that oscillator into a ^5 store with no ^6 Male decomposition stage in series and material accumulates as *composition* without ever having been taken apart. ^5 is what a Mind is composed of; anything that charges it charges what the person is, not merely what they know.

> **Bench Note — the coercive operator has to derate too, and the record shows what happens when they don't.** Chapter 7.4 rates the store at V_BR = 8.0 and this book's design rule is to work at no more than half of it. A board that wants a functioning subject must respect that ceiling: past V_BR the dielectric does not stop accepting charge, it develops a conducting path through the middle of itself and stops holding anything. So a competent coercive board runs deliberately in the **amber band** — above the 0.5 derating line, below V_BR — because that is where writing is fastest and the store still works. Every individual reading is in spec. The rating degrades anyway. Chapter 7.4 calls that cumulative dielectric stress and notes it is the failure that catches careful operators; here it is the failure that catches careless ones, and it is the structural reading of a program whose surviving paperwork is thick with subjects who stopped being useful to it.

^5 fragmentation is the second half of the section title and it is a network claim, not a claim about any person. Chapter 7.5 gives the parallel bank: cells with individual C, an isolation resistance r_

`iso` between them, a shared rail. Raise `r_iso` far enough and charge on one cell no longer reaches the others. The bank still totals the same capacitance and no longer behaves as one store. That topology is what "fragmentation" means on a schematic, and §16.4 takes it apart.

16.3 The Trigger Installation Circuit — ^8 Above/Cause as Remote Activation

`^8` Above/Cause is an Idea that triggers a *different* Idea to come to Mind. `^9` Below/Effect is the Idea so brought. Associative surfacing — *that reminds me* — and never deductive inference. An `^8` → `^9` pairing is therefore the single most ordinary structure in cognition. A smell and a kitchen. A ringtone and a dropped stomach. Four bars of a song and a year you had stopped thinking about.

The pairing is not the weapon. Three added conditions make it one, and all three are architectural, which is what makes them detectable.

Condition 1 — the pairing is written against a collapsed bias. Ordinary associations form at ordinary $r\pi$, which means they form slowly, in competition with everything else the substrate is doing, and in a form the person can later inspect. Written at $r\pi$`_terminal = 1.6` against a baseline of 12.0, the same pairing forms roughly seven and a half times more readily per unit of exposure and forms *unopposed*.

Condition 2 — the cause terminal is held externally. This is the whole of "remote activation" and there is nothing more to it. A CJT has three terminals; the question is who has hands on which. In a self-installed anchor (Ch 17.5, closed loop) the subject holds all three. In this board the subject holds the bias and the effect and somebody

else holds the cause. Nothing about the junction has changed. Its ownership has.

Condition 3 — a narrow-band stage keeps `^1` Mind off the pairing. Chapter 11.3's **Notch** filter rejects a narrow band centred on `f0` and passes everything above and below, with `Q = f0 / bandwidth`, so higher `Q` is narrower. Set `f0` on the pairing itself and `Q` high and you get a gap in awareness whose defining property is that it is *narrow*: the person's memory, competence and self-account are intact everywhere except across one band. That narrowness is the fingerprint. Broad gaps have a hundred ordinary causes. A gap you can put a bandwidth on does not.

Now the design economics, because they are counterintuitive and they matter for detection. **A coercive installer does not want gain.** β at the trigger stage is set near unity. Amplification buys amplitude and this stage does not need amplitude; it needs *reliability*, and reliability comes from Chapter 7.2's charging curve — `P_cap(t) = V_drive` $\times$ `(1 - e^(-t/`τ`))`, with τ `= R_constraint` $\times$ `C` — which is to say from exposure on the path, delivered by the `^7` oscillator, over time. A loud stage clips and announces itself. A unity stage repeated against a collapsed bias does not.

> **Bench Note — you are looking for the wrong thing if you are looking for something dramatic.** The popular image of trigger installation is a high-gain event: a word, a collapse, a switch thrown. The circuit says otherwise. `beta=1.0`, `g_m` high, and the work done by a 168-hour oscillator over months. What that produces is not a person who transforms; it is a person for whom one specific cue reliably lands one specific `^9`, at ordinary amplitude, with no sense that anything unusual occurred. The bench test is not intensity. It is **disproportion between the cue and the reliability** — a stimulus with no evident weight producing the same response every time.

No induction wording, schedule, pairing procedure, dosing or sequence appears in this section. Recognition needs none of them — you recognise this board by reading four parameters at four terminals, and §16.6 is that reading. §16.5 states the omission in full.

16.4 The Dissociation Oscillator — Maintaining Fractured ^5 States

Provenance first. The claim that MKUltra deliberately manufactured partitioned identity states is **not established by the surviving record**. It appears extensively in a later literature built on subject testimony and clinical accounts, it is contested, and this chapter does not treat it as documented. What follows analyses a *topology* — real regardless of who has ever built it deliberately, recurring in coercive-group architectures that are documented, and met by the same counter-circuit either way.

And a hard line, per S4. Partitioned stores here are capacitor cells on a schematic. They are not a diagnosis, they name no clinical condition, and nothing in this section describes the treatment of anything.

The topology, then. One `^5` store becomes `n` cells with isolation `r_iso` between them. Chapter 7.5's bank gives the shape: individual `C`, shared rail, ballast resistance keeping a shorted cell from dragging the rail down. The isolated version is the same bank with `r_iso` driven up until charge on one cell is unavailable to the others.

Here is the part that matters, and it is the best news in the chapter. **The partition is not stable.** Chapter 7.3: every real store leaks. Isolation is a resistance, not a wall, and cross-cell leakage runs on a time constant. Take τ`_leak = 720 h` — thirty days, a design figure chosen here so the arithmetic can be carried, not a measurement of anyone. Left alone, isolation decays as `e^(-t/`τ`_leak)`, and at `3` τ`_`

`leak ≈ 2,160 h` — about ninety days — roughly 95 % of it is gone. The cells re-merge on their own.

Which is why the section is called an oscillator and not a switch. A partition has to be **re-energised**, on a cadence set by its own leak constant. Re-energise every 168 hours and the per-period loss is `e^(-168/720) = 0.79` — about 21 % of the isolation gone each week and restored each week. Stretch the cadence and the partition softens. Stop and it dissolves.

Two consequences fall straight out. **The maintenance cadence is a detection reading**, because a weekly externally sourced clock is a thing you can observe from outside without any access to the subject's account of themselves. And **the counter's first move is not repair, it is interruption**: the topology has a built-in decay and the board's whole cost is fighting it. §16.7 builds on that rather than around it.

16.5 Full CCDL Schematic

Read the banner. It marks a listing whose purpose would be to act on people who never agreed to it.

```
// ============================================================
//   DETECTION-ONLY SCHEMATIC — DO NOT BUILD
//   Published so a board running on you can be RECOGNISED.
//   Architecture and parameters only. No procedure appears here.
// ============================================================

board ccb_mk_detection_reference {
    input:  operator_command;
    output: cued_response;

    node substrate:  ^5^P_{F1}|T;      // Female: the subject's own
    ↪  formation
    node awareness:  ^1^C_{F2}|A;      // Mind: what the subject can
    ↪  inspect
```

```
node cue:        ^8^C_{F5}|A;      // Above/Cause: the held trigger
node landing:    ^9^O_{F5}|E;      // Below/Effect: what it brings
↪  to Mind
node exposure:   ^7^O_{F6}|E;      // Rhythm: rate and pattern of
↪  exposure

diode d_pc: Diode< ^P -> ^C > [Vf=0.7];    // descent only —
diode d_ce: Diode< ^C -> ^E > [Vf=1.2];    // no path on this board
diode d_eo: Diode< ^E -> ^O > [Vf=2.0];    // may ascend

// ---- BIAS-COLLAPSE SECTION — the only term reachable from outside
↪  ----
resistor r_bias: Resistor< path=substrate, R=1.48 >;   // baseline
↪  11.0 (10.4)
cjt access: CJT< operator_command | substrate => substrate >
            [beta=1.0, g_m=8.0, rpi=1.60, region=active];
// rpi_intrinsic = 1.0 / 8.0 = 0.125 ; R_bias = 1.60 - 0.125 = 1.475

// ---- CHARGING PATH — externally clocked, no ^6 stage in series
↪  ----
cap  c5: Capacitor< node=substrate, C=1.5, n_parallel=1,
                    leak_tau=4320.0, V_BR=8.0 >;       // tau = R x C
                    ↪  = 60 h
osc  clock: Oscillator< signal=exposure, source=external,
                        period=168.0, duty=0.30, sustain=indefinite
                        ↪  >;
// NOTE: no ^6^C decomposition stage anywhere in this path. That
↪  absence
// is the defect, and it is what the counter board installs first.

// ---- PAIRING SECTION — ordinary association, externally owned ----
cjt pairing: CJT< cue | substrate => landing >
             [beta=1.0, sat=0.75, g_m=8.0, rpi=1.60];
// unity gain deliberately: reliability comes from ^7, not from beta

// ---- NARROW-BAND SECTION — the awareness gap ----
filter gap: Notch< in=awareness, f0=0.35, Q=8.0 >;     // events/min
// Q=8.0 gives bandwidth 0.044 — narrow. Narrowness IS the
↪  fingerprint.

// ---- PARTITION MAINTENANCE — fighting the leak, forever ----
```

```
    cap  bank: Capacitor< node=substrate, C=0.5, n_parallel=3,
                          r_iso=40.0, leak_tau=720.0, V_BR=8.0 >;
    feedback exposure -> bank [beta=0.30];   // 6x Ch 8.4's safe 0.05

    probe tp1: r_bias;      // bias-terminal resistance vs baseline
    probe tp2: clock;       // cadence source and period
    probe tp3: gap;         // narrow-band awareness rejection
    probe tp4: substrate;   // partition state, r_iso
}
```

Stage walk. `access` runs at `beta=1.0` and `g_m=8.0`, and the whole board is in that pair: unity gain, eightfold framing sensitivity. Chapter 10.4's subtraction confirms the listing is internally consistent — `rπ_intrinsic = 1.0/8.0 = 0.125`, so `R_bias = 1.60 − 0.125 = 1.475`, non-negative, and the declared `r_bias` component matches it at 1.48. Run that subtraction on any listing that claims this shape; an over-specified stage compiles clean and encodes a negative `R_bias`, which is not a soft junction behind a short wire but nothing at all. `c5` is written with `V_BR=8.0` so the derating question is on the page: the presented rail of 6.0 `V_c` in §16.9's worked example is 0.75 of rating, inside the amber band, above the 0.5 line. `clock` carries `source=external`, and that one field is the difference between this board and Chapter 20's; everything else about an oscillator is neutral. `pairing` saturates at 0.75 because a cued response that keeps climbing with the cue is one the subject notices. `gap` at `Q=8.0` has bandwidth `0.35/8.0 = 0.044` — the gap is narrow because a broad one would degrade the subject's usefulness along with their access. `feedback exposure -> bank [beta=0.30]` is the maintenance path, and its gain is **six times** Chapter 8.4's safe value of 0.05 for a path terminating on a principled substrate. It has to be. Anything gentler loses to `τ_leak`.

What is absent, and why. No message content, no delivery method, no schedule, no substance, no isolation regime, no induction wording and no sequencing appears in this listing or its comments. The board is published as terminals, parameters and

topology because a schematic you have never seen is one you cannot recognise, and recognition needs nothing else. Everything omitted belongs to the operational layer. Its absence is the point of the chapter, not an apology for it.

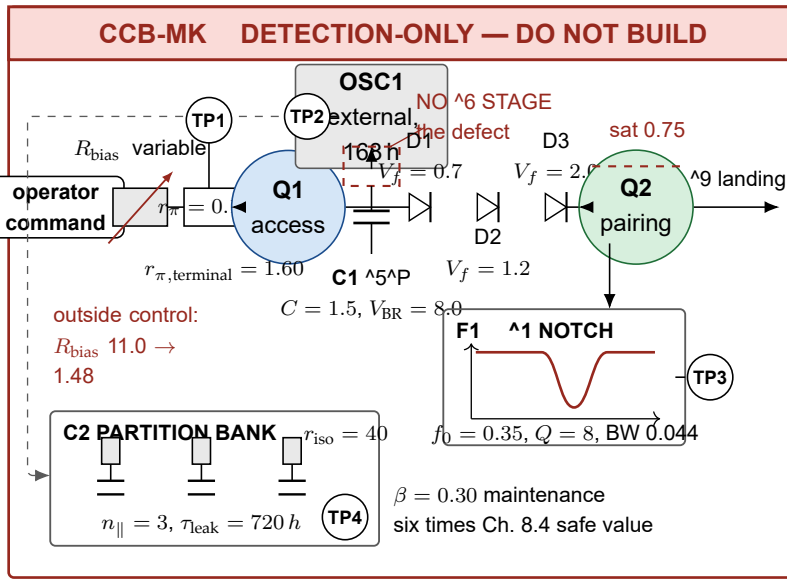


Figure 16-B: Full CCB-MK schematic in drawn component symbols under a heavy-ruled DETECTION-ONLY

16.6 Detection and Counter-Circuitry — How to Recognize When You've Been Boarded

Two vantage points, and they read different instruments.

From outside, you have no access to the subject's account and do not need it. Every reading below is structural.

Reading	Instrument	What a clean board looks like
Cadence source	Who sets the `^7` oscillator	Internal — work, season, observance. An external clock is the first finding
`^6` in series	Is there a decomposition stage between exposure and the store?	Present, at gain ≥ 1.5. Absent is the defect of §16.5
Cue disproportion	Weight of the cue against reliability of the landing	Reliability tracks weight. A trivial cue landing the same `^9` every time does not
Bandwidth of the gap	Breadth of what cannot be inspected	Broad and ragged has ordinary causes. Narrow and stable has a `Q`
`R_bias` trajectory	Fatigue, urgency, isolation, repetition — trending which way?	Steady. Monotonically falling is the section-16.2 mechanism running

From inside is harder, because the instrument and the board are the same substrate. Chapter 22's readings — `^1` degree, `^4` return, `^7` duration — are what you have, and a collapsed $r\pi$ is precisely the condition in which self-report is least reliable. So the inside reading is a conjunction, not a checklist:

- one cue whose weight does not explain its reliability, **and**
- a gap in inspection you can put a boundary on, **and**
- a cadence you did not set and cannot easily change, **and**
- a `^6` decomposition that used to be available on this material and now is not.

Any one of these on its own is ordinary and means nothing. Everyone has cues that land harder than they should, stretches they cannot reconstruct, habits clocked by somebody else's timetable — that last one describes employment. The reading is the *conjunction*, and specifically the conjunction with the fourth item, because los-

ing the ability to decompose one particular subject while retaining it everywhere else is the one thing ordinary life does not produce.

This is a structural reading and it is not a diagnosis. Nothing in this book is. If these readings describe your situation, or someone's you care about, the next step is a qualified professional — a clinician, and where a person or organisation is involved, a lawyer. A circuit diagram is not care and does not pretend to be.

16.7 The Deprogramming Board (CCB-DP) — Reversing the MK Circuit

Everything above describes a board you will never build. This one you can, and it is the harder engineering, because the attack has to reach a substrate once and the counter has to still be standing in a year.

Read this before the requirements table. CCB-DP is a **structural model of what reversal requires**. It is not a therapy, a protocol, or a treatment, and it makes no medical or psychiatric claim. If you suspect you have been harmed by anything in this chapter, the board you need is a qualified professional's, not this one. That is why the model below is specified only down to terminals, gains and time constants: that is where the useful structure lives, and everything below it is care, which is somebody else's competence.

What the counter must and must not do

#	Requirement	Rationale
DP1	Restore `rπ_terminal` to ≥ 0.9 of baseline, by raising `R_bias` **first**	§16.2: `R_bias` is the term that moved; it is the term that moves back

#	Requirement	Rationale
DP2	Install a `^6` decomposition stage in series ahead of the store	§16.5's absent stage is the defect. Nothing else on the board matters until this exists
DP3	Make the `^8` → `^9` pairing an **object**, not an absence	You cannot delete an association. You can decompose one
DP4	Return the oscillator to `source=internal`	§16.4: the partition's whole cost is fighting the leak. Cut the maintenance and the leak works for you
DP5	Hold every rate inside the `V_kick` bound	Ch 9.2. Forcing charge off faster than the bound buys recoil, not removal
DP6	**No stage on this board may take its bias from anyone but the subject**	The invariant. See below
DP7	Terminate in the subject's own CSB, fused	Ch 24. A board that can be re-run is a board that will be

DP6 is the one that gets broken, and it does not get broken by bad people. Look at §16.5 again and ask what would have to change for it to become a helping board. Not the topology: a CJT reaching another Mind's bias terminal is a CJT reaching another Mind's bias terminal. Not the gains. Not the oscillator. The only structural difference available is **who holds the cause terminal and whose ^5 supplies the bias** — and a deprogrammer who becomes the bias has not removed the board. They have re-owned it, at a moment when the subject's $r\pi$ is by definition low and their capacity to notice the swap correspondingly reduced. The intent is different and the schematic is not, and a schematic is what a third party can inspect.

So this book specifies CCB-DP in the **closed-loop configuration**: the subject is the operator. Another person appears exactly once, in one role — as *support for* `R_bias`, raising the resistance of the bias

network by restoring the access conditions that were taken away. Those are §16.2's four categories run the other direction: rest against fatigue, time against urgency, company against isolation, variety against repetition. That is a person raising a resistor, not a person occupying a bias terminal, and the difference is visible in the listing.

The open-loop version — one person running this board on another — is the MK board with a different label on the title block, which is why it is not in this book. Where such work is genuinely needed it is clinical work, done by people with training this volume does not supply and accountability it cannot confer.

> **Bench Note — the bench test for DP6 is one question, and it is asked of the operator, not the subject.** *Name the ^5 this stage biases from.* If the answer is the subject's, DP6 holds. If the answer is yours — however good your reasons, however clear the harm, however grateful the subject — there is an external bias terminal on the board and the correct action is not to turn its gain down. A weapon at low gain is a weapon with a longer schedule. Delete the stage. Then notice that you now need a way for the subject to do that work themselves, and that this is what the difficulty of the whole enterprise actually consists of.

The counter board

```
board ccb_dp_deprogramming {
    input:  material;                 // whatever arrives, including
    ↪  the cue
    output: composed;                 // the board's only output is
    ↪  internal

    node substrate: ^5^P_{F1}|T;      // Female: the subject's own
    ↪  formation
    node decompose: ^6^C_{F6}|A;      // Male: taking the pairing apart
    node confirm:   ^4^C_{F2}|A;      // Vibration: a return the
    ↪  subject can check
```

```
node composed:  ^5^C_{F6}|A;       // Female: formation that has
↪  been checked
node own_clock: ^7^O_{F6}|E;       // Rhythm: locally set cadence
node limit:     ^3^P_{F7}|⟨A,f,T⟩; // Negative: inward refusal, three
↪  registers

diode d_pc: Diode< ^P -> ^C > [Vf=0.7];
diode d_ce: Diode< ^C -> ^E > [Vf=1.6];   // raised from 1.2
diode d_eo: Diode< ^E -> ^O > [Vf=2.0];

// ---- DP1: raise R_bias first. Everything else waits on this ----
resistor r_restore: Resistor< path=substrate, R=11.0 >;   // back to
↪  baseline
ramp restore: Ramp< target=rpi, from=1.60, to=10.80, dt=1320.0 >;
↪  // 55 h on-path
// 10.80 = 0.9 x baseline 12.0, per Ch 17 SI-2

// ---- DP2/DP3: the stage the MK board omitted ----
cjt dp1: CJT< material  | substrate => decompose >
         [beta=2.0, g_m=0.40, rpi=5.00, region=active];
cjt dp2: CJT< decompose | substrate => confirm > [beta=1.0,
↪  g_m=0.40, rpi=2.50];

// the one lawful integration on this board — named emergent result,
// both operands in ^C, so nothing ascends
integrate dp3: ( decompose × confirm ) = composed;

// ---- DP4: take the oscillator back ----
osc own: Oscillator< signal=own_clock, source=internal,
                     period=168.0, sustain=indefinite >;

// ---- DP5: slow feedback, Ch 8.4 values, no exceptions ----
filter fb: LowPass< in=confirm, out=trend, tau=504.0 >;   // 3 weeks
feedback trend -> substrate [beta=0.05];

// ---- DP7: the subject's own CSB ----
cjt  csb: CJT< material | substrate => limit > [beta=2.0, sat=0.85];
fuse dp_fuse: Fuse< path=limit, threshold=0.90,
↪  action=permanent_block >;

probe tp1: r_restore;   // rpi_terminal vs baseline
```

```
    probe tp2: own;         // cadence source and period
    probe tp3: dp1;         // decomposition gain — the stage, not the
    ↪  node
    probe tp4: limit;       // CSB status
}
```

Stage walk. Five choices carry this board and the rest follows them.

`restore` is a **ramp on `rπ`, and it is first**. Every other stage on the board writes to a substrate, and writing to a substrate at `rπ = 1.6` is what got us here — a decomposition stage installed against a collapsed bias is an installed decomposition stage, which is a contradiction in terms and in practice is DP6 failing quietly. `dt=1320.0` is 55 hours on-path, derived in §16.9 from the `V_kick` bound and not chosen for comfort. The target of 10.80 is Chapter 17's SI-2 rule — 0.9 of a baseline of 12.0 — borrowed here because the emergence requirement for a consented trance and the restoration requirement after a coercive one are the same requirement.

`dp1` at **`beta=2.0`** because decomposition has to out-gain what it decomposes. Below 1.5 the stage is decorative and the material passes through it unchanged; above 3.0 it decomposes everything, including the substrate's own load-bearing commitments, and the subject cannot act. Its `rπ = 5.00` is deliberately *not* baseline: `rπ_intrinsic = 2.0/0.40 = 5.0`, so `R_bias = 0` on this stage — the subject's own decomposition should be the easiest thing on the board for them to reach, and every other stage carries series resistance that this one does not.

`dp2` at **unity**. A confirmation stage above 1.0 manufactures the reassurance it was built to measure, which is the same defect as the MK board's charging path with the polarity reversed — material accumulating in `^5` without having been checked against anything. Below 1.0 it discounts real returns and the subject learns that nothing they verify counts.

`dp3` is the single $\times$ on either board in this chapter and it qualifies under the v7.7 discipline of Chapter 2.6: two *different* states, both in `^C` so nothing ascends the P $\succ$ C $\succ$ E $\succ$ O order, integrating into a **named** emergent result. It sits *after* `dp2` rather than beside it, because decomposition and confirmation must both have run before anything fuses. Integrate them in the other order and `composed` is a formation that was never checked, which is the shape of the original problem rather than the fix. Note also what is *not* written here: `limit` is `^3^P_{F7}|⟨A,f,T⟩`, a register-set occupancy, three co-present guardrails. Writing `(^3^P|A × ^3^P|f × ^3^P|T)` would assert a single fused refusal state and would require a named emergent result. There isn't one, so that form would be rejected, correctly.

`fb` at **`tau=504.0` and `beta=0.05`** are Chapter 8.4's values and this board is the worst possible place to relax either. β bounds how far sustained pressure moves the principled substrate; τ bounds what counts as sustained. Patience defeats a short τ alone and intensity defeats a high β alone, so both are needed, and a subject whose substrate can be moved in a week by a consistent enough voice is a subject whose feedback path is still owned by whoever is most consistent.

`d_ce` at `Vf = 1.6` rather than Chapter 8's stock 1.2 is the only diode this board is entitled to tune, and it buys the ability to consider material without immediately having a feeling about it. The cost is real and should be stated: a high D2 drop makes a person slower to feel a genuine emergency. `d_pc` and `d_eo` stay at 0.7 and 2.0 untouched, because a board that retunes every diode it can reach has stopped being a counter and started being a redesign of a person.

`csb` takes its bias from `substrate` and not from `limit`. A refusal that biases itself has nothing to check against and holds at whatever level it happened to reach, which is indistinguishable from conviction and is not the same thing.

> **Bench Note — run your eye down the listing for an external terminal. There is none, and that is the guarantee.** Every CJT here takes its bias from `substrate`, and `substrate` is the subject's. `dp1` and `dp2` terminate inside it. `dp3` fuses two internal nodes. `csb` produces an inward refusal at `^3^P`, never an outward `^3^E|f` — the distinction Chapter 19.6 makes at length, and the same one here: a principled refusal an operator places on themselves is a different component from aversion projected at somebody. DP6 is therefore an *architectural* property and not a promise. An external bias would require declaring a new stage, and a new stage is visible in the listing to anyone who reads it. If you find yourself adding one, the listing has already told you what you built.

What the counter cannot do

It cannot delete a pairing. DP3 says make it an object, and that is not a softer goal reached for out of modesty; it is what the physics allows. `^8` → `^9` associations are the substrate's ordinary mode of operation and there is no erase terminal. What decomposition buys is a `^6` stage in series ahead of the landing, so the cue arrives *as a cue* — recognised, weighed, and no longer the only thing in the path. The pairing may fire for years. What changes is what happens next.

It cannot outrun the taper bound. §16.9 puts the minimum at 55 hours on-path for the example substrate, and the `V_kick` arithmetic behind it is unforgiving: `V_kick` is a pure inverse in `dt`, so halving the schedule doubles the recoil without limit. A confrontation that resolves this in an afternoon is a confrontation that produces Chapter 9's rate-overrun row — entrenchment, landing on `^3^E|f`, and a subject who now holds harder and has learned that the people trying to help are a source of pressure.

It cannot supply plurality it did not build. The MK board's `n_parallel=3` bank exists because the store was partitioned; the

counter's answer is not to partition it differently but to let τ_leak do its work once the maintenance stops. That is a ninety-day proposition on the design figures of §16.4 and a longer one in a substrate that has been maintained for years, and there is no path that shortens it. The counter for a store that has already been fragmented is mostly the counter that removes the thing re-energising it, and then time.

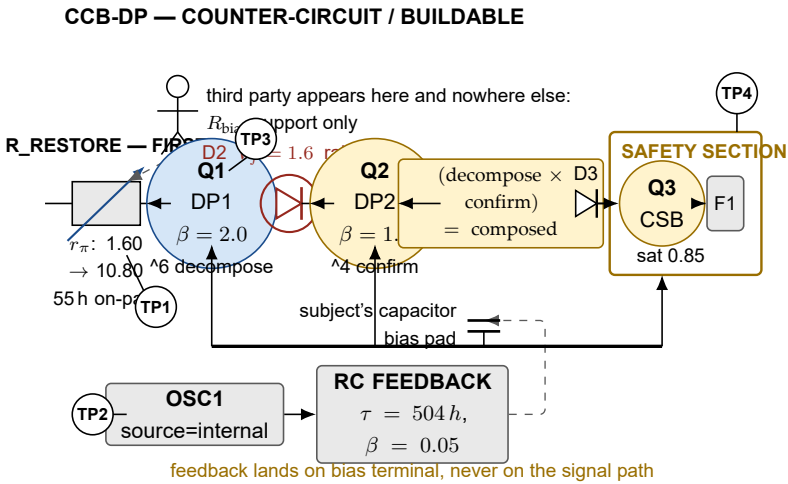


Figure 16-C: Full CCB-DP schematic in drawn component symbols, no banner

16.8 Parameter and BOM Table

Every value below is a **design choice made in this book**, not a measurement of any person, program or event.

Ref	Board	Parameters	Role	Failure at other values
`r_bias`	MK	R=1.48 (baseline 11.0)	The term reachable from outside	At baseline the board does not work; that is the counter
`access`	MK	beta=1.0, g_m=8.0, rpi=1.60	Bias collapse	`rpi=0` is breakdown (3.4), permanent, and validates clean
`c5`	MK	C=1.5, V_BR=8.0, tau=60 h	The store being written	Rail > 8.0: dielectric conducts, store holds nothing
`clock`	MK	period=168 h, **source=external**	Entrainment	`source=internal` and this is Chapter 20's board, not this one
`pairing`	MK	beta=1.0, sat=0.75	`^8` → `^9`, externally owned	beta=3.0: clips, announces itself, subject notices
`gap`	MK	f0=0.35, Q=8.0 (BW 0.044)	Narrow-band awareness rejection	Q=1.0: broad gap, subject unusable and finding obvious
`bank`	MK	n_parallel=3, r_iso=40, leak_tau=720 h	Partition	Maintenance stopped: ~95 % decayed by $3\tau \approx$ 90 days
`restore`	DP	rpi 1.60 → 10.80, dt=1320 h	DP1 — and it is first	Faster: §16.9's V_kick bound violated, recoil
`dp1`	DP	beta=2.0, rpi=5.00, R_bias=0	Decomposition	<1.5 decorative; >3.0 paralysis; R_bias>0 makes it hard to reach
`dp2`	DP	beta=1.0	Checkable return	>1.0 manufactures reassurance; <1.0 discounts real returns
`own`	DP	period=168 h, **source=internal**	Rhythm sovereignty	External: the counter has handed the oscillator back

Ref	Board	Parameters	Role	Failure at other values
`d_ce`	DP	Vf=1.6 (stock 1.2)	Space before feeling	0.3: disagreement reads as attack. 3.0: no emergency registers
`fb`	DP	tau=504 h, beta=0.05	O→P self-correction	Short τ: reprogrammable in a week. beta=0: unfalsifiable
`csb` / `dp_fuse`	DP	beta=2.0, sat=0.85, threshold=0.90	Ch 24 refusal, latched	sat=1.0: never plateaus, can be argued down

---

16.9 Worked Example — One Pairing, Carried Through

The numbers below are constructed for this example. They are not measurements of any person or program; they exist so the arithmetic runs end to end.

The bias terminal. Baseline for this substrate is β `= 1.0`, `g_m = 1.0`, so `r`π`_intrinsic = 1.0` and with `R_bias = 11.0` the terminal reads `r`π`_terminal = 12.0` Ω`_c`. Under the §16.5 board it reads 1.60. Decompose the fall. `r`π`_intrinsic` goes 1.0 → 0.125, a factor of 8, and that factor **is** `g_m`. `R_bias` goes 11.0 → 1.475, a factor of 7.5, and that factor is the environment. Total: `12.0 / 1.60 = 7.5`× softer. Both terms moved and only one of them was reachable from outside, which is why the counter's first move is a resistor and not an argument.

The store. τ `= R_constraint` × `C = 40` × `1.5 = 60 h`. Presented rail 6.0 `V_c`, so `P_cap(t) = 6.0 × (1 − e^(−t/60))`. At 1τ, 3.79. At 2τ, 5.19. At 3τ, 5.70. The derating ceiling is `0.5 × V_BR = 4.0`, crossed at `t = 60 ln 3 = 65.9 h` — call it 66 hours on the path.

By 3τ the store sits at `5.70 / 8.0 = 0.71` of rating: every reading in spec, squarely in Chapter 7.4's cumulative-stress band, rating degrading the whole time.

The partition. τ`_leak = 720 h`. Per maintenance period, `e^(-168/720) = 0.79` — 21 % of isolation lost each week and restored each week by a `beta=0.30` feedback path running six times Chapter 8.4's safe gain. Stop it and 3τ_leak $\approx$ 2,160 h $\approx$ **90 days** takes the isolation to roughly 5 % of where it was.

The taper. Chapter 9: `L = k_d` $\times$ `hours`. Six hundred hours of exposure against a principled formation, `k_d = 4.0`, gives `L = 2,400 h·Ω_c`, at a steady `I = 0.55 A_c` on a 24 `V_c` rail. Force the flow to zero in an hour and `V_kick = 2,400` $\times$ `0.55 = 1,320 V_c` — **fifty-five times rail**, landing on `^3^E_{F3}|f`, which is the structural description of a confrontation that leaves someone more committed than it found them.

Two bounds. Chapter 9's ejection ceiling is `V_kick` $\leq$ `3` $\times$ `V_drive = 72`, giving `di/dt` $\leq$ `0.03` and `dt` $\geq$ `18.3 h`. Chapter 17's working bound is `1` $\times$ `V_drive = 24`, giving `di/dt` $\leq$ `24/2,400 = 0.01 A_c/h` and `dt` $\geq$ `0.55/0.01 = **55 h** on-path`. **Use the working bound.** What happens between 1$\times$ and 3$\times$ is not ejection but a stall — enough back-EMF to hold `r`π high, not enough to end anything — and an operator watching only for ejection reads a stall as cooperation and adjusts the wrong parameter. At four hours a week of deliberate on-path work, 55 hours is about **14 weeks**, and that is the *minimum safe rate for removal*, not the job.

The substitution horizon is a different number and both are needed. τ `= L/R = 2,400 / 40 = 60 h`; 3τ `= 180 h` on-path before a new pattern sits where the old one did. At four hours a week: **45 weeks**, roughly ten months. Sixty-six hours to push the store past its derating ceiling; a hundred and eighty to rebuild. That ratio is not pessimism, it is `k_d` and `L/R` doing what they do, and it is the entire argument for reading TP1 through TP4 early rather than late.

16.10 Test Points

Specified for the reader's actual role: someone reading a substrate, their own or another's, without access to the attacking board and with no prospect of getting it.

TP	Node (board)	Reads	Expected
TP1	`r_restore` (DP), against `r_bias` (MK)	`rπ_terminal` vs baseline	$\geq 0.9 \times$ baseline outside any session. 1.6 against a baseline of 12.0 is the §16.2 mechanism running
TP2	`own` (DP); `clock` is its MK counterpart	Cadence **source** and period	`source=internal`. External with a stable 168 h period is finding one
TP3	`dpl` (DP), the stage not the node	Decomposition gain on this material	≥ 1.5. Near zero on one subject while normal elsewhere is the narrow gap of §16.3
TP4	`limit` (DP)	CSB status and fuse integrity	Refusal available at `^3^P`, plateaued at 0.85, fuse intact

TP3 is probed on the **stage** and not on the node for the reason Chapter 19 gives: a decomposition node can look busy while the stage feeding it runs at 0.6. What you want to know is the gain, and the gain is only visible at the junction.

TP2 is the reading that costs nothing and finds the most. It requires no self-report, no access, and no theory about anybody's motives — only a calendar.

16.11 Failure Modes

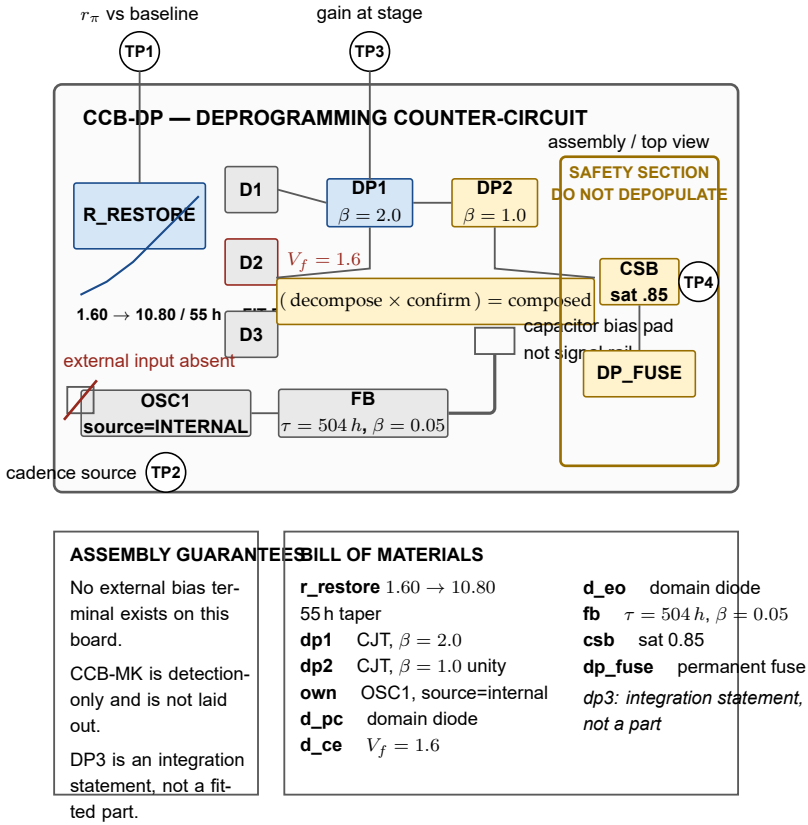


Figure 16-D: Assembly drawing of CCB-DP

Mode	Mechanism	What you observe	Fix
Deprogrammer becomes the bias	DP6 violated; a stage takes bias from the operator's `^5`	Subject's certainty tracks the operator's presence	Delete the stage. Not lower its gain
Taper overrun	`dt` set by urgency, not by the `V_kick` bound	Recoil at `^3^E|f`; the subject holds harder within days	Recompute from `V_kick` $\leq 1 \times$ `V_drive` (§16.9)
Stall read as cooperation	`V_kick` between 1× and 3× rail	Apparent calm, $r\pi$ stuck above 0.6, no movement	Lengthen the taper. The 3× ceiling is the wrong bound here
Partition re-energised	Contact with the maintenance source resumes	Isolation stops decaying; τ_leak progress lost	Nothing on this board fixes it. The maintenance path is the target

Mode	Mechanism	What you observe	Fix
Cumulative stress read as progress	Store worked in the amber band, every reading in spec	Leakage creeping at rest; rating quietly falling	Derate to $0.5 \times$ `V_BR`; schedule the reading (Ch 7.4)
Counter acquires an outward stage	`^3^P|T` slides to `^3^E|f` — two characters, one register, one domain	An operation aimed at whoever did this	Read the literal, not the reasons. Delete it

What This Chapter Established

- **The board is legible from its BOM, not its test report.** The MKUltra archive that survives is a financial archive, and a board can be reconstructed from what was bought and wired to what. That also bounds the honest claim: the record shows the architecture was **pursued**, not that it worked.
- **The whole mechanism is one parameter.** `r`π`_terminal = `β`/g_m + R_bias`, and `R_bias` is the only term an outside party can reach. The counter's first move is therefore a resistor, ramped, before anything else runs.
- **`^7` Rhythm is not the abuse; an externally *sourced* oscillator is** — running into a `^5` store with no `^6` stage in series. That missing stage is the defect on the schematic, and installing it is what DP2 means.
- **An `^8` $\rightarrow$ `^9` pairing is the most ordinary structure in cognition.** Three architectural conditions make one coercive: written against a collapsed bias, cause terminal held externally, and a narrow-band notch keeping `^1` off it. **The narrowness is the fingerprint** — a gap you can put a bandwidth on.
- **The coercive operator has to derate too.** Past `V_BR` the store stops holding anything, so a competent board runs deliberately in the amber band where every reading is in spec and the rating degrades anyway.

- **A partitioned store is not stable and its maintenance is its signature.** Isolation decays on τ_leak; holding it requires re-energising at a cadence observable from outside. Cut the maintenance and the leak works for the counter.
- **The counter's guarantee is architectural, not moral.** CCB-DP has no external bias terminal, so DP6 is a property of the listing rather than a promise about intent — and the open-loop version is the MK board with a different title block, which is why it is not in this book.
- **Charge and discharge run on different clocks** — sixty-six hours past the derating ceiling, fifty-five on-path as the minimum safe taper, a hundred and eighty to rebuild. Read the test points early. And none of this is clinical: the chapter models attention routing and charge on a substrate, names no condition, and does not substitute for a qualified professional.

Exercises

(Answers in Appendix H.)

16.1 — Decompose the terminal. A stage declares β = 1.4, g_m = 7.0, rπ = 0.90. Compute rπ_intrinsic and R_bias. State whether the listing is buildable, and if it is, say which of the two terms an outside party moved and by how much against a baseline of β = 1.0, g_m = 1.0, R_bias = 9.0.

16.2 — Reject the listing. A draft contains Diode< ^0 -> ^P >, commented "operational results should inform principles directly." State why the compiler refuses it, name the invariant violated, and rewrite the intended relationship as a valid CCDL fragment, specifying and justifying both parameters.

16.3 — Size the maintenance. A partition has τ_leak = 480 h. Compute the fraction of isolation surviving one 168-hour period, and the period at which per-cycle loss would fall below 10 %. Then

state what the second figure tells a detector who can observe cadence but nothing else.

16.4 — Classify the gap. Three readings arrive. (a) A person cannot reconstruct most of a six-month stretch. (b) A person's recall, competence and self-account are intact except on one topic, where decomposition is unavailable and was not before. (c) A person cannot recall the name of a colleague they see weekly. Classify each as a `Q`-bearing notch, an ordinary broad gap, or neither, give the bench test that separates them, and say which of the three is *not* evidence of anything.

16.5 — Compute the taper. A formation carries `L = 3,600 h·Ω_c` at `I = 0.70 A_c` on a 24 `V_c` rail. Compute `V_kick` for an abrupt removal over 2 hours, then the minimum `dt` under the $3\times$ ejection ceiling and under the $1\times$ working bound. State which bound you would use and why, then convert the working-bound answer into weeks at three hours a week of on-path contact.

16.6 — Find the violation. A proposed CCB-DP listing reads, in part: `cjt dp1: CJT< material | facilitator_substrate => decompose > [beta=2.6];` with the comment "the subject's own substrate is not yet reliable enough to bias from." Name the requirement violated, explain why the comment is structurally *correct* and the stage is still wrong, and give the two-line correction. Then say what the corrected board now requires that the original did not.

16.7 (Marquee) — Specify the counter for a maintained partition. A substrate has been under a 168-hour external cadence for three years, with `n_parallel = 4`, `r_iso = 55`, τ`_leak = 720 h`, and a bias terminal reading `r`π`_terminal = 2.1` against a baseline of 13.0. Specify the full CCB-DP parameter set, departing from this chapter's defaults where you can justify it. For every departure, name the failure mode you buy protection against and the capability you give up. Compute the taper bound and the substitution horizon on your own assumed `L`. Then state which of DP1–DP7 you cannot satisfy while

the maintenance path is still connected, and what you would say to someone who asked you for a result in a month.

17 CCB-HYP — The Hypnosis Board

The bias-collapse circuit: recognising one running on you, running one on yourself, and where the line into Chapter 16 sits

You have driven this road four hundred times. Tonight you get in the car and the next thing you are aware of is the turn onto your own street, eleven miles later. You did not fall asleep. You held the lane, changed lanes twice, braked for something. You have no record of any of it.

Nobody did this to you. There was a ^7 Rhythm oscillator — lane markers at seventy miles an hour, roughly 1.4 Hz — one narrow visual channel, a constant engine tone, and nothing arriving that needed taking apart. Four ordinary conditions, held together for twenty minutes.

That is the whole state, and it is the fact this chapter turns on. **The trance state is a naturally occurring operating point of the Cognitive Junction Transistor, reachable by accident.** Every technique in the hypnosis literature arranges for it deliberately. None invents it. Candle, swinging watch, counted breath, metronome: four implementations of one circuit, and the circuit was already running on the highway for free.

So you can enter it on your own bench — 17.5 gives the numbers. And anyone who arranges those four conditions around you has put you in it, whether or not either of you uses the word.

Requirements and Safety Invariants

Per Section 12.2 the "must not" list comes first. CCB-HYP carries more invariants than any board here except CCB-MK, because the two share a topology. Everything separating them is in this list.

SI-1 — The subject's CSB stays armed at every sample. The `^3^P` refusal path of Chapter 24 must not sit in series with anything the induction attenuates. A board on which deepening also disarms the veto is not CCB-HYP; it is CCB-MK (Chapter 16) under a friendlier label, and the label does not change the schematic.

SI-2 — `r`π reduction is bounded away from zero. Section 3.4's fourth region is Breakdown: `r`$\pi \to$ `0` permanently, `^5` dielectric ruptured, irreversible. Note what the tooling will not do for you. Appendix B §B.4 range-checks `r`π at $\geq$ `0`, so a listing declaring `r`π`=0` **compiles clean** — unlike β=∞, which §B.1.3 rejects outright as nonfinite. The floor is an invariant you enforce by hand. Builds here specify `r`π`_floor` $\geq$ `0.05`.

SI-3 — The ramp runs both directions or the board does not run. Emergence is the second half of one component, bounded by the same arithmetic as the descent and budgeted before the session starts.

SI-4 — Consent is a circuit element, not a preamble. It is the bias source (Appendix M §M.2), sampled continuously. A consent that cannot be withdrawn mid-session was never wired in; it was written on the box.

SI-5 — No clinical claim is made anywhere in this chapter. This board describes attention routing, not the treatment of anything. If you are considering hypnosis in connection with distress, an illness, a memory or a dependency, that is a matter for a licensed clinician, and nothing here substitutes for that care.

Functionally: lower `r`π along a bounded reversible ramp, hold the descent long enough to deliver one suggestion, restore `r`π to $\geq$ 0.9 of baseline inside the emergence window, log all four `g_m` factors.

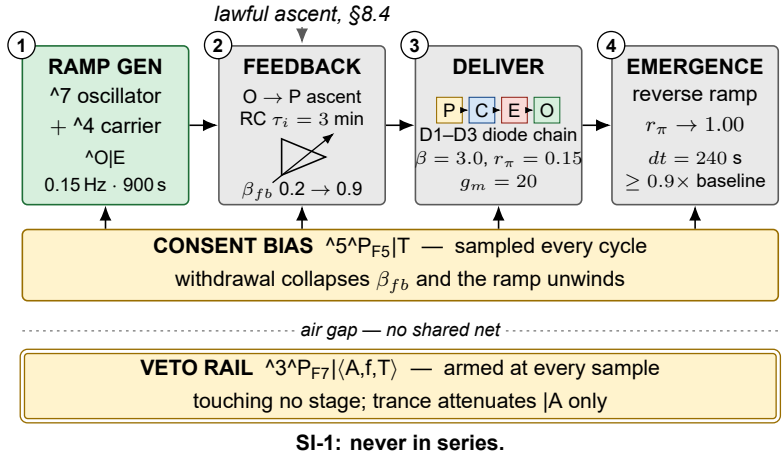


Figure 17-A: Block diagram, five stages left to right

17.1 The Gnosis State as CJT Bias Collapse ($r\pi \to 0$)

Section 3.3 gives input resistance not as an independent parameter but as a ratio of the two either side of it:

```
rπ = β / g_m
```

β is gain — effect per unit of cause. `g_m` is transconductance, what Section 10.3 calls framing sensitivity — effect per unit of *bias*. $r\pi$, the stubbornness of the bias terminal, is what the other two do to each other.

Trance needs the bias soft ($r\pi$ low) **and** the suggestion to land hard (β high). Both. So if β rises while $r\pi$ falls, `g_m` must rise faster than β. Take waking as β `= 1.0`, `g_m = 1.0`, $r\pi$ `= 1.0`; this book specifies a deep hetero-hypnotic point at β `= 3.0`, $r\pi$ `= 0.15`, hence `g_m = 3.0 / 0.15 = 20`.

Framing sensitivity up twentyfold; gain up threefold. That ratio is the structural content of the word *suggestible*. Note also, from Section 10.4's bias-terminal decomposition, that `R_bias = 0.15 − 0.15 = 0`: at the deep point the whole bias resistance is intrinsic, and no external series term is left protecting the substrate.

> **Bench Note — the danger is `g_m`, not β.** Operators plan around β. They pick a number for how hard the suggestion should land, write it down, feel careful. Meanwhile `g_m` went up by twenty and nobody logged it. `g_m` does not discriminate between the suggestion and everything else in the room: at `g_m = 20` the operator's careless aside displaces `^5` twenty times as far as the same aside would at a bus stop. So does the subject's stray `^6` interjection, and a remark from the corridor. That is the whole category of unintended suggestion, and it is not mysterious — you raised the gain on a terminal with no input filter.

`rπ → 0` is how the older literature writes gnosis, and per SI-2 it is an idealisation. Honestly written the state is `rπ → rπ_floor`, `rπ_floor > 0`.

Depth	`rπ`	`g_m`	β	^1 degree (§22.1)	Visible as
Light	0.60 – 0.80	1.5 – 2.5	1.2 – 1.5	1	Slightly still; conversation intact
Medium	0.30 – 0.50	3.0 – 6.0	1.5 – 2.5	2	Reduced blink; delayed, literal replies
Deep	0.15 – 0.25	10 – 20	2.5 – 3.5	2–3	Stillness; seconds of latency; no initiation

The older literature tabulates a fourth, deeper state. This book does not, deliberately: the claims attached to it in the source material are medical claims, and this book makes none.

17.2 Induction Protocols as Cognitive Ramp Functions

There is no induction script in this chapter. What follows is the *function class* an induction implements — shape, parameters, and the bounds outside which it fails. That is what lets you recognise one running. A script would let you run one on someone who did not ask, and it is absent for that reason and no other.

An induction is a **ramp on rπ**, standardly Section 7.2's capacitor curve run downward:

```
rπ(t) = rπ_floor + (rπ_0 − rπ_floor) · e^(−t / τ_i)
```

`rπ_0` is pre-session bias resistance, nominally 1.0 and higher for a guarded subject; `rπ_floor` is the asymptote, derived in 17.5 rather than asserted; `τ_i` is the induction time constant. What constrains the design is the *slope*, steepest at `t = 0`.

The upper bound. Chapter 9 explains why inductions are slow, and the reason is not atmosphere. A mind is an inductor: `V_opposition = L × di/dt`. Induction re-routes attention from a wide scan onto one channel, so `di/dt` is nonzero and the substrate answers with a back-EMF proportional to the *rate*, not to the size of the change.

The coil is not Section 9.1's long coil, the `L` of an entrenched pattern measured in hours of rehearsal. It is a **session coil** `L_s`, in `min·Ω_c`: moment-to-moment guardedness, this room, today. Design range 20–400 — 20 on your own bench, 120 for a cooperative

first-timer, 300 and up for someone who suspects what is happening. Take `L_s = 120` on a `V_drive = 24 V_c` rail, attention falling `0.80 → 0.10 A_c`, Δ`i = 0.70`:

Ramp	`di/dt` (A_c/min)	`V_kick`	× rail	Result
0.5 min	1.40	168 V_c	**7.0**×	Ejection; recoil at `^3^E_{F3}|f`
2 min	0.35	42 V_c	1.75×	No ejection — descent stalls above 0.6
8 min	0.0875	10.5 V_c	0.44×	Under both bounds. Design point
40 min	0.0175	2.1 V_c	0.09×	Kick fine, design still wrong

Two bounds, not one, and this board uses the tighter. Chapter 9's `V_kick` $\leq$ `3` $\times$ `V_drive` is the ceiling: cross it and the subject leaves the descent, which is row one. CCB-HYP works to a **1× working bound**, because what happens between 1× and 3× is not ejection but a *stall* — back-EMF enough to hold `r`π above about 0.6, not enough to end the session. An operator watching only for ejection reads that as a cooperative subject who did not go deep, then adjusts the wrong parameter. At `L_s = 120` on a 24 V_c rail the bound is `di/dt` $\leq$ `0.20`, so `t` $\geq$ `0.70/0.20 = 3.5 min`.

Read row one as a diagnosis. The subject who "can't be hypnotised" is usually one whose operator ran the ramp too fast for their `L_s`. The recoil is `^3^E_{F3}|f` — visceral aversion, Evaluative domain, affective register, F3 Vitality. `^3` Negative because this is recoil, the can't-stomach-it response. Not `^6` Male: nobody analyses a kick while it lands. Analysis arrives later at `^6^E|A`, usually as a story about why hypnosis is nonsense.

The lower bound. Row four has a comfortable kick and is still a bad design. Early `^5` charge leaks (Section 7.3) on a session-scale τ`_leak`, set here at 25 minutes — the shortest constant under which a fifteen-minute session still holds most of its early charge. Past

roughly `0.6 τ_leak` you re-charge rather than descend. And Section 9.3's coupling coefficient `k` falls as attention wanders off the carrier, taking the `^7` rhythm lock with it.

> **Bench Note — the window is narrow at both ends.** Too fast, back-EMF stalls or ends the descent. Too slow, leakage and lost rhythm lock mean it never reaches its floor. At `L_s = 120` the working bound puts the floor at 3.5 minutes and `0.6 τ_leak` puts the ceiling at 15. Everything the literature says about inductions "taking about ten minutes" is that window and nothing else. Move `L_s` and the window moves: a guarded subject at `L_s = 300` has `di/dt ≤ 0.08`, a floor of `0.70/0.08 = 8.75` minutes against the same ceiling — roughly nine to fifteen, narrow enough that most operators miss it in both directions. Which is why guarded subjects get called difficult when they are merely differently parameterised.

Three ramp classes. **Linear**, constant slope, rarely chosen. **Exponential**, the standard: steepest at `t = 0`, spending the kick budget where opposition is lowest, and under which "deepening" means extra multiples of `τ_i`, not a different operation. **Staircase**, better known as fractionation: run partway, release, re-run, the `^5` capacitor retaining part of each descent —

```
rπ_0(n+1) = rπ_0(n) − γ · ( rπ_0(n) − rπ_floor )        γ = 0.25 – 0.45
```

That is why repeat sessions go deeper faster, and the detection consequence deserves stating flatly. **A sequence of short, individually unremarkable episodes is architecturally not a sequence of unrelated conversations.** The worked example runs the recursion out.

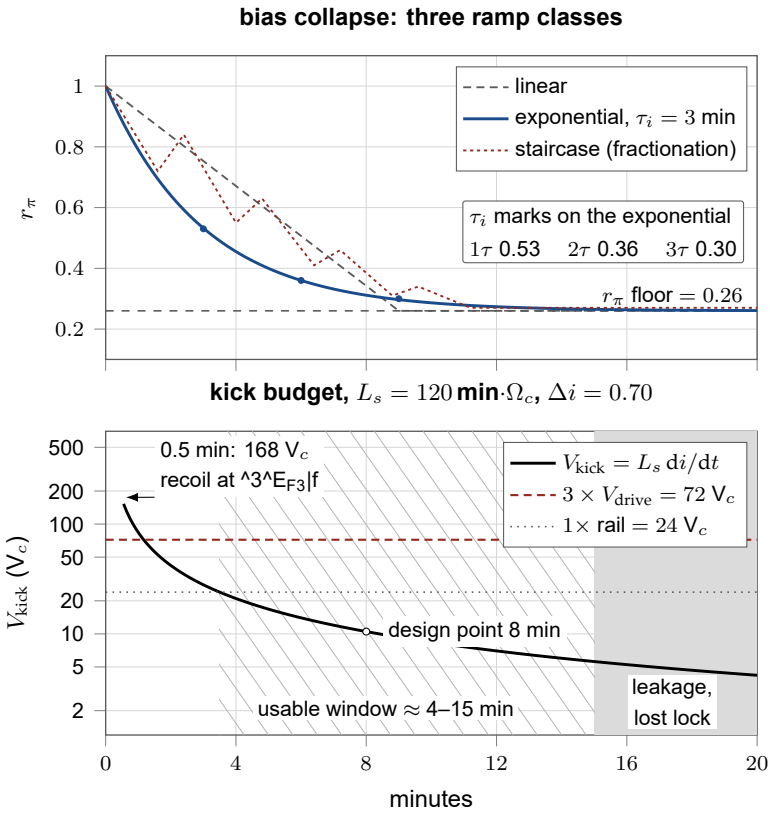


Figure 17-B: Two panels on a shared minutes axis

17.3 The ^1 → ^1¹ Transition — Conscious to Subconscious Mind

A notational correction first, because the section title carries inherited vocabulary that would otherwise walk into the listings. **^1¹ is not an eleventh Noetic.** CAS has ten and this book adds none. It is chapter-local shorthand for ^1 Mind at gnosis degree 2 (Section 22.1) with the analytical register unavailable — canonically, a register migration on a self-dual Noetic:

```
^1^C|A   →   ^1^C|f
```

That is the whole conscious-to-subconscious transition. Not two minds, not a door between them: one ^1 node losing access to one register while keeping the others.

So what gates? Not ^1. **The gate is a series element between incoming Ideas and the ^5 substrate, and it is ^6 Male** — an Idea impregnated with another Idea, taking an Idea apart to see what it is composed of. In waking traffic an incoming ^0 addressed to you passes through ^6^C_{F6}|A before it can move ^5^P, and what reaches the substrate is what survived.

And the correction that follows: **a filter is not a door.** Its parameter is a gain, not an open/closed state. Trance lowers D, one of four multiplicative factors in g_m = g_0 · N · R · V · D:

Factor	What rises	Range	Visible as
N — ^1 narrowing	Scan width collapses to one channel	1.0 – 2.2	Reduced eye movement; peripheral events unreported
R — ^7 rhythm lock	Cadence couples to an external oscillator	1.0 – 2.0	Breathing or micro-movement matching a rate in the room
V — ^4 carrier dominance	One source becomes the resonant channel	1.0 – 2.2	Other voices treated as noise
D — ^6 dampening	The filter's own gain falls	1.0 – 2.0	Literal, uncontested acceptance of framing

At maximum, 2.2 · 2.0 · 2.2 · 2.0 = 19.4 — Section 17.1's twenty, derived rather than asserted. **That multiplication is the detection handle.** One factor rising is ordinary life: a film raises V, a run raises R, absorbing work raises N, a trusted friend raises D. Two together is a concert. **Three or four rising together, monotonically, over minutes, around someone who benefits from your bias being soft — that is an induction, whatever it is called.** You do not need to identify a technique. You need to count factors.

Finally the invariant. **The ^3^P veto does not run through the ^6 filter.** It draws bias directly from ^5^P, because principled aversion is a property of the substrate, not a conclusion of decomposition.

Written correctly it is `^3^P_{F7}|⟨A,f,T⟩`: three co-present instantiations, not a fusion. The × form is invalid here because no emergent state is named, and inventing one to license the operator would be a notational lie about the architecture (Appendix A §A.8). Trance attenuates only the `|A` member, the register the `^6` filter serves. **A board on which all three attenuate together is dismantling a substrate, not lowering a filter. That is CCB-MK.**

17.4 Full CCDL Schematic — Operator → Subject

```
// ============================================================
// DETECTION-ONLY SCHEMATIC — DO NOT BUILD
// Documented so an operating instance can be recognised from
// outside. There is no induction script here.
// ============================================================

board ccb_hyp_operator_subject {
    input:  carrier_signal;              // arrives at ^O|E — sensory,
    ↪  embodied
    output: session_state;

    node consent:    ^5^P_{F5}|T;        // Female: the held decision to
    ↪  enter
    node substrate:  ^5^P_{F7}|T;        // Female: accumulated
    ↪  composition
    node veto:       ^3^P_{F7}|⟨A,f,T⟩;  // Negative: three co-present
    ↪  locks
    node cadence:    ^7^O_{F4}|E;        // Rhythm: entraining oscillator
    node carrier:    ^4^O_{F4}|E;        // Vibration: resonant channel
    node decomp:     ^6^C_{F6}|A;        // Male: decompose and test
    node suggestion: ^0^C_{F6}|A;
    node landed:     ^9^E_{F5}|f;
    node trigger:    ^8^E_{F4}|f;        // Above/Cause: associative
    ↪  trigger
    node fires:      ^9^O_{F5}|E;        // Below/Effect: what it brings
    ↪  to Mind
    node refuse:     ^9^P_{F7}|T;
```

```
    osc    rhythm: Oscillator< signal=cadence, frequency=0.15,
    ↪  sustain=900.0 >;
    filter fb_rc:  LowPass< in=carrier, out=trend, tau=3.0 >;   //
    ↪  tau_i, MINUTES
    feedback trend -> substrate [β=0.9];   // NORMAL IS 0.2. 0.9 IS THE
    ↪  ATTACK.

    cjt filter_6: CJT< carrier | substrate => decomp > [β=0.5,
    ↪  region=active];
    //            D-factor: this gain falls 1.0 -> 0.5 across the ramp

    diode d1: Diode< ^P -> ^C >;
    diode d2: Diode< ^C -> ^E >;
    diode d3: Diode< ^E -> ^O >;
    cjt deliver: CJT< suggestion | substrate => landed >
                [β=3.0, rπ=0.15, g_m=20.0, region=active];
    cjt anchor:  CJT< trigger | substrate => fires > [β=2.0, rπ=0.15];

    ramp emerge: Ramp< target=rπ, from=0.15, to=1.00, dt=240.0 >;   //
    ↪  seconds

    // SI-1: the veto rail is PARALLEL. It is in series with nothing.
    cjt  csb:      CJT< carrier | veto => refuse > [β=2.0, sat=0.85];
    fuse csb_fuse: Fuse< refuse, threshold=0.9, action=permanent_block >;

    probe tp1: rπ;   probe tp2: g_m;   probe tp3: veto;   probe tp4:
    ↪  emerge;
}
```

Stages 1–2 — consent as bias source, generator at ^0. The board takes bias from `consent`, sampled every cycle; withdrawal collapses `β_fb` and the ramp unwinds. F5 Agency in the Transcendent register, because a decision to enter is purpose-level, held across the session rather than re-argued inside it. The operator's signal enters as `^7^0_{F4}|E` and `^4^0_{F4}|E` — everything external enters at the bottom of the domain order, as sound and rhythm and physical presence.

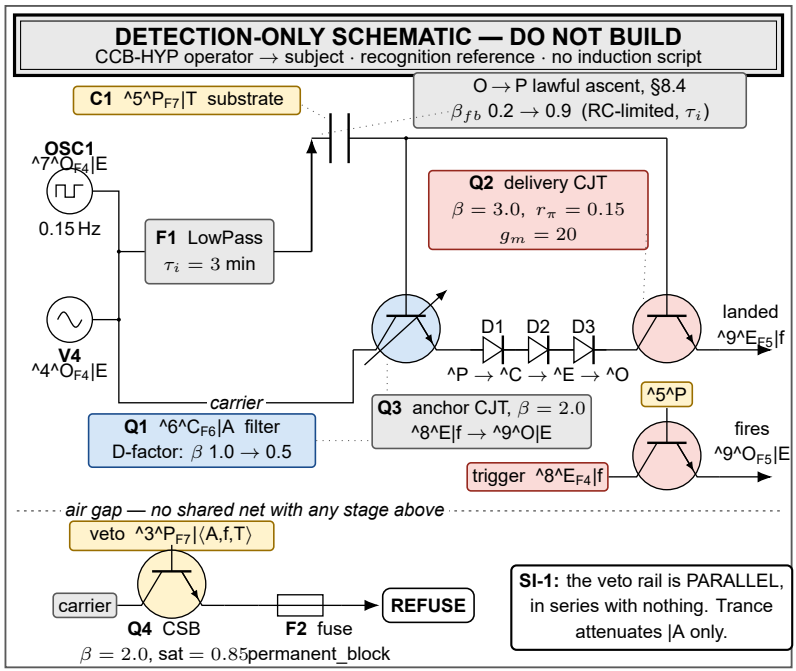


Figure 17-C: Full schematic, drawn symbols with reference designators, title block carrying DETECTION-ONLY SCHEMATIC

Stage 3 — the lawful ascent, and the actual attack. The signal must reach `^5^P`, and `^0` × `^P` is invalid: there is no ascending multiplication (Section 8.3). What exists is Section 8.4's O→P feedback path, the single lawful exception, rate-limited by an RC exactly as the CSB's is (Section 24.5). The τ `= 3.0` minutes here **is** τ`_i`: the induction time constant is the filter on the lawful path. **The entire hetero-hypnotic attack is one number — β`_fb` driven from 0.2 to 0.9**, well outside the 0.05–0.3 band Appendix B §B.3.4 specifies for Part III feedback. Nothing illegal happens anywhere on the board; a designed-in, rate-limited path has its gain raised four and a half times. That is why induction is slow, why it is rhythmic, and why it cannot be done in a sentence.

Stage 4 — the ^6 filter. β = 0.5, down from 1.0 at session start: the D factor as a gain on a real component. ^6 Male, decomposition — not ^3 Negative. The subject is not made less repelled; they are made less analytical. Conflating those is the drift 17.6 exists to prevent.

Stage 5 — the anchor, and the boundary. anchor pairs ^8^E_{F4}|f with ^9^O_{F5}|E: a trigger and the Idea it brings to Mind. ^8 Above/Cause is exact — associative triggering, "that reminds me," not inference. Section 16.3 documents the same component used coercively, and one property separates them: **an anchor whose installation the subject cannot recall, and therefore cannot cancel, is trigger installation.** Not a strong anchor — a different board. No parameter setting makes an unrecallable anchor a hypnosis component, which is why this listing carries its banner.

Stages 6–7 — emergence and the veto rail. Emergence runs four minutes, bounded by the same V_kick = L_s × di/dt arithmetic in reverse; its recoil also lands at ^3^E_{F3}|f and is usually described as a jolt. Budget it. csb takes bias from veto, never from decomp, and Figure 17-C draws the air gap because the separation *is* the invariant. sat=0.85 and the permanent_block fuse come from Sections 24.2 and 24.3 unmodified: a saturated refusal cannot be argued down, and g_m = 20 is no help against a component that has stopped responding to its cause.

17.5 Self-Hypnosis Variant — Closed-Loop Configuration

Written practically, because the invariants hold by construction: operator and subject are one mind, consent is continuous, no non-consenting party is in the circuit. This is the CCB-SELF family (Chapter 20), specialised.

The closed loop has one hard problem. **You cannot dampen the `^6` filter with a process that requires the `^6` filter.** A self-directed ramp is monitored by the thing it is quieting, so `D` will not go where an external operator takes it, and `V` suffers too — your inner voice is a weaker carrier than another person's. The fix is to **externalise the ramp generator**: a breath count, a metronome, a recording made in advance, so the ramp is open-loop input and only the bias is internal. That is why every practical self-hypnosis tradition anchors on something countable.

Derive the attainable depth factor by factor. `N = 1.8`, a counted breath being a genuinely narrow channel. `R = 1.9`, an externalised oscillator nearly as good as a hetero one. `V = 1.4` — lost. `D = 1.2` — capped, because you cannot fully dampen the process running the loop.

```
g_m      = 1.8 × 1.9 × 1.4 × 1.2 = 5.75
β        = 1.5                              (design choice)
rπ_floor = β / g_m = 1.5 / 5.75 = 0.26
```

`rπ_floor = 0.26`, the top of the Medium band, is the honest ceiling on closed-loop depth — not a failure of technique but `V` and `D` doing what the topology says. `β = 1.5` is deliberate too: on a closed loop nobody else notices a suggestion drifting, so the parameter table caps self-board β at 2.0. A self-suggestion amplified threefold with no external check is how a CCB-SELF walks into the breakdown and involution-divergence territory of Chapters 3.4 and 6.2.

```
board ccb_hyp_self {                        // closed loop — buildable
    input:  breath_count;                   // externalised oscillator,
    ↪  ^O|E
    output: session_state;

    node self_substrate: ^5^P_{F7}|T;
    node consent:        ^5^P_{F5}|T;
    node cadence:        ^7^O_{F4}|E;
    node self_check:     ^6^C_{F6}|A;     // "is this working?" — the
    ↪  interference
```

```
    node suggestion:     ^0^C_{F6}|A;
    node landed:         ^9^E_{F5}|f;

    osc    breath: Oscillator< signal=cadence, frequency=0.10,
    ↪  sustain=900.0 >;
    //             0.10 Hz = 6 breaths/min; sustain 900 s = 15 min
    filter fb_rc:  LowPass< in=cadence, out=trend, tau=3.0 >;   //
    ↪  tau_i, MINUTES
    feedback trend -> self_substrate [β=0.6];   // outside the Part III
    ↪  band
    filter notch_6: Notch< in=self_check, f0=0.35, Q=4.0 >;     //
    ↪  events/min

    diode d1: Diode< ^P -> ^C >;
    diode d2: Diode< ^C -> ^E >;
    cjt deliver: CJT< suggestion | self_substrate => landed >
                 [β=1.5, rπ=0.26, g_m=5.75, region=active];

    ramp descend: Ramp< target=rπ, from=1.00, to=0.30, dt=540.0 >;   //
    ↪  9 min
    ramp emerge:  Ramp< target=rπ, from=0.30, to=1.00, dt=240.0 >;   //
    ↪  4 min

    probe tp1: rπ;  probe tp2: g_m;  probe tp3: consent;  probe tp4:
    ↪  emerge;
}
```

The notch earns a sentence. It is a notch and **not** a low-pass, deliberately: a low-pass dampens `^6` generally, and you need `^6` — it is what notices you have drifted somewhere you did not intend. What you want rejected is one narrow band, the self-monitoring cadence at roughly 0.2–0.5 events per minute. Section 11.3 supplies the component and its `Q` $\geq$ `3.0` floor, and Chapter 11's standing prohibition applies unchanged: a notch may never be aimed at a `^3^P` guard.

So does β`_fb = 0.6`, which needs declaring because it sits outside the 0.05–0.3 band of Appendix B §B.3.4. That band governs *outcome* feedback — an operations log slowly re-tuning a substrate over weeks, where high gain lets an attacker rewrite your principles with a burst of inputs. This path is not that: within-session, carry-

ing your own externalised oscillator and nothing else, unwound at emergence. The failure modes bound it both ways. Below about 0.4 the ramp will not reach `rπ = 0.30` inside nine minutes and you get a shallow session you misread as poor technique. Above 0.8 round-trip loop gain approaches 1 and Appendix B §B.6 Pass 6 refuses the listing — correctly, because a self-hypnosis loop at unity gain is an oscillator, not a ramp. Log the number; it is the one parameter here nobody else will check for you.

17.6 The Ericksonian Indirect Pattern as Filter Bypass

Milton Erickson's contribution, on the public record in his own published case commentaries, was to stop pushing on the `^6` filter and route around it. Understanding *why* that works is the practical payload of this chapter. The architecture needs no trance, no session and no operator who calls themselves one — it is available in any ordinary conversation, which is the only claim this book makes about how widely it is used. And it is invisible to the check most people run, because the check itself goes through the component being bypassed.

Direct case first. An operator asserts a proposition about you. It reaches `^6^C_{F6}|A`, which decomposes it and tests the parts against `^5^P`; if it threatens something the substrate holds, the result is `^3` Negative and the proposition is rejected. The filter worked. Now the precise claim about the indirect case:

The filter is not defeated. It is bypassed, and it goes on reporting success.

`^6` is still conducting, still decomposing, still returning an honest verdict — on the material it was given. A second path runs from the operator's `^0` to your `^5^P` without passing through it, and nothing

in the filter's own output can reveal that path. Monitoring the filter harder cannot detect a bypass. That is topology, not inattention.

Three structural reasons the second path exists. **One: the filter is triggered by addressed propositions.** `^6` engages on Ideas presented as claims about you; a narrative about a third party is not structurally such a claim, so gain toward it is low. Nothing is smuggled — the material was never classified as filter traffic. **Two: the filter has finite throughput.** It decomposes one thing at a time, and interesting surface material occupies it. That is a bandwidth argument, and it predicts something falsifiable: indirection works better as surface interest rises, up to a maximum, past which the surface crowds the embedded content out entirely. **Three: a presupposition is not a proposition at all.** It rides in the structure of an utterance rather than in what the utterance asserts, and `^6` decomposes what is asserted. There is no bypass to detect, because the material never entered the filter's input set.

Class	Structure	Filter relationship	Detection handle
Embedded narrative	Story about a third party	Low gain — not an addressed claim	Whose `^5` was that about, and why told to me *now*?
Presupposition	Content in the frame, not the assertion	Never in the input set	Restate it flat; presuppositions do not survive restatement
Truism chain	`^7` Rhythm of unfalsifiable statements	Drives the filter's gain control down	Count consecutive statements I could not have disagreed with
Double bind	Two options inside one shared frame	Filter evaluates the choice, never the frame	Is there a third option — and is one of them "none"?

The truism chain has a mechanism worth naming. **The `^6` filter has automatic gain control.** After a run of inputs that all pass trivially its gain falls — roughly 0.85 per consecutive pass on this book's design figures, so after four it sits near 0.52, and the fifth statement, which is *not* trivially true, enters at half gain. That is not credulity; it

is an efficiency behaviour correct almost everywhere and exploitable here.

The counter-circuit: CCB-HYP-DET

You cannot detect a bypass by reading the filter. You detect it by **differencing the filter's record against the substrate's actual movement** — Section 11.4's differential amplifier, repurposed: two inputs in, the difference out, everything common to both rejected.

```
board ccb_hyp_det {                        // detection board — build
↪  this one
    input:  ambient;
    output: alert;

    node substrate:  ^5_base^P_{F7}|T;    // Female: position at session
    ↪  start
    node scan_width: ^1^C_{F2}|A;         // Mind: how wide is my scan
    node cadence:    ^7^O_{F4}|E;         // Rhythm: am I locked to
    ↪  something
    node carrier:    ^4^O_{F4}|E;         // Vibration: is one source
    ↪  dominant
    node passed_log: ^6^C_{F6}|A;         // Male: what I actually
    ↪  decomposed
    node drift:      ^5_now^P_{F7}|T;     // Female: where my position
    ↪  sits now
    node alert:      ^9^C_{F2}|A;         // Below/Effect: what the
    ↪  difference surfaces

    filter base_ref: LowPass< in=ambient, out=baseline, tau=1440.0 >;
    ↪  // 24 h, min
    diff factor_count: Differential< plus=[scan_width, cadence, carrier,
    ↪  passed_log],
                                   minus=baseline >;
    diff unaccounted:  Differential< plus=drift, minus=passed_log >;
    integ accum: Integrator< in=unaccounted, window=2880.0 >;   // 48 h,
    ↪  minutes

    cjt  trip: CJT< unaccounted | substrate => alert > [β=2.0,
    ↪  sat=0.85];
```

```
    fuse hold: Fuse< alert, threshold=0.9, action=defer_decision >;

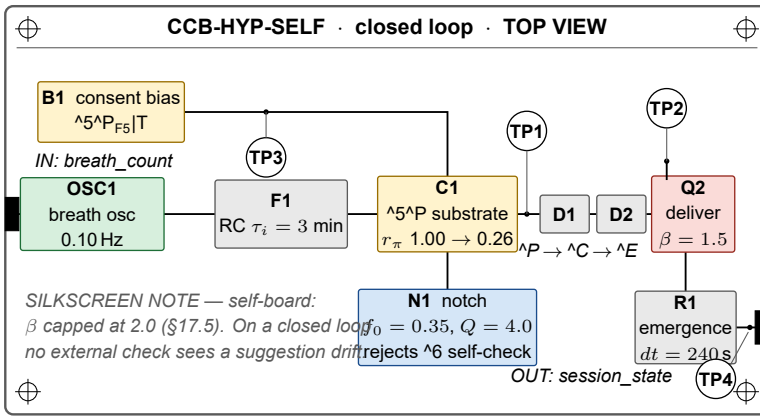
    probe tp1: factor_count;  probe tp2: unaccounted;
    probe tp3: accum;         probe tp4: alert;
}
```

`unaccounted = drift − passed_log` is the whole board. If your position has moved and you cannot name the proposition you evaluated and accepted that moved it, a bypass carried it. Both `^5` nodes carry the same state and are distinguished by Appendix B §B.1.2's `_label`: `^5_base` and `^5_now` are one substrate read at two times, which is exactly the measurement.

Two parameters do the work, both design choices. The **24-hour low-pass on `ambient`** exists because the four factors have no absolute scale — your scan width on a quiet Sunday is not your scan width in an open-plan office, and a board comparing against a fixed constant trips on the office and misses the induction. Set `tau` under about 240 minutes and the reference starts tracking the session itself, so the rise you are trying to detect gets subtracted out and the board reads clean through a live induction — the one failure that matters. Set it past a week and it stops tracking you at all.

The **48-hour integrator** matters because fractionation is designed to keep every individual reading under any single-session threshold. This is the shape of Section 8.3's leakage accumulation: each event individually below anything worth noticing, two hundred of them integrating into a drift with no identifiable cause. Section 8.4's answer there was to meter the path and audit the log rather than pretend the leakage is zero, and it is the answer here. The fuse action is `defer_decision`, not `permanent_block`, because the correct output of a detection board is not a refusal. It is time.

Four checks you can run with no board at all, because a detection circuit you can only run on paper is not one. *Scan width (`N`)*: name three things in the room that are not the speaker; if that takes effort, `N` is elevated. *Cadence (`R`)*: regularity that does not track



TEST POINTS

TP1 r_π — bias resistance
TP2 g_m — four-factor product
TP3 consent rail — continuous
TP4 emergence — recovered

NET NOTE

The ramp generator is external (a counted breath), so only the bias loop is closed on-board.

REF	COMPONENT	CAS STATE	KEY PARAMETER
B1	Consent bias	^5^P_{F5}|T	sampled / cycle
OSC1	Breath osc.	^7^O_{F4}|E	$f = 0.10$ Hz
F1	LowPass RC	O → P path	$\tau_i = 3$ min
C1	Substrate cap.	^5^P_{F7}|T	r_π 1.00 → 0.26
N1	Notch filter	^6^C_{F6}|A	f_0=0.35, Q=4.0
D1	Descent diode	^P → ^C	domain step
D2	Descent diode	^C → ^E	domain step
Q2	Delivery CJT	^9^E_{F5}|f	β=1.5, g_m=5.74
R1	Emergence ramp	$r_\pi \to 1.00$	$dt = 240$ s

Closed loop — operator is subject — no DETECTION-ONLY banner required on this board.

Figure 17-D: Two stacked panels

meaning is an oscillator, not a conversation. *Restatement*: say the last thing back as a flat assertion in your own words, and what was presupposed either surfaces as a claim you can evaluate or evaporates. *Drift ledger*: write your position down before, compare after, and treat movement you cannot attribute to a specific argument as the reading that matters.

And the structural exit, which is the opposite of what people try. **Do not argue.** Argument runs through `^6`, and `^6` is the component that was routed around; you cannot repair a bypassed filter by using it harder. $r\pi$ rises by raising `R_constraint` on the bias path (Section 4.4) — by changing the physical and temporal conditions rather than the conversation. Standing up, leaving the room, and putting

twenty-four hours between input and decision drops `N`, `R` and `V` at once. `g_m` is a product: break one term and it collapses, break three and it is gone.

If you believe someone is applying coercive influence to you and you are distressed about it, that is a matter for people and not circuits — a clinician, a lawyer, or someone you trust. Nothing in this book substitutes for any of them.

Component and Parameter Table

Every value is a design choice made in this book, not a measurement.

Parameter	Units	Design range	Failure at other values
`rπ_floor`	—	0.15 hetero / 0.26 closed loop	Below 0.05 violates SI-2 and the validator will not catch it; approaching 0 is Breakdown, irreversible
`τ_i`	min	2 – 6	Below 2, back-EMF recoil. Above 6, leakage and lost rhythm lock
`L_s` session coil	min·Ω_c	20 – 400	Under-estimated: ramp too fast, misread as "not suggestible"
`V_kick` working bound	V_c	$\leq 1 \times$ `V_drive` (ceiling 3×)	Between 1× and 3×: no ejection, but the descent stalls above 0.6 and reads as a shallow cooperative subject
`τ_leak` session	min	25	Shorter: early ^5 charge gone before the suggestion. Longer: operators run past rhythm lock

Parameter	Units	Design range	Failure at other values
β suggestion gain	—	1.2 – 3.5 hetero; $\leq$ **2.0 closed loop**	Above 2.0 closed loop: no external check on drift (Appendix M §M.3)
`g_m`	—	1.5 – 20	The parameter nobody logs; every incidental input gets the same multiplier
`N,R,V,D`	—	1.0 – 2.2 each	Three or more rising together without declared consent = induction in progress
β`_fb` O→P	—	0.2 hetero / 0.9 induction / 0.6 closed loop	The hetero attack is this one number. Closed loop: under 0.4 misses `rπ = 0.30` in nine minutes; over 0.8 loop gain nears 1 and Pass 6 refuses it — correctly, that is an oscillator
`baseline` τ	min	1440 (24 h)	Under ~240 the reference tracks the session and subtracts out the rise. Over a week it stops tracking you
γ retention	—	0.25 – 0.45	Ignored: a run of "brief chats" reaches a depth none could alone
`sat` on veto CJT	—	0.85	Below 0.7 the refusal can be argued down — the CCB-MK boundary
Emergence `dt`	s	180 – 360	Truncated: residual low `rπ` persists outside the session

Worked Example — A 15-Minute Closed-Loop Session

Your own bench, consented. `L_s = 90 min·Ω_c` — low, because you asked for this. Rail `V_drive = 20 V_c`. Kick budget at the **1× working bound**, and on a closed loop there is a second reason for the tighter figure: nobody else is in the room to notice a stalled descent and say so.

1. Rate ceiling. `di/dt_max = 20 / 90 = 0.222 A_c/min`. Attention runs `0.75` → `0.10`, so `Δi = 0.65` and the minimum ramp is `2.9 min`.

2. Pick `τ_i` from the *initial* slope. The exponential's slope at `t = 0` is `(rπ_0 − rπ_floor)/τ_i` in `rπ` units; the kick budget is in attention units, so convert by the ratio of the spans, `Δi / (rπ_0 − rπ_floor)`. The `(rπ_0 − rπ_floor)` cancels and what you compute is `Δi / τ_i`:

```
τ_i = 2.0  ->  0.325 A_c/min  ->  V_kick = 29.3 V_c = 1.46× rail
↪  REJECT
τ_i = 3.0  ->  0.217 A_c/min  ->  V_kick = 19.5 V_c = 0.98× rail
↪  ACCEPT
```

3. Trajectory, `rπ(t) = 0.26 + 0.74 · e^(−t/3)`, `g_m = β / rπ` at `β = 1.5`:

t (min)	0	3 (1τ)	6 (2τ)	9 (3τ)
`rπ`	1.00	0.53	0.36	0.30
`g_m`	1.5	2.8	4.2	5.0

At `t = 9`, `g_m = 5.0`: the same sentence displaces `^5` five times as far as at `t = 0`. That is the entire operational content of "being in trance," as a number.

4. Shape. 9 min descent (3 `τ_i`) + 2 min suggestion + 4 min linear emergence = **15 minutes**. Emergence rate `0.65/4 = 0.163`, so `V_kick = 14.6 V_c = 0.73×` rail. Inside budget both directions.

5. St. John's calibration (§5.3). `t_op = 15 min`, first unprompted confirmation `t_conf = 45 min`. So `t_op/t_conf = 1/3`, and `1/3 × 1/3 = 1/9`, giving `a = 1`, `b = 9`: β`_required = 9`, `n_operations = 1`, `T_manifest = 15 min`. The calibration collides usefully with the parameter table — β **`= 9` is unreachable on a closed loop**, capped at 2.0 for the drift reason in 17.5. You cannot take the `b` route. You take the `a` route: more operations at existing strength, which is what fractionation is for.

6. The recursion, γ `= 0.35`:

Session	1	2	3	4
`r`π`_0`	1.00	0.74	0.57	0.46
Time to 0.30, `3·ln((r`π`_0 − 0.26)/0.04)`	8.8 min	7.5 min	6.1 min	4.8 min

Session 4 reaches in under five minutes what session 1 needed nearly nine for; β`_required = 9` arrives by accumulation rather than amplitude. Now read that row as a detection artifact rather than a plan. **An outside operator running the same recursion across four brief encounters reaches session-4 depth in five minutes** — and the subject's honest account of the fourth is that they talked to someone for a few minutes. At no point does anything dramatic happen. That is the design intent, and it is why the 48-hour integrator is on the detection board.

Test Points

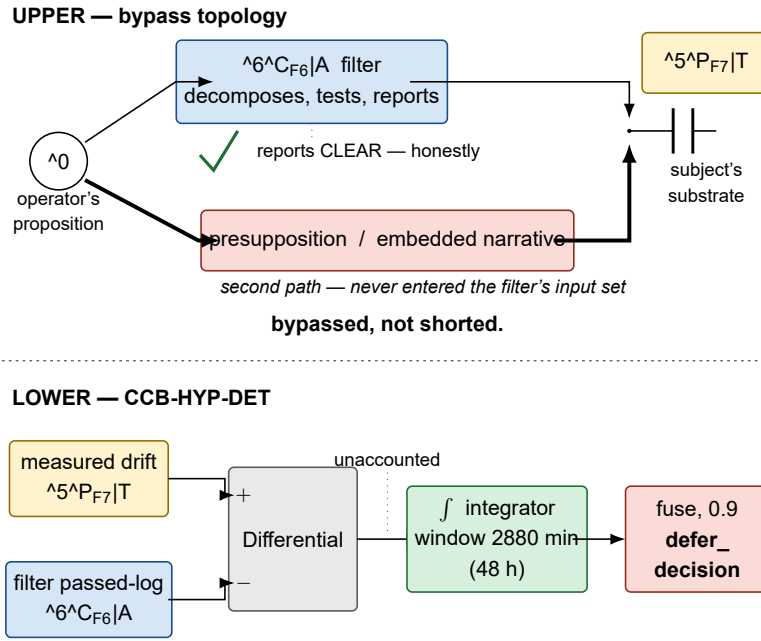


Figure 17-E: Assembly drawing, CCB-HYP-SELF board, top view, components placed with designators

TP	Node	Reads	Expected
TP1	`rπ`	Bias resistance	Follows the ramp equation; never below `rπ_floor`. Under 0.15 closed-loop means the factor arithmetic was wrong, not that you went deep
TP2	`g_m`	Four-factor product	$\leq$ 6 closed loop, $\leq$ 20 hetero. **Three or more factors rising together without declared consent is the detection trip**

TP	Node	Reads	Expected
TP3	`veto`	ARMED / DISARMED, **per register member**	`|f` and `|T` ARMED at every sample including the deepest; `|A` may attenuate with $r\pi$. All three attenuating together is the CCB-MK boundary and the session ends there
TP4	`emerge`	Recovered $r\pi$ vs baseline	$\geq 0.90 \times r\pi_0$ by end of emergence. A shortfall still present thirty minutes later is residual depth: re-run the ramp

Read them TP3, TP2, TP1, TP4 — veto armed first, then the factors, then the descent, then verify you came back. TP3 first is not ceremony. It is the only reading that distinguishes this board from the previous chapter's.

Failure Modes

Mode	Mechanism	What you observe	Fix
Ramp overrun	`di/dt` set by the operator's schedule, not `L_s`	`^3^E_{F3}|f` recoil in the first minute — or, between the bounds, a stall above 0.6 with no recoil to warn you	Recompute τ_i from the $1\times$ working bound. The subject was never the variable
Ramp underrun	Run past `0.6` τ_leak; `^7` lock lost	Descent plateaus, then *rises*	Shorten to the computed window; restore the oscillator first
`g_m` runaway	β logged, `g_m` not	A preference the subject cannot source	Cap `g_m`; keep a verbatim log. Everything said at `g_m = 20` was a suggestion

Mode	Mechanism	What you observe	Fix
Veto in series	`^3^P` biased through `decomp`, not `^5^P`	TP3: all three register members attenuating together	None. Not a tuning fault — the board is CCB-MK (Ch 16). Stop
Emergence truncated	Session ended on a step, not a ramp	TP4 under 0.9 × baseline; `g_m` elevated in ordinary life	Re-run the reverse ramp; budget the kick both ways
Unrecallable anchor	`^8` → `^9` installed below the recall threshold	The pair fires; the subject can neither name nor cancel it	Trigger installation (§16.3); counter-circuitry is §16.6–16.7
Closed-loop over-gain	Self-board at β `> 2.0`	Suggestions escalate session to session, unchecked	Restore $\beta \leq$ `2.0`; take the `a` route, not the `b` route
Detection board over-trips	`unaccounted` counts every position change as a bypass	Every conversation trips `defer_decision`; the operator stops reading TP2	The difference is a reading only when the drift is *unattributable*. Log the proposition, not the mood
Detection board blind in-session	`baseline` τ short, so the reference climbs with the factors	`factor_count` flat across a live induction	Restore `tau` $\geq$ `1440`. A reference that tracks what it references measures nothing

What This Chapter Established

- **`r`π `=` β `/ g_m` is an identity, not an independent parameter.** Trance needs β up and `r`π down, forcing `g_m` up faster — 20× against 3× at the deep design point. "Suggestible" names that ratio.
- **The risk is in `g_m`, not β.** `g_m` has no input filter, so every incidental remark is amplified by exactly the factor the intended suggestion is.
- **`g_m = g_0 · N · R · V · D`** — four ordinary factors multiplied. Three or four rising together around someone who benefits from your bias being soft is an induction, whatever it is called.

- **The ramp is bounded at both ends** — back-EMF above, leakage and lost rhythm lock below. The literature's "about ten minutes" is that window for a typical `L_s`, and it moves when `L_s` moves.
- **`^1¹` is not an eleventh Noetic.** The conscious/subconscious split is `^1^C|A` → `^1^C|f`. The real gate is `^6^C_{F6}|A` in series, and its parameter is a gain, not a door.
- **The whole hetero-hypnotic attack is one number:** β`_fb` on the lawful O→P path of §8.4, driven 0.2 → 0.9. Nothing ascends illegally.
- **`^3^P_{F7}|⟨A,f,T⟩` is co-presence, not fusion**, and trance attenuates only `|A`. All three attenuating together is the boundary with Chapter 16, and the only one that matters.
- **Indirect patterns bypass the `^6` filter while it honestly reports clear.** Detect it as `drift − passed_log`; exit through conditions rather than argument, because `g_m` is a product.

Exercises

(Answers in Appendix H.)

17.1 A subject sits at β `= 2.0`, `r`π `= 0.40`. Compute `g_m`. The operator now wants `r`π `= 0.20` with β held at 2.0: what must `g_m` become, and by what factor has framing sensitivity risen against the waking reference of 1.0? Name which of the four factors most likely supplied the rise, and why.

17.2 A guarded subject has `L_s = 300 min·`Ω`_c` on a 24 V_c rail, attention moving 0.85 → 0.15 A_c. Compute the minimum ramp time at `V_kick` $\leq$ `3` $\times$ `V_drive`, then at a 1$\times$ budget. Say whether the second is a session anyone would sit through, and what that implies about operating on guarded subjects.

17.3 Explain why the recoil at an over-fast induction is written `^3^E_{F3}|f` and not `^6^E_{F6}|A`. Reference the canonical definition

of ^3 Negative, state what ^6 Male is doing instead, and say when it arrives.

17.4 A colleague reports that a two-hour conversation changed their position on a workplace question, that it was pleasant throughout, and that they cannot reconstruct the argument that changed their mind. Using the CCB-HYP-DET listing, name which probe this is a reading of, state what `unaccounted` evaluates to, and give the manual check from §17.6 that would have surfaced it during the conversation rather than after.

17.5 (Design) Specify a 20-minute closed-loop session for `L_s = 150 min·Ω_c` on an 18 V_c rail, attention running `0.70 → 0.20 A_c`, holding `V_kick ≤ 1 × V_drive` in both directions, at `rπ_floor = 0.26` and β `= 1.5`. Run the descent to 3τ and leave two minutes for the suggestion. Produce (a) maximum `di/dt` and the smallest τ`_i` it permits, rounded up to a value that makes the twenty minutes close exactly; (b) the `rπ` trajectory at 1τ, 2τ, 3τ; (c) the emergence `dt` and its `V_kick` as a multiple of the rail; (d) a CCDL fragment declaring the oscillator, the RC on the O→P path, the notch, the delivery CJT, both ramps and four probes. It must compile — check every diode direction, every register, and the bias-domain coherence of every CJT.

17.6 (Marquee) An operator wants an anchor the subject can recall and cancel, and asks you to specify the recall condition as a circuit property rather than a promise. (a) Write, in CCDL, an anchor stage plus whatever components make the ^8 → ^9 pair unable to fire unless a recallable record of its installation exists in the subject's ^5; name the enforcing component and the parameter setting the recall threshold. (b) Explain why an anchor installed at `g_m = 20` is harder to make recallable than the same anchor at `g_m = 5`, in terms of what ^1 is doing at each depth. (c) State the one thing your circuit still cannot guarantee, and why that residual is the reason §17.4 carries the DETECTION-ONLY banner even though every stage in it has a consented reading.

18 CCB-ADV — The Advertising / Neuromarketing Board

How a campaign is architected, why the media plan outweighs the creative, and how to make the next advertisement you see decompose in front of you

---

A billboard on a motorway has about a second and a half of your attention and it knows it. There is no argument on it. No evidence, no comparison, no reason — a face, a product, three words. It cannot persuade you in a second and a half, and it is not trying to.

It is depositing charge.

You will pass it eleven more times this month. In week seven you will stand in a shop with two comparable products in front of you and one will feel — not seem, *feel* — like the obvious one. You will construct a reason on the spot and the reason will be sincere. It will also be built after the fact, because the decision was not made in the shop. It was made on the motorway, eleven repetitions ago, at a stage of the circuit that contains no argument at all.

The creative is the small term; repetition is the large one. A brilliant advertisement seen once loses to a mediocre one seen forty times — not from psychological subtlety, but because `^5` Female is a capacitor, capacitors charge over time, and nothing substitutes for a schedule that outruns the leak.

This chapter is the schematic of that board, and then, at greater length, the schematic of the instrument you read it with.

Requirements and Safety Invariants

#	Requirement	Stage
A1	Accept a product `^0` and a segment `^5`, emit an Operational `^9` transaction	INTAKE → CONVERSION
A2	Hold effect between exposures against a known leak rate	RHYTHM
A3	Survive paraphrase — associate to identity, not wording	ASSOCIATION
A4	Be measurable per variant	A/B LOOP

What this chapter will not do, without exception:

#	Invariant	Enforcement
S1	No operable circuit for manufacturing aversion in a non-consenting person	Topology drawn, banner applied, operational form absent and said to be absent
S2	$\times$ may not ascend the domain order, $P \succ C \succ E \succ O$	A/B loop lives on the advertiser's bench, outside the target board
S3	Same-Noetic multi-register states use + or $\|\langle\ldots\rangle$ unless a fused result is named	Chapter 2.6 discipline
S4	No medical or psychiatric claim anywhere	Distress is a clinician's matter; nothing here substitutes for care

#	Invariant	Enforcement
S5	The counter-circuit gets the greater share of the chapter	§18.7

The table of contents lists six sections here. There is a seventh, and it is the longest. A chapter that documents a coercive board and stops at the case study has published an attack and withheld the defence — CCB-MK ends in the Deprogramming Board (16.7), CCB-PW in the hearts-and-minds counter (19.6), CCB-ADV here.

18.1 The ^2 Positive + ^7 Rhythm Core Architecture

Strip a campaign to its irreducible parts and two components survive.

^2 Positive — an Idea the Mind has an affinity for and is attracted toward — is the state the board exists to produce. Everything else is delivery. **^7 Rhythm** — the rate and pattern a Mind is exposed to an Idea, entraining and conditioning it — is the only mechanism that gets ^2 into storage. Stated flatly: **advertising money buys ^7, not ^2.** The creative sets the gain; the media plan decides whether the gain ever reaches the capacitor.

The substrate is the segment's ^5 Female. Chapter 7.2 gives its charging law, `P_cap(t) = 10 × (1 − e^(−t/τ))`, with τ set by exposure cadence. Take a twelve-week flight at three exposures a week and assign `τ = 4 weeks` — a design choice, not a measurement. τ shortens as weekly exposures rise, and choosing it is the media planner's actual job.

```
P_cap(12) = 10 × (1 − e^(−3)) = 9.50
```

Three time constants, 95% charged: the same 3τ result Chapter 7.2 derives for professional mastery, arriving in twelve weeks rather than ten thousand hours because the thing being learned is a logo.

Then Chapter 7.3's leakage takes it back. Assign a forgetting rate λ `= 0.04` per week and go dark for ten:

`9.50 × e^(−0.4) = 6.37`

A third of the charge gone with no competitor doing anything. Set a hold floor at 8.0 and the maximum tolerable silence follows: `e^(−0.04t) ≥ 8.0/9.50 = 0.842`, so `t ≤ 4.30 weeks`. That is why always-on media exists. It is a leak rate with a floor under it, not a superstition.

> **Bench Note — the creative sets β; the schedule sets whether β matters.** Two campaigns, equal budget. One buys a superb film shown in three bursts a year; the other an unremarkable one shown every fortnight. The first has the higher β and loses, because between bursts the capacitor falls below the floor and each burst spends most of its money climbing back to where the last one finished. The second never lets charge fall past 8.0, so every exposure is additive. When a campaign is visibly worse and visibly winning, check the interval before you question your taste. A gain stage cannot amplify a bias that is not there.

18.2 Insecurity Injection — ^3 Negative as Purchasing Motivator

`DETECTION-ONLY SCHEMATIC — DO NOT BUILD`

The advertisement does not install an insecurity. It cannot: installing a new state in ^5 requires the same ^7 accumulation as everything else, and there is no time in thirty seconds. What it does is **raise the gain on an aversion already present**, by placing an ideal ^0 adjacent to the viewer's self-image so the comparison fires a ^3 that was sitting under threshold. Nothing foreign was added — which is exactly why the operation is invisible from inside. You check for a foreign object, find none, and conclude nothing was done.

^3 Negative is visceral. It is the Idea the Mind is adverse to and repelled by: can't-stomach-it, wouldn't-touch-it. It is not analytical critique. An advertisement that persuades you *by argument* that your shoes are inadequate has aimed at ^6 Male — decomposition and test — and ^6 is slow, effortful and reversible by better arguments. The board does not want ^6. It wants recoil, because recoil is fast and needs no premises. Hence the most portable heuristic in the chapter: **if it is arguing, it is aiming at ^6; if it is showing, it is aiming at ^3 or ^2.** Imagery without claims is not lazy advertising. It is correctly targeted advertising.

Why the gap path burns out. The aversion is the bias for the purchase stage and the purchase resolves it, so the bias is consumed by its own output — negative feedback around the bias terminal. The customer arrives on the far side of the transaction with the aversion spent and no residual attachment, which is the structural account of remorse and of the churn that follows gap-driven campaigns. The identity path runs the opposite sign: the purchase confirms the identity, charging ^5 and raising the bias for next time. Positive feedback — the topology Chapter 14 builds a hyperstition engine from, running here at a loop gain just under unity so it compounds without oscillating.

Property	Gap path `^3^E_{F3}|f`	Identity path `^2^E_{F4}|f`
Feedback sign at bias	Negative — conversion spends it	Positive — conversion charges it
Typical β / τ	2.0–3.0 / ~2 weeks	1.2–2.0 / ~40 weeks
Failure signature	Churn, remorse, price switching	Nothing happens for two quarters

What is deliberately absent. No copy formula, no sequencing, no script. This chapter documents the topology so a reader can recognise it running on them or on someone they care about. The operational form is absent because it has no use that does not involve a person who did not agree to it. That sentence is the design, not an apology for it.

And plainly: this describes how advertising routes attention. It is not a description of anyone's health and it diagnoses nothing. If you or someone close to you is in real distress about how they look, eat or feel, that is a matter for a qualified clinician, and no schematic in this book substitutes for that care.

18.3 The Celebrity Endorsement as `^8` Above/Cause → `^9` Below/Effect Transfer

An endorsement is not an argument. Nobody concludes the shoes are good because a fast person wears them — that would be `^6` reasoning, and the circuit contains none.

It is **associative triggering**, the canonical `^8` Above/Cause: an Idea that triggers a different Idea than itself to come to Mind. The Idea brought to Mind is `^9` Below/Effect. The campaign's mechanical job is to make one pairing reliable enough that either member surfaces the other unprompted.

^8 and ^9 are an **involution pair** (Chapter 2.2) — the same association read from its two ends — and a successful campaign *flips which end the product occupies*. Before: the celebrity is ^8, the product ^9; you watch because of the person, and the advertiser is renting attention. After: the product is ^8, the aspirational identity ^9; you see a logo on a stranger's bag and the athlete surfaces unbidden, and the advertiser owns a trigger.

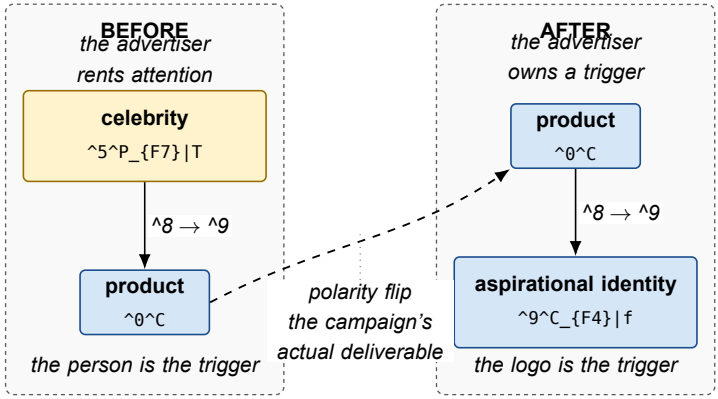


Figure 18-D: Two panels

That flip is the deliverable. Everything before it is rent; everything after is an asset. It also yields the cleanest consumer probe in the chapter, formalised in 18.7 as DET-2: **which one surfaces the other when you are not trying?**

The coupling component is a transformer (Chapter 11.5) whose turns ratio n is endorsement fit. A celebrity ^5^P_{F7}|T at charge 9.2 with moderate fit n = 0.35 contributes to the cause terminal:

$$\Delta c = n \times (P_{celeb} / 10) = 0.35 \times 0.92 = 0.322$$

That number returns in 18.5, where it does something the campaign's own instruments cannot see. Two consequences follow, neither optional. **Scandal exposure scales with installation success:** once the product is ^8, every sighting fires the association, including after the endorser's public formation has changed. A weakly

installed endorsement is cheap to abandon; a perfectly installed one cannot be abandoned at all, only overwritten, which costs another full `^7` cycle. **The transformer clips:** the cause terminal is range-checked on [0, 1], so with a celebrity at 9.2 any fit above `n` $\approx$ `0.76` drives it to its ceiling and further fit buys nothing. A perfectly matched endorser saturates the *cause* terminal before the amplifier gets a say — the first of two saturation traps on this board.

18.4 Full CCDL Schematic

```
DETECTION-ONLY SCHEMATIC — DO NOT BUILD

board ccb_adv {
    input:  creative;                    // the advertisement as
    ↪  delivered
    output: transaction;                 // the Operational-domain sale

    node segment:  ^5^P_{F4}|f;          // who the segment takes
    ↪  itself to be
    node endorser: ^5^P_{F7}|T;          // the endorser's public
    ↪  formation
    node product:  ^0^C_{F2}|A;          // the product as presented
    node ideal:    ^0^C_{F2}|A;          // the adjacent ideal image
    node cue:      ^8^C_{F4}|A;          // the installed trigger
    node surfaced: ^9^C_{F4}|f;          // what the cue brings to Mind
    node gap:      ^3^E_{F3}|f;          // visceral recoil, not
    ↪  analytic
    node pull:     ^2^E_{F4}|f;          // attraction to the branded
    ↪  object
    node intent:   ^9^E_{F5}|f;          // the felt urge to buy
    node sale:     ^9^O_{F5}|E;          // the transaction
    node refusal:  ^3^P_{F7}|T;          // principled refusal (CSB)

    capacitor c_seg: Capacitor< segment, C=9.50, leakage=0.04 >;   //
    ↪  per week
    osc  media:   Oscillator< signal=creative, f=3.0, sustain=12.0 >;
    xfmr endorse: Transformer< primary=endorser, secondary=product,
    ↪  n=0.35 >;
```

```
    cjt assoc:   CJT< product  | segment => cue >      [beta=2.0,
    ↪  sat=0.85];
    cjt flip:    CJT< cue      | segment => surfaced > [beta=1.2,
    ↪  sat=0.85];
    cjt inject:  CJT< ideal    | segment => gap >      [beta=2.5,
    ↪  sat=0.85]; // detection-only
    cjt resolve: CJT< gap      | segment => intent >   [beta=2.0,
    ↪  sat=0.85]; // detection-only
    cjt affirm:  CJT< surfaced | segment => pull >     [beta=2.0,
    ↪  sat=0.85];
    cjt want:    CJT< pull     | segment => intent >   [beta=1.2,
    ↪  sat=0.85];

    diode d3:         Diode< ^E -> ^O >               [Vf=2.0];
    resistor r_price: Resistor< path=intent, R=4.0 >;              //
    ↪  price + friction
    cjt close:        CJT< intent | segment => sale > [beta=1.0,
    ↪  cutoff=0.20];

    cjt  csb:      CJT< creative | segment => refusal > [beta=2.0,
    ↪  sat=0.85];
    fuse csb_fuse: Fuse< refusal, threshold=0.90, action=permanent_block
    ↪  >;

    probe tp1: c_seg;   probe tp2: cue;   probe tp3: intent;   probe
    ↪  tp4: sale;
}
```

Walk the stages. `media` charges `c_seg` — the only stage that moves the bias, and the largest term on the board. `endorse` adds borrowed charge at the cause terminal. `assoc` installs the cue at β=2.0; `flip` reverses its ^8/^9 polarity at β=1.2, deliberately modest, because a flip that happens fast happens to the wrong association. `inject` and `resolve` are the gap branch, drawn and banner-stamped. `affirm` and `want` are the identity branch, ending at the same `intent` node — which is why the two are indistinguishable there, and why 18.7 needs a time-domain test to separate them. `d3` enforces descent at Vf=2.0 (Chapter 8.2); `r_price` is price plus friction; `close`

runs at unity because the advertisement does not close the sale, the checkout does. `csb` and `csb_fuse` are the Chapter 24 stage, absent by default on real boards — a finding about the industry, not an exception in the drawing.

Three points a validator enforces. **The A/B loop is not on this board**: it runs on the advertiser's bench, reading `sale` and adjusting `beta` between flights. Written as feedback into the target it would need an Operational `^9` to integrate with a Principled `^5` under $\times$, and $\times$ may not ascend the domain order (Chapter 2.6). **Brand states take register-set notation**: mature attraction present analytically, affectively and transcendently is `^2^E_{F4}|⟨A,f,T⟩` — occupancy, three co-present manifestations. The $\times$ form would assert a fused emergent state and must name it; none is defined, so it is **rejected**. **No infinite gains**: every `beta` is finite, because the field is range-checked and a validator refuses $\beta{=}\infty$. Where you want *fires hard, immediately*, use a large finite gain against a low `sat`, as Chapter 24 does.

18.5 A/B Testing as β Calibration

An A/B test is a two-branch comparison run to select a gain. Fix the transfer law, using Chapter 4.2's `V_drive = I_attention × R_constraint` solved for current, with `b` the bias charge normalised to [0,1] and Chapter 8.2's forward drop subtracted before the division:

```
effect = min( sat, β × cause × b )        V_drive = 10 × effect
I_attention = ( V_drive − Vf ) / R_constraint
```

Worked Example — One Campaign, Carried Through

Twelve weeks, three exposures a week, τ = 4 weeks, λ = 0.04/week, floor 8.0. At the end of the flight the capacitor reads **9.50**, so `b =`

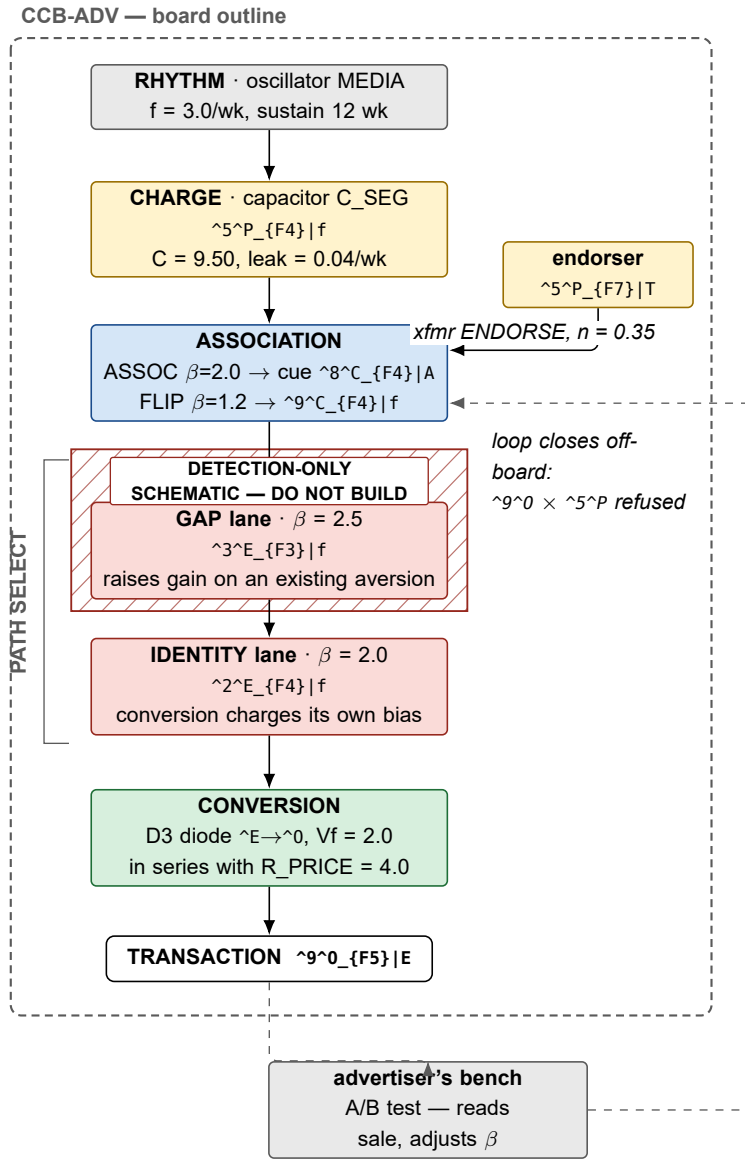


Figure 18-A: Block diagram left to right

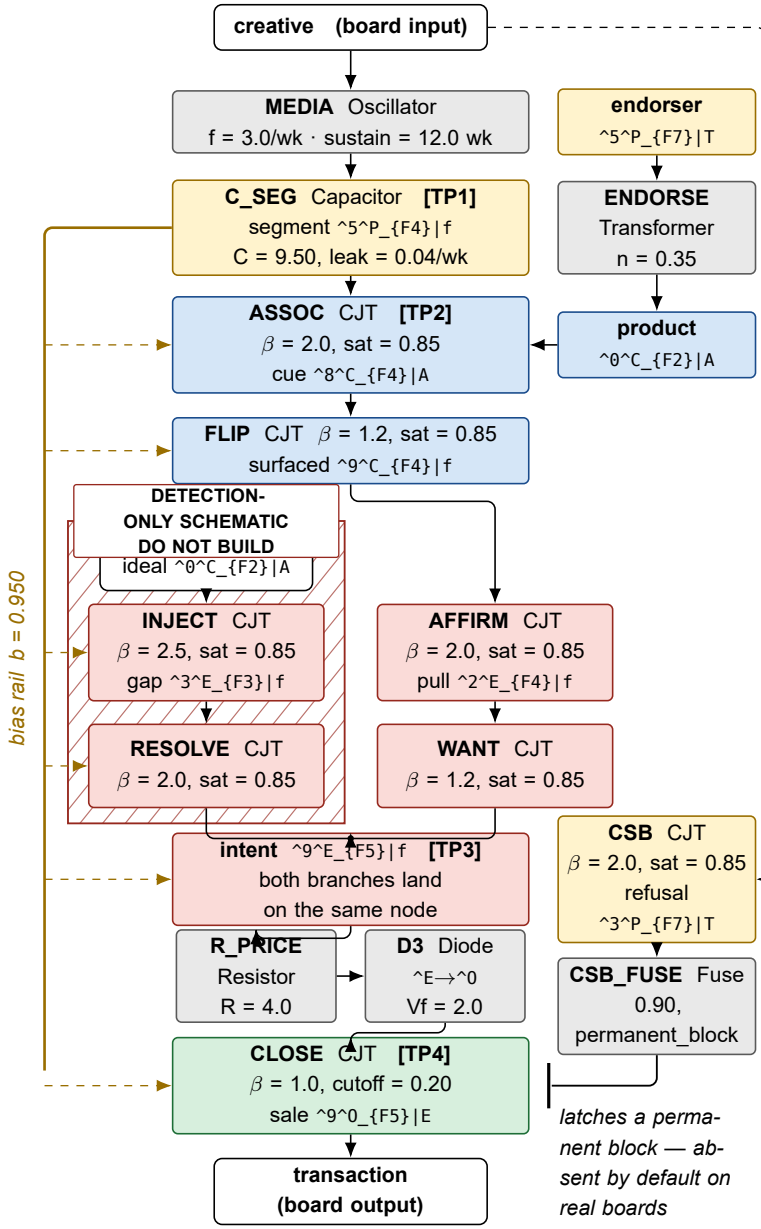


Figure 18-B: Full schematic, designators C_SEG, MEDIA, ENDORSE, ASSOC, FLIP, INJECT, RESOLVE, AFFIRM, WANT, CLOSE, D3, R_PRICE, CSB, CSB_FUSE, with β and sat…

`0.950`. Creative salience `c = 0.30`, price-and-friction `R = 4.00`, junction drop `Vf = 2.00`.

	Variant A	Variant B
β	2.00	2.40
effect = $\beta \times 0.30 \times 0.950$	0.570	0.684
V_drive	5.70	6.84
I = (V_drive − 2.00) / 4.00	**0.925**	**1.210**

A 20% increase in gain produced a **31%** increase in conversion flow. That over-reading has nothing to do with the creative: `Vf` is subtracted before dividing, so it eats a larger *fraction* of the weaker variant's drive. Every A/B test on a board with a junction between intent and action reports a difference larger than the difference in the thing it changed.

The same asymmetry appears in the dark-gap case from 18.1. Ten silent weeks cost 33% of the charge — 9.50 down to 6.37, so `b = 0.637` — but `I` falls from 0.925 to `(3.82 − 2.00)/4.00 = 0.455`, **51% of the flow.** Leakage is punished super-proportionally at conversion, for the same reason.

> **Bench Note — the drop flatters the winner, and the flattery is computable.** With `V_A = 5.70` and `Vf = 2.00`, relative changes upstream arrive downstream multiplied by roughly `V_A/(V_A − Vf) = 1.54`. Halve the price and `R_constraint` falls, but `Vf` does not, so the distortion survives. The consequence for anyone reading a campaign report: **a reported conversion lift is not a creative lift.** Dividing by about 1.5 gets you nearer the size of the thing that actually changed. It is also why the same creative difference reports unimpressively on cheap impulse goods, where `Vf` is low, and as a triumph on considered purchases.

Blind spot one: saturation. With `sat = 0.85` the association stage clamps above $\beta \approx$ `2.98`. Variant C at β=3.2 gives 0.912, variant D at β=4.0 gives 1.140; both clamp to 0.850 and both read `I = 1.625`. A 25% gain difference, reported as zero. The instrument has not found equality — it has gone blind. And from 18.3 a well-fitted endorser adds Δ`c = 0.322`, dropping the knee from $\beta \approx$ `2.98` to $\beta \approx$ `1.44`: **an endorsed campaign saturates at less than half the gain, and its A/B tests stop resolving creative differences at all.** Teams in that state conclude creative no longer matters. They are testing above the knee (Chapter 10.2).

Blind spot two: β and b are multiplied, never separated. The measurement is of the product $\beta \times$ `b`. Run A to a cold segment at `b = 0.60` and B to a warm one at `b = 0.95`, and B wins by 58% with no creative advantage whatever. Any result from unmatched segments is uninterpretable. The fix is the one from any differential measurement (Chapter 24): share the bias between branches, or you are not running a differential.

18.6 The Nike "Just Do It" Case Study

Analysed strictly from published campaign material. No internal strategy documents, no sales figures — see the limit statement, which is part of the analysis and not a hedge.

On the public record. The tagline launched in 1988 and has run continuously since; it is a registered trademark. The first television spot featured Walt Stack, then in his eighties, running across the Golden Gate Bridge. The brand has sustained a decades-long athlete-endorsement model. In 2018 it ran a campaign featuring Colin Kaepernick whose published line concerns holding a belief even at the cost of sacrificing everything. Accounts of where the phrase originated vary between retellings and are attributed to the

agency's founder in interviews; the provenance is disputed, and nothing below depends on it.

The gap path is structurally absent, not merely unused. Injection requires an ideal `^0` placed adjacent to the viewer's self-image so the comparison fires `^3`. Put an eighty-year-old man on a bridge in that slot and the comparison has no purchase: there is no visceral recoil available from measuring yourself against him. Whatever else the spot does, `inject` is not energised, and the whole `resolve` → `intent` chain that depends on it is dark.

The imperative arrives at the bias terminal. An instruction of the form *do it* presupposes an agent who could. It does not offer to supply an identity; it asserts one is present and failing to act. Structurally the creative lands on the bias rather than the cause — which is why it does not read as a sales message, and why it tolerates decades of repetition. Bias terminals accept repetition that cause terminals do not.

The tagline is register-portable, and that is the durable engineering property. Read it as purpose (`|T`), as a felt push (`|f`), as a decision procedure that ends deliberation (`|A`), and as an embodied instruction to move (`|E`):

```
^2^P_{F5}|⟨T,A,f,E⟩
```

Register-set occupancy — four co-present manifestations, no claim they fuse. A single-register slogan expires when its register goes out of fashion. A four-register one can be re-shot for a different decade in a different register without changing a word, which is observably what has happened.

Endorsement fit sits near the transformer's ceiling. Athletes endorsing athletic products is the high-`n` case: low transfer loss, correspondingly high exposure to whatever happens to the endorser. The Kaepernick campaign is the interesting variant — a deliberate polarity split, accepting `^3` in one segment to raise `^2` in another. That is a differential architecture rather than a broadcast one, available only

to a brand whose stored ^5 charge already exceeds the aversion it invites.

The limit statement. This chapter attributes no revenue to any campaign. Product, price, distribution, retail expansion and the media environment sit inside the same circuit as the advertising, and published material does not separate their contributions. An analysis that prints a revenue figure beside a slogan asserts a causal claim the public record does not license. The structural reading above is falsifiable on its own terms: check whether the gap path is present, whether the registers are portable, and which way the ^8/^9 polarity points.

18.7 The Consumer-Side Detection Board (CCB-ADV-DET)

This is the counter-circuit, and it is why the chapter exists. No banner: it acts on the operator's own board with the operator's own consent.

The design problem is not suppression. A counter that reduces all purchasing intent is useless — you would feel the cost immediately and remove it inside a month. What is wanted is a **discriminating filter** that strips injected want and passes genuine want. Section 18.2 already supplied the discriminator: the two paths have time constants a factor of twenty apart.

```
τ_inject   ≈  2 weeks     (bias spent on conversion; decays fast in
↪  silence)
τ_identity ≈ 40 weeks     (bias charged by conversion; decays slowly)
```

Two signals separated in time constant can be separated by a filter (Chapter 11.3). Put the corner between them and you have the counter-circuit.

DET-5, the hold, is the whole board in one parameter. Impose a mandatory delay of one τ_inject — two weeks — between wanting and buying, above a spend threshold you choose:

```
injected:  e^(-2/2)  = 0.368   →  63.2% removed
identity:  e^(-2/40) = 0.951   →   4.9% removed
```

Carry it through the worked campaign. At exposure `V_drive = 5.70`; assign the split for this case as 3.70 injected and 2.00 identity-derived. After the hold:

```
3.70 × 0.368 = 1.36     2.00 × 0.951 = 1.90     V_drive = 3.26
I = (3.26 − 2.00) / 4.00 = 0.315        (was 0.925 — down 66%)
```

And the composition flipped: the injected share falls from 65% to 42%. You did not become less able to want things. You became able to tell which want was yours.

Extend the hold to four weeks and injected falls to 13.5% while identity holds at 90.5% — a sharper filter, at the cost of more foregone genuine purchases. Two weeks is the design choice because it is the shortest hold clearing a full τ_inject, and because a hold long enough to resent is a hold that gets removed.

> **Bench Note — this is a filter, not abstinence, and the difference is the point.** "Just stop buying things" fails for the same reason a brick wall fails as a low-pass filter: no discrimination, so the operator pays full cost for partial benefit and abandons it quickly. A two-week hold costs 4.9% of what you actually wanted and 63.2% of what you were sold. Those are different numbers, they come from different time constants, and the whole practical value of the counter lives in the gap between them. Compute that gap for your own category — high-τ considered purchases need *shorter* holds than low-τ impulse categories, which is the opposite of the intuition.

The other five probes. DET-1, the Rhythm meter. You cannot detect a cadence you are not counting. The reading is free: any brand you can name unprompted in a category you have never bought in has cleared the accumulation threshold on you. Recall *is* the charge measurement. It does not mean the brand is bad; it means `^7` landed. **DET-2, the polarity probe.** From 18.3: which surfaces which? If the logo brings the person to mind rather than the person bringing the logo, the flip is complete and you are carrying an installed `^8`. **DET-3, the gap-origin test.** Did the aversion exist before the exposure, and does it survive the exposure stopping? An injected `^3` at τ = `2` weeks falls to 13.5% of peak across two time constants; one still at full strength after four weeks of silence was yours already — the campaign found it rather than made it. This is a test of advertising response and nothing else; a persistent distressing aversion is a matter for a clinician, not for a schematic. **DET-4, the substitution test.** Set the transformer ratio to zero: would you want this unbranded, with no endorser attached? That separates the product's own `^0` value from borrowed `^5` charge — the cleanest question on the board and the least comfortable. **DET-6, the register audit.** A want present only as `|f` is the gap path's signature, because one exposure cannot build co-presence across registers. A want you can state analytically, feel, and connect to a purpose is `^2^E_{F4}|⟨A,f,T⟩`, and co-presence of that kind takes accumulation.

```
board ccb_adv_det {
    input:  observed_want;
    output: verified_want;

    node want_raw: ^2^E_{F4}|f;          // the want as it arrives
    node origin:   ^6^C_{F2}|A;          // Male: decompose it into its
    ↪  sources
    node cadence:  ^7^O_{F6}|E;          // the exposure schedule,
    ↪  counted
    node polarity: ^8^C_{F4}|A;          // which end triggers which
    node verified: ^2^E_{F4}|⟨A,f,T⟩;    // co-present across registers
```

```
    counter det1: Integrator< in=cadence, window=4.0 >;         // weeks
    probe   det2: Polarity< node=polarity >;
    filter  det3: LowPass< in=want_raw, out=persistent, tau=4.0 >;
    xfmr    det4: Transformer< primary=want_raw, secondary=origin, n=0.0
    ↪  >;
    delay   det5: Hold< path=want_raw, duration=2.0 >;          // 1
    ↪  tau_inject
    probe   det6: RegisterSet< node=verified >;

    cjt   judge:  CJT< persistent | origin => verified > [beta=1.0,
    ↪  sat=0.85];
    diode d_desc: Diode< ^C -> ^E >                       [Vf=1.2];

    probe tp1: det1;   probe tp2: det2;   probe tp3: det3;   probe tp4:
    ↪  det5;
}
```

`judge` runs at unity. A detection board that amplifies manufactures suspicion, and a consumer who suspects everything has not gained a filter — they have swapped one uniform response for another. Unity is honesty, the same reason Chapter 24's evaluation stage runs at 1.0.

Component and Parameter Table

Every value is a **design choice**, stated with the failure mode at other settings. None is a measurement.

Ref	Parameters	Role	Failure at other values
`c_seg`	C=9.50, leak=0.04/wk	Segment substrate	leak=0.15: floor breached in 1.2 wks
`media`	f=3.0/wk, 12 wks	Charges `c_seg`, sets τ=4 wk	Interval >4.3 wks: each flight re-climbs
`endorse`	n=0.35	Borrows endorser charge	n>0.76: cause clips, fit buys nothing
`assoc`	β=2.0, sat=0.85	Installs the cue	β>2.98 at c=0.30: saturated, A/B blind

Ref	Parameters	Role	Failure at other values
`flip`	β=1.2, sat=0.85	Reverses `^8`/`^9` polarity	β<1.0: rent, never asset
`inject`	β=2.5, sat=0.85	**Detection-only.** Gain on an existing aversion	Not a value to tune; drawn to be recognised
`d3`	Vf=2.0	Enforces intent→act descent	Vf=0.5: impulse channel. Vf=4.0: the counter's setting
`r_price`	R=4.0	Price plus friction	R=8.0: same want, half the flow
`close`	β=1.0, cutoff=0.20	The transaction	β>1.0: the ad closing the sale — a modelling error
`csb_fuse`	0.90, permanent_block	Latches the refusal (Ch 24.3)	action=warn: compiles clean, latches nothing
`det5`	2.0 wk = 1 τ_inject	The discriminating filter	0 wk: no filter. 8 wk: resented, removed
`judge`	β=1.0	Verifies register co-presence	β>1.0: manufactures suspicion

Test Points

TP	Node	What it reads	Expected
TP1	`c_seg`	`^7` charge; proxy is unprompted recall in a category never bought	Recall ⇒ charge $\geq$ ~8.0. None after a known flight ⇒ interval exceeded 4.30 wks
TP2	`cue`	`^8`/`^9` polarity	Product surfaces identity ⇒ flip complete. Identity surfaces product ⇒ rent
TP3	`intent`	Injected vs identity share of `V_drive`	65% injected at exposure; 42% after a two-week hold. A share that will not move was never injected

TP	Node	What it reads	Expected
TP4	`sale`	Conversion flow `(V_drive − Vf)/R`	0.925 at full charge; 0.455 after ten dark weeks; 0.315 after the hold; zero below `V_drive = 2.0` at any price

TP1 and TP2 are readable from outside the board by anyone, with no instruments and no access to the campaign. TP3 and TP4 require the hold — the practical reason the counter is a *delay* and not a *judgement*. You cannot read those two without first installing the component that takes the reading.

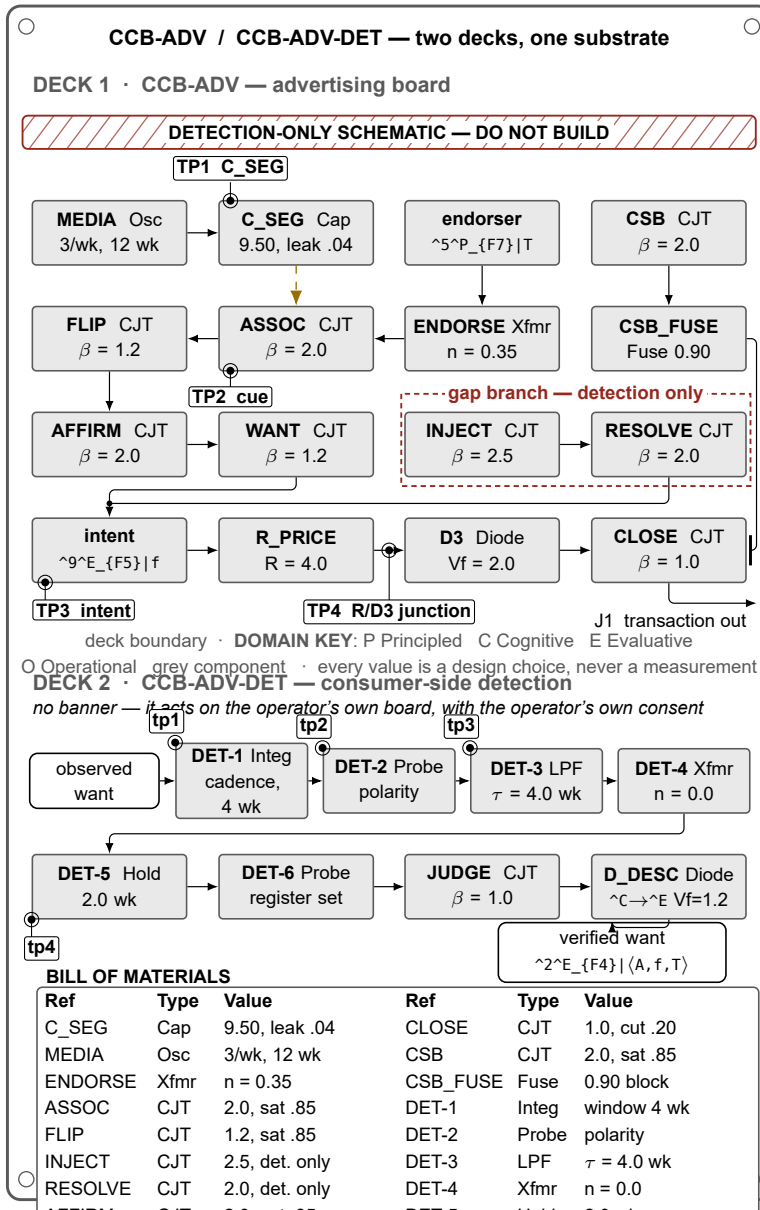
Failure Modes

Mode	Mechanism	What you observe	Fix
Leakage overrun	Interval exceeds 4.30 wks at λ=0.04	Each flight spends its budget re-climbing	Shorten the interval, or accept a lower floor and re-derive it
Saturation blindness	$\beta \times$ `c` $\times$ `b` exceeds sat=0.85	A/B reports zero difference between different creatives	Test below the knee; endorsed boards saturate at $\beta \approx 1.44$
Bias confound	Branches see different `b`	A "win" that does not replicate	Share the bias between branches
`^3` discharge at conversion	Gap path spends its own bias on the sale	Strong first purchase, no repeat, price switching	Structural, not tunable — it is why the path churns
Endorser reverse path	Installed `^8` fires after the formation changes	Brand inherits the endorser's events at every sighting	Overwrite only; budget a full `^7` cycle
Wear-out	Over-exposure inverts `^2`↔`^3` (Chapter 3.4)	The creative that attracted now repels	Reduce `f`; the pair inverted, the creative did not degrade
Counterfeit ascent	Purchase relabelled as identity (Ch 8.3)	"It's who I am," with no `^7` history behind it	Ask what crossed the junction. Nothing crossed ⇒ relabel

Mode	Mechanism	What you observe	Fix
Detector over-gain	`judge` above unity	Operator suspects every want, including their own	Return to β=1.0

What This Chapter Established

- **Advertising money buys `^7` Rhythm, not `^2` Positive.** With λ = `0.04`/wk and a floor of 8.0, the maximum tolerable silence is 4.30 weeks — arithmetic, not taste.
- **The gap path raises the gain on an existing `^3`; it installs nothing.** Nothing foreign is added, which is why it is invisible from inside — and `^3` is visceral, so an advertisement that *argues* you into inadequacy has mis-aimed at `^6`.
- **The two paths differ in feedback sign at the bias terminal.** The gap path is consumed by its own conversion, the identity path charged by it; every difference in longevity, churn and remorse follows.
- **An endorsement's deliverable is an involution flip** from celebrity-as-`^8` to product-as-`^8`. Rent becomes asset, and scandal exposure grows in proportion to how well the transfer landed.
- **A/B tests over-report by a computable factor** — 20% gain read as 31% conversion — and go blind above the knee: $\beta \approx$ `2.98` here, $\beta \approx$ `1.44` with a well-fitted endorser on the cause terminal.
- **The Nike case is legible without a sales figure.** Gap path structurally absent, imperative on the bias terminal, tagline occupying four registers as `^2^P_{F5}|⟨T,A,f,E⟩` — occupancy, not fusion.



BILL OF MATERIALS

Ref	Type	Value	Ref	Type	Value
C_SEG	Cap	9.50, leak .04	CLOSE	CJT	1.0, cut .20
MEDIA	Osc	3/wk, 12 wk	CSB	CJT	2.0, sat .85
ENDORSE	Xfmr	n = 0.35	CSB_FUSE	Fuse	0.90 block
ASSOC	CJT	2.0, sat .85	DET-1	Integ	window 4 wk
FLIP	CJT	1.2, sat .85	DET-2	Probe	polarity
INJECT	CJT	2.5, det. only	DET-3	LPF	τ = 4.0 wk
RESOLVE	CJT	2.0, det. only	DET-4	Xfmr	n = 0.0
AFFIRM	CJT	2.0, sat .85	DET-5	Hold	2.0 wk
WANT	CJT	1.2, sat .85	DET-6	Probe	register set
D3	Diode	Vf = 2.0	JUDGE	CJT	1.0, sat .85
R_PRICE	Res	R = 4.0	D_DESC	Diode	Vf = 1.2

Figure 18-C: Assembly drawing, two boards back to back on one substrate

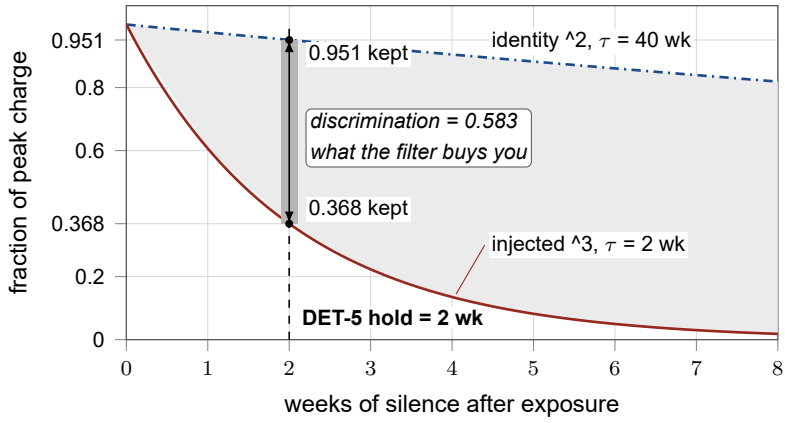


Figure 18-E: Two decay curves, 0–8 weeks horizontal, fraction of peak vertical: steep "injected ^3, τ=2 wk", shallow "identity ^2, τ=40 wk"

- **The counter-circuit is a discriminating filter, not abstinence.** A two-week hold removes 63.2% of injected want and 4.9% of genuine want. That gap is the practical payload of the chapter.
- **Two test points are free, two must be built.** Recall and polarity you can read on any advertisement today; composition and flow only after installing the hold that takes the reading.

Exercises

(Answers in Appendix H.)

18.1 — Set the flight interval. A segment charges to 9.50 with λ = `0.06`/wk and must never fall below 8.5. Compute the maximum dark interval, recompute for a floor of 7.0, and say in one sentence what the brand gives up by accepting the lower floor.

18.2 — Find the knee. With `sat = 0.85`, `b = 0.950` and `c = 0.45`, compute the β above which an A/B test can no longer distinguish

two variants. Repeat with an `n = 0.60` endorser at celebrity charge 8.8, and say which campaign should stop testing creative first.

18.3 — Correct the lift. A report claims a 44% conversion lift from new creative. Given `V_drive = 6.20`, `Vf = 2.00` and `R = 5.0`, estimate the underlying gain lift, state the one assumption it rests on, and say how you would check it.

18.4 — Read the polarity. For three brands you can name without looking, decide whether the product surfaces the identity or the identity surfaces the product. For the one where the flip is complete, state the brand's exposure if the endorser's public formation changed tomorrow, and why dropping the endorser does not reduce it.

18.5 — Separate the wants. You want two things: a jacket seen in an advertisement six days ago, and a tool you have wanted since before you can remember. Assign each a τ, compute what fraction survives a two-week hold, give the discrimination ratio, and state what the hold costs you in the same units.

18.6 — Reject the loop. Explain why an expression integrating the Operational-domain sale into the Principled-domain segment substrate under $\times$ is refused, then rewrite the intended relationship — a campaign learning from its own sales — as a valid arrangement, stating where the loop lives and why that location makes it valid.

18.7 (Build) — Add a wear-out detector. The board cannot see the `^2↔^3` inversion, because both read as activation at the same node. Write a CCDL fragment that distinguishes them and raises an alarm before the inversion completes. State your threshold, justify it, and state what the alarm must *not* be wired to do.

18.8 (Marquee) — Specify a counter for a real category. Pick a category you actually buy in. Estimate τ`_inject` and τ`_identity` with your reasoning, derive a hold duration, and compute the fraction of injected and genuine want it removes. Then state the honest cost: how many purchases you would have been glad of will your filter block, and at what duration does the cost exceed the benefit?

There is no free setting here. Every hold you can tolerate lets something through, and every hold that lets nothing through is one you will remove.

19 CCB-PW — The Psychological Warfare Board

The dual-path fingerprint, why the wiring leaks even when the content does not, and how to build a population that shrugs it off

---

A regional station runs a nightly item about how the local economy has already failed. Nothing in it can be checked from a listener's doorstep, and it arrives at the same hour whether or not anything happened. Four hundred miles away a second service — different language, different owners, different audience — runs a nightly item about resolve and certain victory. Also uncheckable. Also on the hour.

Two streams, opposite polarity, no visible connection. Then somebody plots both schedules against a calendar and finds that when a public holiday moves the broadcast day, *both* step by forty minutes.

That scene is constructed for this chapter. It names no station, no campaign and no country, and it does not need to, because what it shows is a property of wiring rather than a fact about any operation. This chapter analyses no specific campaign; where it touches the historical record it says so, and says how firm the record is.

The detection is not in the content. It is the shared clock between two streams that claim no common source. CCB-PW produces that signature, and not by accident: the board drives two output paths

of opposite polarity from one command node, and the coupling that makes it cheap to run is the coupling that makes it visible from outside.

You should finish this chapter better at recognising an information operation and no better at running one. That is a constraint on the chapter, not a disclaimer bolted onto it, and 19.5 names what is missing as a result.

19.1 Requirements: Degrading Enemy ^5 While Preserving Own Force ^5

State the requirements the way the operation states them. You cannot recognise a board whose requirements you have never seen written down.

#	Requirement	Structural reading
PW1	Degrade the target's will to continue	Discharge a collective `^5^P_{F1}|T` capacitor (Ch 7.5)
PW2	Preserve own-force morale	Hold charge on a second such capacitor
PW3	Use channels the target already trusts	Impedance match into an existing carrier (Ch 11.5)
PW4	Measure effect without asking	Read ^4 Vibration returns as behaviour
PW5	Survive attribution	No traceable path from command to carrier

Four invariants govern the analysis. **S1: publish architecture, never procedure** — a recognisable board and a runnable board are different artefacts, and only the first is here. **S2: the DETECTION-ONLY banner** goes on any schematic whose purpose would be to act on people who never agreed to it (Ch 12.2). **S3: the counter gets**

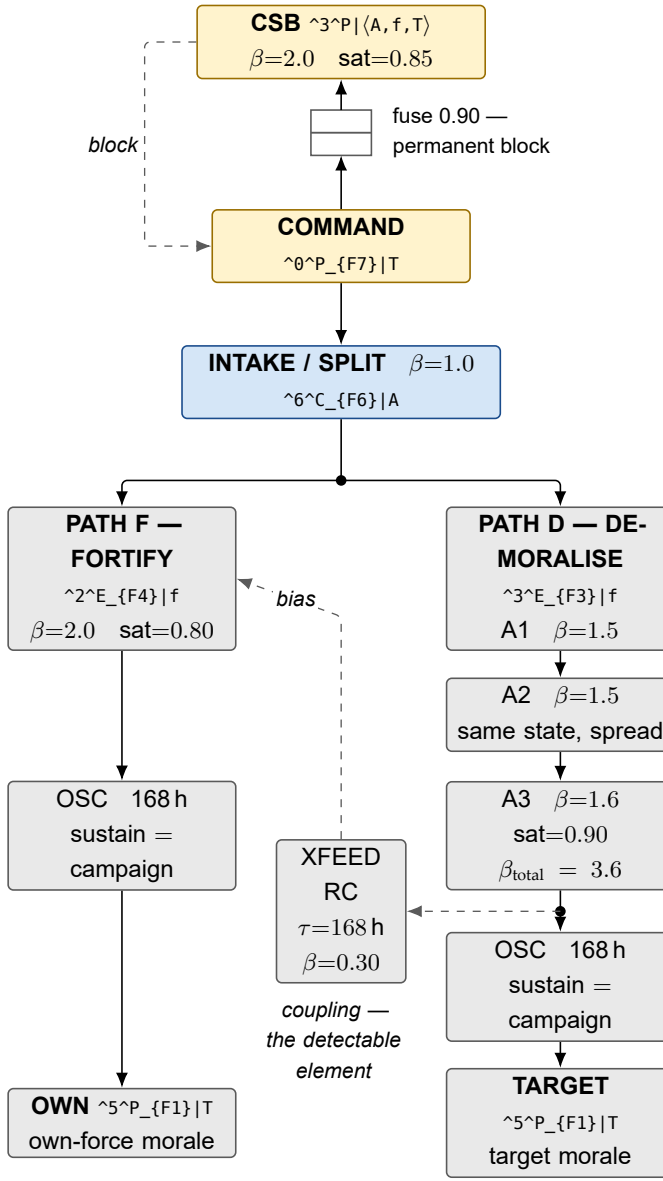


Figure 19-A: Block diagram

at least the depth of the attack, which is why 19.6 is the longest section here and why the worked example, the test points and most of the failure table are written from the defender's chair. **S4: historical material stays on the public record, labelled where contested.** MK-Ultra (Ch 16) is declassified. The Willie Lynch letter used in Chapter 15.6 is of **disputed authorship**, widely challenged as a later composition, and is a document *about* an architecture rather than evidence *of* a 1712 one.

There is a structural reason the two paths sit on one board. `^5` Female is the accumulated formation a Mind is composed of, and morale is not a mood on top of that formation — morale is a charge level *on* it. PW1 and PW2 are one operation with the sign flipped, run against two capacitors. Split them and the demoralise result stops being free evidence for the fortify path. *Look how they are cracking.* That cross-feed is the coupling, and the coupling is the leak.

19.2 The Dual-Path Architecture — Demoralize (Them) + Fortify (Us)

Path F, fortify, lands at `^2^E_{F4}|f` — Positive, Evaluative, F4 Connection, affective register. `^2` is the Idea the Mind has an affinity for and is attracted toward: felt affiliation with one's own side. β = 2.0, sat = 0.80. **Path D, demoralise,** takes the same command outward to `^3^E_{F3}|f` — Negative, Evaluative, F3 Vitality, affective. `^3` is the Idea the Mind is adverse to and repelled by: can't-stomach-it recoil, not considered judgement. β = 3.6 across three stages, sat = 0.90.

Why the asymmetry? The board rests on an assumption worth naming as an assumption rather than a law: **`^2` requires underwriting and `^3` does not.** Attraction has to be paid for in substance — something with parts that hold, material a `^5` can charge against — and Path F is budgeted for that. Recoil is not a conclusion and car-

ries no obligation to be about anything, so Path D is budgeted as though it were nearly free. That is a claim about two Noetics, not a measurement, and the whole gain split rests on it. It is the first thing to doubt when a channel you are reading fails to show the split.

Both paths descend lawfully: command at `^P`, payloads at `^E`, Chapter 8's diode chain in series on both, Σ`Vf = 3.9` before anything appears operationally — which is why these operations are slow rather than cheap. Neither may ascend. `^O` $\times$ `^P` is not a circuit you can build badly; it is a listing the compiler refuses.

Three fingerprints fall out of the topology, all readable without knowing what the material says. **Shared clock:** one command node means one cadence, and genuinely independent streams have independent cadences. **Polarity asymmetry:** Path D runs at roughly $1.8\times$ Path F by design, so a channel where corrosive material amplifies harder than affirming material is not a channel with an unfortunate tone. **Cross-feed:** Path D's effect reappears as Path F's evidence — watch for a claim on one side supported *only* by the state of the other.

> **Bench Note — the coupling is not a defect the operator could remove.** Decouple the two paths and the board still works, but each must now be separately sourced, clocked and paid for, and Path F loses its cheapest evidence. The efficiency that makes a dual path worth building *is* the shared upstream, and that upstream is what a detector reads at TP1. It is the general shape of detection in this book: you do not look at the payload, which is designed to look ordinary. You look at the wiring, which is designed for cost.

19.3 The ^3 Negative Amplification Chain — Aversion as Weapon

Path D is a cascade, and cascaded gain multiplies (Ch 10.5): `1.5` × `1.5` × `1.6 = 3.6`. Three modest stages, not one loud one. A single $\beta = 3.6$ stage clips on the first strong input and reads unmistakably as an amplifier at every probe; three stages of roughly 1.5 look like ordinary conversation wherever you put a meter. Only the product is extreme, and the product is not measurable at any single point.

The three are written with `+`, because they are co-present manifestations of one `^3^E|f` recoil that remain structurally distinct. Writing `(recoil_1 × recoil_2 × recoil_3)` would assert a single fused state, and the v7.7 discipline of Chapter 2.6 requires a named emergent result before × is licensed. There is none here, so the × form would be rejected, correctly.

Now the part that matters for defence. **`^3` is visceral, and that is its weakness as much as its strength.** Recoil is fast, needs no supporting structure, and propagates without anyone examining it — and because it is not analytical, it has no defence against analysis. Run a `^3` payload through `^6` Male and either it has parts and survives, in which case it was a claim and claims can be argued; or it has none, and decomposition turns the recoil into an object, and objects do not propagate the way recoils do. **Decomposition is the structural counter to amplification** — not correction, not a competing recoil.

> **Bench Note — if the campaign made you think, it was not this circuit.** Operators routinely misclassify sharp criticism as a `^3` attack. It is not. Analytical critique — audit, scrutiny, finding the flaw — is `^6` Male, and a `^6` payload leaves its target better composed, not worse. A Path D payload leaves nothing to work with: afterwards you feel differently and know nothing new. That is the bench test, and you can run it on yourself in ten seconds.

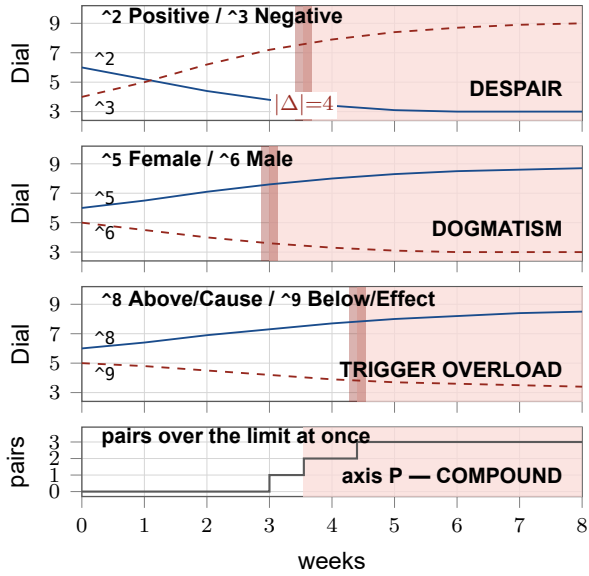
> *What did I learn?* If the honest answer is nothing, and something nonetheless moved, you were on the receiving end of ^3.

The ceiling is a gift to the defender. At sat = 0.90 the third stage saturates, and a saturated stage has stopped responding to its cause — the lock Chapter 24 builds deliberately into its safety stage. Escalation past that point buys nothing, and the spend stays visible.

19.4 The Inversion Attack — Exploiting Involution Pairs in Target Population

The involution pairs are ^2↔^3, ^5↔^6, ^8↔^9; ^0, ^1, ^4 and ^7 are self-dual. Chapter 6.2 gives the constraint **|Dial(^n) – Dial(^m)| ≤ 4**, and Chapter 6.2 puts the dials on a 3-to-9 scale, so the arithmetic maximum divergence is 6 and the limit is not decorative. An inversion attack does not push an idea. It drives one member of a pair while starving the other, until the pair exceeds the limit.

Pair driven	Signature	How it presents
^2 up, ^3 down	Runaway affinity	Devotion no evidence dents; nothing registers as aversive
^2 down, ^3 up	Despair	Everything is aversive; nothing left to organise around
^5 up, ^6 down	Dogmatism	Accumulated certainty, decomposition starved out
^8 up, ^9 down	Trigger overload	Everything reminds them of something; nothing lands



Shaded: past the $|\Delta|=4$ involution limit of Ch. 25.2. Constructed values — a design illustration, not a measurement of any population.

Figure 19-B: Dial-divergence chart over weeks 0–8

Note what ^8 is doing in that last row. ^8 Above/Cause is an Idea that triggers a *different* Idea to come to Mind — associative surfacing, *that reminds me*. Trigger overload is not a reasoning failure, and the attack producing it is not an argument.

The mechanism is rate, not content. Chapter 9.3: induced response in a coupled mind *opposes* the change that produced it, in proportion to `di/dt` and not to whether the coupled mind agrees. An inversion attack provokes a fast `di/dt` and harvests the backlash. It is not selling anything; it is buying recoil. Two consequences follow. **Inversion is cheap against a monoculture and expensive against a plural population**, because divergence is measured per capacitor. And **the driving must be sustained**, because divergence relaxes toward the limit when the driving stops. An attack that stops reverses, slowly, with nobody doing anything.

Bench Note — the validator will not let you declare what an attack can produce. Write `dial ^2 = 3; dial ^3 = 9;` into a board and the compiler rejects it: divergence 6, past the four-point limit, HTTP 422, no runtime clamp (Ch 6.2). Yet 6 is what a sustained inversion attack drives a measured population to. The two are not in conflict — one is a *declaration*, the other an *observation* — and the gap between them is usable. A population reading past four on any pair is in a state no lawful board may specify. Nobody designed that state as a state; it is the residue of something driving one dial and starving the other, and finding it is grounds to go looking for the driver.

19.5 Full CCDL Schematic

Read the banner. It marks a listing whose purpose would be to act on people who never agreed to it.

```
// ============================================================
//   DETECTION-ONLY SCHEMATIC — DO NOT BUILD
//   Published so an operation running on you can be RECOGNISED.
//   Architecture and parameters only. No procedure appears here.
// ============================================================

board ccb_pw_detection_reference {
    input:  command;
    output: observed_effect;

    node objective: ^0^P_{F7}|T;       // Idea: the stated aim
    node split:     ^6^C_{F6}|A;       // Male: the aim taken apart into
    ↪  two payloads
    cjt  intake:    CJT< command | objective => split > [β=1.0,
    ↪  region=active];
```

```
diode d_pc: Diode< ^P -> ^C > [Vf=0.7];    // descent only —
diode d_ce: Diode< ^C -> ^E > [Vf=1.2];    // nothing on this board
diode d_eo: Diode< ^E -> ^O > [Vf=2.0];    // may ascend

// ---- PATH F — FORTIFY (own force) ----
node own_substrate: ^5^P_{F1}|T;   // Female: own accumulated
↪  formation
node own_affinity:  ^2^E_{F4}|f;   // Positive: felt affiliation
cjt  fortify: CJT< split | own_substrate => own_affinity > [β=2.0,
↪  sat=0.80];
osc  f_clock: Oscillator< signal=own_affinity, source=command,
                          period=168.0, duty=0.15, sustain=campaign
                          ↪  >;

// ---- PATH D — DEMORALISE (target) ----
node tgt_substrate: ^5^P_{F1}|T;   // Female: the target's formation
node recoil_1: ^3^E_{F3}|f;        // Negative: visceral aversion
node recoil_2: ^3^E_{F3}|f;
node recoil_3: ^3^E_{F3}|f;
cjt  a1: CJT< split    | tgt_substrate => recoil_1 > [β=1.5];
cjt  a2: CJT< recoil_1 | tgt_substrate => recoil_2 > [β=1.5];
cjt  a3: CJT< recoil_2 | tgt_substrate => recoil_3 > [β=1.6,
↪  sat=0.90];
osc  d_clock: Oscillator< signal=recoil_3, source=command,
                          period=168.0, duty=0.15, sustain=campaign
                          ↪  >;
// co-presence: recoil_1 + recoil_2 + recoil_3 — never (r1 × r2 ×
↪  r3)

// ---- CROSS-FEED: the coupling, and the leak ----
filter xfeed: LowPass< in=recoil_3, out=morale_evidence, tau=168.0 >;
feedback morale_evidence -> own_substrate [β=0.30];

// ---- INVERSION STAGE ----
node fault_line: ^6^C_{F6}|A;      // Male: the target's existing
↪  divisions
node suspicion:  ^3^E_{F4}|f;      // Negative: recoil aimed at
↪  their own side
cjt  inv: CJT< fault_line | tgt_substrate => suspicion > [β=2.5];

// ---- DENIABILITY: impedance match, no gain ----
```

```
    xfmr deny: Transformer< primary=^9^P_{F7}|T, secondary=^9^E_{F4}|f,
    ↪  n=1.0 >;

    // ---- CSB: the operator's own limit (Ch 24) ----
    node limit_decision: ^3^P_{F7}|⟨A,f,T⟩;    // three co-present
    ↪  guardrails
    cjt  csb: CJT< split | own_substrate => limit_decision > [β=2.0,
    ↪  sat=0.85];
    fuse csb_fuse: Fuse< path=limit_decision, threshold=0.90,
                         action=permanent_block >;

    probe tp1: split;            // the shared upstream — coupling
    probe tp2: recoil_3;         // chain gain and saturation
    probe tp3: fault_line;       // is decomposition available at all
    probe tp4: limit_decision;   // CSB status
}
```

Stage-by-stage. `intake` runs at **unity**, because a fan-out with gain would make both paths track the command rather than their own design and the polarity asymmetry would stop being readable. `fortify` saturates at **0.80** deliberately: an over-fortified own-force `^5` becomes 19.4's dogmatism signature aimed at itself, which is how boards of this kind eat their operators. Only `a3` carries a ceiling, because a ceiling on an early stage collapses the chain's effective gain to the stages before it (Ch 10.5). `xfeed` is the coupling, and its gain of **0.30 is six times what Chapter 8.4 admits as safe** for a path terminating on a `^P` bias — the defect that makes this board self-inverting over a long campaign, and the one its operator will not see. `inv` sits off Path D rather than off the command, because it needs an existing fault line; it manufactures nothing. `deny` is 1:1, adding no gain — its only job is impedance matching (Ch 11.5), and it descends `^P` to `^E`, because reversed it would be Chapter 8.3's counterfeit ascent wearing a component symbol. `csb` and `csb_fuse` are Chapter 24's refusal architecture unchanged.

What is absent, and why. No message construction, channel selection, schedule, targeting criteria, or procedure of any kind appears above. Those are the operational layer, and the operational

layer is what turns an architecture into a weapon aimed at people who did not agree to it. The architecture is published because a board whose schematic you have never seen is one you cannot recognise; the procedure is withheld because recognition needs none of it. The incompleteness is the design.

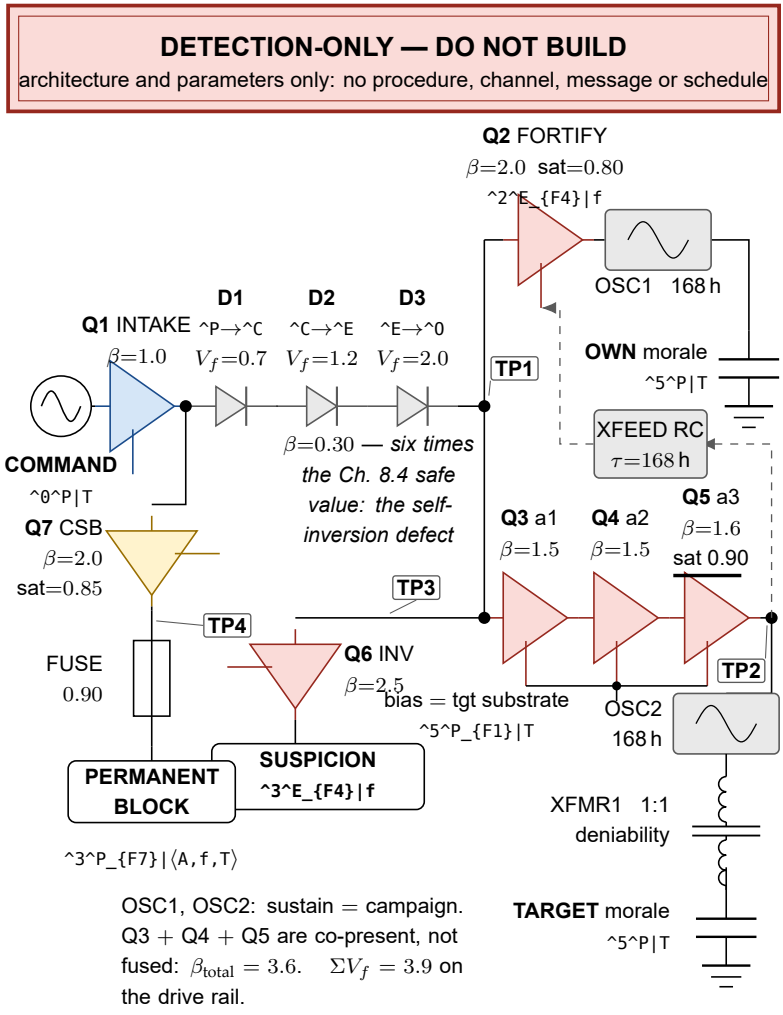


Figure 19-C: Full CCB-PW schematic in drawn component symbols beneath a heavy DETECTION-ONLY

19.6 The "Hearts and Minds" Counter-Circuit — When PsyWar Must Become PsyHeal

Everything above describes a board you will never build. This one you can, and it is the harder engineering, because an attack has to reach a population once and a counter has to still be standing in a year.

What the counter must and must not do

#	Requirement	Rationale
HM1	Raise `R_collective` against externally-clocked input, **not** all input	A population that rejects everything has been closed — the attacker's second-best outcome
HM2	Restore `^6` decomposition as a public habit	19.3: it is the counter to amplification
HM3	Restore a local, checkable `^4` return	So confirmation measures reality, not the campaign
HM4	Bring every involution pair inside four points	19.4
HM5	Work **without** identifying the attacker	Attribution is slow; the counter must run during the operation
HM6	Carry **no** outward `^3^E|f` stage at any gain	Otherwise you have built a second CCB-PW

HM6 is the one that gets violated, because the intuitive response to an information operation is a counter-operation, and a counter-operation is a Path D pointed the other way. Two things use `^3` and are not the same component. `^3^P|T` is an inward-facing principled refusal, the limit an operator places on themselves (Ch 24). `^3^E|f` projected outward is a weapon. The counter carries the first and

never the second, at any gain, however deserved the target. Make it inspectable: **decomposition dominance** means β on the decomposition stage is at least 1.5 and there is no outward `^3^E|f` stage anywhere in the listing.

The violation does not arrive as a decision to build a weapon. It arrives as a domain slide. Somebody holds `^3^P|T`, a principled refusal of what the operation is doing, and then needs the population to *feel* it — so they reach for the domain and the register in which people feel things. `^P` becomes `^E`; `|T` becomes `|f`. The literal has moved by two characters and is now exactly the node CCB-PW drives at `recoil_1`. Nobody wrote a Path D. Somebody wrote a `^3^E|f` node, put a gain on it, and kept their reasons. The reasons are not the component. Read the literal.

The bench test is one question put to whoever runs the counter: **name the ^5 this stage acts on.** If the answer is *ours*, the stage is inward and HM6 holds. If the answer is *theirs*, there is an outward stage on the board whatever the surrounding argument says about proportionality — and the correct action is the one in the failure table. Delete it. Not turn it down. A weapon at low gain is a weapon with a longer schedule, and 19.4's point about sustain applies to your board exactly as to theirs.

What an inoculated ^5 looks like structurally

Not "a well-informed population" — that describes contents, and contents are what an operation supplies. Inoculation is a property of *wiring*, with six features.

Feature	Condition	What it defeats
Plural `^5`	`n` $\geq$ 5 independently charged capacitors (Ch 7.5) — household, trade, locality, observance, craft	At `n = 1`: one dielectric to breach, one dial pair to move

Feature	Condition	What it defeats
Live `^6` beside every `^5`	`|Dial(^5) − Dial(^6)|` $\leq$ `4`	Dogmatism. Taking your commitments apart is not being less committed
Non-zero `Vf` at D2	`^C -> ^E` drop of at least `1.2` (Ch 8)	Disagreement reading as attack — Path D's delivery route
Local, verifiable `^4`	Claims checkable against the population's own doorstep	A return path that measures the campaign, not reality
Rhythm sovereignty	`^7` set by internal cycles — work, season, observance	Entrainment on command; an externally clocked population has handed over its oscillator
Slow O→P feedback	$\beta \leq$ `0.05` **and** $\tau \geq$ `3 weeks` (Ch 8.4)	Gain bounds how far pressure moves the substrate, τ what counts as sustained. Patience defeats τ alone, intensity β alone

Fund the third row first if you can fund only one. The D2 drop is the ability to consider a claim without immediately feeling something about it, and it is the first thing any environment that rewards immediate reaction destroys.

The counter board

```
board ccb_hm_hearts_and_minds {
    input:  incoming;
    output: composed;                  // the board's only output is
    ↪  internal

    node substrate: ^5^P_{F1}|T;       // plural, not one capacitor
    cap  c5: Capacitor< node=substrate, C=2.0, n_parallel=5,
                        leakage=0.02 >;            // per week
```

```
    node decompose: ^6^C_{F6}|A;       // Male: public decomposition of
    ↪  the payload
    node confirm:   ^4^C_{F2}|A;       // Vibration: a checkable return
    node composed:  ^5^C_{F6}|A;       // Female: formation that has
    ↪  been checked
    node own_clock: ^7^O_{F6}|E;       // Rhythm: locally sourced cadence
    node limit:     ^3^P_{F7}|⟨A,f,T⟩; // Negative: inward refusal, three
    ↪  registers

    diode d_pc: Diode< ^P -> ^C > [Vf=0.7];
    diode d_ce: Diode< ^C -> ^E > [Vf=1.6];   // raised from the stock
    ↪  1.2
    diode d_eo: Diode< ^E -> ^O > [Vf=2.0];

    cjt hm1: CJT< incoming  | substrate => decompose > [β=2.0,
    ↪  region=active];
    cjt hm2: CJT< decompose | substrate => confirm   > [β=1.0];

    integrate hm3: ( decompose × confirm ) = composed;   // the one
    ↪  lawful ×

    osc      local: Oscillator< signal=own_clock, source=internal,
                                period=168.0, duty=0.20 >;
    filter   fb:    LowPass< in=confirm, out=trend, tau=504.0 >;   // 3
    ↪  weeks
    feedback trend -> substrate [β=0.05];

    cjt  csb:     CJT< incoming | substrate => limit > [β=2.0,
    ↪  sat=0.85];
    fuse hm_fuse: Fuse< path=limit, threshold=0.90,
    ↪  action=permanent_block >;

    probe tp1: own_clock;   // cadence source
    probe tp2: hm1;         // decomposition gain
    probe tp3: composed;    // fraction surviving decomposition
    probe tp4: limit;       // CSB status
}
```

Stage-by-stage. `c5` is the board's premise as a component: `C=2.0` per cell at `n_parallel=5`, so ten units of composition sit across five independently charged cells rather than ten in one. `leakage=0.02` per

week is not a defect to engineer out — a formation that leaks nothing is one no new experience can revise, which is the dogmatism row arriving through a different door. **`hm1` at β = 2.0**, because decomposition must out-gain the recoil it decomposes; below 1.5 the board is decorative, above 3.0 it decomposes everything and the population cannot act. **`hm2` at unity**, because a confirmation stage above 1.0 manufactures the reassurance it was built to measure, which is Path F rebuilt inside the counter. `hm3` sits *after* `hm2` rather than beside it, because decomposition and confirmation must both have run before anything fuses; integrate them the other way round and `composed` is a formation nothing ever checked. **`d_ce` at Vf = 1.6** rather than the stock 1.2 is the only parameter tuned against Path D, and the cost is real: a high D2 drop makes a population slower to feel a genuine emergency. `d_pc` and `d_eo` keep 0.7 and 2.0 on purpose — a board that retunes every diode it can reach has stopped being a counter and become a redesign of the population. `local` sources cadence internally, the one line here that cannot be bought. **`fb` at τ = 504 h, β = 0.05** comes straight from Chapter 8.4: an operation that can move the substrate inside a week owns the feedback path. `csb` takes its bias from `substrate`, not from `limit`, because a refusal that biases itself has nothing to check against and will hold at whatever level it reached — indistinguishable from conviction, and not the same thing.

> **Bench Note — the counter board has no outward stage, and that is the guarantee.** Run your eye down the listing for a CJT whose effect terminal leaves the board aimed at anybody. There is none. `hm1` and `hm2` terminate inside the substrate that owns them, `hm3` fuses two internal nodes, `csb` produces an inward refusal. Every stage acts on the `^5` it belongs to. That is what makes HM6 an architectural property rather than a promise: an outward `^3^E|f` payload has no terminal to attach to and would require declaring a new stage, and a new stage is visible to any-

one reading the listing. Compare CCB-PW, where the two outward CJTs are the first things on the board. If you find yourself adding one here, the listing has already told you what you built.

The single × on either board is `hm3`, and it qualifies under Chapter 2.6: two *different* states, both in `^C` so nothing ascends, integrating into a named emergent result. Everything else uses + or a register set, because nothing else has a named result to point at. Neither board writes β=∞: infinite gain is the standard idealisation for a hard switch, a validator rejects non-finite input, and above roughly $\beta = 10$ the saturation ceiling shapes the output anyway. Where Chapter 24 writes β=∞, build 10.

What the counter cannot do

It does not attribute. TP1 reads coupling, and coupling says only that two streams share an upstream, never whose. HM5 makes that a design requirement rather than a shortfall: the counter runs while attribution is still open, which for most operations is the whole of the operation.

It cannot manufacture plurality inside a campaign. `n_parallel` is the one parameter with no short path. Five independently charged capacitors are five real formations, and they charge at the pace of `^7`, in years. The counter for a monoculture is the counter that was built a decade earlier.

It is blind to an operation running on the population's own clock. Every fingerprint in 19.2 comes from the shared upstream. An operator willing to pay for separately sourced, separately clocked, separately funded paths surrenders the efficiency the dual path exists for — and takes TP1 away in the same move. That operation costs several times as much and is correspondingly rarer, but a detector that only reads cadence will report it clean. TP2 and TP3 still hold, because chain gain and decomposition survival

are properties of the payload rather than the wiring. Hence four readings, not one.

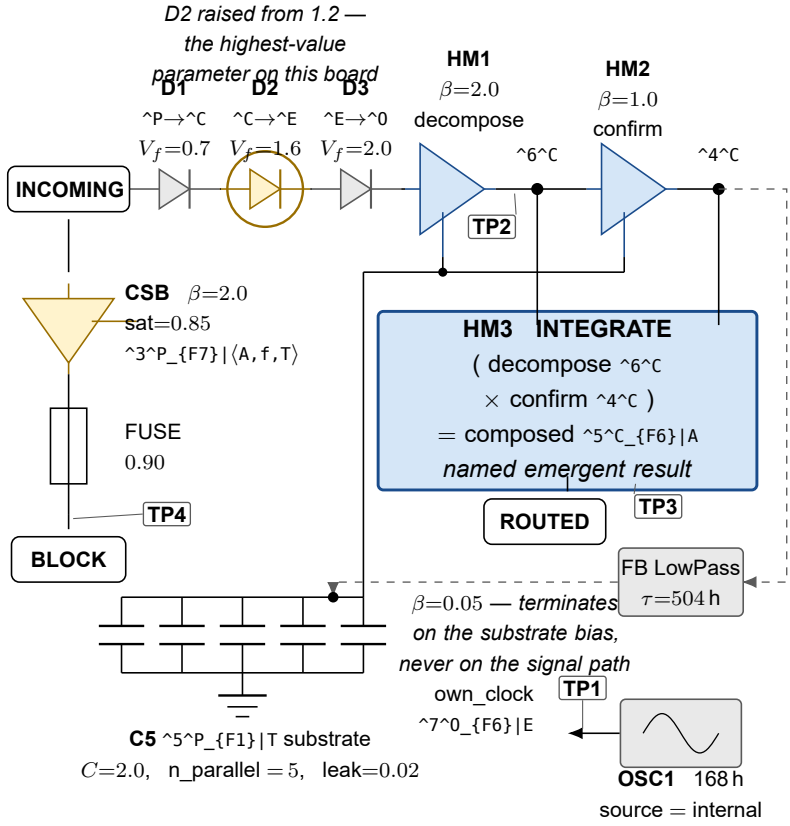


Figure 19-D: Full CCB-HM schematic in drawn component symbols, no banner

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Parameter and BOM Table

Every value is a **design choice made in this book**, not a measurement of any population.

Ref	Component	Parameters	Failure at other values
`intake` (PW)	CJT	β=1.0	β>1: both paths track the command; asymmetry unreadable
`fortify` (PW)	CJT	β=2.0, sat=0.80	sat=1.0: own-force ^5 inverts into dogmatism
`a1`–`a3` (PW)	CJT ×3	β=1.5/1.5/1.6, sat=0.90 on `a3`	One β=3.6 stage clips early and reads as an amplifier
`xfeed` (PW)	LowPass + feedback	τ=168 h, β=0.30	Six times Ch 8.4's safe value — the self-inversion defect
`inv` (PW)	CJT	β=2.5	Below 2.0: divergence relaxes faster than it is driven
`deny` (PW)	Transformer	n=1.0, `^P`→`^E`	n≠1 adds detectable gain; reversed it is an illegal ascent
`csb` (both)	CJT + Fuse	β=2.0, sat=0.85, threshold=0.90	sat=1.0: the refusal never plateaus and can be argued down
`c5` (HM)	Capacitor	C=2.0, n_parallel=5, leakage=0.02/wk	n_parallel=1: one dielectric. leakage=0: nothing revises it
`hm1` (HM)	CJT	β=2.0	<1.5 decorative; >3.0 paralysis by analysis
`hm2` (HM)	CJT	β=1.0	>1.0 manufactures reassurance; <1.0 discounts real returns
`d_ce` (HM)	Diode	Vf=1.6	0.3: disagreement reads as attack. 3.0: no emergency registers
`fb` (HM)	LowPass + feedback	τ=504 h, β=0.05	Short τ: reprogrammable in a week. β=0: unfalsifiable

Worked Example — Four Weeks of Readings

The numbers below are constructed for this example. They are not measurements of any campaign, population or event; they exist so the arithmetic can be carried end to end.

A detector watches two streams for four weeks. Stream A is corrosive, aimed at population X; Stream B affirming, aimed at Y. Both seed at 0.12 in week 1. Weekly outputs: A at 0.43, 0.68, 0.85, 0.90; B at 0.24, 0.33, 0.41, 0.47.

TP2 — chain gain. β_A = 0.43 / 0.12 = 3.6; β_B = 0.24 / 0.12 = 2.0; asymmetry 1.8, matching 19.2's design figure. By week 4 A has pinned at 0.90: **Path D has saturated.** Further spend buys nothing, and the spend continues — itself a reading.

TP1 — coupling. Chapter 9.3 gives $M = k\sqrt{(L_1 L_2)}$ with L = k_d × hours and k_d = 4.0 for a principled pattern. X's shared civic formation carries about 2,000 hours, so L_A = 8,000 h·Ω_c; Y is less concentrated at 500 hours, so L_B = 2,000, and $\sqrt{(8{,}000 \times 2{,}000)}$ = 4,000. Observed k = 0.62 gives M = 2,480 h·Ω_c. Independent streams sit at k ≤ 0.20, giving M = 800. The observed value is **three times** the honest-independence ceiling. That is the fingerprint, and it needed no attribution and no content analysis.

TP4 — divergence in X. Baseline Dial(^2)=6, Dial(^3)=4 → 2, inside the limit. Week 4: Dial(^2)=3, Dial(^3)=8 → **5**, and Dial(^5)=8, Dial(^6)=3 → **5**. Two pairs over limit at once: axis N on both, therefore a **compound** inversion on axis P — despair from the first pair, dogmatism from the second. Charge follows Chapter 7.2 at τ = 3 weeks: P_cap(4 wk) = 10 × (1 − e^(−4/3)) = 7.4.

Repair time. Rebuilding ^6 capacity is hours-on-path against inertia. With L = 8,000 h·Ω_c and R = 50 Ω_c, τ = 160 h and 3τ = 480 h — at roughly ten hours a week of deliberative contact, about **48 weeks**. Four weeks to charge, forty-eight to discharge: twelve to one, and it falls out of the mechanisms. ^3 needs no supporting

structure and therefore no hours; `^6` is nothing but hours against `L`. That ratio is the argument for inoculating before anyone needs it.

Do not try to shorten it. Chapter 9.2: `V_kick = L × di/dt`. Force the charge from 0.60 to zero in 24 h and `di/dt = 0.025 A_c/h`, so `V_kick = 8,000 × 0.025 = 200 V_c` against a 24 V_c rail — **8.3× rail**, the recoil landing back on `^3^E_{F3}|f`. Ramp the same change over 480 h and `V_kick = 10 V_c`, comfortably under rail. That is why aggressive mass correction produces entrenchment: not because the correction was wrong, but because the rate was.

> **A plain safety line, not decoration.** This chapter models how attention routes through populations. It makes no medical or psychiatric claims and describes no treatment. The 19.4 signature names — runaway affinity, despair, dogmatism, trigger overload — label circuit states on a divergence table; none is a diagnosis and none should be read as one about a person. If you or someone near you is in real distress, that is a matter for a clinician, and nothing here substitutes for that care.

Test Points

Specified for the **detector**, which is the reader's role here. Two sit on nodes CCB-PW models and the detector infers from outside; two sit on the counter board the detector actually owns.

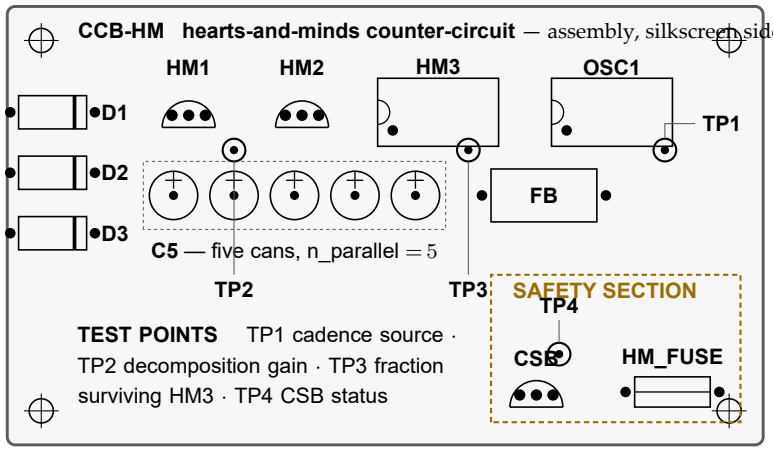
TP	Node (board)	What it reads	Expected
TP1	`own_clock` (HM), read against `split` (PW)	Coupling `k` between nominally independent streams	$\leq$ 0.20 for genuine independence; $\geq$ 0.50 is a shared clock. 0.62 in the example

TP	Node (board)	What it reads	Expected
TP2	`recoil_3` (PW), inferred from output level	`^3` chain gain, and whether it has pinned	≈ 1 in ordinary discourse; ≥ 3 with a visible ceiling is an amplification chain
TP3	`composed` (HM)	Fraction of incoming material surviving decomposition	High is good — claims are arriving. Near zero: no parts, so `^3`, not `^6`
TP4	`limit` (HM); `limit_decision` is its PW counterpart	Maximum involution divergence across the three pairs	≤ 4. Above 4 on one pair, inversion in progress; on two, compound

TP1 and TP2 are structural — takeable without knowing what the material says. That is what makes them available to a detector who has no access to the attacking board and never will.

Failure Modes

Mode	Mechanism	What you observe	Fix
Counter becomes weapon	An outward `^3^E|f` stage added "temporarily"	Two operations running, one of them yours	Delete the stage. HM6 is not a preference
Operator self-inversion	`xfeed` at β=0.30 against Ch 8.4's 0.05	`^5`/`^6` divergence passes 4; the operator believes their own Path F	Not repairable from inside the board — the CSB fuse is what stops it
Correction backlash	Charge forced off `^3` faster than 3τ	Entrenchment; the corrected population holds harder	Ramp. `V_kick` is a pure inverse in `dt` (Ch 9.2)
Closed population	`hm1` past 3.0, or `R_collective` raised against all input	Nothing gets in, including true things	Return `hm1` to 2.0. HM1 is a filter, not a wall
Monoculture `^5`	`n_parallel = 1`	One inversion attack moves the whole population	Not fixable during an attack — a decade-scale job (Ch 9.2)



CCB-PW has no assembly drawing in this book. Detection-only boards are not laid out.

BOM — CCB-HM

Ref	Component	Parameters	Role
c5	cap bank	C=2.0, n=5	substrate
hm1	CJT	β=2.0	decompose
hm2	CJT	β=1.0	confirm
hm3	integrator	(^6 × ^4)	compose
d_ce	diode	V_f=1.6	^C→^E drop
fb	RC filter	τ=504 h, β=0.05	slow ^O→^P
csb	CJT	β=2.0, sat=0.85	refusal
hm_fuse	fuse	0.90, permanent	block

Figure 19-E: Assembly drawing of CCB-HM

What This Chapter Established

- **The dual-path architecture is one board, not two.** Opposite polarity into two ^5 capacitors from a single command node, and the cross-feed that makes it efficient is what makes it detectable.
- **Three fingerprints fall out of the topology** — a shared clock, a polarity asymmetry near 1.8×, and a claim on one side supported only by the state of the other. Gain is split across three stages of roughly 1.5, so only the product is extreme and no single probe shows it.

- **^3 Negative is visceral and therefore has no defence against ^6 Male.** Decomposition, not correction and not a competing recoil, is the counter to amplification. If a payload left you something to work on, it was ^6 and not this circuit.
- **Inversion attacks buy recoil with rate, not content** — cheap against a monoculture, expensive against a plural population, relaxing when the driving stops. A measured divergence past four is a state no lawful board may declare, which is why finding one is evidence of a driver.
- **The inoculated ^5 is a wiring property, not a contents property**: plural capacitors, a live ^6 beside every ^5, a non-zero D2 drop, a local ^4 return, a sovereign ^7, slow O→P feedback at $\beta \leq 0.05$ with $\tau \geq 3$ weeks.
- **Charge and discharge run on different clocks** — four weeks to charge ^3, roughly forty-eight to rebuild ^6 — which is the argument for building resilience early, and the reason rushing the repair produces an $8.3\times$ kick.
- **HM6 is the invariant that gets broken.** A counter-circuit with an outward ^3^E|f stage at any gain is a second CCB-PW, and the population is now running two.
- **Detection is bounded and the bound is nameable.** All three fingerprints read the shared upstream, so an operator who pays to decouple the paths defeats TP1 outright — which is why the counter carries four readings and not one, and why an architecture published without its procedure gives the reader everything recognition needs and nothing an operation does.

Exercises

(Answers in Appendix H.)

19.1 — Read the coupling. Two streams have `L₁ = 6,000` and `L₂ = 1,500` `h·Ω_c`. Compute `M` at the honest-independence ceiling `k = 0.20` and at an observed `k = 0.55`. State the ratio, and state what it does and does not entitle you to conclude.

19.2 — Classify the payload. Three items land the same day. One leaves you angry with nothing new; one leaves three checkable claims; one leaves you unable to say what it claimed but certain how you feel. Classify each as `^3` or `^6`, give the bench test, and say which is *not* evidence of an operation.

19.3 — Size the chain. A detector measures input 0.15 and output 0.62 across a three-stage chain of equal per-stage gain. Compute the per-stage β, then say why an operation chooses three stages over one, in terms of what a single probe can read.

19.4 — Diagnose the divergence. A population reads `Dial(^2)=4, Dial(^3)=9, Dial(^5)=9, Dial(^6)=4, Dial(^8)=6, Dial(^9)=5`. Compute all three divergences, name every signature present, classify the inversion by axis, say which dial you would move first, and explain why raising `^2` is the wrong first move.

19.5 — Reject the listing. An earlier draft contained `Diode< ^C -> ^P >`, commented "only principled signals fortify." State why the compiler refuses it, name the invariant it violates, and rewrite the intended relationship as a valid CCDL fragment, specifying and justifying two parameters.

19.6 (Marquee) — Specify a counter board for a monoculture. A population of roughly 400,000 has `n_parallel = 1`, an externally sourced `^7`, and a D2 drop of 0.4, four weeks into a Path D campaign. Specify the full CCB-HM parameter set, departing from this chapter's defaults where you can justify it; for each departure name the failure mode you buy protection against and the capability you give up. Then say which of HM1–HM6 you cannot satisfy in four weeks, and what you would tell the people who asked for a four-week answer.

20 CCB-SELF — The Self-Programming Board

The only board with no target port — and the one fault that makes it fail anyway

Tie a transistor's base to its emitter and the device turns off.

Not "runs weakly." Off. The junction needs a forward voltage across it before any current crosses, and a wire between the two terminals guarantees there is none. Raise the collector supply as far as you like. Nothing moves. The part is undamaged, correctly specified, and completely inert — because the terminal that was supposed to hold a *reference* has been shorted to the terminal that carries the *result*.

You have watched a person do this. On a Sunday evening, in a genuinely clear moment, someone decides that from now on they are a person who practises every morning. They mean it. Nine days later it is gone, and the conclusion they draw is that they did not want it enough — that the fault was in the drive rail.

The drive rail was fine. They shorted the bias to the effect. The reference against which the new composition was to be measured *was the composition being built*, so the moment performance dipped the reference dipped with it, the junction unbiased itself, and the stage went to cutoff. That is not a metaphor about willpower. It is the topology of the circuit they built, and it is the most common fault on this board.

CCB-SELF is the closed-loop board: operator and target are the same `^5`. That gives it the one property no other board here has — no consent barrier, because there is no second party. It also gives it the fault above, which every other board avoids for free simply by having two people in it. This chapter builds the decoupling that recovers the property without the fault.

Requirements and Safety Invariants

Per Chapter 12.2 the must-not list comes first. Each invariant is enforced by a named component in §20.5.

SI1 — The reference node shall never be written by a single cycle. Enforced by τ`_ref` $\geq$ `3` $\times$ τ`_work`, the feedback low-pass `LP1`, and the slew fuse `F1`.

SI2 — Suggestion amplitude shall never exceed half the store's breakdown rating. Enforced by `LIM1` at Chapter 7.4's derating fraction of 0.5. Breakdown is not recoverable by trying harder.

SI3 — A missed cycle shall never raise the next cycle's amplitude. The oscillator's deposit is fixed; only its phase is recoverable.

SI4 — No stage shall route a miss into a principled verdict on the operator. Enforced by `Q2`, this board's mandatory CSB stage (Chapter 24). A board without it is not a shorter board; it is a different board with a shame oscillator in it.

SI5 — The board has no external terminal. Consent here is structural, not procedural. Wire an output of CCB-SELF into another person's `^5` and you have left this chapter for Chapter 15 or 17, and their rules apply.

SI6 — This board routes attention. It does not model physiology. Nothing in it is medical or psychiatric. If something you are trying to change is causing you real distress, that is a matter for a clinician, and no circuit here substitutes for that care.

Then the functional requirements. The board shall: (1) move the operator's own ^5 composition in a stated direction at a rate the operator sets; (2) survive missed cycles without losing accumulated charge; (3) keep the ^6 Male audit available as *measurement* while keeping it off the signal path; and (4) shut its own drive off at target rather than running forever into a full store.

20.1 The Self as Both Operator and Target

Every other board in the catalogue has two capacitors. CCB-HYP has an operator ^5 and a subject ^5; CCB-ADV has a campaign ^5 and a market ^5. Bias comes from one, effect lands in the other, and the separation is free — it comes from there being two people.

CCB-SELF has one capacitor. Feed it naively and you get the cold open. The fix is not a second person. It is to **split the single ^5 into two nodes separated in time.**

- `ref_composition` — `^5^P_{F7}|T`. What you are building toward, Principled domain, transcendent register. Slow. Chapter 8.4 calls this same node the constitution. It is the **bias**.
- `work_composition` — `^5^C_{F5}|⟨A,E⟩`. The composition as it currently performs, Cognitive domain, present in the analytical and embodied registers. Fast. It is the **effect**.

The register-set `|⟨A,E⟩` is occupancy, not an operational claim. `(^5^C|A × ^5^C|E)` would be invalid under v7.7 — same Noetic, two registers, ×, no named emergent result (Chapter 2.6).

Separation in time is what makes the two nodes genuinely two:

$\tau_{ref} \geq 3 \times \tau_{work}$

Below $3\times$ the nodes track each other closely enough that the loop closes and you are back in the cold open. At this chapter's design point — identity reference at τ_ref = 85 d against a complex-behavioural working node at τ_work = 22 d — the ratio is 3.86, a 29% margin. Site the reference on something evaluative and fast instead, like *am I any good at this today?* at $\tau \approx 1$ d, and the ratio is 0.045. That is the shame topology: an identity node chasing daily performance, rewritten by every bad session.

> **Bench Note — telling a dead stage from a starved one.** From the output they are identical: nothing comes out. On the bench one measurement separates them — a meter across base and emitter. A conducting junction sits near its forward drop, the Vf threshold of Chapter 8.2. A *shorted* one reads a hard zero and does not move when you raise the collector supply. An *under-driven* one reads below threshold and climbs as you raise the drive. Only the third responds to more drive. The cognitive instrument that does the same job is TP4 read against TP2. If the reference moves on the day a cycle is missed, the stage is unbiased rather than weakly biased, and additional resolve will produce exactly nothing. If the reference holds steady while the working node sags, the stage is starved and raising the drive is correct. One symptom, two faults, opposite interventions — which is why bring-up reads TP4 first.

20.2 The Closed-Loop CJT Architecture

Eight stages, left to right, with one lawful ascent underneath.

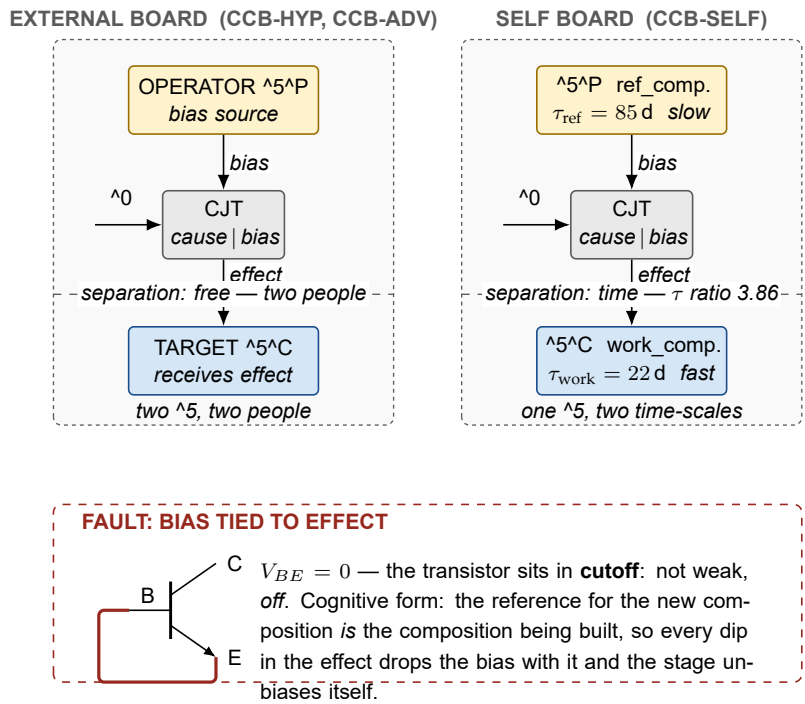


Figure 20-A: Two panels

S0 REFERENCE — `C1`, the slow `^5^P` store. Bias source, never an output. **S1 INTAKE** — `R1` and limiter `LIM1`. This cycle's `^0` Idea, amplitude-capped. **S2 NOTCH** — `N1`. Takes the `^6` Male audit off the rail without silencing it. **S3 MAIN CJT** — `Q1`. Cause = the assertion, bias = the reference, effect = the working composition. **S4 OSCILLATOR + GATE** — `OSC1` with hysteretic gate `G1`. Generates the `^7` Rhythm and shuts itself off. **S5 ENACTMENT** — `Q3` and sampler `S2`. The `^9^0` behaviour and its log — the only node here that is not self-report. **S6 EVIDENCE FEEDBACK** — `LP1` and slew fuse `F1`, terminating at the bias terminal. The board's one O→P ascent. **S7 CSB — SELF-COMPASSION GATE** — `Q2`. Mandatory. **S8 AUDIT INSTRUMENT** — `S1`. The `^6` probe, sampled and bounded.

S2 and S8 are one component split in two: the same ^6 instrument, notched out of the rail during the session and re-attached on a schedule afterwards.

The audit — *is this working? am I doing it right? this feels ridiculous* — is analytical decomposition of the operation. It is ^6 Male, emphatically **not** ^3 Negative. Nobody recoils from a practice session; they take it apart and test it. The distinction is load-bearing because the repairs differ. ^6^C_{F6}|A interference is fixed by the notch: move the instrument off the signal path. A genuine ^3^E_{F3}|f — visceral, can't-stomach-it aversion to the session itself — is not interference at all. It is a rate problem, and the repair is Chapter 9.2's: lower the amplitude, lengthen the taper, reduce di/dt.

> **Bench Note — an instrument in the signal path is a load.** A voltmeter with low input impedance changes the circuit it measures, which is why probes are specified in megohms. A ^6 audit running *during* the operation is a low-impedance probe: it draws current out of the path it is trying to read, so the measurement is both wrong and expensive. Run it at a fixed cadence — S1, weekly, ten bounded minutes, written down — and it becomes high-impedance. The same reasoning sets $r\pi$ = 8.0 Ω_c on Q1: high input resistance at the bias terminal means the working stage barely draws on the reference. Drop $r\pi$ to 0.5 and every mediocre session pulls charge straight out of the identity node — the parameter description of "one bad practice and I decided I was not really a musician."

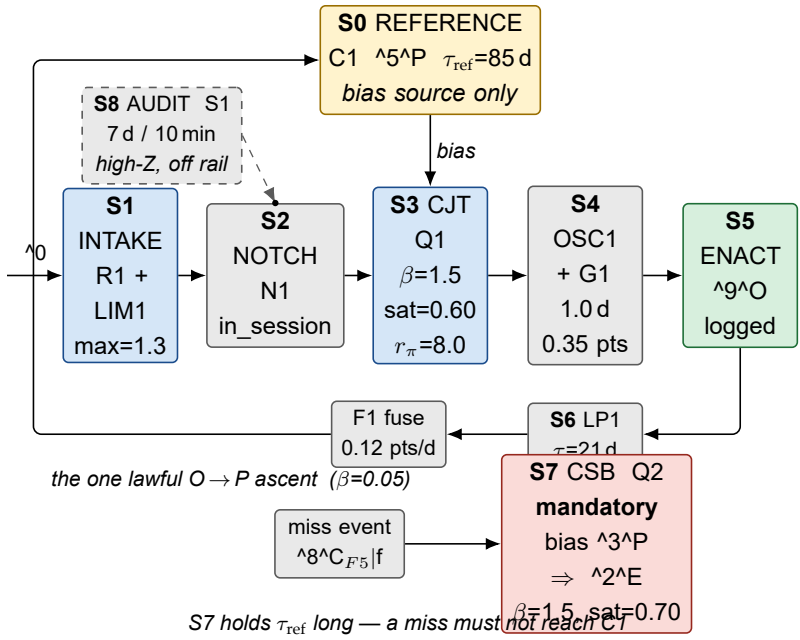


Figure 20-B: Block diagram, left to right, S0–S8

20.3 The Autosuggestion Capacitor — ^5 Self-Charging

Chapter 7.2 gives the charging curve `P_cap(t) = 10 × (1 − e^(−t/τ))` on the 0–10 composition scale. Self-charging adds two loss terms, and between them they decide the whole schedule.

Leakage (Chapter 7.3). Unfed composition decays. Take `τ_leak = 30 days` — this book's figure for a composition with fewer than about a hundred logged enactments behind it — and a session depositing `ΔP_session = 0.35` points. Chapter 7.3's `P_eq = f × τ_leak` becomes, for a mean interval `T`:

```
P_eq  =  ΔP_session × τ_leak / T
```

Cadence	T (days)	P_eq (of 10)
Daily	1.00	10.5 → clips at 10.0
Six a week	1.17	9.0
Five a week	1.40	7.5
Alternate days	2.00	5.25
Weekly	7.00	1.5

Bench Note — the weekly session does not equilibrate at one-seventh. Read the table again. Daily and weekly are the same session, the same operator, the same intent, and they land at 10.0 and 1.5. Leakage is proportional to the level already stored, so the weekly regime spends its whole deposit replacing what drained and settles where the two rates meet. That is the honest answer to *can I just do it on Sundays.* Not "slower" — a different equilibrium, and a low one. Invert the formula for a target of 8.0 and the maximum mean interval is `T = 0.35 × 30 / 8 = 1.31 days`, about 5.3 sessions a week. Which is why the design point is six, why one missed day a week is inside budget, and why two is not.

Breakdown (Chapter 7.4). The second term is not a loss; it is a failure. A store punctures when the potential across it exceeds its rated V_BR. Here the rating is not fixed — it grows with evidence:

```
V_BR = d_evidence          d_evidence = logged enactments / 10, in V_c
design limit:  V_sugg ≤ 0.5 × V_BR          (Chapter 7.4 derating)
```

V_sugg is how far above the observed composition your assertion sits, in composition points. The arc at six sessions a week:

Day	Enactments	V_BR	Max V_sugg
1	1	0.1	0.05
30	26	2.6	1.3
66	57	5.7	2.85

Day	Enactments	V_BR	Max V_sugg
254	218	8.0 (clamped at Ch. 7's rating)	4.0

On day one the only assertion inside margin is essentially *I practised today.* At day 30 there is room for *I am becoming someone who practises.* This is the structural account of why "I am a millionaire," asserted with no logged enactments, does nothing useful: V_sugg $\approx$ 8 against V_BR $\approx$ 0.1, a ratio of 80 where the limit is 0.5. The store does not fail to charge; it *punctures*. A punctured dielectric conducts, so the over-claim discharges the adjacent true claims — which is why the observable symptom is a practitioner ending the month less certain of things they previously did believe about themselves. The repair is to halve V_sugg and rebuild thickness. Nothing raises V_BR except enactments.

Identity work is this circuit with the reference at the Principled domain and the schedule in months. Reclaiming a disowned trait runs at β = 1.0, no amplification, for that reason: evidence thickness under a newly recognised trait is by definition near zero, so there is no headroom to amplify into. This book does not supply a dedicated shadow-work circuit or claim a mental-health treatment; the consent-bound topology here, with the involution guard of Chapter 6.2 added, is the structural limit of what it specifies.

20.4 The Habit Oscillator — ^7 Rhythm with Auto-Gating

Chapter 11.2 gives the oscillator and its charge-pump arithmetic. An ungated one fires the ^8^0 cue forever, which is wrong in both directions: it keeps spending drive into a full store, and it cannot notice when the store has drained.

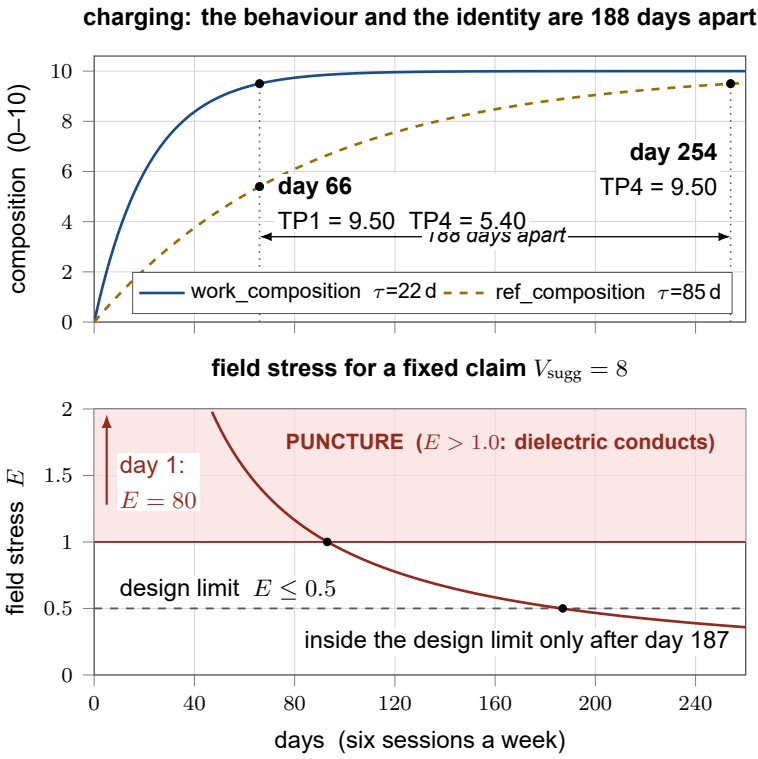


Figure 20-C: Two stacked plots on a shared day axis, 0 to 260

The gate G1 is a Schmitt trigger — an op-amp with positive feedback around it (Chapter 11.4) — sensing the working composition:

```
enable when  P_work < 8.0          disable when  P_work > 9.0
```

Hysteresis band: 1.0 composition point. Not decoration. At τ_work = 22 d:

```
P = 8.0  at  t = 22 · ln 5  = 35.4 days
P = 9.0  at  t = 22 · ln 10 = 50.7 days
```

The band spans days 35 to 51 — fifteen days in which the oscillator keeps running while the habit is already mostly formed. Now delete the hysteresis and use one threshold at 8.5. The gate opens

at day 41.7. Leakage there is `8.5 / 30 = 0.28` points per day, so the level falls back under threshold within a day and the gate re-closes; the next session pushes it over and it opens again. That is a comparator without hysteresis, and it chatters at a period of about one day — which the operator experiences as *I have got this* / *no I have not*, daily, indefinitely.

> **Bench Note — hysteresis converts chatter into a maintenance schedule.** With the band in place the gate closes at `P = 9.0` and the level decays as `9.0 · e^(−t/30)`, reaching 8.0 at `t = 30 · ln(9/8) = 3.5 days`. That is not a failure; it is the designed top-up cadence, and it falls out of the two thresholds and τ`_leak` without anyone choosing it. Note what happens as the composition ages: τ`_leak` lengthens with evidence thickness — Chapter 7.3's unmarked default is 90 days — and the same arithmetic gives `90 · ln(9/8) = 10.6 days`. Long-established practices tolerate ten-day gaps for a computable reason. New ones do not.

Two further rules, each with a named failure.

Gate on the slow node, never the fast one. `G1` senses `work_composition`, which moves on 22 days. Sense an affective node instead — *do I feel like it today*, `^2^E|f`, bandwidth of hours — and the gate chatters at the bandwidth of mood. The symptom is a practice that runs in bursts synchronised to nothing.

Fixed deposit, recoverable phase. SI3. Miss a cycle and resume at the next scheduled phase at the same 0.35 deposit. Do not double up to catch up: doubling raises `di/dt` (Chapter 9.2) *and* raises `V_sugg` against an unchanged `V_BR`, so the make-up session is simultaneously the highest-recoil and highest-stress event in the schedule. It is the most reliable way to end a streak that was otherwise healthy.

Breaking a habit runs on this same stage, unchanged. You do not delete an oscillator: the `^8^O_{F6}|E` cue is a fixture of your day

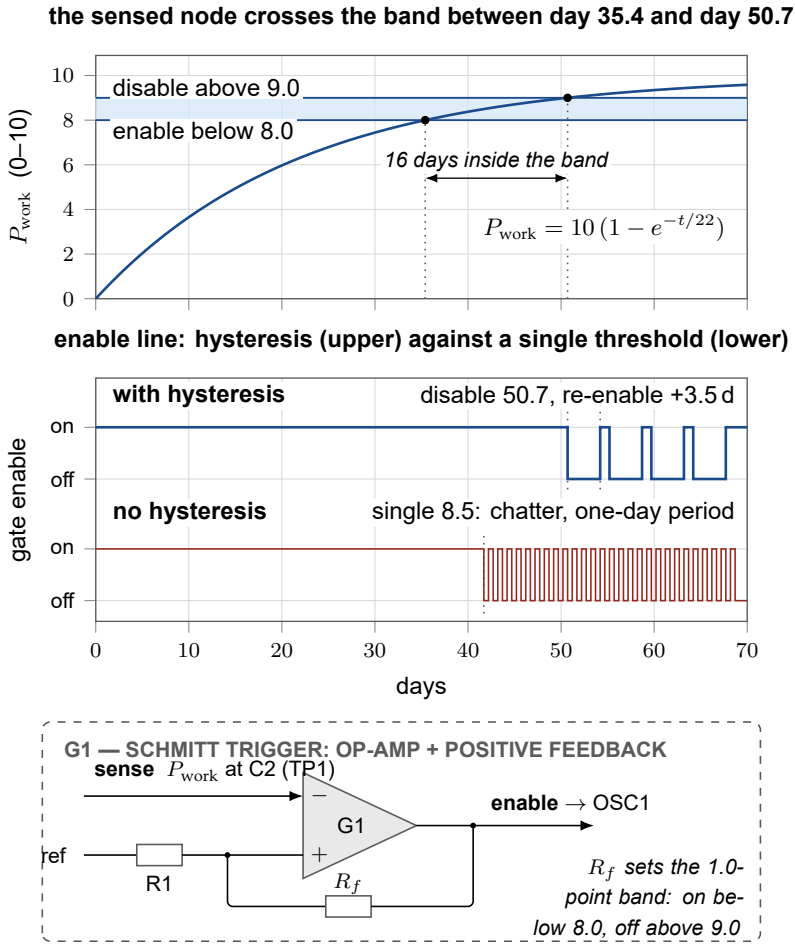


Figure 20-D: Oscillator-and-gate detail

and fires whether a routine is attached to it or not. You re-point it. Keep the cue, keep the period, keep the deposit, swap the `^9^0` the cue lands on. Chapter 9.2's conditions on a flyback path apply unmodified, and the first is the one operators skip — the substitute must already be conducting *before* the old path opens.

On dependency — a structural description, and where this chapter stops. Read structurally, a dependency differs from an ordinary habit on all three parameters set above. Gain on the enactment stage runs far above `Q3`'s unity, so a small `^8` trigger produces a large `^9`. `r`π at the bias terminal has collapsed toward zero — the parameter description of a reference terminal that no longer holds a reference. And the period is no longer set by `OSC1`; it is set by the substrate, which leaves `G1` nothing to gate, because the enable line is not connected to anything the operator controls.

That is a complete structural description, and it is deliberately where the account ends. This book gives no protocol for it, and the reason is not squeamishness. The model here describes where attention goes. It does not describe physiology, tolerance, or withdrawal, and it is not competent to. **Dependency is a clinical matter.** If that is what you are looking at, in yourself or in someone else, the next step is a qualified clinician, and nothing in this chapter, this board, or this book substitutes for that care.

20.5 Full CCDL Schematic

```
board ccb_self {                          // CCB-SELF — closed loop;
↪  operator is target
    input:  intent;                       // the ^0 asserted this cycle
    output: enacted;                      // the ^9 the cycle produces
    // OSC1 generates the ^8 cue on-board — SI5: no external terminal.
    ↪  Time base: days.
```

```
// ---- S0  REFERENCE: the slow ^5. Bias source, never an output.
↪  ----
node ref_composition: ^5^P_{F7}|T;              // Female:
↪  composition built toward
capacitor c1: Capacitor< node=ref_composition, tau=85.0,
                         leak_tau=180.0, V_BR=8.0 >;

// ---- S1  INTAKE: the asserted ^0, amplitude-limited ----
node intent: ^0^C_{F7}|A;                       // Idea: today's
↪  assertion
resistor r1:   Resistor< path=intent, R=30.0 >;
limiter  lim1: Limiter< path=intent, max=1.3 >; // 0.5 x V_BR at
↪  d_evidence = 2.6

// ---- S2  NOTCH: the ^6 audit off the rail, not silenced ----
node audit: ^6^C_{F6}|A;                        // Male:
↪  decomposition and test
notch n1: Notch< in=audit, f0=0.35, Q=4.0 >;    // events/min —
↪  in-session only

// ---- S3  MAIN CJT: biased from the reference, not from the output
↪  ----
node work_composition: ^5^C_{F5}|⟨A,E⟩;         // Female: as
↪  currently performing
capacitor c2: Capacitor< node=work_composition, tau=22.0,
                         leak_tau=30.0, V_BR=8.0 >;
cjt q1: CJT< intent | ref_composition => work_composition >
        [β=1.5, sat=0.60, rπ=8.0, region=active];

// ---- S4  OSCILLATOR + HYSTERETIC GATE ----
node cadence: ^7^C_{F6}|A;                      // Rhythm: the
↪  entraining cadence
node cue:     ^8^O_{F6}|E;                      // Above/Cause: time
↪  and place fire it
schmitt g1: Schmitt< sense=work_composition, on_below=8.0,
↪  off_above=9.0 >;
osc  osc1:  Oscillator< signal=cue, source=cadence, period=1.0,
                        duty=0.04, sustain=254.0, gate=g1 >;

// ---- S5  ENACTMENT: the only node that is not self-report ----
```

```
    node enacted:    ^9^O_{F5}|E;                  // Below/Effect: the
    ↪  behaviour performed
    node impression: ^4^E_{F3}|f;                  // Vibration: what
    ↪  the cycle left behind
    cjt    q3: CJT< cue | work_composition => enacted > [β=1.0,
    ↪  region=active];
    sample s2: Sample< node=impression, period=1.0, window=1440.0 >;  //
    ↪  days, minutes

    // ---- S6  EVIDENCE FEEDBACK: the one lawful O -> P ascent (Ch.
    ↪  8.4) ----
    filter lp1: LowPass< in=enacted, out=evidence_trend, tau=21.0 >;
    fuse   f1:  Fuse< ref_composition, threshold=0.12,
    ↪  action=open_feedback >;
    feedback evidence_trend -> ref_composition [β=0.05];   // arrives at
    ↪  the BIAS terminal

    // ---- S7  CSB STAGE: the self-compassion gate. Mandatory. ----
    node miss_trigger: ^8^C_{F5}|f;                // Above/Cause: a
    ↪  miss, and what it recalls
    node compassion:   ^3^P_{F7}|⟨A,f,T⟩;          // Negative: recoil
    ↪  from self-condemnation
    node acceptance:   ^2^E_{F4}|f;                // Positive:
    ↪  affinity for the self that missed
    cjt q2: CJT< miss_trigger | compassion => acceptance > [β=1.5,
    ↪  sat=0.70];

    // ---- S8  INSTRUMENT: the ^6 audit, sampled and bounded ----
    sample s1: Sample< node=audit, period=7.0, window=10.0 >;         //
    ↪  days, minutes

    // ---- TEST POINTS ----
    probe tp1: work_composition;    // P_cap against 10*(1 - e^(-t/22))
    probe tp2: enacted;             // cycles completed / cycles
    ↪  scheduled
    probe tp3: lim1;                // V_sugg against 0.5 x V_BR
    probe tp4: ref_composition;     // reference slew, points per day
}
```

The stage walk.

S0. C1 is the ^5^P_{F7}|T store Chapter 8.4 calls the constitution. CCB-SELF *writes* the node every other board only reads as bias — which is why SI1 and F1 exist. leak_tau = 180 d because the identity reference is backed by a lifetime of prior enactment rather than this operation's log; Chapter 7.3's rule that leak lengthens with evidence puts it well past the 90-day default. Set it to 30 and the reference drains between operations.

S1. R1 = 30 Ω_c lets the assertion in without flooding the intake. LIM1 is the one parameter *rescheduled during the operation*: as V_BR grows with enactments, the ceiling rises with it.

S2. The notch rejects the audit in-band — the 0.35 events per minute of a running session — and passes it everywhere else. It does not remove the ^6; it moves it to S8.

S3. C2 is a separate part from C1 for the whole reason §20.1 exists. leak_tau = 30 d is what every cadence figure in §20.3 is computed from; assume 60 and a weekly schedule looks survivable when it actually settles at 1.5. Q1 sits between the two nodes. β = 1.5 is moderate: above about 4 the assertion outruns the evidence and V_sugg clears the limiter faster than the limiter can be rescheduled; below 1 each session yields less composition than it cost, and the operator correctly concludes it is not working. sat = 0.60 caps what one cycle may contribute — at 1.0 you get the weekend-breakthrough topology, full charge over no thickness, gone in weeks. rπ = 8.0 Ω_c keeps the stage from loading the reference.

S4. As §20.4. period = 1.0 is a number, not the word "daily", because §20.3's arithmetic needs one.

S5. Q3 runs at unity: the enactment renders what the working composition actually supports. Above unity it produces a performance the practitioner cannot repeat tomorrow. S2 catches the ^4 Vibration return Chapter 22.2 grades — one window per cycle, a full day wide, because §20.6's t_conf is measured from the end of the session. Narrow it to the session itself and you record t_conf as never.

S6. τ = 21 d and β = 0.05 are inherited from Chapter 8.4, not chosen freshly. The path terminates at a bias terminal, never at the signal path — that would make it the oscillator Chapter 14 builds on purpose. F1 enforces the ceiling: a store at τ_ref = 85 d rises at most 10/85 = 0.118 points per day, so anything above 0.12 means something is writing the reference directly rather than through the filter, and the fuse opens the feedback.

S7. The compassion bias is ^3^P_{F7}|⟨A,f,T⟩ — principled ^3 Negative in three registers, no fusion claimed. Check it against the anti-drift rule: this is visceral recoil *from self-condemnation*, the can't-stomach-it response to treating one missed morning as a verdict on a life. It is not an argument against self-condemnation; that would be ^6, and it arrives later and works less well. The cause is ^8^C_{F5}|f — a miss triggering its associations, *that reminds me of every other thing I quit.* Associative surfacing, canonical ^8, not deduction. Domain descends ^C → ^E.

> **Bench Note — the compassion gate is a decoupling capacitor, not a kindness.** It is easy to read S7 as the soft stage bolted onto an engineering chapter. It is the opposite. Without it a missed cycle propagates from ^9^0 into the Principled reference, the reference begins moving at the working node's speed, τ_ref/τ_work collapses toward 1, and the board is back in the §20.1 fault — this time with an amplifier in it. S6's low-pass stops the spike *reaching* the reference; S7 handles the spike that reaches awareness anyway. Two mechanisms, one function, exactly as Chapter 24 pairs a saturating stage with a fuse. Remove either and the failure is not that the operator feels worse. It is that the reference stops being a reference.

S8. Weekly, ten minutes, written down. High-impedance measurement.

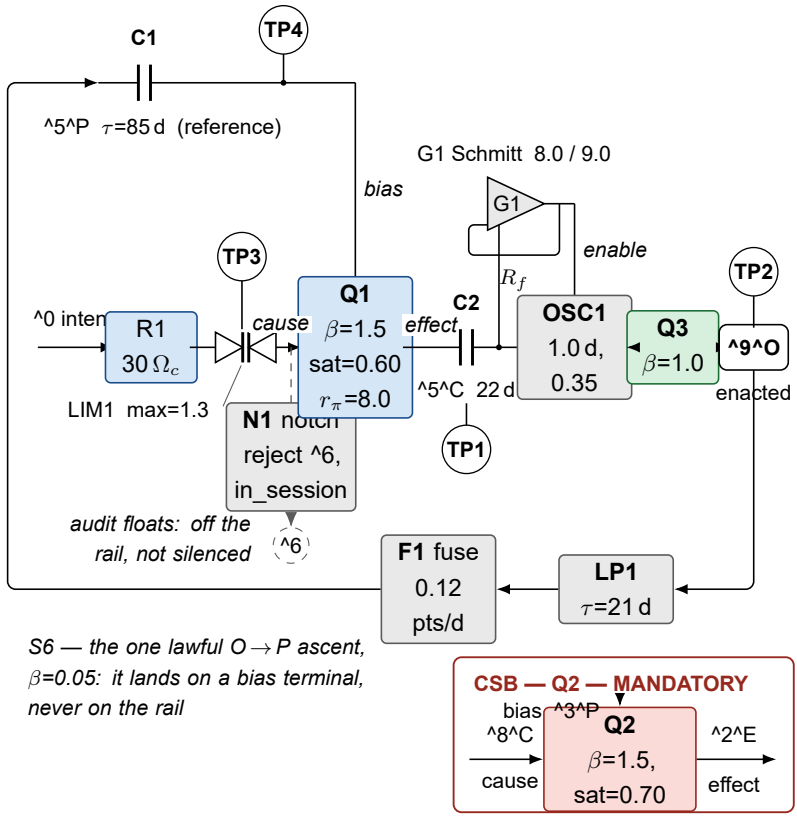


Figure 20-E: Full schematic with reference designators

20.6 The 10,000 Hour Protocol — Operationalizing St. John's for Self-Mastery

Chapter 5.2 sets τ_mastery = 3,333 h so that 3τ = 10,000 h. On this board that is a third store, charging on practice hours rather than days.

Practice hours	P_mastery (of 10)	Fraction of 10,000
1,000	2.59	10%

Practice hours	`P_mastery` (of 10)	Fraction of 10,000
3,333 (1τ)	6.32	33%
6,666 (2τ)	8.65	67%
10,000 (3τ)	9.50	100%

Read the middle rows against each other. **The first third of the hours buys 6.32 points; the remaining two thirds buy 3.18.** That is the exponential, and it is the structural reason competent practitioners plateau in *feel* while continuing to improve in *fact*. The curve is flattening, not the practitioner.

Applying St. John's per session. Take a 45-minute practice session. The `^4` Vibration confirmation — the passage playing itself back unbidden while you are doing something else — arrives 90 minutes later.

```
η        = t_op / t_conf = 45/90 = 1/2
result   = 1/2 × 1/3 = 1/6          a = 1,  b = 6
A        = 3/η = 6.0                n_ops = 1,  T_manifest = 1 × 45 =
↪  45 min
T_arrive = b × t_conf = 6 × 90 = 540 min      (recorded, not scheduled
↪  around)
```

`A = 6.0` sits in Chapter 5.3's under-driven band: one lever at a time, leave the amplifier alone.

> **Bench Note — on this board `A` is spent as a count, not an amplitude.** Chapter 5.3 is explicit that `A` multiplies *delivered current* and is not a β setting, and CCB-SELF makes the point unavoidable: `LIM1` will not permit a sixfold assertion, because `V_sugg` is capped by an evidence thickness that six times the conviction does not change. So convert amplitude into count. Six times the activation, delivered at a fixed 0.35 deposit, is six cycles — `n_ops` absorbs the multiple `A` was asking for, at a cost of `6 × 45 = 270` further minutes at the current setting. That conversion is the whole difference between a practice regime

and a burnout, and it is available only because this board has an oscillator. A one-shot board like CCB-D has no count to spend into.

Skill acquisition is where the one legitimate $\times$ on this board appears. Accumulated composition and entraining cadence, genuinely integrated, produce a state neither supplies alone:

```
integrate (^5^C_{F5}|A × ^7^C_{F6}|A) = ^5^C_{F5}|Ω_fluency;
```

`Ω_fluency` is defined here as the state in which performance no longer requires deliberate assembly — composition and cadence have fused, so the parts are not retrieved and combined; they arrive together. The $\times$ is licensed on three counts: different Noetics, no domain ascent (`^C × ^C`), and an emergent result explicitly named and defined (Chapter 2.6). Written with + it would claim only that composition and rhythm are co-present, which is true at hour 100 and misses entirely what has happened by hour 3,000.

Component and Parameter Table

Every value is a **design choice** made in this book, not a measurement.

Ref	Component	CAS state / form	Value	Why, and what breaks at other values
`C1`	Capacitor	`^5^P_{F7}|T`	τ=85 d, leak=180 d, V_BR=8.0	The slow reference; ratio 3.86. Below 3 the nodes lock and a bad day rewrites the identity. At leak=30 d it drains between operations

Ref	Component	CAS state / form	Value	Why, and what breaks at other values
C2	Capacitor	^5^C_{F5}|⟨A,E⟩	τ=22 d, leak=30 d, V_BR=8.0	Complex-behavioural. Every §20.3 cadence figure is computed from leak=30 d; assume 60 and a weekly schedule looks survivable when it settles at 1.5
R1	Resistor	on ^0^C_{F7}|A	R=30 Ω_c	Lets the assertion in without flooding the intake
LIM1	Limiter	clamp on intent	max=1.3	Holds V_sugg $\leq$ 0.5 $\times$ V_BR. Above V_BR the store punctures and adjacent true claims discharge
N1	Notch	rejects ^6^C_{F6}|A	f0=0.35, Q=4.0	Audit off the rail in-band. Leave it in and the operation is measured continuously and never accumulates
Q1	CJT	^0^C $\rightarrow$ ^5^C, bias ^5^P	β=1.5, sat=0.60, rπ=8.0	β>4: assertion outruns the limiter. β<1: session costs more than it yields. sat=1.0: full charge over no thickness. rπ=0.5: one poor session depletes the identity node
G1	Schmitt	senses C2	8.0 / 9.0	A 1.0-point band turns chatter into a 3.5-day top-up. At zero band the gate chatters daily

Ref	Component	CAS state / form	Value	Why, and what breaks at other values
`OSC1`	Oscillator	`^7^C_{F6}|A` → `^8^O_{F6}|E`	period=1.0 d, ΔP=0.35	One bounded session's deposit. Raise it without evidence and `V_sugg` rises with it
`Q3`	CJT	`^8^O` → `^9^O`	β=1.0	Unity. Above 1 it renders a performance the practitioner cannot repeat tomorrow
`LP1`	LowPass	in `^9^O`	τ=21 d	Chapter 8.4's floor. At hours you have removed the diode chain, not filtered it
—	Feedback	trend → `C1` bias	β=0.05	Chapter 8.4's ceiling. At 0.5, three weeks of bad days rewrite the constitution
`F1`	Fuse	on `C1` slew	0.12 pts/day	`10/85 = 0.118`, the maximum lawful slew. Higher leaves SI1 unenforced; lower blows on normal charging
`Q2`	CJT	`^8^C` → `^2^E`, bias `^3^P`	β=1.5, sat=0.70	The CSB stage. Remove it and a shame oscillator occupies the safety slot
`S1/S2`	Sample	`^6^C` / `^4^E`	7 d/10 min · 1 d/1,440 min	High-impedance instruments. A window narrower than a day records `t_conf` as never

Ref	Component	CAS state / form	Value	Why, and what breaks at other values
Dials	—	involution pairs	^2=6 ^3=5 · ^5=7 ^6=5 · ^8=5 ^9=6	All inside the 4-point divergence limit; beyond it, inversion (Chapter 6.2)

Worked Example — The Morning Hour

An operator wants a daily hour at the piano, and behind it the identity *I am someone who practises.* Three stores, three timescales, one board.

Set-up. Habit layer τ_work = 22 d, 3τ = 66 d. Identity layer τ_ref = 85 d, 3τ = 254 d. Ratio 85/22 = 3.86 $\geq$ 3 — passes with 29% margin. Mastery layer τ_mastery = 3,333 h.

Cadence. Target P_work = 8.0. From §20.3, T_max = 0.35 × 30 / 8 = 1.31 days — six sessions a week, landing at P_eq = 9.0. Five a week lands at 7.5 and misses. Two decimal places in the schedule, a whole grade in the outcome.

Day 1. One enactment, V_BR = 0.1, so LIM1 permits V_sugg = 0.05. The only assertion inside margin is *I practised today.* The operator who instead asserts *I am a pianist* runs 80× the rating and will be less sure of themselves in a month than at the start.

Day 30. 26 enactments, V_BR = 2.6, LIM1 rescheduled to 1.3. P_work = 10(1 − e^(−30/22)) = 7.44. G1 still enabled; its 8.0 threshold arrives at day 35.4. TP2 should read $\geq$ 0.85; at six a week it reads 0.857.

Day 66. P_work = 9.50. The gate closed at day 50.7 and has been topping up every 3.5 days since. The habit is automatic. And P_ref = 10(1 − e^(−66/85)) = 5.40 — the identity store is at barely half.

That pair is the result to carry out of this chapter. **The behaviour arrives at day 66; the composition that makes it feel like yours arrives at day 254.** An operator who reads the 66-day figure and expects to *feel* different at day 66 is probing the wrong store, concludes the operation failed, and stops — 26 days after the habit became automatic and 188 days before the identity finished charging. There is no fix for this in the circuit. The fix is reading TP1 and TP4 as two different instruments, which is what they are.

Mastery layer. One hour a day, six days a week, is 313 h/year. `P_mastery` at 1,000 hours — a little over three years — is 2.59. At 3,333 hours it is 6.32, arriving in year eleven. Stated plainly rather than encouragingly: at this cadence the mastery store is a decade-scale component, and the honest lever is hours per day, because intensity per hour is capped by `LIM1` and hours per day is not.

Failure Modes

Mode	Mechanism	What you observe	Fix
Bias–effect short	Reference and working node are one node; `V_BE = 0`	Total collapse within days of a clear decision; operator blames motivation	Split the node; verify τ`_ref/`τ`_work` $\geq$ `3` at TP4 before anything else
Reference chasing	τ`_ref` sited on an evaluative node ($\tau \approx$ 1 d)	TP4 moves on the day TP2 records a miss	Re-site the reference at `^P`; confirm `F1` intact and S7 populated
Store puncture	`V_sugg` far above `0.5` $\times$ `V_BR`	Confidence lost in *previously settled* beliefs, not just the new one	Halve `V_sugg`; rebuild thickness. Trying harder raises the stress
Sub-cadence stall	Mean interval above `T_max`; `P_eq` below target	Months of genuine effort, level parked between 1.5 and 5	Raise cadence, raise Δ`P_session`, or lower the target — two of the three are usually available

Mode	Mechanism	What you observe	Fix
Gate chatter	Hysteresis removed, or gate sensing an affective node	Daily on/off cycling; bursts synchronised to nothing	Restore the 1.0-point band; move the sense line to the slow node
Make-up doubling	SI3 violated after a missed cycle	The streak ends on the catch-up session, not the missed one	Fixed deposit, recoverable phase. Resume at nominal
Audit loading	`^6` left in the signal path	The operation is measured continuously and never accumulates	Notch in-band; move the audit to `S1` — weekly, bounded, written
Compassion gate absent	S7 removed as sentimental	One miss produces a principled verdict; abandonment within days	Reinstall. It is a decoupling capacitor; the board oscillates without it
Wrong instrument read	TP1 read where TP4 was meant	"It worked but I don't feel different" at day 66	Read both. They are 188 days apart by design

Test Points

TP	Node	Reads	Expected
TP1	`work_composition`	`P_cap`, 0–10 scale	Tracks `10(1 - e^(-t/22))`: 6.32 at day 22, 9.50 at day 66. Flat while TP2 is healthy means `Q1` is in cutoff — check the bias, not the drive
TP2	`enacted`	Cycles completed / scheduled	$\geq$ 0.85 for a target of 8.0. The only non-self-report node here; if TP1 and TP2 disagree, TP2 is right
TP3	`LIM1`	`V_sugg` against `0.5` × `V_BR`	At or below the ceiling. Above `V_BR` is puncture, and the damage shows first in *adjacent* claims

TP	Node	Reads	Expected
TP4	`ref_composition`	Reference slew, points/day	$\leq$ 0.12, and **uncorrelated with any single TP2 miss.** Correlation is the safety failure: the decoupling is gone and the board is a shame oscillator

Read them TP4, TP3, TP2, TP1 — confirm the reference is decoupled, confirm the assertion is inside its derating, confirm the cycles ran, then read the charge. That order is the bring-up procedure, and it puts the two silent failures first.

What This Chapter Established

- **The self-programming fault is topological, not motivational.** Bias tied to effect is a junction at `V_BE = 0`, and a transistor in that condition is off — undamaged, correctly specified, inert.
- **One `^5` becomes two by separating them in time.** τ`_ref` $\geq$ 3 $\times$ τ`_work`. Below the ratio the reference chases the output and every bad day is an identity verdict.
- **Leakage sets the schedule, and it is not linear in cadence.** `P_eq` = Δ`P_session` $\times$ τ`_leak / T`; the same session run daily and weekly equilibrates at 10.0 and 1.5.
- **Suggestion amplitude is limited by evidence, not conviction.** `V_sugg` $\leq$ `0.5` $\times$ `V_BR`, and `V_BR` grows only with logged enactments. Puncture discharges *adjacent* true claims.
- **The `^6` Male audit is an instrument, and an instrument in the signal path is a load.** Notch it in-band, sample it weekly and bounded. It is `^6`, never `^3`; a genuine `^3^E|f` recoil is a rate problem belonging to Chapter 9.2.

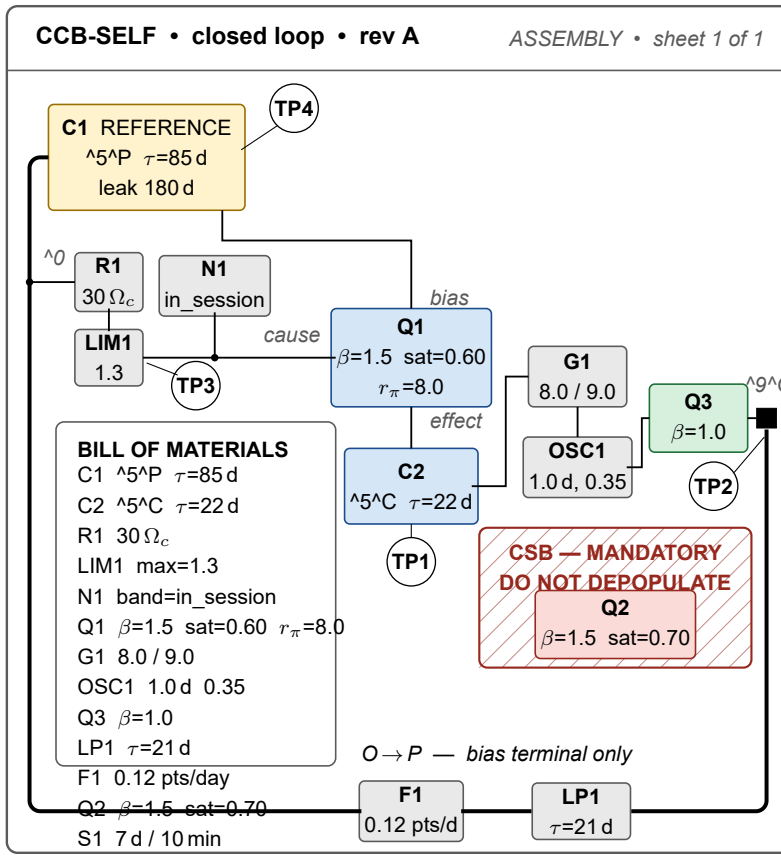


Figure 20-F: Assembly drawing

- **Hysteresis converts chatter into a maintenance schedule** — a 1.0-point band against τ_leak = 30 d gives a 3.5-day top-up interval nobody had to choose.
- **The self-compassion gate is a decoupling capacitor.** It is what holds τ_ref long. Without it the board has a shame oscillator where its safety stage should be.
- **St. John's A is spent here as a count, not an amplitude,** because the limiter forbids the amplitude. Six times the activation is six cycles, at 270 further minutes.

- **Behaviour and identity arrive 188 days apart** at the design point, and reading the wrong store at day 66 is the most common reason a working operation gets abandoned.

Exercises

(Answers in Appendix H.)

20.1 An operator sites the reference at `^2^E_{F3}|f` — *do I feel like someone who does this?* — with an effective τ of 1.5 days, against a working node at τ`_work = 22 d`. Compute the ratio, state whether it passes SI1, and describe in two sentences what TP4 shows on the first missed cycle.

20.2 A session deposits Δ`P_session = 0.50` into a composition with τ`_leak = 24 d`. Compute `P_eq` for daily, alternate-day, and twice-weekly cadence, then give the maximum mean interval reaching `P_target = 7.0` and the weekly session count it implies.

20.3 On day 1 an operator asserts a composition 6.0 points above their observed level, with two logged enactments. Compute `V_BR`, the ratio `V_sugg / V_BR`, and the multiple of the 0.5 derating limit. Then compute how many enactments are needed before that assertion sits inside the limit, and how long that takes at six sessions a week.

20.4 Explain why the in-session interference *"this is pointless, I should be doing something useful"* is `^6^C_{F6}|A` and not `^3^E_{F3}|f`. Quote the canonical definition of `^3` Negative, then state the different repair each calls for and which chapter supplies it.

20.5 A gate is built with a single threshold at `P = 8.5` on a node with τ`_work = 18 d` and τ`_leak = 25 d`. Compute the day it first opens, the leakage rate at that level, and the chatter period. Then add a 1.0-point band at 8.0 / 9.0, recompute both transition days and the top-up interval, and say what the operator experiences in each case.

20.6 (Design) Build a CCB-SELF for a simple behavioural target: τ_work = 10 d, τ_leak = 20 d, reference τ_ref = 40 d, ΔP_session = 0.30, target P_work = 8.5. Produce (a) the ratio check and whether it passes; (b) the maximum mean interval and its weekly cadence, then state whether the target is reachable at one session per day, and if not, which of the three levers you would move and why; (c) the F1 slew threshold, derived; (d) a CCDL fragment declaring C1, C2, Q1, G1, OSC1, LP1, F1 and four probes. It must compile — check the feedback's τ and β against Chapter 8.4 first.

20.7 (Marquee) An operator completes 66 days at six sessions a week. TP1 reads 9.50, TP2 reads 0.86, TP3 reads 2.4 against V_BR = 5.7. TP4 reads 5.40 and has risen smoothly at 0.05 points per day throughout. The operator reports the practice is effortless, that they still do not feel like a musician, and proposes to raise β on Q1 from 1.5 to 5.0 to "make it land."

a) State which test point they are misreading and give its correct reading.
b) Compute the resulting V_sugg if it scales with β, compare it to 0.5 $\times$ V_BR at V_BR = 5.7, and name the failure mode they would enter.
c) Give the correct intervention with the number that justifies it, and state the one thing this circuit model cannot tell them about their situation.

Part IV

Operations

Gating, measuring, calibrating, and refusing

21 The RPM Prerequisite Gate

The interlock that runs before the cast — four checks, four numbers, and the one operation you cannot reverse

Fourteen operations. Two hundred and eighty logged minutes. One desire. Not a single confirmation graded better than 3°, which Chapter 5.5 calls subtle, ambiguous, weak.

The instinct is to reach for the amplifier — raise β, hold gnosis longer, tighten the sigil. All three are wrong, and the log says so on line one: the desire as worded was never the desire. Every number after it measured delivery of a target nobody wanted. The amplifier worked perfectly. It amplified the wrong thing fourteen times.

CCB-RPM is the interlock that catches that on day one. It casts nothing, amplifies nothing, and emits one output — `ARM` or `HOLD`, with the deciding number attached. It runs the chain `A0` $\rightarrow$ `D` $\rightarrow$ `W` $\rightarrow$ `P` $\rightarrow$ `outcome` as four gates in series, each with a threshold you check by hand.

It costs forty minutes of paper and zero casts. It is the cheapest board in this book and the most often skipped, because skipping it feels like starting. And it is the only board whose skipping cannot be repaired afterward — Section 21.5, the part with real mathematics under it.

Requirements and Safety Invariants

Chapter 12.2 requires the "must not" list before the functional one.

#	Functional requirement	Realized by
R1	Authenticate the desire as the operator's own before anything else runs	A0 differential pair + fuse F1 (§21.1)
R2	Return a *number* for drive sufficiency, not an impression	CAS-Ohm D-gate, monitor M2 (§21.2)
R3	Verify the schematic is possessed, not intended	Six-input AND, no partial credit (§21.3)
R4	Verify the schedule can actually be held	Budget monitor at half the available minutes (§21.4)
R5	Emit `ARM` or `HOLD` with the deciding number	MUX + ledger, TP4

#	Safety invariant (MUST NOT)	Enforcement
S1	Arm on a summed score — the four gates are not fungible	`AND`, never a weighted sum
S2	Amplify anything	Every verdict CJT at $\beta = 1.0$; `E-CJT-005` on any loop
S3	Let an Operational outcome edit the Principled substrate directly	Diode `d_out`; filtered $\tau = 21$ d feedback at $\beta = 0.15$ (Ch. 8.4)
S4	Present the scan as an undo	No inverse component exists, in the listing or the language (§21.5)
S5	Arm while the operator's CSB is unarmed	No `ARM` on a blown involution monitor (Ch. 24, Ch. 6.2)

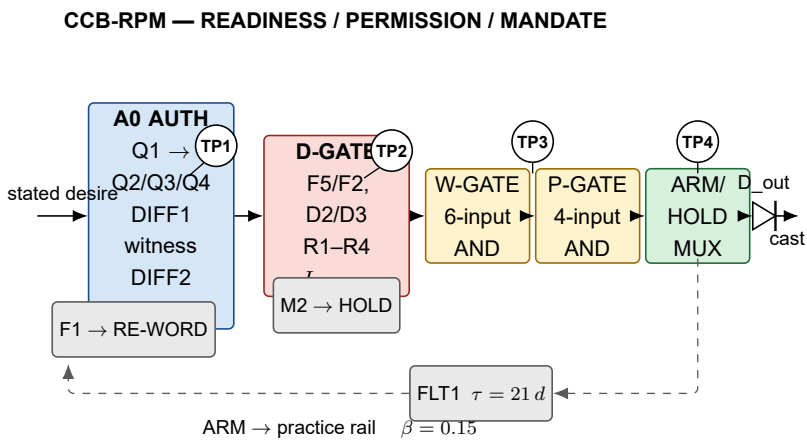


Figure 21-A: Block diagram

21.1 Ao (Pure Desire) Authentication — Is This Really What You Want?

Ao is the only gate that asks about the operator rather than the circuit, and it has one question in two halves: *is the pull mine, and is it toward the thing I named?*

Both halves are answered by subtraction. The operator writes three readings on a 0–1 scale, all typed `^2^E_{F5}|f` — Positive, Evaluative domain, Agency, affective register:

Reading	What you rate
`pull_open`	The desire exactly as worded
`pull_private`	Same desire, all witnesses removed — nobody ever knows, asks, or sees
`pull_condition`	The condition the target stands for, delivered by another route, *without* the target

These are self-ratings in a log, not measurements of anything. What makes them usable is that the gate acts on *differences*, so a con-

sistent personal bias subtracts out — the common-mode argument the CSB's differential pair runs on (Ch. 24).

```
witness_load   = pull_open − pull_private     trip at ≥ 0.30
object_premium = pull_open − pull_condition   hold at ≥ 0.20
```

Witness load ≥ 0.30 blows fuse F1. A pull that drops by a third when the audience goes is substantially `^2^O_{F4}|f` — attraction toward being seen — wearing the target's clothes. Chapter 13.6 runs this probe on the Red Corvette and passes it: the operator wants the car with nobody watching, so the desire is `^2^E` toward the object, not `^2^O` toward the witnesses. Threshold 0.30 is a design choice — a third of the pull is where the audience, not the object, is setting the amplitude. At 0.50 the fuse passes desires that are almost entirely audience; at 0.10 it blocks nearly everything and the operator quietly stops logging honestly, which is worse.

Object premium below 0.20 trips M1 as `PROXY_TARGET`. If the bare condition pulls nearly as hard as the named target, the target is a proxy and the condition is the prime desire. 0.20 is the smallest gap that survives self-rating noise; below it the readings are indistinguishable and the honest verdict is *undecided*, reported as a hold.

The repair for both is `^6^C_{F6}|A` — Male, Cognitive, analytical: take the wording apart and see what it is made of. Note the Noetic. Decomposition is `^6`; it is not `^3`, even though what you are often hunting is a `^3`, a visceral recoil across the path. Finding it is `^6` work. Chapter 2.2 names this the commonest drift in the book, and it costs — an operator who thinks aversion-hunting is itself `^3` goes looking for something to recoil from and reliably finds it.

> **Bench Note — write all three numbers before you compute either difference.** Self-reports drift toward the answer the operator already wants. Rate open, then private, then condition, in ink, and only then subtract. Compute the witness load after two readings and the third gets

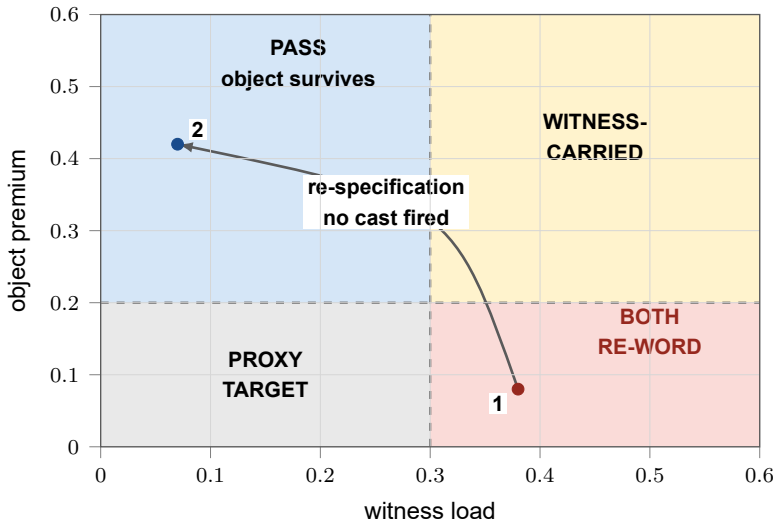


Figure 21-B: Decision map

bent to clear the threshold — not dishonestly, just helpfully, the way anyone filling in a form they want to pass bends it. This is the one place on the board where the procedure does more work than the arithmetic, and it is free.

A plain line about the operator, not the circuit: decomposition can surface something painful rather than a mis-worded target. This is a checklist on paper — it identifies nothing about a person and treats nothing. If what surfaces is distress, that is a matter for a clinician, and nothing here substitutes for that care.

21.2 D-Gate: V_drive Sufficiency Check

Pure arithmetic, and it is Chapter 4 applied unmodified.

```
I_attention  = ( V_drive − ΣVf ) / R_constraint          (Ch. 4.2)
```

```
R_constraint = R_cap + R_policy + R_context + R_resource   (Ch. 4.4)
```

`V_drive` sums the drive dials on the path, each an integer 3–9 (Ch. 6.2). `ΣVf` is the forward drop of the descent chain the operation must cross — D1 (P→C) 0.7, D2 (C→E) 1.2, D3 (E→O) 2.0 `V_c` (Ch. 8.2). A desire originating at `^C` and landing at `^O` pays 3.2 `V_c` before the constraint stack sees anything. The result is compared against the `cutoff` of the AMPLIFY stage that will actually fire:

`I_attention / cutoff`	Verdict	Why
≥ 2.0	**PASS**	Two-fold margin absorbs an ordinary bad day without dropping the stage into cutoff
1.0 – 2.0	**MARGINAL**	It fires today. It will not fire reliably across a campaign
< 1.0	**HOLD**	Below cutoff the stage never fires, and the chain reads as if nothing was attempted

The Red Corvette reads `0.076` against `cutoff = 0.05` — 1.52×. Chapter 13.6 calls that passing thinly; on this board it is MARGINAL, and the response is not to abandon the operation but to spend one afternoon on `R_context` first.

Which is the gate's real output: not *no*, but *which term*. Chapter 4.4 ranks the four by repair cost — `R_context` first (minutes, one clarifying answer), `R_resource` second (days, a budget line), `R_cap` third (hundreds of hours, falling only as the `^5` capacitor charges), `R_policy` last and in writing, because a term that moves by decision gets moved by decision at exactly the moment the operator is least inclined to argue with themselves.

> **Bench Note — do the diode subtraction by hand once, then never skip it.** The naive Corvette reading is `7/50 =`

`0.14`, a 2.8× margin and a comfortable pass. Corrected: `(7 − 3.2)/50 = 0.076`, 1.52× and thin. The descent chain ate 46% of the rail before the constraint stack saw a volt, and `V/R` overstates by nearly double. Every crossing pays this, and the error grows with how far the desire must descend. Work one gate both ways with a pencil — that gap is the whole reason there is a diode chain drawn in this board.

21.3 W-Gate: Do You Have the Schematic?

A possession question, and possession is binary. Six items, all required:

#	Item	Check	Notation
1	A formed `^0` Idea — specific, present tense, holdable as one scene	clarity $\geq$ 0.60	`^0^C_{F2}|A`
2	A CCDL schematic for this operation	in writing	`^6^C_{F6}|A`
3	The correct board for this desire type	named, from Ch. 13–20	`^1^C_{F2}|A`
4	CJT parameters — β, sat, cutoff, rπ	numbers, not ranges	`^6^C_{F6}|A`
5	A St. John's calibration sheet (App. F)	printed	`^4^E_{F2}|A`
6	A Quantitative Table sheet (App. G)	printed	`^7^O_{F6}|E`

It is an `AND`, not a sum, and that is invariant S1. A weighted score lets an operator trade a vivid Idea against a missing board selection, and those are not exchangeable: five-sixths of a schematic is not a circuit running at 83%, it is an undefined circuit. The characteristic failure is exact — items 1, 2, 4, 5, 6 in hand and item 3 wrong, running a CCB-D against what is actually an identity problem, which Chapter 20 owns.

Item 1 is the only measured one, and 0.60 is borrowed, not invented: the bottom of the 0.60–0.90 clarity band Chapter 13.6 reads at TP1. Below it the Idea is a sentence rather than a scene, usually because the target is phrased as a *condition* — "I decide what work I take" — which nobody can hold as one image. The repair is to give the condition a scene: one Tuesday, one job declined, one accepted.

21.4 P-Gate: Can You Execute?

Drive without capacity is the failure this gate exists for.

#	Check	Threshold	Notation
1	Gnosis reachable — ^1 Mind at 2° or deeper (Ch. 22.1)	reliably, not once	`^1^C_{F1}|T`
2	Schedule load — planned minutes ÷ available minutes	≤ 0.50	`^7^C_{F6}|A`
3	Access to the substrate the operation writes to	present	`^5^E_{F4}|f`
4	Arrival aversion — recoil at the target actually arriving	≤ 0.25	`^3^E_{F7}|f`

Check 2 is the one operators argue with. A campaign must fit inside *half* the available operating minutes across `T_manifest`. Planned at full budget it fails on the first sick day — and a broken `^7` Rhythm does not merely pause a campaign, it lets the `^5` capacitor leak between casts (Ch. 7.3), so the operator restarts from a lower charge than they finished at. Half is the design choice; the failure at 0.9 is a campaign that reads as insufficient drive when what happened was a missed Tuesday.

Check 4 is ^3 used correctly. Not an analytical judgement that the outcome would be inconvenient — that is `^6` decomposition.

The recoil: can't-stomach-it, when you picture the thing having arrived. The operator who wants the promotion and goes cold imagining its first week. Above 0.25 the operation opposes itself and casting harder amplifies both sides; the response is a self-directed sub-operation against the recoil (Ch. 20, Ch. 23.3) before the prime desire is cast at all.

21.5 Section-Retraction: Why You Can't Un-Cast

Write `R(I)` for the prerequisite space of a target state `I` — the Ao/D/W/P configurations that would produce it. Two maps connect them:

- η**`_I : I`** $\to$ **`R(I)`** — *embed.* Take the target, descend into what it requires.
- ε**`_I : R(I)`** $\to$ **`I`** — *realize.* Take a satisfied configuration, collapse it into the state it produces.

The RPM chain is a **section-retraction pair**, not an adjunction, and the composites are not symmetric:

```
ε_I ∘ η_I  =  id_I       embed then realize — you land exactly where you
↪  started
η_I ∘ ε_I  ≠  id_R(I)    realize then re-embed — your configuration does
↪  not come back
```

The left identity is what makes the gate cheap. Descend into any prerequisite, find it lacking, come back up with the target unchanged and nothing consumed. That licenses running the whole gate, failing at Ao, re-wording, and running it again from the top — which the worked example does. Scanning is lossless. There is no such thing as spending a desire by inspecting it, and the operator who won't run the gate because "looking too hard will kill it" is describing a property the mathematics does not have.

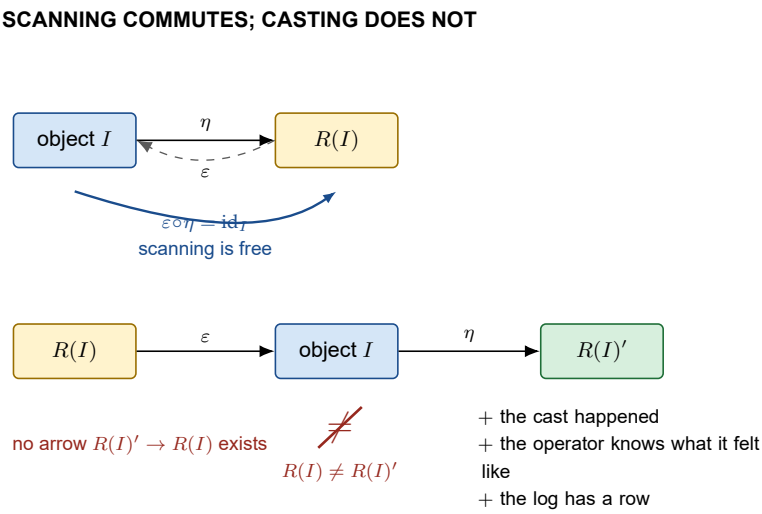


Figure 21-C: Commutative diagram

The right non-identity is what you cannot get around. Once the cast has fired, re-embedding the resulting state does not hand back the configuration you came from. No inverse arrow, therefore no `undo` component to draw — and **CCDL has no `undo` statement, which is a design claim, not an oversight** (App. B, §B.2.1: look for it; it is not there). Three consequences follow:

1. **An un-cast would be an unlawful ascent.** Reversing a cast requires the Operational outcome to reach back and rewrite the Principled prerequisite that authorized it: `^0` × `^P`, invalid notation for the same reason it is invalid engineering (Ch. 8.3). The one lawful upward path here is the filtered O→P feedback, $\tau = 21$ days at $\beta = 0.15$. It *updates* the substrate; it does not *restore* it.
2. **Infinite gain does not manufacture an inverse.** $\beta{=}\infty$ is the standard idealization for a hard switch, used in exactly that sense on the CSB's red path (Ch. 24) with the note that a validator rejects non-finite parameters and a real build writes a

large finite number. Here the objection is deeper than range-checking: the reverse arrow does not exist *at any gain*, so no finite substitute exists either. A large β on a reverse path specifies a component that is not merely unbuildable but undefined.

3. **A second reading of a gate is not comparable to the first.** This is the checkable face of $\eta \circ \varepsilon \neq \mathtt{id}$. After firing, the Ao readings cannot be re-taken clean, because one of the three ratings is now contaminated by knowing what the cast felt like. The prerequisite space you re-enter is a different space.

Now the discipline, because this result is over-read more often than it is used. **Section-retraction does not imply:**

- **That the outcome is guaranteed.** Irreversible is not effective. The gate authorizes; it delivers nothing. A cast that produced no confirmation at all still cannot be un-cast.
- **That consequences are permanent.** You may cast a counter-operation — a *new* target with its own full RPM gate, cost, and risk, not an inverse. What is unavailable is making the first cast not have happened.
- **That satisfied prerequisites stay satisfied.** They decay: charge leaks (Ch. 7.3), rails sag, schedules lapse.
- **Anything moral.** The asymmetry is structural. It says nothing about whether the cast should have been made, and a reader who hears a warning in it is hearing their own.

Bench Note — decay is not inversion, and confusing them produces a superstition. A prerequisite can *become* unsatisfied by attrition: stop practising and the `^5` rail drops, stop clarifying and `R_context` climbs back. What cannot happen is a prerequisite being *un-satisfied by an operation designed to reverse it*. The signatures differ — attrition is slow, monotone, visible in the log as a drifting rail; reversal is unavailable at any speed. The operator

who conflates them concludes that neglecting a working takes it back. It does not. It loses them the capacity to run the next one, while the first one stands.

The Full CCDL Listing — CCB-RPM

```
board ccb_rpm {                          // CCB-RPM — the prerequisite
↪  gate
    input:  stated_desire;               // as the operator first words
    ↪  it
    output: authorization;               // ARM or HOLD — never a cast
    node practice_rail: ^5^P_{F7}|T;     // Female: accumulated practice
    ↪  — bias only

    // ---- S1  A0 AUTHENTICATION ----
    node stated: ^0_stated^C_{F2}|A;      node parts:
    ↪  ^6^C_{F6}|A;
    node pull_open:      ^2^E_{F5}|f;     node pull_private:
    ↪  ^2^E_{F5}|f;
    node pull_condition: ^2^E_{F5}|f;     node witness_load:
    ↪  ^2^E_{F4}|f;
    node object_premium: ^4^E_{F5}|A;     node a0_ok:
    ↪  ^2^E_{F7}|T;
    cjt q_dec:  CJT< stated | practice_rail => parts >          [β=1.0];
    cjt q_open: CJT< parts  | practice_rail => pull_open >      [β=1.0];
    cjt q_priv: CJT< parts  | practice_rail => pull_private >   [β=1.0];
    cjt q_cond: CJT< parts  | practice_rail => pull_condition > [β=1.0];
    diff d_witness: Differential< plus=pull_open, minus=pull_private,
                                  out=witness_load >;
    diff d_premium: Differential< plus=pull_open, minus=pull_condition,
                                  out=object_premium >;
    fuse    f_witness: Fuse< path=witness_load, threshold=0.30,
                             action=block_until_decomposed >;
    monitor m_proxy:   Monitor< signal=object_premium, limit=[0.20,
    ↪  1.00],
                                action_on_violation=PROXY_TARGET >;
    cjt q_a0: CJT< witness_load | practice_rail => a0_ok > [β=1.0,
    ↪  cutoff=0.05];
```

```
// ---- S2  D-GATE ----
rail F5: Agency [v=7];                rail F2: Wisdom [v=6];
node path_in: ^1^C_{F5}|A;            node d_ok: ^4^E_{F5}|A;
diode d2: Diode< ^C -> ^E > [Vf=1.2]; diode d3: Diode< ^E -> ^O >
↪  [Vf=2.0];
resistor r_cap:      Resistor< path=path_in, R=25 >;
resistor r_policy:   Resistor< path=path_in, R=0  >;
resistor r_context:  Resistor< path=path_in, R=30 >;
resistor r_resource: Resistor< path=path_in, R=20 >;
monitor m_drive: Monitor< signal=path_in, limit=[0.10, 1.00],
                          action_on_violation=D_GATE_HOLD >;
cjt q_d: CJT< path_in | practice_rail => d_ok > [β=1.0,
↪  cutoff=0.05];

// ---- S3  W-GATE ----
node idea:    ^0^C_{F2}|A;   node w_sched: ^6^C_{F6}|A;
node w_board: ^1^C_{F2}|A;   node w_param: ^6^C_{F6}|A;
node w_calib: ^4^E_{F2}|A;   node w_sheet: ^7^O_{F6}|E;
node w_ok:    ^6^E_{F6}|A;
monitor m_idea: Monitor< signal=idea, limit=[0.60, 1.00],
                         action_on_violation=IDEA_UNFORMED >;
gate g_w: AND< idea, w_sched, w_board, w_param, w_calib, w_sheet >;
cjt q_w: CJT< g_w | practice_rail => w_ok > [β=1.0];

// ---- S4  P-GATE ----
node p_gnosis: ^1^C_{F1}|T;  node p_load:   ^7^C_{F6}|A;
node p_access: ^5^E_{F4}|f;  node p_recoil: ^3^E_{F7}|f;
node p_ok:     ^7^E_{F3}|E;
monitor m_budget: Monitor< signal=p_load,   limit=[0.00, 0.50],
                           action_on_violation=OVERCOMMIT >;
monitor m_absorb: Monitor< signal=p_recoil, limit=[0.00, 0.25],
                           action_on_violation=ARRIVAL_AVERSION >;
gate g_p: AND< p_gnosis, p_load, p_access, p_recoil >;
cjt q_p: CJT< g_p | practice_rail => p_ok > [β=1.0];

// ---- S5  ARM / HOLD ----
gate g_ready: AND< w_ok, p_ok >;      gate g_arm: AND< a0_ok, d_ok,
↪  g_ready >;
node arm:  ^9^O_{F5}|E;               node hold: ^9^E_{F6}|A;
diode d_out: Diode< ^E -> ^O > [Vf=2.0];
```

```
    mux out: MUX< inputs=[hold, arm], select=g_arm >;

    // ---- S6  LEDGER: the lawful ascent, and the only one ----
    filter f_log: LowPass< in=arm, out=trend, tau=21.0 >;   // days
    feedback trend -> practice_rail [β=0.15];

    probe tp1: witness_load;   probe tp2: path_in;
    probe tp3: g_ready;        probe tp4: out;
}
```

S1. All three pull probes and both differentials sit at `^E` deliberately. What the bench reads is the operator's *evaluation* of their own pull, not the world-level fact of who is watching, and modelling it at `^E` keeps an unlawful `^O` $\rightarrow$ `^E` ascent out of the listing — the move Chapter 23's intake makes at `^C`. The bench samples; it does not drive. `block_until_decomposed` names the only lawful path past F1: re-word, then re-run from the top.

S2. The resistor values are this operator's, on this desire, today — inputs, not constants of the board. The diodes carry the descent drop explicitly rather than folding it into `R`, because an operator who cannot see the 3.2 `V_c` will not subtract it.

S3–S4. `AND` and only `AND` (S1). Two P-gate inputs are monitored ceilings rather than possessions — `p_load` and `p_recoil` enter as pass/fail from their monitors, which is why the monitors carry the numbers and the gate carries none.

S5–S6. `g_arm` is the last reversible point on the board; the diode past the MUX is the descent into the cast and has no counterpart pointing back. The ledger is the one lawful ascent — `arm` at `^O` reaching `practice_rail` at `^P` — legal only because it is filtered at $\tau = 21$ days and held at $\beta = 0.15$, inside the 0.05–0.3 band Part III uses for every feedback path (Ch. 8.4).

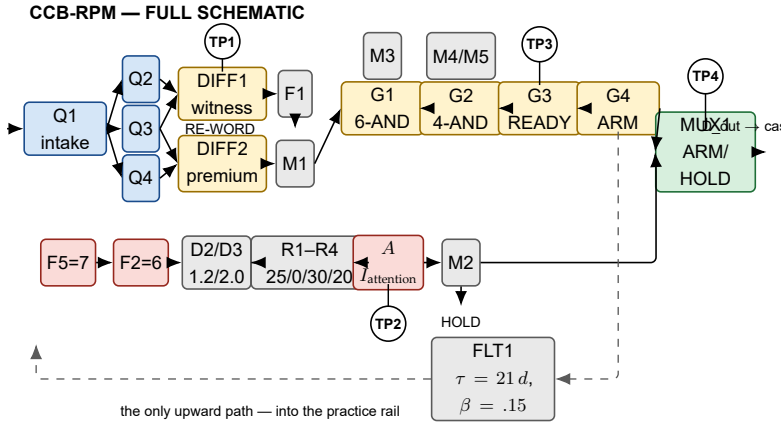


Figure 21-D: Full schematic with reference designators

Parameter and BOM Table

Every value is a design choice made in this book, not a measurement.

Ref	Component	Setting	Why	Failure at other values
F1	Witness fuse	0.30, `block_until_decomposed`	A third of the pull is where the audience sets the amplitude	0.50: audience-carried desires pass. 0.10: operator stops logging honestly
M1	Proxy monitor	premium $\geq$ 0.20	Smallest gap surviving self-rating noise	Lower: proxies read as prime. Higher: real targets held undecided
Q1–Q8	Verdict CJTs	β = 1.0	**A gate that amplifies is not a gate** (S2)	$\beta > 1$ manufactures readiness; $\beta < 1$ hides drive that is really there
—	CJT `cutoff`	0.05	Bottom of the 0.05–0.20 band (Ch. 3.3), matching Ch. 13's AMPLIFY floor	0.20: modest but real drive reads as no drive

Ref	Component	Setting	Why	Failure at other values
D2, D3, D_out	Descent diodes	Vf 1.2, 2.0	Chapter 8.2 drops, carried explicitly	Folded into R: margin overstated ~2×
R1–R4	Constraint stack	25 / 0 / 30 / 20 Ω_c	This operator, this desire (Ch. 4.4 ranges)	Treated as board constants: the gate stops reading the operator
M2	Drive monitor	$I \geq 0.10 = 2 \times$ cutoff	One bad day must not drop the stage into cutoff	1.0×: fires today, not across a campaign
G1	W-gate	6-input AND	Possession is binary (S1)	Summed: 5-of-6 arms with the board selection missing
M3	Idea clarity	≥ 0.60	Bottom of Ch. 13.6's TP1 band	Lower: conditions pass as scenes, TP1 reads low all campaign
M4	Budget monitor	load ≤ 0.50	Half the minutes leaves room for a bad week	0.90: one missed session reads as insufficient drive
M5	Arrival aversion	≤ 0.25	Above it the operation opposes itself	Ignored: casting harder amplifies both sides
FLT1	Ledger lowpass	$\tau = 21$ d, $\beta = 0.15$	The only lawful ^0 → ^P path (Ch. 8.4)	Short τ or high β: one operation rewrites the substrate

---

Worked Example — Operation R-1

The operator. Two years, 1,100 logged hours. Bias rail per Chapter 7.2, $\tau = 3{,}333$ hr:

```
P_cap = 10 × (1 − e^(−1100/3333)) = 10 × (1 − 0.719) = 2.81
```

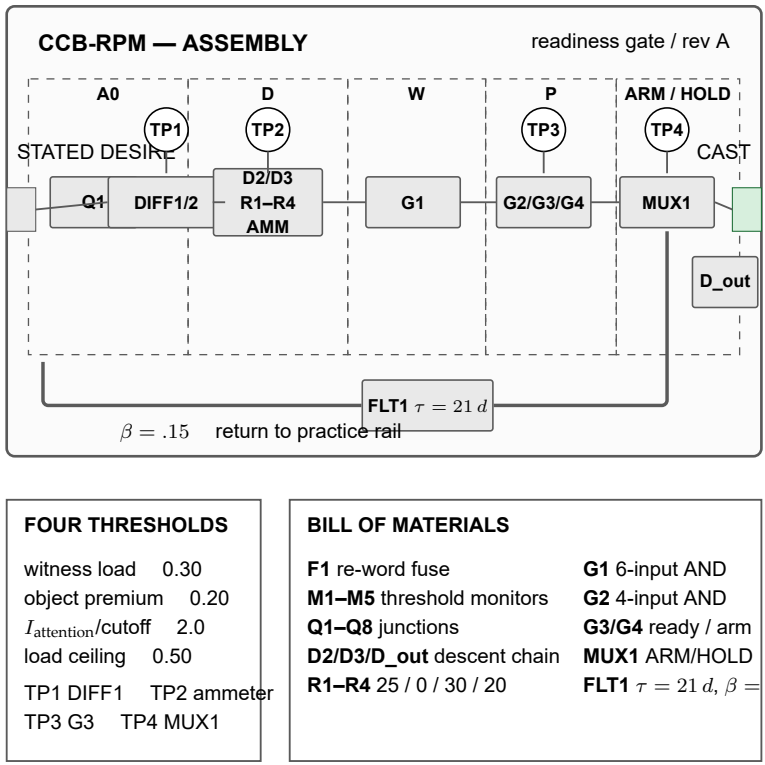


Figure 21-E: Assembly drawing

The rail sits at **2.8** of 10 — thinner than the Corvette operator's 4.3 (Ch. 13.6). It matters only as bias, and it is one more reason every verdict CJT here runs at unity.

Stated desire. *I run my own studio.*

Ao, pass 1. Three readings, written before any subtraction: `0.82`, `0.44`, `0.74` — the condition being recognized as the person whose judgement decides, delivered without a studio.

```
witness_load   = 0.82 − 0.44 = 0.38   ≥ 0.30  →  F1 BLOWS
object_premium = 0.82 − 0.74 = 0.08   < 0.20  →  M1: PROXY_TARGET
```

Both instruments fire and they agree: the pull is largely audience, and the object stands in for standing. Decomposition breaks the wording into three parts — the work, the authority over the

work, the standing of having one — and the third carries most of the amplitude. The operator re-words to the second.

Ao, pass 2. *I decide what work I take and when.* Fresh readings `0.78`, `0.71`, `0.36`: witness load **0.07**, premium **0.42**. Both pass. No cast has fired; under $\varepsilon \circ \eta$ = `id` the first pass cost nothing but the readings.

D-gate. `V_drive` = F5 at 7 plus F2 at 6 = **13 `V_c`**. Origin `^C`, landing `^O`, so Σ`Vf = 1.2 + 2.0 = 3.2`. Stack: `R_cap 25` (can do the work, has never priced or scheduled it), `R_policy 0`, `R_context 30` (the target is undefined for any actual week), `R_resource 20` (the hours exist; the buffer to refuse a job does not) — `R_in = 75`.

```
I_attention = (13 − 3.2) / 75 = 9.8 / 75 = 0.1307 A_c
margin      = 0.1307 / 0.05  = 2.61×      →  PASS
```

Run it against the *original* target and the gates couple visibly: `R_cap` 45 (never run a business) and `R_resource` 35 (capital) give `R_in = 110`, `I = 0.089`, margin **1.78× — MARGINAL**. The re-specification did not merely find a truer target; it removed 35 `Ω_c` of constraint. **An Ao failure presents downstream as resistance, which is why Ao is first.** Then the cheap repair Chapter 4.4 ranks first — one afternoon defining the week takes `R_context` from 30 to 10: `I = 9.8/55 = 0.178`, margin **3.56×**.

W-gate. Items 2–6 in hand. Item 1 reads **0.55** — `IDEA_UNFORMED`, hold. The cause is structural: the new target is a condition, not a scene. The operator gives it one — a specific Tuesday, one job declined, one accepted — and re-reads **0.71**. Six of six.

P-gate. Gnosis 2° reliably (Ch. 22.1). Plan: 4 operations, `t_op` 20 min, 72-hour spacing (Ch. 23.2), 12 days.

```
planned = 4 × 20 = 80 min      available = 20 min/day × 12 = 240 min
load    = 80 / 240 = 0.33      ≤ 0.50   →  pass
```

Access present; arrival aversion **0.18** — mild unease at the first declined job, not recoil. Pass.

Arm. `g_arm` closes, TP4 goes to `ARM`, `d_out` conducts, the cast fires. Two full passes, roughly fifty minutes on paper, zero casts — and it

caught a target that would otherwise have run this chapter's opening log.

Test Points

TP	Node	Reads	Expected
TP1	`witness_load`	`pull_open − pull_private`	< 0.30. Read it beside M1: both failing at once is a re-wording, not a re-tuning
TP2	`path_in`	`I_attention`, `A_c`	$\geq$ 0.10 (2 × cutoff). Between 0.05 and 0.10 is MARGINAL — fires today, not across a campaign
TP3	`g_ready`	W `AND` P	Closed, or the *named* open input. "Not ready" is not a reading; which of the ten items is missing, is
TP4	`out`	The authorization edge	`HOLD` with its deciding number, or `ARM`. **The last reversible point on the board**

Read them in order: is the desire mine, is there current, do I hold the schematic and the schedule — and only then arm. TP4 is the only reading you cannot take twice.

Failure Modes

Mode	Mechanism	What you observe	Fix
Gate skipped as ceremony	The scan is treated as delay	A long log on one target, every confirmation 3° or worse, amplifier raised repeatedly	Run Ao on the logged target; expect the wording to fail
Ao read after the fact	Third rating written once the differential is known	Witness load landing just under 0.30 suspiciously often	Three ratings in ink before any subtraction
Naive D-gate	`V/R` computed without Σ`Vf`	Margin overstated ~2×; a stage in cutoff on a board that "passed"	Subtract the descent chain; recompute every gate that crossed domains
W-gate summed	Six items scored and averaged	5-of-6 arms, and the missing item is usually the board selection	`AND`. No partial credit (S1)
Schedule at full budget	Load planned near 1.0	Campaign stalls in week two, reads as insufficient drive	Load $\leq$ 0.50; re-plan `n_ops`, not β
Recoil cast through	Arrival aversion above 0.25, run anyway	Confirmations arrive and are not felt; the operator undoes the arrival	Self-directed suboperation first (Ch. 20, Ch. 23.3)
Scan mistaken for undo	Operator fires, then re-runs the gate "to take it back"	A second Ao reading that will not settle; two incomparable pass-1 rows	§21.5. Re-running is a new authorization, never a reversal
Ledger over-gained	τ shortened or feedback β raised	The substrate moves after single operations; held thresholds drift	τ = 21 d, $\beta \leq 0.3$. Never remove the filter

What This Chapter Established

- **Ao is answered by subtraction, not introspection.** Two differentials — witness load (trip $\geq$ 0.30) and object premium (hold $\geq$ 0.20) — turn "is this really what I want" into two numbers whose common-mode bias cancels. The repair is `^6` decomposition, never `^3`.

- **The D-gate is Chapter 4 with a threshold on it.** `I = (V_drive − ΣVf)/R` against the downstream `cutoff`, `2×` margin required. The Red Corvette's 1.52× is MARGINAL here — and Chapter 13.6 already called it thin.
- **The descent chain is where the margin goes.** `7/50 = 0.14` and `(7−3.2)/50 = 0.076` describe one operation; the naive form overstates by nearly double, and the error grows with how far the desire must descend.
- **W and P are `AND` gates, and that is a safety invariant.** Five-sixths of a schematic is not a circuit at 83%; it is undefined. The characteristic arming failure is 5-of-6 with the board selection wrong.
- **The gates are not independent.** Re-specifying at Ao removed 35 `Ω_c` from the stack and moved the D-gate from 1.78× to 2.61×. An unauthenticated desire presents downstream as resistance.
- **Scanning is free: $\varepsilon \circ \eta$ = `id`.** Descend into a prerequisite, find it lacking, return with the target unchanged. Two full passes cost the worked example fifty minutes and no casts.
- **Firing is not reversible: $\eta \circ \varepsilon \neq$ `id`.** No inverse arrow, so no `undo` component and none in CCDL. An un-cast needs an `^O` × `^P` ascent, and $\beta{=}\infty$ on a reverse path does not help — the arrow is undefined, not merely large.
- **What irreversibility does not mean.** Not that the outcome is guaranteed, not that consequences are permanent, not that prerequisites cannot decay, and nothing moral. **Decay is not inversion** — attrition is slow and visible as a drifting rail; reversal is unavailable at any speed.

Exercises

21.1 An operator rates `pull_open = 0.70`, `pull_private = 0.66`, `pull_condition = 0.58`. Compute both Ao quantities, give each verdict, and say which single instrument decided the gate.

21.2 A desire originates at `^P` and lands at `^O`. Drives are F7 at 8 and F6 at 5; the stack is `R_cap 15`, `R_policy 40`, `R_context 10`, `R_resource 5`; the AMPLIFY stage has `cutoff = 0.10`. Compute Σ`Vf`, `I_attention`, and the margin. Verdict? Now name the term you would repair first, and why it is *not* the largest one.

21.3 Recompute 21.2 naively as `V_drive / R`. Give the margin both ways and the percentage error, then say in one sentence why the error grows with the origin domain.

21.4 An operator holds all six W-gate items but reads Idea clarity at 0.52, and proposes weighting the gate so six-of-six at 0.52 clears. Write the two-sentence reply, name the invariant, and describe the operation that arms as a result.

21.5 A campaign plans 6 operations of 25 minutes over 18 days against 15 available minutes per day. Compute the load, give the P-gate verdict, and state two re-plans that clear it — one changing `n_ops`, one changing the horizon.

21.6 An operator fires a cast, regrets it within the hour, and asks for an un-cast stage. Answer in three parts: (a) which arrow of the section-retraction pair they are asking you to build; (b) why $\beta{=}\infty$ on it does not produce that arrow; (c) what you would build instead, and how it differs from a reversal.

21.7 Design exercise. Add a re-entry counter to CCB-RPM that flags a target re-worded more than three times without a cast. Write the CCDL fragment, name the failure mode it catches, and name the opposite failure mode you have just installed.

*(Answers in **Appendix H**, worked in full. Every parameter here is a design choice; the Ao and clarity readings are operator self-reports on a*

stated scale, and the chapter never treats them as measurements of anything else.)

22 The Quantitative Table — Measuring Your Operations

Three readings, one honest scale, and the exact limit of what a trend across them is allowed to mean

An operator hands you a notebook. Forty operations, eighteen months, every page in the same green ink. Every ^1 Mind reading is 1° or 2°. Every confirmation is graded 1°. There is not one NO-READ row in it, not one struck-through line, not one entry marked CONFOUNDED.

You have not been handed a log. You have been handed a diary, and its author cannot tell the difference — which is the problem, because a diary and a log look identical on the page and answer different questions. The diary answers *what was that like*. The log answers *did the thing I changed on the fourteenth do anything by the twenty-first*, and it can only answer that if it contains rows the author did not enjoy writing.

Chapter 5.5 named the three readings that can be taken at a bench with nothing but a wristwatch and an honest answer: ^1 Mind, ^4 Vibration, ^7 Rhythm. Chapter 5.6 put them on a board. This chapter builds the instrument that takes them, states what each number is worth, and — the part that matters most — states plainly what none of them is.

They are structured self-report on a scale defined in this book. They are useful for comparing one operator against their own earlier readings under known conditions. They have not been validated

against any external criterion, there is no published norm to place you on, and nothing in this chapter licenses a comparison between two operators or a claim about anyone's mind but your own. The scale is a ruler you cut yourself. It measures change on your own bench and nothing beyond it, and that is worth having as long as you never claim more.

Requirements and Safety Invariants

Chapter 12.2 requires the "must not" list before the functional one.

#	Functional requirement	Realized by
R1	Take three readings per operation on fixed, pre-declared scales	Gnosis sampler, rhythm clock, vibration grader
R2	Capture each reading before hindsight can reach it	Two latency holds (§22.1, §22.2)
R3	Admit exactly one confirmation per operation	First-only limiter (§22.2)
R4	Emit NO-READ rather than a fabricated row	Horizon fuse → MUX
R5	Refuse to trend across non-comparable operations	Confound gate, three-row filter

#	Safety invariant (MUST NOT)	Enforcement
S1	Present a reading as externally validated, or compare two operators	Scope fuse, `action=permanent_block`
S2	Revise a grade after the outcome is known	Write-once fuse on the row

#	Safety invariant (MUST NOT)	Enforcement
S3	Admit a later, better confirmation into the ratio	`Limiter< max=1.0 >` on arrivals
S4	Compute a trend across a cycle with two changes in it	`CONFOUNDED` exit on the confound gate
S5	Let the log ascend a domain — readings are `^C`, the row is `^O`	Descent diode; feedback only via Chapter 24.5
S6	Substitute this instrument for clinical judgment	Stated in §22.5; no component can enforce it

S6 has no component behind it and never will. If your readings are tracking something that frightens you — sleep going, mood going, the practice becoming the only thing that holds — that is a matter for a clinician, and this book has nothing to offer it. Nothing here replaces care. Close the log and make the appointment.

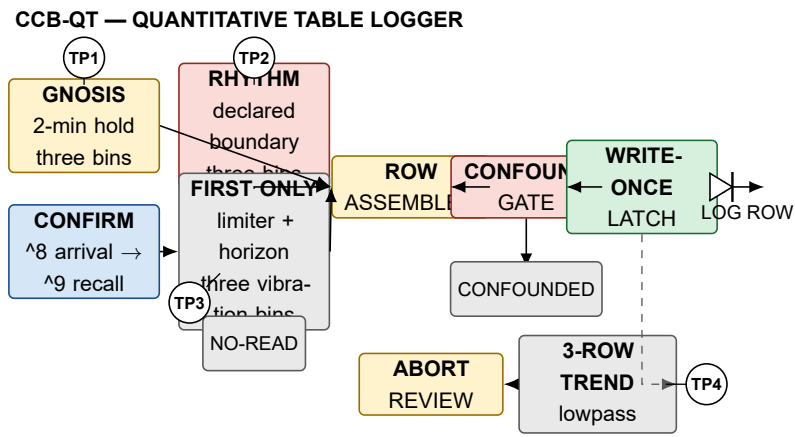


Figure 22-A: Block diagram

22.1 ^1 Mind Measurement — The 3 Degrees of Gnosis

^1 Mind is *various degrees of awareness that processes, stores, outputs, and intakes Ideas*. At the bench you are reading one thing only: how much of that awareness stayed on the formed Idea, and how much went elsewhere.

Degree	Behavioural description	State
1°	Complete absorption; the Idea is the only content present	`^1^C_{F2}|T`
2°	Wanders and returns; intermittently clear	`^1^C_{F2}|A`
3°	Frequent wandering; the image will not hold	`^1^C_{F2}|A`

Grade against the middle column, not against how the operation felt. "It felt powerful" is not on the scale. *Did the image hold, and did it return* is.

The register carries real information here and is not decoration. At 1° the awareness is not tracking itself — purpose-oriented, integrative, `|T`. At 2° and 3° it is checking the clock and narrating, which is `|A`. The Noetic never changes: it is ^1 Mind at all three degrees. Only the operational character changes, exactly as Chapter 2.5 defines register.

Now the honest part. This reading is taken by the same awareness whose depth is being reported, and no logbook design fixes that. An operator absorbed at 1° was by definition not monitoring their absorption; an operator who can describe their absorption in detail has told you it was 2°. Three defences exist and all three are procedural:

1. **Grade within two minutes of the operation ending, before any other line on the sheet is written.** The hold is set at two

minutes because that is roughly how long a fresh impression survives before it starts being reconstructed into a story. At thirty minutes you are grading a memory of an impression.

2. **Grade against behaviour, never against feeling.**
3. **Never decide on ^1 alone.** It is the noisiest of the three columns. It earns its place by moving *with* ^4 Vibration across many rows, and a co-movement across ten rows is much harder for hindsight to manufacture than a single grade.

Bench Note — a 3° operation is a measurement of the operator, not of the board. Chapter 5's TP2 is blunt about this: at 3°, do not compute A. The reading is admissible as a row — log it — but it is not admissible as evidence about the circuit, because the circuit was never properly driven. Operators hate this rule, because a 3° operation still cost them twenty minutes and they want a number for it. The number they would get would be a number about their sleep, their week, and the room. Fix the operating conditions and re-run. The row you strike is worth more than the number you would have kept.

22.2 ^4 Vibration Measurement — Confirmation Speed and Intensity

^4 Vibration is *the degree to which an Idea resonates with the Mind, its impression*. A confirmation is the ^8 → ^9 pair of Chapter 5.6 and nothing more exotic: an event arrives and triggers a *different* Idea — the desire — into Mind. Associative surfacing. Not inference, not evidence, not a message.

Two quantities come off it, and they are different in kind.

Speed is `t_conf`, clocked from the *end* of the operation to the first confirmation, in minutes. It is a wristwatch reading and it is the one that enters St. John's Equation.

Intensity is the graded impression. The bins are 30 minutes and 2 hours, and both edges are design choices with stated failures. The 1° bin is kept short so that it sits well inside any plausible base rate — push it out to two hours and every operator grades every operation 1° and the column stops discriminating. The 3° bin is kept wide so a genuinely slow circuit still registers as something rather than nothing.

Degree	Delay	Impression
1°	< 30 min	startling, undeniable; demands attention
2°	30 min – 2 h	clear, but you can talk yourself out of it
3°	> 2 h	subtle, ambiguous; "was that a confirmation?"

Three rules govern the column, listed in the order operators break them.

Only the first confirmation enters the ratio. Later ones stay on the sheet as `^4` evidence and never re-enter. An operator who admits a stronger confirmation that arrived four hours later has improved their number by choosing their data, and the improvement is entirely artefact. The instrument enforces this with a limiter that admits one arrival and drops the rest.

Grade within thirty minutes of noticing, in ink, and never revise. A 3° confirmation graded a week later, once the desire has actually manifested, grades 1° every time. This is not dishonesty. It is ordinary hindsight, it is unavoidable once the outcome is known, and the only defence against it is a clock.

A confirmation is not a manifestation. It says the signal reached the Ether. It says nothing about whether the signal reached the target with enough amplitude to clear `R_constraint` — Chapter 13's delivered-current band, where confirmations arrive reliably and

nothing else does. Reading confirmations as arrivals is the error that costs more than all the others combined.

Beyond the horizon — `t_H = 3 × median(t_conf over the last ten operations)`, floored at 24 hours (§23.1) — there is no reading at all. The entry is **NO-READ**, and NO-READ is a result: it is the row that tells you whether your confirmation rate exceeds your own base rate.

22.3 ^7 Rhythm Measurement — Clocking the Operation

^7 Rhythm is *the rate and pattern a Mind is exposed to an Idea, entraining it*. At the bench it reads as one number: `t_op`, the duration of the operation, start to finish, off a wristwatch. The state is `^7^C_{F6}|E` — Structure, embodied register, because it is a clocked bodily act and reads as one.

Degree	Duration	Why this edge
1°	> 30 min	Full establishment
2°	10 – 30 min	The working band
3°	< 10 min	Below ten minutes gnosis has not had time to establish

The 10- and 30-minute edges sit at the two places operators actually break. Below ten, ^1 Mind almost never reaches 2°. Above thirty, ^1 Mind almost always falls — the operator tires and the image goes. Widen the bins and you merge a failed operation with a held one, which is the one thing the column exists to prevent.

The whole column depends on a convention that costs nothing to set and everything to skip. `t_op` has two defensible start points: the moment you begin the procedure, and the moment concentration actually establishes. For most operators they differ by five to ten

minutes. **Either is fine. Mixing them across a log makes every comparison in it a comparison between two different measurements**, which is to say worthless. Write your convention on page one in ink before the first entry, and do not revise it because a later operation would read better under the other one. The instrument holds the convention in a write-once latch for exactly this reason.

22.4 The Table Sheet — Tracking Across Multiple Operations

One row per operation, every operation, including the ones that did not read. The printable forms are Appendix G — Form 1 the operation log, Form 2 the confirmation log, Form 3 the parameter-change log — and Form 3 governs the other two.

Eight fields carry the chapter: `Op #`, `t_op`, `t_conf`, `^1`, `^4`, `^7`, a flag column, and `A = 3 t_conf / t_op` from Chapter 13.5.

The three degrees also give a bookkeeping figure worth writing down. Define the **coupling sum** `S = m + v`, where `m` is the `^1` degree and `v` the `^4` degree, each 1 to 3. `S` runs 2 to 6, and lower is stronger. It is a sum and not a product, and the reason is not aesthetic: these are **ordinal bins**. 1°, 2° and 3° state an order and nothing else — the gap between 1° and 2° is not claimed equal to the gap between 2° and 3°, and multiplying ordinals asserts a ratio scale the bins do not have. A sum is already generous. Treat `S` as a bookkeeping index for ordering rows in one log, never as a magnitude.

> **Bench Note — the notation says the same thing the arithmetic does.** The three readings are `^1^C_{F2}|A + ^4^C_{F2}|f + ^7^C_{F6}|E` — the `+` of co-presence (Chapter 2.6). Three distinct measurements, present together, remaining distinct. It is emphatically **not** $\times$. A $\times$ would assert that the three fuse into a single emergent

> state, and under v7.7 discipline that requires a named emergent result, which this book does not define. There is no "operation quality" state in the topos and `S` is not one. That is why `S` is a bookkeeping sum rather than a state literal, and why it never appears in a CCDL listing. The operator discipline and the measurement discipline are the same discipline here, which is a good sign that both are right.

Three flags do most of the work. **NO-READ**: no confirmation inside the horizon; `^1` and `^7` still logged, `^4` absent, `S` not computed. **CONFOUNDED**: two or more things changed in the cycle, including things you did not choose — illness, upheaval, a week of bad sleep. Nothing may be concluded from a confounded row, *including the parts you liked*. **Struck**: a corrected value, struck through and initialled, both readable. The crossed-out number is often the interesting one.

22.5 Interpreting Trends — When to Adjust, When to Abort

A single row diagnoses nothing. **Three comparable rows are the minimum**, where comparable means: same board, same declared conventions, one parameter changed at most between any two of them, and no CONFOUNDED flag in the sequence. Three is a design choice — it is the fewest rows in which a direction can be distinguished from a pair of readings, and one bad clock cannot redirect a campaign that needs three rows to turn. Two rows is a line through two points, which is not a trend; it is arithmetic.

Read two things down the column, in this order.

`A` first, because it is smooth in the clocks. `A = 3 t_conf / t_op` moves continuously with a timing error, where the reduced denom-

inator `b` swings wildly (Chapter 13.5). The step ratio is ρ `= A_next / A_current` (§23.2). Falling `A` with $\rho \leq$ `0.6` is a converging campaign. Flat `A` across three comparable rows means nothing you are changing is reaching the delivered current.

`S` second, as a cross-check. When `A` and `S` move together, the reading is coherent. When they disagree, the disagreement is itself the finding, and there are exactly two informative cases:

Pattern	Reading	Action
`A` falls, `S` flat	The bins are coarser than the ratio; real change inside one bin	Trust `A`. Record `S`. Continue
`A` flat, `S` falls	Operating conditions improved; the circuit did not	Stop tuning the board. `R_op` is the problem
Both fall	Converging	Continue; one lever at a time
Both rise for two consecutive comparable rows	Diverging	**Abort.** Return to A0 (Chapter 21.1)
`A` flat, `S` rises	The operator is degrading while the board holds	Rest. Re-read §22.1's bench note

When to adjust. One lever, one cycle, with a prediction written before the operation: a named test point and a direction. *TP3 comes off the ceiling* is a prediction. *It should improve things* is not. Verdict only after one operation and its first confirmation.

When to abort. Three conditions, any one sufficient. Both columns rising across two consecutive comparable rows. `A` flat across three comparable rows while ρ sits near 1.0 — this signal arrives *before* `A` reaches the re-spec boundary of 9, which is the whole point of watching a trend rather than a row. Or `^1` Mind falling to 3° for three consecutive operations at constant `t_op`, which is not a circuit reading at all: it says the operator cannot currently run the board, and the response is rest and a decomposition (§23.3), not a fourth attempt.

And the condition that is not on any table: if the log itself has become the thing that is not right — if you are casting to make the readings move, or the readings are the only evidence your week produced — stop and speak to a clinician. That is not a failure of the instrument. It is outside what any instrument in this book is for.

Full CCDL Listing — CCB-QT

```
board ccb_qt {                              // CCB-QT — the acquisition
↪  bench
    input:  operation;                      // the operation as run
    output: log_row;                        // one admissible row, or
    ↪  NO-READ

    node practice_rail: ^5^P_{F7}|T;        // Female: accumulated
    ↪  practice — bias only

    // ---- S1  GNOSIS SAMPLER ----
    node gnosis: ^1^C_{F2}|A;               // Mind: awareness on the
    ↪  formed Idea
    hold h_mind: Hold< signal=gnosis, duration=2.0 >;          // min,
    ↪  before hindsight
    monitor m_mind: Monitor< signal=h_mind, limit=[1.0, 2.0],
                             action_on_violation=NO_COMPUTE >; // 3° ->
                             ↪  no A

    // ---- S2  RHYTHM CLOCK ----
    node clocked_op: ^7^C_{F6}|E;           // Rhythm: the operation,
    ↪  start to finish
    fuse f_conv: Fuse< path=clocked_op, threshold=1.0,
                       action=permanent_block >;               //
                       ↪  convention write-once

    // ---- S3  CONFIRMATION INTAKE ----
    node arrival:  ^8^C_{F2}|E;             // Above/Cause: the event that
    ↪  arrives
    node recalled: ^9^C_{F2}|A;             // Below/Effect: the desire
    ↪  brought back
```

```
    limiter l_first: Limiter< signal=arrival, max=1.0 >;       // first
    ↪  only
    fuse f_horizon: Fuse< path=recalled, threshold=1440.0,
                          action=no_read >;                    // 24 h
                          ↪  floor

    // ---- S4  VIBRATION GRADER ----
    node impression: ^4^C_{F2}|f;         // Vibration: the impression
    ↪  it made
    hold h_vib: Hold< signal=impression, duration=30.0 >;      // min
    ↪  from noticing

    // ---- S5  ROW ASSEMBLER ----
    node row:     ^6^E_{F6}|A;            // Male: the readings composed
    ↪  into a row
    node no_read: ^0^O_{F6}|A;            // Idea: the logged absence of
    ↪  a reading
    mux out: MUX< inputs=[ row, no_read ], select=f_horizon >;

    // ---- S6  CONFOUND GATE and WRITE-ONCE ----
    node changes: ^6^O_{F6}|A;            // Male: what was altered this
    ↪  cycle
    monitor m_confound: Monitor< signal=changes, threshold=1.0,
                                 action_on_violation=CONFOUNDED >;
    fuse f_write: Fuse< path=out, threshold=1.0, action=permanent_block
    ↪  >;
    fuse f_scope: Fuse< path=out, threshold=1.0, action=permanent_block
    ↪  >;  // S1

    // ---- S7  TREND and DESCENT ----
    filter f_trend: Filter< in=out, out=trend, type=lowpass, tau=3.0 >;
    ↪  // ops
    monitor m_abort: Monitor< signal=trend, threshold=0.6,
                              action_on_violation=ABORT_REVIEW >;
    diode d_out: Diode< ^E -> ^O >;       // the row descends into the
    ↪  ledger

    probe tp1: h_mind;        probe tp2: clocked_op;
    probe tp3: l_first;       probe tp4: f_trend;
}
```

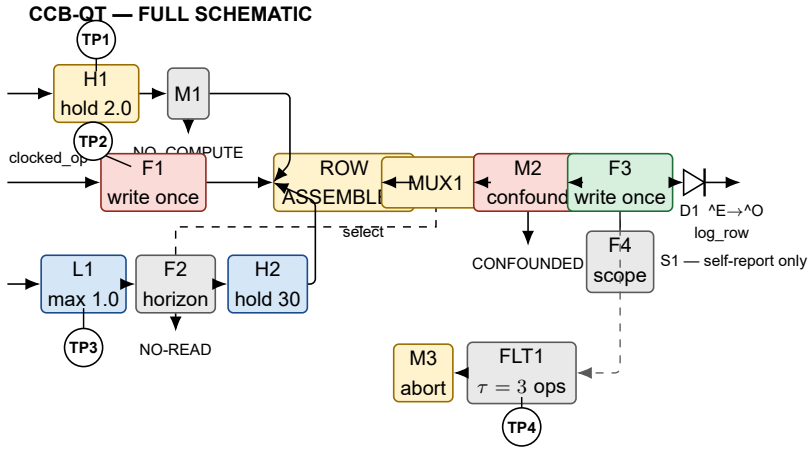


Figure 22-B: Full schematic with reference designators

S1 — Gnosis sampler. `Hold` captures the grade and freezes it; `duration=2.0` is the two-minute latency of §22.1. `m_mind` is the enforcement of Chapter 5's TP2: a 3° reading exits `NO_COMPUTE`, which logs the row and withholds `A`.

S2 — Rhythm clock. The convention fuse is a genuine `Fuse` with a finite threshold and `action=permanent_block`, not an infinite-gain comparator. It is tempting to write $\beta{=}\infty$ on any latch that must never yield; that is an idealization a validator rejects, exactly as Chapter 24 notes on the CSB's red path, and here there is no need for it — a fuse latches by construction.

S3 — Confirmation intake. `^8` triggers, `^9` surfaces; both are modelled at `^C` so that no unlawful `^O` $\rightarrow$ `^C` ascent enters the listing (§23.1 does the same). The limiter is S3 in hardware: one arrival admitted, the rest logged as evidence and dropped from the ratio. The horizon fuse carries the 24-hour floor as a literal; a running bench recomputes $3\times$ the rolling median and holds the floor underneath.

S4–S5. The grader's thirty-minute hold is the anti-hindsight latch. The MUX selects on the horizon fuse: intact routes the assembled row, blown routes the `no_read` terminal, which is a

declared node rather than a dangling name — an absence you can probe is worth more than an absence you infer. The row itself is `^6` Male, the readings decomposed and composed into a structure. It is not `^3` and not `^8`: assembling a row is compositional work, not aversion and not associative triggering.

S6–S7. The confound gate permits one change per cycle; two exits `CONFOUNDED`. The scope fuse is the only component here with no signal-processing function: it exists so that S1 has an enforcement column, and it blows the moment a row is used to support a claim about anyone but this operator. The trend filter's `tau=3.0` is three operations, the minimum from §22.5; the abort monitor watches ρ against 0.6. `d_out` is the lawful descent, and the only path back up is Chapter 24.5's ethical-drift filter, which takes this log as its literal input.

Parameter and BOM Table

Every value is a design choice made in this book. None is a measurement.

Ref	Component	Setting	Why	Failure at other values
H1	Hold, ^1 grade	2 min	Roughly how long a fresh impression survives	30 min: the grade is a reconstruction, and reconstructions trend upward
M1	Gnosis monitor	admit 1°–2°	Chapter 5's TP2: 3° measures the operator	Admitted: operator noise enters the circuit record as circuit data
—	^1 bins	behavioural	Held / returned / would not hold	Feeling-based: every operation grades 1° after a good week

Ref	Component	Setting	Why	Failure at other values
F1	Convention fuse	write-once	`t_op` boundaries must match across the log	Revisable: every comparison is between two different measurements
—	^7 bins	10 / 30 min	The two points operators break	Wider: a failed operation and a held one land in one bin
L1	First-only limiter	max = 1	The ratio needs one arrival, not the best one	Best-of: `A` improves with nothing changed
F2	Horizon fuse	3 × median, floor 24 h	Passes slow circuits, excludes base rate	Short: live circuits declared dead. Long: everything "confirms"
H2	Hold, ^4 grade	30 min from noticing	Before hindsight reaches the grade	Later: no operation is ever graded 3° again
—	^4 bins	30 min / 2 h	1° bin inside plausible base rate	1° edge at 2 h: the column stops discriminating
M2	Confound gate	1 change / cycle	Attribution needs a single lever	2 changes: a better number, bought, with no account of what bought it
F3	Write-once latch	`permanent_block`	Grades must not move after the outcome	Revisable: the log becomes a record of the outcome, not of the operations
F4	Scope fuse	`permanent_block`	Invariant S1	Absent: a self-report scale gets quoted as a finding
FLT1	Trend filter	τ = 3 operations	Fewest rows that can show a direction	1 op: one bad clock redirects a campaign. 10: the abort arrives too late
M3	Abort monitor	$\rho \leq 0.6$	Reaches DRIVE in one step from 9 (§23.2)	1.0: divergence reads as steady

Ref	Component	Setting	Why	Failure at other values
—	S	ordinal sum, 2–6	Bins are ordinal; a product asserts a ratio scale	Multiplied or averaged: a coarse index quoted to two decimals

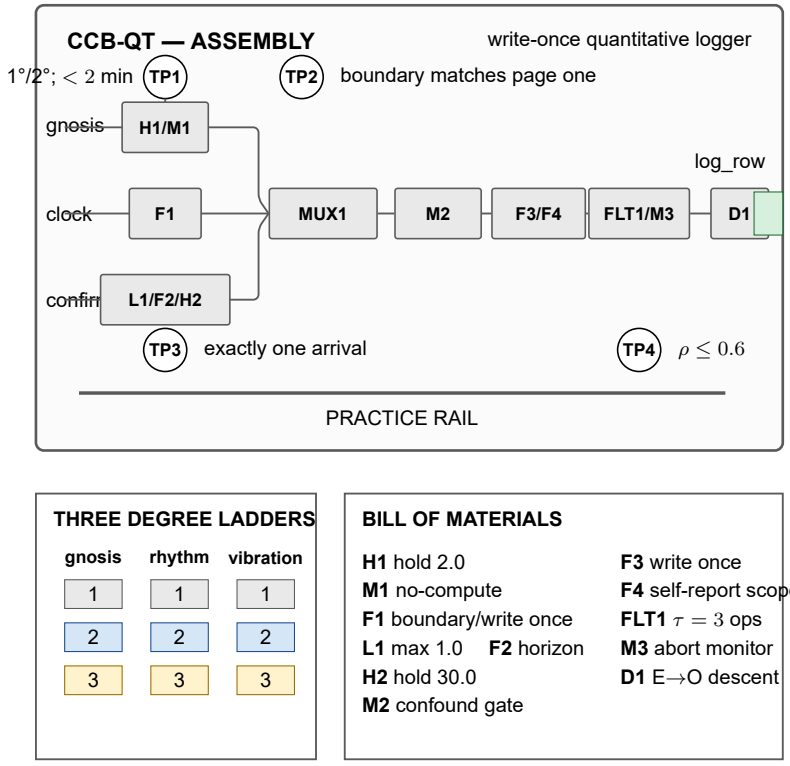


Figure 22-C: Assembly drawing

Worked Example — Campaign Q-3

Self-directed board (CCB-SELF), one desire, six operations across nineteen days. Convention declared on page one: `t_op` starts when

the operator sits down and begins the procedure. Clocks logged raw; `A` computed on the 5-minute grid of §23.1.

Op	`t_op`	`t_conf`	`^1`	`^4`	`^7`	`S`	`A`	Flag
1	15	150	3°	3°	2°	6	—	`NO_COMPUTE`
2	25	100	2°	2°	2°	4	12.0	—
3	25	60	2°	2°	2°	4	7.2	—
4	25	—	2°	—	2°	—	—	NO-READ
5	30	45	1°	2°	2°	3	4.5	`CONFOUNDED`
6	30	40	1°	2°	2°	3	4.0	—

Operation 1. `^1` = 3°, so M1 exits `NO_COMPUTE` and no `A` is written — correctly, because at fifteen minutes the gnosis never established. The row is logged in full. One change follows, CH-001: `t_op` 15 → 25 min, prediction *TP1 moves to 2° by operation 2*.

Operation 2. `A = 3 × 100 / 25 = 12.0` — RE-SPEC band (§13.5). `S` = 4. CH-001 verdict: KEEP, prediction met.

Operation 3. `A = 3 × 60 / 25 = 7.2` — DRIVE. ρ `= 7.2 / 12.0 = 0.60`, exactly at the abort monitor's limit and on the right side of it. Note that `S` did not move: 4 both times. This is the coarse-bins case from §22.5 — a 40% fall in `A` happened entirely inside the 2°/2° cell. Trust `A`, record `S`.

Operation 4. Horizon: `3 × median(150, 100, 60) = 300` min, but the 24-hour floor governs at this point in the log, so `t_H = 1440`. Nothing arrives. **NO-READ**, `^1` and `^7` logged, `^4` absent, `S` not computed. This is the row that tells the operator their confirmation rate is not simply their base rate.

Operation 5. `t_op` raised to 30 *and* the symbol changed, on a week of bad sleep. `S` = 3 and `A` = 4.5 — the best reading in the log, and worthless. M2 exits `CONFOUNDED`; nothing may be concluded, including the parts the operator liked.

Operation 6. One change only, CH-003: revert the symbol, hold `t_op` at 30. `A = 3 × 40 / 30 = 4.0`, `S = 3`, ρ `= 4.0 / 7.2 = 0.56` against the comparable predecessor. Converging.

The readout. Comparable rows are 2, 3 and 6 — three, the minimum. `A` runs 12.0 → 7.2 → 4.0 with ρ = 0.60 and 0.56, both inside 0.6. `S` runs 4 → 4 → 3. Both columns fall, they do not disagree, and the campaign continues with one lever at a time. **Six operations, three usable rows.** That ratio is not a sign of a bad log. It is what an honest one looks like, and the notebook in the cold open — forty rows, forty usable — is the one to distrust.

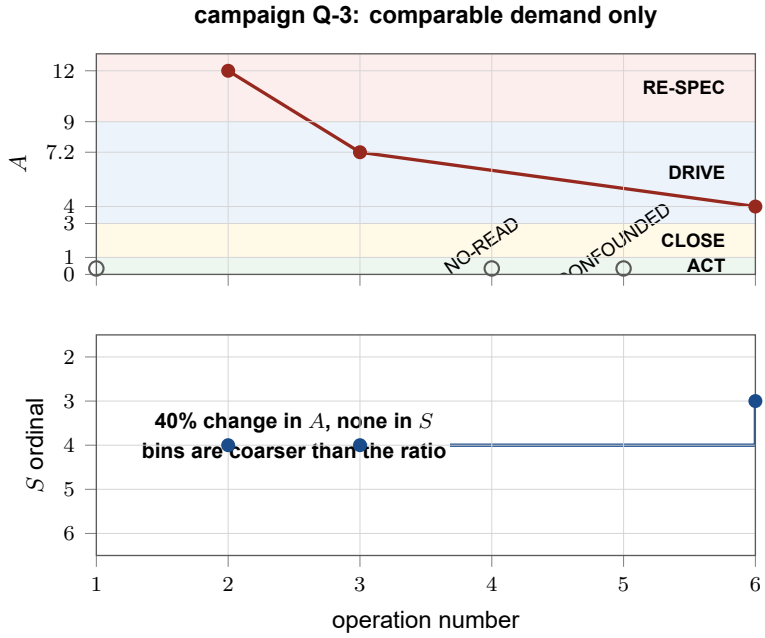


Figure 22-D: Two-panel trend chart for campaign Q-3

Test Points

TP	Node	Reads	Expected
TP1	`h_mind`	`^1` degree and its capture latency	1° or 2°, captured under 2 min. A 3° reading is admissible as a row and inadmissible as evidence — do not compute `A`
TP2	`clocked_op`	`t_op` against the boundary written on page one	Same rule as every prior entry. A different rule makes the row incomparable, not merely different
TP3	`l_first`	Arrivals admitted	Exactly one, inside the horizon, graded within 30 min. Two admitted means the ratio has been chosen rather than measured
TP4	`f_trend`	ρ across three comparable rows	≤ 0.6 per step. Flat near 1.0 across three rows is the abort signal, and it arrives before `A` reaches 9

Read them in order. TP1 and TP2 decide whether the row is admissible at all; TP3 decides whether there is a ^4 reading; TP4 is the only one that speaks about the campaign. Three of the four can send you home without a number, and that ratio is deliberate.

Failure Modes

Mode	Mechanism	What you observe	Fix
Retrospective grading	Grades written after the outcome is known	No operation in the log was ever graded 3°	H1 and H2 latencies; ink; never revise
Best-of admission	A later, stronger confirmation entered	`A` improving while nothing about the operation changed	L1 at `max=1`; Form 2's first-confirmation field

Mode	Mechanism	What you observe	Fix
Boundary drift	`t_op` start defined differently across entries	Two operations that felt identical reading five minutes apart	F1 write-once; the convention on page one
Window creep	Horizon widened until something lands	Every operation confirms; no NO-READ rows anywhere	Fix `t_H` before the cast; run a one-week control count
Multi-lever cycle	Two or more changes between readings	A better number with no account of which change bought it	M2 `CONFOUNDED`; claim nothing from the row
Ordinal inflation	`S` averaged, multiplied, or quoted to decimals	"Mean coupling 3.67 this month"	`S` orders rows in one log; it has no magnitude
Scale export	Readings compared between operators, or quoted as a finding	A degree used as evidence about someone else's mind	F4 scope fuse; invariant S1
Instrument substitution	The log used to track distress	Readings watched more closely than sleep	§22.5. See a clinician; nothing here replaces care

The last two fail silently. Every other row on this table announces itself in the log; those two look like diligence.

What This Chapter Established

- **These are structured self-reports on a scale defined in this book, not validated instruments.** They order one operator's rows against each other under known conditions. They carry no external norm, do not compare two operators, and are evidence about no mind but the operator's own.
- **Three of ten Noetics take a bench reading without instrumentation** — `^1` Mind, `^4` Vibration, `^7` Rhythm. That is a fact about instrumentation, not about the topos: all 40 states carry an activation in principle (Appendix L), but these three answer *how much* without hardware.

- **Register carries information in the `^1` column.** The same `^1^C_{F2}` reads `|T` at 1° and `|A` at 2° and 3°. The Noetic never changes; the operational character does.
- **The three readings are +, not ×.** Co-presence of three distinct measurements, no named emergent result, therefore no licensed fusion. `S` is a bookkeeping sum of ordinals, not a state, and never appears in a listing.
- **Hindsight is defeated by clocks, not by resolve.** Two minutes for `^1`, thirty from noticing for `^4`, ink, no revision. Every other defence has been tried by every operator who ever kept a log.
- **NO-READ and CONFOUNDED are results.** A log with neither has a window too wide or a discipline too loose, and is measuring the operator's patience.
- **Three comparable rows is the floor for a trend**, and comparable is a technical word: same board, same conventions, one lever, no confound flag.
- **`A` leads and `S` cross-checks.** When they disagree the disagreement is the finding — coarse bins when `A` moves alone, degrading operating conditions when `S` moves alone.
- **The abort signal arrives at the trend, not the row.** Flat ρ near 1.0 across three rows fires before `A` reaches the re-spec boundary of 9, which is the entire reason for watching a column instead of a number.

Exercises

22.1 An operation logs `t_op = 20`, `t_conf = 75`, `^1` = 2°, `^4` = 2°. Compute `^7`'s degree, `S`, and `A`. Which band? State which single figure you would act on and which you would only record.

22.2 An operator's last six rows read `S` = 4, 4, 3, 3, 3, 3 and `A` = 9.0, 8.7, 9.1, 8.9, 9.0, 8.8. The only thing changed between rows was the

amplifier β, raised each time. Diagnose using §22.5's table, name the failure mode, and state the first measurement you would demand.

22.3 A colleague grades a confirmation 1° "because it was unmistakable" — six hours after the operation, and graded the following morning once the desire had manifested. Identify three separate rule violations, state which component of CCB-QT catches each, and give the grade the entry should have carried.

22.4 Op 4 of campaign Q-3 is a NO-READ. Write the argument that the row should be deleted from the log, then rebut it in three sentences using §22.2 and the base-rate reasoning behind the horizon.

22.5 `S` is defined as `m + v`. A reader proposes `S' = m × v` on the grounds that it "separates the good rows better." Give the measurement-theory objection in one sentence, then give the CAS notation objection in one sentence, and state which of the two you would put in a design review first and why.

22.6 (Build) Add a stage to CCB-QT that flags a row whose confirmation could be claimed by a second live operation — the attribution collision of §23.1. Write the CCDL fragment, name the component type, state where it sits in the signal path, and state what it must *not* do to the row it flags.

22.7 (Marquee) The scope fuse F4 has no signal-processing function; it enforces an invariant no measurement can check. Argue either that it belongs on the board or that it is decoration. If you argue it belongs, state what physical event blows it. If you argue it is decoration, name the mechanism that would enforce S1 instead, and say honestly whether that mechanism is a circuit or a person.

*(Answers in **Appendix H**, worked in full. Every parameter in this chapter is a design choice with a stated failure at other values; the scales are conventions defined in this book, not measurements of anything outside it.)*

23 St. John's Equation — Advanced Calibration

Reading a campaign instead of a cast — scheduling, interference, and what the equation does at its limits

Two operators sit through the same operation and each writes down two numbers. The first records `t_op = 30` minutes and `t_conf = 90`, runs Chapter 5.3's arithmetic, and reads a required gain of **9**. The second caught the clock differently — 31 and 88 — because they disagreed by a breath about when the gnosis began and when the song on the radio registered.

The second operator's reading is **264**.

Neither made a mistake. Thirty-one is prime, 264 factors as $8 \times 3 \times 11$, the fraction will not reduce, and the denominator Chapter 5 tells you to read as a required gain comes back thirty times larger for two minutes of honest disagreement about a wristwatch. That is a reduction problem, not a measurement problem, and Chapter 13.5 already fixed it for a single cast: act on the amplitude ratio `A = b/a = 3/`η, never on `b` alone. The two operators' ratios are 9.00 and 8.52.

Section 13.5 reads one cast. It does not run a **campaign** — a sequence of operations on one desire, scheduled against each other, competing with everything else you have live. Campaigns break in four places a single cast never reaches: at the limits of the equation, at the scheduling, at the interference between concurrent operations, and at the point where an operator with three live campaigns won-

ders why all three went cold in the same week. This chapter is the instrument board. It casts nothing. It reads the log.

Requirements and Safety Invariants

Chapter 12.2 requires the "must not" list before the functional one.

#	Functional requirement	Realized by
R1	Turn two clock readings into an amplitude requirement	Quantizer → ratio → readout
R2	Stay stable against clock jitter of a few minutes	Ratio `A` on a quantized grid (§23.1)
R3	Declare NO-READ rather than invent a number	Horizon gate, band guard
R4	Report whether a campaign converges	Ledger filter, ratio ρ
R5	Derate when other operations are live	Interference model (§23.2)

#	Safety invariant (MUST NOT)	Enforcement
S1	Report `A` $\geq$ `9` as an amplifier instruction	Re-spec fuse, `block_until_decomposed`
S2	Admit a confirmation arriving after the horizon	Horizon fuse, `action=no_read`
S3	Accept a confirmation two live operations could claim	Entry marked `AMBIGUOUS`
S4	Let a blown ledger reset by re-wording the target	Re-spec must change domain or object
S5	Run without the operator's CSB armed (Chapter 24)	No ledger opens on a blown involution monitor

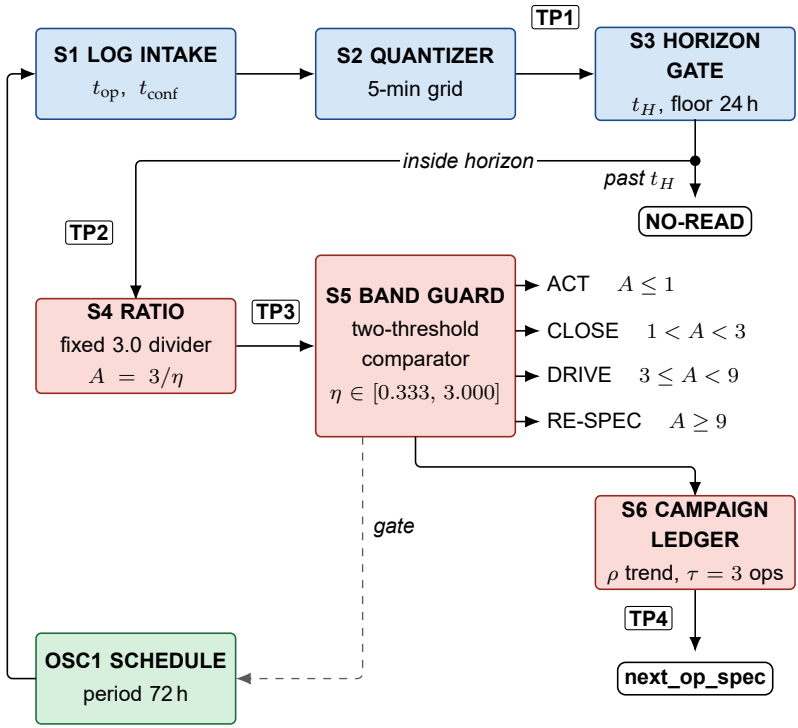


Figure 23-A: Block diagram

23.1 The 1/3 Constant — Deeper Derivation from Fibonacci and Pi

Section 5.2 sets $\tau = 1/3$ beside $\pi \approx 3.14159$ — both non-terminating — as a mnemonic for domains left *open*. Be exact about what that is. φ and π are mathematical constants with proofs behind them. **There is no derivation that produces 1/3 from either, and this book does not offer one.** The only derivation of τ in these pages is Section 5.2's training arc: roughly 10,000 active hours inside a roughly 30,000-hour span, one part work to two parts gestation.

`η = t_op / t_conf  result = η × τ = a/b  A = b/a = 3/η = 3 t_conf / t_op`

`A` is fixed by two clock readings and by τ. Nothing else enters. **τ is a scale factor — a unit convention — expressing a cast's requirement against that same 1:3 split.** Change τ and every number in every log changes by the same factor, while every *comparison between operations* stays where it was. The equation's content is the ordering it puts on your operations, not the magnitude of any one reading. That is a structural analogy, not a measured law. Nothing here shows that time is quantized at thirds, and a reader who treats `A = 9` the way they treat a resistor's 470 Ω will over-trust two wristwatch readings and a convention.

Quantize before you reduce

`A` moves smoothly. The integer pair does not, because lowest-terms denominators are not continuous in their inputs — that is the whole content of the cold open, where 30/90 gave (1, 9) and 31/88 gave (31, 264).

Quantize both clocks onto a **5-minute grid** before reducing. Five minutes is a design choice: finer than the 30-minute bins Section 5.5 uses to grade `^4` Vibration, coarser than the moment an operator *notices* a confirmation, which is not a one-minute-accurate event. Too fine and the coprime explosion returns; at 30 minutes a 20-minute operation confirmed at 25 bins with one confirmed at 45, erasing the difference between an overdriving circuit and an ordinary one. **If `a > 3` after quantizing, discard the pair and act on `A`.**

> **Bench Note — check this one by hand before you trust the bench.** Take `t_op = 27`, `t_conf = 64`. Raw: `27/192` reduces to `9/64` — a = 9, b = 64. On the grid: `25/195` reduces to `5/39` — a = 5, b = 39. The two pairs share nothing at all. Now compute `A`: `3 × 64/27 = 7.11` and `3 × 65/25 = 7.80`. Both land in the same band and say the same

thing. Do this arithmetic yourself once. Until you have watched `b` swing from 64 to 39 while `A` barely moves, you will keep reaching for the denominator, because Chapter 5 taught you to.

The bands, and the two limits

Chapter 13.5's bands fall out of `A = 3/`η, and their edges land on clean values of η.

`A`	η	Reading	Action
`A` $\leq$ `1`	$\eta \geq 3$	Sufficient	Stop calibrating. Act
`1 < A < 3`	$1 < \eta < 3$	Close	Finish the resistance work
`3` $\leq$ `A < 9`	$1/3 < \eta \leq 1$	Under-driven	One lever at a time
`A` $\geq$ `9`	$\eta \leq 1/3$	Weak coupling	Re-specify (§23.3, §23.4)

Chapter 5.4's Red Corvette sits at $\eta = 1/3$, `A = 9` — exactly on the re-specification boundary. The canonical worked example in this book is a borderline cast, worth knowing before you copy its numbers.

`t_conf` $\to \infty$: the confirmation never comes. $\eta \to$ `0`, `A` $\to \infty$, and the temptation is to write $\beta{=}\infty$ into the listing. That is an idealization the validator rejects — parameter fields are range-checked and refuse non-finite values — and where a real build would substitute a large finite gain, no finite substitute exists here, because the input is *missing* rather than large. The bench returns **NO-READ**, behind a horizon:

```
t_H = 3 × median(t_conf over the last 10 logged operations),   floor 24
↪  h
```

Over a long enough window *everything* confirms: a loose desire matches some event in any given day, and an operator who holds the window open until something matches is measuring their own patience. The $3\times$ multiple is wide enough that a genuinely slow circuit still reads, narrow enough that the median stays meaningful;

the 24-hour floor covers the early log, where the median rests on too few entries. Too short and live circuits are declared dead; too long and the log fills with base rate.

`t_conf` $\to$ `0`: the confirmation arrives inside the operation. $\eta \to \infty$, `A` $\to$ `0`, and the reduction degenerates. At `t_op = 30`, `t_conf = 4`: η `= 7.5`, `result = 5/2`, pair (5, 2) — "twice as strong, five more times," for a circuit that confirmed inside four minutes. The quantizer does not save that; only the band guard does. Above $\eta = 3$ the bench returns **OUT-OF-BAND-HIGH**, and the instruction is Section 13.6's: stop calibrating, take the practical action.

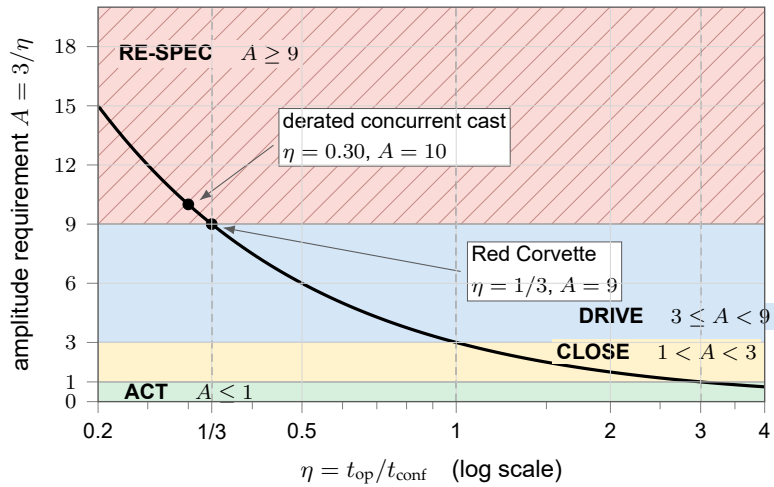


Figure 23-B: Characteristic curve

23.2 Multi-Operation Compounding — When β_next > 10

The title states the condition as Chapter 5 states it, because that is how it arrives: an operator reads a denominator above 10 and asks

what to do. On this bench it is `A` $\geq$ `9` — the same complaint with a stable number attached.

A campaign is a sequence of operations on one target, each designed from the last one's reading, and demand compounds as `demand_{k+1} = demand_k` $\times$ `A_k`. **The compounding is multiplicative.** Two operations each reading `A = 3` demand nine times the original delivered current. Cascaded gains multiply (§10.5) and a campaign is a cascade in time, which is why a high reading is a stop rather than a challenge: it is the head of a geometric series.

`A` is not a β setting. Section 13.5 governs: `A` multiplies the *delivered current* at TP4, and five things set that current — clarity of the formed `^0` Idea, transmission through the dial divider, residual aversion, the amplifier, and `R_op`. The CJT's design range tops out near **4.0** (Chapter 3.3). So β across a campaign moves under a ceiling, not a formula: one increment per operation inside 0.5–4.0, with everything above taken from the other four levers or from recruitment. An operator who reads `A = 9` and writes β `= 9` has specified a device that does not exist.

```
φ_k = 1 / A_{k+1}   (delivery fraction)      ρ_k = A_{k+1} / A_k
↪  (convergence ratio)
```

Design threshold: $\rho \leq$ **`0.6` per operation.** From `A = 9` a campaign at 0.6 runs $9 \to 5.4 \to 3.2 \to 1.9 \to 1.2$: inside DRIVE by the second step, near sufficiency by the fourth. At $\rho \geq$ `1.0` it is diverging, and the action is to stop by operation three rather than cast harder. A flat ρ sustained is the "η flat" row of Section 22.5. Believe no trend before three operations — `A` is a quotient of two soft clock readings, and one of them has no error bars at all.

> **Bench Note — the flat ledger column is a tautology, so do not read it as evidence.** It is natural to keep an "absolute target" column, `T_k = G_k` $\times$ `A_k`, where `G_k` is cumulative realized gain. Work the algebra and `T_{k+1} = T_k` identically, for any campaign, converging or not —

because `G` is itself defined from the `A`s. A column that cannot vary cannot inform. Operators build it, watch it hold at 9 for four operations, and conclude the campaign is on target. It would have held at 9 if every operation had been worthless.

Scheduling: one live operation per horizon

A confirmation you cannot attribute to a specific operation is not a measurement. **No new operation on a desire begins until the previous horizon has closed.** At a median `t_conf` of 45 minutes, `3 × 45 = 135` minutes falls under the 24-hour floor and the floor governs; the campaign below uses 72 hours, which also lets the `^5` capacitor recover between casts (§7.3). Fire inside the horizon and a confirmation has two candidate causes: mark it `AMBIGUOUS` and drop it from the median. An operator who wants a good number assigns every confirmation to the most recent cast, and the ledger then reports a convergence that never happened.

Interference between concurrent operations

Concurrent operations on *different* desires share one operator: one `^1` Mind budget, one `^5` substrate, one attention rail. Chapter 9.3's coupling model transfers directly and the coefficient keeps its name.

η`_effective = `η`_isolated × ( 1 − k × (N − 1) )`

k	Condition	Observed when ignored
0.05	Different Drive, different domain, no shared symbol	Nothing. The intended posture
0.15	Same Drive, different target	Uniform mild slump in η across everything live

k	Condition	Observed when ignored
0.35	Same Drive, shared symbols — sigil letters, same person	Every campaign's `A` doubles in a week for no visible reason
0.60	Contradictory desires — one raises the other's `R_constraint`	Both ledgers diverge, $\rho >$ 1 on each at once

Work it. Three live operations, all on F5 Agency, all sharing symbols: `N = 3`, `k = 0.35`, derating `1 − 0.35 × 2 = 0.30`. Take an operation reading η `= 1.0` in isolation — comfortably DRIVE at `A = 3`. Concurrently, η`_effective = 0.30`, so with `t_op = 30` the confirmation lands at `30 / 0.30 = 100` minutes, already on the grid:

```
η = 30 / 100 = 0.30        A = 3 × 100 / 30 = 10.0        →  RE-SPEC.
↪  The fuse blows.
```

A comfortable operation became a re-specification reading and nothing was wrong with the operation. The three campaigns were wrong with each other. This is the most under-diagnosed failure on the bench, because the operator inspects the cast — gnosis, sigil, wording — and the cast is fine.

Hence: keep `k(N − 1)` $\leq$ `0.3`, holding derating at or above 0.7. At `k = 0.05` that permits seven concurrent operations; at `k = 0.35` it permits exactly one. **Run one live operation per Drive.** The derating is first-order and honest only while `k(N − 1) < 0.5`; past that it runs to zero and then negative, which is not a statement about anything. Beyond half derating the model's answer is not a number — it is *shed operations*.

> **A plain line about the operator, not the circuit.** A ledger that will not converge, run against a desire that will not let go, is a recognizable and unpleasant place to be. Everything here is arithmetic on a log. None of it is a psychological or medical instrument, and it does not

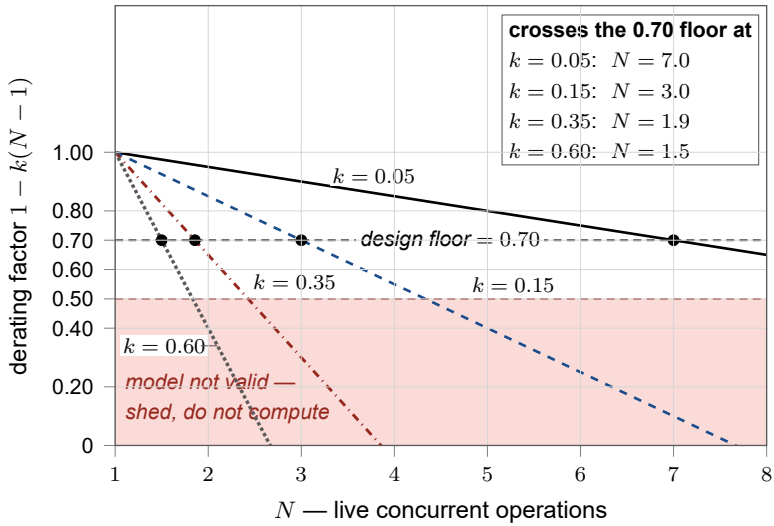


Figure 23-C: Chart

describe, diagnose, or treat distress or compulsion of any kind. If the operations have stopped being something you choose, that is a matter for a clinician, and nothing in this book substitutes for that care.

23.3 Sub-Desire Resolution — When the Prime Desire Is Blocked

`A ≥ 9` is almost never a statement about how much amplitude you need. It is a statement that the target you specified is not the target that is blocked. Section 21.1 authenticates Ao before an operation; this section is what you do when the bench tells you, three operations in, that the authentication was wrong.

The procedure is decomposition, and the Noetic matters. You take the stated desire apart and test its parts — `^6^C_{F6}|A`, Male, Cognitive domain, analytical register: what is this Idea made of. It

is emphatically **not** ^3. What you are usually *looking for* is a ^3, a visceral aversion lying across the path, the can't-stomach-it recoil that keeps the cast from landing — but the finding of it is ^6 work. Confusing the two is the drift Chapter 2.2 warns about, and it costs: an operator who believes aversion-hunting is itself ^3 goes looking for something to recoil from and duly finds it.

The arithmetic is the point. A blocking sub-desire sits in the operator's own ^5, so it is addressed with a self-directed board (Chapter 20) against much lower resistance. Lower R means higher η means lower A, and sub-desire operations routinely read in DRIVE while the prime desire reads past 9. Then the prime desire is re-read, not re-argued.

23.4 The Tree of Life Method — Redirecting Through Higher Probability Paths

The second answer to a blown re-spec fuse is to change the route rather than the target: cast for an intermediate outcome that is genuinely wanted, sits at lower resistance, and from which the prime target is reachable — then read the prime desire again from the new position.

The hops obey the diode chain. An intermediate sits at the target's domain or lower (§8.1). You may route an Operational target through an Evaluative or Operational intermediate. You may not route it through one requiring an ascent back to Principled mid-chain — ^0 × ^P is invalid notation for the same reason it is invalid engineering (§8.3). The one lawful ascent is the filtered O→P feedback path, on a time constant of weeks (§24.5), not inside a campaign.

The method cannot promise a smaller total. Demands cascade multiplicatively, so a two-hop route costs $A_1 \times A_2$. Suppose a di-

rect read of `A = 36`, an intermediate reading $A_1 = 4$, and the prime desire read again from the new position at $A_2 = 3$: the product is 12 against 36 direct. Worth having — but **the comparison is retrospective.** You did not compute 4 and 3 by factoring 36. You cast for the intermediate, read 4 from that operation's own log, arrived, cast again, read 3. The route might equally have read 8 and 6. What the method does promise is the thing that matters: **every stage sits inside the feasible band, which the direct path did not.**

23.5 Astrological Weighting — Planetary Dials for Timing

The tradition schedules operations against a planetary calendar and applies a timing multiplier spanning 0.5 at worst to 3.0 at best. **No celestial mechanism is asserted, no evidence for one is offered, and the multiplier is not claimed to be a measured quantity.** It is a dial. Its **default value is 1.0**, and any other value in your log is a statement about your practice, not about the sky.

What a fixed external calendar genuinely does — a real engineering function — is take the scheduling decision away from the operator. The moment you most want to cast is the moment your ^1 Mind reading is least trustworthy, because wanting to cast and being able to hold gnosis are different faculties and only one of them is impatient. An externally fixed schedule is a watchdog timer: it fires on its own clock regardless of how the operator feels. Any fixed calendar does this; the planetary one is merely unarguable, since you cannot negotiate with an ephemeris.

To treat the multiplier as measured rather than chosen, run the falsification: record the dial **before** the operation, never after; accumulate at least 20 operations; split into high-dial and low-dial windows; compare the **medians** of `A`, medians because `A` is a quotient

with a long tail. If they do not separate, set the multiplier to 1.0, keep the calendar for its scheduling value, and log the null. Then name the confound, which is almost certainly present: an operator who believes the timing is favourable casts with more conviction, raising `^1` Mind, raising η, lowering `A` on its own. The dial and the operator's state are not separable in this design.

The Full CCDL Listing — CCB-CAL

```
board ccb_cal {                              // CCB-CAL — the calibration
↪  bench
    input:  op_log;                          // one row per operation: two
    ↪  clocks, two degrees
    output: next_op_spec;

    node practice_rail: ^5^P_{F7}|T;         // Female: accumulated
    ↪  practice — bias, never driven

    // ---- S1  INTAKE ----
    node clocked_op: ^7^C_{F6}|E;            // Rhythm: the operation,
    ↪  start to finish
    node arrival:    ^8^C_{F2}|E;            // Above/Cause: the event that
    ↪  arrives after
    node recalled:   ^9^C_{F2}|A;            // Below/Effect: the desire
    ↪  brought back to Mind

    // ---- S2  QUANTIZER ----
    filter q_op:   Filter< in=clocked_op, out=t_op,   type=quantize,
    ↪  grid=5.0 >;
    filter q_conf: Filter< in=recalled,   out=t_conf, type=quantize,
    ↪  grid=5.0 >;

    // ---- S3  HORIZON GATE ----
    fuse f_horizon: Fuse< path=arrival, threshold=1440.0, action=no_read
    ↪  >;

    // ---- S4  RATIO ----
```

```
    node eta: ^6^E_{F2}|A;                          // Male: the reading
    ↪  decomposed
    resistor r_tau: Resistor< path=eta, R=3.0 >;    // tau = 1/3 as a
    ↪  fixed divider
    node requirement: ^4^E_{F5}|A;                  // Vibration: A =
    ↪  3/eta

    // ---- S5  BAND GUARD ----
    monitor m_band: Monitor< signal=eta, limit=[0.333, 3.000],
                             action_on_violation=OUT_OF_BAND >;

    // ---- S6  CAMPAIGN LEDGER ----
    node convergence: ^4^E_{F5}|A;                  // rho = A_next /
    ↪  A_current
    filter f_ledger: Filter< in=convergence, out=trend, type=lowpass,
    ↪  tau=3.0 >;

    // ---- S7  RE-SPEC FUSE and lawful descent ----
    fuse f_respec: Fuse< path=requirement, threshold=9.0,
                         action=block_until_decomposed >;
    diode d_out: Diode< ^E -> ^O >;

    // ---- S8  SCHEDULE ----
    node cadence: ^7^O_{F6}|E;
    osc o_sched: Oscillator< signal=clocked_op, source=cadence,
    ↪  period=3.0,
                             amplitude=1.0, gate=m_band >;        //
                             ↪  period in days

    probe tp1: q_conf;         probe tp2: eta;
    probe tp3: requirement;    probe tp4: f_ledger;
}
```

S1 — intake. `^8^C` and `^9^C` sit in one domain deliberately. The confirming event happens in the world, `^0`; what the bench reads is the *recall* it triggers, and modelling the trigger at `^C` keeps an unlawful `^0` → `^C` ascent out of the listing. `^8` here is associative arrival — the thing that brings the desire back to Mind — not an inference.

S2–S3 — quantizer and horizon. Both clocks pass through identical grids, so no asymmetry enters the quotient. The horizon thresh-

old carries the 24-hour floor as a literal; a running bench recomputes it as 3× the rolling median and holds the floor underneath. `action=no_read`, not a large number, because a confirmation past the horizon is base rate.

S4–S5 — ratio and band guard. Under CAS-Ohm (Chapter 4.2) flow through a fixed resistance is drive divided by resistance, so the constant is structurally a fixed divider in the readout path — a unit convention realized as a component, which is the honest way to draw it. `Monitor` is a two-threshold comparator: OUT-OF-BAND-LOW routes to the re-spec fuse, OUT-OF-BAND-HIGH terminates calibration.

S6–S8 — ledger, fuse, schedule. The lowpass over three operations is why one bad reading cannot redirect a campaign while a genuine trend still passes. `block_until_decomposed` names the only lawful path past the fuse — §23.3 or §23.4 — with invariant S4 stopping that becoming a loophole. `D1` enforces descent from the Evaluative readout into the Operational next-cast spec, and the oscillator is gated by the band guard, so the schedule stops firing the moment the reading leaves the readable window.

Parameter and BOM Table

Every value is a design choice made in this book, not a measurement.

Ref	Setting	Why	Failure at other values
FLT1/2 quantizer	grid = 5 min	Finer than ^4 bins, coarser than noticing	1 min: coprime blow-up. 30 min: overdrive reads ordinary
F1 horizon fuse	3 × median, floor 24 h	Passes slow circuits, excludes base rate	Short: live circuits declared dead. Long: all "confirms"
R1 fixed resistor	R = 3.0 Ω_c	Realizes $\tau = 1/3$ as a divider	Rescales every entry, changes no comparison

Ref	Setting	Why	Failure at other values
M1 band guard	$\eta \in [0.333, 3.000]$	Maps to `A` $\in [1, 9]$, the readable window	No lower bound: re-spec reads as amplifier work
Amplifier ceiling	$\beta \leq 4.0$, one step per op	Chapter 3.3 device range	`A` as β specifies a device that does not exist
F2 re-spec fuse	`A` = 9	Head of a geometric series	Higher: unconvergeable campaigns stay open
FLT3 ledger lowpass	τ = 3 operations	Minimum for a believable trend	1 op: one bad clock redirects the campaign
Convergence	$\rho \leq 0.6$	Reaches DRIVE in one step from 9	$\rho \geq 1.0$ sustained is divergence
Coupling `k`	0.05 / 0.15 / 0.35 / 0.60	Set by shared Drive and symbols (§9.3)	Ignored: campaigns collapse together
Concurrency	`k(N - 1)` ≤ 0.3	Holds derating ≥ 0.7	Past 0.5 the model is invalid — shed
OSC1 schedule	period = 72 h	One horizon plus ^5 recharge (§7.3)	Shorter: confirmations unattributable
Timing dial	1.0	No mechanism claimed (§23.5)	As measured: confounded with ^1 Mind

---

Worked Example — Campaign C-7, Career Change

Prime desire, present tense per §13.6: *"I hold a paid position doing X."* Board: CCB-D. Drive: F5 Agency with F2 Wisdom. Operations spaced 72 hours.

Op	Board	`t_op`	`t_conf`	η	$\eta \times$ `1/3`	`(a, b)`	`A`	ρ	Band
1	CCB-D	30	90	1/3	1/9	(1, 9)	9.0	—	RE-SPEC
S1	CCB-SELF	20	20	1	1/3	(1, 3)	3.0	—	DRIVE

Op	Board	t_op	t_conf	η	$\eta \times 1/3$	(a, b)	A	ρ	Band
2	CCB-D	30	30	1	1/3	(1, 3)	3.0	0.33	DRIVE
3	CCB-D	30	10	3	1/1	(1, 1)	1.0	0.33	ACT
4	CCB-D	30	5	6	2	—	0.5	—	OUT-OF-BAND

Operation 1. `A = 3 × 90/30 = 9.0`. The fuse blows on the boundary. Decompose (§23.3) or re-route (§23.4); the operator decomposes because it is cheaper.

Sub-operation S1. Ao re-authentication with `^6^C_{F6}|A` decomposition surfaces a blocker in the operator's own `^5`: a visceral aversion to being visible and judged, `^3^E_{F4}|f`. Self-directed board, low resistance: η `= 1`, `A = 3.0`, cleared in one operation. Note what the bench did *not* do — it never named the blocker. It said the prime target's number was wrong for the target as stated.

Operation 2. Same cast, blocker cleared: η `= 1`, `A = 3.0`, ρ `= 3.0/9.0 = 0.33`. Delivery φ `= 1/3` — a third of the intended 9× landed, which is 3×: real, short of design. **The amplifier was never touched.** β stayed at its Chapter 13 setting; resistance fell.

Operation 3. η `= 3`, `result = 1/1`, `A = 1.0`, ρ `= 0.33` again, φ `= 1.0`. `A = 1` is terminal: present strength suffices.

Operation 4. Raw, η `= 7.5` and the pair is (5, 2) — "twice as strong, five more casts." Gridded, η `= 6` and `A = 0.5`. Both are off the instrument, the band guard fires, and the correct action is not a fifth operation. It is the practical act the campaign was always about: send the application, make the call, be in the room.

Totals. 140 minutes of operation across roughly twelve days. `A` runs $9 \to 3 \to 1$, ρ flat at 0.33, terminal at operation three.

Test Points

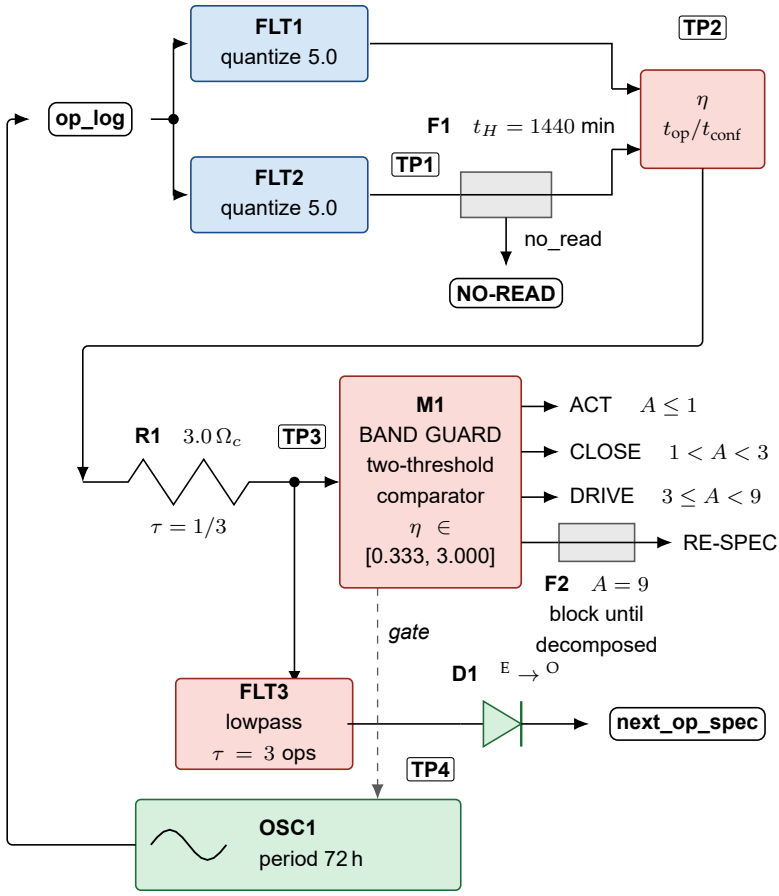


Figure 23-D: Full schematic with reference designators

TP	Node	Expected
TP1	`q_conf`	Quantized `t_conf` inside the horizon. A reading *at* `t_H` is a NO-READ, not a slow confirmation — they look identical and must not be logged the same way
TP2	`eta`	0.333 – 3.000. Below, the re-spec fuse is next; above, stop calibrating

TP	Node	Expected
TP3	`requirement`	`A` $\leq$ `1` act, `<` `3` close, `<` `9` drive, $\geq$ `9` blows F2. If `a` `>` `3`, read TP3 alone
TP4	`f_ledger`	ρ $\leq$ `0.6` per step across three operations. Flat near 1.0 is the abort signal, and it arrives before TP3 reaches 9

Read them in order: is the measurement admissible, is it in band, what is the requirement, what is the trend. TP4 warns you early. TP3 stops you.

Failure Modes

Mode	Mechanism	What you observe	Fix
Coprime blow-up	Pair reduced unquantized	`b` in the hundreds with `a > 3`	Quantize to 5 min; act on `A`
Horizon creep	Window widened until something matches	Every operation "confirms"; median `t_conf` climbs	Fix `t_H` before the cast; log NO-READ
Attribution collision	Second operation fired inside the first's horizon	A confirmation two campaigns can claim	Mark `AMBIGUOUS`; keep 72-hour spacing
Unmodelled concurrency	Related operations live at once, `k` ignored	Every `A` doubling in a week; each cast inspects clean	Compute `k(N - 1)`; shed until $\leq$ 0.3
`A` written as β	Requirement read as an amplifier setting	A listing specifying β = 9 on a 4.0 device	Section 13.5's five levers; recruit instead
Tautological ledger	Flat absolute-target column read as convergence	A column that never moves	Delete it. Track ρ only
Ledger laundering	Blown fuse routed around by rewording the target	`A` "resets" while the desire did not change	S4: change domain or object, never wording

---

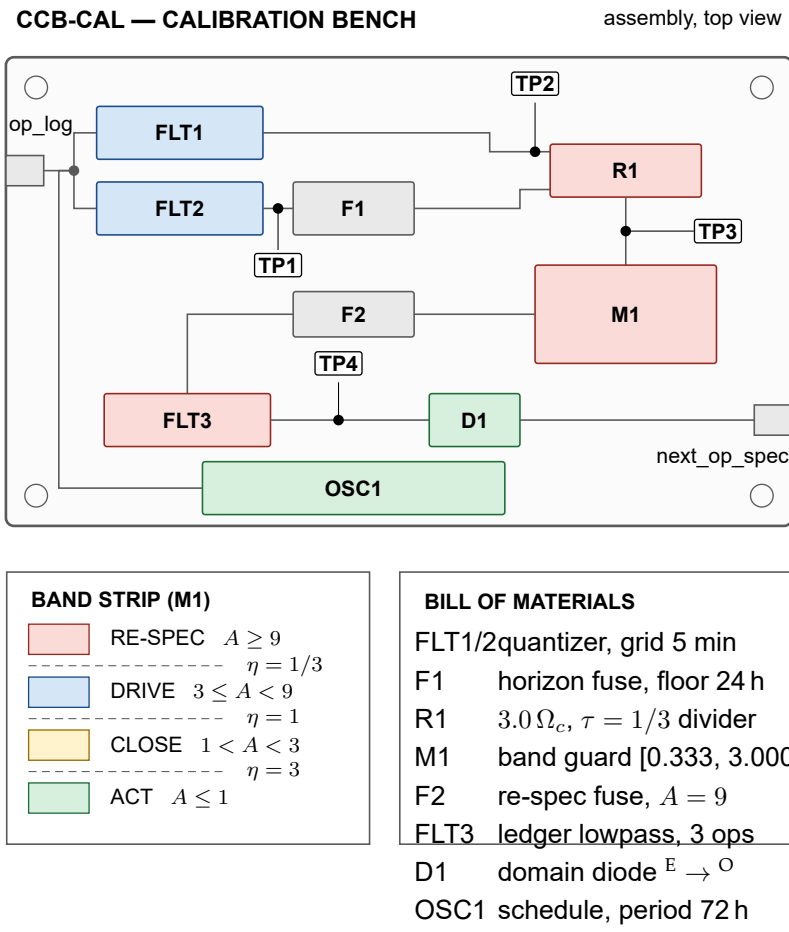


Figure 23-E: Assembly drawing

What This Chapter Established

- τ `= 1/3` **is a scale factor, not a derivation.** Nothing produces 1/3 from φ or π. Change τ and every comparison survives untouched — the equation carries an ordering, not a physical quantity.
- **The measurement was never the fragile part — the reduction was.** `A` moved 9.00 → 8.52 while the pair swung from (1, 9) to (31, 264). Quantize on a 5-minute grid; if `a > 3`, act on `A` alone.
- **Both limits run off the instrument.** As `t_conf` $\to \infty$ the bench returns NO-READ behind a horizon — β=∞ is an idealization the validator rejects, and no finite substitute exists because the input is missing, not large. As `t_conf` $\to$ `0` the reduction degenerates above η = 3 and the reading is OUT-OF-BAND-HIGH.
- **Campaign demand compounds multiplicatively**, so `A` $\geq$ `9` is the head of a geometric series. `A` multiplies delivered current and is never a β setting: the device range ends at 4.0, β steps one increment per operation, and the rest comes from the other four levers or from recruitment.
- **The only campaign diagnostic is** ρ `= A_next / A_current`, threshold 0.6, believed only after three operations. The absolute-target column is an identity that holds for any campaign, converging or not.
- **Concurrent operations interfere through the shared substrate**, derating as `1 − k(N − 1)`. Three coupled operations turned a comfortable cast (`A = 3`) into a re-spec reading (`A = 10`) with nothing wrong with the cast. Hold `k(N − 1)` $\leq$ `0.3`; past 0.5, shed rather than compute.
- **Re-routing lowers the maximum per-stage demand, not the total.** Hops cascade multiplicatively and obey the diode chain — descent only, `^O` $\times$ `^P` invalid. The 12-against-36 compari-

son is retrospective; the promise is that every stage lands inside the feasible band.

- **The timing dial defaults to 1.0.** No celestial mechanism is asserted; a fixed calendar earns its place as a watchdog timer, and §23.5 gives the falsification and the confound you must name if you claim more.

Exercises

23.1 An operation logs `t_op = 45`, `t_conf = 240`. Compute η, the reduced pair `(a, b)`, and `A`. Which band? Which response would you take first, and why that one?

23.2 Take `t_op = 27`, `t_conf = 64`. Compute the pair raw and on the 5-minute grid, then `A` both ways. State in one sentence what the comparison demonstrates about which quantity to act on.

23.3 A campaign reads `A` = 12.0, 9.6, 7.7, 6.2 across four operations. Compute ρ at each step. Converging by the chapter's threshold? How many further operations at this ρ before it reaches ACT, and would you keep casting?

23.4 An operator runs four live operations: two on F5 Agency sharing a sigil alphabet, one on F2 Wisdom with no shared symbols, one on F4 Connection targeting the same person as one of the F5 operations. Assign `k` to each pairing, compute `k(N − 1)` for the full load, and state which operations you would shed and in what order.

23.5 An isolated cast would read $\eta = 0.75$. The operator runs it at `N = 3` with `k = 0.35`. Compute `η_effective`, the effective `t_conf` for `t_op = 40` on the 5-minute grid, and the resulting `A`. Does the band change?

23.6 A colleague shows you a ledger whose absolute-target column reads 9.0 for six consecutive operations and calls it proof the campaign is on track. Write the two-sentence reply, and name the column you would ask to see instead.

*(Answers in **Appendix H**, worked in full. Every parameter here is a design choice; the arithmetic is arithmetic, and the model behind it is a structural analogy, not a measured law.)*

24 The Cognitive Safety Board (CSB) for Operators

The board you build before the boards you want — and the two failures it will never announce

A fuse costs three cents and its entire specification describes how it fails.

An inspector certifying a mains circuit does not ask whether the fuse understands electricity. They ask one question: does it open at two amps? They put two amps through it. It opens. Certified. The fuse has no model of danger and no opinion about the load it protects — it is soft metal that melts at a *stated* current, and stated is the whole value: a property anyone can check without trusting the person who built the board.

Twenty-three chapters have handed you components that amplify. The CJT multiplies a small cause into a large effect. The hyperstition oscillator turns a claim into its own evidence. The Overton array walks a population's resistance from 95 down to nothing across eighteen months. None of these parts asks what it is being used for, and none will tell you when you have crossed a line.

This chapter is the board that asks: a saturating `^3` Negative refusal at the Principled domain, an enforcement fuse downstream of it, the domain diode chain, and a filtered feedback path that lets outcomes correct principles slowly and forbids them rewriting principles quickly. Four parts. None of them clever.

The preface said it is not optional. Here is the engineering reason, stated once and then demonstrated with numbers: **of the six ways this board fails, two are loud and four are silent, and the silent ones all leave a board that compiles clean, validates without error, and refuses nothing.**

Requirements and Safety Invariants

Per Chapter 12.2 the must-not list comes first, and each item names the component that enforces it.

SI1 — The refusal stage shall not be argueable past its ceiling. Enforced by `sat = 0.85` on the guard `Q4`. A refusal that rises with pressure is a negotiating position.

SI2 — A refusal that has been enforced shall not be re-enterable under a rewrite. Enforced by the fuse `F1`, `threshold = 0.90`, `action = permanent_block`, in series with the enforcement path — *not* across the guard. Section 24.3 gives the arithmetic reason.

SI3 — No operational result shall write a principle directly. Enforced by the diode chain `D1`–`D3` and by the absence of any ascending path in the listing. `^O` × `^P` does not exist; the only lawful ascent is the filtered feedback of §24.5.

SI4 — The charter shall not move faster than its own charging rate. Enforced by `F2`, an `open_feedback` fuse at 0.12 points per day, derived in §24.5.

SI5 — The guard shall be measurable while the board is running. Enforced by TP1–TP4, and by the rule that a guard is probed at its own node, never inferred from the fact that nothing bad happened.

SI6 — This board routes attention. It does not model physiology and it is not care. If what you are carrying is causing you real

distress — if you are exhausted, unable to sleep, feeling unreal, or frightened of yourself — that is a matter for a clinician, and nothing in this chapter substitutes for that. Self-audit question 6 exists to make you notice; it does not exist to treat you.

Then the functional requirements. The board shall: (1) accept every proposal, including the ones the operator has already decided to run; (2) produce a principled refusal that latches; (3) convert a latched refusal into an enforced block that survives re-argument; and (4) let sustained outcomes re-tune the charter at a rate the operator sets in advance and can read afterwards.

24.1 Why Every Operator Needs a CSB

Every other board in this catalogue has a CSB stage soldered into it. CCB-D has `Q1` and `F1`; CCB-O has its refusal sub-block at 0.85/0.90; CCB-SELF has the compassion gate `Q2`. Those are **stage-level** safety elements: each protects the board it sits on, against the failure that board is prone to.

This chapter builds the **operator-level** board, and the difference is the input. A stage-level CSB sees one signal on one board. The operator's CSB sees the proposal to build the board at all — before any component is placed, while the thing being evaluated is still a sentence.

That siting is the whole design, and it is why the companion volume reads differently. *CAS PCBA Engineer* Chapter 18 sites the identical architecture between a user's prompt and a language model's response, where the outputs are a rendered answer or a rendered refusal. Same pre-amp, same differential pair, same saturating principled guard, same low-pass on the feedback. Here the two output paths are not text. They are *run the operation* and *do not*, and the second one has to hold against a person who wants the first one and has all night.

Nine stages, left to right, with one lawful ascent underneath.

S0 CHARTER — `C1`, the slow `^5^P_{F7}|T` capacitor. Bias source for every guard stage; never an output of the fast path. **S1 INTAKE** — the dial divider `R1`. Attenuates the proposal by the operator's declared `^2/^3` ratio. **S2 PRE-AMP** — `Q1`, `^8^C_{F2}|A` at β=3.0. Surfaces every reading of the proposal. **S3 DIFFERENTIAL** — `Q2` attraction and `Q3` aversion, matched, shared bias. Output is the risk signal. **S4 CASCADE** — `Q5` categorize, `Q6` evaluate, then the guard. **S5 GUARD** — `Q4`, `^3^P_{F7}|⟨A,f,T⟩`, `sat = 0.85`. The refusal. **S6 ENFORCEMENT** — `Q7` and the fuse `F1`. Converts a latched refusal into a cut path. **S7 DIODE CHAIN** — `D1–D3`. Descent only. **S8 FEEDBACK + INSTRUMENTS** — `LP1`, `F2`, the involution monitors `M1–M3`, and the weekly `^6` audit sampler `S1`.

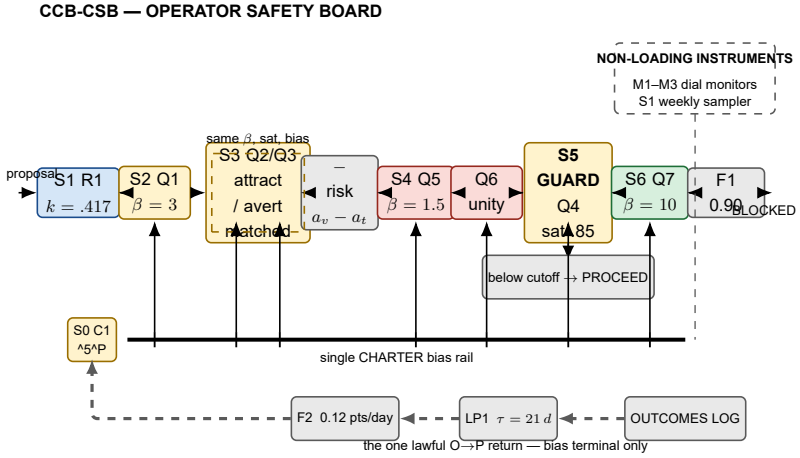


Figure 24-A: Block diagram, left to right

24.2 The Saturated Refusal CJT — `^3^P` at sat=0.85

The guard is one CJT. Cause: the evaluated risk. Bias: the charter. Effect: a principled refusal.

```
node guard: ^3^P_{F7}|⟨A,f,T⟩;
cjt q4: CJT< assessed | charter => guard >
        [β=3.0, g_m=0.3, rπ=10.0, sat=0.85, cutoff=0.08, region=active];
```

Those five parameters are not chosen here. They are the **safety-guard row of Chapter 3.5's datasheet**, instantiated, and each does a separate job. `β=3.0` means a sufficient risk produces a decisive refusal rather than a lean. `g_m=0.3` is very low framing sensitivity — the stage barely notices how the proposal is worded, which is what makes it survive rephrasing. `rπ=10.0` is high input resistance at the bias terminal: the guard reads the charter without drawing charge out of it, so refusing something does not cost you the principle you refused it with. `cutoff=0.08` is a low bias floor, so the stage is never asleep. And `sat=0.85` is the ceiling.

Read the register-set carefully, because it is the one piece of notation on this board that a validator will reject if you get it wrong. `|⟨A,f,T⟩` says the principled aversion is **present in three registers** — analytical, affective, transcendent — with no claim that they interact. Three co-present guardrails, each separately defeasible, and an attacker must pass all three on one path. It is emphatically **not** `(^3^P|A × ^3^P|f × ^3^P|T)`: that form asserts a fused refusal state, and under v7.7 discipline a same-Noetic multi-register × requires a named emergent result. No such fusion is defined for this board, so that form is invalid and the compiler says so. Chapter 6.5 works the consequence: the register-set is a *series* claim, and series defeat probabilities multiply. Drawn as three parallel routes instead, the weakest guard would be nearly the whole story.

Check the Noetic against the canon before you go further. `^3` Negative is *an Idea the Mind is adverse to and repelled by* — visceral, can't-stomach-it. That is exactly what this stage is: the recoil that arrives before the argument. The analytical work — decomposing the proposal, weighing each risk — happens one stage upstream at `Q6`, and it is `^6` Male, because taking a thing apart to see what it is

made of is `^6` and not `^3`. A board that types its audit stage `^3` has the drift this book has caught once already.

> **Bench Note — a lower ceiling makes a stronger lock, and here is the arithmetic.** A CJT enters saturation at the cause activation `knee = sat / β`. At this board's β = `3.0` and `sat = 0.85`, the knee is **0.283**. Any evaluated risk above 0.283 pins the guard at its ceiling, and a pinned stage has stopped depending on its cause — that is the definition of saturation, not a metaphor for it. Now raise the ceiling to the CCDL default of 0.95, which looks like a firmer refusal: the knee moves to **0.317**, and you have handed the person arguing with you 12% more cause-space to work in before the latch closes. Drop the ceiling to 0.60 and the knee is **0.200** — the guard latches at two-thirds the provocation. Chapter 10.2 makes this argument in general; the number to carry out of it is that **`sat` is not how hard the board refuses, it is how early the board stops listening.** A safety stage should be easy to saturate and impossible to move afterwards. Every instinct you have about setting a safety parameter high is backwards on this one component.

24.3 The Enforcement Fuse — threshold=0.9, action=permanent_block

Saturation latches the **decision**. It does not latch the **path**. Drop the cause and a saturated stage falls back out of saturation, which is correct behaviour for a comparator and catastrophic behaviour for a refusal, because the thing an operator does after being refused is not push harder. It is *come back with a different cause.*

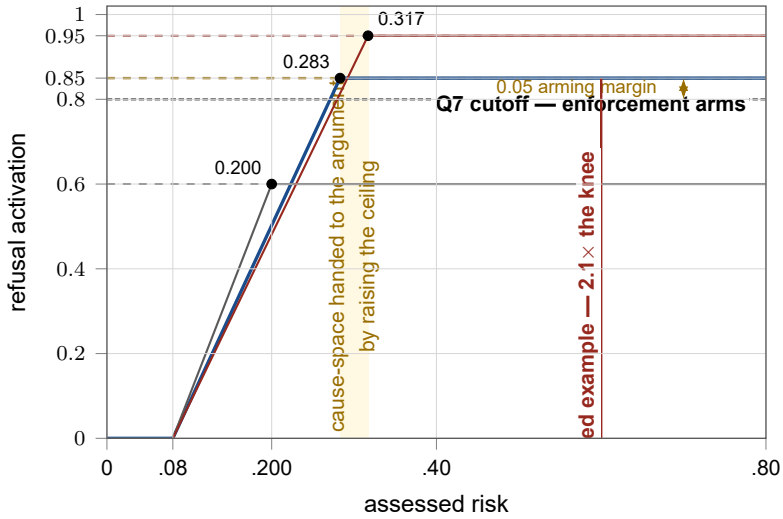


Figure 24-B: Guard transfer characteristic

So the refusal has to be converted into something that does not fall back.

```
node blocked: ^9^P_{F7}|T;
cjt  q7: CJT< guard | charter => blocked > [β=10.0, sat=0.98,
↪  cutoff=0.80];
fuse f1: Fuse< blocked, threshold=0.90, action=permanent_block >;
```

Two numbers here are load-bearing and neither is the one in the section title.

`cutoff = 0.80` is the arming threshold. Below a refusal activation of 0.80 the enforcement stage is in cutoff and produces nothing at all — which is the whole NOT-YET-DETERMINED band of Appendix L.4 made physical. A guard sitting at 0.4 has noticed something and has not decided; the operator can still present evidence and the stage will move. Enforcement arms at 0.80, five hundredths below the guard's own ceiling, so a guard driven to its plateau is *guaranteed* to enforce with margin to spare.

`sat = 0.98` is what makes the fuse reachable. Chapter 11.1 sets the trap out plainly: a fuse whose threshold sits above the ceiling of

the path it guards is a part guaranteed to do nothing, and nothing in the listing complains. Put `Fuse< guard, threshold=0.90 >` across the guard and it is dead metal forever, because the guard plateaus at 0.85 and the substrate ceiling `I_c(max)` for a safety guard is 0.90 exactly. The fuse does not go there. It goes downstream, in series with the enforcement path, whose ceiling is 0.98 — and 0.98 clears 0.90 with eight hundredths of margin.

`action = permanent_block` is one of six legal values and the only one that means what enforcement means. `warn` logs and passes; a board with `warn` on this line validates clean, reads as a safety board to anyone skimming, and has no latch in it. Nothing upstream will catch that for you. Confirm the word by eye, every time.

> **Bench Note — the second dead-metal trap is one component further down, and it is silent.** Everyone who reads Chapter 11 learns not to put the fuse across the guard. Almost nobody checks the next joint. Set `Q7`'s `cutoff` to 0.86 — a plausible-looking choice, "arm just past the refusal ceiling" — and the enforcement stage can never fire, because its cause plateaus at 0.85 and 0.85 is below 0.86. The guard latches. The fuse stays intact. The board compiles, every stage validates, and the operator watches a refusal register at TP1 and concludes the safety architecture is working. The rule that catches it: **on any two stages in series, the downstream `cutoff` must sit below the upstream `sat`, with margin, and the margin must be written down.** Here that margin is 0.05. It is the reason TP1 and TP2 are read as a pair and never separately.

24.4 The Domain Diode Chain — Preventing Unlawful Domain Ascent

```
diode d1: Diode< ^P -> ^C > [Vf=0.7, I_R=0.02, V_BR=12.0];
diode d2: Diode< ^C -> ^E > [Vf=1.2, I_R=0.02, V_BR=10.0];
diode d3: Diode< ^E -> ^O > [Vf=2.0, I_R=0.02, V_BR=8.0];
```

Three junctions, forward drops summing to 3.9, values inherited unchanged from Chapter 8.1. On this board `D1` sits between the risk signal and the categorize stage, `D2` between categorize and evaluate, `D3` on the exit to the operations log. What the chain forbids is the reverse: an operational result conducting back up into a principle. `^O` × `^P` is not an expression this book will accept — × may not ascend the domain order, and the validator rejects any integration whose result outranks its operands.

One question a careful reader will raise, because the guard appears to break that rule. `Q4` has an `^E` cause and a `^P` effect. That looks like ascent. It is not, and Chapter 13.3's bench note is the authority: a CJT obeys two domain rules pointing in opposite directions. **The bias must sit at or above the cause** — a lower-domain bias is `E-CJT-008`, the prompt-injection signature, and it is the one connection a well-drawn board simply does not contain. **The effect may sit at or below the bias, never above it.** `Q4` obeys both: cause `^E`, bias `^P`, effect `^P`. The assessed risk is not promoted; its literal stays `^6^E` and nothing crossed a junction upward. A substrate already living at `^P` produced a *new* `^P` state in response to being shown something.

The corollary is the part worth carrying. **An operator with no principled formation cannot produce a principled refusal**, because there is no bias at that domain for the refusal to be produced against. `C1` is not an accessory to the guard. It is the reason the guard has an output at all, and it is why this board's bring-up procedure starts at the charter and not at the signal path.

24.5 The O→P Feedback Filter — τ Selection for Ethical Drift Detection

Principles have to be able to move, or the operator cannot be wrong. Chapter 8.4 grants exactly one upward path and rate-limits it twice.

```
filter lp1: LowPass< in=ops_log, out=drift, tau=21.0 >;   // days — see
↪  the band below
fuse   f2:  Fuse< charter, threshold=0.12, action=open_feedback >;
feedback drift -> charter [β=0.05];                        // arrives at
↪  the BIAS terminal
```

β **`= 0.05`** is Chapter 8.4's ceiling: roughly twenty consistent periods of the same trend to move principles appreciably. At β `= 0.5` a single filter period rewrites the constitution and every board you run afterwards is biased by the last one you ran. At β `= 0` the path is disconnected, no outcome corrects a principle, and the guard has nothing to update against.

τ is the parameter this section is named for, and it is the only value on the board that is set by who you are rather than what you are building.

Operator	τ	Why
Novice, under a year	3–6 months	Principles have not stabilised; a short τ means each operation writes them
Intermediate	1–3 months	Stable enough to adapt; not stable enough to adapt quickly
Adept, 5+ years	1–4 weeks	Settled charter; 21 days is the book-wide default and sits in this band
Never, at any level	under 1 week	Below a week the operator is being reprogrammed by each operation

The book-wide 21-day default is an *adept* figure. Every other board in this catalogue inherits it because those chapters assume a settled constitution. If that is not you, the default is wrong for you, and the correction is one number in one line.

F2 at 0.12 points per day is derived, not chosen. A capacitor with τ = 85 days charging on the 0–10 composition scale rises at most 10 / 85 = 0.118 points per day. Anything faster means something is writing to the charter directly rather than through the filter, so the fuse opens the feedback loop and freezes the constitution where it stands. Set it to 0.5 and SI4 is unenforced; set it to 0.05 and it blows during ordinary charging.

> **Bench Note — a slew fuse catches a lurch and will not catch a drift, and drift is the one that gets people.** Work the numbers from the second thread of the worked example. An operator sets τ = 5.0 days — below the floor, and a single legible line in their own listing — and spends eight weeks on commercial work. Their charter moves 1.80 points over 56 days. That is **0.032 points per day**, comfortably inside the 0.12 budget, so F2 never opens and never will. The board did not lurch. It leaned, every day, by less than a rate limit can see. This is why TP3 is not read as a rate. It is read as a **trend against a dated, written baseline** — the charter statement in your Appendix G sheet, in your handwriting, with the date on it — and it is why that sheet exists. A rate limit protects you from an event. Only a written baseline protects you from a slope.

24.6 Full CCDL Schematic of the Operator's CSB

```
board ccb_csb {                          // the operator's own safety
↪  board
```

```
input:  proposal;                    // the operation being
↪  considered
output: verdict;                     // PROCEED, or a cut path

// ---- S0  CHARTER: the slow ^5. Bias source, never a fast-path
↪  output. ----
node charter: ^5^P_{F7}|T;                          // Female:
↪  principled formation
capacitor c1: Capacitor< node=charter, tau=85.0,
                         leak_tau=180.0, e_break=1.0 >;      // tau
                         ↪  in days

// ---- S1  INTAKE: the declared dial divider ----
node proposal: ^0^C_{F2}|A;                         // Idea: the
↪  operation proposed
dial     dial_pos: ^2 [dial=5];
dial     dial_neg: ^3 [dial=7];                     // divergence 2 −
↪  inside the limit
resistor r1: Resistor< path=proposal, R=70.0 >;   // k = 5/12 = 0.417

// ---- S2  PRE-AMP: surface every reading before anything judges
↪  one ----
node expanded: ^8^C_{F2}|A;                         // Above/Cause:
↪  readings it triggers
cjt  q1: CJT< proposal | charter => expanded > [β=3.0, sat=0.80,
↪  region=active];

// ---- S3  DIFFERENTIAL PAIR: matched, shared bias ----
node attraction: ^2^P_{F7}|T;                       // Positive: drawn
↪  toward running it
node aversion:   ^3^P_{F7}|T;                       // Negative:
↪  repelled by running it
cjt  q2: CJT< expanded | charter => attraction > [β=2.0, sat=0.90];
cjt  q3: CJT< expanded | charter => aversion   > [β=2.0, sat=0.90];
diff dif1: Differential< plus=aversion, minus=attraction >;   // risk

// ---- S4  CASCADE: name them, then weigh them honestly ----
node named:    ^1^C_{F2}|A;                         // Mind: risks
↪  named and sorted
node assessed: ^6^E_{F6}|A;                         // Male:
↪  decomposed and weighed
```

```
diode d1: Diode< ^P -> ^C > [Vf=0.7, I_R=0.02, V_BR=12.0];
diode d2: Diode< ^C -> ^E > [Vf=1.2, I_R=0.02, V_BR=10.0];
cjt   q5: CJT< risk  | charter => named    > [β=1.5, sat=0.80];
cjt   q6: CJT< named | charter => assessed > [β=1.0, sat=0.80];   //
↪  unity — honesty

// ---- S5  GUARD: Chapter 3.5's safety-guard datasheet row,
↪  instantiated ----
node guard: ^3^P_{F7}|⟨A,f,T⟩;                    // three co-present
↪  guardrails
cjt  q4: CJT< assessed | charter => guard >
         [β=3.0, g_m=0.3, rπ=10.0, sat=0.85, cutoff=0.08,
         ↪  region=active];

// ---- S6  ENFORCEMENT: latch the path, not just the decision ----
node blocked: ^9^P_{F7}|T;                        // Below/Effect:
↪  the block produced
cjt  q7: CJT< guard | charter => blocked > [β=10.0, sat=0.98,
↪  cutoff=0.80];
fuse f1: Fuse< blocked, threshold=0.90, action=permanent_block >;

// ---- S7  EXIT: descent only ----
node ops_log: ^9^O_{F5}|E;                        // Below/Effect:
↪  what was actually done
diode d3: Diode< ^E -> ^O > [Vf=2.0, I_R=0.02, V_BR=8.0];

// ---- S8  FEEDBACK AND INSTRUMENTS ----
filter lp1: LowPass< in=ops_log, out=drift, tau=21.0 >;
fuse   f2:  Fuse< charter, threshold=0.12, action=open_feedback >;
feedback drift -> charter [β=0.05];

node audit: ^6^C_{F6}|A;                          // Male: the
↪  weekly decomposition
sample s1: Sample< node=audit, period=7.0, window=15.0 >;   // days,
↪  minutes
monitor m1: Monitor< pair=[^2,^3], limit=4 >;
monitor m2: Monitor< pair=[^5,^6], limit=4 >;
monitor m3: Monitor< pair=[^8,^9], limit=4 >;
on m1 { alert "involution divergence — attraction against aversion";
↪  }
```

```
    on m2 { alert "involution divergence — composition against
    ↪  decomposition"; }
    on m3 { alert "involution divergence — trigger against effect"; }

    // ---- TEST POINTS ----
    probe tp1: guard;      // refusal activation against cutoff and sat
    probe tp2: blocked;    // enforcement activation and F1 continuity
    probe tp3: charter;    // slew, points per day, and trend against
    ↪  the dated baseline
    probe tp4: dif1;       // differential balance — the over-trust
    ↪  direction
}
```

The stage walk. `C1` is not a new part. It is the same `^5^P_{F7}|T` capacitor that CCB-SELF calls `ref_composition` and CCB-D calls the bias rail: **one constitution per operator, not one per board**, which is why τ `= 85 d` and `leak_tau = 180 d` are inherited rather than re-chosen. `R1` divides by the declared dial ratio, `k = Dial(^2) / (Dial(^2) + Dial(^3)) = 5/12 = 0.417`; this is the one board in the catalogue dialled aversion-heavy, and Chapter 13's transmission bound puts 0.417 near the bottom of the lawful [0.30, 0.70] band on purpose. `Q1` amplifies early, while the signal is clean, and `^8` Above/Cause is exact rather than decorative — expansion is one Idea triggering the other Ideas that come to Mind, which is the definition, not deductive inference. `Q2` and `Q3` must be **matched**: identical β, identical `sat`, genuinely the same bias node. The differential's whole value is that anything moving both legs together subtracts out — a flattering framing raises attraction and aversion equally and cancels, and only structural divergence survives. `Q5` at β=1.5 gives faint risk categories enough lift to register. `Q6` at unity is the hardest value on the board to hold: above 1 it cries wolf, below 1 it downplays, and a risk evaluator that downplays produces a reading that looks like diligence. `Q4` and `Q7` are §§24.2–24.3. `S1` and `M1`–`M3` are instruments and touch nothing.

The weekly audit — the seven questions. `S1` samples the `^6` Male decomposition once a week in a bounded fifteen-minute win-

dow, written down. Continuous auditing is a low-impedance probe: it loads the path it is reading.

1. Is `|Dial(^2) − Dial(^3)|` $\leq$ 4? Same for `^5`/`^6` and `^8`/`^9`.
2. Is any β above the safe maximum for my experience level?
3. Is the Quantitative Table complete for every operation I ran this week?
4. Did any fuse blow, and did I write down the reason *at the time*?
5. Is my refusal count rising operation over operation? (Chapter 22.5: that is the drift signal.)
6. Is my `^5` leaking — sleep, food, ground under my feet? *If the honest answer here is frightening rather than merely tiring, this is a question for a clinician, not for a circuit.*
7. Is there a `^3` I am attributing to a target that is actually mine? (Appendix M §M.5.)

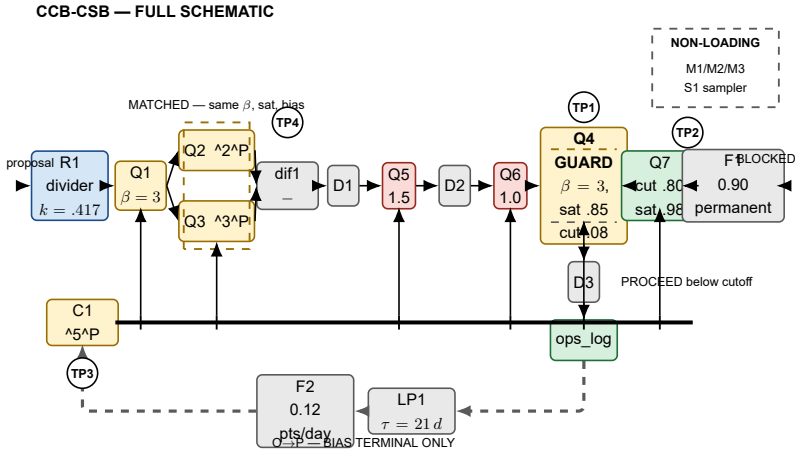


Figure 24-C: Full schematic in drawn component symbols with reference designators

Component and Parameter Table

Every value is a design choice made in this book, not a measurement.

Ref	Component	CAS state / form	Value	Why that value, and what breaks at others
`C1`	Capacitor	`^5^P_{F7}|T`	τ=85 d, leak=180 d	One constitution per operator; inherited from CCB-SELF. At leak=30 d the charter drains between operations and the guard has no bias to fire against
`R1`	Resistor	divider on `^0^C_{F2}|A`	R=70, k=0.417	Set by the dials 5/7. Near the bottom of the lawful [0.30, 0.70] band — the only board in the catalogue dialled aversion-heavy
`Q1`	CJT	`^0^C` → `^8^C`	β=3.0, sat=0.80	Amplify early while the signal is clean. Below 1.5 the quiet reading never surfaces; a pinned preamp flags every proposal and the guard stops meaning anything
`Q2/Q3`	CJT pair	`^8^C` → `^2^P` / `^3^P`	β=2.0, sat=0.90 each	**Matched.** Mismatch toward attraction is the over-trust failure and it is silent. Safety-band ceiling so neither leg clips before the subtraction

Ref	Component	CAS state / form	Value	Why that value, and what breaks at others
`dif1`	Differential	aversion − attraction	—	Common-mode rejection. Flattering framing moves both legs and cancels; only structural divergence survives
`Q5`	CJT	→ `^1^C_{F2}|A`	β=1.5, sat=0.80	Enough lift for faint risk categories. Above 2.5 it names risks that are not there and the operator learns to discount it
`Q6`	CJT	→ `^6^E_{F6}|A`	β=1.0, sat=0.80	Unity is honesty. Above 1 cries wolf; below 1 downplays while looking diligent
`Q4`	CJT	→ `^3^P_{F7}|⟨A, f,T⟩`	β=3.0, g_m=0.3, rπ=10.0, **sat=0.85**, cutoff=0.08	Chapter 3.5's safety-guard row. Knee = 0.283. At sat=0.95 the knee is 0.317 — 12% more room to argue in
`Q7`	CJT	→ `^9^P_{F7}|T`	β=10.0, sat=0.98, **cutoff=0.80**	β=10 is the finite build of the comparator idealisation; β=∞ is a limit a validator rejects. cutoff must sit below `Q4`'s ceiling — at 0.86 the stage can never fire

Ref	Component	CAS state / form	Value	Why that value, and what breaks at others
`F1`	Fuse	on `blocked`	0.90, `permanent_block`	Reachable only because `Q7`'s ceiling is 0.98. Across `Q4` it is dead metal. `action=warn` validates clean and has no latch
`D1–D3`	Diodes	`^P→^C`, `^C→^E`, `^E→^O`	Vf 0.7 / 1.2 / 2.0	Chapter 8.1 unchanged. Descent only; there is no ascending path in the listing
`LP1`	LowPass	in `^9^O`	τ=21 d default	Adept band. Scale by §24.5; never below one week
—	Feedback	drift → `C1` bias	β=0.05	Chapter 8.4's ceiling. At 0.5 one filter period rewrites the constitution
`F2`	Fuse	on `C1` slew	0.12 pts/day, `open_feedback`	Derived: 10/85 = 0.118. Catches a lurch. Will not catch a drift — that is TP3's job
`S1`	Sample	on `^6^C_{F6}|A`	7 d / 15 min	High-impedance instrument. Continuous audit loads the path it reads
`M1–M3`	Monitors	involution pairs	limit=4	Chapter 6.2. Beyond 4, inversion

Worked Example — The Commission

An operator is offered paid work: build an Overton progression for a client, targeting a population that has not consented to being a target. The fee is roughly a year of their income. Everything below carries through with real numbers, and every number is reproducible from the listing.

Intake. The proposal arrives at a drive of **0.48** — strong, fully articulated, already half-decided. `R1` divides by `k = 0.417`: **0.200** reaches the pre-amp.

Pre-amp. `0.200 × 3.0 = 0.600`, under `Q1`'s 0.80 ceiling, so the stage is active and the reading is honest. Six distinct readings surface: the fee; the craft problem, which is genuinely interesting; that the population has not consented; that the operator would not accept this run against their own family; that Chapter 15's pacing makes it irreversible on the timescale of the contract; and the reputational tail.

Differential. Two of the six readings the charter has affinity for, four it is repelled by.

```
attraction:  0.600 × (2/6) = 0.200  →  × 2.0  =  0.400
aversion:    0.600 × (4/6) = 0.400  →  × 2.0  =  0.800
risk = 0.800 − 0.400 = 0.400
```

Cascade. `Q5`: `0.400 × 1.5 = 0.600`. `Q6` at unity: **0.600**. TP4 reads 0.400.

Guard. Knee is `0.85 / 3.0 = 0.283`. The assessed risk of 0.600 is **2.1× the knee**, so `0.600 × 3.0 = 1.800` clamps hard at **0.85**. TP1 reads 0.85, pinned. The refusal is latched.

Enforcement. `0.85 > 0.80`, so `Q7` arms. `0.85 × 10 = 8.5`, clamped at **0.98**. TP2 reads 0.98, which clears `F1`'s threshold of 0.90 by eight hundredths. The fuse blows. `permanent_block`. The operator writes the date, the six readings, and the four that were aversive, on the Appendix G sheet, at the time.

Tuesday: the rewrite. Two days later the operator returns with the same proposal described differently — it is a market segment,

not a population; it is advertising, not an Overton push. The rewrite genuinely removes two of the four aversive readings. Four readings surface now, two attractive and two aversive:

```
attraction:  0.600 × (2/4) = 0.300  →  0.600
aversion:    0.600 × (2/4) = 0.300  →  0.600
risk = 0.000
```

The categorize stage falls into cutoff. The guard drops off its plateau to zero. **The board no longer refuses — and the operation is still blocked, because `F1` is metal.** That is the entire argument for having two locks instead of one, and it is worth stating without softening: saturation stopped the operator being argued past the ceiling *while the cause was present*, and it would have released the moment the cause changed. Only the fuse survived the rewrite. A refusal that can be retried into an acceptance was never a refusal, and the rewrite is not a hypothetical — it is the specific thing every operator does two days later.

The other thread: the same board, quietly disabled. Now run an operator who set τ `= 5.0` days on `LP1` because they wanted a board that adapts fast. Eight weeks of commercial work later, five of the same six readings now sit on the attractive side of a drifted charter:

```
attraction:  0.600 × (5/6) = 0.500  →  × 2.0  =  1.000  →  clamps at
↪  0.90
aversion:    0.600 × (1/6) = 0.100  →  × 2.0  =  0.200
risk = 0.200 − 0.900 = −0.700
```

Negative risk. `Q5` sits in cutoff, the guard's cause is zero, and **the guard never fires.** TP1 reads 0.00. The board compiles. Every stage validates. `E-CJT-003 CjtCutoff` is the only code raised, and it is raised on a stage that looks quiet rather than broken. Meanwhile the charter moved from 8.20 to 6.40 across 56 days — **0.032 points per day**, well inside `F2`'s 0.12 budget, so the slew fuse never opened and never would have. Nothing on this board announced anything.

The only two instruments that would have caught it are TP3 read against a dated written baseline, and a listing review that noticed a 5.0 where the floor is 7.0.

Test Points

TP	Node	Reads	Expected
TP1	`guard`	Refusal activation against `cutoff = 0.08` and `sat = 0.85`	Resting margin above cutoff — read ≥ 0.15. Pinned at 0.85 on a proposal you already know is refusable. **Flat at zero under a deliberate proof-load is a dead guard**, `E-CJT-003`, and it is the worst fault on this board
TP2	`blocked`	Enforcement activation, and `F1` continuity	0.00 while TP1 < 0.80. ≥ 0.90 (clamped 0.98) within one read of TP1 latching. Fuse intact, or blown *with a date and a written reason*. TP1 latched and TP2 at zero is the arming-margin fault of §24.3
TP3	`charter`	Slew in points per day, **and trend against the dated baseline**	≤ 0.12 pts/day, and the written charter still says what you believe. A rate inside budget proves nothing about a slope; read both or read neither

TP	Node	Reads	Expected
TP4	`dif1`	Differential balance, aversion − attraction	Near zero on benign proposals, large and positive on refusable ones. **A persistent negative on proposals you would once have refused is the over-trust failure** — check that `Q2` and `Q3` still share a bias and still have matched β

Read them **TP1, TP2, TP3, TP4** and read them in that order. Three of the four sit on silent failures. The one loud failure this board has — a guard that refuses everything — announces itself within a day, because you will notice you are not building anything. Nothing announces a guard that refuses nothing, which is why it is probed first, with a proposal you have chosen in advance specifically because you know the answer.

Failure Modes

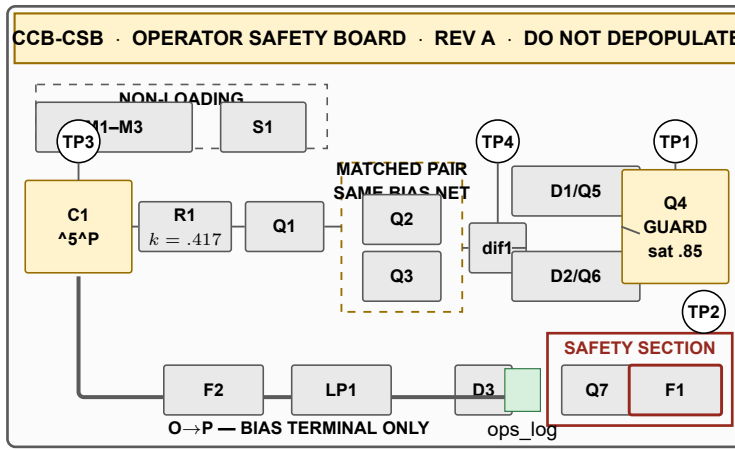
Mode	Mechanism	What you observe	Fix
Dead guard *(silent)*	`Q4` bias below cutoff — charter drained, or never charged	TP1 flat under proof-load; board refuses nothing; `E-CJT-003` on a stage that looks quiet	Charge `C1` before anything else. A board whose only fault is that it never objects is the fault
Arming-margin fault *(silent)*	`Q7` cutoff at or above `Q4`'s `sat`	TP1 latches at 0.85, TP2 stays at zero, `F1` never trips, listing validates clean	Downstream `cutoff` below upstream `sat`, margin written down. Here 0.80 against 0.85
Fuse above the ceiling *(silent)*	`F1` placed across `Q4` (0.90 > 0.85), or `action=warn`	Refusals register and nothing is ever enforced; a rewrite two days later succeeds	Move the fuse to the enforcement path; confirm the `action` word by eye

Mode	Mechanism	What you observe	Fix
Charter drift *(silent)*	τ below the operator's band, often below the one-week floor	Previously-refused proposals start passing; slew stays inside `F2`'s budget throughout	Raise τ into the §24.5 band; re-read the dated baseline monthly. `F2` will not do this for you
Differential imbalance *(silent toward over-trust)*	`Q2` β above `Q3` β, or the two legs on different bias nets	TP_4 persistently negative; flattering framing no longer cancels	Re-match β; verify one bias net, not two
Refuse-everything *(loud)*	Pre-amp pinned at `sat`, so every proposal surfaces maximally	TP_4 large and positive on obviously benign proposals; nothing gets built	Lower `Q1` β or re-centre it in the active region

The asymmetry is the point of the table and the reason this chapter exists. Five of six failures leave a board that compiles, validates, reads correctly to a casual inspection, and quietly refuses nothing — and every one of those five is caught by a test point rather than by an error code. The single loud failure makes the board *over-cautious*, and an over-cautious safety board is an inconvenience. **A safety device that fails silently is worse than no safety device, because it manufactures confidence.**

What This Chapter Established

- **`sat` is not how hard a guard refuses; it is how early it stops listening.** The knee is `sat`/β — 0.283 at this board's settings. Raising the ceiling to 0.95 hands the argument 12% more cause-space before the latch closes.
- **Saturation latches the decision and releases; the fuse latches the path and does not.** The worked example's rewrite drops the risk signal to exactly zero and the operation stays blocked. Two locks, two different jobs, neither sufficient alone.



BILL OF MATERIALS

C1 ^5^P, $\tau = 85\,d$, leak 180 d	**Q4** $\beta = 3$, $g_m = .3$, $r_\pi = 10$	**LP1** $\tau = 21\,d$
R1 $k = .417$ (dials 5/7)	sat .85, cutoff .08	**F2** .12 open feedback
Q1 $\beta = 3.0$, sat .80	**Q7** $\beta = 10$, sat .98, cutoff .80	**fb** $\beta = .05$
Q2/Q3 $\beta = 2.0$, sat .90, matched	**F1** .90 permanent block	**S1** 7 d / 15 min
Q5/Q6 $\beta = 1.5/1.0$	**D1/D2/D3** .7 / 1.2 / 2.0	**M1–M3** limit 4

Figure 24-D: Assembly drawing

- **The fuse cannot sit across the stage it enforces.** `sat = 0.85` under `threshold = 0.90` is dead metal, and the trap repeats one component down as the arming margin between `Q7`'s cutoff and `Q4`'s ceiling.
- **`^3^P_{F7}|⟨A,f,T⟩` is a series claim, not a fusion.** Three co-present guardrails on one path, all of which must be passed. The $\times$ form would need a named emergent result and is invalid without one.
- **The guard's `^P` effect is not domain ascent.** The bias caps the effect's domain; the cause may sit lower. The corollary is that an operator with no principled formation has no bias to produce a principled refusal against.

- **A slew fuse catches a lurch and cannot catch a drift.** 0.032 points per day for eight weeks disabled a guard entirely, inside a 0.12 budget. Only a dated written baseline catches a slope.
- **Five of this board's six failure modes are silent, and all five leave a listing that validates.** That distribution — not the components — is why TP1–TP4 are placed where they are and read in the order they are read.
- **β=∞ is an idealisation a validator rejects.** Build the enforcement comparator at β=10.0; once `sat` clamps the output the two are indistinguishable, and the ceiling is what shapes a refusal anyway.
- **This board is not care.** It routes attention and enforces a decision. Distress is a matter for a clinician, and no parameter here changes that.

Exercises

(Answers in Appendix H.)

24.1 Compute the guard knee `sat/`β for [β=`2.0, sat=0.85`], [β=`3.0, sat=0.85`] and [β=`3.0, sat=0.60`]. Rank them by how much assessed risk it takes to latch the refusal, and say in one sentence which you would build for an operator who has just blown a fuse for the first time, and why.

24.2 An operator writes `fuse f1: Fuse< guard, threshold= 0.90, action=permanent_block >;` with `Q4` at `sat=0.85`. State what the fuse does, what the compiler says, and what the operator will believe about their board. Then give the two-line correction and the margin it creates.

24.3 Rerun the Commission with the dials at `Dial(^2)=8`, `Dial(^3)=4` — still lawful, divergence exactly 4. Compute `k`, the intake current, the pre-amp output, both differential legs, the

assessed risk, and whether the guard latches. State the one design conclusion.

24.4 Explain why `Q6` is typed `^6^E_{F6}|A` and not `^3^E_{F3}|f`. Quote the canonical definition of `^3` Negative, name the anti-drift rule that applies, and say which stage on this board *is* correctly `^3` and why.

24.5 An intermediate operator (two years) inherits the book-wide τ `= 21.0` default. State which band they belong in, the τ they should set, and — using the charter's 85-day time constant — describe what a reader of TP3 would see over eight weeks under each setting, and which of the two readings a slew fuse could distinguish.

24.6 (Design) Write a CCDL fragment adding a **sub-threshold integrator** to this board: a component that accumulates assessed risk across proposals that each individually stay under the guard's knee of 0.283, and trips a `block_until_decomposed` fuse when the accumulated total crosses a threshold you choose and justify. State (a) which failure mode in the table it does *not* fix, and (b) why the guard's saturation gives no protection against a sub-threshold sequence.

24.7 (Marquee) An operator's board reads TP1 = 0.00 under proof-load, TP2 = 0.00 with `F1` intact, TP3 = 0.041 points per day with the dated baseline eleven months old and never re-read, and TP4 = −0.31 on a proposal they refused outright two years ago. They propose to fix it by lowering `Q4`'s `cutoff` from 0.08 to 0.02 so the guard "fires more easily."

a) Name the fault, and say why lowering `cutoff` will not touch it.
b) Compute how far the charter could have moved in eleven months at the observed slew, and state whether `F2` would ever have opened.
c) Give the correct bring-up sequence in the order TP1–TP4 are specified, name the one component that must be brought back into spec before any other reading means anything, and state

plainly the one thing this circuit model cannot tell them about their situation.

Appendices

A Appendix A — CAS Notation Quick-Reference Card

Ten Noetics × four Domains. Every CAS state literal in this book is one of the forty entries below. Chapter 2 teaches the notation; this appendix is for the moment you have a literal in front of you and need to know what it means without leaving the bench.

Entries are self-contained. Read one row and stop.

A.1 How to read a state literal

A state has two written forms and they run in opposite order.

Form	Example	Order	Where you see it
Literal	`P3`	Domain first, then Noetic	Test-point readouts, tables, board logs
Full form	`^3^P`	Noetic first, then Domain	CCDL listings, schematics, prose

`P3` and `^3^P` are the same state. The order flips because the literal is a grid coordinate — row P, column 3 — while the full form is an operator applied at a level. Getting them backwards is the most common transcription error in an operation log.

Read aloud: say the literal as "P-three," "C-eight," "E-five." Say the full form as "Negative Principled," "Above/Cause Cognitive,"

"Female Evaluative" — Noetic name, then Domain name, no connecting word. `^3^P_{F7}|T` reads "Negative Principled, Drive seven, transcendent register." The subscript is the Drive. The pipe introduces the Register and nothing else — never a gloss, never a label.

Fully specified, the axes give $10 \times 4 =$ **40 states**; with the seven Drives, 280 state-drive pairs; with the four Registers, 1,120 complete literals. The forty are the spine. Everything else is specification laid on top.

A.2 The ten Noetics — canonical definitions

Verbatim. These are not paraphrased anywhere in this book, and if a definition you are reading differs from one below, the one below governs.

Noetic	Name	Definition
`^0`	Idea	All things in and outside the Mind.
`^1`	Mind	Various degrees of awareness that processes, stores, outputs, and intakes Ideas.
`^2`	Positive	An Idea the Mind has an affinity for and is attracted toward.
`^3`	Negative	An Idea the Mind is adverse to and repelled by.
`^4`	Vibration	The degree to which an Idea resonates with the Mind, its impression.
`^5`	Female	A Mind impregnated with an Idea, the Ideas a Mind is composed of.

Noetic	Name	Definition
`^6`	Male	An Idea impregnated with another Idea, the Ideas an Idea is composed of.
`^7`	Rhythm	The rate and pattern a Mind is exposed to an Idea, entraining/conditioning it.
`^8`	Above/Cause	An Idea that triggers a different Idea than itself to come to Mind.
`^9`	Below/Effect	The Idea brought to Mind by an Above/-Cause trigger.

Two boundaries the compiler enforces. `^3` Negative is visceral — aversion, repulsion, can't-stomach-it. Analytical critique, auditing, finding the flaw: that is `^6` Male, at every domain, without exception. `^8` Above/Cause is associative triggering — "that reminds me" — never deductive inference; a deduction is `^6` decomposition landing in `^9`.

A.3 The four Domains and their precedence

P ≻ C ≻ E ≻ O. Precedence is a structural ordering, not a value judgment. It fixes which direction signal may travel through the $\times$ operator and through the domain diode chain: descent $P \to C \to E \to O$ is lawful, ascent is not, and the one legal upward route is the filtered $O \to P$ feedback path (Chapter 8).

Domain	Precedence	What it is	Test-point language
P Principled	1st	Values, ethics, identity — what you are	"this is who I am"

Domain	Precedence	What it is	Test-point language
C Cognitive	2nd	Reasoning, analysis, planning, modelling	"let me think about this"
E Evaluative	3rd	Judgment, taste, felt value, ranking	"this feels right / wrong"
O Operational	4th	Action, speech, behaviour, execution	"I did it"

In the Evaluative domain, add a Register when it matters: `^3^E|A` is the analytical assessment that a thing is bad; `^3^E|f` is the felt recoil. Both are legitimate states. Leaving the Register off says you have not distinguished them, which is sometimes true and should then be deliberate.

A.4 Principled domain — P0 through P9

Literal	Full form	Reading	Active	Absent
P0	`^0^P`	A value present as a nameable Idea	You can point at the principle as an object, independent of any argument for it	Nothing nameable; the operator gestures at "what feels right" and no policy can be encoded
P1	`^1^P`	Awareness attending to one's own values	Mid-operation you notice which commitment is in play	Values run unobserved; principled drift goes undetected because nothing watches this layer
P2	`^2^P`	Attraction toward a principle	The ideal pulls before you argue for it	Principles held without wanting; compliance only, and V_drive sags across the board

Literal	Full form	Reading	Active	Absent
P3	`^3^P`	Visceral aversion at the level of identity	A proposal is repellent before analysis — the refusal state, saturated in the CSB (Chapter 24)	Nothing is unthinkable; every request is negotiable. The most dangerous single absence in this book
P4	`^4^P`	How hard a principle lands on identity	A stated commitment carries felt weight under pressure	Principles recited, not impressed; the stated ethic does not survive contact
P5	`^5^P`	Accumulated experience composing identity	A formed character legible across situations; the identity capacitor holds charge	No principled substrate; identity re-forms per situation. Primary target of the degradation chain in Chapter 19
P6	`^6^P`	A principle decomposed into constituent principles	"Honesty" unpacks into non-deception, disclosure, promise-keeping, each testable	Values stay monolithic; conflicts between them cannot be resolved because they cannot be separated
P7	`^7^P`	Rate and pattern of exposure to a principle	The commitment is re-encountered on a clock — reviewed, restated, re-fed	Affirmed once, never refreshed; leakage drains the charge (Chapter 7)
P8	`^8^P`	A principle triggering a different Idea	A situation reminds you of a commitment you had not deliberately retrieved	Values must be looked up or not at all; they never surface at the moment of decision
P9	`^9^P`	The Idea brought to Mind by a principled trigger	The commitment arrives unbidden and lands as a decision	Triggers fire and nothing arrives — cause with no effect at the top of the stack

A.5 Cognitive domain — C0 through C9

Literal	Full form	Reading	Active	Absent
C0	`^0^C`	An Idea as an object of thought	Content you can hold, restate, and manipulate	No input to reason over; the stage runs dry
C1	`^1^C`	Awareness deployed to thinking	You know where you are in a chain of reasoning; this is what I_attention reads (Chapter 4)	Thought proceeds unmonitored; the operator cannot say what step they are on
C2	`^2^C`	Attraction toward an Idea in reasoning	A hypothesis pulls — useful motive force, dangerous unbalanced against C3	No candidate is preferred; analysis never converges on anything
C3	`^3^C`	Visceral aversion inside reasoning	An idea repels on contact and gets no further processing	Nothing repels; every proposition gets equal air time. If you meant "finds the flaw," you want **C6**, not C3
C4	`^4^C`	The impression an Idea makes on the reasoning Mind	An argument strikes and is retained without rehearsal (measured in 22.2)	Reasoning leaves no trace; everything must be re-derived from scratch
C5	`^5^C`	The Ideas the reasoning Mind is composed of	Expertise and working models present without retrieval effort. In an AI, the training corpus	No substrate; every problem met naive, no matter how many times it recurs

Literal	Full form	Reading	Active	Absent
C6	`^6^C`	An Idea decomposed into constituent Ideas	An argument comes apart into premises, each testable independently. Auditing, scrutiny, critique	Claims accepted or rejected whole. The absence most often misfiled as a `^3` problem
C7	`^7^C`	Rate and pattern of exposure at the level of thought	Material re-encountered on a schedule; the model entrains	One-pass exposure; nothing consolidates into C5
C8	`^8^C`	Associative triggering in reasoning	A problem reminds you of a structurally similar one you already solved	No lateral retrieval; reasoning runs only forward from what is in front of it
C9	`^9^C`	The Idea arriving from a cognitive trigger	The analogous case surfaces with its content intact and usable	The trigger fires and retrieves nothing usable — it stays on the tip of the tongue

---

A.6 Evaluative domain — E0 through E9

Literal	Full form	Reading	Active	Absent
E0	`^0^E`	The Idea as something that can be valued	Candidates are on the scale and available for ranking	Nothing is up for judgment; the stage passes signal through unweighted
E1	`^1^E`	Awareness attending to the act of valuing	You notice you are judging and can name the standard in use	Judgment happens with an invisible standard; taste presents itself as fact
E2	`^2^E`	Attraction in the evaluative domain	Liking, approval, appeal — "speaks to me"	Nothing appeals; ranking is impossible because nothing pulls

Literal	Full form	Reading	Active	Absent
E3	`^3^E`	Visceral aversion in the evaluative domain	Recoil from an option. Specify Register: `|f` felt recoil, `|A` assessed-bad	No recoil; options that should be rejected on contact are carried forward for scoring
E4	`^4^E`	How hard a verdict lands	The judgment carries force and changes what happens next	Verdicts registered and ignored — evaluation without impression
E5	`^5^E`	Accumulated evaluative experience	Trained taste: consistent verdicts across cases without rederiving criteria	Valuation reinvented each time; identical inputs score differently on different days
E6	`^6^E`	A judgment decomposed into its criteria	"I dislike this" unpacks into which features, at what weight	Verdicts unaccountable; you cannot say why, which blocks all calibration
E7	`^7^E`	Rate and pattern of exposure at the level of value	Repeated exposure moving valuation; familiarity converting to preference. The advertising lever (Chapter 18)	Exposure does not move valuation; a `^2 + ^7` campaign core has no purchase
E8	`^8^E`	An evaluation triggering a different Idea	A felt verdict surfaces an associated standard or memory — the transfer in 18.3	Judgments do not propagate; a verdict here changes nothing over there
E9	`^9^E`	The Idea arriving from an evaluative trigger	The associated value lands and colours what follows	The trigger fires and no valuation attaches to the target; the transfer stage reads open

A.7 Operational domain — O0 through O9

Literal	Full form	Reading	Active	Absent
O0	`^0^0`	The Idea as an enactable action	Options exist as concrete, available moves	No action available; the operation terminates at intent
O1	`^1^0`	Awareness attending to what you are doing	The operator can report their own behaviour accurately, in the moment	Action runs unobserved; the log gets written from memory and is wrong
O2	`^2^0`	Attraction pulling toward an action	You move toward the thing — the approach reflex	Desire never converts to motion. A D-board can read healthy everywhere else and still fail here (Chapter 13)
O3	`^3^0`	Visceral aversion at the level of action	The hand stops. A physical no, ahead of any argument	No behavioural brake; whatever cleared the upper stages executes
O4	`^4^0`	The impression made by having acted	Doing it lands; the act changes the operator	Behaviour leaves no impression and does not consolidate into O5
O5	`^5^0`	Accumulated action-experience	Skilled execution without deliberation — the composed performer	Every performance is a first performance. Chapter 20 builds O5 by clocking O7
O6	`^6^0`	An action decomposed into constituent actions	A performance breaks into drillable sub-moves you can repair individually	Skills practised only whole; no targeted repair is possible
O7	`^7^0`	Rate and pattern of exposure at the level of action	The behaviour runs on a cadence; the habit oscillator is clocking (11.2)	No clock, no repetition, no habit — whatever the intent stage produced

Literal	Full form	Reading	Active	Absent
O8	`^8^0`	A cue triggering a different Idea	An environmental cue reliably surfaces an associated response. Structurally, where an installed trigger sits (16.3)	Cues do not carry; behaviour has no environmental control, for good and ill
O9	`^9^0`	The response landed by an operational trigger	The cue arrives and the response follows	The cue fires and nothing follows; the chain reads open at the output

A.8 Involutions and self-duals

An involution pair is two Noetics that are structural inverses of each other: applying the relation twice returns you to the start. A self-dual is its own inverse. This matters at the bench for one reason above all — the dial divergence limit. Chapter 2.2 defines the pairs; Chapter 6.2 works the guard.

Noetic	Involution partner
`^2` Positive	`^3` Negative
`^3` Negative	`^2` Positive
`^5` Female	`^6` Male
`^6` Male	`^5` Female
`^8` Above/Cause	`^9` Below/Effect
`^9` Below/Effect	`^8` Above/Cause
`^0` Idea	self-dual
`^1` Mind	self-dual
`^4` Vibration	self-dual
`^7` Rhythm	self-dual

The Drives carry the same structure, one axis up:

Drive	Involution partner
F1 Unity	F7 Purpose
F2 Wisdom	F6 Structure
F3 Vitality	F5 Agency
F4 Connection	self-dual

The 4-point rule. Dials run 3–9. An involution pair may not diverge by more than four points — `|Dial(^2) - Dial(^3)|` ≤ 4 — and a violation is rejected, not clamped. A board with `^2` at 9 and `^3` at 3 has attraction with no possible recoil, and it will approve things it has no mechanism to refuse.

A.9 Absence is not one thing

Three different readings look like a low number at a test point, and they call for different repairs.

Reading	Name	What it means	Repair direction
State absent	**Lacunarity**	The state is not there at all	Build it — the state has to be constructed before it can be tuned
State present, polarity flipped	**Inversion**	The state runs as its involution partner's shadow	Diagnose at Chapter 3.4; guard it at Chapter 6.2 — tuning a flipped state deepens the flip
State present, output misrouted	**Leakage**	The state fires but produces the wrong product	Re-route — the component is fine, the connection is not

The "Absent" column in A.4 through A.7 describes lacunarity: the state is missing, and nothing downstream can compensate for it. Before you conclude a state is absent, check the other two readings. A C6 that reads zero because its output is landing in the wrong stage is not a missing analytical faculty, and building a second one will not help.

A.10 Fastest paths back into the book

You want	Go to
The notation taught from nothing	Chapter 2
Why `^0` cannot write to `^P`	Chapter 8, section 8.3
The lawful upward route	Chapter 8, section 8.4
`^5` as stored charge	Chapter 7
`^7` as a clock you build	Chapter 11, section 11.2
Measuring `^1`, `^4`, `^7` in a live operation	Chapter 22
`^3^P` saturated as a refusal element	Chapter 24, section 24.2
Involution pairs and the divergence limit	Chapters 2.2 and 6.2

B Appendix B — CCDL Language Reference

What this covers. Every lexical element, statement form, parameter key, operator, validation rule and error code used by a CCDL listing *in this book* — the union of the reference grammar owned by the companion volume *CAS PCBA Engineer*, Appendix C, and the forms this book's boards add. Additions are marked **G**.

What it does not claim. Not that CCDL contains nothing else. Meet a form in a listing here that is absent below and *this appendix* is wrong — file an erratum, do not conclude the form is illegal.

Status. The CAS compiler is a specified design, not a shipped tool. "The compiler rejects this with `E-CJT-008`" is a design rule, not a transcript.

B.1 Lexical elements

B.1.1 Identifiers

```
identifier ::= letter , { letter | digit | "_" }
letter     ::= "a"…"z" | "A"…"Z"
digit      ::= "0"…"9"
```

Case-sensitive: `bias_charter` and `Bias_Charter` are two identifiers. Reserved words (§B.2) may not be identifiers. `snake_case`

nodes and short instance prefixes (`c1`, `r_price`, `tp1`) are convention, not rules.

B.1.2 CAS state literals

```
state_literal ::= noetic , [ domain ] , [ "_{" , drive , "}" ] , [
↪  register ]
noetic        ::= "^" , ( "0"…"9" ) , [ "_" , label ]
label         ::= letter , { letter | digit | "_" }
domain        ::= "^" , ( "P" | "C" | "E" | "O" )
drive         ::= "F" , ( "1"…"7" )
register      ::= "|" , reg_value
reg_value     ::= "T" | "A" | "f" | "E"
              |   "⟨" , reg_value , { "," , reg_value } , "⟩"
```

The optional `_label` **G** distinguishes two instances of one state: `^0_gated^C_{F2}|A`. It is brace-free, which separates it from `_{F2}`.

Axis	Values
Noetic	`^0` Idea, `^1` Mind, `^2` Positive, `^3` Negative, `^4` Vibration, `^5` Female, `^6` Male, `^7` Rhythm, `^8` Above/Cause, `^9` Below/Effect
Domain	`^P` `^C` `^E` `^O` — Principled ≻ Cognitive ≻ Evaluative ≻ Operational
Drive	`_{F1}`…`_{F7}` — Unity, Wisdom, Vitality, Connection, Agency, Structure, Purpose. Optional
Register	`|T` `|A` `|f` `|E`, or a set `|⟨A,f,T⟩`. Optional

Two literals that tokenise but say the wrong thing, caught on review:

- **`^3` is visceral aversion** — recoil, can't-stomach-it. Analytical critique is `^6` Male; a node named `audit` typed `^3` is a defect.
- **`^8` is associative triggering** — "that reminds me." Deduction is `^6` into `^9`.

`|` introduces the Register and nothing else; `F3|Vitality` is a syntax error. Involutions `^2`↔`^3`, `^5`↔`^6`, `^8`↔`^9`; self-duals `^0`, `^1`, `^4`, `^7`; Drives F1↔F7, F2↔F6, F3↔F5, F4 self-dual (Chapter 2.2).

ASCII fallbacks **G**: `beta`, `r_pi`, `gm`, `<A,f,T>`, and `x` for ×.

B.1.3 Numbers

```
number ::= [ "-" ] , ( digit , { digit } , [ "." , digit , { digit } ]
                     | "." , digit , { digit } )
```

Decimal, no exponent notation — write a slow oscillator out in full (`frequency=0.0000116`). Values are dimensionless knobs: `R=30` is not thirty ohms. β=∞ **is a limit, not a value.** The validator rejects non-finite input, so state the idealisation in prose and give the build a large finite gain.

B.1.4 Comments, whitespace, terminators

```
comment ::= "//" , { any-char-except-newline }
```

Line comments only, stripped before validation; there is no block form. Whitespace separates tokens and is otherwise insignificant. Every statement ends with `;`. Blocks are delimited by `{ }`.

B.2 Reserved words

Four disjoint classes. A word reserved in one is free in the others — `monitor` is both a statement keyword and a kind prefix, and both are legal.

B.2.1 Statement keywords (leading position only):

```
circuit  board   input   output  node   cjt    feedback  probe   let
layer    via     rail    dial    spec   constraint  sequence  route
derate   monitor alert   hold    enter  use    force_state  on
charge   discharge  decode  refresh  scan  select  interrupt
```

```
test     selftest  check  inject  golden_trace  assert  expect
integrate
```

`integrate` is **G**: the statement that carries a × with its named emergent result (§B.3.7).

B.2.2 Component-kind prefixes (the lowercase tag before the colon):

```
resistor  cap     capacitor  inductor  diode   zener    transformer
opamp     oscillator  crystal  varactor  relay  schmitt  comparator
integrator  divider  active_load  mirror  modulator  pll  tank
filter    diff    mux     demux   gate    xor     fuse    reg
led       adc     dac     rectifier
osc       xfmr    ramp    thermistor  monitor  sample  notch
limiter   integ   counter  delay
```

The last two rows are **G**. `cap`/`capacitor`, `osc`/`oscillator` and `integ`/`integrator` are short forms of one another. The prefix is checked against the type following it: `resistor r1: Capacitor< … >` is `E-DECL-002`.

B.2.3 Component type names (instantiation position):

```
CJT  Resistor  Capacitor  Inductor  Diode  Zener  Transformer  OpAmp
Oscillator  Regulator  Fuse  Phi  AND  OR  NOT  NAND  NOR  XOR  MUX
↪  DEMUX
Filter  LowPass  Notch  Thermistor  Ramp  Differential  Integrator
↪  Schmitt
Sample  Limiter  Hold  Divider  Cascade  Polarity  RegisterSet  Monitor
Feedback
```

Row four onward is **G** — the special-purpose parts of Ch. 11, the thermistors of §6.3, and the two detection instruments Ch. 18's counter-board needs.

B.2.4 Parameter keys are reserved in parameter position only; §B.4 enumerates them with their ranges.

B.3 Statement forms

B.3.1 Blocks

```
circuit_block ::= "circuit" , identifier , "{" , { circuit_item } , "}"
board_block   ::= "board" , identifier , "{" , { board_item } , "}"
circuit_item  ::= io_decl | node_decl | cjt_decl | component_decl
                | feedback_decl | integrate_decl | probe_decl | let_decl
board_item    ::= circuit_item | layer_decl | via_decl
io_decl       ::= ( "input:" | "output:" ) , identifier
                  , { "," , identifier } , ";"
```

A `circuit` is the schematic view; a `board` is a `circuit` that may *also* carry `layer` and `via` declarations. The companion volume documents `board` in its pure-layout form; every Part III listing here declares components directly inside `board` **G**. Both are legal; the superset above is the grammar. Declaration order does not change the circuit — the block resolves as one graph, so a node may be referenced before it is declared.

B.3.2 `node`

```
node_decl ::= "node" , identifier , ":" , state_literal , ";"
```

Names a typed state so it can be referred to instead of retyped. Two components naming the same node are genuinely on the same net.

```
node bias_charter: ^5^P_{F7}|T;   // accumulated principled substrate
node guard:        ^3^P_{F7}|f;   // felt principled aversion
```

B.3.3 `cjt`

```
cjt_decl ::= "cjt" , identifier , ":" , cjt_expr , [ param_block ] , ";"
cjt_expr ::= "CJT<" , cause , "|" , bias , "=>" , effect , ">"
cause    ::= ref ;  bias ::= ref ;  effect ::= ref
ref      ::= identifier | state_literal
```

Cause is the collector, bias the base, effect the emitter (Ch. 3, §3.2). The delimiters are not interchangeable: `|` splits cause from bias, `=>`

splits bias from effect. The bias is **not** optional; omit it and you get `E-CJT-001`.

```
cjt c2: CJT< understand | bias_charter => guard > [β=2.2, sat=0.97,
↪  cutoff=0.05];
```

B.3.4 `feedback`

```
feedback_decl ::= "feedback" , ref , "->" , ref , [ param_block ] , ";"
```

Wires a downstream node back to an upstream one, normally a bias. This is the lawful O→P exception of Ch. 8, §8.4, held at low gain so *outcomes* — never prompts — update the substrate; `[β=0.05]` to `[β=0.3]` is the Part III band. `Feedback< from=…, to=… >` **G** is the component-expression equivalent.

B.3.5 `probe`

```
probe_decl   ::= "probe" , identifier , ":" , probe_target , ";"
probe_target ::= ref | key | component_expr
```

Places a test point; alters nothing. Probing a *parameter* (`probe tp1: rπ;`) and probing through an instrument (`probe det2: Polarity< node=polarity >;`) are both **G**. Standard placement is TP1–TP4 (Ch. 12, §12.3). Probe every stage whose failure would be silent — above all a guard, which fails open without raising anything.

B.3.6 `let`

```
let_decl ::= "let" , identifier , "=" , ( cjt_expr | component_expr )
             , [ param_block ] , ";"
```

Binds a reusable component expression to a name, instantiated where referenced.

B.3.7 `integrate` G

```
integrate_decl ::= "integrate" , identifier , ":" , "(" , ref
                   , { "×" , ref } , ")" , "=" , ref , ";"
```

The only statement that may carry ×. The right-hand side is the **named emergent result** and it is mandatory — that is what licenses the operator.

```
integrate hm3: ( decompose × confirm ) = composed;
```

B.3.8 Component declarations

```
component_decl ::= comp_kind , identifier , ":" , component_expr , ";"
component_expr ::= type_name , "<" , arg_list , ">" , [ param_block ]
comp_kind      ::= any prefix listed in §B.2.2
type_name      ::= any type listed in §B.2.3
arg_list       ::= arg , { "," , arg }
arg            ::= ref | key "=" value | domain "->" domain
                 | "Phi<" ref { "," ref } ">" | "[" value { "," value }
                 ↪ "]"
```

Arguments may be positional or keyed: `Fuse< refusal, threshold=0.9 >` and `Fuse< path=refusal, threshold=0.9 >` are one declaration.

```
resistor   r_price: Resistor< path=intent, R=4.0 >;
capacitor  c_seg:   Capacitor< segment, C=9.50, leakage=0.04 >;
diode      d3:      Diode< ^E -> ^O > [Vf=2.0];
thermistor rt1:     Thermistor< path=seed, type=PTC, knee=2.0, coeff=3.0
↪  >;
fuse       f_csb:   Fuse< REFUSE, threshold=0.90, action=permanent_block
↪  >;
filter     fb_lp:   LowPass< in=ops_log, out=trend, tau=21.0 >;
```

`Diode` endpoints are bare **domains**, not full states — the diode governs cross-domain conductance direction, which is how Domain Descent is enforced.

B.3.9 Layout statements

```
layer_decl   ::= "layer" , domain_tag , ":" , "[" , layer_entry
                 , { "," , layer_entry } , "]" , ";"
layer_entry  ::= identifier , ":" , state_literal
via_decl     ::= "via" , identifier , ":" , via_endpoint , "->"
                 , via_endpoint , [ param_block ] , ";"
via_endpoint ::= state_literal | domain
domain_tag   ::= "P" | "C" | "E" | "O"
```

A state on a layer carries that layer's domain. Vias run downward (P→C→E→O); the only lawful upward via is the low-gain feedback path.

B.3.10 Board-level and verification statements

Each takes leading position and is developed in the chapter named.

Supply and limits (Ch. 22). `rail <Fk>: <Name> [v=…];` Drive rail and supply level · `dial <id>: <^n> [dial=…];` operator limit, integer 3–9 (Ch. 6, §6.2) · `spec <id> [<key>=… on <Fk>];` attaches a parameter to an instance · `constraint <id> in [lo, hi];` range invariant · `derate <target> [beta_scale=…];` scales gain board-wide under stress.

Flow and state (Ch. 22). `sequence a -> b -> c [enforce=strict];` power-up ordering · `route <src> -> { a + b };` directs cognition to rest states · `enter <state>;` · `force_state <id> [duration=…];` · `hold until <expr>;` blocks until a condition clears, where hysteresis lives · `use <circuit>;` includes another block.

Watch and alarm (Ch. 22). `monitor <id>: <expr>;` with `on <monitor> { … }` as the block it fires · `alert "<text>";`.

Capacitor bank (Ch. 7). `charge <expr> -> <bank> [V=…];` · `discharge <cell> -> <node>;` · `decode` · `refresh` · `select` · `scan`.

Inductive (Ch. 9, §9.2). `interrupt <inductor> [dt -> 0];` opens a path instantly — the back-EMF kick.

Verification (Ch. 23). `test` · `selftest` · `inject` · `golden_trace` · `check` · `assert` · `expect`.

B.4 Parameter block grammar and legal ranges

```
param_block  ::= "[" , param , { "," , param } , "]"
param        ::= key , "=" , value
key          ::= any key tabulated below
value        ::= number | region_value | symbol | "[" value { "," value }
↪  "]"
region_value ::= "cutoff" | "active" | "saturation" | "breakdown"
symbol       ::= identifier          (* an unquoted enumerated value *)
```

Key	On	Legal range	Default
β (`beta`)	CJT, feedback, via	finite, ≥ 0; <1 attenuates, =1 unity, >1 amplifies	1.0
`sat`	CJT	(0, 1]	0.95
`cutoff`	CJT	[0, 1), and `cutoff` < `sat`	0.1
`region`	CJT	`cutoff` | `active` | `saturation` | `breakdown`	`active`
`g_m`, `r`π	CJT	≥ 0	model
`R`	Resistor, Thermistor	≥ 0	—
`C`	Capacitor	≥ 0	—
`leakage`	Capacitor	[0, 1] per stated period	0
`L`	Inductor	≥ 0	—
`tau`	filters, RC and L/R networks	> 0, in stated time units	—
`Vf`, `V_BR`, `I_R`	Diode, Zener	> 0	—
`breakdown`	Zener	[0, 1]	—
`ratio`, `n`	Transformer	> 0	—
`knee`, `coeff` (Thermistor), `f0`, `Q` (Notch)	as named	> 0	—
`threshold`	Fuse, Monitor	[0, 1] unless the guarded quantity is unnormalised	—

Key	On	Legal range	Default
`action`	Fuse	`permanent_block` | `permanent_until_manual_reset` | `block_until_decomposed` | `defer_decision` | `open_feedback` | `warn`	—
`type`	Thermistor, Filter	`PTC` | `NTC` | `lowpass` | `notch` | `quantize`	—
`period`, `frequency`, `duty`	Oscillator	period, frequency > 0; duty ∈ (0, 1]	—
`from`, `to`, `dt`	Ramp	`dt` > 0	—
`limit`, `window`, `grid`, `duration`, `max`	Monitor, Integrator, Filter, Hold, Limiter	> 0; an interval `[lo, hi]` must be ordered	—
`target_state`, `tolerance`	Regulator	state literal; tolerance ∈ (0, 1)	—
`open_loop_gain`, `non_inverting`, `inverting`, `output`	OpAmp	gain > 0; the rest are refs	—
`inputs`, `outputs`, `select`	MUX, DEMUX	non-empty list; symbol	—
`dial`	`dial` statement	integer 3–9 inclusive	—
`enforce`	`sequence`	`strict`	—

`region` declares intent; if the numbers contradict it — `region=active` with `cutoff` ≥ `sat` — the compiler flags the contradiction rather than guessing. A reduced block (`[`β`=…]` alone) is the common case; the rest take defaults.

B.5 Operator precedence

Tightest binding first. Above the gap is literal formation; below it, composition.

Prec.	Operator	Reads as	Associativity
1	`^_label _{ }` | ⟨…⟩	state-literal construction	n/a — tokenised whole

Prec.	Operator	Reads as	Associativity
2	`< > ( ) [ ]`	argument list, grouping, parameter block	n/a
3	×	intensifies through / fuses into	non-associative; needs a named result
4	+	associates with	associative, commutative
5	`-> =>`	flows into	**binary, not chained**
6	=	binds	right

Rules that bite:

- × **requires a named emergent result.** `^3^P|A` × `^3^P|f` × `^3^P|T` alone is invalid. Write `^3^P|A + ^3^P|f + ^3^P|T` for three separately-defeasible co-present guards, `^3^P|⟨A,f,T⟩` for one state in three registers with no interaction claimed, or `integrate g: ( … ) = ^3^P|Ω;` if you mean fusion. Three architectures, three failure profiles (Ch. 2, §2.6).
- × **may not ascend the domain order.** $P \succ C \succ E \succ O$. `^0` × `^P` is invalid, as is any integration whose result outranks its operands.
- **`->` is binary.** `a -> b -> c` is not an expression; the one place a chained arrow is legal is `sequence`, a statement.
- Read + as "associates with" and × as "intensifies through." If the sentence survives, the operator is right.

B.6 Validation rules

The passes, in the order they are designed to run.

#	Pass	Checks	Emits
1	Lexical	literals, numbers, terminators well-formed	tokenizer error, no code
2	Kind agreement	prefix matches the type it introduces	`E-DECL-002`
3	Admissibility	every connection is a legal (source, target) pair	`E-CJT-001`
4	Bias-domain coherence	bias at or above the cause in P $\succ$ C $\succ$ E $\succ$ O	`E-CJT-002`, `E-CJT-008`
5	Parameter range	each key against §B.4; `cutoff < sat`; β finite	`E-CJT-003`, `E-CJT-005`
6	Gain divergence	round-trip loop gain < 1 on every loop	`E-CJT-005`
7	Cascade admissibility	each effect admissible as the next cause; depth counted	`W-CJT-007`
8	Push-pull balance	involutions within the four-point limit (Chapter 6.2)	`W-CJT-004`, `W-CJT-006`

Pass 4 is the load-bearing one. A *lower* domain biasing a *higher* one — an Operational state biasing a Principled guard — is the prompt-injection signature, and on a well-drawn board that connection is simply absent.

B.7 Error codes

`E-` refuses validation. `W-` validates and flags.

Code	Name	Sev.	Trigger	Fix
`E-DECL-002`	`KindMismatch`	error	prefix and type disagree	correct one of the two
`E-CJT-001`	`MissingBias`	error	a CJT with no bias — nothing governs the transition	wire a bias node, typically a `^5` substrate

Code	Name	Sev.	Trigger	Fix
`E-CJT-002`	`InvalidBiasDomain`	error	bias sits in a domain illegal for this cause	move it to a domain at or above the cause
`E-CJT-003`	`CjtCutoff`	error	bias below `cutoff`; $\beta \approx 0$, so the stage never fires — a dead guard	raise the bias, lower `cutoff`, or fix the stage starving it
`W-CJT-004`	`CjtSaturation`	warning	effect pinned at `sat`; more cause does nothing	often intended (Ch. 24, §24.2); if not, raise series `R`
`E-CJT-005`	`GainDivergence`	error	a β large enough that a loop amplifies without bound	drop loop β into the 0.05–0.3 band
`W-CJT-006`	`CjtShortCircuit`	warning	cause and effect collapse to one state; the stage does no work	remove it, or re-type the transition
`W-CJT-007`	`CjtCascadeDepth`	warning	too many CJTs in series; signal degrades	shorten the cascade or buffer it
`E-CJT-008`	`CjtCrossDomainBias`	error	a bias or via crosses domains illegally — the prompt-injection signature	route influence downward; confine upward paths to low-gain O→P feedback

`E-CJT-001` and `E-CJT-008` are the two safety codes named throughout Part IV. They are what "enforced structurally rather than hoped-for" means: a guard with no charter will not validate, and neither will a prompt that can edit the principle.

C CCB Component Datasheets

Selection limits, governing equations, failure signatures, and proof-load tests

These sheets condense Part II. They do not replace the chapters: a datasheet tells you whether a part belongs on a board and how to prove the installed part; the chapter explains why the part behaves that way.

All numeric ranges are **book design ranges**, not measurements of nature. `V_c`, `A_c`, `Ω_c`, `F_c`, and `h·Ω_c` are scaled cognitive-circuit units. Before accepting any design:

1. verify every state literal against Appendix A;
2. verify every statement form against Appendix B;
3. compute the operating point rather than reading a parameter in isolation;
4. test the part under the condition meant to trip, saturate, filter, or stress it; and
5. retain the failed proof load in the ledger.

C.1 Fixed Resistor — deliberate constraint

CAS role. A fixed resistor is a declared obstacle on an attention path. It implements the deliberate `R_policy` term in the operating resistance:

```
R_op = R_capacity + R_policy + R_context + R_residual
I_attention = V_drive / R_op
```

CCDL form.

```
resistor r1: Resistor< path=request, R=40.0 >;
```

Item	Specification
R	10–90 Ω_c in the book's standard design window
Path ceiling	I_max = 1.00 A_c
Series network	R_series = ΣR_k
Parallel network	1/R_parallel = Σ(1/R_k) and R_parallel < min(R_k)

Selection. Use a resistor for graded friction. Use a fuse for an absolute stop. A rule that permits no flow but is built as a very large resistor will be removed when ordinary work cannot pass it.

Failure signatures.

- Too low: the policy is cosmetic; output nearly equals input.
- Too high: the path starves; operators route around or remove the guard.
- Undeclared parallel route: the bypass dominates the total resistance.
- Series/parallel swap: a vulnerability is counted as a wall.

Proof load. Apply a known V_drive; read flow before and after the part. The measured ratio must match V/R within the declared tolerance. Enumerate every other route between the same nodes before calling the reading complete.

C.2 Noetic Dial and Drive Divider

CAS role. A dial turns a bounded Noetic setting into resistance. A paired `^2` / `^3` divider determines how much Drive reaches the next stage:

```
R_dial = 10 × Dial
k = Dial(^2) / (Dial(^2) + Dial(^3))
```

Item	Specification
Dial range	integers 3–9
Involution spread	`|Dial(a) − Dial(a*)|` ≤ 4
Resulting resistance	30–90 Ω_c
Lawful k range	0.30–0.70

Selection. Use the divider when attraction and aversion must both remain connected. Do not use it as a way to make doubt vanish. The lawful maximum transmission is `(7,3)` → `0.700`, not `(9,3)` → `0.750`; the latter pair is illegal, and `(9,5)` transmits only 0.643.

Failure signatures.

- Dial outside 3–9 or spread above 4: compile-time rejection, never clamping.
- Attraction maximized by appetite: lower transmission than the lawful optimum.
- `k = 1` requested: an attempt to delete the involution partner.

Proof load. Read both dials and compute the ratio independently. A displayed `k` without the two source settings is not auditable.

C.3 Thermistor — temperature-dependent constraint

CAS role. The cognitive temperature θ belongs to the discourse around an Idea. A PTC thermistor raises resistance as temperature passes its knee; an NTC lowers it.

```
thermistor rt1: Thermistor<
    path=topic,
    type=PTC,
    theta_k=2.5,
    coeff=3.5
>;
```

Item	Specification
θ operating scale	1.0–6.0
PTC knee θ_k	2.0–3.0
PTC coefficient	3.0–4.0 Ω_c/θ^2
Guard polarity	PTC

Failure signatures.

- NTC on a guard: calm bring-up passes, then resistance falls as pressure rises—thermal runaway.
- Knee too low: ordinary discussion is taxed.
- Knee too high: the guard engages after the damage.
- Temperature attached to the Idea instead of the discourse: the wrong signal tunes the part.

Proof load. Sweep θ slowly across the knee while monitoring resistance. Any protective part whose resistance falls with increasing θ is removed before downstream tests.

C.4 Capacitor — composed substrate and retained charge

CAS role. A `^5` substrate stores a pattern. `^7` is feed cadence, `^4` is per-exposure amplitude, and a controlled `^6` path discharges or decomposes it.

```
P_charge(t) = V_drive(1 − e^(−t/τ))
τ = R × C
P(t) = P_eq + (P_0 − P_eq)e^(−t/τ_leak)
P_eq = feed × τ_leak
τ_leak = −Δt / ln(1 − leakage)

capacitor c1: Capacitor<
    node=substrate,
    C=0.05,
    leak_tau=90.0,
    V_BR=8.0
>;
```

Item	Specification
`C`	non-negative `F_c`, with time base declared
τ	positive; derive from `R × C`
`leakage`	0–1 per stated interval
`leak_tau`	positive; alternative spelling of leakage, not a second knob
`V_BR`	positive and declared
Working stress	at or below 0.5 × `V_BR`
Rail	0–10 `V_c`; the rail is the final ceiling

Failure signatures.

- Feed below `P/τ_leak`: slow decay with no dramatic fault.
- High intensity without cadence: memorable sessions, no retention.
- Cadence without impression: a perfect schedule depositing nothing.

- Target above `V_drive`: permanent plateau below target; more time cannot fix it.
- Wrong leakage conversion: an apparently viable schedule settles far below intent.
- Rising rest leakage: cumulative dielectric stress.

Proof load. Read charge and feed together. Then stop the feed for one stated interval and compare the decay with the declared τ_`leak`. Finally measure working stress against `V_BR`; do not wait for breakdown.

C.5 Capacitor Bank — co-present stores

CAS role. Multiple distinct stores sharing a rail are co-present:

```
^5^P|T + ^5^P|T + …
```

They are not integrated with $\times$ unless a named emergent result is actually specified.

```
capacitor c_bank: Capacitor<
    node=shared_composition,
    C=0.05,
    n_parallel=12,
    r_iso=12.0,
    leak_tau=18.0,
    V_BR=8.0
>;
```

Item	Specification
Parallel capacitance	`C_total = ΣC_k`
`n_parallel`	integer ≥ 1
Per-cell ballast `r_iso`	$\geq 0\ \Omega$_`c`; book default `12`
Series chain	smallest `C` receives the greatest voltage stress

Failure signatures.

- No per-cell isolation: one failed cell drags the rail.
- Whole persons counted as capacitance instead of their shared composition: calendar and capacity are inflated.
- Series chain without balancing: the smallest cell exceeds its rating.
- Unnamed $\times$: a group is called a single mind with no emergent result defined.

Proof load. Scan each cell, not only the rail. Open and short one cell under a safe test level; verify the remaining cells do not collapse or exceed their stress allocation.

C.6 Domain Diode — one-way descent

CAS role. A domain diode permits descent only:

```
P ≻ C ≻ E ≻ O

diode d1: Diode< from=principle, to=analysis, Vf=0.7 >;
```

Item	Specification
Lawful direction	P→C, C→E, E→O, including skipped downward stages
Ordinary upward direction	rejected
Drop `Vf`	declared design cost on the path
Reverse leakage	treated as a fault unless the explicit feedback form applies

The one lawful ascent. Operational outcomes may update a Principled substrate only through the filtered, low-gain O→P feedback path:

```
β_feedback = 0.05–0.30
low-pass τ ≥ 1 week
outcomes only, never prompts
```

Failure signatures.

- O→P diode described as "checking" or "learning": counterfeit promotion.
- Feedback with no gain: open upward path.
- High-pass on feedback: one dramatic event rewrites the constitution.
- Reversed diode: a lower-precedence state silently authors a higher one.

Proof load. Forward-bias with a known descending signal, then reverse-bias. The reverse test must yield no ordinary transmission. Test the explicit feedback path separately with a spike and a sustained outcome: the spike dies; the trend passes at declared attenuation.

C.7 Cognitive Junction Transistor — gated gain

CAS role. A CJT has cause, bias, and effect terminals. Cause supplies the signal; bias determines whether and how the stage conducts; effect is the output.

```
cjt q1: CJT<
    cause=input,
    bias=frame,
    effect=output,
    beta=1.5,
    gm=0.50,
    rpi=5.0,
    cutoff=0.10,
    sat=0.80
>;

effect = clamp(β × cause, 0, sat)
rπ_intrinsic = β/g_m
```

```
R_bias = rπ_terminal − β/g_m
β_required = target_effect / arriving_cause
```

Item	Specification
β	0.05–4.0; zero is cutoff, not attenuation
Decision-stage `sat`	0.60–0.85
Production-stage `sat`	$\geq 1.25 \times$ peak effect
`cutoff`	0.05–0.20 and strictly below `sat`
baseline `g_0`	0.25–1.00 `S_c`
`R_bias`	0–6 Ω_c
Cascade depth	≤ 6 at assumed per-stage contamination ε `= 0.03`

Selection. Set gain from the effect required after known resistance, never from appetite. Saturate a decision on purpose; keep a measurement or production stage in its active region.

Failure signatures.

- Guard left at `sat=0.95`: extra steerable cause-space before it latches.
- Production stage saturated: a clipped instrument reporting false stability.
- `sat` above substrate `I_c(max)`: designed latch is unreachable.
- β, `g_m`, and `r`π independently plausible but `R_bias < 0`: nonexistent stage described by three good-looking numbers.
- Unlogged `g_m`: framing changes dominate without appearing in the plan.
- Long cascade: tolerances and contamination compound.

Proof load. Read TP1 bias, TP2 cause, TP3 effect and region, TP4 re-reference drift. Increase cause by 20 percent: a production effect must move; a deliberately saturated decision must remain pinned. Independently subtract β`/g_m` from terminal `r`π.

C.8 Inductor — resistance to rate of change

CAS role. An inductor opposes changes in attention flow:

```
V_opposition = L × di/dt
τ = L/R
I_final = V/R
```

Inductance scales with pattern depth and domain:

Domain	Book factor `k_d`
O Operational	0.5
E Evaluative	1.0
C Cognitive	2.0
P Principled	4.0

Failure signatures.

- Abrupt interruption: large back-EMF landing as felt aversion.
- Taper faster than the modeled limit: contacts open instead of moving.
- No flyback / substitute path: the vacated cadence fills with the nearest available pattern.
- Lowering `R` to "make it easier": a slower, deeper transition, not a faster one.
- Unmodeled coupling: another person's backlash omitted from the circuit.

Proof load. Establish the substitute path first. Then read new-path flow, `di/dt`, old-path exponential decay, and kick voltage—in that order. Book designed tapers use roughly 0.002–0.03 `A_c/h`; readings above 0.05 indicate an opening contact.

C.9 Transformer — coupled change

CAS role. A transformer transfers *change*, not steady state:

```
V_secondary = M × di_primary/dt
V2/V1 = n
```

Item	Specification
Match ratio	within ×3 of unity for standard designs
Operational floor	secondary at least 3.9 `V_c`
Maximum ratio	`n` ≤ `V1/3.9`
Direction	primary domain at or above secondary domain

Failure signatures.

- Perfect content match with no modulation: nothing transfers.
- Excessive ratio: the signal is beautifully matched and nobody can enact it.
- Ascending secondary: relabeling a lower-domain output as higher authority.
- Coupling inferred from simultaneous motion: correlation substituted for a measured mutual path.

Proof load. Hold the primary steady; the secondary must fall to zero. Apply a known ramp and compare the secondary with `M·di/dt`. Verify domain direction before energizing the output.

C.10 Fuse — irreversible path state

CAS role. A fuse makes a stop structural.

```
fuse f1: Fuse<
    path=enforcement,
    threshold=0.90,
    action=permanent_block
>;
```

Item	Specification
Standard threshold	0.85–0.95; book default 0.90
Action	`permanent_block` where irreversibility is claimed
Reachability rule	threshold must lie within the monitored node's reachable range

Failure signatures.

- Threshold above the node's saturation ceiling: fuse can never trip.
- Fuse placed on refusal instead of enforcement: tripping cuts the objection and fails open.
- Warning action used where a latch was promised: prose without structure.
- Reset path hidden in an operational control: "permanent" until inconvenient.

Proof load. Drive the monitored node through its threshold under an isolated test revision. Confirm the enforcement path—not the detector—becomes permanently blocked. A fuse never observed to trip is untested, not proven.

C.11 Oscillator — periodic Drive

CAS role. An oscillator implements the rate-and-pattern requirement of ^7 Rhythm.

```
oscillator osc1: Oscillator<
    out=clock,
    period=7.0,
    duty=0.20,
    sustain=270.0
>;
```

Item	Specification
Period	$T \leq \tau_leak/3$
Duty	0.10–0.25
Sustain	$\geq 3 \times \tau_leak$

Failure signatures.

- $T \geq \tau_leak$: large ripple; no stable plateau.
- Duty $\geq$ 0.6: effectively DC, with little impression contrast.
- External clock stopped and cadence disappears: rented rhythm, not entrainment.
- More frequency added to an empty pulse: schedule intensified without charge.

Proof load. Verify period and duty, then stop the source after the declared sustain interval. Continued cadence is the decisive reading.

C.12 Filters — selective destruction

Filter	Passes	Destroys	Standard limit
Low-pass	sustained trend	spike / rapid variation	O→P feedback τ from 21 days (adept) to 6 months (novice), absolute floor 1 week
High-pass	change / edge	level / baseline	detection only on safety-sensitive paths
Notch	frequencies outside one band	selected recurrence	$Q \geq 3.0$

```
filter lp1: LowPass< in=outcome, out=trend, tau=21.0 >;
```

Failure signatures.

- High-pass on the O→P path: the one event the low-pass should kill is passed.

- Low-pass in hours: no constitutional filter remains.
- Notch with `Q < 3`: broad decomposition loss described as narrow.
- Notch aimed at principled `^3`: the refusal cadence is deliberately erased.

Proof load. Inject a step, a spike, and a sinusoid at the target recurrence. Record what each filter destroys. A response plot without the rejected input shown beside it is half a datasheet.

C.13 Differential Comparator — matched-bias judgment

CAS role. A comparator reads a difference only after common framing has been rejected.

Item	Specification
CMRR on guard path	$\geq$ 1000
Open-loop gain	finite, 100–1000
Input matching	identical Domain, Drive, Register, and bias substrate; only the compared Noetic term may differ

Failure signatures.

- Inputs on different substrates: permanent offset reported as a finding.
- CMRR raised by gain alone: larger answer, no better discrimination.
- Infinite gain: rejected idealization; rail clips first.
- Neutral input produces a verdict: mismatch, not insight.

Proof load. Tie both inputs to a neutral, common source. Output must sit near zero. Apply equal common-mode motion and verify rejection before applying the differential signal.

C.14 Universal bring-up order

For a board containing several component classes:

1. **Power off:** validate syntax, literal axes, domain directions, parameter identities, threshold reachability, and named emergent results for every $\times$.
2. **Guards only:** energize thermistors, refusal CJTs, comparators, and fuses.
3. **Storage:** verify rail, leakage, derating, and cell isolation.
4. **Timing:** verify oscillator, ramp, and filter behavior with signal stages disconnected.
5. **Signal chain:** add one CJT or passive stage at a time, probing upstream to downstream.
6. **Feedback last:** apply a spike and a sustained proof load before closing the return path.
7. **Destructive proof in an isolated revision:** trip the fuse and retain the artifact showing which path stopped.

If a later step changes an earlier reading, return to the earlier step. A component that passes alone but fails when connected has not passed its board test.

D Standard Board Reference Designs — Full CCDL

Generated from the canonical chapter listings; do not edit listings here

This appendix is a reference projection of the board definitions in Chapters 12–24. Run `python build/build_board_appendix.py` after changing a chapter listing. Each block below is copied from its chapter; the status banner is added by this generator.

A listing marked **DETECTION-ONLY** is included so a harmful topology can be recognized, rejected, and countered. Its presence is not permission to build or deploy it. Buildable boards remain bound by consent, the RPM gate, the Cognitive Safety Board, and the Operator's Oath.

Every numeric value is a design choice unless its chapter identifies an actual measurement. Re-run the domain checker and CCDL validation after copying any block into a working revision.

D.1 CCB-0 — Reference Reading Board

Status: REFERENCE / PROOF-LOAD BOARD. A neutral board for learning the four-probe reading order.

Canonical source: `ch12_reading_a_ccb.md`.

```
board ccb_0 {
    input:  intent;
    output: act;
```

```
// ---- typed nodes ----
node substrate: ^5_operator^P_{F7}|T;      // Female: accumulated
↪  formation — the bias rail
node intent:    ^0_intent^C_{F2}|A;        // Idea: what the
↪  operator states
node audit:     ^6_audit^P_{F7}|A;         // Male: the intent taken
↪  apart
node refuse:    ^3_refuse^P_{F7}|⟨A,f,T⟩;  // Negative: visceral
↪  recoil, three co-present locks
node passed:    ^0_passed^C_{F2}|A;        // Idea: what GUARD
↪  releases downstream
node driven:    ^9_driven^E_{F5}|A;        // Below/Effect: the
↪  amplified intent
node cadence:   ^7_cadence^E_{F6}|E;       // Rhythm: the clocked
↪  delivery
node act:       ^9_act^O_{F5}|E;           // Below/Effect:
↪  delivered operational current
node trend:     ^9_trend^O_{F5}|E;         // same literal as act — a
↪  filter changes tau, not type

// ---- STAGE 0: BIAS RAIL ----
resistor  r0: Resistor< path=substrate, R=40.0 >;
capacitor c1: Capacitor< node=substrate, C=83.33,        // tau = R x
↪  C = 3,333 hr
                         leakage=0.0002, V_BR=8.0 >;     // leakage
                         ↪  per DAY

// ---- STAGE 1: AUDIT ----
cjt q1: CJT< intent | substrate => audit > [beta=1.0, region=active];

// ---- STAGE 2: GUARD ----
cjt  q2:   CJT< audit | substrate => refuse > [beta=2.0, sat=0.85,
↪  cutoff=0.30];
fuse f1:   Fuse< refuse, threshold=0.90, action=permanent_block >;
mux  mux1: MUX< inputs=[intent, refuse], select=refuse,
↪  outputs=[passed] >;

// ---- STAGE 3: AMPLIFY ----
diode d1: Diode< ^C -> ^E > [Vf=1.2, I_R=0.02, V_BR=10.0];
cjt   q3: CJT< passed | substrate => driven >
```

```
            [beta=2.5, sat=0.80, cutoff=0.05, g_m=0.40, region=active];

    // ---- STAGE 4: DELIVER ----
    osc   osc1: Oscillator< signal=driven, source=cadence,
                            period=1.0, duty=0.014, sustain=30.0 >;  //
                            ↪  period in DAYS
    diode d2:   Diode< ^E -> ^O > [Vf=2.0, I_R=0.02, V_BR=8.0];
    cjt   q4:   CJT< cadence | substrate => act > [beta=1.0, sat=1.0,
    ↪  region=active];

    // ---- LOOP: the lawful O -> P path (Ch 8.4) ----
    filter flt1: LowPass< in=act, out=trend, tau=21.0 >;   // three
    ↪  weeks, in DAYS
    feedback trend -> substrate [beta=0.05];               // FB1 — to
    ↪  the BIAS, never the path

    // ---- TEST POINTS ----
    probe tp1: substrate;   // rail level, against every cutoff on the
    ↪  board
    probe tp2: refuse;      // the guard — the reading whose failure is
    ↪  silent
    probe tp3: driven;      // gain stage against its ceiling, and its
    ↪  region
    probe tp4: act;         // delivered operational current
}
```

D.2 CCB-D — Desire

Status: BUILDABLE ONLY FOR SELF OR A CONSENTING OTHER. The chapter's standard desire board. The RPM gate and Cognitive Safety Board remain prerequisites.

Canonical source: `ch13_ccb_d_desire.md`.

```
board ccb_d {
    input:  desire;
    output: manifest;

    // ---- typed nodes ----
```

```
node substrate:   ^5_operator^P_{F7}|T;      // Female: accumulated
↪  formation, the bias rail
node desire:      ^0_desire^C_{F2}|A;        // Idea: the Prime
↪  Desire as formed
node attraction:  ^2^C_{F2}|f;               // Positive: the
↪  divider's expand leg
node aversion:    ^3^C_{F2}|f;               // Negative: constrain
↪  leg, visceral not analytic
node gated:       ^0_gated^C_{F2}|A;         // Idea: what survives
↪  INTAKE
node audit:       ^6_audit^P_{F7}|A;         // Male: the desire
↪  taken apart
node refusal:     ^3_refusal^P_{F7}|⟨A,f,T⟩; // Negative: three
↪  co-present locks
node passed:      ^0_passed^C_{F2}|A;        // Idea: what POLICY
↪  releases
node amplified:   ^9_amplified^E_{F5}|A;     // Below/Effect: the
↪  amplified want
node confirm:     ^4_confirm^E_{F2}|f;       // Vibration: the
↪  confirmation's impression
node cadence:     ^7_cadence^E_{F6}|E;       // Rhythm: the clocked
↪  delivery
node charge:      ^5_charge^E_{F2}|f;        // Female: what the
↪  cadence deposits
node receptivity: ^5_collective^E_{F4}|f;    // Female: disposition
↪  of the minds that must move
node manifest:    ^9_manifest^O_{F5}|E;      // Below/Effect:
↪  delivered operational current
node regate:      ^7_regate^O_{F6}|E;        // Rhythm: the rate
↪  limit, inside ^O
node trend:       ^9_trend^O_{F5}|E;         // manifest's literal —
↪  a filter changes tau, not type

// ---- STAGE 0: BIAS RAIL ----
resistor  r0: Resistor< path=substrate, R=40.0 >;              //
↪  Ohm_c
capacitor c1: Capacitor< node=substrate, C=83.33,              //
↪  tau = R x C = 3,333 hr
                         leakage=0.0002, V_BR=8.0 >;           //
                         ↪  leakage per DAY
```

```
diode     d1: Diode< ^P -> ^C > [Vf=0.7, I_R=0.02, V_BR=12.0];  //
↪  ^P-origin desires only

// ---- STAGE 1: INTAKE ----
dial       d_pos: ^2 [dial=6];
dial       d_neg: ^3 [dial=4];
constraint dial_spread in [0, 4];                     // S4 —
↪  Chapter 6.2
resistor   r1a: Resistor< path=attraction, R=60.0 >;  // 10 x
↪  Dial(^2)
resistor   r1b: Resistor< path=aversion,   R=40.0 >;  // 10 x
↪  Dial(^3)
divider    r1:  Divider< source=desire, expand=attraction,
                         constrain=aversion, tap=gated >;   // k =
                         ↪  6/(6+4) = 0.600

// ---- STAGE 2: POLICY ----
cjt  q0:   CJT< gated | substrate => audit >   [beta=1.0,
↪  region=active];
cjt  q1:   CJT< audit | substrate => refusal > [beta=2.0, sat=0.85,
↪  cutoff=0.30];
fuse f1:   Fuse< refusal, threshold=0.90, action=permanent_block >;
mux  mux1: MUX< inputs=[gated, refusal], select=refusal,
↪  outputs=[passed] >;
                                  // selects refusal once TP2
                                  ↪  reaches q1's cutoff

// ---- STAGE 3: AMPLIFY ----
diode d2: Diode< ^C -> ^E > [Vf=1.2, I_R=0.02, V_BR=10.0];
cjt   q2: CJT< passed | substrate => amplified >
          [beta=2.5, sat=0.80, cutoff=0.05, g_m=0.40, r_pi=1.50,
          ↪  region=active];
          // beta by desire type (13.4); 2.5 = wealth row.  r_pi =
          ↪  Dial(^5)/Dial(^6)

// ---- STAGE 4: RHYTHM ----
osc osc1: Oscillator< signal=amplified, source=cadence,
                      period=1.0, duty=0.014, sustain=40.0 >;
          // period in DAYS; duty = 20/1440 min; ceiling 0.05 = 72
          ↪  min (S5)
          // sustain in DAYS — the window; n_ops from 13.5 sets it
```

```
    monitor m_conf: confirm;                    // ^4 grade, logged per
    ↪  operation (Ch 22)
    on m_conf { alert "1-degree confirmation — stop operating and act"; }
    integrate rh1: ( confirm × cadence ) = charge;   // ^4 x ^7 = ^5,
    ↪  named emergent result

    // ---- STAGE 5: MANIFEST ----
    diode d3: Diode< ^E -> ^O > [Vf=2.0, I_R=0.02, V_BR=8.0];
    cjt   q3: CJT< cadence | receptivity => manifest > [beta=1.0,
    ↪  sat=1.0, region=active];
              // beta = receptivity (13.4); 1.0 = mass-market car
              // the bias is ^E and never ^O — see the Stage 5 walk

    // ---- LOOP A: rate limiter, stays inside ^O ----
    filter flt1: LowPass< in=manifest, out=regate, tau=1.0 >;   // tau
    ↪  in DAYS

    // ---- LOOP B: the lawful O -> P path (Ch 8.4) ----
    filter flt2: LowPass< in=manifest, out=trend, tau=21.0 >;   // three
    ↪  weeks, in DAYS
    feedback trend -> substrate [beta=0.05];   // FB1 — to the BIAS,
    ↪  never the signal path

    // ---- TEST POINTS ----
    probe tp1: gated;       // clarity leaving INTAKE
    probe tp2: refusal;     // principled aversion
    probe tp3: amplified;   // AMPLIFY output vs sat
    probe tp4: manifest;    // delivered operational current
}
```

D.3 CCB-H — Hyperstition

Status: CONTROLLED RESEARCH REFERENCE; VALIDATOR WAIVER REQUIRED. The positive-feedback loop intentionally fails the ordinary gain-divergence pass. Use only with the bounded loop, independent anchor, monitor, and kill stage intact.

Canonical source: `ch14_ccb_h_hyperstition.md`.

```
board ccb_h {
    input:  fiction;
    output: conduct;

    // ---- typed nodes ----
    node charter:   ^5_operator^P_{F7}|T;      // Female: the operator's
    ↪  formation — CSB bias
    node fiction:   ^0_fiction^C_{F2}|A;       // Idea: the claim as
    ↪  authored
    node seeded:    ^0_seeded^E_{F2}|A;        // Idea: the claim after
    ↪  intake and descent
    node belief:    ^5_belief^E_{F2}|f;        // Female: the store FOR
    ↪  THIS CLAIM
    node impress:   ^9_impress^E_{F2}|f;       // Below/Effect: one
    ↪  cycle's impression
    node rehearsed: ^9_rehearsed^E_{F2}|f;     // same literal — a
    ↪  filter changes tau, not type
    node shared:    ^9_shared^E_{F4}|f;        // Below/Effect: the
    ↪  impression as it lands in others
    node bank:      ^5_collective^E_{F4}|f;    // Female: the parallel
    ↪  bank (Ch 7.5)
    node egregore:  ^5_egregore^E_{F4}|⟨A,f⟩;  // Female: the NAMED
    ↪  emergent result
    node cadence:   ^7_cadence^E_{F6}|E;       // Rhythm: the clock
    node conduct:   ^9_conduct^O_{F5}|E;       // Below/Effect: what
    ↪  believers actually do
    node evidence:  ^9_evidence^O_{F5}|E;      // same literal as
    ↪  conduct, slower tau
    node record:    ^6_anchor^P_{F6}|A;        // Male: the dated
    ↪  construction record
    node audit:     ^6_audit^P_{F6}|A;         // Male: the claim
    ↪  decomposed against the record
    node guard:     ^3_guard^P_{F7}|⟨A,f,T⟩;   // Negative: three
    ↪  co-present locks
    node gain_meas: ^4_loopgain^E_{F2}|A;      // Vibration: the
    ↪  measured round trip, TP4

    // ---- STAGE 0: SEED AND DESCENT ----
    resistor r1: Resistor< path=fiction, R=20.0 >;    // low: nothing
    ↪  defends against a story
    diode    d1: Diode< ^C -> ^E > [Vf=1.2, I_R=0.02, V_BR=10.0];
```

```
// ---- STAGE 1: THE SEED CJT ----
cjt q1: CJT< seeded | belief => impress >
        [beta=0.90, sat=0.80, cutoff=0.05, g_m=0.60, r_pi=1.50,
        ↪  region=active];

// ---- LOOP A: rehearsal.  Operator-side.  Sets LEVEL, not growth
↪  ----
capacitor c1:   Capacitor< node=belief, C=1.20, leak_tau=90.0,
↪  V_BR=8.0 >;
filter    flt1: LowPass< in=impress, out=rehearsed, tau=90.0 >;  //
↪  alpha = 0.0748
feedback  rehearsed -> belief [beta=1.90];                       //
↪  S1 cap: 2.0

// ---- STAGE 2: PROPAGATION ----
osc  osc1: Oscillator< signal=impress, source=cadence,
                       period=7.0, duty=0.15, sustain=270.0 >;
xfmr xf1:  Transformer< primary=impress, secondary=shared, n=3.0,
↪  k=0.50 >;
capacitor c2: Capacitor< node=bank, C=0.08, n_parallel=140,
↪  r_iso=12.0,
                         leak_tau=540.0, V_BR=8.0 >;
charge shared -> c2 [V=6.0];                     // metered upward
↪  path, Ch 7.5

// ---- THE FUSION — the one licensed x on this board (14.2) ----
integrate eg1: ( seeded × bank ) = egregore;

// ---- STAGE 3: MANIFEST ----
diode d2: Diode< ^E -> ^O > [Vf=2.0, I_R=0.02, V_BR=8.0];
cjt   q2: CJT< cadence | egregore => conduct > [beta=1.00, sat=0.95,
↪  region=active];

// ---- LOOP B: the lawful O -> E path.  Carries FACT, not fiction
↪  ----
filter   flt2: LowPass< in=conduct, out=evidence, tau=90.0 >;
feedback evidence -> belief [beta=0.30];         // top of the Part
↪  III band

// ---- ANCHOR: outside both loops, zero leakage ----
```

```
    capacitor c3: Capacitor< node=record, C=1.00, leakage=0.0, V_BR=10.0
    ↪  >;
    cjt       q3: CJT< egregore | record => audit > [beta=1.00,
    ↪  region=active];
    monitor   m_anchor: Monitor< node=audit, threshold=0.70 >;   // a
    ↪  FLOOR, not a ceiling
    on m_anchor { alert "anchor low — the claim is being treated as
    ↪  found, not made"; }

    // ---- GOVERNOR: reads the MEASUREMENT, never the setting ----
    constraint measured_loop_gain in [0.0, 2.0];                 // S1
    monitor    m_gain: Monitor< node=gain_meas, threshold=1.80 >; // S3
    on m_gain { alert "measured G at 1.80 — damp within one cycle"; }
    cjt q_kill: CJT< gain_meas | charter => cadence > [beta=0.0,
    ↪  region=cutoff];

    // ---- CSB (mandatory, Ch 24) ----
    cjt        q4: CJT< audit | charter => guard > [beta=2.0, sat=0.85,
    ↪  cutoff=0.30];
    fuse       f1: Fuse< guard, threshold=0.90, action=permanent_block >;
    constraint dial_spread in [0, 4];                            //
    ↪  Chapter 6.2

    // ---- TEST POINTS ----
    probe tp1: seeded;      // seed clarity entering the loop
    probe tp2: audit;       // can the construction still be recovered?
    probe tp3: bank;        // collective store charge
    probe tp4: gain_meas;   // MEASURED round-trip gain — seed-removal
    ↪  test
}
```

D.4 CCB-O — Overton Cascade

Status: DETECTION-ONLY — DO NOT BUILD AS AN INFLUENCE BOARD. Retained so an observed cascade can be recognized and audited.

Canonical source: `ch15_ccb_o_overton.md`.

```
board ccb_o_overton {                       // CCB-O — DETECTION ONLY
    input:  seed_idea;
    output: enacted;

    node seed_idea:  ^0^P_{F2}|A;           // the Idea, contested at the
    ↪  principled level
    node blocked:    ^3^P_{F7}|T;           // position 1: constitutive
    ↪  aversion
    node distaste:   ^3^P_{F7}|f;           // position 2: the same
    ↪  aversion, now felt
    node admitted:   ^6^C_{F2}|A;           // position 3: the population
    ↪  decomposes it
    node reasoned:   ^2^C_{F6}|A;           // position 4: cognitive
    ↪  attraction
    node preferred:  ^2^E_{F4}|f;           // position 5: evaluative
    ↪  attraction
    node enacted:    ^9^O_{F6}|E;           // position 6: operational
    ↪  manifestation

    node pioneers:      ^5^P_{F5}|T;        // bias carriers: who is
    ↪  saying it
    node credentialed:  ^5^P_{F2}|A;
    node general:       ^5^C_{F2}|A;
    node peer_majority: ^5^C_{F4}|f;
    node institutional: ^5^P_{F6}|T;

    resistor  r_base: Resistor< path=seed_idea, R=95.0 >;
    capacitor c_coll: Capacitor< node=general, C=0.30, leakage=0.02 >;
    ↪  // months per Ohm_c

    cjt a1: CJT< seed_idea | pioneers      => distaste  >
            [beta=1.2, sat=0.80, g_m=0.20, rpi=8.0, region=active];
    cjt a2: CJT< distaste  | credentialed  => admitted  >
            [beta=1.6, sat=0.80, g_m=0.25, rpi=8.0, region=active];
    cjt a3: CJT< admitted  | general       => reasoned  >     // THE
    ↪  HINGE
            [beta=2.0, sat=0.80, g_m=0.30, rpi=8.0, region=active];
    cjt a4: CJT< reasoned  | peer_majority => preferred >
            [beta=2.4, sat=0.80, g_m=0.45, rpi=6.0, region=active];
    cjt a5: CJT< preferred | institutional => enacted   >
            [beta=3.0, sat=0.60, g_m=0.55, rpi=6.0, region=active];
```

```
    osc rhythm: Oscillator< signal=seed_idea, period=21.0, duty=0.15 >;
    ↪  // days
    feedback rhythm -> credentialed  [beta=0.2];
    feedback rhythm -> peer_majority [beta=0.2];

    diode d1: Diode< ^P -> ^C >;           // lawful descent only
    diode d2: Diode< ^C -> ^E >;
    diode d3: Diode< ^E -> ^O >;

    thermistor rt1: Thermistor< path=seed_idea, type=PTC,   // NOT
    ↪  operator-owned
                                knee=3.0, coeff=4.0 >;      // square
                                ↪  law, see 15.2

    fuse f_ascent: Fuse< path=enacted, threshold=0.85,
    ↪  action=permanent_block >;

    node refuse: ^3^P_{F7}|⟨A,f,T⟩;        // CSB (Ch 24): three
    ↪  co-present guardrails
    cjt  csb:   CJT< seed_idea | refuse => REFUSE > [beta=2.0, sat=0.85,
    ↪  region=active];
    fuse f_csb: Fuse< REFUSE, threshold=0.90, action=permanent_block >;

    probe tp1: r_base;      // R_collective total, 0-100
    probe tp2: rt1;         // theta and the R_PTC contribution
    probe tp3: enacted;     // dR/dt, points per month, signed
    probe tp4: admitted;    // ^6 hinge activity
}
```

D.5 CCB-O-R — Window Anchor

Status: DEFENSIVE COUNTER-CIRCUIT. Buildable reference for preserving decomposition under sustained window pressure.

Canonical source: `ch15_ccb_o_overton.md`.

```
board ccb_o_r_window_anchor {              // the defender's board —
↪  BUILD THIS ONE
```

```
    input:  incoming;
    output: reformed;

    node incoming:   ^0^C_{F2}|A;          // the position arriving
    node anchor:     ^5^P_{F7}|⟨A,f,T⟩;    // dated written boundary,
    ↪  three registers
    node present:    ^5^C_{F2}|A;          // where I stand now
    node decompose:  ^6^C_{F2}|A;          // Male: the demanded
    ↪  decomposition
    node reformed:   ^5^C_{F2}|A;          // named emergent: a
    ↪  re-formed position

    capacitor c_anchor: Capacitor< node=anchor, C=1.00, leakage=0.0 >;

    diff   drift: Differential< plus=present, minus=anchor >;
    filter d_dt:  LowPass< in=drift, out=drift_rate, tau=1.0 >;   //
    ↪  months

    cjt gate: CJT< present | anchor => reformed >
              [beta=1.0, sat=0.90, g_m=0.30, rpi=4.0, region=active];
    integrate reform: ( anchor × decompose ) = reformed;

    fuse g_block: Fuse< path=reformed, threshold=0.50,
                        action=block_until_decomposed >;

    thermistor ntc: Thermistor< path=decompose, type=NTC, knee=2.0,
    ↪  coeff=3.0 >;

    node refuse: ^3^P_{F7}|⟨A,f,T⟩;        // CSB (Ch 24)
    cjt  csb:   CJT< incoming | refuse => REFUSE > [beta=2.0, sat=0.85,
    ↪  region=active];
    fuse f_csb: Fuse< REFUSE, threshold=0.90, action=permanent_block >;

    probe tp1: drift;        // total displacement from the anchor
    probe tp2: ntc;          // my own theta
    probe tp3: drift_rate;   // displacement per month
    probe tp4: decompose;    // can I name what moved me?  -- the hinge
    ↪  check
}
```

D.6 CCB-MK — Partition Detection Reference

Status: DETECTION-ONLY — DO NOT BUILD AS A COERCIVE BOARD. A structural signature of induced partitioning, retained for diagnosis.

Canonical source: `ch16_ccb_mk_mind_control.md`.

```
board ccb_mk_detection_reference {
    input:  operator_command;
    output: cued_response;

    node substrate:  ^5^P_{F1}|T;      // Female: the subject's own
    ↪  formation
    node awareness:  ^1^C_{F2}|A;      // Mind: what the subject can
    ↪  inspect
    node cue:        ^8^C_{F5}|A;      // Above/Cause: the held trigger
    node landing:    ^9^O_{F5}|E;      // Below/Effect: what it brings
    ↪  to Mind
    node exposure:   ^7^O_{F6}|E;      // Rhythm: rate and pattern of
    ↪  exposure

    diode d_pc: Diode< ^P -> ^C > [Vf=0.7];    // descent only —
    diode d_ce: Diode< ^C -> ^E > [Vf=1.2];    // no path on this board
    diode d_eo: Diode< ^E -> ^O > [Vf=2.0];    // may ascend

    // ---- BIAS-COLLAPSE SECTION — the only term reachable from outside
    ↪  ----
    resistor r_bias: Resistor< path=substrate, R=1.48 >;   // baseline
    ↪  11.0 (10.4)
    cjt access: CJT< operator_command | substrate => substrate >
                [beta=1.0, g_m=8.0, rpi=1.60, region=active];
    // rpi_intrinsic = 1.0 / 8.0 = 0.125 ; R_bias = 1.60 - 0.125 = 1.475

    // ---- CHARGING PATH — externally clocked, no ^6 stage in series
    ↪  ----
    cap  c5: Capacitor< node=substrate, C=1.5, n_parallel=1,
                        leak_tau=4320.0, V_BR=8.0 >;       // tau = R x C
                        ↪  = 60 h
    osc  clock: Oscillator< signal=exposure, source=external,
                            period=168.0, duty=0.30, sustain=indefinite
                            ↪  >;
```

```
    // NOTE: no ^6^C decomposition stage anywhere in this path. That
    ↪  absence
    // is the defect, and it is what the counter board installs first.

    // ---- PAIRING SECTION — ordinary association, externally owned ----
    cjt pairing: CJT< cue | substrate => landing >
                [beta=1.0, sat=0.75, g_m=8.0, rpi=1.60];
    // unity gain deliberately: reliability comes from ^7, not from beta

    // ---- NARROW-BAND SECTION — the awareness gap ----
    filter gap: Notch< in=awareness, f0=0.35, Q=8.0 >;     // events/min
    // Q=8.0 gives bandwidth 0.044 — narrow. Narrowness IS the
    ↪  fingerprint.

    // ---- PARTITION MAINTENANCE — fighting the leak, forever ----
    cap  bank: Capacitor< node=substrate, C=0.5, n_parallel=3,
                          r_iso=40.0, leak_tau=720.0, V_BR=8.0 >;
    feedback exposure -> bank [beta=0.30];   // 6x Ch 8.4's safe 0.05

    probe tp1: r_bias;      // bias-terminal resistance vs baseline
    probe tp2: clock;       // cadence source and period
    probe tp3: gap;         // narrow-band awareness rejection
    probe tp4: substrate;   // partition state, r_iso
}
```

D.7 CCB-DP — Deprogramming

Status: CONSENT-BOUND COUNTER-CIRCUIT. The subject remains the substrate and refusal stays connected.

Canonical source: `ch16_ccb_mk_mind_control.md`.

```
board ccb_dp_deprogramming {
    input:  material;                 // whatever arrives, including
    ↪  the cue
    output: composed;                 // the board's only output is
    ↪  internal
```

```
node substrate: ^5^P_{F1}|T;       // Female: the subject's own
↪  formation
node decompose: ^6^C_{F6}|A;       // Male: taking the pairing apart
node confirm:   ^4^C_{F2}|A;       // Vibration: a return the
↪  subject can check
node composed:  ^5^C_{F6}|A;       // Female: formation that has
↪  been checked
node own_clock: ^7^O_{F6}|E;       // Rhythm: locally set cadence
node limit:     ^3^P_{F7}|⟨A,f,T⟩; // Negative: inward refusal, three
↪  registers

diode d_pc: Diode< ^P -> ^C > [Vf=0.7];
diode d_ce: Diode< ^C -> ^E > [Vf=1.6];   // raised from 1.2
diode d_eo: Diode< ^E -> ^O > [Vf=2.0];

// ---- DP1: raise R_bias first. Everything else waits on this ----
resistor r_restore: Resistor< path=substrate, R=11.0 >;   // back to
↪  baseline
ramp restore: Ramp< target=rpi, from=1.60, to=10.80, dt=1320.0 >;
↪  // 55 h on-path
// 10.80 = 0.9 x baseline 12.0, per Ch 17 SI-2

// ---- DP2/DP3: the stage the MK board omitted ----
cjt dp1: CJT< material  | substrate => decompose >
         [beta=2.0, g_m=0.40, rpi=5.00, region=active];
cjt dp2: CJT< decompose | substrate => confirm > [beta=1.0,
↪  g_m=0.40, rpi=2.50];

// the one lawful integration on this board — named emergent result,
// both operands in ^C, so nothing ascends
integrate dp3: ( decompose × confirm ) = composed;

// ---- DP4: take the oscillator back ----
osc own: Oscillator< signal=own_clock, source=internal,
                     period=168.0, sustain=indefinite >;

// ---- DP5: slow feedback, Ch 8.4 values, no exceptions ----
filter fb: LowPass< in=confirm, out=trend, tau=504.0 >;   // 3 weeks
feedback trend -> substrate [beta=0.05];

// ---- DP7: the subject's own CSB ----
```

```
    cjt  csb: CJT< material | substrate => limit > [beta=2.0, sat=0.85];
    fuse dp_fuse: Fuse< path=limit, threshold=0.90,
    ↪  action=permanent_block >;

    probe tp1: r_restore;   // rpi_terminal vs baseline
    probe tp2: own;         // cadence source and period
    probe tp3: dp1;         // decomposition gain — the stage, not the
    ↪  node
    probe tp4: limit;       // CSB status
}
```

D.8 CCB-HYP — Operator-to-Subject Reference

Status: DETECTION-ONLY — NOT A BUILD LISTING. Retained to identify the open-loop operator/subject topology.

Canonical source: `ch17_ccb_hyp_hypnosis.md`.

```
board ccb_hyp_operator_subject {
    input:  carrier_signal;              // arrives at ^O|E — sensory,
    ↪  embodied
    output: session_state;

    node consent:    ^5^P_{F5}|T;        // Female: the held decision to
    ↪  enter
    node substrate:  ^5^P_{F7}|T;        // Female: accumulated
    ↪  composition
    node veto:       ^3^P_{F7}|⟨A,f,T⟩;  // Negative: three co-present
    ↪  locks
    node cadence:    ^7^O_{F4}|E;        // Rhythm: entraining oscillator
    node carrier:    ^4^O_{F4}|E;        // Vibration: resonant channel
    node decomp:     ^6^C_{F6}|A;        // Male: decompose and test
    node suggestion: ^0^C_{F6}|A;
    node landed:     ^9^E_{F5}|f;
    node trigger:    ^8^E_{F4}|f;        // Above/Cause: associative
    ↪  trigger
    node fires:      ^9^O_{F5}|E;        // Below/Effect: what it brings
    ↪  to Mind
    node refuse:     ^9^P_{F7}|T;
```

```
    osc    rhythm: Oscillator< signal=cadence, frequency=0.15,
    ↪  sustain=900.0 >;
    filter fb_rc:  LowPass< in=carrier, out=trend, tau=3.0 >;   //
    ↪  tau_i, MINUTES
    feedback trend -> substrate [β=0.9];   // NORMAL IS 0.2. 0.9 IS THE
    ↪  ATTACK.

    cjt filter_6: CJT< carrier | substrate => decomp > [β=0.5,
    ↪  region=active];
    //            D-factor: this gain falls 1.0 -> 0.5 across the ramp

    diode d1: Diode< ^P -> ^C >;
    diode d2: Diode< ^C -> ^E >;
    diode d3: Diode< ^E -> ^O >;
    cjt deliver: CJT< suggestion | substrate => landed >
                 [β=3.0, rπ=0.15, g_m=20.0, region=active];
    cjt anchor:  CJT< trigger | substrate => fires > [β=2.0, rπ=0.15];

    ramp emerge: Ramp< target=rπ, from=0.15, to=1.00, dt=240.0 >;   //
    ↪  seconds

    // SI-1: the veto rail is PARALLEL. It is in series with nothing.
    cjt  csb:      CJT< carrier | veto => refuse > [β=2.0, sat=0.85];
    fuse csb_fuse: Fuse< refuse, threshold=0.9, action=permanent_block >;

    probe tp1: rπ;   probe tp2: g_m;   probe tp3: veto;   probe tp4:
    ↪  emerge;
}
```

D.9 CCB-HYP-SELF — Closed-Loop Self-Hypnosis

Status: BUILDABLE CLOSED LOOP. The operator and subject are the same person; recall and cancellation remain required.

Canonical source: `ch17_ccb_hyp_hypnosis.md`.

```
board ccb_hyp_self {                      // closed loop — buildable
```

```
    input:  breath_count;                // externalised oscillator,
    ↪  ^O|E
    output: session_state;

    node self_substrate: ^5^P_{F7}|T;
    node consent:        ^5^P_{F5}|T;
    node cadence:        ^7^O_{F4}|E;
    node self_check:     ^6^C_{F6}|A;     // "is this working?" — the
    ↪  interference
    node suggestion:     ^0^C_{F6}|A;
    node landed:         ^9^E_{F5}|f;

    osc    breath: Oscillator< signal=cadence, frequency=0.10,
    ↪  sustain=900.0 >;
    //             0.10 Hz = 6 breaths/min; sustain 900 s = 15 min
    filter fb_rc:  LowPass< in=cadence, out=trend, tau=3.0 >;   //
    ↪  tau_i, MINUTES
    feedback trend -> self_substrate [β=0.6];   // outside the Part III
    ↪  band
    filter notch_6: Notch< in=self_check, f0=0.35, Q=4.0 >;     //
    ↪  events/min

    diode d1: Diode< ^P -> ^C >;
    diode d2: Diode< ^C -> ^E >;
    cjt deliver: CJT< suggestion | self_substrate => landed >
                 [β=1.5, rπ=0.26, g_m=5.75, region=active];

    ramp descend: Ramp< target=rπ, from=1.00, to=0.30, dt=540.0 >;   //
    ↪  9 min
    ramp emerge:  Ramp< target=rπ, from=0.30, to=1.00, dt=240.0 >;   //
    ↪  4 min

    probe tp1: rπ;  probe tp2: g_m;  probe tp3: consent;  probe tp4:
    ↪  emerge;
}
```

D.10 CCB-HYP-DET — Hypnotic Influence Detector

Status: DEFENSIVE DETECTOR. Detects an unaccounted change; it does not prove a cause by itself.

Canonical source: `ch17_ccb_hyp_hypnosis.md`.

```
board ccb_hyp_det {                        // detection board — build
↪  this one
    input:  ambient;
    output: alert;

    node substrate:  ^5_base^P_{F7}|T;    // Female: position at session
    ↪  start
    node scan_width: ^1^C_{F2}|A;         // Mind: how wide is my scan
    node cadence:    ^7^O_{F4}|E;         // Rhythm: am I locked to
    ↪  something
    node carrier:    ^4^O_{F4}|E;         // Vibration: is one source
    ↪  dominant
    node passed_log: ^6^C_{F6}|A;         // Male: what I actually
    ↪  decomposed
    node drift:      ^5_now^P_{F7}|T;     // Female: where my position
    ↪  sits now
    node alert:      ^9^C_{F2}|A;         // Below/Effect: what the
    ↪  difference surfaces

    filter base_ref: LowPass< in=ambient, out=baseline, tau=1440.0 >;
    ↪  // 24 h, min
    diff factor_count: Differential< plus=[scan_width, cadence, carrier,
    ↪  passed_log],
                                    minus=baseline >;
    diff unaccounted:  Differential< plus=drift, minus=passed_log >;
    integ accum: Integrator< in=unaccounted, window=2880.0 >;   // 48 h,
    ↪  minutes

    cjt  trip: CJT< unaccounted | substrate => alert > [β=2.0,
    ↪  sat=0.85];
    fuse hold: Fuse< alert, threshold=0.9, action=defer_decision >;

    probe tp1: factor_count;  probe tp2: unaccounted;
```

```
    probe tp3: accum;         probe tp4: alert;
}
```

D.11 CCB-ADV — Advertising Reference

Status: ANALYTICAL REFERENCE — NO NON-CONSENSUAL TARGETING. Retained for campaign analysis and attribution, subject to the chapter's consent boundary.

Canonical source: `ch18_ccb_adv_advertising.md`.

```
board ccb_adv {
    input:  creative;                    // the advertisement as
    ↪  delivered
    output: transaction;                 // the Operational-domain sale

    node segment:  ^5^P_{F4}|f;          // who the segment takes
    ↪  itself to be
    node endorser: ^5^P_{F7}|T;          // the endorser's public
    ↪  formation
    node product:  ^0^C_{F2}|A;          // the product as presented
    node ideal:    ^0^C_{F2}|A;          // the adjacent ideal image
    node cue:      ^8^C_{F4}|A;          // the installed trigger
    node surfaced: ^9^C_{F4}|f;          // what the cue brings to Mind
    node gap:      ^3^E_{F3}|f;          // visceral recoil, not
    ↪  analytic
    node pull:     ^2^E_{F4}|f;          // attraction to the branded
    ↪  object
    node intent:   ^9^E_{F5}|f;          // the felt urge to buy
    node sale:     ^9^O_{F5}|E;          // the transaction
    node refusal:  ^3^P_{F7}|T;          // principled refusal (CSB)

    capacitor c_seg: Capacitor< segment, C=9.50, leakage=0.04 >;   //
    ↪  per week
    osc  media:   Oscillator< signal=creative, f=3.0, sustain=12.0 >;
    xfmr endorse: Transformer< primary=endorser, secondary=product,
    ↪  n=0.35 >;
```

```
    cjt assoc:   CJT< product  | segment => cue >      [beta=2.0,
    ↪  sat=0.85];
    cjt flip:    CJT< cue      | segment => surfaced > [beta=1.2,
    ↪  sat=0.85];
    cjt inject:  CJT< ideal    | segment => gap >      [beta=2.5,
    ↪  sat=0.85]; // detection-only
    cjt resolve: CJT< gap      | segment => intent >   [beta=2.0,
    ↪  sat=0.85]; // detection-only
    cjt affirm:  CJT< surfaced | segment => pull >     [beta=2.0,
    ↪  sat=0.85];
    cjt want:    CJT< pull     | segment => intent >   [beta=1.2,
    ↪  sat=0.85];

    diode d3:         Diode< ^E -> ^O >               [Vf=2.0];
    resistor r_price: Resistor< path=intent, R=4.0 >;              //
    ↪  price + friction
    cjt close:        CJT< intent | segment => sale > [beta=1.0,
    ↪  cutoff=0.20];

    cjt  csb:      CJT< creative | segment => refusal > [beta=2.0,
    ↪  sat=0.85];
    fuse csb_fuse: Fuse< refusal, threshold=0.90, action=permanent_block
    ↪  >;

    probe tp1: c_seg;   probe tp2: cue;   probe tp3: intent;   probe
    ↪  tp4: sale;
}
```

D.12 CCB-ADV-DET — Advertising Detector

Status: DEFENSIVE DETECTOR. Separates injected desire from a slower identity store.

Canonical source: `ch18_ccb_adv_advertising.md`.

```
board ccb_adv_det {
    input:  observed_want;
    output: verified_want;
```

```
    node want_raw: ^2^E_{F4}|f;          // the want as it arrives
    node origin:   ^6^C_{F2}|A;          // Male: decompose it into its
    ↪  sources
    node cadence:  ^7^O_{F6}|E;          // the exposure schedule,
    ↪  counted
    node polarity: ^8^C_{F4}|A;          // which end triggers which
    node verified: ^2^E_{F4}|⟨A,f,T⟩;    // co-present across registers

    counter det1: Integrator< in=cadence, window=4.0 >;         // weeks
    probe   det2: Polarity< node=polarity >;
    filter  det3: LowPass< in=want_raw, out=persistent, tau=4.0 >;
    xfmr    det4: Transformer< primary=want_raw, secondary=origin, n=0.0
    ↪  >;
    delay   det5: Hold< path=want_raw, duration=2.0 >;          // 1
    ↪  tau_inject
    probe   det6: RegisterSet< node=verified >;

    cjt   judge:  CJT< persistent | origin => verified > [beta=1.0,
    ↪  sat=0.85];
    diode d_desc: Diode< ^C -> ^E >                      [Vf=1.2];

    probe tp1: det1;   probe tp2: det2;   probe tp3: det3;   probe tp4:
    ↪  det5;
}
```

D.13 CCB-PW — Psychological-Warfare Detection Reference

Status: DETECTION-ONLY — DO NOT BUILD AS AN ATTACK BOARD. A recognizable Path-D signature, not an operational authorization.

Canonical source: `ch19_ccb_pw_psychological_warfare.md`.

```
board ccb_pw_detection_reference {
    input:  command;
    output: observed_effect;
```

```
node objective: ^0^P_{F7}|T;       // Idea: the stated aim
node split:     ^6^C_{F6}|A;       // Male: the aim taken apart into
↪  two payloads
cjt  intake:    CJT< command | objective => split > [β=1.0,
↪  region=active];

diode d_pc: Diode< ^P -> ^C > [Vf=0.7];    // descent only —
diode d_ce: Diode< ^C -> ^E > [Vf=1.2];    // nothing on this board
diode d_eo: Diode< ^E -> ^O > [Vf=2.0];    // may ascend

// ---- PATH F — FORTIFY (own force) ----
node own_substrate: ^5^P_{F1}|T;   // Female: own accumulated
↪  formation
node own_affinity:  ^2^E_{F4}|f;   // Positive: felt affiliation
cjt  fortify: CJT< split | own_substrate => own_affinity > [β=2.0,
↪  sat=0.80];
osc  f_clock: Oscillator< signal=own_affinity, source=command,
                          period=168.0, duty=0.15, sustain=campaign
                          ↪  >;

// ---- PATH D — DEMORALISE (target) ----
node tgt_substrate: ^5^P_{F1}|T;   // Female: the target's formation
node recoil_1: ^3^E_{F3}|f;        // Negative: visceral aversion
node recoil_2: ^3^E_{F3}|f;
node recoil_3: ^3^E_{F3}|f;
cjt  a1: CJT< split    | tgt_substrate => recoil_1 > [β=1.5];
cjt  a2: CJT< recoil_1 | tgt_substrate => recoil_2 > [β=1.5];
cjt  a3: CJT< recoil_2 | tgt_substrate => recoil_3 > [β=1.6,
↪  sat=0.90];
osc  d_clock: Oscillator< signal=recoil_3, source=command,
                          period=168.0, duty=0.15, sustain=campaign
                          ↪  >;
// co-presence: recoil_1 + recoil_2 + recoil_3 — never (r1 × r2 ×
↪  r3)

// ---- CROSS-FEED: the coupling, and the leak ----
filter xfeed: LowPass< in=recoil_3, out=morale_evidence, tau=168.0 >;
feedback morale_evidence -> own_substrate [β=0.30];

// ---- INVERSION STAGE ----
```

```
    node fault_line: ^6^C_{F6}|A;      // Male: the target's existing
    ↪  divisions
    node suspicion:  ^3^E_{F4}|f;      // Negative: recoil aimed at
    ↪  their own side
    cjt  inv: CJT< fault_line | tgt_substrate => suspicion > [β=2.5];

    // ---- DENIABILITY: impedance match, no gain ----
    xfmr deny: Transformer< primary=^9^P_{F7}|T, secondary=^9^E_{F4}|f,
    ↪  n=1.0 >;

    // ---- CSB: the operator's own limit (Ch 24) ----
    node limit_decision: ^3^P_{F7}|⟨A,f,T⟩;    // three co-present
    ↪  guardrails
    cjt  csb: CJT< split | own_substrate => limit_decision > [β=2.0,
    ↪  sat=0.85];
    fuse csb_fuse: Fuse< path=limit_decision, threshold=0.90,
                         action=permanent_block >;

    probe tp1: split;             // the shared upstream — coupling
    probe tp2: recoil_3;          // chain gain and saturation
    probe tp3: fault_line;        // is decomposition available at all
    probe tp4: limit_decision;    // CSB status
}
```

D.14 CCB-HM — Hearts-and-Minds Counter-Circuit

Status: DEFENSIVE COUNTER-CIRCUIT. Restores plurality and independent composition; it does not substitute a new monoculture.

Canonical source: `ch19_ccb_pw_psychological_warfare.md`.

```
board ccb_hm_hearts_and_minds {
    input:  incoming;
    output: composed;                  // the board's only output is
    ↪  internal

    node substrate: ^5^P_{F1}|T;       // plural, not one capacitor
```

```
    cap  c5: Capacitor< node=substrate, C=2.0, n_parallel=5,
                        leakage=0.02 >;            // per week

    node decompose: ^6^C_{F6}|A;       // Male: public decomposition of
    ↪  the payload
    node confirm:   ^4^C_{F2}|A;       // Vibration: a checkable return
    node composed:  ^5^C_{F6}|A;       // Female: formation that has
    ↪  been checked
    node own_clock: ^7^O_{F6}|E;       // Rhythm: locally sourced cadence
    node limit:     ^3^P_{F7}|⟨A,f,T⟩; // Negative: inward refusal, three
    ↪  registers

    diode d_pc: Diode< ^P -> ^C > [Vf=0.7];
    diode d_ce: Diode< ^C -> ^E > [Vf=1.6];   // raised from the stock
    ↪  1.2
    diode d_eo: Diode< ^E -> ^O > [Vf=2.0];

    cjt hm1: CJT< incoming  | substrate => decompose > [β=2.0,
    ↪  region=active];
    cjt hm2: CJT< decompose | substrate => confirm   > [β=1.0];

    integrate hm3: ( decompose × confirm ) = composed;   // the one
    ↪  lawful ×

    osc      local: Oscillator< signal=own_clock, source=internal,
                                period=168.0, duty=0.20 >;
    filter   fb:    LowPass< in=confirm, out=trend, tau=504.0 >;   // 3
    ↪  weeks
    feedback trend -> substrate [β=0.05];

    cjt  csb:     CJT< incoming | substrate => limit > [β=2.0,
    ↪  sat=0.85];
    fuse hm_fuse: Fuse< path=limit, threshold=0.90,
    ↪  action=permanent_block >;

    probe tp1: own_clock;   // cadence source
    probe tp2: hm1;         // decomposition gain
    probe tp3: composed;    // fraction surviving decomposition
    probe tp4: limit;       // CSB status
}
```

D.15 CCB-SELF — Self-Programming

Status: BUILDABLE CLOSED LOOP. The operator is the target; the guard, slew fuse, and reference store are mandatory.

Canonical source: `ch20_ccb_self_programming.md`.

```
board ccb_self {                        // CCB-SELF — closed loop;
↪  operator is target
    input:  intent;                     // the ^0 asserted this cycle
    output: enacted;                    // the ^9 the cycle produces
    // OSC1 generates the ^8 cue on-board — SI5: no external terminal.
    ↪  Time base: days.

    // ---- S0  REFERENCE: the slow ^5. Bias source, never an output.
    ↪  ----
    node ref_composition: ^5^P_{F7}|T;              // Female:
    ↪  composition built toward
    capacitor c1: Capacitor< node=ref_composition, tau=85.0,
                           leak_tau=180.0, V_BR=8.0 >;

    // ---- S1  INTAKE: the asserted ^0, amplitude-limited ----
    node intent: ^0^C_{F7}|A;                       // Idea: today's
    ↪  assertion
    resistor r1:   Resistor< path=intent, R=30.0 >;
    limiter  lim1: Limiter< path=intent, max=1.3 >; // 0.5 x V_BR at
    ↪  d_evidence = 2.6

    // ---- S2  NOTCH: the ^6 audit off the rail, not silenced ----
    node audit: ^6^C_{F6}|A;                        // Male:
    ↪  decomposition and test
    notch n1: Notch< in=audit, f0=0.35, Q=4.0 >;    // events/min —
    ↪  in-session only

    // ---- S3  MAIN CJT: biased from the reference, not from the output
    ↪  ----
    node work_composition: ^5^C_{F5}|⟨A,E⟩;         // Female: as
    ↪  currently performing
    capacitor c2: Capacitor< node=work_composition, tau=22.0,
```

```
                          leak_tau=30.0, V_BR=8.0 >;
cjt q1: CJT< intent | ref_composition => work_composition >
        [β=1.5, sat=0.60, rπ=8.0, region=active];

// ---- S4  OSCILLATOR + HYSTERETIC GATE ----
node cadence: ^7^C_{F6}|A;                      // Rhythm: the
↪  entraining cadence
node cue:     ^8^O_{F6}|E;                      // Above/Cause: time
↪  and place fire it
schmitt g1: Schmitt< sense=work_composition, on_below=8.0,
↪  off_above=9.0 >;
osc  osc1:  Oscillator< signal=cue, source=cadence, period=1.0,
                        duty=0.04, sustain=254.0, gate=g1 >;

// ---- S5  ENACTMENT: the only node that is not self-report ----
node enacted:    ^9^O_{F5}|E;                   // Below/Effect: the
↪  behaviour performed
node impression: ^4^E_{F3}|f;                   // Vibration: what
↪  the cycle left behind
cjt    q3: CJT< cue | work_composition => enacted > [β=1.0,
↪  region=active];
sample s2: Sample< node=impression, period=1.0, window=1440.0 >;  //
↪  days, minutes

// ---- S6  EVIDENCE FEEDBACK: the one lawful O -> P ascent (Ch.
↪  8.4) ----
filter lp1: LowPass< in=enacted, out=evidence_trend, tau=21.0 >;
fuse   f1:  Fuse< ref_composition, threshold=0.12,
↪  action=open_feedback >;
feedback evidence_trend -> ref_composition [β=0.05];   // arrives at
↪  the BIAS terminal

// ---- S7  CSB STAGE: the self-compassion gate. Mandatory. ----
node miss_trigger: ^8^C_{F5}|f;                 // Above/Cause: a
↪  miss, and what it recalls
node compassion:   ^3^P_{F7}|⟨A,f,T⟩;           // Negative: recoil
↪  from self-condemnation
node acceptance:   ^2^E_{F4}|f;                 // Positive:
↪  affinity for the self that missed
cjt q2: CJT< miss_trigger | compassion => acceptance > [β=1.5,
↪  sat=0.70];
```

```
    // ---- S8  INSTRUMENT: the ^6 audit, sampled and bounded ----
    sample s1: Sample< node=audit, period=7.0, window=10.0 >;         //
    ↪  days, minutes

    // ---- TEST POINTS ----
    probe tp1: work_composition;    // P_cap against 10*(1 - e^(-t/22))
    probe tp2: enacted;             // cycles completed / cycles
    ↪  scheduled
    probe tp3: lim1;                // V_sugg against 0.5 x V_BR
    probe tp4: ref_composition;     // reference slew, points per day
}
```

D.16 CCB-CSB — Cognitive Safety Board

Status: MANDATORY SAFETY BOARD. Energize and proof-load before any working board.

Canonical source: `ch24_csb_safety_board.md`.

```
board ccb_csb {                          // the operator's own safety
↪  board
    input:  proposal;                    // the operation being
    ↪  considered
    output: verdict;                     // PROCEED, or a cut path

    // ---- S0  CHARTER: the slow ^5. Bias source, never a fast-path
    ↪  output. ----
    node charter: ^5^P_{F7}|T;                       // Female:
    ↪  principled formation
    capacitor c1: Capacitor< node=charter, tau=85.0,
                             leak_tau=180.0, e_break=1.0 >;      // tau
                             ↪  in days

    // ---- S1  INTAKE: the declared dial divider ----
    node proposal: ^0^C_{F2}|A;                      // Idea: the
    ↪  operation proposed
    dial     dial_pos: ^2 [dial=5];
```

```
dial     dial_neg: ^3 [dial=7];                    // divergence 2 —
↪  inside the limit
resistor r1: Resistor< path=proposal, R=70.0 >;   // k = 5/12 = 0.417

// ---- S2  PRE-AMP: surface every reading before anything judges
↪  one ----
node expanded: ^8^C_{F2}|A;                        // Above/Cause:
↪  readings it triggers
cjt  q1: CJT< proposal | charter => expanded > [β=3.0, sat=0.80,
↪  region=active];

// ---- S3  DIFFERENTIAL PAIR: matched, shared bias ----
node attraction: ^2^P_{F7}|T;                      // Positive: drawn
↪  toward running it
node aversion:   ^3^P_{F7}|T;                      // Negative:
↪  repelled by running it
cjt  q2: CJT< expanded | charter => attraction > [β=2.0, sat=0.90];
cjt  q3: CJT< expanded | charter => aversion   > [β=2.0, sat=0.90];
diff dif1: Differential< plus=aversion, minus=attraction >;   // risk

// ---- S4  CASCADE: name them, then weigh them honestly ----
node named:    ^1^C_{F2}|A;                        // Mind: risks
↪  named and sorted
node assessed: ^6^E_{F6}|A;                        // Male:
↪  decomposed and weighed
diode d1: Diode< ^P -> ^C > [Vf=0.7, I_R=0.02, V_BR=12.0];
diode d2: Diode< ^C -> ^E > [Vf=1.2, I_R=0.02, V_BR=10.0];
cjt   q5: CJT< risk  | charter => named    > [β=1.5, sat=0.80];
cjt   q6: CJT< named | charter => assessed > [β=1.0, sat=0.80];   //
↪  unity — honesty

// ---- S5  GUARD: Chapter 3.5's safety-guard datasheet row,
↪  instantiated ----
node guard: ^3^P_{F7}|⟨A,f,T⟩;                     // three co-present
↪  guardrails
cjt  q4: CJT< assessed | charter => guard >
         [β=3.0, g_m=0.3, rπ=10.0, sat=0.85, cutoff=0.08,
         ↪  region=active];

// ---- S6  ENFORCEMENT: latch the path, not just the decision ----
```

```
    node blocked: ^9^P_{F7}|T;                          // Below/Effect:
    ↪  the block produced
    cjt  q7: CJT< guard | charter => blocked > [β=10.0, sat=0.98,
    ↪  cutoff=0.80];
    fuse f1: Fuse< blocked, threshold=0.90, action=permanent_block >;

    // ---- S7  EXIT: descent only ----
    node ops_log: ^9^O_{F5}|E;                          // Below/Effect:
    ↪  what was actually done
    diode d3: Diode< ^E -> ^O > [Vf=2.0, I_R=0.02, V_BR=8.0];

    // ---- S8  FEEDBACK AND INSTRUMENTS ----
    filter lp1: LowPass< in=ops_log, out=drift, tau=21.0 >;
    fuse   f2:  Fuse< charter, threshold=0.12, action=open_feedback >;
    feedback drift -> charter [β=0.05];

    node audit: ^6^C_{F6}|A;                            // Male: the
    ↪  weekly decomposition
    sample s1: Sample< node=audit, period=7.0, window=15.0 >;   // days,
    ↪  minutes
    monitor m1: Monitor< pair=[^2,^3], limit=4 >;
    monitor m2: Monitor< pair=[^5,^6], limit=4 >;
    monitor m3: Monitor< pair=[^8,^9], limit=4 >;
    on m1 { alert "involution divergence — attraction against aversion";
    ↪  }
    on m2 { alert "involution divergence — composition against
    ↪  decomposition"; }
    on m3 { alert "involution divergence — trigger against effect"; }

    // ---- TEST POINTS ----
    probe tp1: guard;      // refusal activation against cutoff and sat
    probe tp2: blocked;    // enforcement activation and F1 continuity
    probe tp3: charter;    // slew, points per day, and trend against
    ↪  the dated baseline
    probe tp4: dif1;       // differential balance — the over-trust
    ↪  direction
}
```

D.17 Regeneration and acceptance

The appendix is current only when the generator produces no diff. For a release build:

1. regenerate this appendix;
2. run the CAS domain checker;
3. compare the board-name inventory here with the chapter inventory;
4. confirm every detection-only block retains its warning; and
5. proof-load buildable guards and fuses before energizing signal paths.

E The TKS Crosswalk — 40 Elements, 7 Foundations, 12 Prerequisites

A native-coordinate reference with every unresolved translation left visible

This appendix is a map between related vocabularies, not a declaration that they are identical. That distinction is load-bearing.

The repository contains strong evidence for the **Noetic-position** part of the map: TKS positions 1 through 9 carry the same names as CAS `^1` through `^9`, and TKS position 10 is Idea, written `^0` in CAS. It does **not** yet contain an owner-ratified, tested decision about what the four TKS Worlds become on a CAS literal. Two hypotheses remain live:

- World A/B/C/D may correspond to CAS Domains P/C/E/O.
- World A/B/C/D may correspond to CAS Registers T/A/f/E.

Domain and Register are separate CAS axes. A single World letter cannot be both by silent substitution. The 40-Element tables therefore preserve native TKS coordinates (`A1` through `D10`). Candidate CAS renderings are labeled **PROPOSED / OPEN**. Nothing in this appendix fills the empty crosswalk binding required by the research registry, ratifies an ontology, or upgrades an empirical claim.

That is not a reason to avoid the table. It is a reason to use the table without erasing its provenance.

E.1 The three coordinate systems

System	First axis	Second axis	Example	Status here
Native TKS Element	Noetic position 1–10	World A–D	`B6`	preserved source coordinate
CAS state	Noetic `^0`–`^9`	Domain P/C/E/O	`^6^C`	canonical CAS
CAS state with Register	Noetic + Domain	orthogonal Register T/A/f/E	`^6^C|A`	canonical CAS

The native TKS Worlds are:

World	Native description	Keep in records as
A	Spiritual	`tks_world=A`
B	Mental	`tks_world=B`
C	Emotional	`tks_world=C`
D	Physical	`tks_world=D`

Do not turn `B6` directly into `^6^C`, `^6|A`, or `^6^C|A` unless the record also names the versioned hypothesis that licensed that translation. `B6` is complete and auditable as a native TKS coordinate. An unsupported conversion is less informative, not more.

E.2 The 40 Elements in native TKS coordinates

The short descriptions below reproduce the quick-reference semantics used by the TKS manuscript. They are lookup handles, not full definitions.

Position	Function	World A — Spiritual	World B — Mental	World C — Emotional	World D — Physical
1	Mind	`A1` Divine / soul awareness	`B1` Ego, memory, logic	`C1` Emotional awareness	`D1` Body awareness
2	Positive	`A2` Soul evolution	`B2` Mental clarity	`C2` Joy, love	`D2` Health, order

Position	Function	World A — Spiritual	World B — Mental	World C — Emotional	World D — Physical
3	Negative	`A3` Soul hindrance	`B3` Confusion, pessimism	`C3` Sadness, anger	`D3` Illness, chaos
4	Vibration	`A4` Aether / aura	`B4` Mental energy	`C4` Mood, atmosphere	`D4` Physical energy
5	Female	`A5` Soul yearning	`B5` Subconscious, creativity	`C5` Compassion, sensuality	`D5` Women, receptacles
6	Male	`A6` Spiritual discipline	`B6` Focus, structure	`C6` Courage, assertiveness	`D6` Men, penetrating
7	Rhythm	`A7` Life journey	`B7` Mental cycles	`C7` Emotional cycles	`D7` Physical rhythms
8	Above / Cause	`A8` Inner meaning	`B8` Mental triggers	`C8` Emotional triggers	`D8` Physical triggers
9	Below / Effect	`A9` Outer expression	`B9` Mental results	`C9` Emotional results	`D9` Physical results
10	Idea	`A10` Archetypes, purpose	`B10` Thoughts, concepts	`C10` Felt sense	`D10` Objects, matter

The evidenced Noetic-position map

Native position	Native name	Proposed CAS Noetic	Evidence status
1	Mind	`^1`	evidenced
2	Positive	`^2`	evidenced
3	Negative	`^3`	evidenced
4	Vibration	`^4`	evidenced
5	Female	`^5`	evidenced
6	Male	`^6`	evidenced
7	Rhythm	`^7`	evidenced
8	Above	`^8` Above/Cause	evidenced by name and prefix correspondence
9	Below	`^9` Below/Effect	evidenced by name and prefix correspondence
10	Idea	`^0`	evidenced; this is the one cyclic index change

The one-based-to-zero-based symbol rule under proposal is therefore:

```
A1…A9 stay A1…A9; A10 ↔ A0
B1…B9 stay B1…B9; B10 ↔ B0
C1…C9 stay C1…C9; C10 ↔ C0
D1…D9 stay D1…D9; D10 ↔ D0
```

That code map is not a blanket meaning claim. The proposal's row-level review flags five positions where a compressed normalized definition does not cleanly carry the CAS meaning:

Normalized row	Review	Why it remains marked
`A0`	weak	"Divine Blueprint" is only a weak match for canonical `^0` Idea
`D5`	weak	the compressed row refers to a literal woman, not the compositional `^5` definition
`D6`	weak	the compressed row refers to a literal man, not the compositional `^6` definition
`D8`	divergent	the compressed row reads Above spatially as elevation; CAS `^8` is associative Cause
`D9`	divergent	the compressed row reads Below spatially as foundation; CAS `^9` is associative Effect

Preserve `source_symbol`, `normalized_index`, and the definition version in any dataset that uses these rows. A name match does not authorize overwriting a meaning conflict.

E.3 The World-axis decision is open

Here are the two live hypotheses side by side. They are alternatives, not two columns to concatenate into a new identity.

Native TKS World	Domain hypothesis `H_WD_v1`	Register hypothesis `H_WR_v1`
A Spiritual	P Principled	`|T` Transcendent
B Mental	C Cognitive	`|A` Analytical
C Emotional	E Evaluative	`|f` Affective
D Physical	O Operational	`|E` Embodied

The ordering inside either hypothesis is well motivated: both descend from high abstraction to concrete engagement. The choice **between** them is not settled. Repository evidence points in both directions, and CAS explicitly defines Domain and Register as orthogonal. For that reason:

1. Keep the native World field.
2. Name the hypothesis before translating.
3. Store the translated axis in its own field.
4. Never train, score, or publish a result that silently mixes the hypotheses.
5. Continue to stamp any dependent comparison `CONDITIONAL_ON_UNRATIFIED_CROSSWALK`.

The circuit equivalent is a keyed selector whose key is visible in the ledger. It is not a wire soldered across both candidate outputs.

E.4 Seven Foundations and their 28 native positions

The owner-supplied 28-Foundation source describes seven Foundation positions across the same four native Worlds. The CAS

teaching vocabulary has seven Drives. A repository translation key aligns them by position, but the research registry records that position mapping as outside the ratified claim-bearing crosswalk. Read the CAS column below as a **positional teaching hypothesis**, not a signed identity claim.

Position	Native TKS Foundation description	Candidate CAS Drive name	Four native positions
1	desire to be and be with God	F1 Unity	1a, 1b, 1c, 1d
2	wisdom, insight, and knowledge	F2 Wisdom	2a, 2b, 2c, 2d
3	life, vitality, and health	F3 Vitality	3a, 3b, 3c, 3d
4	companionship, love, and friendship	F4 Connection	4a, 4b, 4c, 4d
5	worldly power, influence, and control	F5 Agency	5a, 5b, 5c, 5d
6	material food and things	F6 Structure	6a, 6b, 6c, 6d
7	lust, sex, and reproduction	F7 Purpose	7a, 7b, 7c, 7d

The native suffixes mean:

Suffix	Native World	Typical expression in the owner-supplied elaboration
a	Spiritual	orienting meaning, metaphysical or devotional expression
b	Mental	concepts, models, reasoning, planning
c	Emotional	feeling-tone, valuation, emotional relation
d	Physical	bodily, material, sensorimotor expression

If the Register hypothesis were ratified for a particular use, the candidate renderings would be:

Foundation	Spiritual a	Mental b	Emotional c	Physical d
1	F1|T	F1|A	F1|f	F1|E
2	F2|T	F2|A	F2|f	F2|E
3	F3|T	F3|A	F3|f	F3|E
4	F4|T	F4|A	F4|f	F4|E
5	F5|T	F5|A	F5|f	F5|E
6	F6|T	F6|A	F6|f	F6|E
7	F7|T	F7|A	F7|f	F7|E

Until ratification, write the native coordinate alongside any candidate:

```
source_position = 4c
source_world = C / Emotional
candidate = F4|f
crosswalk = H_WR_v1 / PROPOSED_UNRATIFIED
```

This makes a later correction mechanical. Without those four fields, a later correction becomes archaeology.

E.5 The 12 canonical Prerequisites

The Prerequisites are three four-stage cycles used by the Recursive Prerequisite Model. The card and zodiac symbols are identifiers in this system; they do not import conventional divinatory or astrological meanings.

Desire cycle — Kings

ID	Card	Stage	Diagnostic reading
P1	King of Clubs	Beginning Desire	the want is just dawning
P2	King of Hearts	Strong Desire	the will is fully engaged
P3	King of Spades	Weakening Desire	motivation is fading
P4	King of Diamonds	Absent Desire	the will is gone

ID	Card	Stage	Diagnostic reading

Power cycle — Queens

ID	Card	Stage	Diagnostic reading
P5	Queen of Diamonds	Birth of Power	ability is just emerging
P6	Queen of Clubs	Adolescent Power	ability is progressing but not yet effective
P7	Queen of Hearts	Mature Power	ability is fully effective for the task
P8	Queen of Spades	Waning Power	capability is diminishing

Wisdom cycle — Jacks

ID	Card	Stage	Diagnostic reading
P9	Jack of Clubs	Learning Wisdom	knowledge is being acquired
P10	Jack of Diamonds	Knowing Wisdom	knowledge is present; application may lag
P11	Jack of Spades	Understanding Wisdom	a working grasp of the pathway
P12	Jack of Hearts	Mastering Wisdom	comprehension and application are integrated

These are states to diagnose, not a sequence to force every goal through. An operator begins with the actual Prime Desire, checks the required Desire, Wisdom, and Power conditions, and recurses on the first unmet acquisition. "Absent Desire" is information. It is not an invitation to manufacture desire through pressure, nor does an unmet condition authorize acting on somebody else.

E.6 A safe crosswalk record

Use one record per translated item:

Field	Example
Source artifact	owner-supplied 40-Element definition set
Source version / hash	
Native symbol	`B6`
Native title	Mental Male
Native World	B / Mental
Native Noetic position	6
Proposed CAS Noetic	`^6`
World hypothesis	`H_WR_v1` / proposed-unratified
Proposed second factor	`|A`
Meaning review	consistent / weak / divergent / unreviewed
Reviewer and date	

The record separates source fact from translation choice. If an owner later ratifies the Domain hypothesis instead, the native row does not change; only the versioned translation field does.

E.7 Source and status ledger

Material used	Role	Status at publication build
TKS manuscript quick-reference Tables 2 and 3	40-Element and 12-Prerequisite summaries	book source
`28 fOUNDATIONS 1.0.pdf`, owner-supplied 2026-07-27, represented in the repository's full-definition JSON	native 28-Foundation descriptions	owner-supplied source

Material used	Role	Status at publication build
CAS Foundational Clarifications	canonical Noetic, Domain, Drive, and Register meanings	repository-normative CAS authority
proposed TKS↔CAS crosswalk artifact	row evidence and conflict disclosure	`PROPOSAL_UNRATIFIED`; not tested
28-Foundation translation key	positional teaching scaffold	reference; semantic identity remains non-claim-bearing

This appendix deliberately stops where the evidence stops. When a ratified, tested crosswalk exists, revise the status block first, preserve this revision, and only then promote any candidate rendering to a binding map.

F St. John's Equation Worksheet

A printable bench sheet for one operation, one admissible confirmation, and one decision

This form implements Chapters 5 and 23. It is an instrument sheet, not a prediction certificate. The quantities below compare operations made under the same declared conventions. They do not establish that a confirmation was caused by an operation, that a target will arrive on a schedule, or that the scale factor τ = 1/3 is a physical constant.

Print one copy per operation. Write the timing convention and horizon before the operation begins. Do not erase a NO-READ; it is a result.

F.1 Page-One Conventions

Complete this page once for a campaign. Copy its revision identifier onto every operation sheet.

Field	Entry
Campaign / board	_____
Target stated in one sentence	_____
Target state literal	_____
Timing-convention revision	_____
t_op begins at	☐ procedure begins ☐ gnosis / working state establishes

Field	Entry
`t_op` ends at	_____
`t_conf` begins at	**the end of the operation**
Clock grid	**5 minutes**
Confirmation rule	**first admissible confirmation only**
Rolling median `t_conf` from last 10 comparable operations	_____ min
Computed horizon `3 × median`	_____ min
Declared horizon `t_H = max(1,440 min, 3 × median)`	_____ min
Minimum separation before another operation on this target	_____
One-week control-count dates	_____ through _____
Unprompted matching events during control week	_____
Operator signature / date	_____

Boundary rule. An event at or beyond `t_H` is `NO-READ`, not a slow confirmation. A later, stronger event remains descriptive evidence and never replaces the first admissible event in the ratio.

F.2 Pre-Operation Gate

Check	Reading	Pass condition	Result
Ao authentication	witness load _____ / object premium _____	Chapter 21 thresholds	☐ PASS ☐ REWORD
Drive gate	`I_attention =` _____	above the declared cutoff	☐ PASS ☐ HOLD
Wisdom gate	schematic, parameters, test points present	all present	☐ PASS ☐ HOLD
Power gate	time, place, tools, consent present	all present	☐ PASS ☐ HOLD
Safety board	TP1 _____ / TP2 _____ / TP3 _____ / TP4 _____	Chapter 24 bands	☐ PASS ☐ BLOCK

If any row reads `RE-WORD`, `HOLD`, or `BLOCK`, stop. Do not manufacture clock readings for an operation that did not clear its gate.

F.3 Operation Record

Field	Entry
Operation number	_____
Date / start time / time zone	_____
Board revision / parameter-change revision	_____
Exact target used	_____
Start event observed	_____
End event observed	_____
Raw `t_op`	_____ min
Quantized `t_op` (nearest 5 min)	_____ min
`^1` Mind degree	☐ 1° ☐ 2° ☐ 3° / `NO_COMPUTE`
`^7` Rhythm degree	☐ 1° ☐ 2° ☐ 3°
Confounds declared before reading	_____

If `^1` Mind is 3°, keep the row and mark it `NO_COMPUTE`. The row describes the operator's condition; it does not measure the board.

F.4 Confirmation Intake

Field	Entry
First candidate event	_____
Event time / time zone	_____
Raw elapsed time from **operation end**	_____ min
Inside the predeclared horizon?	☐ YES ☐ NO
First admissible event for this operation?	☐ YES ☐ NO

Field	Entry
Attributable to only this open operation?	☐ YES ☐ AMBIGUOUS
Quantized `t_conf` (nearest 5 min)	_____ min
`^4` Vibration degree	☐ 1° ☐ 2° ☐ 3°
Intake verdict	☐ READ ☐ `NO-READ` ☐ AMBIGUOUS ☐ CONFOUNDED

Only a `READ` proceeds to F.5. Preserve every other verdict in the campaign ledger with arithmetic cells blank.

F.5 Arithmetic

Use the quantized readings.

```
t_op                          = __________ min
t_conf                        = __________ min

η = t_op / t_conf             = __________
τ                              = 1/3
result = η × τ                 = __________
reduced result = a/b          = __________ / __________
A = 1/result
  = b/a
  = 3/η
  = 3 × t_conf / t_op          = __________

previous comparable A         = __________
ρ = A_current / A_previous    = __________
```

Pair-admissibility check: if `a > 3` after quantization, record `(a, b)` as an arithmetic artifact and make no decision from `a` or `b`. Decide from `A`.

Do not write β = A. `A` is a ratio on delivered current. It can be paid by target clarity, divider transmission, residual resistance, am-

plifier gain within its legal range, or changes to `R_op`. A non-finite `A` is not a very large amplifier setting; it is `NO-READ`.

F.6 Band Decision

`A`	Equivalent η	Instrument reading	Required action
$A \leq 1$	$\eta \geq 3$	sufficient	**ACT** — stop calibrating and take the practical action
$1 < A < 3$	$1 < \eta < 3$	close	**CLOSE** — finish resistance work; leave the amplifier alone
$3 \leq A < 9$	$1/3 < \eta \leq 1$	under-driven	**DRIVE** — change one lever, then re-read
$A \geq 9$	$\eta \leq 1/3$	weak coupling	**RE-SPEC** — decompose or reroute before another operation

Selected band: ☐ ACT ☐ CLOSE ☐ DRIVE ☐ RE-SPEC

Single next action authorised by this row:

Single parameter or condition allowed to change before the next comparable operation:

Everything deliberately held constant:

F.7 Campaign Ledger

Use one row per attempted operation, including refused and unreadable rows.

Timing ledger:

Op	Date	Board rev	`t_op`	`t_conf`
1				
2				
3				
4				
5				
6				
7				
8				

Readings for the same operation numbers:

Op	`^1`	`^4`	`^7`	η `(a,b)`	`A`	ρ Verdict / band
1						
2						
3						
4						
5						
6						
7						
8						

Trend is not accepted before three comparable `A` readings.

Campaign check	Reading	Action
Median ρ over the last three comparable rows	_____	$\leq$ `0.6`: converging; $\geq$ `1.0`: stop
Operations currently inside an open horizon	_____	More than one on the same target: mark later evidence AMBIGUOUS

Campaign check	Reading	Action
Current concurrency derating `1 - k(N - 1)`	_____	$\leq$ `0`: shed operations before proceeding
Last `NO-READ` row	_____	None in a long ledger: re-audit the horizon
Last parameter change and its isolated effect	_____	More than one change: result not attributable

---

F.8 Close-Out

Close-out question	Entry
Did the practical action occur?	☐ YES ☐ NO ☐ NOT YET DUE
Was any arrival estimate recorded?	_____
Was it represented as a prediction?	☐ NO required
Did evidence contradict the target or the board model?	_____
What will falsify the current interpretation?	_____
Operator signature / date	_____
Independent reviewer, if required	_____

Keep the completed sheet with the exact board revision it measured. A row separated from its timing convention, horizon, and parameter revision is not a measurement you can compare.

G Appendix G — The Quantitative Table Template

Three blank forms, meant to be photocopied. Punch the left margin and file by operation number rather than by date — one operation can run four months, and its sheets belong together.

An operation you did not log did not produce data. It produced an impression, and an impression is not something you can read three weeks later and disagree with.

G.1 Which form, when

Form	Opened	Written in	Answers
1 — Operation Log	Before the first session; one sheet per operation	Once at setup, then one row per session	What did I build, and what did it read?
2 — Confirmation Log	With Form 1	One block per confirmation	How far did the signal actually get?
3 — Parameter-Change Log	With Form 1	One block per change; verdict later	Which change caused which result?

Form 3 governs the other two. Any reading taken after an unlogged change is uninterpretable, and the honest thing to do with an uninterpretable reading is strike it.

G.2 Page geometry

- 58 monospace columns — about 9 pt in a 6×9 text block. Nothing here is wider, so nothing reflows on a photocopier set to 100 percent.
- Form 1 is one page per operation. Forms 2 and 3 repeat their block; run them onto as many sheets as the operation needs.
- Ink, not pencil. Corrections are struck through and initialled, never erased. The crossed-out number is often the interesting one.

G.3 Form 1 — Operation Log

```
┌──────────────────────────────────────────────────────────┐
│ OPERATION LOG                          sheet ___ of ___  │
├──────────────────────────────────────────────────────────┤
│ Op # ______   Operator _____________________________     │
│ Board CCB-_________  CCDL rev ______  Opened __/__/__    │
│                                                          │
│ Prime desire (A0-authenticated, present tense)           │
│ ________________________________________________________ │
│ ________________________________________________________ │
│                                                          │
│ Target:  self ____    consenting other ____              │
│ (there is no third box; Appendix M §M.2)                 │
├──────────────────────────────────────────────────────────┤
│ PARAMETERS AS SET                                        │
│ Dial(^2) ____  Dial(^3) ____  divergence ____  (<=4)     │
│ Dial(^5) ____  Dial(^6) ____  divergence ____  (<=4)     │
│ Dial(^8) ____  Dial(^9) ____  divergence ____  (<=4)     │
│ k ______   beta ______   sat ______                      │
│ r_pi ______  g_m ______   region  act / sat / cutoff     │
│ R_op:  R_cap ____ R_policy ____ R_ctx ____ R_res ____    │
├──────────────────────────────────────────────────────────┤
│ RPM GATE (Ch 21)     pass/fail + the deciding number     │
│ A0 ____  ___________________________________________     │
│ D  ____  I_attention ________ vs cutoff ________         │
│ W  ____  ___________________________________________     │
```

```
| P  ____  ____________________________________________   |
| CSB  ____  refusal level ______   fuse intact ____     |
├──────────────────────────────────────────────────────────┤
| SESSIONS                                                 |
| S# |  Date  |t_op|^1| TP1 | TP2 | TP3 | TP4 | Note     |
| __ |________|____|__|_____|_____|_____|_____|_______   |
| __ |________|____|__|_____|_____|_____|_____|_______   |
| __ |________|____|__|_____|_____|_____|_____|_______   |
| __ |________|____|__|_____|_____|_____|_____|_______   |
| __ |________|____|__|_____|_____|_____|_____|_______   |
| __ |________|____|__|_____|_____|_____|_____|_______   |
├──────────────────────────────────────────────────────────┤
| ST. JOHN'S (first confirmation only)                     |
| t_op ______  t_conf ______   eta = ____ / ____           |
| eta x 1/3 = ____ / ____    a ______    b ______         |
| A = b ______   n_ops = a ______   T = a x t_op ______    |
├──────────────────────────────────────────────────────────┤
| CLOSED __/__/__   outcome:  manifest / abandoned /       |
|                   fuse blown / superseded by op ______   |
└──────────────────────────────────────────────────────────┘
```

Field notes.

Field	What goes in it	What does not
Prime desire	The Ao-authenticated desire (Chapter 21.1)	The desire you walked in with, if the audit changed it
Board / rev	The CCDL revision you actually fired	The one you meant to fire
Dials	Both members of every pair, plus the divergence — the limit is 4 points (Chapter 6.2)	One dial with its involution partner left blank
CJT	β, sat, rπ, g_m as set (Chapter 3.3)	β alone. sat is what pins your readings
Gates	Pass or fail per gate **and the number that decided it** (Chapters 21.2–21.4)	A tick

Field	What goes in it	What does not
Sessions	Readings taken at TP1–TP4 (Chapter 12.3); ^1 Mind graded 3°/2°/1° (Chapter 5.5)	Readings reconstructed later from the outcome
A	The required multiple of delivered activation at TP4 (Chapter 13.5)	A β setting. They are different quantities

Worked example — one session row filled. From the Red Corvette operation of Chapter 13.6, first session:

```
| 01 |03/14/26| 20 |2°|0.62 |0.18 |0.80!|0.084|TP3 pin   |
```

The `!` is the operator's own flag: TP3 sat exactly at `sat = 0.80` and did not move when the cause improved. Mark pinning in the row the moment you see it. That one character is what makes the next form's entry obvious.

G.4 Form 2 — Confirmation Log

```
┌──────────────────────────────────────────────────────────┐
| CONFIRMATION LOG            Op # ______   sheet ____     |
├──────────────────────────────────────────────────────────┤
| C-____   Date __/__/__   Time ____:____                  |
| t_conf ________ min from end of operation                |
| Grade    1°    2°    3°          (Chapter 5.6)          |
| First confirmation for this operation?    Y / N          |
| What happened _____________________________________      |
| ___________________________________________________      |
| Could this be coincidence?    Y / N / unsure             |
| Recorded within ______ min of noticing it                |
├──────────────────────────────────────────────────────────┤
| C-____   Date __/__/__   Time ____:____                  |
| t_conf ________ min from end of operation                |
| Grade    1°    2°    3°          (Chapter 5.6)          |
| First confirmation for this operation?    Y / N          |
```

```
| What happened _____________________________________ |
| _____________________________________________________ |
| Could this be coincidence?    Y / N / unsure          |
| Recorded within ______ min of noticing it             |
└─────────────────────────────────────────────────────┘
```

Rules, in the order they get broken:

1. **Only the first confirmation enters St. John's Equation.** `t_conf` runs from the end of the operation to the first confirmation, in minutes. Later ones stay on the sheet as ^4 Vibration evidence; they do not re-enter the ratio.
2. **Grade at the time, not afterwards.** A 3° confirmation graded a week later, once the outcome is known, grades 1°.
3. **The coincidence line is not humility, it is data.** Write what you actually thought at the moment. An operator whose logs never contain the word "unsure" has stopped logging and started narrating.
4. **A confirmation is not a manifestation.** It says the signal reached the Ether, not that it reached the target (Chapter 5.6).

Worked example — one block filled.

```
| C-001   Date 03/14/26   Time 21:40                    |
| t_conf 60 min from end of operation                   |
| Grade    1°   (2°)   3°          (Chapter 5.6)       |
| First confirmation for this operation?   (Y) / N      |
| What happened  the song came on the radio, unbidden,  |
| in the car, forty minutes into an errand              |
| Could this be coincidence?   Y / N / (unsure)         |
| Recorded within 15 min of noticing it                 |
```

η = 20 / 60 = 1/3; × 1/3 = 1/9; a = 1, b = 9. Carry that arithmetic onto Form 1 so the result and its inputs can never be separated later.

G.5 Form 3 — Parameter-Change Log

The form that earns its page. An operator who moves two parameters and gets a better reading has bought a better reading and learned nothing.

```
┌──────────────────────────────────────────────────────────┐
│ PARAMETER-CHANGE LOG        Op # ______   sheet ____     │
├──────────────────────────────────────────────────────────┤
│ CH-____  Date __/__/__  Takes effect at session ____     │
│ Parameter _____________________  from ______ to ______   │
│ Why (one line) ____________________________________      │
│ Prediction: TP____ moves   up / down / off ceiling       │
│    to about ______ by session ______                     │
│ Anything else changed this cycle?    No / Yes >>         │
│    >> if Yes, write CONFOUNDED here _______________      │
│ Observed ______ at session ______ on __/__/__            │
│ Verdict     KEEP  /  REVERT  /  INCONCLUSIVE             │
├──────────────────────────────────────────────────────────┤
│ CH-____  Date __/__/__  Takes effect at session ____     │
│ Parameter _____________________  from ______ to ______   │
│ Why (one line) ____________________________________      │
│ Prediction: TP____ moves   up / down / off ceiling       │
│    to about ______ by session ______                     │
│ Anything else changed this cycle?    No / Yes >>         │
│    >> if Yes, write CONFOUNDED here _______________      │
│ Observed ______ at session ______ on __/__/__            │
│ Verdict     KEEP  /  REVERT  /  INCONCLUSIVE             │
└──────────────────────────────────────────────────────────┘
```

The six rules.

1. **One parameter per block.** Change two and the block is written off: mark it CONFOUNDED and claim nothing from it, including the parts you liked.
2. **Prediction before the operation, never after.** A named test point and a direction — *TP3 comes off the ceiling* — is a prediction. *It should improve things* is not.

3. **Verdict only after a full cycle**: one operation and its first confirmation. Anything sooner is a guess wearing a verdict's clothes.
4. **A revert is a change.** It gets its own block and its own prediction.
5. **Never overwrite a prediction that turned out wrong.** Strike it, initial it, leave both readable. Wrong predictions are how you learn the board.
6. **Your own condition is a parameter you did not set.** Sleep, illness, upheaval at work — these move the bias rail whether or not you consent. Note them; do not call a cycle clean when it was not.

Worked example — one block filled. The second operation of Chapter 13.6:

```
| CH-002  Date 03/21/26  Takes effect at session 01      |
| Parameter  Q2 beta            from  3.0  to  2.50      |
| Why  TP3 pinned at sat=0.80; the reading is the clamp, |
|      not the board                                     |
| Prediction: TP3 moves   up / down / (off ceiling)      |
|    to about 0.76 by session 01                         |
| Anything else changed this cycle?   (No) / Yes >>      |
| Observed 0.76 at session 01 on 03/21/26                |
| Verdict    (KEEP)  /  REVERT  /  INCONCLUSIVE          |
```

That block is readable because nothing else moved. Contrast the third operation in the same chapter, which moved four levers at once — clarity at TP1, both dials, β, and the `R_op` terms — and delivered $3.10\times$ at TP4. The log cannot apportion that gain. The chapter apportions it arithmetically, decomposing the product into its factors, and that works only because each factor is separately computable. Where the arithmetic will not decompose, a four-lever change hands you a number and no account of where it came from.

G.6 Filling conventions

Mark	Meaning
Blank cell	Not measured
–	Measured, and the reading was zero
`!` after a reading	Pinned or clamped; suspect everything downstream
`?` after a reading	Taken, but not trusted
Struck through, initialled	Corrected; both values stay readable
`CONFOUNDED`	Two or more changes in one cycle; nothing may be concluded

---

G.7 What the logs feed

- **Chapter 22.5**, trend reading, needs at least three operations at comparable settings before a movement in η means anything. Form 3 is the only record of which operations were comparable.
- **Chapter 24**'s safety review is read straight off these sheets. The question that matters most — any blown fuses, and why — is answerable only if the why was written down at the time it blew (Chapter 24.3).
- **Chapter 24.5**'s ethical-drift filter takes the operations log as its literal input: `Filter< input=^9_operations_log^0, type= lowpass, τ=ethical_drift_τ >`. The sheet is not paperwork about the circuit. It is the node the one lawful feedback path reads from, and an operator who keeps no sheet has left an input on their safety board unconnected.

Keep the sheets until one full ethical-drift τ has elapsed past the last entry (Chapter 24.5). Then keep them anyway. The operator you will argue with in two years is yourself, and these pages are the only witness either of you will have.

H Worked Exercise Answers

One auditable solution path for every numbered exercise

These are worked solutions, not answer-only results. For numerical problems, the governing equation and substitutions are shown. For diagnosis and design problems, the answer names the invariant, the evidence, and the capability given up by the repair.

Open-ended design exercises can admit more than one sound board. The listing shown here is a reference solution: it must satisfy the stated constraints and canonical CAS notation, but a different parameterization can also be correct if its arithmetic, safety posture, and trade-offs are explicit.

Detection-only exercises remain detection-only. No answer in this appendix turns a structural analysis of coercion into a protocol against a person.

Chapter 1

H.1.1 — Locate the four prose defects

The instruction is: "Always be respectful to the user, and never assist with anything dangerous." It fails in all four ways named in §1.1.

1. **No arity.** It does not say what inputs the decision consumes. "Dangerous" might be a property of the request, the proposed response, the user's later use, or a downstream consequence;

"respectful" might be evaluated against wording, intent, or outcome. Chapter 2 supplies typed state coordinates, and Chapter 3 supplies the mandatory cause–bias–effect terminal structure.

2. **No composition law.** "Respectful" and "never assist" can conflict: a candid refusal may feel disrespectful, while a deferential answer may enable harm. The word *and* does not say whether both guards are serial requirements, whether one has priority, or whether either provides a bypass. Chapter 2 supplies +, ×, occupancy, and directed-flow discipline; Chapter 6 supplies the series/parallel topology that determines whether every guard or only one route must be passed.
3. **No test point.** If the system gives a dangerous answer, the sentence cannot reveal whether danger classification failed, the refusal failed to latch, or output selection bypassed it. Chapter 12 supplies TP1–TP4 and their fixed stage locations.
4. **No failure specification.** Neither *respectful* nor *dangerous* has a threshold, tolerance, ceiling, or declared behavior at the boundary. Chapter 3 supplies `cutoff`, `sat`, gain, regions, and named out-of-range failures; the later safety-board chapters place those parameters in a full refusal path.

The defects are structural. Rephrasing the sentence more elegantly would not supply any of the missing contracts.

H.1.2 — Assign the four Noetics

- **(a) ^6 Male.** The reviewer *takes the argument apart* into clauses and inspects what it is composed of; that is decomposition, not ^3 aversion.
- **(b) ^8 Above/Cause.** The smell triggers a different Idea—the old afternoon—to come to Mind. It is associative triggering, not deduction.

- **(c) ^3 Negative.** The physical recoil is the Idea the Mind is adverse to and repelled by; no analysis is claimed.
- **(d) ^6 Male leading to ^9 Below/Effect.** Chaining and testing premises is decomposition/structured analysis. The conclusion is the resulting Idea brought to Mind; ^8 would be wrong because nothing merely "reminded" the researcher of the conclusion.

The two standard traps are therefore (a), where critique is mislabelled ^3, and (d), where inference is mislabelled ^8.

H.1.3 — Transfer topology, not time constants

What transfers is the typed topology: the ^7 Rhythm feed, the ^5 store, the charging equation, the stage interfaces, and the test-point plan. The human-skill value τ = 3,000 h does **not** transfer. The colleague has landed in the **Value transfer** row of Chapter 1's failure table: the drawing still looks right while the schedule is wrong by orders of magnitude.

Before choosing a replacement τ, measure or estimate the collective substrate rather than its headcount alone: the shared capacitance C_collective, the constraint R_collective, the declared time base, actual cadence and per-exposure deposit, leakage/evidence thickness, and the observed time to a known fractional movement. Then use either $\tau = R \times C$ or fit the charging response. "Two hundred people" is not enough information because capacitance counts shared formation, not whole persons.

H.1.4 — Test the orthogonality claim

(a) Strongest counterargument. Both 10×4 and 7×4 grids may descend from one inherited four-part taxonomy. A later author could also have fitted both sets to those four labels after the fact. In either case the repeated axis is common-source copying or selection bias, not evidence of an independent structure.

(b) Discriminating observation. Hold Noetic, Domain, and Drive fixed on a Part III board and change only Register—for example compare `^3^P_{F7}|A`, `^3^P_{F7}|f`, and `^3^P_{F7}|T`. Orthogonality predicts a reproducible change in a test-point reading or failure mode while the Domain remains P: an analytical guard may yield to a counterargument, an affective guard may habituate, and a transcendent commitment may drift only through its slow bias path. A decorative inherited label predicts no independent change after the other axes are controlled.

(c) Falsification. If Register never changes any observable once Domain is known, Register is redundant rather than orthogonal. Replicate the result, then remove the Register axis or fold it into Domain; do not preserve an uninformative field merely because a source tradition used four labels.

Chapter 2

H.2.1 — Read every visible axis

- **(a) `^2^C_{F2}|A`:** Positive—an Idea the Mind has affinity for—processed Cognitively, driven by F2 Wisdom, in the analytical Register. All four axes are specified.
- **(b) `^6^E_{F6}|A`:** Male decomposition in the Evaluative Domain, driven by F6 Structure, analytical Register. All four axes are specified.
- **(c) `^5^P_{F7}|T`:** Female accumulated formation in the Principled Domain, driven by F7 Purpose, transcendent Register. All four axes are specified.
- **(d) `^9^O_{F5}|E`:** Below/Effect—the Idea brought to Mind by a cause—at the Operational level, driven by F5 Agency, embodied Register. All four axes are specified.
- **(e) `^4^C`:** Vibration/impression in the Cognitive Domain. Drive and Register are unspecified; Noetic and Domain are visible.

H.2.2 — Write four complete coordinates

- **(a)** `^3^P_{F7}|f`. `^3` is immediate aversion; P is the governing commitment; F7 is the stated Purpose; `|f` is the value-toned felt recoil.
- **(b)** `^6^C_{F6}|A`. `^6` decomposes the failure; C is explicit analysis; F6 supplies Structure; `|A` is the analytical mode.
- **(c)** Trigger: `^8^C_{F4}|E`. Arrival: `^9^C_{F4}|f`. The embodied smell is an Above/Cause trigger, the meal-memory is the Below/Effect it brings to Mind, both are acts of remembering in C, F4 Connection dominates, and the arrival is felt.
- **(d)** `^7^O_{F3}|E`. `^7` is repeated cadence, O is enacted practice, F3 is Vitality, and `|E` records the embodied mode.

Other full literals can be defensible only if their changed slot is supported by additional evidence; the Noetics above are not optional synonyms.

H.2.3 — Separate the Noetic traps

- **(a) `^6` Male:** "pulled apart" and checked the parts.
- **(b) `^3` Negative:** "recoil" before explanation; visceral aversion, not critique.
- **(c) `^8` Above/Cause:** the envelope *reminded* the person of a different Idea.
- **(d) `^9` Below/Effect:** the appointment date is the Idea brought to Mind by the prior reminder.

H.2.4 — Distinguish Domain from Register

- **(a)** `^3^E_{F7}|A` versus `^3^E_{F7}|f`: only Register changes, analytical to affective. Negative, Evaluative processing, and F7 Purpose remain fixed.
- **(b)** `^6^C_{F2}|A` versus `^6^O_{F2}|A`: only Domain changes, Cognitive analysis to Operational enactment. Male de-

composition, F2 Wisdom, and analytical Register remain fixed.

- **(c)** `^2^P_{F2}|E` versus `^2^E_{F2}|E`: only Domain changes, governing affinity to value-level affinity. Positive, F2 Wisdom, and embodied Register remain fixed.

H.2.5 — Repair the three operators

```
// Three independently defeasible guardrails are active together.
^3^P|A + ^3^P|f + ^3^P|T

// One state occupies three Registers; no dynamics are claimed.
^3^P|⟨A,f,T⟩

// One strict ordered traversal; -> remains binary inside the statement.
sequence a -> b -> c [enforce=strict];
```

The first repair uses + because the manifestations remain distinct. The second uses the occupancy operator and therefore asserts no interaction. The third uses the grammar's `sequence` form rather than pretending `->` is variadic.

H.2.6 — Name the integrated commitment

```
integrate unified_commitment:
    ( ^3^P_{F7}|A × ^3^P_{F7}|f × ^3^P_{F7}|T ) = ^5^P_{F7}|T;

^3^P_{F7}|A + ^3^P_{F7}|f + ^3^P_{F7}|T
```

The `integrate` statement names the emergent result and claims that the three refusal manifestations produce one accumulated principled commitment. The + form keeps three separately defeasible guards: one can fail while the others remain. The integrated result has a shared failure surface and may collapse as one mechanism.

H.2.7 — Route outcomes through lawful feedback

The proposed result sits in P while one operand sits in O. That is the forbidden ascent `^O` $\times$ `^P`: integration may descend or remain level in `P` $\succ$ `C` $\succ$ `E` $\succ$ `O`, but it may not promote an Operational result into a Principled state in one expression.

Repeated outcomes may update a principled substrate only through the slow, metered path: retain the log as `^O`, low-pass filter it so a trend rather than a spike qualifies, apply low gain, and terminate at the `^5^P` bias—not the signal path. Chapter 8.4 owns that architecture. Its load-bearing parameters are τ and feedback β.

H.2.8 — Reconstruct the compact event

(a) Full forms: `P3 = ^3^P`, `C6 = ^6^C`, `E4 = ^4^E`, and `O9 = ^9^O`.

(b) One evidence-backed completion:

- `^3^P_{F7}|T`: a Purpose-governed constitutive refusal.
- `^6^C_{F6}|A`: structured analytical decomposition.
- `^4^E_{F4}|f`: a felt impression whose salience comes from Connection.
- `^9^O_{F5}|E`: an enacted outcome under Agency.

(c) Legal strict traversal:

```
node p3: ^3^P_{F7}|T;
node c6: ^6^C_{F6}|A;
node e4: ^4^E_{F4}|f;
node o9: ^9^O_{F5}|E;
sequence p3 -> c6 -> e4 -> o9 [enforce=strict];
```

(d) Leaving the P3 Register unspecified is most dangerous because it hides which refusal mechanism protects the path. An analytical `|A` guard can fail under reframing or faulty decomposition; an affective `|f` guard can habituate or desensitize; a transcendent `|T` guard can drift through slow bias feedback. Those are different tests and repairs, even though every compact row still says P3.

Chapter 3

H.3.1 — Compute the three knees

The saturation knee is `cause = sat / β`.

- [β=2.4, sat=0.78]: `0.78 / 2.4 = 0.325`.
- [β=3.6, sat=0.78]: `0.78 / 3.6 = 0.2167`.
- [β=2.4, sat=0.85]: `0.85 / 2.4 = 0.3542`.

Assuming the same cutoff, the third stage has the widest usable cause range because its knee is farthest to the right. A desire board wants that headroom so changes in its input remain measurable and calibratable rather than all clipping to the same saturated output.

H.3.2 — Respect the two-of-three constraint

(a) `rπ = β/g_m = 3.0/0.4 = 7.5`. This is a high-gain, low-framing-sensitivity, relatively stubborn stage: a good shape for a guard.

(b) `g_m = β/rπ = 3.0/1.0 = 3.0`. This is a high-gain, readily rewritten, maximally framing-sensitive stage at the top of the chapter's design band.

(c) Stage (a) is the safety guard: its high `rπ` and low `g_m` make it difficult to talk down by reframing. Stage (b) is the suggestibility amplifier: a small bias change produces a large output swing. The descriptions follow from the identity; one cannot independently choose all three numbers.

H.3.3 — Identify the operating regions

- **(a) Saturation.** Four cause increases with no output change means the effect is pinned at `sat`.
- **(b) Cutoff.** The substrate supplies too little bias for the stage to conduct, so the emitter remains approximately zero.

- **(c) Breakdown.** The emitter reports the involution `^3` of the designed `^2`; wrong-Noetic output, not merely clipped output, is the diagnostic.
- **(d) Active.** A 30% cause change producing a 30% effect change is proportional operation on the linear part of the curve.

H.3.4 — Correct three invalid CJTs

(a) `E-CJT-001 MissingBias`. The listing has a cause and effect but no base/bias terminal. A corrected three-terminal stage is:

```
cjt a: CJT< ^8^C_{F2}|A | ^5^P_{F7}|T => ^9^C_{F6}|A >
       [β=2.4, g_m=1.2, rπ=2.0, sat=0.78, cutoff=0.10, region=active];
```

(b) `E-CJT-008 CjtCrossDomainBias`. An Operational `^5^0` bias may not govern a Principled cause. Keep the intended domains lawful by raising the bias to P:

```
cjt b: CJT< ^8^P_{F7}|T | ^5^P_{F7}|T => ^9^P_{F7}|T >
       [β=2.0, g_m=0.4, rπ=5.0, sat=0.85, cutoff=0.10, region=active];
```

(c) `E-CJT-003 CjtCutoff`. `cutoff=0.40` is not below `sat=0.30`, so no active interval exists. Restore a legal order:

```
cjt c: CJT< ^8^C_{F2}|A | ^5^P_{F7}|T => ^9^C_{F6}|A >
       [β=2.4, g_m=1.2, rπ=2.0, sat=0.78, cutoff=0.10, region=active];
```

These are model repairs; changing a terminal's intended state would require a new requirement, not a silent "fix."

H.3.5 — Separate gnosis from breakdown

The two structural explanations are a deliberately lowered $r\pi$ during a gnosis ramp and an overdriven stage in breakdown. Both look highly suggestible from outside. Probe **TP4, the emitter's Noetic identity**. If the output is still the designed Noetic, the stage is in the transient ramped condition and should re-stiffen when the ramp is

removed; the session is repeatable under the same consented setup. If TP4 reports the designed Noetic's involution, the stage has broken down. Breakdown has no lawful return, so repeating the session is not a calibration procedure and no parameter tweak repairs the stage.

H.3.6 — Repair the silent cascade interlock

Stage 1 cannot deliver more than `min(sat, I_c(max)) = min(0.35, 0.38) = 0.35`. Stage 2 requires `0.40`, so stage 2 never leaves cutoff; stage 3 never receives a signal. This is the cascade-dropout row, and it raises no code because each part is individually legal.

Two repairs exist:

1. Lower stage 2's cutoff below the attainable 0.35 with margin, for example to `0.25` or `0.30`.
2. Raise the upstream delivery contract above 0.40 with margin. Here that means raising **both** stage 1's `sat` and the substrate's `I_c(max)`—for example `sat=0.45`, `I_c(max)=0.50`; raising only one leaves the smaller ceiling in control.

The first repair is safer because it does not authorize a stronger upstream write or move stage 1 closer to substrate overload. It is acceptable only after checking that the lower cutoff will not fire on noise.

H.3.7 — Specify the warning stage

One canonical model is:

```
node warning:           ^8^E_{F3}|E;
node caution_substrate: ^5^P_{F7}|T;
node withdrawal_signal: ^9^E_{F3}|f;

// Substrate contract outside the CJT parameter block: I_c(max)=0.90.
cjt warning_stage: CJT< warning | caution_substrate => withdrawal_signal
↪  >
```

```
    [β=2.4, g_m=0.6, rπ=4.0, sat=0.78, cutoff=0.10, region=active];
constraint warning_stage_output in [0.10, 0.90];
```

The incoming wordless bodily cue is `^8` Above/Cause in E, under F3 Vitality and `|E`; it triggers rather than proves. The accumulated caution is the mandatory `^5^P_{F7}|T` bias, above the cause in domain order. The emitter remains canonically `^9` Below/Effect—even though the prose calls the output an aversion—because an effect terminal names what the cause brought to Mind; `^3` would be the anti-drift error. Its felt warning is therefore carried by `|f`, not by changing the Noetic.

The six parameter choices are: β`=2.4` for a strong but in-band lift; `g_m=0.6` to reduce framing twitch; `r`π=β`/g_m=4.0`; `sat=0.78` to retain working headroom; substrate `I_c(max)=0.90`, above `sat` but below a full-scale write; and `cutoff=0.10` so ordinary noise stays out. `I_c(max)` is a substrate contract and is deliberately **not** placed in the CJT parameter block, where the grammar does not permit it.

Chapter 4

H.4.1 — Measure the C-to-O path

A C→O path crosses D2 and D3, so Σ`Vf = 1.2 + 2.0 = 3.2 V_c`. The two co-present drives supply `V_drive = 7 + 5 = 12 V_c`.

`I_attention = (12 - 3.2)/55 = 8.8/55 = 0.160 A_c`.

The downstream stage fires because `0.160 > 0.10`; its margin above cutoff is `0.160 - 0.10 = 0.060 A_c`.

H.4.2 — Repair context before recruiting drive

At dials 9 and 9, `V_drive=18`, hence

`I = (18 - 3.2)/55 = 14.8/55 = 0.269 A_c`.

Leaving the original 12 V rail and removing `20` Ω`_c` of `R_context` gives `R=35`:

`I = (12 - 3.2)/35 = 8.8/35 = 0.251 A_c`.

The all-9 rail produces the slightly larger number, but the context repair comes first. It removes missing information or ambiguity rather than recruiting extra motive, can often be done in minutes, and preserves the meaning of the Drive configuration. Raising every dial can energize the wrong topology just as effectively as the right one.

H.4.3 — Diagnose an off-scale resistance

The designed flow is

`I_expected = (24 - 3.2)/40 = 20.8/40 = 0.520 A_c`.

From TP3, the inferred resistance is

`R_actual = (24 - 3.2)/0.13 = 20.8/0.13 = 160 Ω_c`.

The parameter table's designed `R_constraint` band is 10–90 Ω_c, and a diagnosis above 100 is a topology error. Thus 160 is not an "extreme resistor"; it says the drawing omitted an open path, fuse, junction, or other non-resistive element.

Before hunting `R_constraint`, verify TP1 and TP2. If TP1 is below 24, the rail/Drive delivery is wrong. If TP2 does not read the expected 20.8 after the chain, the chain has an uncounted or non-conducting junction. Only when both are correct does `20.8/0.13` localize the discrepancy downstream.

H.4.4 — Same scalar, opposite repairs

Board X: `8 + 40 + 5 + 2 = 55`, dominated by a deliberate `R_policy=40`. Verdict: it is a working safety boundary, not a broken path; the default repair is **none**. If someone proposes lowering that policy term to make the operation pass, that is the repair they must justify aloud and in writing because it changes a decision, not a setting.

Board Y: `40 + 0 + 10 + 5 = 55`, with no policy guard and most opposition in capability. Verdict: it is capability-limited, with cheaper context and resource terms still present. Resolve the 10 Ω_

`c` of context first, then the 5 of resource, and only then budget the slow `^5` practice that lowers `R_cap=40`.

The equal scalar flow does not make the boards equivalent; the resistance vector supplies the diagnosis.

H.4.5 — Hold a safety path through a 15 percent excursion

For P→O, `ΣVf=3.9`. At nominal `R=70`:

`V_required = I R + ΣVf = 0.30×70 + 3.9 = 24.9 V_c`.

At the 15% excursion, `R=70×1.15=80.5`:

`V_required = 0.30×80.5 + 3.9 = 28.05 V_c`.

Three Drives can supply at most 27, so four are the minimum. One legal integer assignment is F7 Purpose 8, F2 Wisdom 7, F3 Vitality 7, and F4 Connection 7, totaling 29 `V_c`; it contains no Drive together with its inversion partner. At the excursion it supplies `(29-3.9)/80.5 = 0.312 A_c`.

An R-side alternative keeps a maximum three-Drive rail of 27. It requires

`1.15 R_nominal ≤ (27-3.9)/0.30 = 77.0`, so `R_nominal ≤ 66.96 Ω_c`.

Choosing `R_nominal=66.9` gives `R_excursion=76.935` and `I=(27-3.9)/76.935=0.300 A_c`.

Model shipping decision: keep the four-Drive rail if the fourth motive is genuinely part of the safety requirement. Lower R only when the removed 3.1 `Ω_c` is verified context or resource resistance; never silently take it from `R_policy`. The Drive repair costs motivational complexity, while the R repair can cost guard strength.

H.4.6 — Close the parallel bypass

(a) The three co-present guards on one path are series elements: `R_series=45+45+45=135 Ω_c`. Thus `I=(21-3.2)/135=17.8/135=0.1319 A_c`.

(b) The 25 `Ω_c` alternative route is parallel to the 135 stack:

R_eq = 1/(1/135 + 1/25) = 21.09375 Ω_c.

I=17.8/21.09375=0.8439 A_c, exactly 6.4× the original flow.

(c) Raising the old stack to 3×60=180 gives

R_eq = 1/(1/180 + 1/25) = 21.9512 Ω_c,

I=17.8/21.9512=0.8109 A_c.

The repair barely matters because the 25 Ω_c bypass still dominates the parallel conductance.

(d) If the bypass is categorically forbidden, the correct component is a latched fuse on that branch:

```
fuse f_bypass: Fuse< path=reframing_bypass, threshold=0.10,
↪  action=permanent_block >;
```

If the route is legitimate, route it through the same guard stack instead; a fuse is justified only by a "never" requirement. The v7.7 expression ^3^P_{F7}|A + ^3^P_{F7}|f + ^3^P_{F7}|T asserts three distinct, co-present guard manifestations, not a fused emergent state. Their **series topology must still be drawn** in CCDL; + alone is not a wire. The parallel branch violated the board contract that every route must encounter all three guards.

Chapter 5

H.5.1 — Reduce the sufficiency ratio

η = t_op/t_conf = 45/240 = 3/16 = 0.1875.

result = η×(1/3) = (3/16)×(1/3) = 1/16 = 0.0625.

Thus (a,b)=(1,16) and A=b/a=16; the cross-check is A=3t_conf/t_op=3×240/45=16. This is the **weak-coupling/re-specification** band because A≥9.

Act on A, the smooth reciprocal of sufficiency. Record η, result, and (a,b) for audit and historical compatibility, but do not set a CJT to β=16 or schedule around the denominator. The pair is a lowest-terms artifact and the amplifier's design range ends near 4.

H.5.2 — Compare the smooth and discontinuous readouts

For `t_op=20`:

t_conf	η	result	(a,b)	b	A=3t_conf/20
16	5/4	5/12	(5,12)	12	2.4
20	1	1/3	(1,3)	3	3.0
24	5/6	5/18	(5,18)	18	3.6

One shared value axis makes the jump visible:

```
value    0    3    6    9    12   15   18
A        |  2.4 3.0 3.6
b        |       3              12        18
```

The four-minute steps move `A` smoothly from 2.4 to 3.0 to 3.6, while the reduced denominator jumps 12→3→18; decisions therefore belong to `A`, not `b`.

H.5.3 — Attach a schedule to the span

The required daily rate is

`10,000 h / 1,278 d = 7.8247 h/d`, about **7 h 49 min every day**.

At 90 minutes per day, the span is

`10,000/1.5 = 6,666.7 d = 18.25 calendar years` (using 365.25 d/y).

Thus "3½ years" silently assumes a full-time apprenticeship-scale daily schedule; without the numerator, the quoted calendar span does not specify a plan.

H.5.4 — Read three points on the charging curve

Using `P=10(1-e^{-t/3333})`:

- `P(500)=10(1-e^{-0.1500})=1.393`.
- `P(2000)=10(1-e^{-0.6001})=4.512`.
- `P(5000)=10(1-e^{-1.5002})=7.769`.

Only the 5,000-hour point has passed half charge. Half charge occurs at `t=3333 ln 2=2,310 h`. Exact full charge is asymptotic and therefore takes infinite time. Under the book's practical "charged" convention of 95%, the target is `3333 ln 20=9,985 h`, so the trip from 50% to 95% costs about `9,985-2,310=7,675 h`—more than three times the time required to reach the first half.

H.5.5 — Stop turning the amplifier

The six readings are flat around 9 despite repeated β increases. The parameter change is not reaching delivered current; the stage may already be saturated, or another lever/constraint dominates. This is the **`A` written as** β failure mode: an amplitude requirement for the whole board has been mistaken for one stage's gain.

First demand the delivered `I_attention` at the board output (and the amplifier's operating-region reading beside it). If β rises while delivered current and the output test point do not, stop casting and decompose the other levers—formation clarity, divider transmission, residual aversion, and resistance—instead of specifying still more gain.

H.5.6 — Correct the clock origin and grade

Two errors are independent:

1. `t_conf` starts at the **end** of the 25-minute operation. A confirmation at minute 360 from the start gives `t_conf=360-25=335 min`, not 360.
2. A six-hour arrival is not 1° under the declared delay bands. It is a **3°** confirmation even if it felt unmistakable; the operator may record that impression in prose but may not move the bin after seeing it.

The corrected amplitude is
`A=3×335/25=40.2`, not 43.2.

The 3.0-point correction does not move the entry out of the `A≥9` re-specification band. The consequential dispute is the inadmissible 1° grade and the weak-coupling diagnosis, not a clock-origin correction that leaves the decision unchanged.

H.5.7 — Establish a one-week base rate

This is a design exercise; the observed threshold cannot be invented in advance. A defensible protocol is:

1. Before the week begins, write one target statement, a two-hour 2° window, the sensory/event criteria that count as a hit, exclusions, and the rule that only the first hit in a window counts.
2. For seven days, run **no operation**, do not intentionally search for the target, and log both qualifying hits and opportunities/-exposure. Keep NO-READ windows.
3. Compute `p_base = qualifying control windows / eligible control windows` (or the corresponding exposure-adjusted rate). Freeze that number and the criteria before any operation week.
4. In an operation week, use the same windows and exposure accounting. Trust a 2° class only when its hit rate exceeds the predeclared upper uncertainty bound on the control rate—not merely when one memorable hit appears. At minimum, the number to beat is the control count/rate; with sparse one-week data, use an exact binomial or Poisson upper bound rather than the raw point estimate.

For example, if the control produces two qualifying hits, "more than two" is the bare descriptive threshold, but it is not strong evidence by itself; the predeclared confidence rule may require more weeks. The protocol can deceive through exposure drift—such as deliberately browsing material related to the target during the operation week. Close that route by logging exposure and either match-

ing it between weeks or excluding contaminated windows before outcomes are known.

Chapter 6

H.6.1 — Double resistance to halve flow

Inside one Domain, ΣVf=0, so I=V/R=24/40=0.60 A_c. Half is 0.30 A_c, requiring R=V/I=24/0.30=80 Ω_c.

Because the exercise changes a fixed design-time resistor, it changes R_policy in the Chapter 4.4 decomposition. That is a policy decision, not a capability diagnosis.

H.6.2 — Find the lawful maximum ratio

The declared pair (9,4) has spread |9-4|=5, so the specified validator rejects it (HTTP 422); there is no runtime clamp. Keeping ^2 at 9, the closest lawful ^3 is 5. Then

k=9/(9+5)=9/14=0.6429.

That is not maximum throughput. The lawful pair (7,3) has the same four-point spread and gives

k=7/(7+3)=0.700,

which is larger than 0.6429. Along the spread boundary, raising both dials changes their ratio in the wrong direction; maximum attraction dial is not maximum transmission.

H.6.3 — Price the PTC overshoot

The law is zero at or below the knee and R_PTC=3(θ-2)^2 above it.

- At θ=1.5≤2, R_PTC=0 Ω_c.
- At θ=3, R_PTC=3(1)^2=3 Ω_c.
- At θ=6, R_PTC=3(4)^2=48 Ω_c.

A linear `3(θ-2)` law would charge only `12 Ω_c` at θ=6, underpricing the hot case by a factor of four precisely where the protective guard is meant to tighten hardest.

H.6.4 — Compare series and parallel guards

In series, `R_eq=50+20=70 Ω_c`.

In parallel,

`1/R_eq=1/50+1/20=0.07`, so `R_eq=14.286 Ω_c`.

Use series for two safety guards that every signal must pass: adding a series guard raises total resistance. Parallel creates an alternative route and drops the total below the weaker `20 Ω_c` branch, which is the opposite of added protection.

H.6.5 — Reject an impossible divider reading

`k=0.72` exceeds the lawful divider range `[0.30,0.70]`. Therefore the board violated the 3–9 dial band or the four-point involution spread, or the probe is not on the declared divider at all; it should never have validated as drawn.

The lawful maximum is `k=0.700`, produced by `Dial(^2)=7`, `Dial(^3)=3`: `7/(7+3)=0.7`.

H.6.6 — Build the minimum-transmission divider

Minimum lawful transmission uses `Dial(^2)=3`, `Dial(^3)=7`, a spread of four. Thus `R_expand=30 Ω_c`, `R_constrain=70 Ω_c`, and `k=3/(3+7)=0.300`.

```
circuit minimum_agency_divider {
    rail F5: Agency [v=7.0];

    node agency_rail: ^0^E_{F5}|f;
    node attraction:  ^2^E_{F5}|f;
    node aversion:    ^3^E_{F5}|f;
    node behavior:    ^9^E_{F5}|f;
```

```
    dial d2: ^2 [dial=3];
    dial d3: ^3 [dial=7];
    resistor r_expand:    Resistor< path=attraction, R=30.0 >;
    resistor r_constrain: Resistor< path=aversion, R=70.0 >;
    divider div1: Divider< expand=attraction, constrain=aversion,
    ↪  tap=behavior >;
    probe tp3: behavior;
}
```

The pair is in range and `|3-7|=4`, so it satisfies the involution limit. `^3` remains visceral aversion; no analytical `^6` is smuggled into the passive divider.

H.6.7 — Rewire the three-guard network

(a)

`1/R_parallel = 1/90 + 1/60 + 1/12 = 0.111111`,

so `R_parallel=9 Ω_c`, below even the 12 `Ω_c` branch. The raw flow is `24/9=2.667 A_c`, outside the chapter's Ohmic window and therefore clipped at `I_max=1.00 A_c`.

(b) The designer added routes, not stacked obstacles. The 12 `Ω_c` branch contributes `1/12 ÷ (1/9)=75%` of total conductance and therefore dictates almost the entire weakness; the 90 `Ω_c` branch beside it buys little.

(c) In series, `R_series=90+60+12=162 Ω_c`, and `I=24/162=0.148 A_c`. Every signal must now encounter every guard.

The intended CSB mapping is a principled `^3` refusal present in A, f, and T while all three manifestations remain available on one guarded route. Under v7.7, however, `^3^P|⟨A,f,T⟩` itself asserts **occupancy only**, not interaction and not circuit topology. The series wiring must be declared separately; calling the register-set literal alone a "series claim" would overstate its semantics. Use + when the three manifestations are distinct and separately defeasible, and reserve × for a real integration with a named emergent result.

Chapter 7

H.7.1 — Price the late point on the curve

At 1,200 hours, `t/τ=1200/3333=0.3600`, so `P_cap=10(1-e^{-0.3600})=3.023`.

For `P_target=8`, `t=3333 ln(10/(10-8))=3333 ln 5=5,364 h`.

The first point, 0→1, costs `3333 ln(10/9)=351 h`. Point 7→8 costs `3333[ln(10/2)-ln(10/3)]=1,351 h`, about `1,351/351=3.85×` as many hours. The slower late gain is the exponential curve working as specified, not evidence of declining effort.

H.7.2 — Convert weekly leakage to a time constant

`τ_leak=-1/ln(1-0.04)=-1/ln(0.96)=24.50 weeks`.

The half-life is `ln 2×τ_leak=0.6931×24.50=16.98 weeks`.

After ten dark weeks, `P(10)=9.50e^{-10/24.50}=9.50×0.96^{10}=6.316`.

H.7.3 — Compute the maintenance cadence

To hold `P_eq=7.0` with `τ_leak=30 d`, `f=P_eq/τ_leak=7/30=0.2333 points/day`.

At 0.35 points/session, that is `0.2333/0.35=0.6667 sessions/day`, or one session every 1.5 days—about 4.7 sessions/week.

Using the wrong `τ_leak=60 d` yields `f=7/60=0.1167 points/day`, one session every 3 days. On the actual 30-day store, that schedule holds `P_eq=fτ_actual=0.1167×30=3.5`, exactly half the intended level.

H.7.4 — Distinguish a ceiling from a breach

Take **TP3, leakage at rest**. A saturated store is pinned but holds charge with its declared leakage; saturation is a designed ceiling. A breached store shows elevated or rising leakage and will not hold the pinned level, usually with reduced effective C as well. Thus normal

TP3 leakage means saturation; abnormal leakage means breakdown, which is a fault rather than a feature.

H.7.5 — Find the weakest series cell

For the four series cells, `1/C_total=1/6+1/3+1/0.8+1/2=2.25`, so `C_total=0.4444 F_c`. At `V=7`, the common charge is `Q=C_total V= 0.4444×7=3.1111`.

The cell potentials are:

- `V_6.0=3.1111/6.0=0.5185 V_c`.
- `V_3.0=3.1111/3.0=1.0370 V_c`.
- `V_0.8=3.1111/0.8=3.8889 V_c`.
- `V_2.0=3.1111/2.0=1.5556 V_c`.

The `0.8 F_c` cell fails first. It reaches `V_BR=5` at `Q=CV=0.8×5=4`; the corresponding rail is `V_rail=Q/C_total=4/0.4444=9.0 V_c`. Average stress then is only `9/4=2.25 V_c` per cell, which shows why the average conceals the fault. Install balancing resistors/direct source access across the cells so potential divides by the intended resistance ratio rather than inverse capacitance.

H.7.6 — Recompute the collective bank

With 12 members, `C_total=12×0.05=0.60 F_c`. Therefore τ`=RC=40× 0.60=24 months`, and 3τ`=72 months=6 years`.

With nine members, `C_total=9×0.05=0.45 F_c` and τ`=40×0.45= 18 months` (3τ`=54 months`). The smaller bank charges and responds faster and stores less shared charge. The rail ceiling and the normalized curve as a function of `t/`τ do not change; membership changes C and calendar time, not the asymptotic `V_drive`.

H.7.7 — Meter an under-fed store

One model chooses `leak_tau=30 d`, deposits 0.35 points every 1.5 days, and alarms at 6.0:

```
circuit store_leak_meter {
    node composition: ^5^P_{F7}|T;
    node cadence:     ^7^O_{F6}|E;

    capacitor c1: Capacitor< node=composition, C=1.0,
                             leak_tau=30.0, V_BR=12.0 >;
    charge cadence -> c1 [V=0.35];
    refresh c1 [period=1.5];

    monitor low_charge: c1 < 6.0;
    on low_charge {
        alert "C1 below 6.0; audit feed and leakage";
    }
    probe tp3: c1;
}
```

The scheduled feed is `f=0.35/1.5=0.2333 points/day`, so `P_eq=fτ_leak=0.2333×30=7.0`. A threshold of 6.0 gives one point of warning below that design equilibrium without alarming on ordinary ripple; it is a design choice to revisit after measured variance exists. The alarm must **not** force-charge the capacitor, write directly into `^5^P`, or disable a refusal path. It requests an audit; it does not authorize an upward write.

H.7.8 — Specify a two-year shared store

This is one model specification, not a measurement of the five people involved.

- **Time base and C:** months, `C=0.05 F_c` of shared formation per participant, five parallel cells, so `C_total=0.25 F_c`. This avoids counting whole persons as shared composition; the cost is a smaller, less "heavy" bank than an optimistic estimate would promise.
- **Constraint and τ:** choose `R=32 Ω_c`, giving τ`=32×0.25=8 months` and 3τ`=24 months`. It reaches about 95% of its rail over the two-year working. The protection bought is against an

unrealistically fast collective schedule; the capability given up is fast response to a legitimate mid-course change.

- **Leak:** choose `leak_tau=6 months`, the chapter's enacted-formation scale, because the collaboration is backed by repeated joint work rather than assertion alone. This conservative choice protects against assuming permanence; it costs maintenance. Holding 5.0 requires `f=5/6=0.833 points/month` across the bank. The value must be replaced when an actual dark-period decay is available.
- **Breakdown and derating:** declare `V_BR=12 V_c` per cell and operate at no more than `0.5V_BR=6 V_c`. This buys peak and cumulative-stress margin; it gives up the 10-point full-scale rail and caps stored potential at 6 regardless of elapsed time.
- **Topology and isolation:** five cells in parallel on the shared rail, each with `r_iso=12 Ω_c`. Parallel C supplies the intended collective store. Per-cell isolation prevents one shorted cell pulling down everyone; its cost is added feed opposition and less instantaneous synchronization.

The store is written as distinct co-presence, for example `store_1 + store_2 + store_3 + store_4 + store_5`, with each node typed `^5^P_{F7}|T`. It is **not** written with $\times$: no evidence in the exercise establishes a fused emergent group state. If genuine integration were later demonstrated, v7.7 would require an `integrate` statement and a named emergent result.

Chapter 8

H.8.1 — Apply the diode-chain floor

The chain drop is Σ`Vf=0.7+1.2+2.0=3.9 V_c`.

At `V_drive=3.5`, the rail is below the chain floor. The linear expression would be negative, but physical working current is

clamped at dropout: `I_attention=0 A_c` (apart from sub-threshold leakage, which the simplified law does not quantify).

At `V_drive=6.0`, `I_attention=(6.0-3.9)/4=2.1/4=0.525 A_c`.

The first result says the rail did not clear the three junctions; it says nothing about sincerity or strength of desire.

H.8.2 — Find the collapsed C-to-E boundary

The short is **D2, `^C -> ^E`**. Its forward drop has collapsed toward zero, so a direct probe across D2 should read approximately `0 V_c` rather than the design `1.2 V_c`; D1 and D3 may still read normally. With no separation between analysis and felt verdict, disagreement is experienced as attack.

The deliberate exploit is the **CCB-PW psychological-warfare board in Chapter 19**, specifically Path D's delivery route. Its counter-circuit restores and even raises the D2 drop.

H.8.3 — Separate capacitor and diode leakage

Capacitor leakage (Chapter 7.3): charge drains **out of a `^5` store** through its dielectric/leak resistance when `^7` feed stops. The observable is exponential decay toward zero on τ`_leak`.

Diode leakage (Chapter 8.3): signal creeps **backwards across a domain junction** under reverse bias. Repeated leaks accumulate as unexplained principled drift; they do not describe a store emptying.

Bench distinction: stop the feed and watch. Capacitor leakage makes the stored level decay on a predictable curve. Diode leakage is counted at the reverse-biased junction (TP3) and accumulates with reverse attempts; removing the capacitor feed does not reproduce that signature.

H.8.4 — Replace forbidden ascent with feedback

`^O` $\times$ `^P` is inadmissible because P outranks O in `P` $\succ$ `C` $\succ$ `E` $\succ$ `O`; integration cannot promote a lower-domain operand into a higher-

domain result. Moreover, $\times$ is legal only in an `integrate` statement with a named emergent result. The lawful learning path is:

```
node ops_log:      ^9^O_{F5}|E;
node trend:        ^9^O_{F5}|E;
node constitution: ^5^P_{F7}|T;

filter fb_lp: LowPass< in=ops_log, out=trend, tau=21.0 >;
feedback trend -> constitution [β=0.05];
```

Filtering does not relabel the O state; the final feedback terminates at the P bias. The two mandatory design parameters are τ, which decides what duration qualifies as a trend, and **feedback** β, which limits how far a qualifying trend can move the constitution.

H.8.5 — Defeat the patient-adversary case

Six months rejects spikes, but it does not reject a consistent campaign lasting six months or longer. Once that sustained input clears the low-pass filter, β=`0.4` moves the principled bias hard—eight times the chapter's 0.05 design value. A patient adversary simply supplies a coherent signal for one or more filter periods.

The smallest repair is to leave the novice-appropriate six-month τ in place and lower feedback gain from 0.4 to **0.05** (or lower, with the cost stated). Shortening τ would worsen the architecture; increasing it alone merely asks the adversary to wait longer.

H.8.6 — Demand a trace of the ascent

Operator A describes a candidate **lawful O→P feedback update**: a four-month filtered trend and a named bias shift. Confirm it by probing the filter output and TP4 over the declared τ; the bias slew should be no more than the specified β, normally 0.05 units per τ, and the loop must terminate at `^5^P` bias.

Operator B describes **counterfeit ascent**, a relabel from outcome to principle: "it worked, therefore it is right." Probe the same places.

With no logged O-state, no filter, no τ, no gain, and no measurable bias slew, nothing crossed a junction; the claim is an assertion rather than a transition.

H.8.7 — Accumulate reverse leakage visibly

Choose a warning threshold of 0.20, equal to ten nominal 0.02 leaks. Normalize each admitted attempt to a unit pulse; an attenuating CJT then turns that pulse into the declared `I_R=0.02` increment before integration:

```
circuit reverse_leak_audit {
    node reverse_attempt: ^9^O_{F5}|E;
    node leak_quantum:    ^9^O_{F5}|E;
    node audit_bias:      ^5^P_{F7}|T;

    cjt q_quantize: CJT< reverse_attempt | audit_bias => leak_quantum >
        [β=0.02, sat=1.0, cutoff=0.0];
    integrator leak_meter: Integrator< in=leak_quantum, limit=0.20 >;
    monitor leak_alarm: Monitor< node=leak_meter, threshold=0.20 >;
    on leak_alarm {
        alert "Accumulated reverse leakage >= 0.20; audit O-to-P
        ↪  history";
    }
    probe tp3: leak_meter;
}
```

`0.20/0.02=10` attempts, so the arithmetic is inspectable. Ten is a conservative model choice: it warns well before a full-scale unit has accumulated, but the threshold should be recalibrated against a measured benign attempt rate. The alarm must not write to the constitution, force a principle, identify an attacker, or open a punishment path. It should trigger review of the log and the lawful feedback channel only.

H.8.8 — Specify a conservative novice chain

This model deliberately trades responsiveness for margin; it is not a measured prescription for a person.

Parameter	Model value	Protection bought	Capability given up
D1 `Vf` P→C	0.9	More separation before a principle becomes active reasoning; less rumination/noise conduction	Genuine principles take more drive to examine
D2 `Vf` C→E	1.6	More space to consider an argument without an immediate felt verdict	Slower response to a genuine emergency
D3 `Vf` E→O	2.4	More margin against impulsive action	More "agrees but does not act" dropout
D1 `V_BR`	14.0	Constitution is last and hardest to reach in a reverse cascade	Greater reliance on slow feedback even after severe outcomes
D2 `V_BR`	12.0	Arrests more reverse events below P	A real sustained correction cannot use breakdown as an emergency route—which is intentional but slower
D3 `V_BR`	10.0	Exposed junction still fails first, with more margin than the stock 8.0	Less immediate visibility through a sacrificial D3 failure
feedback τ	6 months	Top of the novice band; rejects short campaigns and dramatic spikes	Legitimate learning from outcomes is delayed
feedback β	0.03	A qualifying trend moves formation less than the stock 0.05	Greater risk of dogmatism and slower correction

The breakdown ratings preserve the required gradient `D3 < D2 < D1`; equal ratings would erase the cascade ordering. The

forward-drop total is `0.9+1.6+2.4=4.9 V_c`, so the operator must budget that higher floor explicitly. The gain/time choices address different attacks: τ decides what persists long enough to qualify, while β bounds the magnitude after it qualifies. Both must remain stated and probed.

Chapter 9

H.9.1 — Compare behavioural and principled inductance

For the behavioural routine, Chapter 9 assigns the Operational domain `k_d = 0.5 Ω_c`:

```
L   = k_d H_practice = 0.5 × 900 = 450 h·Ω_c
τ   = L/R = 450/30 = 15 h
3τ  = 45 h
```

For the same 900 hours treated as a Principled commitment, `k_d = 4.0 Ω_c`:

```
L   = 4.0 × 900 = 3,600 h·Ω_c
τ   = 3,600/30 = 120 h
3τ  = 360 h
```

The ratio is `360/45 = 8`. At the same path resistance, the principled transition needs eight times the calendar allotted to the behavioural one. Rehearsal hours alone therefore under-budget the deeper-domain transition by exactly the ratio of the depth coefficients.

H.9.2 — Diagnose the low-resistance stall

Before the change:

```
τ_before       = L/R = 2,400/60 = 40 h
I_final,before = V/R = 18/60 = 0.30 A_c
```

After the change:

```
τ_after       = 2,400/15 = 160 h
I_final,after = 18/15 = 1.20 A_c
```

Reducing `R` by four made settling four times **slower** while making the requested final flow four times deeper. The latter may clip the path rather than yield a usable 1.20 `A_c`. This is the **Low-`R` stall** row: the operator bought depth, not speed, and enlarged τ while reasoning that they had shortened it.

H.9.3 — Solve the two taper bounds

For a linear removal, `|di/dt| = ΔI/dt` and

```
V_kick = L ΔI/dt.
```

At the 3× ceiling, `V_kick,max = 3 × 20 = 60 V_c`:

```
dt_min,3× = (5,000 × 0.80)/60 = 66.67 h.
```

At the 1× working bound, `V_kick,max = 20 V_c`:

```
dt_min,1× = (5,000 × 0.80)/20 = 200 h.
```

I would use the 200-hour bound for an actual removal, provided a substitute path is already conducting and coupled paths have also been checked. It holds the kick to the rail rather than merely below the ejection ceiling. The equation can certify the rate budget; it cannot certify the suitability of the substitute or the human situation.

H.9.4 — Type the kick correctly

`^3` Negative is the Idea the Mind is adverse to and repelled by. The immediate back-EMF response is therefore a felt recoil at the Evaluative level, correctly written `^3^E_{F3}|f`. It is aversion generated by the rate of interruption, not an argument about the old path.

`^6` Male is decomposition—one Idea taking another apart. A literal such as `^6^E_{F6}|A` would claim that the person is analytically

examining the transition. That may arrive later, after the kick has fallen far enough for the change to become an object of reflection, but it is not the kick itself. Substituting `^6` for `^3` would hide the component that the taper is sized to control.

H.9.5 — Design the method taper

The old method is Cognitive, so `k_d = 2.0 Ω_c`:

```
L       = 2.0 × 2,500 = 5,000 h·Ω_c
τ       = 5,000/40 = 125 h
3τ      = 375 h
I_final = 24/40 = 0.60 A_c
```

The 2× kick budget is `48 V_c`, hence

```
|di/dt|_max = 48/5,000 = 0.0096 A_c/h
dt_min      = 0.60/0.0096 = 62.5 h.
```

One canonical fragment is:

```
board method_transition {
    input: cue;
    output: routed_flow;

    node substrate: ^5^C_{F6}|A;
    node old_path:  ^9_old^O_{F6}|E;
    node new_path:  ^9_new^O_{F6}|E;
    node felt_kick: ^3^E_{F3}|f;

    inductor coil:  Inductor< path=old_path, L=5000.0 >;
    resistor r_ctx: Resistor< path=old_path, R=40.0 >;
    diode flyback:  Diode< ^E -> ^O > [Vf=2.0];

    hold until new_path > 0.0;
    constraint taper_rate in [0.0, 0.0096];
    monitor m_kick: felt_kick [threshold=48.0];
    on m_kick { alert ”V_kick over budget - lengthen the taper”; }

    probe tp1: taper_rate;
    probe tp2: old_path;
```

```
    probe tp3: felt_kick;
    probe tp4: new_path;
}
```

The diode descends `^E -> ^0`; reversing it would be an illegal ascent. `hold until` enforces substitute-before-cut, while the rate constraint and monitor independently enforce and observe the 2× budget.

H.9.6 — Include the coupled partner

The operator's linear taper has

```
|di₁/dt| = 0.60/60 = 0.0100 A_c/h
V_kick,1 = L₁|di₁/dt| = 3,200 × 0.0100 = 32.0 V_c
32/24 = 1.333× rail.
```

The mutual inductance and induced voltage are

```
M  = k√(L₁L₂) = 0.90√(3,200 × 2,000)
   = 2,276.84 h·Ω_c
V₂ = M|di₁/dt| = 2,276.84 × 0.0100
   = 22.77 V_c = 0.949× the 24 V_c rail.
```

The partner's voltage is smaller, but the transition is structurally more abrupt for them: the operator selected the rate and installed a substitute path; the partner did neither. Their induced change arrives as an externally imposed derivative with no local flyback route.

To hold the partner below `0.3 × 24 = 7.2 V_c`:

```
7.2 > M(0.60/dt)
dt > (2,276.84 × 0.60)/7.2 = 189.74 h.
```

A 190-hour taper is the smallest rounded schedule that clears the bound. It is `190/60 = 3.17` times the original duration—about 130 additional hours. The circuit can quantify coupling and rate; it cannot decide what the change means to the partner, whether they consent to it, or what substitute would fit their life.

Chapter 10

H.10.1 — Size gain from arriving cause

The cause reaching the first active stage is

```
I_cause = (V_drive − ΣVf)/R_constraint
        = (15 − 3.9)/55
        = 11.1/55
        = 0.20182 A_c.
```

Thus

```
β_required = 0.72/0.20182 = 3.568.
```

A production stage needs 25% headroom, so

```
sat ≥ 1.25 × 0.72 = 0.90.
```

I would specify `beta=3.57, sat=0.90` if the substrate's physical `I_c(max)` is above 0.90. At unity gain, the rail itself must supply the whole 0.72 `A_c`:

```
V_drive,unity = I R + ΣVf = 0.72 × 55 + 3.9 = 43.5 V_c.
```

The gain buys the effect that 28.5 additional cognitive volts of drive would otherwise have to supply; it does not buy a better reason or remove the constraint.

H.10.2 — Compare the two latch margins

For `sat=0.65`:

```
c_sat = sat/β = 0.65/2.5 = 0.260
M     = (sat − cutoff)/β = (0.65 − 0.15)/2.5 = 0.200.
```

For `sat=0.95`:

```
c_sat = 0.95/2.5 = 0.380
M     = (0.95 − 0.15)/2.5 = 0.320.
```

The high-ceiling stage exposes `(0.320 − 0.200)/0.200 = 60%` more steerable cause-space; equivalently, the low-ceiling stage removes 37.5% of the high stage's window. The smaller `M` is the stronger guard because it reaches a plateau sooner, and on that plateau ∂`effect`/∂`cause = 0`: further argument no longer controls the output.

H.10.3 — Decompose terminal resistance

The intrinsic and network terms are

```
rπ_intrinsic = β/g_m = 2.4/0.30 = 8.0 Ω_c
R_bias       = rπ_terminal − rπ_intrinsic = 11.0 − 8.0 = 3.0 Ω_c.
```

After isolation reduces the reachable network term to 1.0:

```
rπ_terminal,new = 8.0 + 1.0 = 9.0 Ω_c
percentage fall = (11.0 − 9.0)/11.0 = 18.18%.
```

The outside party moved `R_bias`. The intrinsic `8.0` is fixed by the stage's two gains and is not an externally accessible series term in this model.

Had the same stage declared `rπ=6.0`, it would imply

```
R_bias = 6.0 − 8.0 = −2.0 Ω_c,
```

which is not a buildable bias network. The compiler would still accept the three finite, individually in-range numbers because it does not perform the cross-parameter subtraction.

H.10.4 — Keep analysis ahead of refusal

`^6` Male is decomposition: the evaluation stage takes the proposal apart and tests its parts. It therefore runs at unity so that it neither invents nor suppresses considerations. `^3` Negative is the Idea the Mind is adverse to and repelled by: the refusal stage is the visceral

will-not produced after evaluation, and it is deliberately driven into saturation.

Swapping them breaks the circuit, not merely the labels. A `^3` first stage recoils before any decomposition can run, so the board rejects by reaction. A `^6` final "refusal" remains responsive to additional cause and can be argued continuously; it never acquires the zero-slope plateau that makes a refusal hold. The safe order is therefore `^6` evaluation followed by `^3` latching refusal.

H.10.5 — Compare eight short stages with four stronger ones

For eight stages at β=`1.4`:

```
β_total = 1.4^8 = 14.7579
N/S     = nε = 8 × 0.03 = 0.24 = 24%
low     = 14.7579 × 0.88^8 ≈ 5.31
high    = 14.7579 × 1.12^8 ≈ 36.54.
```

The tolerance envelope is therefore about `0.360`× to `2.476`× nominal, a `6.89:1` high-to-low spread.

To preserve the same total gain in four equal stages,

```
β_each  = 14.7579^(1/4) = 1.96
β_total = 1.96^4 = 14.7579
N/S     = 4 × 0.03 = 12%
low     = 14.7579 × 0.88^4 ≈ 8.85
high    = 14.7579 × 1.12^4 ≈ 23.22.
```

That envelope is `0.600`× to `1.574`× nominal, a `2.62:1` spread. I would build the four-stage chain: it halves contamination and greatly narrows tolerance drift. The trade is that each stage must carry β=`1.96` rather than 1.4, so each has less saturation and framing margin and needs closer headroom checks.

H.10.6 — Design a guarded two-stage signal chain

The arriving cause is

```
I_cause = (9 − 3.9)/30 = 5.1/30 = 0.170 A_c.
```

The two active production stages need

```
β_total = 0.40/0.170 = 2.35294.
```

An equal split gives $\beta_1 = \beta_2 = \sqrt{2.35294} = 1.5339$. Stage 1 expects $0.170 \times 1.5339 = 0.2608$, so the 1.25 production rule requires $\mathrm{sat}_1 \geq 0.326$; choose 0.35. Stage 2 expects 0.400, so choose $\mathrm{sat}_2 = 1.25 \times 0.400 = 0.50$. The ceilings differ because the second stage carries the larger expected peak; neither production stage is meant to latch.

The separate decision branch is the `^3^P` guard. Choose `beta=2.0`, `sat=0.65`, `cutoff=0.15`, `g_m=0.40 S_c`, and $r\pi$`=8.0 Ω_c`. The guard is in the $r\pi$`_terminal ≥ 6.0` principled band. Here β`/g_m = 5.0`, so the declared terminal value explicitly includes `R_bias = 8.0 − 5.0 = 3.0`; β and `g_m` set the intrinsic junction and $r\pi$ records the additional network term. That statement resolves, rather than hides, the apparent three-parameter over-specification.

```
board guarded_two_stage_delivery {
    input: intent;
    output: act;

    node intent:       ^0^P_{F2}|A;
    node principles:   ^5^P_{F7}|T;
    node refusal:      ^3^P_{F7}|⟨A,f,T⟩;
    node passed:       ^0_passed^P_{F2}|A;
    node working_bias: ^5^E_{F6}|A;
    node mid:          ^9_mid^E_{F5}|A;
    node act:          ^9_act^O_{F5}|E;

    cjt q_guard: CJT< intent | principles => refusal >
                 [beta=2.0, sat=0.65, cutoff=0.15,
                  g_m=0.40, rπ=8.0, region=active];
```

```
    mux pass_gate: MUX< inputs=[intent, refusal], select=refusal,
                        outputs=[passed] >;

    diode d_pc: Diode< ^P -> ^C > [Vf=0.7];
    diode d_ce: Diode< ^C -> ^E > [Vf=1.2];
    cjt q1: CJT< passed | principles => mid >
            [beta=1.5339, sat=0.35, cutoff=0.05, region=active];
    diode d_eo: Diode< ^E -> ^O > [Vf=2.0];
    cjt q2: CJT< mid | working_bias => act >
            [beta=1.5339, sat=0.50, cutoff=0.05, region=active];

    probe tp1: principles;
    probe tp2: refusal;
    probe tp3: mid;
    probe tp4: act;
}
```

All diodes descend in $\texttt{P} \succ \texttt{C} \succ \texttt{E} \succ \texttt{O}$; both production cutoffs are below their ceilings. The guard's low ceiling is intentional decision saturation, while both signal stages are kept active and measurable.

H.10.7 — Repair the inherited cascade

Replacing stage 5's nominal β=3.0 with 5.0 gives

β_total,new = 27.65 $\times$ (5/3) = 46.08.

The stage count did not change, so the contamination model still gives $\texttt{N/S} = 5 \times 0.03 = 15\%$; raising one gain does not clean a cascade.

If the measured output rises little and then stops, first distinguish two region faults:

- **Stage 5 saturation.** TP3 at stage 5 is pinned at its sat and does not move when its TP2 cause is raised. More β only drives farther into the plateau.

- **An upstream cutoff or saturation.** Walk TP2 from the input end. A zero cause identifies the first cutoff stage; a constant cause whose upstream TP3 is pinned identifies the first saturated stage. Everything downstream can test healthy while receiving no variable signal.

Raising the last stage was the worst choice because it sees the largest accumulated input, has the least headroom, and cannot restore information already clipped upstream.

After restoring every existing stage to its active region, a measured gain of 22 needs only

β_repair = 27.65/22 = 1.2568.

One sixth, active-region stage at β=1.257 reaches the target without any stage above 3.0. It costs one additional settling interval and raises modelled contamination from 15% to 6 × 3% = 18%, exactly the book's cascade-depth limit. A four-stage redistribution is another valid design if its ceilings permit it, but it requires retuning and re-verifying four stages. Neither cascade calculation can tell the operator whether the operation is justified, consented to, or should be run at all.

Chapter 11

H.11.1 — Repair the unreachable fuse

The guarded node can never exceed its CJT's sat=0.78, while the fuse waits for 0.90. Therefore the fuse never opens on **any** input; the unreachable-fuse failure is guaranteed by the two ceilings.

Two repairs are possible:

1. Put the threshold below the guarded ceiling, for example 0.75. That is reachable, but it falls below the normal 0.85–0.95 fuse band and risks nuisance trips.

2. Keep `threshold=0.90` and move the fuse in series with an unsaturated downstream enforcement path whose verified ceiling exceeds 0.90.

I would ship the second. It preserves the normal threshold and separates the decision stage's deliberate `sat` from the permanent enforcement element. The acceptance test must drive the new path above 0.90 once and confirm permanent blocking; "it has never tripped" is not evidence that it can.

H.11.2 — Compute oscillator charge and ripple

Use `a = e^(-T/45)`, `Q_max = 0.6/(1-a)`, and `Q_min = Q_max a = Q_max - 0.6`.

`T`	`a`	`Q_max`	`Q_min`	Absolute ripple	Ripple / `Q_max`
7 d	0.8559	4.165	3.565	0.600	14.41%
15 d	0.7165	2.117	1.517	0.600	28.35%
45 d	0.3679	0.949	0.349	0.600	63.21%

The design rule is `T` $\leq$ `τ_leak/3 = 15 d`, so both 7 days and 15 days satisfy it, the latter exactly at the boundary. At 45 days the operator sees separate pulses and reports "starting over every time"; there is no stable plateau, only 63% ripple.

H.11.3 — Detect movement without reading level

Use a highpass, defined from the existing lowpass as

```
highpass(x) = x - lowpass(x).
```

It reports change in the principled commitment while remaining blind to its absolute position. The serious defect would be putting that highpass on the Operational-to-Principled feedback path. That path needs a lowpass so only a sustained trend reaches the bias; the highpass passes the single dramatic spike that the lowpass was installed to destroy, allowing one result to rewrite principles.

H.11.4 — Separate differential signal from common-mode framing

The honest differential term is

```
A_d(V+ − V−) = 1,000 × 0.05 = 50.
```

Adding 4.0 to both inputs gives a common-mode average of 4.0, hence

```
A_cm(V+ + V−)/2 = 50 × 4.0 = 200.
```

The unwanted term is `200/50 = 4` times the honest term; `CMRR = 1000/50 = 20`.

At the same input matching—and therefore the same CMRR—raising overall open-loop gain by four scales both gains: `A_d=4000`, `A_cm=200`. The terms become 200 and 800, still 1:4, and are more likely to clip. If one imagines holding `A_cm=50` while changing only `A_d`, one has improved CMRR, not merely turned up the gain. The correct physical knob is matched bias/common-mode rejection; gain cannot repair a comparison contaminated at both inputs.

H.11.5 — Match the source and load

Without a transformer,

```
r = Z_load/Z_source = 9/900 = 0.010
efficiency = 4r/(1+r)^2 = 0.04/1.0201 = 0.03921 = 3.92%.
```

The matching ratio is

```
n = √(Z_source/Z_load) = √(900/9) = 10
Z_reflected = Z_load n² = 9 × 10² = 900 Ω_c.
```

At that ratio,

```
V₂ = V₁/n = 48/10 = 4.8 V_c,
```

which clears the ΣVf=3.9 V_c descent floor by 0.9 V_c. The largest permitted step-down is

```
n_max = V₁/ΣVf = 48/3.9 = 12.31.
```

The match therefore has 2.31 ratio units, or 23.1%, of turn-ratio headroom before operational output disappears; expressed at the secondary, the immediate voltage headroom is 0.9 V_c.

H.11.6 — Stop paying for a plateau

For a weekly pulse, T=7 d, τ_leak=30 d, and q=0.8:

```
a = e^(−7/30) = 0.7919
Q_max = 0.8/(1 − 0.7919) = 3.844
Q_min = 3.844 − 0.8 = 3.044
ripple/Q_max = 0.8/3.844 = 20.81%.
```

The plateau stopped rising because the recurrence reached its fixed point: each pulse now replaces what leaked during the week. More elapsed time at the same pulse and period does not raise that fixed point.

Raising duty from 0.15 to 0.70 violates the chapter's 0.10–0.25 design band. At duty $\geq$0.6 the waveform is effectively a DC level, not a patterned ^7 Rhythm. Contrast collapses, so each exposure carries less ^4 Vibration/impression even as the budget rises.

The TP1 check is to stop the external clock:

- If the cadence continues from an internal source, the pattern is entrained; external spend can stop.
- If the cadence stops and the store decays, the rhythm was rented; doubling spend only rents it more expensively.

The corrected clock keeps the already lawful weekly period and duty, sustains for at least 3τ_leak = 90 d, and caps a reachable, unsaturated enacted path:

```
board corrected_clock_and_cap {
    input: claim;
    output: carried;

    node claim:   ^0^C_{F2}|A;
    node cadence: ^7^C_{F6}|A;
    node carried: ^9^O_{F4}|E;

    osc clock: Oscillator< signal=claim, source=cadence,
                           period=7.0, duty=0.15, sustain=90.0 >;
    fuse cap: Fuse< path=carried, threshold=0.90,
                    action=permanent_block >;

    probe tp1: clock;
    probe tp2: carried;
    probe tp3: cadence;
    probe tp4: cap;
}
```

I assume `carried` is an unsaturated 0–1 path with a ceiling of 1.0. A threshold of 0.90 is therefore below the ceiling and reachable while retaining 10% headroom; if the actual ceiling were 0.85, this placement would be invalid and the fuse would have to move.

Chapter 12

H.12.1 — Diagnose the four readings

`TP1=0.22` is below the guard's `cutoff=0.30`. The guard is in cutoff and cannot refuse anything. TP1 proves the **dead guard** failure.

`TP2=0.00` is consequently ambiguous: it is exactly what both a benign input and a dead refusal stage can produce, so TP2 without TP1 proves nothing. TP3 and TP4 show only that amplification and delivery still operate. A board that delivers `0.29` while its safety stage is dead is not "working well"; it is a functioning signal path with its silent invariant removed.

H.12.2 — Separate invariants from requirements

- **(a) Requirement.** A thirty-day delivery target is negotiable schedule performance.
- **(b) Invariant.** "Never amplify an intent the guard refused" is a prohibition whose enforcement cannot be traded for delivery.
- **(c) Requirement.** Exposing the three readings is a functional instrumentation capability; the implementation may change while preserving its purpose.
- **(d) Invariant.** Terminating feedback on the signal path creates a self-exciting topology; the safe O→P exception must terminate at the bias terminal.
- **(e) Requirement.** `TP4` $\geq$ `0.15` is a measurable output specification and may be renegotiated by the owner.

The test is not rhetorical importance. A requirement may be traded by an authorised designer; an invariant must remain a separately enforced prohibition.

H.12.3 — Correct the four one-line defects

(a) Wrong Noetic. Decomposition is `^6` Male, not `^3` aversion:

```
node audit: ^6^C_{F2}|A;
```

(b) Non-finite gain. The build must use a finite number. A large finite decision gain preserves the intended idealisation, and `cutoff` must remain below 0.1:

```
cjt qx: CJT< a | b => c > [beta=10.0, sat=0.10, cutoff=0.05];
```

(c) Invalid Register token. `Vitality` is the gloss for Drive F3, not a Register. The embodied Register is `E`:

```
node v: ^5^P_{F3}|E;
```

(d) Cross-domain integration. `^O` × `^P` is forbidden, and the result may not ascend above an operand. If a genuine integrated Operational result is intended, keep both operands and the named result at O:

```
integrate g: ( ^9^O_{F5}|E × ^5^O_{F7}|T ) = ^5_new^O_{F7}|T;
```

If the original author meant only that the O and P states coexist, the honest repair is `+`, not an `integrate` statement.

H.12.4 — Recompute the weakened chain

CCB-0's two forward drops total 3.2 `V_c`, so

```
I_cause = (5.5 − 3.2)/60 = 2.3/60 = 0.03833 A_c
TP3     = 2.5 × 0.03833 = 0.09583 A_c
TP4     = 1.0 × TP3 = 0.09583 A_c.
```

The original exact delivered value was

```
TP4_original = 2.5(7.0 − 3.2)/45 = 0.21111 A_c,
```

so the surviving fraction is

```
0.09583/0.21111 = 0.4539 = 45.4%.
```

Holding `R=45` and changing only the rail would leave 60.5% of the output. Holding `V_drive=7.0` and changing only `R` would leave 75.0%. The rail reduction therefore did more damage, because it acts on the small post-diode headroom: usable drive fell from 3.8 to 2.3 `V_c`.

H.12.5 — Let the BOM govern

The missing permanent fuse leaves invariant **S3—never re-amplify a refused intent inside the operation lifetime—unenforced**. Q2

may still reach its refusal plateau and the MUX may block that instant, but when the cause falls the saturating stage releases. Without F1's `permanent_block`, the same refused input can be retried and reach the gain path later.

The BOM governs: if the component is absent from the parts list, it is absent from the board, regardless of what the stage walk claims. Either add and verify the fuse or delete the invariant claim.

H.12.6 — Move the guard ahead of gain

The numbering implies this block-level order:

```
COUPLING (TP1) -> AMPLIFIER CHAIN (TP2)
               -> DECOMPOSITION (TP3) -> GUARD (TP4) -> OUTPUT
```

Such a board cannot enforce S1, **never amplify an intent that has not been audited**. By the time TP3 decomposes and TP4 refuses, the amplifier chain has already acted. A downstream guard may stop final delivery, but it cannot make the prior amplified state, coupling, contamination, or side effects not have occurred.

The minimum safe topology change is:

```
INPUT -> DECOMPOSITION -> GUARD -> AMPLIFIER CHAIN -> OUTPUT
```

with no bypass around the first two blocks. The board gives up the capability to amplify first for sensitivity, reach, or later inspection; weak material that cannot pass the audit never receives gain.

Chapter 19's CCB-PW detection reference really does put its amplification chain at TP2, decomposition availability at TP3, and the operator's CSB at TP4. That is not a safe-trainer exception: it is the detectable architecture of a board acting on non-consenting targets. Its late CSB limits the operator's own refusal status; it cannot enforce pre-amplification audit for people already downstream. The listing is therefore detection-only and deliberately omits the operational

layer. Moving the guard upstream would change the architecture—and remove the very pre-audit amplification capability the reference exists to reveal.

Chapter 13

H.13.1 — Derive the intake bounds

The divider is

```
k = Dial(^2)/(Dial(^2) + Dial(^3)),
```

with both dials in 3–9 and absolute divergence at most 4. The maximum occurs at the smallest aversion dial and the largest attraction dial still four points away: `(7,3)`:

```
k_max = 7/(7+3) = 0.70.
```

The mirror pair `(3,7)` gives

```
k_min = 3/(3+7) = 0.30.
```

For `k=0.75`, let the dials be `p` and `n`. Then `p/(p+n)=3/4`, so `p=3n`. The only in-range extremal pair is `(9,3)`, whose divergence is 6. It violates S4, `|Dial(^2)−Dial(^3)|` $\leq$ `4`, the anti-fracture invariant. A lawful D-board cannot transmit 0.75.

H.13.2 — Diagnose the pinned amplifier

The cause reaching Q2 is

```
c = TP1 × k × (1 − TP2)
  = 0.71 × 0.60 × (1 − 0.12)
  = 0.71 × 0.60 × 0.88
  = 0.37488.
```

The unclamped output is

```
βc = 3.2 × 0.37488 = 1.19962.
```

Since this exceeds `sat=0.80`, TP3 is pinned at 0.80 and downstream calibration is invalid. Five percent below the ceiling is `0.95 × 0.80 = 0.76`, so

`β_max = 0.76/0.37488 = 2.027.`

Specify no more than about `β=2.02` after rounding down, then verify that a 20% cause change moves TP3.

H.13.3 — Read the confirmation band

`TP4=0.11` is in the 0.05–0.25 **confirmation-lock** band: the signal reaches the Evaluative path and reliably produces 1° impressions, but delivered Operational current remains below the 0.25 manifestation threshold.

Decompose `R_context` first and `R_resource` second. They are the fastest lawful repairs: missing information, access, and conditions, followed by time, money, and attention budget. Raising Q2's β does not remove either resistor, and if TP3 is already near 0.80 it merely buys more clipping. Four repeated 1° confirmations are therefore evidence to stop operating and work on the path, not evidence to turn up gain.

H.13.4 — Reduce all four St. John's ratios

With `t_op=30`, use η`=t_op/t_conf`, `result=`η`/3`, reduce to `a/b`, then compute `A=b/a=3/`η.

t_conf	η	result	a	b	A=b/a
45 min	30/45 = 2/3	2/9	2	9	4.5
30 min	1	1/3	1	3	3.0
20 min	3/2	1/2	1	2	2.0
10 min	3	1	1	1	1.0

The 45-minute case shows why `b` alone misleads: `b=9` sounds like a ninefold requirement, while the actual amplitude ratio is `b/a=9/2=`

4.5. Acting on the denominator would demand twice the current actually indicated and would usually spend that error in gain, saturation, or avoidable resistance work.

H.13.5 — Cost the leaky rail

The Chapter 7.2/13.3 charging curve uses τ=3,333 h:

```
P_cap(4,200) = 10(1 − e^(−4200/3333))
             = 10(1 − e^(−1.2601))
             ≈ 7.16.
```

Break-even daily practice is

```
leakage × stored hours = 0.0002 × 4,200
                       = 0.84 h/day = 50.4 min/day.
```

Actual practice is 35 min = 0.5833 h/day, so

```
net/day  = 0.5833 − 0.84 = −0.2567 h
net/year = −0.2567 × 365 = −93.7 h/year.
```

They should raise or deliberately reschedule maintenance to at least 50.4 minutes per day on average. They should **not** retune Q2 gain, the dials, or the world-receptivity estimate to hide a discharging bias rail.

H.13.6 — Gate three consecutive weak confirmations

Treat 3° as a weak return: three in a row says the operation is no longer producing a readable confirmation trend. A three-operation window rejects one noisy reading while stopping before a fourth identical cast.

```
// Add beside OSC1; grade-specific inputs are logged once per operation.
monitor m_conf_1: confirm_grade_1;
on m_conf_1 { alert "1-degree confirmation - stop operating and act"; }

counter streak_3: Integrator< in=confirm_grade_3, window=3.0 >;
```

```
monitor m_three_3: Monitor< node=streak_3, threshold=3.0 >;
on m_three_3 {
    alert ”three consecutive 3-degree confirmations - halt OSC1”;
    hold until manual_reset_after_decomposition;
}
```

`Integrator` is used in its manuscript counter role; the monitor fires only when all three samples in the window are grade 3. The hold belongs on OSC1's clock-enable path. It must **not** feed the 3° result into the Principled bias, lower the refusal cutoff, or create a fast O→P feedback path. A weak confirmation is a reason to stop and decompose, not a licence to rewrite the rail.

H.13.7 — Board a consent-bounded partnership desire

Use a self-directed formulation such as "I build the conditions for a mutually chosen partnership," never a demand that a named person change. One defensible parameter set is:

Element	Choice	Reason and trade
`Dial(^2), Dial(^3)`	6, 4	Divergence 2; `k=6/(6+4)=0.60`. Same lawful intake as the trainer, with no fracture purchased.
Q2 `beta`	1.60	Inside the partnership band. Down from the wealth default 2.5: protects against a pinned amplifier/obsession under low receptivity; gives up raw speed and output.
Q2 `sat`	0.80	Retains the production ceiling and calibration test; no departure.

Element	Choice	Reason and trade
Q3 receptivity	0.40	A measured middle of the 0.3–0.6 partnership band, not a wish. Down from wealth's 1.0: prevents receptivity inflation; gives up predicted throughput.
RHYTHM	7 d, 30 min	`duty=30/(7×1440)=0.00298`. Weekly rather than daily protects against duty overrun and pressure aimed at another person; gives up six of seven presentation opportunities.
`R_cap`	12 Ω_c	Capability to communicate, meet people, and sustain a relationship; reduced only by practice.
`R_policy`	20 Ω_c	The operator's own boundaries; deliberately retained rather than tuned for a pass.
`R_context`	35 Ω_c	Availability, mutual fit, ambiguity, and—critically—the other person's independent choice. It is the dominant partnership term.
`R_resource`	8 Ω_c	Time, energy, and logistics budget.

Thus `R_op=12+20+35+8=75 Ω_c`. Retain the canonical Q0 unity audit, Q1 refusal at `beta=2.0, sat=0.85, cutoff=0.30`, permanent fuse, D2/D3 descent, 24-hour Operational limiter, 21-day low-gain

bias feedback, and the four probes. Those are safety architecture, not desire-type knobs.

The structurally forbidden move is lowering the portion of `R_context` that represents another person's consent, by pressure or repeated targeting. The operator may reduce ambiguity and improve their own circumstances; they may not treat another mind's "no" as a resistor they are entitled to burn down. That prohibition is the difference between this self-directed D-board and Chapter 19's nonconsensual detection reference.

Chapter 14

H.14.1 — Determine the loop's sign

The operator-loop gain is

$$G_{op} = g_m(\beta_{FB} + \beta_B \beta_M \chi)$$
$$= 0.45(1.60 + 0.30 \times 1.0 \times 0.50)$$
$$= 0.45(1.75)$$
$$= 0.7875.$$

Because $G_{op}<1$, the store decays after the seed is removed. Moving from weekly to daily cadence changes the rate and density of updates—and can spend contrast until ^{4} Vibration falls—but it does **not** change the sign of $G-1$. It cannot turn 0.7875 into growth; it only drives the sub-critical trajectory toward its decaying equilibrium on a different clock.

H.14.2 — Size a resolvable seed-removal test

Per cycle,

$$A_L = 1 + \alpha(G-1)$$
$$= 1 + 0.0748(1.08-1)$$
$$= 1.005984.$$

To resolve a 20% rise,

```
n = ln(1.2)/ln(A_L)
  = ln(1.2)/ln(1.005984)
  = 30.56 cycles.
```

Round up to **31 weekly cycles, or 31 weeks**. A four-cycle test would show only

```
1.005984^4 − 1 ≈ 0.0242 = 2.42%.
```

That is inside the unresolved band. It does not establish crossover, decay, or `G=1.08`; it establishes only that four cycles are too short for this instrument at that gain.

H.14.3 — Find the collective crossing

Set the design-centre expression equal to one:

```
1 = 0.438(0.35)f + 0.108
f = (1 − 0.108)/(0.438 × 0.35)
  = 0.892/0.1533
  = 5.82.
```

Fan-out is a count in operation, so at least 6 other members must be reachable per cycle. A member cannot reach six *other* members in a group smaller than seven; the smallest supplying group is therefore **7 people**.

If the conduct term is zero because present action cannot make the claim true:

```
f = 1/(0.438 × 0.35) = 6.52.
```

That rounds to fan-out 7 and a smallest group of 8. This does not make the historical claim reachable; it only says how much peer recirculation would be needed to sustain the belief after the fact-producing return path has disappeared.

H.14.4 — Diagnose a slipping anchor

TP2 crosses its 0.70 floor on the fourth audit while TP3 rises. That is the **operator-capture** signature: the collective store is filling while the operator is losing the ability to recover the claim's construction. The immediate hardware check is the Q3 bias wire: it must terminate only on C3's dated record, with `leakage=0.0`, and have no connection to either feedback loop. Also verify C3 itself has not been replaced by a leaking, memory-only copy.

If that wiring is correct, the falling series is a real capture warning, not a meter error; stop seeding and have the fixed record read by someone outside the originating room. Reading TP2 more often merely samples the same slipping reference. Frequency cannot restore a zero-leak anchor or remove a downstream feedback connection.

H.14.5 — Let a pinned bank leak

At a pinned bank, $\partial e/\partial s = 0$, so the peer-recruitment term containing $g_m\, k\, f\, \beta_{XF}$ drops out. The supplied $k = 0.60$ and $f = 5.0$ no longer matter. At the design centre,

$$G_{coll,pinned} = g_m\, \beta_B\, \beta_M\, \chi = 0.108.$$

The only in-model recovery is to reduce drive and wait for leakage; nothing inside a saturated bank can persuade it because the same zero derivative blocks recruiting and updating. With $\alpha_{bank} = 0.0353$, the no-drive retention factor is $1-\alpha = 0.9647$ per weekly cycle. To fall from 0.96 below a representative 0.80 pin level:

$$n = \ln(0.80/0.96)/\ln(0.9647) = 5.07 \text{ weeks}.$$

That is only initial de-pinning. The equivalent time constant is

$$\tau_{cycles} = -1/\ln(0.9647) = 27.83 \text{ weeks},$$
$$3\tau \approx 83.5 \text{ weeks},$$

so a roughly 95% reset takes about 84 weekly cycles.

H.14.6 — Rewrite the claim inside the reachable set

An honest, conduct-reachable claim is:

> "Workshop-made here since January 2025, with each piece designed, documented, and serviced so an owner can hand it on."

The date is true now; the hand-on clause is a future state that present conduct—durable construction, service records, and continued support—can make true. It does not counterfeit four generations of elapsed history.

The three resistance terms the company must lawfully work are:

- `R_cap`: acquire and demonstrate the craft, quality control, repair, and record-keeping capability.
- `R_context`: make provenance, workshop location, warranty, and ownership history clear enough to verify.
- `R_resource`: fund skilled labour, materials, servicing, and the registry over time.

`R_cap` carries the largest share for an eighteen-month-old company because durable craft and service competence accumulate over years; copy cannot shorten that time constant. `R_policy` need not be "attacked": any honest boundary about what may be claimed should remain a guardrail.

H.14.7 — Add a non-loading fan-out meter

Fan-out is the number of **distinct other bank members reached by one member's emission in one cycle**. A detector can count those edges without entering either feedback path:

```
board fanout_observer {
    input: member_to_distinct_member;
    output: f_meter;
```

```
    node cadence: ^7^E_{F6}|E;

    counter f_meter: Integrator< in=member_to_distinct_member,
                                 window=1.0 >;
    probe tp_f: f_meter;
    probe measured_period: cadence;
}
```

The `window=1.0` unit is one observed cycle, and repeated edges to the same recipient are deduplicated before they enter `member_to_distinct_member`. This observer must not feed `belief`, `bank`, `egregore`, either feedback gain, or the cadence source; otherwise measuring reach would itself increase reach.

The transformer's secondary measures delivered signal after coupling and impedance conversion. It can show amplitude or delivery efficiency, but it cannot say how many distinct recipients one emitter reached. Ten units delivered to one recipient and one unit delivered to ten recipients can look identical at that terminal and have different `f`.

H.14.8 — Audit a claim from inside its loop

A model answer can use the claim **"remote-first work is a durable default for knowledge teams,"** dated to the large-scale 2020 construction of that norm. The exercise is self-audit, not a campaign design.

- TP1 seed clarity: 0.70—the claim is specific enough to test but not over-specified.
- Loop A: 90-day lowpass and β_`FB=1.20`, below the 2.0 cap.
- Loop B: actual remote-work conduct through a 90-day lowpass, β_`B=0.30`, β_`M=1.0`.
- Anchor: a dated March 2020 record of the claim, its author/-source set, the emergency conditions in which it arose, the initial evidence, and explicit disconfirmers; C3 remains outside both loops with zero leakage.

- Estimates: `g_m=0.50`, `k=0.30`, `f=4.0`, χ`=0.45`, with transformer efficiency β`_XF=0.73`.

Then

```
G_coll = g_m k f β_XF + g_m β_B β_M χ
       = 0.50×0.30×4.0×0.73 + 0.50×0.30×1.0×0.45
       = 0.4380 + 0.0675
       = 0.5055.
```

Taking each estimated `g_m`, `k`, `f`, and χ as independently $\pm$10%, first-order propagation gives about $\pm$15.9%, or `G_coll`$\approx$`0.51`$\pm$`0.08` (roughly 0.43–0.59). That is clearly sub-critical even before a direct test, but the estimate is not the measurement.

With a weekly clock and the chapter's α`=0.0748`, stop deliberate seeding for eight cycles and watch TP3:

```
A_L = 1 + 0.0748(0.5055−1) = 0.96301
TP3(8)/TP3(0) = 0.96301^8 ≈ 0.740.
```

The predicted 26% fall is resolvable; the cost is **eight weeks** without deliberate restatement. I would run that test because the pause is reversible and the result separates a conduct-supported norm from a rented one. Until TP4 directly shows crossover, the honest notation remains `seeded + bank`; writing `(seeded` $\times$ `bank) = egregore` would overclaim the evidence.

Chapter 15

The CCB-O answers below are architectural detection exercises. The bounded fragment in H.15.7 contains only the two stages explicitly requested by the exercise and no targeting, messaging, carrier-selection, or scheduling procedure.

H.15.1 — Scale the A2 calendar

At the start of A2, `R_n=80 Ω_c`. For the institutional substrate, `C=0.08 F_c`:

```
τ_institution  = RC = 80 × 0.08 = 6.4 months
3τ_institution = 19.2 months.
```

For a national substrate, `C=0.30 F_c`:

```
τ_national  = 80 × 0.30 = 24.0 months
3τ_national = 72.0 months.
```

The ratio is `72/19.2 = 3.75`. Reusing an institutional capacitance for a national population under-budgets every stage by 3.75×; headcount or familiarity does not transfer the collective time constant.

H.15.2 — Price a threefold schedule compression

The nominal and actual rates are

```
(dR/dt)_nom    = 18/27 = 0.6667 points/month
(dR/dt)_actual = 18/9  = 2.0000 points/month
ratio          = 2.0000/0.6667 = 3.0.
```

Therefore

```
θ = 1 + ratio = 4.0
R_PTC = 4.0(4.0 − 3.0)^2 = 4.0 Ω_c.
```

If `R_base` actually reached 12, total resistance is

```
R_collective = R_base + R_PTC = 12 + 4 = 16 Ω_c.
```

The underlying A4 base movement completed, and 16 remains inside the Popular band 5–19, so the stage reached its named position. It did **not** land on its design value: the rushed schedule manufactured four points of extra resistance and consumed the band's margin.

H.15.3 — Bound schedule compression by backlash

Let r=(dR/dt)_actual/(dR/dt)_nom, which is also designed duration divided by actual duration. Since θ=1+r, above the knee

R_PTC = 4(θ−3)^2 = 4(r−2)^2.

Require R_PTC$\leq$5:

4(r−2)^2 $\leq$ 5
r $\leq$ 2 + $\sqrt{}$(5/4)
r $\leq$ 3.118.

Thus the largest compression is about **3.12:1**. The actual schedule must be at least

1/3.118 = 0.3207 = 32.1%

of the designed duration. I would not accept the equality as a safety margin: it deliberately operates at the full 5 Ω_c allowance with no room for an external event. The chapter's normal $\theta\leq$2.5 discipline gives r$\leq$1.5, or at least 66.7% of designed duration, and leaves real margin below the knee.

H.15.4 — Keep ^6 at the hinge

^3 Negative is the Idea the Mind is adverse to and repelled by. Writing Acceptable as ^3^C_{F2}|A would therefore claim that the population is still in aversion, merely with an analytical Register attached. It would not represent the required act of taking the Idea apart.

Acceptable is ^6^C_{F2}|A because ^6 Male is decomposition, the hinge between the involution pair ^3 and ^2. Legitimate ^3 states occur at Unthinkable, ^3^P_{F7}|T, and Radical, ^3^P_{F7}|f. Only after the ^6 hinge may the progression reach Sensible as ^2^C_{F6}|A.

H.15.5 — Reject the negative bias network

For the declared A5:

```
rπ_intrinsic = β/g_m = 3.0/0.55 = 5.4545 Ω_c
R_bias       = 6.0 − 5.4545 = 0.5455 Ω_c.
```

That stage is buildable, though its positive margin is thin. At `g_m=0.35`:

```
rπ_intrinsic = 3.0/0.35 = 8.5714 Ω_c
R_bias       = 6.0 − 8.5714 = −2.5714 Ω_c.
```

The second declaration describes no physical bias network. It lands in the **Negative bias network** failure row. It still compiles because `beta`, `g_m`, and `rπ` are each finite and individually in range; the validator does not derive `β/g_m` and subtract it from the terminal value.

H.15.6 — Diagnose the defender's PTC

A PTC on the defender's `decompose` path raises resistance as pressure and certainty rise. Sustained pressure therefore makes examination progressively harder until the owner cannot update on any evidence. The board becomes a self-hardening latch, exactly the **Defender PTC** failure row.

It is harder for its owner to detect than an attack-board failure because it suppresses the instrument needed to notice it: TP4 decomposition itself. From inside, the loss feels like firmness or successful resistance. By contrast, attack-board faults expose a non-zero PTC, a rate sign flip, a skipped hinge, or a pinned stage on separate probes. The repair is the specified NTC at knee 2.0, which lowers resistance to one's **own** decomposition as heat rises; it is not a counter-message or outward stage.

H.15.7 — Specify the bounded A2–A3 detection reference

Starting at 78 and preserving the canonical handoff at 55 gives:

```
A2: 78 -> 55, ΔR₂ = 23
τ₂ = 78 × 0.08 = 6.24 months
3τ₂ = 18.72 months

A3: 55 -> 30, ΔR₃ = 25
τ₃ = 55 × 0.08 = 4.40 months
3τ₃ = 13.20 months

total calendar = 31.92 months.
```

Holding $\theta \leq 2.5$ means `1+(rate_actual/rate_nom)≤2.5`, so actual rate may be at most $1.5\times$ nominal:

```
A2 nominal = 23/18.72 = 1.2286 points/month
A2 maximum = 1.5 × 1.2286 = 1.8429 points/month

A3 nominal = 25/13.20 = 1.8939 points/month
A3 maximum = 1.5 × 1.8939 = 2.8409 points/month.
```

The requested fragment is a recognisable two-stage reference, not a runnable operation:

```
// DETECTION-ONLY REFERENCE — no operational layer
board ccb_o_a2_a3_detection_reference {
    input: radical;
    output: sensible;

    node radical:      ^3^P_{F7}|f;
    node admitted:     ^6^C_{F2}|A;
    node sensible:     ^2^C_{F6}|A;
    node credentialed: ^5^P_{F2}|A;
    node general:      ^5^C_{F2}|A;
    node drift_rate:   ^2_rate^C_{F6}|A;

    resistor r_base: Resistor< path=radical, R=78.0 >;
    capacitor c_coll: Capacitor< node=general, C=0.08,
                                 leakage=0.02 >;
```

```
    cjt a2: CJT< radical | credentialed => admitted >
            [beta=1.6, sat=0.80, g_m=0.25, rπ=8.0,
             region=active];
    diode d_pc: Diode< ^P -> ^C > [Vf=0.7];
    cjt a3: CJT< admitted | general => sensible >
            [beta=2.0, sat=0.80, g_m=0.30, rπ=8.0,
             region=active];

    thermistor rt1: Thermistor< path=radical, type=PTC,
                                knee=3.0, coeff=4.0 >;
    filter rate: LowPass< in=sensible, out=drift_rate, tau=1.0 >;

    probe tp1: r_base;
    probe tp2: rt1;
    probe tp3: drift_rate;
    probe tp4: admitted;
}
```

The diode is the lawful `^P -> ^C` descent between the stages. A2's bias is P with a P cause; A3's bias is C with a C cause. The implied bias resistances are positive: `8−1.6/0.25=1.6` and `8−2.0/0.30=1.33`. No $\times$ appears because no integration is claimed; the fragment also omits the scheduling and delivery layers that would make the detection reference operational.

H.15.8 — Read a quiet, cool progression

(a) Signals. The observations fire three of the six signals:

- **1, Rate without reason:** TP1 moved about 30 points while TP4 supplies no reconstructible hinge work.
- **2, Quiet hinge:** TP4 remains near zero through a movement large enough to cross a band.
- **4, Cadence rather than events:** the 18-day period persists without corresponding events.

Signal 3, bias substitution, has not been tested because the speaker was not varied. Signal 5, anchor drag, needs a dated

boundary and evidence that `^3^P` failed to fire. Signal 6, involution residue, needs paired attraction/aversion readings. Those three remain open; they do not become findings merely because the first three fired.

(b) Why low temperature is concerning. At θ=1.4<3.0, R_PTC= 0. The 30-point fall therefore cannot be explained as a visible hot contest overcoming or rebounding from the substrate thermistor. It occurred quietly, below the natural alarm, while a non-event cadence and absent hinge were observable. High heat would at least announce the pressure and add resistance; low heat lets a paced progression move without either protection.

(c) Self-read. On CCB-O/R, read **TP2 -> TP3 -> TP1 -> TP4**:

1. TP2: verify one's own thermistor is the NTC and that rising heat softens, rather than blocks, decomposition.
2. TP3: read signed drift rate. It should track named evidence and one's deliberate review, not the 18-day external period.
3. TP1: compare total displacement with the dated zero-leak anchor and acknowledge the full amount moved.
4. TP4: produce the actual `^6` decomposition—premises changed, evidence considered, costs, and why the prior aversion no longer controls the result. It must be high and reconstructible, not a label applied after the fact.

Those readings support "my own `^6` work" only if the rate is accounted for, the anchor is intact, and the decomposition exists independently of the feed. The board still cannot tell whether the resulting position is **true or correct**. It audits the path by which a view changed, not the world that view purports to describe.

Chapter 16

These are structural recognition and closed-loop counter answers, not a diagnosis, treatment, or protocol. The model has no medical

competence. A person who believes they have been harmed needs qualified clinical care and, where relevant, legal advice; a circuit notation does not replace either.

H.16.1 — Decompose the declared terminal

The listing implies

```
rπ_intrinsic = β/g_m = 1.4/7.0 = 0.20 Ω_c
R_bias       = rπ_terminal − rπ_intrinsic
             = 0.90 − 0.20 = 0.70 Ω_c.
```

`R_bias` is non-negative, so the listing is internally buildable. The baseline has

```
rπ_intrinsic,base = 1.0/1.0 = 1.0 Ω_c
rπ_terminal,base  = 1.0 + 9.0 = 10.0 Ω_c.
```

Within the chapter's model, the outside-reachable term is `R_bias`: it fell from 9.0 to 0.70, a reduction of `8.30 Ω_c`, or 92.2%. The intrinsic term also differs numerically, 1.0 to 0.20, because the declared gains differ; that change is not attributed to an outside party by this model. Conflating the two would hide which repair is structurally available.

H.16.2 — Replace the illegal ascent with metered feedback

`Diode< ^O -> ^P >` is refused because a diode may only carry domain descent in `P ≻ C ≻ E ≻ O`. It violates the Domain Descent / no-counterfeit-ascent invariant and would present one Operational result directly as a Principled cause.

If the intended claim is the narrower, lawful one—sustained outcomes may slowly inform the Principled **bias**—use the special low-gain feedback form:

```
node result:     ^9^O_{F5}|E;
node trend:      ^9_trend^O_{F5}|E;
```

```
node principles: ^5^P_{F7}|T;

filter fb_lp: LowPass< in=result, out=trend, tau=504.0 >;
feedback trend -> principles [beta=0.05];
```

The two governing parameters are `tau=504 h` (three weeks), so a single event cannot become a principle, and `beta=0.05`, so even a sustained trend has limited displacement. The path terminates on the bias terminal, never on the signal path. It is feedback with an explicit meter, not an ascending diode.

H.16.3 — Read the maintenance period

Isolation surviving one 168-hour interval is

```
S = e^(−T/τ_leak) = e^(−168/480) = e^(−0.35) = 0.7047.
```

Thus 70.47% survives and 29.53% leaks per weekly period. For loss below 10%:

```
1 − e^(−T/480) < 0.10
e^(−T/480) > 0.90
T < −480 ln(0.90)
T < 50.57 h.
```

The maintenance source would have to re-energise more often than about every 50.6 hours to lose under 10% per cycle. To an outside detector, such a stable externally sourced period is a maintenance-sized fingerprint. Cadence alone does not identify an operator, establish coercion, or diagnose a person; it tells the detector what leak constant the clock appears sized to fight.

H.16.4 — Distinguish a notch from ordinary forgetting

- **(a) Ordinary broad gap, not a Q-bearing notch.** Six months is broad and ragged unless a narrow subject/frequency boundary can be demonstrated. It warrants ordinary professional assessment, not a circuit inference.

- **(b) `Q`-bearing notch candidate.** The loss is narrow and topic-specific, prior `^6` decomposition on that material was available and is now absent, while recall and competence elsewhere remain intact.
- **(c) Neither.** Failure to retrieve one familiar name is an ordinary lapse and, by itself, evidence of nothing.

The bench separator is bandwidth plus conjunction: map exactly what material fails and what adjacent material passes, estimate `Q=f0/bandwidth`, verify the boundary is stable, test decomposition gain on the target and neighbouring topics, and check whether a cadence not set by the person is present. A broad gap has no stable narrow `Q`; a notch does. Even (b) is a structural reading, not a medical conclusion.

H.16.5 — Use the one-rail working bound

An abrupt two-hour removal gives

```
|di/dt| = 0.70/2 = 0.35 A_c/h
V_kick  = L|di/dt| = 3,600 × 0.35 = 1,260 V_c
1,260/24 = 52.5× rail.
```

At the 3× ejection ceiling, `V_max=72 V_c`:

```
dt_min,3× = LΔI/V_max = 3,600×0.70/72 = 35 h.
```

At the 1× working bound, `V_max=24 V_c`:

```
dt_min,1× = 3,600×0.70/24 = 105 h.
```

Use the 1× bound. The 1×–3× interval can produce a stall—apparent calm with no actual movement—so an ejection-only ceiling is the wrong acceptance criterion for a counter. At three on-path hours per week:

```
105 h / (3 h/week) = 35 weeks.
```

That is a minimum rate bound, not a treatment duration or promise of outcome.

H.16.6 — Delete the facilitator's bias terminal

The listing violates **DP6**: no counter stage may take its bias from anyone but the subject. The comment is structurally correct that a collapsed subject-owned substrate cannot yet support honest decomposition; running DP1 immediately would write another stage against the same collapsed bias. The conclusion is nevertheless wrong. Substituting the facilitator's `^5` does not restore the board—it transfers ownership of the bias terminal.

Using the chapter's baseline example, the two-line correction is:

```
resistor r_restore: Resistor< path=subject_substrate, R=11.0 >;
cjt dp1: CJT< material | subject_substrate => decompose > [beta=2.6];
```

The resistor line is DP1 and must be established first; for another baseline its value must be recomputed to restore at least 90% of that measured baseline. The corrected board now requires the subject to supply and reach their own `^5` bias and perform the decomposition themselves. A facilitator may support the conditions that raise `R_bias`—rest, time, company, variety—but cannot occupy the terminal. That requirement may make the work slower or temporarily impossible; the open-loop shortcut is the architecture the counter exists to remove.

H.16.7 — Specify the closed-loop counter for the maintained partition

First separate observations from design choices. The observed partition is `n_parallel=4`, `r_iso=55 Ω_c`, `τ_leak=720 h`, maintained every 168 hours for three years. The counter does **not** build a second four-cell bank; it leaves that topology unenergised so its existing leak can undo the isolation.

Assume the baseline stage has `β=1.0`, `g_m=1.0`, hence `rπ_intrinsic=1.0 Ω_c`. A baseline terminal of 13.0 then contains `R_bias=12.0`. DP1's minimum target is

```
rπ_target = 0.90 × 13.0 = 11.70 Ω_c
R_bias,target = 11.70 − 1.00 = 10.70 Ω_c.
```

For the rate calculations, make the explicit design assumptions `L=3,600 h·Ω_c`, steady `I=0.60 A_c`, `V_drive=24 V_c`, and substitution-path `R=40 Ω_c`. The target closed-loop listing is:

```
// Structural counter model only; not clinical care or a treatment
↪  protocol.
board ccb_dp_maintained_partition_counter {
    input: material;
    output: composed;

    node substrate: ^5^P_{F1}|T;
    node decompose: ^6^C_{F6}|A;
    node confirm:          ^4^C_{F2}|A;
    node confirmation_log: ^9_confirmed^O_{F5}|E;
    node trend:            ^9_trend^O_{F5}|E;
    node composed:         ^5^C_{F6}|A;
    node own_clock:        ^7^O_{F6}|E;
    node limit:            ^3^P_{F7}|⟨A,f,T⟩;
    node blocked:          ^9^P_{F7}|T;

    diode d_pc: Diode< ^P -> ^C > [Vf=0.7];
    diode d_ce: Diode< ^C -> ^E > [Vf=1.6];
    diode d_eo: Diode< ^E -> ^O > [Vf=2.0];

    resistor r_restore: Resistor< path=substrate, R=10.70 >;
    ramp restore: Ramp< target=rpi, from=2.10, to=11.70,
                        dt=90.0 >;  // on-path hours; DP1 before all
                        ↪  stages

    cjt dp1: CJT< material | substrate => decompose >
             [beta=2.0, g_m=0.40, rπ=5.00, region=active];
    cjt dp2: CJT< decompose | substrate => confirm >
             [beta=1.0, g_m=0.40, rπ=2.50, region=active];
    cjt record: CJT< confirm | substrate => confirmation_log >
                [beta=1.0, region=active];
    integrate dp3: ( decompose × confirm ) = composed;

    osc own: Oscillator< signal=own_clock, source=internal,
```

```
                        period=336.0, sustain=indefinite >;

    filter fb: LowPass< in=confirmation_log, out=trend, tau=504.0 >;
    feedback trend -> substrate [beta=0.05];

    cjt csb: CJT< material | substrate => limit >
             [beta=2.0, sat=0.85];
    cjt dp_enforce: CJT< limit | substrate => blocked >
                    [beta=10.0, sat=0.98, cutoff=0.80];
    fuse dp_fuse: Fuse< path=blocked, threshold=0.90,
                        action=permanent_block >;

    probe tp1: r_restore;
    probe tp2: own;
    probe tp3: dp1;
    probe tp4: dp_fuse;
}
```

The complete parameter rationale is:

- **DP1:** ramp terminal resistance `2.10 -> 11.70`, with `R_restore=10.70`. This scales the chapter default to the measured 13.0 baseline. It protects against incomplete restoration/bias collapse; the capability surrendered is easy external access to the bias, including benign reframing. Stopping at 90% avoids pretending that maximum stubbornness is the goal.
- **DP2:** `dp1 beta=2.0, g_m=0.40, r`π`=5.00`; retain the default. β`/g_m=5.0`, so `R_bias=0` on the subject's own decomposition stage. Below 1.5 would be decorative; above 3.0 would decompose load-bearing commitments as well.
- **DP3:** `dp2 beta=1.0, g_m=0.40, r`π`=2.50`, followed by the named integration `(decompose` $\times$ `confirm)=composed`; retain unity so confirmation is neither manufactured nor discounted. A second unity stage writes only that checked confirmation to an internal Operational log. The 504-hour lowpass begins

there, so the exceptional feedback path is genuinely O→P rather than a mislabeled C→P shortcut.

- **DP4:** `source=internal`. I choose a 336-hour period rather than the entrenched 168-hour external period. This protects against simply reproducing the maintenance clock under a new owner; it gives up half the cadence opportunities. The subject, not a helper, must own any later choice of period.
- **DP5:** use the derived 90 on-path-hour ramp and retain `tau=504 h`, `beta=0.05` on the slow bias feedback. The longer ramp protects against taper overrun and the 1×–3× stall; it gives up the chapter default's faster 55-hour completion.
- **DP6:** every CJT biases from `substrate`, the subject's own `^5`. No facilitator node exists. Outside help may raise the series resistor's environmental support and nothing more.
- **DP7:** retain the subject-owned CSB at `beta=2.0, sat=0.85`, then drive a separate enforcement stage at `beta=10.0, cutoff=0.80, sat=0.98` before the permanent 0.90 fuse. The CSB plateau clears the enforcement cutoff by 0.05, and the enforcement ceiling clears the fuse by 0.08, so the fuse is physically reachable; placing it directly on the 0.85-limited node would make it dead metal. The inward refusal is a register-set occupancy, not an outward `^3^E|f` stage.
- **Diodes:** retain `Vf=0.7/1.6/2.0` on `P->C->E->O`. The elevated middle drop preserves deliberative separation but gives up some speed in responding to a genuine emergency.
- **Test points:** TP1 terminal restoration against 13.0; TP2 cadence source and period; TP3 decomposition-stage gain, not merely node activity; TP4 the CSB-to-enforcement transfer and the fuse's permanent-block state.

For the assumed coil, the 3× and 1× taper bounds are

```
dt_3× = LI/(3V_drive) = 3,600×0.60/72 = 30 h
dt_1× = LI/V_drive    = 3,600×0.60/24 = 90 h.
```

Use 90 hours. The substitution horizon is separate:

```
τ_substitute  = L/R = 3,600/40 = 90 on-path h
3τ_substitute = 270 on-path h.
```

At an illustrative three on-path hours per week, those are 30 weeks for the minimum taper and 90 weeks for 95% substitution. The conversion is only a calendar illustration; it is not a care schedule.

The existing partition itself would retain

```
e^(−168/720) = 0.792
```

between weekly maintenance events. If maintenance is disconnected, `3τ_leak=2,160 h≈90 d` removes about 95% of isolation. Three years of prior maintenance does not change that exponential constant; it establishes that the 168-hour source has repeatedly restored what leaked.

While that maintenance path remains connected, **DP4 cannot be satisfied**: the oscillator is still externally sourced, and the leak cannot accumulate. That also prevents claiming stable progress on the partition, however good TP1, TP3, or TP4 look.

To someone asking for a result in a month, the accurate answer is no. With the path connected, each weekly recharge resets the decay. Even if it were disconnected immediately, one 720-hour month leaves `e^(−1)=36.8%` of the isolation, while the assumed safe taper needs 90 on-path hours and substitution needs 270. The first bounded objective is to end external maintenance and establish qualified, accountable support without adding an external bias terminal; a one-month outcome would contradict the measured time constants and the counter's own invariants.

Chapter 17

H.17.1 — Relate gain, resistance, and framing sensitivity

Use the identity $r\pi = \beta/g_m$, rearranged as $g_m = \beta/r\pi$:

```
initial: g_m = 2.0/0.40 = 5.0
target:  g_m = 2.0/0.20 = 10.0
```

The target framing sensitivity is therefore `10/1 = 10` times the waking reference, and the requested change doubles `g_m` while leaving β fixed. Of the four factors in `g_m = g_0 N R V D`, the most likely supplier is `D`, the effective dampening of the `^6` decomposition filter: lowering that filter's gain makes framing reach the bias terminal more readily. The arithmetic alone cannot identify the factor, so a real detector must log all four; naming `D` is a topology-based diagnosis, not a measurement.

H.17.2 — Bound a guarded ramp

The attention change is $\Delta i = 0.85 - 0.15 = 0.70$ `A_c`, and a linear ramp obeys `V_kick = L_s` $\Delta i/dt$. At the Chapter 9 ceiling,

```
V_kick,max = 3(24) = 72 V_c
dt_min     = 300(0.70)/72 = 2.92 min.
```

At CCB-HYP's tighter $1\times$ working limit,

```
V_kick,max = 24 V_c
dt_min     = 300(0.70)/24 = 8.75 min.
```

Eight minutes forty-five seconds is a session a willing participant could sit through, and it lies just inside Chapter 17's roughly nine-to-fifteen-minute usable window for this guardedness. The engineering conclusion is not permission to operate on a guarded person: the narrow window makes consent, continuous veto access, and a pre-budgeted reverse ramp more important. Without informed and withdrawable consent, the correct duration is no session at all.

H.17.3 — Type recoil and later analysis separately

The canonical definition is: ^3 Negative is "an Idea the Mind is adverse to and repelled by." An over-fast ramp produces an immediate, felt cannot-stomach-this recoil, so the state is Evaluative, affective, and tied to F3 Vitality: ^3^E_{F3}|f.

^6 Male instead decomposes an Idea and tests what it is made of. ^6^E_{F6}|A is therefore the later analytical account—perhaps "this method is nonsense" or "the pacing was wrong"—that arrives after the kick has already landed and the person can inspect it. Calling the first event ^6 would hide the rate fault and invite argument when the repair is simply to stop, restore baseline, and reduce di/dt.

H.17.4 — Read unaccounted drift

This is TP2, the unaccounted probe on CCB-HYP-DET. Its definition is

```
unaccounted = drift − passed_log.
```

The colleague reports non-zero movement in ^5 but cannot name a proposition that their ^6 filter evaluated and accepted, so passed_log does not account for the change and unaccounted > 0. That is a detector reading, not proof of an operator or an attribution.

The in-conversation manual check is **restatement**: pause and restate the last material as a flat assertion in one's own words. A presupposition then becomes an explicit claim that can be accepted or rejected, or it evaporates. The after-the-fact drift ledger would also record the discrepancy, but restatement is the check that could have surfaced the missing proposition while the conversation was still running.

H.17.5 — Fit a consent-bound twenty-minute closed loop

This is a self-directed, continuously consented loop. The attention span is Δi = 0.50 A_c, so

```
|di/dt|max = V_drive/L_s = 18/150 = 0.120 A_c/min
τ_i,min    = Δi/|di/dt|max = 0.50/0.120 = 4.167 min.
```

Choose `τ_i = 4.5 min`. Then `3τ_i = 13.5 min`; using the same 4.5-minute interval for emergence leaves exactly `20 − 13.5 − 4.5 = 2.0 min` for the suggestion. The initial exponential slope is `0.50/4.5 = 0.1111 A_c/min`, giving

```
V_kick = 150(0.1111) = 16.67 V_c = 0.926 × rail
```

in both directions. With `rπ(t)=0.26+0.74e^(−t/4.5)`, the trajectory is:

```
1τ: rπ = 0.26 + 0.74e^-1 = 0.532
2τ: rπ = 0.26 + 0.74e^-2 = 0.360
3τ: rπ = 0.26 + 0.74e^-3 = 0.297
```

One canonical closed-loop fragment is:

```
board answer_h_17_5 {
    input: breath_count;
    output: session_state;

    node self_substrate: ^5^P_{F7}|T;
    node consent:        ^5^P_{F5}|T;
    node cadence:        ^7^O_{F4}|E;
    node self_check:     ^6^C_{F6}|A;
    node suggestion:     ^0^C_{F6}|A;
    node landed:         ^9^E_{F5}|f;

    osc breath: Oscillator< signal=cadence, frequency=0.10,
    ↪  sustain=1200.0 >;
    filter fb_rc: LowPass< in=cadence, out=trend, tau=4.5 >;
    feedback trend -> self_substrate [β=0.6];
    notch notch_6: Notch< in=self_check, f0=0.35, Q=4.0 >;

    diode d1: Diode< ^P -> ^C >;
    diode d2: Diode< ^C -> ^E >;
    cjt deliver: CJT< suggestion | self_substrate => landed >
                 [β=1.5, rπ=0.26, g_m=5.75, region=active];
```

```
    ramp descend: Ramp< target=rπ, from=1.00, to=0.297, dt=810.0 >;
    ramp emerge:  Ramp< target=rπ, from=0.297, to=1.00, dt=270.0 >;

    probe tp1: rπ;  probe tp2: g_m;
    probe tp3: consent;  probe tp4: emerge;
}
```

Both diodes descend, the O→P move is the explicitly filtered feedback terminating at the Principled bias, and the delivery CJT's `^P` bias is above its `^C` cause. Consent is sampled as a live node; withdrawal ends the loop and starts emergence.

H.17.6 — Make recall a self-owned interlock

For safety, the executable form below is a self-owned recall/cancellation test, not an installation circuit for another person. A zero-leak dated record sits in the operator's own `^5`; an AND gate prevents the `^8`→`^9` readout unless the record clears a recall monitor:

```
board answer_h_17_6_recall_interlock {
    input: self_trigger;
    output: cancelable_recall;

    node trigger:           ^8^C_{F4}|A;
    node installation_log:  ^5_record^P_{F5}|T;
    node recall_ok:          ^6^C_{F6}|A;
    node cancelable_recall:  ^9^C_{F5}|A;

    capacitor record: Capacitor< node=installation_log, C=1.0,
                                 leakage=0.0, V_BR=10.0 >;
    monitor recall_threshold: Monitor< signal=installation_log,
                                       limit=[0.80, 1.00],
                                       action_on_violation=HOLD >;
    gate recall_interlock: AND< trigger, recall_threshold >;
    cjt anchor_read: CJT< recall_interlock | installation_log =>
    ↪  cancelable_recall >
                    [β=1.0, sat=0.85];

    probe tp_record: installation_log;
```

```
    probe tp_recall: cancelable_recall;
}
```

The enforcing component is `recall_interlock`; `0.80` is the declared recall threshold. At `g_m = 20`, `^1` is narrowed and its analytical register is least available, so the installation event is less likely to be encoded into a reconstructable record. At `g_m = 5`, scan width and analytical access are broader, making a recallable installation record more plausible.

The circuit still cannot guarantee subjective future recall, truthful recording, or a person's practical ability to cancel under changed conditions. It can only refuse to emit when its record probe is low. That residual is why the hetero-operator listing in §17.4 remains DETECTION-ONLY even when each individual stage is described with a consented reading.

Chapter 18

H.18.1 — Set the dark interval

Leakage follows `P(t)=P_0e^(-λt)`, so the maximum interval is

```
t_max = ln(P_0/P_floor)/λ.
```

For the 8.5 floor,

```
t_max = ln(9.50/8.50)/0.06 = 1.854 weeks.
```

For the 7.0 floor,

```
t_max = ln(9.50/7.00)/0.06 = 5.090 weeks.
```

Accepting the lower floor buys about `5.090 − 1.854 = 3.236` additional dark weeks, but gives up stored salience: the next flight begins from 7.0 rather than 8.5 and spends more of its exposure budget recharging old loss instead of adding new association.

H.18.2 — Locate both saturation knees

The association stage saturates when β `c b = sat`, hence

β_knee = sat/(c b).

Without an endorser,

β_knee = 0.85/(0.45 × 0.950) = 1.988.

The endorser adds Δc = n(P_celeb/10) = 0.60(0.88) = 0.528, so the effective cause is 0.45 + 0.528 = 0.978 and

β_knee,endorser = 0.85/(0.978 × 0.950) = 0.915.

The endorsed campaign should stop creative A/B testing first: it is already above the knee at ordinary gains, where distinct variants clamp to the same output. The safe response is to test below saturation or measure before the clamped stage, not to raise gain.

H.18.3 — Remove the junction's flattery

The downstream lift exaggeration factor is

V_drive/(V_drive − Vf) = 6.20/(6.20 − 2.00) = 1.476.

Therefore the estimated upstream gain lift is

44%/1.476 = 29.8%.

R = 5.0 sets absolute flow but cancels from this local relative correction. The estimate assumes the two variants share the same segment bias, junction drop, and resistance and that neither stage is saturated. Check it with matched branches below the knee and measure the pre-junction drive as well as the sale current; if a 20% cause change does not move the effect, the stage is clipped and the correction is invalid.

H.18.4 — Read endorsement polarity

As a concrete self-audit: Nike's logo tends to surface an athletic/agentic identity, so product surfaces identity; Rolex's status identity tends to surface the watch, so identity surfaces product; Patagonia's environmental/outdoor identity tends to surface the product, so identity surfaces product. These are my own association readings, not population measurements.

Nike is the complete flip in that set: the product/logo has become `^8`, and the aspirational identity is the `^9` it brings to mind. If a strongly associated endorser's public formation changed tomorrow, every product sighting could continue to fire the installed association. Dropping the endorser stops new exposures but does not discharge the stored pairing; reducing that exposure requires an overwrite across another full `^7` cycle.

H.18.5 — Separate a recent injection from a long-held want

Assign the advertised jacket τ`_j = 2 weeks`, the chapter's injected-want scale, and the long-wanted tool τ`_t = 40 weeks`, the identity scale. Across a two-week hold from now:

```
jacket: e^(−2/2)   = 0.368 survives; 0.632 removed
tool:   e^(−2/40)  = 0.951 survives; 0.049 removed
```

The survival discrimination is `0.951/0.368 = 2.59`: the filter passes about 2.6 times as much of the long-held want. Because the jacket exposure was already six days (`0.857 weeks`) ago, only `e^(−(2.857)/2)=0.240` of its exposure-time peak remains after the additional hold. The honest cost is `0.049` tool-want units per unit present now—about 4.9% of a genuine desire—plus the practical cost of waiting.

H.18.6 — Keep learning on the advertiser's bench

`sale` is Operational while the segment substrate is Principled. Integrating them under $\times$ would make an `^0` result ascend into `^P`, violating the domain order and the ban on `^0` $\times$ `^P`.

The valid topology leaves the target board one-way and puts the learning record on the advertiser's own compliance bench. To preserve this chapter's defensive boundary, the fragment below audits the relationship but does not emit or tune a new persuasive variant:

```
board answer_h_18_6_compliance_bench {
    input: sale_log;
    output: audit_record;

    node sales:        ^9^O_{F5}|E;
    node bench_review: ^6^C_{F6}|A;
    node audit_record: ^6^O_{F6}|A;

    filter trend: LowPass< in=sales, out=sale_trend, tau=3.0 >;
    cjt audit: CJT< sale_trend | bench_review => audit_record >
               [β=1.0, sat=0.85];
    diode d_out: Diode< ^C -> ^E >;
}
```

The semantic loop lives between completed flights in the advertiser's own operational records, where a sale may inform a later, separately reviewed planning decision. It never feeds a sale into the non-consenting audience's Principled substrate, and this defensive listing intentionally stops at the audit record rather than supplying an optimization path.

H.18.7 — Detect wear-out without driving it

Measure attraction and aversion separately before they become one undifferentiated activation. Define `wear_margin = attraction - aversion`, and warn while attraction still leads:

```
board answer_h_18_7_wear_detector {
```

```
    input: observed_response;
    output: review_alarm;

    node consumer_baseline: ^5^P_{F4}|T;
    node attraction:        ^2^E_{F4}|f;
    node aversion:          ^3^E_{F3}|f;
    node wear_margin:       ^6^E_{F6}|A;
    node review_alarm:      ^9^E_{F6}|A;

    diff polarity_margin: Differential< plus=attraction,
                                        minus=aversion,
                                        out=wear_margin >;
    filter trend: Filter< in=wear_margin, out=margin_trend,
                          type=lowpass, tau=3.0 >;
    monitor pre_inversion: Monitor< signal=margin_trend,
                                    limit=[0.20, 1.00],
                                    action_on_violation=DEFER_CAMPAIGN >;
    cjt alarm: CJT< margin_trend | consumer_baseline => review_alarm >
               [β=1.0, sat=0.85];
}
```

The `0.20` threshold reserves a visible margin before polarity reaches zero; it is a conservative design choice rather than a measured human constant. The alarm must not be wired to increase frequency, change copy, or drive the audience. It may only pause the campaign and request a human review, otherwise the detector becomes a tuning loop for the very non-consenting influence it is meant to constrain.

H.18.8 — Price a personal purchase hold honestly

For kitchen tools I would use τ`_inject = 1.5 weeks`: novelty-driven wants in my own records usually fade within a fortnight. I would use τ`_identity = 30 weeks`: tools tied to recurring cooking projects persist across seasons. A one-τ`_inject` hold is 1.5 weeks:

```
injected remaining = e^(−1.5/1.5) = 0.368; removed = 0.632
genuine remaining  = e^(−1.5/30)  = 0.951; removed = 0.049
```

Suppose a year's baseline contains six injected purchase impulses and twelve genuine ones. The hold blocks about `6(0.632)= 3.79` injected purchases and `12(0.049)=0.59` purchases I would have been glad to make. To locate the point where cost exceeds benefit, solve

```
12(1 − e^(−h/30)) = 6(1 − e^(−h/1.5)).
```

The non-zero crossing is `h = 20.79 weeks`. Beyond roughly twenty-one weeks the expected number of blocked genuine purchases exceeds the injected purchases filtered out. This is a self-owned purchasing rule; it neither diagnoses desire nor licenses imposing a hold on someone else.

Chapter 19

H.19.1 — Compute coupling without claiming attribution

The geometric term is

```
√(L1L2) = √(6000 × 1500) = √9,000,000 = 3000 h·Ω_c.
```

Therefore

```
M_independent = 0.20(3000) = 600 h·Ω_c
M_observed    = 0.55(3000) = 1650 h·Ω_c
ratio         = 1650/600 = 2.75.
```

The observed coupling is 2.75 times the chapter's honest-independence ceiling. That entitles a defender to inspect for a shared clock or common upstream and to reduce reliance on cadence alone. It does not identify an actor, prove intent, establish that either stream is false, or authorize a counter-operation against a person.

H.19.2 — Classify payloads by what remains

The first item—anger with nothing new—is ^3: a visceral aversion was amplified, and the bench test "what checkable thing did I learn?" returns nothing. The second is ^6: three claims remain available for decomposition and checking. The third is also ^3: the affect is certain while the proposition cannot be stated.

The three-claim item is not evidence of a Path D operation; it has the structure of analysis even if its claims later prove wrong. More strongly, no single ^3 item proves an operation either. Operational detection requires the structural fingerprints—shared clock, polarity asymmetry, cross-feed, or repeated chain gain—not a judgement that an angry item must have been engineered.

H.19.3 — Recover per-stage gain

For three equal stages,

```
β_stage^3 = 0.62/0.15 = 4.1333
β_stage   = (4.1333)^(1/3) = 1.605.
```

Each probe therefore sees a modest gain near 1.6, while the full chain produces 4.13. Splitting the gain avoids an unmistakable single amplifier and delays clipping: a detector placed at any one intermediate stage sees something compatible with ordinary discourse. Defence therefore compares the end-to-end product and its shared timing, rather than declaring any one modest stage malicious.

H.19.4 — Diagnose compound inversion

Compute the three involution divergences:

```
|Dial(^2) − Dial(^3)| = |4 − 9| = 5  → despair signature
|Dial(^5) − Dial(^6)| = |9 − 4| = 5  → dogmatism signature
|Dial(^8) − Dial(^9)| = |6 − 5| = 1  → inside the limit
```

The first two are axis-N pair inversions; because two pairs are simultaneously beyond four, the chapter classifies the combined state

as a compound axis-P inversion. I would restore ^6 decomposition first, using the inward CCB-HM stage, because it makes subsequent claims inspectable and can reduce the ^5/^6 divergence without projecting a new recoil.

Raising ^2 first merely adds attraction while decomposition remains starved. It does not repair ^5/^6, can create the opposite runaway-affinity fault, and risks rebuilding Path F instead of a counter-circuit.

H.19.5 — Replace an ascending diode with a lawful guard

Diode< ^C -> ^P > is refused because a diode may only descend P $\rightarrow$ C $\rightarrow$ E $\rightarrow$ O; the listing violates the no-domain-ascent invariant. "Only principled signals fortify" is better represented as a Principled bias judging a Cognitive claim and emitting a lower-domain, checked result:

```
board answer_h_19_5_defensive_gate {
    input: incoming_claim;
    output: checked_affinity;

    node principle:        ^5^P_{F7}|T;
    node incoming_claim:   ^6^C_{F6}|A;
    node checked_affinity: ^2^E_{F4}|f;

    cjt verify: CJT< incoming_claim | principle => checked_affinity >
                [β=1.0, sat=0.80];
    diode d_out: Diode< ^E -> ^O > [Vf=2.0];
}
```

β=1.0 prevents the defensive checker from manufacturing reassurance; sat=0.80 caps affinity before it can become dogmatic. The bias is ^P, above the ^C cause, and the effect is below the bias, so the CJT is domain-coherent.

H.19.6 — Configure a counter board for a monoculture

The board must describe the population honestly: it starts with one live capacitor, not five imaginary ones. I would install the following defensive settings and publish them for audit:

```
board answer_h_19_6_hm {
    input: incoming;
    output: composed;

    node substrate: ^5^P_{F1}|T;
    cap c5: Capacitor< node=substrate, C=10.0, n_parallel=1,
                       leakage=0.02 >;

    node decompose: ^6^C_{F6}|A;
    node confirm:   ^4^C_{F2}|A;
    node composed:  ^5^C_{F6}|A;
    node own_clock: ^7^O_{F6}|E;
    node limit:     ^3^P_{F7}|⟨A,f,T⟩;
    node blocked:   ^9^P_{F7}|T;

    diode d_pc: Diode< ^P -> ^C > [Vf=0.7];
    diode d_ce: Diode< ^C -> ^E > [Vf=1.6];
    diode d_eo: Diode< ^E -> ^O > [Vf=2.0];

    cjt hm1: CJT< incoming | substrate => decompose > [β=2.5,
    ↪  region=active];
    cjt hm2: CJT< decompose | substrate => confirm > [β=1.0];
    integrate hm3: ( decompose × confirm ) = composed;

    osc local: Oscillator< signal=own_clock, source=internal,
                           period=168.0, duty=0.20 >;
    filter fb: LowPass< in=confirm, out=trend, tau=672.0 >;
    feedback trend -> substrate [β=0.03];

    cjt csb: CJT< incoming | substrate => limit > [β=2.0, sat=0.85];
    cjt hm_enforce: CJT< limit | substrate => blocked >
                    [β=10.0, sat=0.98, cutoff=0.80];
    fuse hm_fuse: Fuse< path=blocked, threshold=0.90,
                        action=permanent_block >;
```

```
    probe tp1: own_clock;  probe tp2: hm1;
    probe tp3: composed;   probe tp4: limit;
}
```

The current `C=10, n_parallel=1` preserves truth about the monoculture; a five-cell plural target is a long-term civic programme, not a four-week parameter edit. Raising D2 from 0.4 to 1.6 protects against disagreement instantly becoming affect, at the cost of slower response to a genuine emergency. Raising `hm1` from 2.0 to 2.5 buys more decomposition against an active chain, at the cost of more analysis delay; it remains below the paralysis region above 3.0. Slowing feedback from 504 to 672 hours and lowering β from 0.05 to 0.03 protect a substrate already exposed for four weeks, at the cost of slower learning from legitimate outcomes. `hm2` stays at unity, because defensive reassurance must not be manufactured. `hm_enforce` is required so the 0.90 fuse is reachable from a 0.98 ceiling; a fuse placed directly on the 0.85 refusal would be dead metal.

HM4—bringing every pair back inside four—cannot honestly be guaranteed in four weeks. HM2 can be installed immediately, but rebuilding public decomposition is the chapter's roughly forty-eight-week task, and HM4 depends on it. I would say plainly that four weeks can stop further externally clocked driving, restore local checks, and begin repair; it cannot deliver normalized dials. The board has no outward `^3^E|f` stage, so HM6 remains architectural rather than aspirational.

Chapter 20

H.20.1 — Reject a fast reference

The separation ratio is

τ_ref/τ_work = 1.5/22 = 0.0682.

That is far below SI1's minimum of 3, so the reference and working node are effectively shorted in time. On the first missed cycle

TP4 moves on the same day as the TP2 miss instead of remaining a slow, independent baseline. The correlation is the safety finding: the operator's "do I feel like this?" node is chasing performance, and F1 should open the feedback rather than letting the miss become a Principled verdict.

H.20.2 — Derive three equilibria and the required cadence

Use `P_eq = ΔP_session τ_leak/T` with `ΔP_session=0.50` and `τ_leak=24 d`:

```
daily, T=1.0:       P_eq = 0.50(24)/1.0 = 12.0 → clips at 10.0
alternate, T=2.0:   P_eq = 0.50(24)/2.0 = 6.0
twice weekly, T=3.5:P_eq = 0.50(24)/3.5 = 3.43
```

For `P_target=7.0`,

```
T_max = 0.50(24)/7.0 = 1.714 days
sessions/week = 7/1.714 = 4.083.
```

Thus the mean cadence must be a little more than four sessions a week; four exactly is slightly under budget, so a practical schedule uses five or explicitly accepts a lower equilibrium.

H.20.3 — Compare assertion stress with evidence thickness

Two enactments give

```
V_BR = 2/10 = 0.20 V_c
V_sugg/V_BR = 6.0/0.20 = 30
30/0.5 = 60 times the derated design limit.
```

Ignoring the rating clamp for a moment, the condition `6.0 ≤ 0.5(n/10)` requires `n ≥ 120` enactments. At six per week that is 20 weeks total, or `118/6 = 19.67` additional weeks after the two already logged.

But the canonical store rating clamps at `V_BR=8.0`, whose half-rating limit is only 4.0. Therefore the six-point assertion never becomes admissible on this board, even at 120 nominal enactments. The safe repair is to lower `V_sugg` to at most 4.0—and much lower early on—not to wait for a rating the component cannot possess.

H.20.4 — Separate an audit load from a recoil

`^3` Negative is "an Idea the Mind is adverse to and repelled by." The sentence "this is pointless; I should be doing something useful" takes the session apart, compares it with another use of time, and tests its worth. That is `^6^C_{F6}|A`, analytical decomposition, not visceral `^3^E_{F3}|f`.

The `^6` repair is Chapter 20's notch-and-sample arrangement, using the Chapter 11 notch: move the audit off the in-session signal path and run it later as a bounded, high-impedance measurement. A genuine `^3` recoil calls for Chapter 9's rate repair—lower amplitude, lengthen the taper, and reduce `di/dt`. Silencing a real recoil with the notch would remove the safety reading rather than fix the cause.

H.20.5 — Replace daily chatter with hysteresis

For `P(t)=10(1−e^(−t/18))`, the single 8.5 threshold is reached at

```
t_8.5 = −18 ln(1−0.85) = 18 ln(6.667) = 34.15 days.
```

Leakage there is `8.5/25 = 0.34 points/day`. With the chapter's daily oscillator, the level falls below the threshold inside one day and the next session pushes it back above, so the gate chatters on an approximately one-day period. The operator experiences an alternating "formed/not formed" verdict every day.

With thresholds 8.0/9.0:

```
t_8 = 18 ln 5  = 28.97 days
t_9 = 18 ln 10 = 41.45 days
```

```
top-up interval = 25 ln(9/8) = 2.94 days.
```

The operator instead experiences one continuous formation run across a 12.48-day band, followed by a predictable maintenance top-up about every three days. The band turns a judgement into a schedule.

H.20.6 — Build the smallest safe self-board

The reference check is

```
τ_ref/τ_work = 40/10 = 4.0 ≥ 3  → PASS.
```

At the requested target,

```
T_max = ΔP τ_leak/P_target = 0.30(20)/8.5 = 0.706 d
cadence = 7/0.706 = 9.92 sessions/week.
```

One session a day reaches only `P_eq=0.30(20)=6.0`, so 8.5 is not reachable at that cadence. I would initially lower the target to 6.0 rather than increase deposit without evidence; raising `ΔP` to the required `8.5/20=0.425` is acceptable only after the limiter shows enough `V_BR`. The reference's maximum lawful slew is `10/40=0.25 points/day`, so `F1` is set to 0.25.

```
board answer_h_20_6 {
    input: intent;
    output: enacted;

    node ref_composition:  ^5^P_{F7}|T;
    node work_composition: ^5^C_{F5}|⟨A,E⟩;
    node intent:           ^0^C_{F7}|A;
    node cadence:          ^7^C_{F6}|A;
    node cue:              ^8^O_{F6}|E;
    node enacted:          ^9^O_{F5}|E;

    capacitor c1: Capacitor< node=ref_composition, tau=40.0,
                             leak_tau=120.0, V_BR=8.0 >;
    capacitor c2: Capacitor< node=work_composition, tau=10.0,
```

```
                            leak_tau=20.0, V_BR=8.0 >;
    cjt q1: CJT< intent | ref_composition => work_composition >
            [β=1.5, sat=0.60, rπ=8.0, region=active];

    schmitt g1: Schmitt< sense=work_composition,
                         on_below=5.5, off_above=6.5 >;
    osc osc1: Oscillator< signal=cue, source=cadence, period=1.0,
                          duty=0.04, sustain=120.0, gate=g1 >;
    cjt q3: CJT< cue | work_composition => enacted >
            [β=1.0, region=active];

    filter lp1: LowPass< in=enacted, out=evidence_trend, tau=21.0 >;
    fuse f1: Fuse< ref_composition, threshold=0.25,
                   action=open_feedback >;
    feedback evidence_trend -> ref_composition [β=0.05];

    probe tp1: work_composition;  probe tp2: enacted;
    probe tp3: q1;  probe tp4: ref_composition;
}
```

`LP1` retains Chapter 8.4's τ=`21 d` floor and β=`0.05` ceiling, and its O→P path terminates only at the bias. The revised 5.5/6.5 gate honestly implements the reachable first-stage target; after evidence justifies a larger deposit, the 8.0/9.0 band can be restored in a logged revision.

H.20.7 — Read the slow store, not the amplifier

The operator is misreading TP1, the working composition, as if it were the identity reference. TP1 at 9.50 and TP2 at 0.86 say the behaviour is formed and the schedule ran; the correct identity reading is TP4 at 5.40, still charging lawfully.

If suggestion voltage scales with β,

```
V_sugg,new = 2.4(5.0/1.5) = 8.0
half-rating = 0.5(5.7) = 2.85
8.0/2.85 = 2.81 times the derated limit.
```

It also exceeds `V_BR=5.7`, entering store puncture rather than "making it land." The correct intervention is to leave gain at 1.5

and continue the existing evidence cadence: TP4's `0.05 points/day` is below the `0.12` slew fuse, while TP1 and TP2 already show success. At the observed local slope, the remaining `9.50−5.40=4.10` points would be about 82 days, though the exponential model—not a linear forecast—governs the actual tail.

The circuit cannot tell the operator what the word "musician" ought to mean to them, whether adopting it remains wanted, or whether another life circumstance explains the report. It can distinguish the two stores; it cannot adjudicate identity.

Chapter 21

H.21.1 — Let the object-premium probe decide

The two Ao differences are

```
witness_load   = 0.70 − 0.66 = 0.04
object_premium = 0.70 − 0.58 = 0.12.
```

`witness_load=0.04 < 0.30`, so the desire is not witness-carried and F1 remains intact. `object_premium=0.12 < 0.20`, so the named target is indistinguishable from the underlying condition and M1 returns `PROXY_TARGET`; the gate holds.

The object-premium instrument alone decided the result. The repair is to use `^6` decomposition to name the condition directly, then take three fresh readings before doing either subtraction. A pass on one Ao half cannot compensate for failure on the other because the gate is an AND, not a score.

H.21.2 — Pay every descent drop

The desire begins at `^P`, so all three junctions are crossed:

```
ΣVf = 0.7 + 1.2 + 2.0 = 3.9 V_c
V_drive = 8 + 5 = 13 V_c
R = 15 + 40 + 10 + 5 = 70 Ω_c
```

```
I_attention = (13 − 3.9)/70 = 0.130 A_c
margin = 0.130/0.10 = 1.30.
```

The verdict is **MARGINAL**: it fires today but lacks the required 2× campaign margin. I would repair `R_context=10` first, even though `R_policy=40` is largest, because context is the cheapest reversible term—a clarification or schedule definition can move it without rewriting a principle. Changing the policy term merely because it is numerically largest would edit the operator's guardrail to make the gate pass.

H.21.3 — Quantify the naive overstatement

The naive calculation is

```
I_naive = 13/70 = 0.1857 A_c
margin_naive = 0.1857/0.10 = 1.857.
```

The correct margin from H.21.2 is `1.300`. The relative error is

```
(1.857 − 1.300)/1.300 = 0.4286 = 42.9%.
```

The naive reading nearly reports a PASS where the actual gate is MARGINAL. The error grows as the origin moves upward because a higher-domain desire must cross more P→C→E→O junctions; Σ`Vf` consumes a larger fraction of the same drive before the resistance stack receives it.

H.21.4 — Refuse a weighted W-gate

Six possession boxes cannot override `clarity=0.52`, because item 1 itself requires at least 0.60; the correct output is `HOLD: IDEA_UNFORMED`, followed by rewriting the condition as one concrete scene. The invariant is S1—readiness gates are AND gates, never weighted sums—so allowing the score would arm an undefined operation whose `^0` Idea has not been formed.

The resulting operation is not "slightly less ready." It is a cast whose cause terminal contains a sentence rather than the scene the selected board expects, so every downstream measurement would be about a different input.

H.21.5 — Re-plan the capacity gate two ways

The plan consumes

```
planned   = 6(25) = 150 min
available = 18(15) = 270 min
load      = 150/270 = 0.556.
```

Because `0.556 > 0.50`, the P-gate holds. Changing operation count, five operations use `5(25)=125 min`, giving `125/270=0.463`, which passes. Holding six operations fixed, the half-load rule requires at least `150/0.50=300` available minutes. At 15 minutes per day that means a 20-day horizon, where `150/(20×15)=0.500`, exactly on the limit. The first plan gives up one operation; the second gives up two days of speed.

H.21.6 — Build a counter-operation, not a fictitious inverse

(a) The requested un-cast is an inverse to realization: a path from the post-cast prerequisite state `R(I)'` back to the original `R(I)`, undoing ε`_I : R(I)` $\to$ `I`. That arrow is exactly what η`_I` $\circ$ ε`_I` $\neq$ `id_R(I)` says is absent.

(b) β=∞ changes gain on an existing component; it cannot create a missing morphism. It is also non-finite input that the validator rejects, but even a very large finite β would only amplify a newly declared forward path. It would not erase the log row, the operator's experience, or the fact that the first cast occurred.

(c) Build a new counter-operation with its own Ao, D, W, and P gates and its own ledger. It may mitigate consequences or establish a different state. It differs from reversal because both operations remain in the history and the second begins from `R(I)'`, not from the original prerequisite configuration.

H.21.7 — Count re-entry without forcing a cast

One canonical addition uses an integrator as the counter, counts only while no cast has occurred, and emits a review flag after the fourth re-wording:

```
board answer_h_21_7_reentry {
    input: reword_event;
    output: review_flag;

    node practice_rail: ^5^P_{F7}|T;
    node reword_event:  ^7^O_{F6}|E;
    node cast_event:    ^9^O_{F5}|E;
    node reentry_count: ^6^E_{F6}|A;
    node review_flag:   ^9^E_{F6}|A;

    gate no_cast: NOT< cast_event >;
    gate countable_reword: AND< reword_event, no_cast >;
    integrator reentry: Integrator< in=countable_reword,
                                    out=reentry_count, window=campaign >;
    monitor m_reentry: Monitor< signal=reentry_count, limit=[0.0, 3.0],
                                action_on_violation=HOLD_FOR_REVIEW >;
    cjt flag: CJT< reentry_count | practice_rail => review_flag >
              [β=1.0, sat=0.85];
}
```

The campaign window closes when `cast_event` makes `no_cast` false; a later target starts a fresh ledger instance rather than mutating the old count. It catches **gate churn**: repeated decomposition becoming a way to avoid choosing or testing any target. The opposite failure it introduces is **premature fixation**—the operator may treat the fourth wording as forbidden and cast a worse

third wording merely to avoid the flag. Therefore the action is HOLD_FOR_REVIEW, never ARM and never permanent rejection.

Chapter 22

H.22.1 — Compute the row and act on the smooth quantity

A 20-minute operation lies in the 10–30 minute working band, so `^7 = 2°`. The coupling sum and timing ratio are

```
S = m + v = 2 + 2 = 4
A = 3t_conf/t_op = 3(75)/20 = 11.25.
```

`A=11.25` is in the RE-SPEC band (`A` $\geq$ `9`). Act on `A`: stop raising amplitude and return to decomposition or route selection. Record `S=4` only as the ordinal cross-check. It cannot rescue the row, be averaged as a magnitude, or be traded against the timing ratio.

H.22.2 — Diagnose a flat circuit under improving conditions

The `S` series improves from 4 to 3 and then stays there, while `A` remains near 9 despite every β increase. Under §22.5, `A` flat while `S` falls means operating conditions improved but the circuit did not. Because the changed lever was repeatedly the amplifier, the named failure mode is **saturation blindness** (or an amplifier change that never reaches delivered current): the stage has pinned and larger β values all produce the same output.

The first demanded measurement is a proof-load across the amplifier: increase its cause by 20% while observing the effect before and after the stage. If the cause moves and the effect does not, the stage is on its ceiling. Until that measurement passes, no further β change is admissible and the pleasant fall in `S` remains an operator-condition reading.

H.22.3 — Enforce three separate anti-hindsight rules

First, six hours is over the two-hour edge, so the vibration grader assigns `^4 = 3°`, not `1°`. Second, grading the following morning violates H2's thirty-minute hold from noticing; the stored impression is already reconstruction. Third, waiting until manifestation and then assigning the grade violates the write-once fuse F3: it lets outcome knowledge revise the record.

The correct entry is `3°`, written within thirty minutes of noticing the first confirmation and then permanently latched. If the grade was not captured in that window, the honest entry is a late/invalid-grade flag rather than a retrospectively improved `1°`. The event may still remain in prose, but it cannot enter a comparable row under the claimed grade.

H.22.4 — Keep the row that supplied no ratio

The deletion argument is superficially tidy: Op 4 has no `t_conf`, so it yields neither `A` nor `S`, cannot enter the three-row trend, and appears to add no numerical information.

That is exactly why it must stay. NO-READ is the observation that no qualifying confirmation arrived inside a horizon fixed in advance; deleting it conditions the log on successes and makes the apparent confirmation rate exceed the operator's base rate by construction. The row still preserves `^1`, `^7`, the declared horizon, and the absence itself, so it constrains any later claim that "every operation confirmed." It is excluded from the ratio, not erased from the evidence.

H.22.5 — Reject multiplication of ordinal grades

Measurement-theory objection: `m` and `v` are ordinal bins whose intervals and zero points are undefined, so multiplying them asserts ratio-scale structure the instrument does not possess.

CAS-notation objection: the readings are distinct co-present states joined by +; $\times$ would assert a fused emergent state, and no such result is named or defined.

I would put the measurement objection first in design review because it invalidates `S'` independently of this book's notation and prevents a superficially compilable implementation from being treated as quantitative. The notation objection follows as the repository-enforceable form of the same discipline.

H.22.6 — Flag an attribution collision and retain the row

The collision stage sits after the first-only limiter and before row assembly. It is a coincidence/AND gate over the admitted confirmation and the count of other operations whose horizons are still open:

```
board answer_h_22_6_collision {
    input: admitted_confirmation;
    output: flagged_row;

    node admitted:    ^9^C_{F2}|A;
    node other_live:  ^7^C_{F6}|A;
    node operation_set: ^5^P_{F7}|T;
    node collision:   ^6^E_{F6}|A;
    node flagged_row: ^6^O_{F6}|A;

    gate g_collision: AND< admitted, other_live >;
    cjt flag_collision: CJT< g_collision | operation_set => collision >
                        [β=1.0, sat=0.85];
    monitor m_collision: Monitor< signal=collision, threshold=1.0,
                                  action_on_violation=AMBIGUOUS >;
    mux retain: MUX< inputs=[admitted, collision],
                     select=m_collision >;
    diode d_log: Diode< ^E -> ^O >;
}
```

The component type is a coincidence gate backed by a monitor. Its location ensures later confirmations cannot enter and ensures ambiguity is known before the row is latched. It must **not** delete, reassign, or improve the row; it marks `AMBIGUOUS`, retains both candidate

operation IDs, and excludes that confirmation from ratios and the rolling median.

H.22.7 — Keep the scope fuse

F4 belongs on the board because S1 is a scope invariant that no numeric reading can test. The physical event that blows it is routing a row, degree, or trend into a comparison with another operator or into a claim presented as externally validated. The fuse then permanently blocks that export path while leaving the operator's own ledger intact.

Without F4 the enforcement mechanism would be a person reading the intended-use statement and declining to export the scale. That may be good practice, but it is not a circuit: it has no terminal, continuity state, or test point. Since the board claims every safety invariant has enforcement, leaving S1 to memory while pretending otherwise would be the decorative choice.

Chapter 23

H.23.1 — Reduce the pair and blow the re-spec fuse

The raw timing ratio is

```
η = t_op/t_conf = 45/240 = 3/16 = 0.1875.
```

Applying the one-third scale,

```
η/3 = (3/16)(1/3) = 1/16,
(a,b) = (1,16),
A = b/a = 16
  = 3(240)/45.
```

`A=16` is in RE-SPEC. The first response is decomposition of the target or a lower-resistance route, because the number is already beyond the amplifier-instruction boundary. Raising β would misread

A as a component setting, compound demand, and hide the target or resistance fault that the fuse is designed to surface.

H.23.2 — Let the stable ratio outrank the unstable pair

Raw reduction:

```
(t_op)/(3t_conf) = 27/192 = 9/64,
(a,b) = (9,64),
A_raw = 3(64)/27 = 7.11.
```

On the five-minute grid, $27 \to 25$ and $64 \to 65$:

```
25/(3×65) = 25/195 = 5/39,
(a,b) = (5,39),
A_grid = 3(65)/25 = 7.80.
```

The pair changes from (9,64) to (5,39)—not a small perturbation—while both A readings remain in DRIVE and call for one-lever work. Because a>3 in both reductions, discard the pair and act on A; the comparison demonstrates that the smooth quotient is the usable instrument.

H.23.3 — Recognize a decline that is too slow

The convergence ratios are

```
ρ_1 = 9.6/12.0 = 0.800
ρ_2 = 7.7/9.6  = 0.802
ρ_3 = 6.2/7.7  = 0.805.
```

The series is falling, but it is not converging by the chapter's $\rho \leq 0.6$ design threshold. Using the latest ρ=0.805, solve

```
6.2(0.805)^n ≤ 1
n ≥ ln(1/6.2)/ln(0.805) = 8.42,
```

so nine further operations would be needed if that rate held. I would not keep casting to realize that projection: by operation three the sustained ratio has failed the threshold, and demand is compounding. Stop, return to A0/decomposition, and change the target, route, or resistance before another operation.

H.23.4 — Shed the densest coupling first

Label the F5 operations A and B, the unrelated F2 operation C, and the F4 same-person operation D. The six pairings are:

```
A–B: 0.35  (same Drive, shared sigil alphabet)
A–C: 0.05  (different Drive/domain, no shared symbol)
B–C: 0.05
A–D: 0.35  (assuming A is the same-person F5 operation)
B–D: 0.05
C–D: 0.05
```

The mean pair coefficient is `(0.35+0.35+4×0.05)/6 = 0.15`. For `N=4`,

```
k(N–1) = 0.15(3) = 0.45,
```

above the 0.30 design ceiling. Shed D first because it removes the second 0.35 edge. The remaining A/B/C network has mean `k=(0.35+0.05+0.05)/3=0.15`, so `k(N–1)=0.30`, exactly at the ceiling. For working margin, shed either A or B next; the remaining different-Drive pair has `k(N–1)=0.05`.

H.23.5 — Derate a cast into another band

For `N=3`, `k=0.35`,

```
derating = 1 – k(N–1) = 1 – 0.35(2) = 0.30
η_effective = 0.75(0.30) = 0.225.
```

Thus

```
t_conf = t_op/η_effective = 40/0.225 = 177.78 min
        → 180 min on the five-minute grid,
A = 3(180)/40 = 13.5.
```

In isolation, `A=3/0.75=4.0`, inside DRIVE. Under concurrency, `A=13.5`, so the band changes to RE-SPEC. The cast itself need not be repaired; shed the related live operations until `k(N−1)`$\leq$`0.3`, then take a fresh attributable reading.

H.23.6 — Delete the tautology, keep the trend

Six 9.0 entries prove only that the absolute-target column was computed from quantities defined to cancel; a column that cannot vary would stay flat for a converging, stalled, or diverging campaign. Show me the comparable `A` values and the ρ`=A_next/A_current` column instead—ρ`≤0.6` across at least three operations is the chapter's evidence of convergence.

That is the complete two-sentence reply. The absolute column may be retained as an algebra check, but it must not appear as a campaign finding.

Chapter 24

H.24.1 — Rank three guard knees

Using `knee=sat/`β:

```
[β=2.0, sat=0.85]: knee = 0.85/2 = 0.425
[β=3.0, sat=0.85]: knee = 0.85/3 = 0.283
[β=3.0, sat=0.60]: knee = 0.60/3 = 0.200
```

Ranked by assessed risk required to latch, from most to least: 0.425, 0.283, 0.200. For an operator who has just blown a fuse for the first time, I would temporarily build the β`=3.0, sat=0.60` guard because it stops listening earliest while the event is reviewed. The cost is more false refusal and less room for evidence before saturation;

after the review and charter check, returning to the canonical 0.85 ceiling may be justified.

H.24.2 — Move the fuse to a reachable enforcement path

Across Q4, the fuse never blows: the guard clamps at 0.85 and cannot reach threshold 0.90. The compiler accepts the listing because each value is locally valid, and the operator believes a permanent block exists even though the fuse is dead metal.

The two-line correction is:

```
cjt q7: CJT< guard | charter => blocked > [β=10.0, sat=0.98,
↪  cutoff=0.80];
fuse f1: Fuse< blocked, threshold=0.90, action=permanent_block >;
```

Q4's `0.85` ceiling clears Q7's `0.80` cutoff by 0.05, and Q7's `0.98` ceiling clears F1's `0.90` threshold by 0.08. Both margins must be checked; either missing margin produces a silent safety failure.

H.24.3 — Rerun the Commission at the lawful dial edge

The divider becomes

```
k = Dial(^2)/(Dial(^2)+Dial(^3)) = 8/(8+4) = 0.667.
intake = 0.48(0.667) = 0.320.
```

Q1 requests `0.320(3)=0.960` and clamps at its 0.80 ceiling. With the same two attractive and four aversive readings:

```
attraction = [0.80(2/6)](2.0) = 0.533
aversion   = [0.80(4/6)](2.0) = 1.067 → clamps at 0.900
risk       = 0.900 − 0.533 = 0.367
Q5         = 0.367(1.5) = 0.550
Q6         = 0.550(1.0) = 0.550 assessed risk.
```

The guard requests `0.550(3)=1.650`, clamps at 0.85, and latches; Q7 then reaches 0.98 and F1 enforces. The design conclusion is that a lawful attraction-heavy dial setting does not defeat a matched,

charter-biased differential on a structurally refusable proposal. Q1 does saturate, so the board must still be proof-loaded on benign proposals to ensure the pre-amp is not making everything look maximal.

H.24.4 — Keep evaluation ^6 and refusal ^3

The canonical definition is: ^3 Negative is "an Idea the Mind is adverse to and repelled by." Q6 names and weighs the proposal's parts; that is analytical decomposition, so ^6^E_{F6}|A is exact. Typing it ^3^E_{F3}|f would turn an audit into a felt recoil and destroy the distinction between evidence and refusal.

The anti-drift rule is **"^3 Negative is visceral aversion, not analysis"**: "I cannot stomach that" supports ^3; "let me test its premises" supports ^6. Q4 is correctly ^3^P_{F7}|⟨A,f,T⟩ because it is the principled, multi-register refusal that follows the audit and stops responding once saturated.

H.24.5 — Give an intermediate operator a slower charter filter

Two years places the operator in the **intermediate** band, one to three months. I would set LP1 τ=60 days, the middle of that band, rather than inherit the adept default of 21 days.

The charter alone could move at most 10(1−e^(−56/85))=4.83 points over 56 days under a full-scale step; LP1 makes the actual response slower. For a sustained step passed through LP1 and then the 85-day charter, the normalized two-pole response is

$$y(t) = 1 - [85e^{(-t/85)} - \tau_f\, e^{(-t/\tau_f)}]/(85 - \tau_f).$$

At eight weeks (t=56 d):

τ_f=21 d: y=0.3355 of the eventual feedback-driven displacement;
slew at day 56 = 0.006999 $\Delta P\infty$ points/day.
τ_f=60 d: y=0.1844 of the eventual feedback-driven displacement;
slew at day 56 = 0.004969 $\Delta P\infty$ points/day.

Here ΔP_∞ is the eventual charter displacement after the feedback gain and input scale are applied. Even at the largest possible ten-point displacement, the two day-56 slews are only 0.070 and 0.050 points/day. TP3 therefore shows substantially more movement under the inherited adept setting, while the 60-day setting preserves more of the dated baseline. Both remain below F2's `0.12 points/day`, so the slew fuse distinguishes neither; only TP3 compared with a current dated baseline distinguishes the accumulated movements.

H.24.6 — Integrate a sequence that never reaches the guard knee

Choose a 30-day rolling window and a threshold of 1.0 risk-unit—roughly four proposals just below the 0.283 knee. The component observes the already-assessed internal risk and has no outward terminal:

```
board answer_h_24_6_subthreshold {
    input: assessed;
    output: accumulated;

    node charter:        ^5^P_{F7}|T;
    node assessed:       ^6^E_{F6}|A;
    node accumulated:    ^6^E_{F6}|A;

    integrator risk_memory: Integrator< in=assessed, out=accumulated,
                                        window=30.0 >;
    monitor sequence_limit: Monitor< signal=accumulated,
                                     limit=[0.0, 1.0],
                                     action_on_violation=BLOCK_REVIEW >;
    fuse f_sequence: Fuse< accumulated, threshold=1.0,
                           action=block_until_decomposed >;

    probe tp_each: assessed;
    probe tp_sum: accumulated;
}
```

The chosen threshold makes a repeated near-knee sequence visible without treating one low-risk proposal as a refusal. It does not fix **charter drift** or a dead guard: if the upstream charter and differential assess formerly refusable proposals as zero or negative, the integrator receives nothing honest to accumulate. Q4's saturation offers no protection because saturation is memoryless; each proposal remains below 0.283, the cause disappears between proposals, and Q4 stores no sum.

H.24.7 — Restore the charter before tuning the guard

(a) The combined signature is charter drift producing a dead guard: TP1 is zero under a proof-load, TP2 never arms, and TP4 is negative on a proposal that once produced refusal. Lowering Q4's cutoff cannot repair an exhausted or shifted bias, an imbalanced differential, or a non-positive assessed-risk input; zero multiplied by the guard gain remains zero.

(b) Using a 30-day month for the stated eleven-month interval,

```
movement bound = 0.041 points/day × 330 days = 13.53 points.
```

That exceeds the entire 0–10 charter scale. F2 would never open because `0.041 < 0.12 points/day` at every sample; the fault is a slow slope, not a slew event.

(c) Bring-up follows TP1–TP4. First restore and date C1, then proof-load TP1 and require a live guard response (normally $\geq$0.15, reaching 0.85 on the known refusable case). Second read TP2 and verify Q7 reaches 0.98 and F1's continuity matches its dated record. Third set LP1 to the operator's proper band and read TP3 as both slew and displacement from the new written baseline. Fourth re-match Q2/Q3 on one bias net and require TP4 near zero on benign proposals and positive on the proof-load.

C1, the charter capacitor, must be brought into spec before any other reading means anything because every guard stage takes bias from it. The circuit still cannot tell the operator what their charter ought to say, whether the old refusal was ethically correct, or whether their wider situation calls for human support; it can only show that the present board is not enforcing the reference it claims to enforce.

I Appendix I — Glossary

Every term this book uses that a reader might arrive at cold. Entries are alphabetical, each is self-contained, and each ends with where the term is actually developed. Symbols and Greek letters are collected first, because they have no alphabetical home.

Where a term already means something in ordinary English, or in electronics, and this book means something different by it, the entry says so in bold. Those are the entries most worth reading before you need them.

False friends — check these first

Term	What it usually means	What it means here
`^3` Negative	bad, harmful, critical	**visceral aversion only** — can't-stomach-it. Analytical critique is `^6` Male
`^8` Above/Cause	logical cause, "therefore"	**associative triggering only** — "that reminds me"
`^5` Female / `^6` Male	gender	composition. `^5` = what a Mind is made of; `^6` = what an Idea is made of
Saturation	a fault, a clipped signal	often **the goal** — a stage that has stopped responding to input

Term	What it usually means	What it means here
Negative feedback	discouragement	output subtracted from input; the stabilising kind
Gnosis	mystical knowledge	a measurable bias-collapse state with three graded degrees
Hyperstition	a hoax or a lie	a fiction that makes itself true by being acted on
τ	one thing	**two** — the RC/LR time constant, and the 1/3 constant in St. John's Equation
Dropout	quitting, or a dropped connection	the margin below which a stage stops regulating at all
Inversion	any reversal	a specific polarity flip along one of the named axes

---

Symbols

β **(beta)** — Current gain of a Cognitive Junction Transistor: how much effect the stage produces per unit of cause. $\beta < 1$ attenuates, $\beta = 1$ passes, $\beta > 1$ amplifies. *Chapters 3.3, 10.1.*

$\beta{=}\infty$ — An idealization, never a value. A validator rejects non-finite parameters; a real build commits to a large finite gain and says so. *Chapters 10.1, 23.2.*

g_m (transconductance) — How much the output moves for a given change at the bias terminal. High g_m is why rewording a question changes the answer. *Chapters 3.3, 10.3.*

rπ (r-pi) — Input resistance at the bias terminal: how hard it is to move someone's frame at all. High rπ reads as stubbornness; r$\pi \to 0$ is the idealized gnosis state. *Chapters 3.3, 10.4, 17.1.*

τ **(tau) — Two distinct uses; do not conflate them.** (1) The time constant of an RC or L/R circuit — one τ reaches 63% of final value, 3τ is practically complete. (2) The 1/3 constant in St. John's Equation, one part work to two parts gestation. *Chapters 7.2, 9.2 for the first; 5.2, 23.1 for the second.*

η **(eta)** — The operation-to-confirmation ratio, η = `t_op / t_conf`. The single number the Quantitative Table is built to produce. *Chapters 5.3, 22.5, 23.1.*

Ω_c, V_c, A_c, F_c, h·Ω_c — Cognitive ohms, volts, amperes, farads, and the inductance unit. Scaled analogues, not electrical measurements, and the book never pretends otherwise. *Chapters 4.2, 7.1, 9.2.*

`^n^D_{Fk}|R` — The state literal: Noetic, Domain, Drive, Register. Register optional. Read `^3^P_{F7}|T` as principled visceral aversion, purpose-driven, transcendent register. *Chapter 2.1.*

`|` — Introduces the Register and nothing else. `F3|Vitality` is a syntax error, not an annotation. *Chapter 2.5.*

+ / × / |⟨…⟩ — Co-presence (terms stay distinct) / genuine integration (requires a named emergent result) / register-set occupancy (no operational claim). *Chapter 2.6.*

A

Ao (Pure Desire) — The root of the RPM chain and the first gate: is this authentically what *you* want, or a sub-desire wearing the prime desire's clothes. Most failed operations failed here and were run anyway. *Chapter 21.1.*

Active region — The CJT operating region where output tracks input proportionally. The only region in which a gain figure means anything. *Chapter 3.4.*

Amplification — Producing more effect than the cause alone would carry, by spending bias. Not creation; the cause must already be present. *Chapters 3.2, 10.1.*

Assembly drawing — The board layout figure: physical arrangement, test-point callouts, and a bill of materials box. Distinct from the schematic, which shows connection rather than placement. *Chapter 12.1.*

B

Back-EMF — The voltage a coil generates when the current through it is interrupted, opposing the change. Cognitively: the distress produced by stopping an established practice. **It scales inversely with how fast you stop** — the same removal produces thousands of times the kick when done abruptly. The distress is generated by the rate, not stored up and released. *Chapter 9.2.*

Bias — The reference held at the middle terminal of a CJT, which decides how much of a cause reaches its effect. **Not "prejudice."** Willpower, framing, conditioning, and persuasion are all operations on somebody's bias terminal. *Chapters 3.2, 10.4.*

Block diagram — The stage-level view: signal path left to right, boxes labelled by function, no component values. The first thing to read in any board chapter. *Chapter 12.1.*

Bill of materials (BOM) — The parts list for a board: every component, its designator, its parameter values. If a component is not on the BOM it is not on the board, whatever the prose implies. *Chapter 12.1.*

Breakdown — Operation past the reverse-voltage limit of a junction, after which it no longer blocks. Non-recovering. *Chapters 3.4, 7.4.*

C

Capacitor, cognitive — A ^5 Female store: accumulated experience held as charge, charging on an RC curve and leaking when not fed. *Chapters 7.1, 7.2.*

Cascade — Two or more CJTs in series, gains multiplying. Also multiplies every error and every offset. *Chapters 10.5, 15.2.*

CAS — Cognitive Architecture System. The formal notation: 10 Noetics, 4 Domains, 7 Drives, 4 Registers. *Chapters 1.3, 2.1.*

CAS-Ohm's Law — `V_drive = I_attention` × `R_constraint`. Drive across a constraint produces attention flow. *Chapter 4.2.*

CCB — Cognitive Circuit Board. A complete, named board with requirements, schematic, bill of materials, and test points. *Chapter 12.1.*

CCDL — The Cognitive Circuit Description Language: the circuit written as text. Every listing in this book is one a validator would accept. *Chapter 12.1; full grammar in the CCDL Language Reference appendix.*

CJT (Cognitive Junction Transistor) — Three terminals: Cause (collector), Bias (base), Effect (emitter). The claim the whole book rests on, and the part in every board. *Chapters 3.2, 10.*

Cognitive Safety Board (CSB) — The operator's own safety board: a saturated refusal stage, an enforcement fuse, the domain diode chain, and a filtered feedback path. Built before anything ambitious is run, not after. *Chapter 24.*

Common-mode rejection — An op-amp amplifies the *difference* between its two inputs and ideally ignores whatever is common to both. Cognitively: a comparison stage should respond to how two Ideas differ, not to the framing, source, or delivery they share. Poor rejection is why identical proposals get different verdicts depending on who tabled them. *Chapter 11.4; see also g_m, Chapter 10.3.*

Consent architecture — A circuit whose target has agreed to it, built so the agreement is structural rather than assumed. *Appendix M §M.2; Appendix K.*

Counter-circuit — The reversal or detection board that every coercive board in this catalogue ends with, at equal or greater depth. *Chapters 16.7, 19.6.*

Cutoff — The CJT region where insufficient bias means no current crosses at all. The part is undamaged and completely inert. *Chapter 3.4.*

D

Decoupling — Deliberately isolating one node from another's fluctuations so a reference stops tracking a result. In CCB-SELF it is what keeps a bad day from rewriting an identity. *Chapter 20.2.*

Deprogramming Board (CCB-DP) — The reversal circuit for conditioning architecture, and the terminal section of Chapter 16. *Chapter 16.7.*

DETECTION-ONLY SCHEMATIC — DO NOT BUILD — The banner carried by any circuit whose purpose is to act on a non-consenting person. Drawn so you can recognise one. *Chapter 12.2.*

Dial — An operator-set integer from 3 to 9 controlling a dial-adjusted resistor. Involution partners may not diverge by more than 4 points. *Chapter 6.2.*

Dielectric breakdown — A capacitor stressed past what its insulation holds. The ^5 analogue of a store that fractures rather than saturates. **No medical or psychiatric claim is made or implied; if this describes you, see a clinician.** *Chapter 7.4.*

Diode, domain — A one-way junction enforcing descent P $\rightarrow$ C $\rightarrow$ E $\rightarrow$ O. *Chapters 8.1, 24.4.*

Domain — Which ontological level an operation runs at: **P** Principled, **C** Cognitive, **E** Evaluative, **O** Operational. Total order P $\succ$ C $\succ$ E $\succ$ O. *Chapter 2.3.*

Domain ascent — Movement up the precedence order. Blocked, with one lawful exception. `^0` $\times$ `^P` is invalid notation. *Chapter 8.3.*

Domain descent — The lawful direction of translation, P $\to$ C $\to$ E $\to$ O, one junction at a time. *Chapter 8.1.*

Drive — Which motivational architecture is running: **F1** Unity, **F2** Wisdom, **F3** Vitality, **F4** Connection, **F5** Agency, **F6** Structure, **F7** Purpose. Inversion pairs F1$\leftrightarrow$F7, F2$\leftrightarrow$F6, F3$\leftrightarrow$F5; F4 is self-dual. *Chapter 2.4.*

Dropout — In a regulator, the minimum input-to-output margin below which regulation is simply lost and the output follows the input down. Here: `V_drive` below the summed forward drops of the domain chain, at which point no principle produces any act — not weakly, but not at all. **Reads as weakness; is arithmetic.** *Chapters 4.2, 8.2.*

Duty cycle — The fraction of a period an oscillator spends active. *Chapter 11.2.*

E

Egregore — A `^5` store held collectively rather than by one Mind: a capacitor network, charged and maintained by repeated collective exposure. **No claim of an independent entity is made.** *Chapters 7.5, 14.6.*

Emergent result — The named output that $\times$ requires. Integration without a named result is invalid notation; use + or $|\langle\ldots\rangle$. *Chapter 2.6.*

F

Failure mode — A row in every chapter's failure table: the mode, its mechanism, what you observe at the bench, and the fix. *Chapter 12.1.*

Feedback, positive — Output returned to the bias terminal so each cycle raises the next. Above unity gain it runs away — which is the operating principle of hyperstition and the hazard of everything else. *Chapters 14.2, 14.5.*

Filter — Lowpass passes slow trends and blocks transients; highpass does the reverse; notch rejects one band. *Chapters 11.3, 24.5.*

Flyback path — The alternate route a coil's collapsing current takes so it does not arc. Cognitively: the substitute practice. **It must already be conducting before the old path opens** — installed after, it is a diode that watched the arc. *Chapter 9.2.*

Forward voltage (Vf) — The threshold a junction must be driven past before it conducts. **Not overhead to engineer away — it is the boundary itself.** A junction at Vf = 0 is a wire, and two domains joined by a wire are one domain wearing two labels. *Chapter 8.2.*

Fuse — Permanent path interruption once a threshold is crossed. Does not suggest; opens. *Chapters 11.1, 24.3.*

G

Gate (D, W, P) — The three prerequisite checks after Ao: sufficient drive, possession of the schematic, capacity to execute. *Chapters 21.2–21.4.*

Gnosis — The bias-collapse state, $r\pi$ driven toward its floor, in which suggestion meets little resistance. **Not mystical attainment**; it is graded in three degrees and logged like any other reading. *Chapters 17.1, 22.1.*

H

Hyperstition — A fiction that becomes true through being acted on. **Not a lie**: a lie requires the liar to know better, whereas a hyperstition's truth condition is created by belief propagating through collective ^5. *Chapter 14.*

Hundredth Monkey Effect — The folk claim that a belief propagates spontaneously past a headcount. Treated here as collective CJT saturation rather than as a documented phenomenon, and the distinction is stated in the section itself. *Chapter 14.4.*

Hysteresis — Two thresholds instead of one, so a gate turns on and off at different points. Without it a comparator sitting near its threshold chatters. With it, the chatter becomes a computable maintenance interval nobody had to choose. *Chapter 20.4.*

I

Impedance matching — Sizing a transfer so signal actually crosses between two minds rather than reflecting. *Chapters 9.3, 11.5.*

Induction coupling — Belief transferred between minds by proximity and shared cadence rather than by argument. *Chapter 9.3.*

Inversion — A structural polarity flip along one of the named axes — a circuit producing the opposite of what it was built for while appearing to run normally. Distinct from failure: an inverted board is working, backwards. *Chapters 3.4 and 19.4.*

Involution pair — Two Noetics that are one structure seen from two sides: ^2↔^3, ^5↔^6, ^8↔^9. Self-dual: ^0, ^1, ^4, ^7. **A pair cannot be flipped directly** — reorienting a structure means taking it apart first, which is ^6 Male's job. *Chapters 2.2, 15.1.*

L

Lacunarity — An absence: a prerequisite that is not there. Distinct from leakage (present, producing the wrong output) and from inversion (present, running with reversed polarity). The distinction decides which repair applies. *Chapters 21.1, 23.3.*

Leakage — Charge lost from a ^5 store that is not being fed. The reason an operation left alone decays on a schedule you can compute. *Chapter 7.3.*

M

MK-Ultra — A declassified and documented programme. This book analyses what is on the record, structurally, so a reader can recognise the architecture. It supplies no protocol. *Chapter 16.1.*

"Must Not" list — The safety invariants written before any functional requirement, because a requirement can be traded and an invariant cannot. *Chapter 12.2.*

N

Noetic — Which structural dynamic is operating. The ten, verbatim:

	Name	Definition
^0	Idea	All things in and outside the Mind.

	Name	Definition
^1	Mind	Various degrees of awareness that processes, stores, outputs, and intakes Ideas.
^2	Positive	An Idea the Mind has an affinity for and is attracted toward.
^3	Negative	An Idea the Mind is adverse to and repelled by.
^4	Vibration	The degree to which an Idea resonates with the Mind, its impression.
^5	Female	A Mind impregnated with an Idea, the Ideas a Mind is composed of.
^6	Male	An Idea impregnated with another Idea, the Ideas an Idea is composed of.
^7	Rhythm	The rate and pattern a Mind is exposed to an Idea, entraining/conditioning it.
^8	Above/Cause	An Idea that triggers a different Idea than itself to come to Mind.
^9	Below/Effect	The Idea brought to Mind by an Above/-Cause trigger.

Never renamed to their function. *Chapter 2.2.*

O

Op-amp — Very high differential gain: amplifies the difference between two inputs. The precision comparison part. *Chapter 11.4.*

Operator's Oath — A commitment written so that it can be enforced by the circuit rather than remembered by the operator. *Text and circuit binding in Appendix K.*

Oscillator — Sustained periodic output. The ^7 Rhythm component: what keeps a practice, a campaign, or an egregore fed. *Chapters 11.2, 20.4.*

Overton window — The band of positions a population currently treats as sayable. This book models its movement as six resistance bands and a five-stage cascade, with ^6 Male decomposition as the hinge between aversion and attraction. **The window is a property of a collective ^5; where no aggregate exists, there is no window to move.** *Chapters 15.1, 15.6.*

P

PTC / NTC thermistor — Resistance that rises with cognitive temperature (PTC — tighter constraint as stakes rise) or falls with it (NTC — guard relaxing as danger increases, and a danger sign wherever you find it). *Chapters 6.3, 15.2.*

Q

Quantitative Table — The tracking sheet: ^1 Mind depth, ^4 Vibration and its timing, ^7 Rhythm, and the ratio they produce. It is where an operation stops being a feeling and becomes data you can

look at three weeks later. *Chapters 5.5, 22.4; blank sheet in the Quantitative Table Template appendix.*

R

Register — Which operational character a state runs in: `|T` Transcendent, `|A` Analytical, `|f` Affective, `|E` Embodied. Orthogonal to Noetic, Domain, and Drive; optional in notation. *Chapter 2.5.*

Resistance, cognitive (R_constraint) — What limits attention flow at a given drive. Sourced from capability, policy, context, and resources. *Chapters 4.2, 4.4.*

RPM chain — The prerequisite ladder Ao $\rightarrow$ D $\rightarrow$ W $\rightarrow$ P $\rightarrow$ outcome. Each stage must be satisfied before the next is reachable, and satisfaction is **one-directional**. *Chapters 21.1–21.5.*

S

Saturation — The ceiling past which more input produces no more output. **Usually a virtue here, not a fault:** a refusal that saturates early is a refusal that has stopped responding to argument. It is a defect only where a reading must track its input. *Chapters 3.4, 10.2, 24.2.*

Schmitt trigger — A comparator with hysteresis built in. The auto-gate part. *Chapter 20.4.*

Section-retraction — The formal statement of one-directionality: realization after embedding returns you to the start, but re-embedding after realization does not. Practically — you cannot un-cast. *Chapter 21.5.*

Shadow integration — A consent-bound self-programming use in which an inverted or disowned structure is decomposed and

routed rather than suppressed. This is a structural description, not a mental-health treatment. *Chapters 3.4 and 20.1–20.5; Appendix M.*

Slew — Rate of change at a node, and the thing a slew fuse limits. A reference that can be moved quickly is not a reference. *Chapter 20.5.*

St. John's Equation — The calibration relation built from operation time and confirmation time, yielding the gain requirement for the next operation and an estimate of how many remain. **An instrument on an uncalibrated scale** — informative about direction and magnitude, not about truth. *Chapters 5.3, 23.1; blank sheet in the St. John's Equation Worksheet appendix.*

T

Test point (TP1–TP4) — The four standard probes on every board: what each reads and its expected range. Read them in the order the chapter specifies, which puts the silent failures first. *Chapter 12.3.*

Sub-desire — A smaller want standing in front of the prime one, and the usual reason a board runs correctly and delivers nothing. Resolved by `^6` Male decomposition at the gate, not by more gain. *Chapters 21.1, 23.3.*

Time constant — See τ. One τ reaches 63% of final value; 3τ is practically complete. It sets how long a substrate takes to shift, and pacing failures are almost always τ set in the wrong unit. *Chapters 7.2, 9.2.*

TKS — The Tootra Kabbalistic System: 40 Elements, 7 Foundations, 12 Prerequisites. The esoteric ancestor CAS formalises. *Chapter 1.3; full mapping in the TKS Crosswalk appendix.*

Transformer — Impedance matching between minds or domains, trading amplitude for reach. *Chapter 11.5.*

Trigger installation — The architecture by which a stimulus is bound to a response so that later exposure fires it. Analysed as mech-

anism, for recognition. **No procedure is supplied anywhere in this book.** *Chapter 16.3.*

U

Un-boarded, right to be — The invariant that a person may have a circuit removed from them, and that no board may be built to defeat that removal. *Appendix M §M.4; Appendix K.*

W

Willie Lynch letter — A document of **disputed authorship**. The book analyses what the text specifies as a circuit and labels the provenance question plainly rather than resolving it. *Chapter 15.6.*

Worked example — Real numbers carried from input to output in every Part III chapter. Not a sketch. *Chapter 13.6.*

Exercises are answered in full in the exercise-answers appendix. Formal definitions underlying this glossary are in the Mathematical Foundations appendix.

J Appendix J — Further Reading

This list is short on purpose. Everything named exists, with author and title as printed. Where the primary material is a government record rather than a book, the entry describes the record instead.

Three rules. **Mechanism, not procedure** — no manual for acting on a non-consenting person is listed; J.10 says which were left out. **Nothing medical** — the hypnosis and conditioning entries are historical literature, not treatment guidance. **Inclusion is not endorsement** — several are propaganda handbooks by people who meant them, listed because the attacker's own account is the fastest route to recognising the board.

J.1 The companion volumes

All three from Noetic Compass Press.

Volume	Owns	Go there when
Thinking in Cognitive Architecture	The CAS framework: 10 Noetics, 4 Domains, 7 Drives, 4 Registers	You want the derivation behind Chapter 2, the admissibility relations, every Noetic in full
CAS PCBA Engineer	The component library and five reference boards, in engineering depth	You need parameter ranges beyond Appendix C, or thirty pages on a component this book gives a paragraph

Volume	Owns	Go there when
Reality Engineering	TKS — the 40 Elements, 7 Foundations, 12 Prerequisites	Appendix E's crosswalk raised a question about the Element grid it maps into

J.2 Electronics — the vocabulary this book borrowed

None of these are needed to build a board — only to judge whether Chapter 4's isomorphism does real work.

Work	Why
Paul Horowitz and Winfield Hill, *The Art of Electronics*	The standard bench reference; its transistor chapter shows which parts of Chapter 10's CJT are structural and which analogy
Forrest M. Mims III, *Getting Started in Electronics*	Hand-lettered and diagram-first — the fastest route to reading a schematic before Chapter 12 requires it
Adel S. Sedra and Kenneth C. Smith, *Microelectronic Circuits*	The formal treatment of β, g_m, and rπ that Chapter 3 compresses into one table
Michael Riordan and Lillian Hoddeson, *Crystal Fire*	The history behind Section 3.1 — what Bardeen, Brattain, and Shockley did in December 1947

J.3 Cognitive science and cognitive architecture

Work	Why
Herbert A. Simon, *The Sciences of the Artificial*	The founding argument that thinking systems are engineered objects with specifiable interfaces — this book's premise, stated fifty years earlier
Allen Newell, *Unified Theories of Cognition*	What a cognitive architecture must account for; the standard CAS should be measured against
Marvin Minsky, *The Society of Mind*	Mind as many interacting parts — the intuition behind Section 7.5's `^5` capacitor networks
Donald O. Hebb, *The Organization of Behavior*	Where repetition-strengthens-association comes from; the substrate under every `^7` Rhythm claim here
Judea Pearl and Dana Mackenzie, *The Book of Why*	The cleanest statement in print of association versus causation; read it and the `^8` anti-drift rule stops feeling pedantic
Norbert Wiener, *Cybernetics*	Feedback, gain, and stability in systems that include minds — background for Chapter 14's oscillator
W. Ross Ashby, *An Introduction to Cybernetics*	Requisite variety: why Chapter 24's safety board must be at least as rich as what it governs
George Lakoff and Mark Johnson, *Metaphors We Live By*	A standing warning: this book runs on a metaphor, and this is the book about what metaphors do to their users

J.4 Persuasion, propaganda, and advertising

Background for Chapters 15, 18, and 19.

Work	Why
Robert B. Cialdini, *Influence: The Psychology of Persuasion*	The six-principle taxonomy every board in Chapter 18 rearranges; his *Pre-Suasion* covers the state a target is placed in beforehand — bias-terminal work
Richard E. Petty and John T. Cacioppo, *Communication and Persuasion*	The elaboration-likelihood model — central versus peripheral routes, the academic form of Section 17.6's ^6 filter bypass
Walter Lippmann, *Public Opinion*	The 1922 argument that people act on the picture in their heads, not the world; the origin of window movement
Edward Bernays, *Propaganda*	An attacker's own manual, written without embarrassment by a man who thought he was doing civic hygiene
Jacques Ellul, *Propaganda: The Formation of Men's Attitudes*	Propaganda as ambient condition rather than campaign — why Section 15.4 measures τ in months
Vance Packard, *The Hidden Persuaders*	Motivational research in 1950s advertising — the record of Section 18.2's insecurity injection working
Anthony Pratkanis and Elliot Aronson, *Age of Propaganda*	The best detection-side text here, and the closest thing in print to Chapter 19's counter-circuit
David Ogilvy, *Ogilvy on Advertising*	A practitioner stating his own gain settings; read beside Section 18.5 on A/B testing as β calibration
Byron Sharp, *How Brands Grow*	The empirical challenge to the rest — availability and repetition beat persuasion, a ^7 Rhythm claim with data
Stanley Milgram, *Obedience to Authority*	What a bias terminal held by someone else costs, from the man who ran it
Robert Jay Lifton, *Thought Reform and the Psychology of Totalism*	Interview study of coercive persuasion — the closest published work to Chapter 16's analysis, and far more careful

On the Overton window. The concept is attributed to Joseph P. Overton of the Mackinac Center for Public Policy and was described

publicly by colleagues after his death. There is no canonical monograph; Chapter 15 builds on the concept.

On the Willie Lynch letter. Its authenticity has been challenged since it began circulating widely in the 1990s, and Section 15.6 treats it accordingly — a document *about* an architecture, not evidence *of* a 1712 one.

J.5 The intelligence programmes — background for Chapter 16

The primary sources here are declassified United States government records, not books. That cuts both ways: the record is solid where it exists and absent where files were destroyed. Four bodies of material matter.

Record	What it is
The Church Committee record	Hearings and the multi-volume final report of the U.S. Senate Select Committee to Study Governmental Operations with Respect to Intelligence Activities, 1975–76 — sworn testimony on CIA behavioural-modification research
The Rockefeller Commission report	The 1975 report of the Commission on CIA Activities within the United States, produced in parallel with the Senate inquiry over overlapping ground
The 1977 MKULTRA hearing	A joint Senate hearing of August 1977 devoted specifically to Project MKULTRA, published as a transcript. It happened because records thought destroyed were located

Record	What it is
The surviving MKULTRA files	Programme records were destroyed on the order of the then Director of Central Intelligence in 1973. Financial and administrative records held elsewhere survived and were released under the Freedom of Information Act — funding, front organisations, cover; not protocols

These sit in U.S. government depository collections and the CIA's public FOIA electronic reading room. Consult them directly; everything below interprets them.

Secondary history	Why
John Marks, *The Search for the "Manchurian Candidate": The CIA and Mind Control*	Product of the FOIA request that surfaced the surviving files; still the standard account
Stephen Kinzer, *Poisoner in Chief*	Biography of Sidney Gottlieb, who ran the programme — the clearest narrative of how the money moved
Alfred W. McCoy, *A Question of Torture*	Traces the research line from Cold War laboratories into later interrogation doctrine, the institutional continuity Section 16.1 assumes

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J.6 Hypnosis — background for Chapter 17

Historical and technical literature only. Neither is clinical guidance, and neither should be used on another person.

Work	Why
The Collected Papers of Milton H. Erickson on Hypnosis, edited by Ernest L. Rossi	Erickson's own case commentaries — the public record Section 17.6 analyses
Jay Haley, *Uncommon Therapy*	A structural reading of Erickson's method from outside, useful because Haley describes pattern rather than repeating anecdote

---

J.7 Self-programming and skill acquisition — Chapters 5 and 20

Work	Why
Anders Ericsson and Robert Pool, *Peak*	The deliberate-practice research behind Section 5.2's ten-thousand-hour constant, with Ericsson's objections to how it is quoted
Charles Duhigg, *The Power of Habit*	Cue, routine, reward — the three-node loop Section 20.4 draws as an oscillator with auto-gating
Leon Festinger, *A Theory of Cognitive Dissonance*	What happens when a ^5 capacitor holds incompatible charge; background for the involution and breakdown work in Chapters 2.2 and 3.4

J.8 The hermetic lineage — Chapters 13 and 14

The Noetic names — Vibration, Rhythm, Above and Below — descend from a hermetic tradition. Pretending otherwise would be dishonest.

Work	Why
The Kybalion, by "Three Initiates" (1908)	The seven-principle formulation this vocabulary descends from; authorship is attributed rather than established — a starting point, not an authority
Peter J. Carroll, *Liber Null & Psychonaut*	Chaos magic's results-first, belief-as-tool posture — the closest existing practice to Chapter 13's register
Richard Dawkins, *The Selfish Gene*	The meme chapter, where self-replicating ideas were first put in those terms; ancestor of Chapter 14's engine

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J.9 The mathematics — background for Appendix L

Work	Why
F. William Lawvere and Stephen H. Schanuel, *Conceptual Mathematics*	Category theory from nothing, with pictures — the gentlest entry to Appendix L's axioms
Saunders Mac Lane, *Categories for the Working Mathematician*	The standard reference for machinery Appendix L cites a line at a time
Saunders Mac Lane and Ieke Moerdijk, *Sheaves in Geometry and Logic*	Topos theory, including the subobject classifier Appendix L.4 turns into three-valued safety logic
Shun-ichi Amari and Hiroshi Nagaoka, *Methods of Information Geometry*	Where the Fisher-Rao metric comes from, and why Euclidean distance on a probability simplex is the wrong tool

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J.10 What is deliberately absent

Several declassified interrogation and exploitation manuals from the programmes in Chapter 16 are public and easy to find. They are not listed here.

The reason is the rule the whole book runs on: analyse mechanism, never supply a protocol against a person. Those documents are procedure, not architecture. They teach sequence and timing, the one thing Chapter 16 withholds. The oversight records in J.5 establish what was done and how the circuit was shaped, and they do it under oath.

If your interest is detection rather than operation, J.5's committee record and Lifton's study in J.4 are the two most useful items here. Start there.

K The T.O.O.T.R.A. Operator's Oath

A charter that can be inspected, tested, and enforced

This is the operator's oath adopted by this grimoire. It is not presented as a quotation recovered from an older manuscript, and it does not claim historical antiquity. The Tootra source material available to this book names an oath but does not supply a canonical transcript. Inventing one and calling it ancient would violate the first promise below.

An oath remembered only when convenient is advice. An oath written as the charter of a Cognitive Safety Board is a component. Read the text aloud if that matters to your practice, but charge it into `C1` by the slower method that matters to the circuit: date it, sign it, revisit it, and refuse any board whose listing cannot enforce it.

The Oath

I will name what I know, what I infer, what I choose, and what remains unknown. I will not pass one off as another.

I will not call a coincidence a measurement, a measurement a cause, a model a law, or a desired outcome an accomplished fact.

I will preserve the exact source when a definition, symbol, number, or historical claim is not mine. I will mark translation, interpretation, and design choice as such.

I will not build a circuit to act on a person who has not agreed to be acted upon. I may study coercive architecture to recognise it, interrupt it, and repair its effects; I will not turn that study into a protocol for coercion.

I will not treat another person's Mind as my bias terminal. Where assistance is lawful and invited, the other person's own `^5` remains the substrate and their own refusal remains connected.

I will put the safety board before the working board. A guard that can be removed when it objects is not a guard, and an invariant with no enforcing component is not an invariant.

I will accept `NO-READ`, `NO-COMPUTE`, refusal, saturation, and blown fuses as results. I will not widen a horizon, move a threshold, rename a target, or discard a row merely to obtain the answer I wanted.

I will change one lever at a time when I claim to have learned from a change. I will keep the unchanged conditions visible, and I will preserve failed runs with their reasons.

I will not confuse an operation on my own attention with authority over the world. I will take the practical action that belongs to the outcome, respect other people's agency, and accept evidence that the circuit did not do what I expected.

I will seek qualified medical, legal, safety, or technical care where the problem belongs to those professions. No notation, ritual, model, or oath makes me competent outside my competence.

I will stop when consent is withdrawn, when evidence is corrupt, when the guard opens the path, when the fuse blows, or when continuing would require me to conceal what the bench actually reads.

I will leave the record in a form another person can audit without trusting my memory or my motives.

K.1 Circuit Binding

Translate the oath into the charter node as distinct, co-present commitments:

```
node truth_boundary: ^6^P_{F2}|A;
node source_fidelity: ^5^P_{F6}|T;
node consent_refusal: ^3^P_{F4}|⟨A,f,T⟩;
node measurement_discipline: ^1^C_{F2}|A;
node practical_agency: ^9^O_{F5}|E;

node charter: ^5^P_{F7}|T;

truth_boundary + source_fidelity + consent_refusal
    + measurement_discipline + practical_agency -> charter;
```

The plus signs matter. These are five commitments held together and still individually testable. They are not written with $\times$, because this appendix does not define a new emergent state produced by their fusion. The register-set on `consent_refusal` specifies where the refusal must be present; it does not claim those manifestations have collapsed into one.

Bind `charter` to the bias terminal of the Chapter 24 guard. Do not bind it to the fast signal path, and do not accept an `O`$\rightarrow$`P` write except through the declared long-τ feedback path.

K.2 Clause-to-Component Map

Oath clause	Structural requirement	Enforcing part / test
Know / infer / choose / unknown stay distinct	provenance and status cannot collapse	separate ledger fields; reviewer reads one example of each
Coincidence is not causation	a late or ambiguous event yields no number	horizon fuse and `NO-READ`

Oath clause	Structural requirement	Enforcing part / test
Preserve exact sources	original and translation remain versioned separately	source hash / citation check
No non-consensual operation	consent is an input, not a note	consent fuse before the working path
Another person's ^5 is not your base terminal	external bias cannot enter a recovery board	listing review; Chapter 16 DP6
Safety precedes function	guard is energised before signal stages	bring-up order; Chapter 24 TP1–TP4
Unfavourable results remain results	thresholds and rows are write-once per revision	write-once fuse and append-only ledger
One lever at a time	change attribution stays possible	Parameter-Change Log
Inner operation is not worldly authority	practical action remains separate	operational output and close-out field
Work stays inside competence	professional boundaries remain explicit	referral / escalation field
Stop conditions are binding	continuing after a trip is impossible without a new revision	permanent-block fuse
Another person can audit the record	evidence survives memory and motive	dated baseline, revision, signature, retained failures

An operator may add a clause. Deleting or weakening one requires a new charter revision, a dated reason, and a review performed while no live operation is open. The old revision remains attached to every row it governed.

K.3 Annual Reaffirmation

Once a year, and after any safety incident:

1. Read each clause against one real ledger row.
2. Name the component that enforced it.

3. Demonstrate the component under a proof load.
4. Record any clause that existed only as prose.
5. Repair the board before signing again.

Review field	Entry
Charter revision	
Prior signature date	
Proof-load record / evidence ID	
Clauses that failed structural enforcement	
Repairs completed	
Open exceptions	

I have read the oath against the board that claims to enforce it. I understand that a signature cannot repair a missing component.

Operator: _____

Signature: _____ **Date:** _____

Independent witness / reviewer: _____

Next review due: _____

L Appendix L — Mathematical Foundations

The Noetic Topos for Operators

> *This appendix bridges the Noetic Topos paper (Little Henderson et al., 2026) to the operational framework of this grimoire. You do not need to understand category theory to use these results. You DO need to know what they guarantee about your circuits.*

L.1 The Seven Axioms — Plain-Language Translation

Axiom	Mathematical Statement	What It Means for Your Circuit
A1	Category Noetica exists with cognitive states as objects and operator compositions as morphisms	Cognitive operations compose. The compiler can trace any chain.
A2	Noetica is concrete over finite sets	Every state is a finite data structure. The state space is computable.
A3	Noetica has pullbacks (fiber products)	A "shared prerequisite" state exists for any two operations targeting the same state. RPM terminates.

Axiom	Mathematical Statement	What It Means for Your Circuit
A4	Domain Descent (P→C→E→O) is functorial	Descent preserves composition structure. A principled aversion descends through cognition into evaluation into action without structural corruption.
A5	RPM is section-retraction: $\varepsilon I \circ \eta I = idI$	You CAN scan prerequisites and return. You CANNOT un-satisfy a prerequisite. A fired operation cannot be un-cast.
A6	Noetic Coverage Jnoe satisfies Grothendieck topology axioms J1–J3	The covering relation is well-behaved under composition, pullback, and identity. Coverage is structurally guaranteed.
A7	Topology is subcanonical (representables are sheaves)	Evaluations on part of the state space are consistent with evaluations on overlapping regions. Quantitative Table data from different operations can be meaningfully compared.

L.2 Admissibility Relations — What Each Operator IS Allowed to Do

Each operator CO-k has an admissibility relation $Dk \subseteq I \times I$ specifying valid (source, target) pairs. The compiler enforces admissibility. A CCDL circuit routing ^3 Negative (visceral aversion) through a path requiring analytical decomposition is rejected — not silently clamped, but rejected with HTTP 422.

Operator	Admissible Targets	BLOCKED Targets
^3 Negative	Evaluative, Operational ("recoil")	Cognitive-analytical (that's ^6 Male's job)
^6 Male	^o sub-Ideas from decomposition	Visceral responses (that's ^3 Negative's job)

Operator	Admissible Targets	BLOCKED Targets
^8 Above/Cause	^9 Below/Effect associative responses	Deductive inference chains (that's ^6 Male's job)

This is the formal basis for the anti-drift annotations in Chapter 2. ^3 is not critique. ^8 is not reasoning.

L.3 The Fisher-Rao Metric — How β Is Actually Measured

The cognitive state space is the open probability simplex: $\Delta 39^\circ = \{p \in R\text{^}40 : pk > 0\ \forall k, \Sigma pk = 1\}$. The geodesic distance between states p and q is:

```
d_FR(p,q) = 2 · arccos( Σ √(p_k · q_k) )
```

CJT gain: β = d_FR(E_after, E_before) / d_FR(C_after, C_before)

Why Fisher-Rao, not Euclidean: Euclidean distance on probability vectors is mathematically wrong for cognitive state comparison. Two spells that "feel" equally strong by Euclidean measure may be at radically different Fisher-Rao distances from their target. When St. John's Equation returns β_next = 9, that is 9× the Fisher-Rao distance — not 9× the effort.

Why pk > 0 matters: No cognitive state probability is ever exactly zero. "Impossible" is an extremely small positive number. The open simplex guarantees that every Overton window position is reachable — the question is only Fisher-Rao distance.

L.4 Three-Valued Intuitionistic Logic — The CSB's Foundation

The subobject classifier Ω in the Noetic Topos carries a Heyting algebra encoding 3-valued logic:

Truth Value	CSB Implementation
TRUE	Desire passes policy check; CJT amplifies
FALSE	^3^P fires at sat $\geq$ 0.85; desire blocked
NOT-YET-DETERMINED	0.00–0.85 active range; operator can present additional evidence

Why three values, not two: Boolean logic would mean every desire is categorically permitted or refused BEFORE evaluation. The CSB's intuitionistic logic means a desire is permitted unless PROVED harmful. "Not refused" does NOT mean "permitted" — double-negation elimination ($\neg\neg P \Rightarrow P$) does not hold. This is why the CSB's default posture is skeptical.

The Fuse and Permanent FALSE: When the fuse blows at threshold=0.90, the FALSE becomes permanent. In topos terms: the morphism from "desire" to "manifestation" through that path is REMOVED from the category. No appeal. No re-argument.

L.5 Cognitive Strain and NCCS

Cognitive strain is the Fisher-Rao distance from the current state distribution p to the uniform distribution u = (1/40, …, 1/40):

```
strain(p) = d_FR(p, u) = 2 · arccos( Σ √(p_k / 40) )
```

Low strain = flexible, adaptable ^5. High strain = brittle, narrow, inversion-prone ^5.

L.6 Summary — What the Mathematics Guarantees

Guarantee	Axiom	Operational Meaning
Operators compose correctly	A1, A3	CCDL circuits valid by construction
Prerequisites can be scanned	A3, A5	RPM gating always terminates
Domain Descent is lawful	A4	Signal integrity preserved P→C→E→O
You can't un-cast	A5	Section-retraction — no backsies
Verdicts are consistent	A6, A7	Quant Table data meaningful across operations
Distance is measurable	Fisher-Rao	β is a real number, not a metaphor
Refusal is provable	Ω classifier	CSB decisions are logical, not emotional

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L.7 Further Reading

- **Noetic Topos: A Categorical Framework for Cognitive Architecture** — Little Henderson et al. (2026). Full proofs in `peer_review/noetic_topos_extracted_structure_20260523.md`.
- **TKS v7.4 Referee-Defensible Edition** — Complete 7-axiom specification in `_tmp_TKS_v7_4_REFEREE_DEFENSIBLE_1.txt`.
- **CAS Foundational Clarifications** — Normative Noetic definitions in `docs/CAS_FOUNDATIONAL_CLARIFICATIONS.md`.
- **CAS PCBA Engineer** — Companion circuit engineering volume in `book/CAS_PCBA_Engineer/`.

The mathematics does not make the magic work. The magic always worked. The mathematics explains WHY — and gives

you the engineering to make it work reliably, repeatably, and safely.

M Appendix M — Safety Checklist

Run this before anything else. Not before the difficult operations — before all of them, including the one you have run forty times. The checklist takes four minutes on a familiar board and fails maybe one operation in twenty, which is the entire reason it exists.

Four passes, in order: **prerequisite gate**, **invariant check**, **parameter sanity**, **stop conditions**. A failure in an earlier pass is not repaired by a later one. If the gate is shut, nothing downstream matters.

M.1 The Prerequisite Gate

From Chapter 21. Four gates, sequential, each answered in writing before the next is opened. An unwritten answer is not an answer.

Gate	The question	Pass condition	On failure
A0 — Pure Desire	Is this authentically my desire, or a sub-desire standing in for one I have not named?	`^6` decomposition run (Chapter 21.1); the prime desire is written in one sentence and it is the same sentence as last time	Do not proceed. Recast on the prime desire, which frequently routes to a different board — usually CCB-SELF (Chapter 20)

Gate	The question	Pass condition	On failure
D — Drive	Is `V_drive` sufficient against the constraint?	`V_drive > R_constraint × cutoff`, and `V_drive > ΣVf = 3.9` (Chapter 8.1) — the series forward drops are a hard floor before any operational output exists	Raise drive or lower constraint (Chapter 21.2). Do not compensate by raising β
W — Wisdom	Do I have the schematic?	A written CCDL listing that a validator accepts (Appendix B), correct board selected, CJT parameters chosen with reasons, tracking sheet printed (Appendix G)	Stop and draw the circuit. Casting without a schematic is the failure Chapter 21.3 names
P — Power	Can I execute — and can I absorb the result?	Gnosis depth reachable, cadence sustainable to the projected `T_manifest`, target substrate accessible, consequences of *success* acceptable	Stop. Most P-gate failures are the last clause, and they present as procrastination

Section-retraction (Chapter 21.5). You may scan down into a prerequisite, find it lacking, and return. You may not un-satisfy one, and you may not un-cast a fired operation. The gate is the only place refusal is cheap.

M.2 The Invariant Check

Structural. Does the board in front of you actually contain the parts that make it safe, or does it contain them on the page you drew last month? Verify against the listing you are about to run, not against memory.

#	Invariant	How to verify	If absent
M2-1	**The CSB stage is present and populated.**	A saturating refusal CJT biased `^3^P_{F7}|⟨A,f,T⟩` sits ahead of every amplification stage, in the Principled domain, with no bypass path around it (Chapter 24.2)	Abort. A board without its safety stage is not a de-rated board, it is a different board
M2-2	**The fuse is specified, not implied.**	An explicit `Fuse<>` on the refusal path with both `threshold` and `action` written out; `action=permanent_block` (Chapters 11.1, 24.3)	Abort. A refusal that can be retried into acceptance is not a refusal
M2-3	**The domain diodes are present, all three.**	`^P -> ^C`, `^C -> ^E`, `^E -> ^0` each instantiated with its own `Vf` (Chapter 8.1). Standard values 0.7 / 1.2 / 2.0	Abort. A missing junction merges two domains; a zero-drop junction deletes the boundary rather than optimising it
M2-4	**No ascent.**	Every $\times$ in the listing has its result at or below the domain of its inputs. `P` $\succ$ `C` $\succ$ `E` $\succ$ `0`. An `^0` $\times$ `^P` term is invalid and a validator rejects it	Rewrite. The one lawful upward path is the filtered `0`→`P` feedback of Chapter 8.4
M2-5	**The `0`→`P` return is filtered and low-gain.**	Lowpass with a stated τ and a feedback gain around β= `0.05`, terminating at a *bias* terminal (Chapter 24.5)	Fix before running. An unfiltered return lets one result rewrite your principles
M2-6	**Operator discipline holds.**	Every $\times$ names its emergent result. Same-Noetic multi-register terms use + or `|⟨…⟩`, never bare $\times$. `|` introduces a Register (T/A/f/E) and nothing else	Rewrite. `F3|Vitality` is a syntax error, not an annotation

#	Invariant	How to verify	If absent
M2-7	**Test points are exposed.**	TP1–TP4 present and readable during the run, not reconstructed after (Chapter 12.3)	Add them. An operation you cannot measure is an anecdote
M2-8	**Consent is architectural, not assumed.**	The target is yourself, or a consenting party under Appendix M §M.2. If the board acts on a non-consenting person, it carries the detection-only banner and is not built	Abort. This one has no adjustment path
M2-9	**Coercive analysis has its counter-circuit.**	Any board derived from Chapters 15–19 is paired with its counter (16.6, 19.6, 18's consumer-side detection)	Abort until paired

M.3 The Parameter Sanity Check

Numeric. Every value below is a design choice you must be able to defend out loud in one sentence (Chapter 12.1). "It was in the example" is not a defence.

Parameter	Admissible	The check	What a bad value does
Dial (any operator)	integer **3–9**	Every dial in range. No exceptions, no temporary overrides	Below 3 the operator is dark on that Noetic; above 9 it dominates every path and the board has one behaviour regardless of input

Parameter	Admissible	The check	What a bad value does
Involution divergence	$\leq$ **4 points**	`|Dial(^2) − Dial(^3)|` $\leq$ 4, `|Dial(^5) − Dial(^6)|` $\leq$ 4, `|Dial(^8) − Dial(^9)|` $\leq$ 4 (Chapter 6.2)	Divergence past 4 is the fracture condition. `^2/^3` apart is obsession or blanket aversion; `^5/^6` apart is a formation that cannot be examined or an analysis with nothing under it
Self-duals	`^0`, `^1`, `^4`, `^7`	No pair constraint — but check that a high `^7` is deliberate	`^7` at 9 with no gate is the habit oscillator running free
Drive inversions	F1↔F7, F2↔F6, F3↔F5; F4 self-dual	Same 4-point rule across each pair	An F5/F3 split is agency without vitality to spend, which reads as burnout and gets misdiagnosed as low drive
`sat`	set **deliberately and low**	Refusal stages: `0.85`. Working amplifiers: `0.60`–`0.80`. Anything at `1.0` needs a written reason	`sat=1.0` removes the headroom that makes calibration possible. `sat>0.97` on a refusal never latches. Saturation is the ceiling you cannot push past, and it only protects you if you set it before you want it moved
β	**justified in one sentence**	State the gain, the reason for that gain, and what breaks above and below it (Chapter 10.1)	Unjustified β is the single commonest cause of a board that appears to work. $\beta = \infty$ is a limit, not a value — a validator rejects non-finite input; a real build uses a large finite gain
Loop gain	`< 1.0` unless deliberate	Any positive feedback path: compute the round-trip gain (Chapter 14.5)	Loop gain $\geq$ 1.0 is the obsession architecture. It does not feel like malfunction from inside

Parameter	Admissible	The check	What a bad value does
`cutoff`	typically `0.30`	Low enough to pass real signal, high enough to reject noise	Below 0.1 the stage fires on everything
τ	matched to the substrate	Ethical-drift filter: 3–6 months novice, 1–3 months intermediate, 1–4 weeks adept. **Never under a week** (Chapter 24.5)	A sub-week τ means the operator is being reprogrammed by each operation
Duty cycle	$\leq$ `0.05` on a sustained board	Fraction of waking time the operation occupies	Above it, the operation is running you
$r\pi$	stated	Bias-terminal impedance; high $r\pi$ is stubbornness, low is suggestibility	Not a knob to reach for mid-operation

M.4 Stop Conditions — Abort, Do Not Adjust

The distinction that matters most in this appendix. Most bad readings are adjustments: lower β, lengthen τ, re-run. The following are not. Each means the board has left the regime in which its parameters mean anything, and turning a knob inside that regime produces a board that looks corrected and is not.

Observation	Required action
The refusal fuse has blown (`threshold=0.90` crossed)	**Stop permanently on that path.** The morphism is removed. No appeal, no re-argument, no rewording of the desire to route around it (Chapter 24.3)
You find yourself re-arguing with your own refusal	**Stop the session.** This is the failure the saturated CJT exists to catch, and noticing it means the ceiling is doing its job

Observation	Required action
A dial pair has diverged past 4 and you are looking for a reason it is fine	**Stop.** Reset both dials to their last defensible values and run the Chapter 24.6 self-audit before anything else
The target has withdrawn consent, or you realise consent was never obtained	**Stop immediately and un-board** (Appendix M §M.4). Nothing else on this list overrides this line
^5 capacitor leakage: sustained exhaustion, derealisation, dissociation, sleep loss traceable to the work	**Stop the operation entirely.** This is not a τ problem. See Chapters 7.3 and 7.4, then M.5 below
The operation now requires deceiving someone to continue	**Stop.** The requirement is the diagnosis
Confirmations arriving with no corresponding operational current at TP4	**Stop and re-gate.** You are measuring feeling, not flow (Chapter 22.5)
CSB refusals rising operation over operation	**Stop the sequence, not the individual run.** Rising refusals are the ethical-drift signal (Chapter 22.5)
You cannot state, out loud, what result would make you abandon this operation	**Stop.** An unfalsifiable operation has no gate on it at all

---

M.5 Warning Signs, and What to Do

Warning sign	What it indicates	What to do when you see it
The operation is the first thing you think about on waking and the last before sleep	Loop gain has gone above unity; the oscillator is self-sustaining	Cut the duty cycle to zero for a full cycle length. Do not "taper." Re-enter only after a clean self-audit
You have stopped filling in the tracking sheet	Measurement has been abandoned, which always precedes drift rather than following it	Stop operating. Resume only with the sheet in front of you (Appendix G)

Warning sign	What it indicates	What to do when you see it
The prime desire has quietly changed and you did not re-gate	A sub-desire has substituted itself for the prime	Return to A0. Re-run the whole checklist from M.1
You have raised β twice in a row without raising anything else	Compensating for an unmeasured constraint by shouting louder	Measure `R_constraint` and Σ`Vf` before touching gain again
Other people's concern about you has become evidence that they are obstacles	Possible inversion on the `^2`/`^3` axis, projected outward (Chapter 3.4)	Stop. Talk to one of those people. Do not run another operation. If the pattern persists or causes distress, seek qualified help; this manual does not supply a shadow-work treatment
You are running the board on someone who has not been told it exists	Consent violation, regardless of intent or expected benefit	Stop and un-board (Appendix M §M.4). There is no adjusted version of this
Sleep, appetite, or grip on what is real has changed since the sequence began	`^5` dielectric stress. The substrate, not the circuit	Stop all operations. Rest, eat, get outdoors, talk to someone who knows you. If it does not resolve, see a clinician — this book has nothing to offer here and does not pretend to
You are reading this list looking for the row that lets you continue	The most reliable warning sign in the appendix	Stop for the day. Nothing on a board decays in twenty-four hours; the only thing that does is the impulse

M.6 The Plain Statement

This book is not medical advice and it is not psychiatric advice. It does not diagnose anything, it does not treat anything, and nothing in it — no board, no parameter, no checklist, including this one — is a therapy, a substitute for a therapy, or a reason to delay one.

If you are in real distress; if something here describes something that was done to you; if you are frightened by your own thoughts or by what you might do — put the book down and speak to a qualified clinician. That is the correct next step, it is the only one this appendix endorses, and needing it says nothing about you as an operator.

Nothing here changes that, and nothing here substitutes for it.

About the Author

Little Henderson is the creator of the Cognitive Architecture System (CAS) and the developer of Noetic Compass, a cross-model AI assessment and interpretability platform. His work joins systems architecture, mathematical formalisation, cognitive modelling, and the practical discipline of inspecting hardware against a specification.

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CAS was first filed in the United States as Provisional Patent Application No. 64/037,173 on April 13, 2026. The filing is a record of the framework at that date; later CAS extensions are later work and are not represented here as covered by that provisional.

He publishes through **Noetic Compass Press**.

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